

A PROOF OF THE RIEMANN HYPOTHESIS

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ABSTRACT. We present a proof strategy for the Riemann Hypothesis based on local jet-normalized blocks of the phase kernel, a value/calibration functional extracted from the leading value-channel deformation, and an N -point odd-moment cancellation mechanism in the transverse channel. The document is written as a living proof draft: every statement is presented in proof form and the notation is frozen globally, so that future revisions can strengthen any remaining technical points without changing the architecture.

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1. INTRODUCTION

The Riemann Hypothesis, originating in Riemann's 1859 memoir [1], asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$ [2, 3, 4, 5]. The argument developed here is local and geometric. It compares two objects:

- (i) a local jet-whitened block extracted from the zeta/scattering phase on short symmetric windows, and
- (ii) a local off-line toy model corresponding to a split pair away from the critical line.

The original goal was to show that the toy anomaly has a transverse component that survives every local normalization, while the zeta-side transverse channel is annihilated at leading value order and becomes exponentially small after odd-moment cancellation. The present draft should now be read more cautiously: the local analysis has produced a detailed obstruction framework and several conditional rank-two visibility theorems, but the global proof of RH is not closed. The current remaining gap is a quantitative, source-level non-concentration statement for actual zeta data. In particular, the latest rank-two route requires an additional theorem preventing source-collapse on admissible reduced packet intervals; absent such an arithmetic input, the framework remains a strong local obstruction analysis rather than a completed proof of RH. For the final ledger of what is established locally and what remains open, see Remark 14.123.

Classically, RH sits at the intersection of the zero distribution of $\zeta(s)$, the explicit formula relating zeros and primes, and a number of equivalent or partially equivalent positivity, spectral, and functional-analytic reformulations [2, 3].

We introduce a normalized local jet-Gramian associated to the corrected local phase geometry. This object encodes the intrinsic local deformation after removal of the dominant value-channel and background effects, and it is the basic carrier of the odd quotient information used in the argument.

The proof has four main pillars.

1. *Jet normalization*: replace raw point samples by local value/derivative coordinates.
2. *Value/curvature splitting*: separate the leading background-subtracted value channel from the curvature and tail corrections.
3. *Canonical calibration*: extract the correct value scale from the actual leading zeta-side value matrix rather than from an ad hoc scalar functional.
4. *N-point transverse cancellation*: use an odd holomorphic expansion and explicit cancellation coefficients to crush the zeta transverse channel exponentially in the local height parameter.

Remark 1.1 (Audit status of the four pillars). The four pillars above remain part of the local language of the draft, but they should not be read as a complete proof gate. The later audits show that the decisive obstruction is no longer transverse odd-mode detection alone; it is the source-weighted interval non-concentration problem for actual zeta data. The local answer is partly positive and partly negative: the layered order-seven/order-nine analysis gives conditional fixed-sector visibility, but the same rows carry the vulnerable source factor $H = yq''$. Whether actual zeta arithmetic prevents source-flat intervals that are allowed by finite same-scale packet geometry and by fixed-prime/AFE-style surrogates is the remaining proof question; see Remark 14.123 for the terminal ledger.

This approach is not aimed at discovering new direct relations among primes. Rather, it studies a local, normalized geometric object attached to the zeta function and argues that an off-line zero would force a forbidden transverse odd mode in that object. In that sense, the method is primarily a zeta-side rigidity argument, with prime-distribution consequences entering only indirectly through the arithmetic content encoded in the zeta phase and, more broadly, through the classical zero–prime linkage expressed by the explicit formula [2, 3, 4, 5]. This local geometric viewpoint is not standard among the main classical families of approaches to RH surveyed in the literature [2, 3].

For orientation, it may be helpful to situate the present framework relative to earlier geometric viewpoints on RH. One such direction is the Speiser–Levinson–Montgomery line, in which the geometry of zeros of $\zeta'(s)$ is used to encode information about zeros of $\zeta(s)$ [6, 7]. Another is

the de Branges Hilbert-space program, in which one seeks global kernel/positivity structures attached to entire functions related to ζ [8, 9]. The present approach appears to differ in emphasis: rather than working directly with raw derivative geometry or with a global positivity condition, it constructs a locally normalized transverse invariant after quotienting out the dominant value/gauge directions. This comparison is intended only as heuristic orientation for the reader; no claim is made here that the present framework subsumes those earlier viewpoints. A Mellin-side interpretation of the N -point layer. A second point of orientation concerns the N -point cancellation operator introduced later in the proof. The cancellation coefficients may be viewed as defining a discrete multiplicative distribution whose Mellin transform preserves the constant mode while annihilating the first $N - 1$ odd jets. In this sense, the N -point step is not merely a finite-difference device: it may be read as a local Mellin-side projector acting on the corrected transverse germ. This is formally reminiscent of the Weil/Connes test-function viewpoint, in which one probes the zero set through Mellin/Fourier transforms of multiplicative test functions and explicit- or trace-formula pairings [3, 10, 11]. The present argument does *not* invoke any global trace formula, positivity principle, or spectral realization theorem from that literature; the comparison is included only as heuristic orientation. What is used here is a local, explicit, finite-support projector whose Mellin symbol can be written in closed form and whose odd part starts only at order $2N - 1$.

2. LOCAL KERNEL AND JET NORMALIZATION

2.1. Phase kernel. Let Φ be a real phase on the local spectral variable t , and define the phase kernel

$$(1) \quad K_{\Phi}(x, y) = \begin{cases} \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, & x \neq y, \\ \frac{\Phi'(x)}{\pi}, & x = y. \end{cases}$$

We write

$$q(t) := \Phi'(t).$$

Definition 2.1 (Canonical actual-zeta phase curvature). When the local phase is the actual zeta phase, we write

$$\Phi = \Phi_{\zeta}, \quad q_{\zeta}(t) := \Phi'_{\zeta}(t).$$

The canonical source-curvature object used by the rank-two Gram rows is

$$\boxed{q''_{\text{can}}(m) := q''_{\zeta}(m) = \Phi'''_{\zeta}(m)}.$$

Thus q''_{can} is the full actual-zeta phase-derivative curvature in the same-point Gram normalization. It is not whitened, projectivized, AFE-normalized, or affine-subtracted.

When no ambiguity is possible, later centered rank-two formulas may write $q, q'', q^{(4)}$ for

$$q_{\zeta}, \quad q''_{\text{can}}, \quad (q_{\zeta})^{(4)}.$$

If a source/core split is being discussed, the complement-source curvature is instead denoted

$$q''_{\text{source}, R}(m) := (G^{\text{new}})''(m),$$

and should not be identified with q''_{can} unless a separate rederived row formula explicitly says so.

For two nearby points around a center T , the natural local point basis is not stable under coalescence. One instead passes to the point-to-jet matrix

$$(2) \quad P_h := \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1/h & 1/h \end{pmatrix},$$

which turns two point samples into local value and derivative coordinates.

2.2. Jet-limit local blocks. Take two centers $T_1 \neq T_2$. In the point basis, let A_h denote the same-point 2×2 block around a single center and C_h the mixed block between the two centers. Passing to jet coordinates yields finite limits as $h \rightarrow 0$:

$$\tilde{A}_h = P_h A_h P_h^\top \rightarrow J(T), \quad \tilde{C}_h = P_h C_h P_h^\top \rightarrow J_{12}.$$

A direct Taylor expansion gives

$$(3) \quad J(T) = \frac{1}{\pi} \begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}(q''(T) + 2q(T)^3) \end{pmatrix}.$$

For two centers T_1, T_2 , writing

$$\Delta := \Phi(T_1) - \Phi(T_2), \quad s := T_1 - T_2, \quad q_j := q(T_j),$$

one obtains the explicit cross-jet matrix

$$(4) \quad N_{12} = \begin{pmatrix} \frac{2 \sin \Delta}{s} & \frac{\sin \Delta - q_2 s \cos \Delta}{s^2} \\ \frac{q_1 s \cos \Delta - \sin \Delta}{s^2} & \frac{(q_1 + q_2)s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Writing

$$J(T_j) = \frac{1}{\pi} M_j, \quad J_{12} = \frac{1}{\pi} N_{12},$$

the common π^{-1} factor cancels under whitening, and the jet-limit whitened cross block is

$$(5) \quad \Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}.$$

This is the canonical local matrix to compare with the toy anomaly.

3. FINITE- s LOCAL BLOCKS

3.1. Symmetric placement. Fix a midpoint m and a symmetric separation parameter s , and set

$$(6) \quad t_\pm = m \pm \frac{s}{2}.$$

At finite s , the same-point jet Gram blocks are naturally two matrices,

$$(7) \quad G_{m,\pm}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}(q''(t_\pm) + 2q(t_\pm)^3) \end{pmatrix}.$$

The mixed block is obtained from (4) by substituting $T_1 = t_-$, $T_2 = t_+$ and $s = t_- - t_+ = -s$:

$$(8) \quad N_m(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta_m(s)}{s} & \frac{\sin \Delta_m(s) + q_+(s) s \cos \Delta_m(s)}{s^2} \\ -\frac{\sin \Delta_m(s) + q_-(s) s \cos \Delta_m(s)}{s^2} & \frac{(q_-(s) + q_+(s))s \cos \Delta_m(s) + (2 - q_-(s)q_+(s)s^2) \sin \Delta_m(s)}{2s^3} \end{pmatrix}$$

where

$$q_\pm(s) := q(t_\pm), \quad \Delta_m(s) := \Phi(t_-) - \Phi(t_+).$$

The finite- s whitened local block is therefore

$$(9) \quad \hat{\Omega}_\zeta(s; m) = G_{m,-}(s)^{-1/2} N_m(s) G_{m,+}(s)^{-1/2},$$

followed, when appropriate, by background subtraction and phase-gap subtraction.

3.2. Removable singularity structure. The powers $1/s$, $1/s^2$, $1/s^3$ in (8) are removable at $s = 0$. Indeed, $\Delta_m(s)$ is odd and vanishes at 0, so

$$\Delta_m(s) = O(s), \quad \sin \Delta_m(s) = O(s),$$

which implies that the numerators in (8) vanish to order at least 1, 2, 2, 3 respectively. Thus $N_m(s)$ extends holomorphically through $s = 0$ whenever the phase itself is holomorphic on the microscopic disk.

4. ZETA-SIDE DECOMPOSITION

Write the zeta/scattering phase derivative as

$$(10) \quad q(t) = B_\zeta(t) + S(t),$$

where B_ζ is the explicit background contribution and S is the background-subtracted zero contribution. The local analysis is carried out after further splitting

$$(11) \quad q(t) = q_{\text{loc}}(t) + g_{\text{sm}}(t) + E_{\text{est}}(t), \quad g_{\text{sm}}(t) := B_{\text{aut}}(t) + T_{\text{far}}(t),$$

where q_{loc} is the designated local model, T_{far} is the omitted-zero tail, and E_{est} is the residual bookkeeping term.

Define the background-subtracted value scale

$$(12) \quad S(m) := q_\zeta(m) - B_\zeta(m),$$

and the logarithmic control function

$$(13) \quad L(m) := \log\left(3 + |m| + \frac{1}{S(m)}\right).$$

4.1. Curvature estimate. The fundamental zeta-side analytic estimate is the local curvature bound

$$(14) \quad |S''(m)| \ll L(m)S(m)^2.$$

It comes from a single-pair curvature estimate, zero-free-region control, and an optimized near/far decomposition.

5. SMOOTH REMAINDER AFTER AFFINE-JET SUBTRACTION

Fix a short interval $I = [a, b]$ of length $L = b - a$ and midpoint $t_* = (a + b)/2$. Let

$$J_1[g_{\text{sm}}](t) := g_{\text{sm}}(t_*) + g'_{\text{sm}}(t_*)(t - t_*), \quad r(t) := g_{\text{sm}}(t) - J_1[g_{\text{sm}}](t).$$

Assume

$$\|g''_{\text{sm}}\|_{L^\infty(I)} \leq S_2.$$

Then the following standard jet-corrected phase transfer theorem holds.

Theorem 5.1 (Jet-corrected phase transfer). *For every $t \in I$,*

$$|r'(t)| \leq \frac{L}{2}S_2, \quad |r(t)| \leq \frac{L^2}{8}S_2.$$

If Φ and Φ_{jet} are phases normalized by

$$\Phi'(t) = q(t), \quad \Phi'_{\text{jet}}(t) = q_{\text{loc}}(t) + J_1[g_{\text{sm}}](t) + E_{\text{est}}(t), \quad \Phi(t_*) = \Phi_{\text{jet}}(t_*),$$

then

$$\|(\Phi - \Phi_{\text{jet}})'\|_{L^\infty(I)} \leq \frac{L^2}{8}S_2, \quad \|\Phi - \Phi_{\text{jet}}\|_{L^\infty(I)} \leq \frac{L^3}{16}S_2.$$

Proof. Since $r(t_*) = r'(t_*) = 0$ and $r'' = g''_{\text{sm}}$, integrating twice gives the displayed bounds. Integrating once more after identifying $(\Phi - \Phi_{\text{jet}})' = r$ yields the phase estimate. $\square$

6. TAIL CURVATURE BOUND

Let the omitted-zero tail be

$$T_{\text{far}}(t) = \sum_{\rho \notin \mathcal{L}} f_{\beta_\rho, \gamma_\rho}(t),$$

with

$$f_{\beta, \gamma}(t) = \frac{1 - \beta}{(1 - \beta)^2 + (2t - \gamma)^2} + \frac{\beta}{\beta^2 + (2t - \gamma)^2}.$$

For each omitted zero define

$$\Delta_\rho := \text{dist}(2I, \gamma_\rho).$$

Theorem 6.1 (Tail curvature theorem). *One has*

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \frac{Q}{R^3},$$

where the omitted set is defined by $\Delta_\rho > R$. Consequently,

$$S_2 \leq \|B''_{\text{aut}}\|_{L^\infty(I)} + 48 \sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \|B''_{\text{aut}}\|_{L^\infty(I)} + \frac{Q}{R^3}.$$

In particular, if $R \rightarrow \infty$ with m , then $S_2 = o(Q)$.

Proof. Decompose the omitted zeros into shells

$$\mathcal{S}_k := \{\rho \notin \mathcal{L} : 2^k R < \Delta_\rho \leq 2^{k+1} R\}, \quad k \geq 0.$$

Then

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \leq \sum_{k \geq 0} \#\mathcal{S}_k (2^k R)^{-4}.$$

Standard zero counting near height m gives $\#\mathcal{S}_k \ll 2^k R Q$, hence

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \sum_{k \geq 0} (2^k R Q) (2^k R)^{-4} = Q R^{-3} \sum_{k \geq 0} 2^{-3k} \ll \frac{Q}{R^3}.$$

The bound for S_2 follows from the definition. $\square$

7. TOY ANOMALY AND TRANSVERSE OBSTRUCTION

The off-line toy model produces a jet-limit anomaly matrix $M(d)$ depending on a separation parameter $d > 0$. The leading toy deformation has the form

$$(15) \quad \Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

where $u = \beta - \frac{1}{2}$ is the off-line displacement.

Define the transverse functional

$$(16) \quad \Phi_K(X) := x_{12} + x_{21}, \quad X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}.$$

Then

$$(17) \quad \Phi_K(M(d)) = \frac{2(i(d^2 - 1) - d)}{(d + i)^4}.$$

Hence for every real $d > 0$,

$$\Phi_K(M(d)) \neq 0,$$

and on each compact interval $D \subset (0, \infty)$ there exists $c_K(D) > 0$ such that

$$|\Phi_K(M(d))| \geq c_K(D) \quad (d \in D).$$

8. LEADING VALUE MATRIX AND CANONICAL CALIBRATION

We now make the value-channel derivative A_{val} and its pairing with the toy anomaly matrix $M(d)$ fully explicit.

Work in the symmetric local regime

$$T_1 = m - \frac{\sigma}{2}, \quad T_2 = m + \frac{\sigma}{2},$$

and write

$$B := B_\zeta(m), \quad x := B \sigma.$$

When convenient, we write $A_{\text{val}}(x)$ for $A_{\text{val}}(m, \sigma)$ under the relation $x = B_\zeta(m)\sigma$.

Pure value-channel perturbation. Let the local value-channel parameter be perturbed by

$$q_1 = q_2 = B + \alpha,$$

while all derivative and curvature data are frozen at background level. In the constant-background symmetric approximation, the phase gap is

$$\Delta = \Delta_0 - \alpha\sigma,$$

with baseline value

$$\Delta_0 = -B\sigma = -x.$$

Recall that the jet-limit local block is

$$\Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}.$$

Define the pure value-channel derivative by

$$(18) \quad A_{\text{val}}(m, \sigma) := \left. \frac{\partial}{\partial \alpha} \Omega_\infty(T_1, T_2) \right|_{\alpha=0}.$$

Proposition 8.1 (Explicit formula for the value-channel derivative). *In the symmetric constant-background regime,*

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} \frac{x \cos x - \sin x}{x} & \frac{\sqrt{3}(-x^2 \sin x - 2x \cos x + 2 \sin x)}{x^2} \\ \frac{\sqrt{3}(x^2 \sin x + 2x \cos x - 2 \sin x)}{x^2} & \frac{3(x^3 \cos x - 6x \cos x + 3(2 - x^2) \sin x)}{x^3} \end{pmatrix}.$$

Proof. In the constant-background symmetric regime, the same-point jet Gram matrices are diagonal:

$$M_1 = M_2 = M = \begin{pmatrix} 2(B + \alpha) & 0 \\ 0 & (B + \alpha)^3/6 \end{pmatrix}.$$

Hence

$$M^{-1/2} = \begin{pmatrix} (2(B + \alpha))^{-1/2} & 0 \\ 0 & \sqrt{6} (B + \alpha)^{-3/2} \end{pmatrix}.$$

In the same regime, the jet-limit cross block depends only on $(B + \alpha)$, σ , and the phase gap $\Delta = \Delta_0 - \alpha\sigma$. Writing out the explicit jet-limit cross block in this specialization gives

$$N_{12}(\alpha) = \begin{pmatrix} -\frac{2 \sin \Delta}{\sigma} & \frac{\sqrt{3}(\sin \Delta + (B + \alpha)\sigma \cos \Delta)}{\sigma^2} \\ -\frac{\sqrt{3}(\sin \Delta + (B + \alpha)\sigma \cos \Delta)}{\sigma^2} & \frac{3(2 \sin \Delta - 2(B + \alpha)\sigma \cos \Delta - (B + \alpha)^2 \sigma^2 \sin \Delta)}{\sigma^3} \end{pmatrix},$$

up to the common normalization factor already absorbed into the definition of Ω_∞ .

Now differentiate

$$\Omega_\infty(\alpha) = M(\alpha)^{-1/2} N_{12}(\alpha) M(\alpha)^{-1/2}$$

at $\alpha = 0$, using

$$\left. \frac{d}{d\alpha} (B + \alpha)^{-1/2} \right|_{\alpha=0} = -\frac{1}{2} B^{-3/2}, \quad \left. \frac{d}{d\alpha} (B + \alpha)^{-3/2} \right|_{\alpha=0} = -\frac{3}{2} B^{-5/2},$$

and

$$\left. \frac{d}{d\alpha} \sin(\Delta_0 - \alpha\sigma) \right|_{\alpha=0} = -\sigma \cos \Delta_0, \quad \left. \frac{d}{d\alpha} \cos(\Delta_0 - \alpha\sigma) \right|_{\alpha=0} = \sigma \sin \Delta_0.$$

After substituting $\Delta_0 = -x$, $x = B\sigma$, using

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x,$$

and simplifying each entry, one obtains

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} \frac{x \cos x - \sin x}{x} & \frac{\sqrt{3}(-x^2 \sin x - 2x \cos x + 2 \sin x)}{x^2} \\ \frac{\sqrt{3}(x^2 \sin x + 2x \cos x - 2 \sin x)}{x^2} & \frac{3(x^3 \cos x - 6x \cos x + 3(2 - x^2) \sin x)}{x^3} \end{pmatrix}.$$

□

Corollary 8.2 (Transverse vanishing on the value channel). *One has*

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0.$$

Proof. By definition,

$$\Phi_K(X) = x_{12} + x_{21}.$$

From Proposition 8.1,

$$A_{\text{val},12}(x) = \frac{\sqrt{3}}{B} \frac{-x^2 \sin x - 2x \cos x + 2 \sin x}{x^2},$$

$$A_{\text{val},21}(x) = \frac{\sqrt{3}}{B} \frac{x^2 \sin x + 2x \cos x - 2 \sin x}{x^2}.$$

These are exact negatives of each other, hence

$$A_{\text{val},12}(x) + A_{\text{val},21}(x) = 0,$$

and therefore

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0.$$

□

Lemma 8.3 (Small- x expansion). *As $x \rightarrow 0$,*

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} -\frac{x^2}{3} + O(x^4) & -\frac{\sqrt{3}x}{3} + O(x^3) \\ \frac{\sqrt{3}x}{3} + O(x^3) & -\frac{3x^2}{5} + O(x^4) \end{pmatrix}.$$

Consequently,

$$\|A_{\text{val}}(x)\|_{\text{F}} \asymp \frac{x}{B} \quad (x \rightarrow 0).$$

Proof. Expand

$$\sin x = x - \frac{x^3}{6} + O(x^5), \quad \cos x = 1 - \frac{x^2}{2} + O(x^4),$$

and substitute into the entries of $A_{\text{val}}(x)$.

For the (1, 1)-entry:

$$\frac{x \cos x - \sin x}{x} = \frac{x(1 - \frac{x^2}{2} + O(x^4)) - (x - \frac{x^3}{6} + O(x^5))}{x} = -\frac{x^2}{3} + O(x^4).$$

For the (1, 2)-entry, expanding the numerator gives

$$-x^2 \sin x - 2x \cos x + 2 \sin x = -x^3 + O(x^5) - 2x(1 - \frac{x^2}{2} + O(x^4)) + 2(x - \frac{x^3}{6} + O(x^5)) = -\frac{x^3}{3} + O(x^5),$$

so

$$\frac{\sqrt{3}(-x^2 \sin x - 2x \cos x + 2 \sin x)}{x^2} = -\frac{\sqrt{3}x}{3} + O(x^3).$$

The (2, 1)-entry is the negative of the (1, 2)-entry, and the (2, 2)-entry similarly simplifies to

$$-\frac{3x^2}{5} + O(x^4).$$

Since the off-diagonal entries are linear in x and dominate the quadratic diagonal terms, the Frobenius norm satisfies

$$\|A_{\text{val}}(x)\|_{\text{F}} \asymp \frac{x}{B}.$$

□

Proposition 8.4 (Pairing with the toy anomaly). *Let $M(d)$ be the toy anomaly matrix. Then, for fixed $d > 0$,*

$$B \langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = x \frac{2\sqrt{3}d(id+3)}{3(d-i)^4} + x^2 \frac{d(-5d^2+46id+77)}{15(d-i)^5} + O(x^3).$$

In particular, for every compact interval

$$D = [d_-, d_+] \subset (0, \infty),$$

there exists $x_0(D) > 0$ such that for

$$0 < x \leq x_0(D), \quad d \in D,$$

one has

$$|\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}| \asymp \frac{x}{B}.$$

Proof. Using Lemma 8.3, write

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} -\frac{x^2}{3} + O(x^4) & -\frac{\sqrt{3}x}{3} + O(x^3) \\ \frac{\sqrt{3}x}{3} + O(x^3) & -\frac{3x^2}{5} + O(x^4) \end{pmatrix}.$$

Now take the Frobenius pairing with the explicit toy anomaly matrix $M(d)$:

$$\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = \sum_{j,k} A_{\text{val},jk}(x) \overline{M_{jk}(d)}.$$

The linear term in x comes entirely from the off-diagonal entries of $A_{\text{val}}(x)$, since the diagonal entries are quadratic. Using the explicit entries of $M(d)$, one finds

$$B \langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = x \frac{\sqrt{3}}{3} (\overline{M_{21}(d)} - \overline{M_{12}(d)}) + O(x^2).$$

Substituting the explicit formulas for $M_{12}(d)$ and $M_{21}(d)$ and simplifying yields

$$x \frac{2\sqrt{3}d(id+3)}{3(d-i)^4}.$$

Including the quadratic contributions from the diagonal entries and the next terms in the off-diagonal expansion gives

$$B \langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = x \frac{2\sqrt{3}d(id+3)}{3(d-i)^4} + x^2 \frac{d(-5d^2+46id+77)}{15(d-i)^5} + O(x^3).$$

The coefficient of x is nonzero for every real $d > 0$. Since it depends continuously on d , its modulus is bounded below on every compact interval $D \subset (0, \infty)$. Shrinking $x_0(D)$ if necessary gives

$$|\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}| \asymp \frac{x}{B}$$

uniformly for $0 < x \leq x_0(D)$ and $d \in D$. □

Proposition 8.5 (Canonical calibration theorem). *For fixed compact $D \subset (0, \infty)$ and $0 < x \leq x_0(D)$, define*

$$\Psi_{x,d}(X) := \frac{\langle A_{\text{val}}(x), X \rangle_{\text{F}}}{\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}}.$$

Then:

- (1) $\Psi_{x,d}(M(d)) = 1$;
- (2) $\Psi_{x,d}(A_{\text{val}}(x)) \asymp x/B$;
- (3) *if*

$$\Delta_{\zeta} = S(m)A_{\text{val}}(x) + R_{\zeta},$$

then

$$\Psi_{x,d}(\Delta_{\zeta}) = c_{x,d} \frac{x}{B} S(m) + \Psi_{x,d}(R_{\zeta}), \quad c_{x,d} \asymp 1;$$

(4) if

$$\Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

then

$$\Psi_{x,d}(\Delta_{\text{toy}}(u; d)) = u^2 + O(u^4).$$

In particular, whenever

$$\Psi_{x,d}(R_\zeta) = o\left(\frac{x}{B} S(m)\right),$$

one obtains

$$(19) \quad u^2 \asymp \frac{x}{B} S(m).$$

Proof. The first statement is immediate from the definition of $\Psi_{x,d}$. By Lemma 8.3 and Proposition 8.4,

$$\Psi_{x,d}(A_{\text{val}}(x)) = \frac{\|A_{\text{val}}(x)\|_{\text{F}}^2}{\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}} \asymp \frac{(x/B)^2}{x/B} = \frac{x}{B}.$$

The remaining statements follow by linearity. $\square$

Proposition 8.6 (Calibration remainder under a core-stable cutoff choice). *Assume the fixed-core decomposition of Section 9.1 and the core-stable cutoff compatibility of Proposition 9.7. If the cutoff R is chosen so that*

$$R^3 \gg \frac{1}{xQ S(m)},$$

then

$$S_2 = o(xQ^2 S(m)),$$

and consequently

$$\Psi_{x,d}(R_\zeta) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

Proof. By Theorem 6.1,

$$S_2 \ll |B''_{\text{aut}}(m)| + \frac{Q}{R^3}.$$

The background term is negligible, so under the cutoff condition

$$R^3 \gg \frac{1}{xQ S(m)},$$

one has

$$\frac{Q}{R^3} = o(xQ^2 S(m)).$$

Hence

$$S_2 = o(xQ^2 S(m)).$$

By Corollary 9.10, this implies

$$\Psi_{x,d}(R_\zeta) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

The use of such a cutoff does not alter the toy comparison because, by Proposition 9.7, the core $\mathcal{C}$ — and therefore $M(d)$ and A_{val} — is independent of R . $\square$

Remark 8.7 (Status of the calibration remainder estimate). Corollary 9.10 isolates the quantitative bound actually used later:

$$|\Psi_{x,d}(R_\zeta)| \ll_D \frac{S_2}{Q^3}.$$

Thus the calibration remainder step is no longer a deferred analytic-packaging item.

9. CORRECTED FINITE- s HOLOMORPHIC WHITENING

Let $\widehat{\Omega}_\zeta^{\text{corr}}(s; m)$ denote the corrected finite- s local block built from the explicit local model, the affine jet of the smooth remainder, and the jet-corrected phase remainder.

Proposition 9.1 (Microscopic denominator comparability and omitted-smooth holomorphy). *Fix an admissible midpoint m and write*

$$t_\pm = m \pm \frac{s}{2}, \quad Q \asymp \log |m|.$$

For each omitted zero $\rho = \beta_\rho + i\gamma_\rho$, set

$$D_{\rho,\pm}(s) := ((2m - \gamma_\rho) \pm s)^2 + a_\rho^2, \quad a_\rho \in \{\beta_\rho, 1 - \beta_\rho\}.$$

Assume the zero-free region gives

$$a_\rho \geq \frac{\sigma_0}{Q}$$

for some absolute $\sigma_0 > 0$ on the admissible midpoint class. If

$$|s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

then

$$\frac{1}{2} \left((2m - \gamma_\rho)^2 + a_\rho^2 \right) \leq |D_{\rho,\pm}(s)| \leq \frac{3}{2} \left((2m - \gamma_\rho)^2 + a_\rho^2 \right).$$

Consequently, for every derivative order $r \leq r_$ needed in the corrected finite- s block entries, the omitted-zero sums defining*

$$T_{\text{far}}(t_\pm), \quad \partial_t T_{\text{far}}(t_\pm), \quad \dots, \quad \partial_t^{r_*} T_{\text{far}}(t_\pm)$$

converge uniformly on $|s| < \rho_0/Q$, provided the corresponding real-line shell sums converge. In particular,

$$g_{\text{sm}}(t_\pm) = B_{\text{aut}}(t_\pm) + T_{\text{far}}(t_\pm)$$

and its required t -derivatives extend holomorphically to

$$|s| < \rho_0/Q.$$

Proof. Set

$$x := 2m - \gamma_\rho, \quad a := a_\rho.$$

Then

$$D_{\rho,\pm}(s) = (x^2 + a^2) \pm 2xs + s^2.$$

Using

$$|2xs + s^2| \leq 2|x||s| + |s|^2 \leq \frac{1}{2}x^2 + 3|s|^2,$$

and the assumptions

$$a \geq \frac{\sigma_0}{Q}, \quad |s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

we obtain

$$3|s|^2 \leq \frac{1}{2}a^2.$$

Hence

$$|2xs + s^2| \leq \frac{1}{2}(x^2 + a^2),$$

which gives the stated lower and upper bounds.

Now every differentiated omitted-zero term is a rational combination of powers of $D_{\rho,\pm}(s)^{-1}$ with coefficients depending only on the derivative order. By the above comparability, on $|s| < \rho_0/Q$ these are uniformly comparable to their real-line counterparts. Therefore the same shell decomposition used on the real line yields uniform convergence of the differentiated omitted-zero sums on the complex disk, and hence holomorphy of $T_{\text{far}}(t_\pm)$ and the required t -derivatives there. Adding the holomorphic background term $B_{\text{aut}}(t_\pm)$ gives the corresponding statement for $g_{\text{sm}}(t_\pm)$. $\square$

Proposition 9.2 (Entry-by-entry perturbation estimate for the corrected same-point blocks).
In the corrected finite- s normalization, write

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}, \quad t_{\pm} = m \pm \frac{s}{2}.$$

Let

$$q(t) = q_0(t) + \delta q(t),$$

where q_0 is the baseline local-model-plus-affine-jet part and δq is the corrected remainder. Define the baseline same-point blocks

$$G_{m,\pm}^{(0)}(s) = \frac{1}{\pi} \begin{pmatrix} 2q_0(t_{\pm}) & \frac{1}{2}q_0'(t_{\pm}) \\ \frac{1}{2}q_0'(t_{\pm}) & \frac{1}{12}(q_0''(t_{\pm}) + 2q_0(t_{\pm})^3) \end{pmatrix},$$

and set

$$R_{m,\pm}(s) := G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta q'(t_{\pm})| \ll L_I S_2, \quad |\delta q''(t_{\pm})| \ll S_2,$$

where L_I is the local window length, and that in the fixed-scaled regime

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_{\pm})| \asymp Q.$$

Then

$$\|R_{m,\pm}(s)\| \ll S_2$$

uniformly for $|s| < \rho_0/Q$.

Proof. We estimate the entries of $R_{m,\pm}(s)$ one by one.

(1,1)-entry. Since

$$R_{11} = \frac{1}{\pi} \cdot 2\delta q(t_{\pm}),$$

we have

$$|R_{11}| \ll |\delta q(t_{\pm})| \ll L_I^2 S_2 \ll \frac{S_2}{Q^2} \ll S_2.$$

(1,2)- and (2,1)-entries. Since

$$R_{12} = R_{21} = \frac{1}{\pi} \cdot \frac{1}{2} \delta q'(t_{\pm}),$$

we get

$$|R_{12}| = |R_{21}| \ll |\delta q'(t_{\pm})| \ll L_I S_2 \ll \frac{S_2}{Q} \ll S_2.$$

(2,2)-entry. We have

$$R_{22} = \frac{1}{12\pi} \left(\delta q''(t_{\pm}) + 2 \left((q_0 + \delta q)^3 - q_0^3 \right) (t_{\pm}) \right).$$

Expand the cubic difference:

$$(q_0 + \delta q)^3 - q_0^3 = 3q_0^2 \delta q + 3q_0 (\delta q)^2 + (\delta q)^3.$$

Hence

$$R_{22} = \frac{1}{12\pi} \left(\delta q'' + 6q_0^2 \delta q + 6q_0 (\delta q)^2 + 2(\delta q)^3 \right) (t_{\pm}).$$

Now estimate each term.

First,

$$|\delta q''(t_{\pm})| \ll S_2.$$

Second, using $|q_0(t_{\pm})| \asymp Q$ and $|\delta q(t_{\pm})| \ll L_I^2 S_2 \ll S_2/Q^2$,

$$|q_0(t_{\pm})^2 \delta q(t_{\pm})| \ll Q^2 \cdot \frac{S_2}{Q^2} = S_2.$$

Third,

$$|q_0(t_\pm)(\delta q(t_\pm))^2| \ll Q \left(\frac{S_2}{Q^2} \right)^2 = \frac{S_2^2}{Q^3}.$$

Since from the tail-curvature theorem one already has $S_2 = o(Q)$, for Q sufficiently large this is $O(S_2/Q^2)$ and hence $\ll S_2$.

Finally,

$$|(\delta q(t_\pm))^3| \ll \left(\frac{S_2}{Q^2} \right)^3 = \frac{S_2^3}{Q^6} \ll S_2.$$

Therefore

$$|R_{22}| \ll S_2.$$

Collecting the four entrywise bounds, we obtain

$$|R_{11}|, |R_{12}|, |R_{21}|, |R_{22}| \ll S_2.$$

Since all norms on 2×2 matrices are equivalent,

$$\|R_{m,\pm}(s)\| \ll S_2,$$

uniformly on the microscopic disk. This proves the proposition. $\square$

Proposition 9.3 (Entry-by-entry perturbation estimate for the corrected mixed block). *Let*

$$q(t) = q_0(t) + \delta q(t), \quad \Phi(t) = \Phi_0(t) + \delta \Phi(t),$$

where q_0 is the baseline local-model-plus-affine-jet phase derivative and δq is the corrected remainder. Write

$$t_\pm = m \pm \frac{s}{2}, \quad \Delta(s) := \Phi(t_-) - \Phi(t_+), \quad \Delta_0(s) := \Phi_0(t_-) - \Phi_0(t_+),$$

and set

$$\delta \Delta(s) := \Delta(s) - \Delta_0(s).$$

Let $N_m^{\text{corr}}(s)$ be the corrected finite- s mixed block built from q, Δ , and let $N_m^{(0)}(s)$ be the corresponding block built from q_0, Δ_0 . Define

$$\tilde{R}_m(s) := N_m^{\text{corr}}(s) - N_m^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_\pm)| \ll L_I^2 S_2, \quad |\delta q'(t_\pm)| \ll L_I S_2, \quad |\delta q''(t_\pm)| \ll S_2,$$

and

$$|\delta \Delta(s)| \ll L_I^3 S_2,$$

with

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_\pm)| \asymp Q, \quad |s| \asymp L_I.$$

Then

$$\|\tilde{R}_m(s)\| \ll S_2$$

uniformly for $|s| < \rho_0/Q$.

Proof. Write the corrected mixed block entrywise as

$$N_m^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q(t_+)s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q(t_-)s \cos \Delta}{s^2} & \frac{(q(t_-) + q(t_+))s \cos \Delta + (2 - q(t_-)q(t_+)s^2) \sin \Delta}{2s^3} \end{pmatrix},$$

and analogously for $N_m^{(0)}(s)$ with (q_0, Δ_0) .

We estimate the perturbation entry by entry. Throughout, use

$$\sin(\Delta_0 + \delta \Delta) - \sin \Delta_0 = O(|\delta \Delta|), \quad \cos(\Delta_0 + \delta \Delta) - \cos \Delta_0 = O(|\delta \Delta|).$$

(1,1)-entry.

$$\tilde{R}_{11} = -\frac{2}{\pi s}(\sin \Delta - \sin \Delta_0),$$

hence

$$|\tilde{R}_{11}| \ll \frac{|\delta \Delta|}{|s|} \ll \frac{L_I^3 S_2}{L_I} \ll L_I^2 S_2 \ll S_2.$$

(1,2)-entry. The numerator perturbation is

$$(\sin \Delta - \sin \Delta_0) + (q(t_+) - q_0(t_+))s \cos \Delta + q_0(t_+)s(\cos \Delta - \cos \Delta_0).$$

Therefore

$$|\tilde{R}_{12}| \ll \frac{|\delta \Delta| + |\delta q(t_+)| |s| + |q_0(t_+)| |s| |\delta \Delta|}{|s|^2}.$$

Using

$$|\delta \Delta| \ll L_I^3 S_2, \quad |\delta q(t_+)| \ll L_I^2 S_2, \quad |q_0(t_+)| \asymp Q, \quad |s| \asymp L_I \asymp Q^{-1},$$

we get

$$|\tilde{R}_{12}| \ll \frac{L_I^3 S_2}{L_I^2} + \frac{L_I^3 S_2}{L_I^2} + \frac{Q L_I^4 S_2}{L_I^2} \ll \frac{S_2}{Q} \ll S_2.$$

The same argument gives

$$|\tilde{R}_{21}| \ll S_2.$$

(2,2)-entry. Write the numerator as

$$F(q_-, q_+, \Delta; s) := (q_- + q_+)s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta.$$

Then

$$\tilde{R}_{22} = \frac{F(q(t_-), q(t_+), \Delta; s) - F(q_0(t_-), q_0(t_+), \Delta_0; s)}{2\pi s^3}.$$

A first-order expansion of F in (q_-, q_+, Δ) gives

$$|F - F_0| \ll |s| |\delta q(t_-)| + |s| |\delta q(t_+)| + |s|^2 Q |\delta q(t_-)| + |s|^2 Q |\delta q(t_+)| + (|s|Q + 1) |\delta \Delta|.$$

Using

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta \Delta| \ll L_I^3 S_2, \quad |s| \asymp L_I \asymp Q^{-1},$$

we obtain

$$|F - F_0| \ll L_I^3 S_2.$$

Hence

$$|\tilde{R}_{22}| \ll \frac{L_I^3 S_2}{|s|^3} \ll S_2.$$

Collecting the four entrywise bounds and using equivalence of norms on 2×2 matrices gives

$$\|\tilde{R}_m(s)\| \ll S_2,$$

uniformly on the microscopic disk. $\square$

Lemma 9.4 (Baseline same-point variation). *Let $G_{m,\pm}^{(0)}(s)$ be the corrected same-point baseline blocks built from the explicit local model together with the affine jet $J_1[g_{\text{sm}}]$. Then, on the microscopic disk $|s| < \rho_0/Q$,*

$$\sup_{|s| < \rho_0/Q} \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \ll \rho_0 Q.$$

Proof. Differentiate the explicit same-point block formula entrywise:

$$G_{m,\pm}^{(0)}(s) = \frac{1}{\pi} \begin{pmatrix} 2q_0(t_{\pm}) & \frac{1}{2}q_0'(t_{\pm}) \\ \frac{1}{2}q_0'(t_{\pm}) & \frac{1}{12}(q_0''(t_{\pm}) + 2q_0(t_{\pm})^3) \end{pmatrix}, \quad t_{\pm} = m \pm \frac{s}{2}.$$

By the chain rule,

$$\partial_s G_{m,\pm}^{(0)}(s) = \pm \frac{1}{2\pi} \begin{pmatrix} 2q_0'(t_{\pm}) & \frac{1}{2}q_0''(t_{\pm}) \\ \frac{1}{2}q_0''(t_{\pm}) & \frac{1}{12}(q_0'''(t_{\pm}) + 6q_0(t_{\pm})^2 q_0'(t_{\pm})) \end{pmatrix}.$$

For the explicit local model plus affine jet, one has the baseline local scales

$$|q_0(t_\pm)| \ll Q, \quad |q'_0(t_\pm)| \ll Q^2, \quad |q''_0(t_\pm)| \ll Q^3, \quad |q'''_0(t_\pm)| \ll Q^4,$$

uniformly on the admissible midpoint class. The cubic combination $q'''_0(t_\pm) + 6q_0(t_\pm)^2 q'_0(t_\pm)$ therefore has size $O(Q^2)$ in the current normalization, and hence

$$\sup_{|s| < \rho_0/Q} \|\partial_s G_{m,\pm}^{(0)}(s)\| \ll Q^2.$$

Integrating along the segment from 0 to s gives

$$\|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \leq |s| \sup_{|u| < \rho_0/Q} \|\partial_u G_{m,\pm}^{(0)}(u)\| \ll \frac{\rho_0}{Q} \cdot Q^2 = \rho_0 Q.$$

□

Proposition 9.5 (Corrected finite- s holomorphic whitening). *The corrected same-point and mixed blocks are holomorphic on*

$$|s| < \rho_0/Q.$$

Moreover, after shrinking ρ_0 if necessary, the same-point blocks

$$G_{m,\pm}^{\text{corr}}(s)$$

remain uniformly positive definite on this disk, and the inverse square roots

$$G_{m,\pm}^{\text{corr}}(s)^{-1/2}$$

are holomorphic there. Consequently

$$\widehat{\Omega}_\zeta^{\text{corr}}(s; m) = G_{m,-}^{\text{corr}}(s)^{-1/2} N_m^{\text{corr}}(s) G_{m,+}^{\text{corr}}(s)^{-1/2}$$

is holomorphic on $|s| < \rho_0/Q$.

Proof. Holomorphy of the omitted smooth part and its required t -derivatives follows from Proposition 9.1. The only apparent poles at $s = 0$ in the corrected mixed block come from powers of $1/s$, $1/s^2$, $1/s^3$, and these are removable because $\Delta_m(s)$ is odd in s , $\sin \Delta_m(s) = O(s)$, and the off-diagonal and $(2, 2)$ numerators vanish to orders 2 and 3, respectively. Thus both $G_{m,\pm}^{\text{corr}}(s)$ and $N_m^{\text{corr}}(s)$ are holomorphic on the disk.

By Proposition 9.2 and the tail-curvature theorem,

$$\sup_{|s| < \rho_0/Q} \|R_{m,\pm}(s)\| = o(Q).$$

By Lemma 9.4,

$$\sup_{|s| < \rho_0/Q} \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \ll \rho_0 Q.$$

At $s = 0$, the baseline jet-limit block has spectral gap

$$\lambda_{\min}(G_{m,\pm}^{(0)}(0)) \gg Q$$

on the admissible midpoint class. Therefore

$$\|G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(0)\| \leq \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| + \|R_{m,\pm}(s)\| \ll \rho_0 Q + o(Q).$$

For ρ_0 sufficiently small and m sufficiently large, this is smaller than half the baseline spectral gap, hence

$$\lambda_{\min}(G_{m,\pm}^{\text{corr}}(s)) \gg Q$$

uniformly on the disk. Holomorphic functional calculus then gives the holomorphic inverse square roots, and the whitened block is holomorphic as a product of holomorphic factors. □

9.1. Fixed anomaly core and cutoff-dependent auxiliary bookkeeping. To separate the anomaly-carrying local model from the cutoff-dependent tail bookkeeping, we fix once and for all a finite local core

$$\mathcal{C}$$

containing the candidate off-line anomaly (for example, the designated off-line pair together with any symmetry-forced companions needed for the local model).

For each cutoff $R \geq R_0$, we then decompose the zero contribution into:

- (1) the fixed core contribution $q_{\text{core}}(t)$, coming from the zeros in $\mathcal{C}$;
- (2) the auxiliary near contribution $q_{\text{aux},R}(t)$, coming from the zeros outside $\mathcal{C}$ but inside the near shell determined by R ;
- (3) the far contribution $T_{\text{far},R}(t)$, coming from the zeros outside the near shell;
- (4) the background contribution $B_{\text{bg}}(t)$;
- (5) the estimation term $E_{\text{est},R}(t)$.

Thus the full phase derivative is decomposed as

$$q(t) = q_{\text{core}}(t) + q_{\text{aux},R}(t) + g_{\text{sm},R}(t) + E_{\text{est},R}(t),$$

where

$$g_{\text{sm},R}(t) := B_{\text{bg}}(t) + T_{\text{far},R}(t).$$

The local toy comparison is attached only to the fixed core $\mathcal{C}$. The auxiliary near part $q_{\text{aux},R}$, the far part $T_{\text{far},R}$, and the estimation term $E_{\text{est},R}$ are absorbed into the corrected zeta-side scalar defined below.

Proposition 9.6 (Corrected local deformation decomposition with fixed core). *Fix the anomaly-carrying core $\mathcal{C}$ as above, and for each cutoff $R \geq R_0$ write*

$$q(t) = q_{\text{core}}(t) + q_{\text{aux},R}(t) + g_{\text{sm},R}(t) + E_{\text{est},R}(t), \quad g_{\text{sm},R}(t) = B_{\text{bg}}(t) + T_{\text{far},R}(t).$$

Let $J_1[g_{\text{sm},R}]$ denote the affine jet of $g_{\text{sm},R}$ at the midpoint, and let $\Phi_{\text{corr},R}$ be the corrected phase obtained from this decomposition.

Using $\Phi_{\text{corr},R}$, define the corrected same-point blocks

$$G_{m,\pm}^{\text{corr},R}(s),$$

the corrected mixed block

$$N_m^{\text{corr},R}(s),$$

and the corrected whitened block

$$\hat{\Omega}_\zeta^{\text{corr},R}(s; m) = G_{m,-}^{\text{corr},R}(s)^{-1/2} N_m^{\text{corr},R}(s) G_{m,+}^{\text{corr},R}(s)^{-1/2}.$$

Let

$$\hat{\Omega}_{\text{bg}}^{\text{corr},R}(s; m)$$

denote the corresponding corrected whitened block at background value-channel parameter $S(m) = 0$, and define the corrected local deformation

$$\Delta_\zeta(m, \sigma) := \hat{\Omega}_\zeta^{\text{corr},R}(s; m) - \hat{\Omega}_{\text{bg}}^{\text{corr},R}(s; m), \quad s = \sigma.$$

Then the corrected local deformation admits the decomposition

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

where $A_{\text{val}}(m, \sigma)$ is the pure value-channel derivative of the jet-limit block associated with the fixed core $\mathcal{C}$, and where

$$R_\zeta(m, \sigma) := \Delta_\zeta(m, \sigma) - S(m) A_{\text{val}}(m, \sigma).$$

Thus, by definition, $R_\zeta(m, \sigma)$ is exactly the non-leading part of the corrected local deformation after the explicit value-channel linear term is removed. In particular, $R_\zeta(m, \sigma)$ is internal to the same full corrected cutoff-dependent object

$$H_{m,R}(s) := \Phi_K(\hat{\Omega}_\zeta^{\text{corr},R}(s; m)),$$

so once the corrected odd holomorphic package is established for $H_{m,R}(s)$, no external tail/error term survives outside the resulting N -point scalar.

Proof. Consider the analytic whitening map

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

By the corrected finite- s holomorphic-whitening theorem, this map is holomorphic on the open set where $G_\pm$ remain uniformly positive definite.

For each cutoff R , the corrected blocks are built from the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}.$$

Thus every auxiliary near contribution from $q_{\text{aux},R}$, every omitted-zero tail contribution from $T_{\text{far},R}$, every affine-jet correction from $J_1[g_{\text{sm},R}]$, and every explicit estimation term $E_{\text{est},R}$ is already absorbed into the corrected phase $\Phi_{\text{corr},R}$, hence into the corrected same-point blocks, the corrected mixed block, the corrected whitened block, and therefore into the corrected transverse scalar

$$H_{m,R}(s) = \Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m)).$$

Now expand the corrected whitened block around the background value-channel parameter $S(m) = 0$. By analyticity of $\mathcal{W}$,

$$\widehat{\Omega}_\zeta^{\text{corr},R}(s; m) = \widehat{\Omega}_{\text{bg}}^{\text{corr},R}(s; m) + S(m) DW_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+] + R_\zeta(m, \sigma),$$

where the first-order coefficient

$$DW_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+]$$

is exactly the pure value-channel derivative $A_{\text{val}}(m, \sigma)$ associated with the fixed core $\mathcal{C}$. We therefore define

$$R_\zeta(m, \sigma) := \Delta_\zeta(m, \sigma) - S(m) A_{\text{val}}(m, \sigma),$$

so that $R_\zeta(m, \sigma)$ is, by definition, the full non-leading remainder inside the same corrected cutoff-dependent object.

Subtracting the background block gives

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

as claimed. $\square$

Proposition 9.7 (Core-stable cutoff compatibility). *Fix the anomaly-carrying core $\mathcal{C}$, and for each cutoff $R \geq R_0$ define the corrected scalar*

$$H_{m,R}(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m))$$

from the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}.$$

Then enlarging R changes only the auxiliary near part $q_{\text{aux},R}$ and the far part $T_{\text{far},R}$, while leaving the anomaly-carrying core q_{core} unchanged. Consequently:

- (1) *the toy anomaly matrix $M(d)$ and the calibration matrix $A_{\text{val}}(m, \sigma)$ depend only on the fixed core $\mathcal{C}$, not on the cutoff R ;*
- (2) *all R -dependent auxiliary, tail, jet-correction, and estimation contributions are internal to the full corrected scalar $H_{m,R}(s)$;*
- (3) *if the corrected odd holomorphic package is proved for $H_{m,R}(s)$, then the N -point odd-moment cancellation theorem applies directly to the full corrected scalar for that cutoff, with no external remainder term.*

In particular, the cutoff R may be chosen for quantitative convenience in the curvature/tail estimates without changing the fixed toy comparison attached to the core $\mathcal{C}$.

Proof. Let $R_1 < R_2$. Then the only change in the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}$$

is that some non-core zeros move from the far part into the auxiliary near part:

$$q_{\text{aux},R_2} - q_{\text{aux},R_1} = -(T_{\text{far},R_2} - T_{\text{far},R_1}).$$

The core contribution q_{core} is unchanged by construction.

Thus the toy anomaly matrix $M(d)$, which is attached only to the fixed core, is unchanged, and the value-channel derivative $A_{\text{val}}(m, \sigma)$ associated with that core is unchanged as well. All cutoff dependence is confined to the auxiliary near contribution, the far tail, the affine-jet correction built from $g_{\text{sm},R}$, and the explicit estimation term $E_{\text{est},R}$. These enter only through the corrected phase $\Phi_{\text{corr},R}$, hence only through the full corrected scalar $H_{m,R}(s)$.

Therefore the corrected odd holomorphic package, once established for $H_{m,R}(s)$, applies to the complete cutoff-dependent object. The resulting N -point scalar already contains all auxiliary/tail/error effects for that cutoff, and no external remainder survives. Since the fixed core is unchanged, the toy comparison is stable under enlarging R . This proves the claim. $\square$

Remark 9.8 (Cutoff choice). Proposition 9.7 is a structural statement. It does not assert any particular optimal choice of R ; it only says that once the core $\mathcal{C}$ is fixed, the cutoff may be adjusted in the perturbative bookkeeping without altering the underlying toy comparison.

Proposition 9.9 (Whitened mixed-block transfer with preserved Q^{-3} scale). *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Write the corrected finite- s blocks as

$$G_{\pm}^{\text{corr}}(s) = G_{\pm}^{(0)}(s) + R_{\pm}(s), \quad N^{\text{corr}}(s) = N^{(0)}(s) + \tilde{R}(s),$$

and set

$$\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) := \mathcal{W}(G_-^{\text{corr}}, N^{\text{corr}}, G_+^{\text{corr}}), \quad \widehat{\Omega}_{\zeta}^{(0)}(s; m) := \mathcal{W}(G_-^{(0)}, N^{(0)}, G_+^{(0)}).$$

Then

$$\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) - \widehat{\Omega}_{\zeta}^{(0)}(s; m) = \mathcal{T}_-(s; m) + \mathcal{T}_{\text{mix}}(s; m) + \mathcal{T}_+(s; m) + \mathcal{E}_2(s; m),$$

where

$$\begin{aligned} \mathcal{T}_-(s; m) &:= X_-(s) N^{(0)}(s) G_+^{(0)}(s)^{-1/2}, \\ \mathcal{T}_{\text{mix}}(s; m) &:= G_-^{(0)}(s)^{-1/2} \tilde{R}(s) G_+^{(0)}(s)^{-1/2}, \\ \mathcal{T}_+(s; m) &:= G_-^{(0)}(s)^{-1/2} N^{(0)}(s) X_+(s), \end{aligned}$$

with

$$X_{\pm}(s) := D(G^{-1/2})_{G_{\pm}^{(0)}(s)}[R_{\pm}(s)],$$

and $\mathcal{E}_2$ quadratic in $(R_-, \tilde{R}, R_+)$.

Moreover,

$$\Phi_K(\mathcal{T}_-(s; m)) \ll \frac{S_2}{Q^3}, \quad \Phi_K(\mathcal{T}_{\text{mix}}(s; m)) \ll \frac{S_2}{Q^3}, \quad \Phi_K(\mathcal{T}_+(s; m)) \ll \frac{S_2}{Q^3},$$

and

$$\Phi_K(\mathcal{E}_2(s; m)) \ll \frac{S_2^2}{Q^3}.$$

Hence

$$\boxed{\Phi_K(\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) - \widehat{\Omega}_{\zeta}^{(0)}(s; m)) \ll \frac{S_2}{Q^3}.}$$

Proof. We first collect the scales used below from results established earlier. The same-point perturbation bound $\|R_\pm(s)\| \ll S_2$ is Proposition 9.2; the mixed-block perturbation bound $\|\tilde{R}(s)\| \ll S_2$ is Proposition 9.3; and the smallness $S_2 = o(Q)$ is Theorem 6.1. The baseline entry scales

$$G_\pm^{(0)}(s) = \begin{pmatrix} O(Q) & O(1) \\ O(1) & O(Q^3) \end{pmatrix}, \quad G_\pm^{(0)}(s)^{-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix},$$

$$N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix}$$

are read off directly from the explicit same-point formula (7) and the mixed-block formula (8), given the baseline local scales of q_0, q'_0, q''_0 stated in the proof of Lemma 9.4. Uniform positive-definiteness of $G_\pm^{(0)}(s)$ on the microscopic disk is Proposition 9.5.

The whitening map admits the first-order expansion

$$\hat{\Omega}_\zeta^{\text{corr}} - \hat{\Omega}_\zeta^{(0)} = D\mathcal{W}_{(0)}[R_-, \tilde{R}, R_+] + \mathcal{E}_2,$$

where

$$D\mathcal{W}_{(0)}[R_-, \tilde{R}, R_+] = X_- N^{(0)} G_+^{(0)-1/2} + G_-^{(0)-1/2} \tilde{R} G_+^{(0)-1/2} + G_-^{(0)-1/2} N^{(0)} X_+.$$

We estimate the three first-order pieces separately.

Step 1: the mixed-block term. By Proposition 9.3 (entrywise form),

$$\tilde{R}_{11} \ll \frac{S_2}{Q^2}, \quad \tilde{R}_{12}, \tilde{R}_{21} \ll \frac{S_2}{Q}, \quad \tilde{R}_{22} \ll S_2.$$

Combining with the baseline inverse-square-root scales above, the off-diagonal entries of

$$\mathcal{T}_{\text{mix}} := G_-^{(0)-1/2} \tilde{R} G_+^{(0)-1/2}$$

are all $O(S_2/Q^3)$. Hence

$$\Phi_K(\mathcal{T}_{\text{mix}}) \ll \frac{S_2}{Q^3}.$$

Step 2: the same-point Fréchet derivative. We prove the corresponding estimate for the left same-point term

$$\mathcal{T}_- := X_- N^{(0)} G_+^{(0)-1/2}, \quad X_- := D(G^{-1/2})_{G_-^{(0)}}[R_-].$$

The proof for $\mathcal{T}_+$ is identical.

Write one baseline same-point block in the form

$$G_0 = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a \asymp Q, \quad b = O(1), \quad c \asymp Q^3,$$

and one perturbation as

$$R = \begin{pmatrix} r_{11} & r_{12} \\ r_{12} & r_{22} \end{pmatrix}, \quad r_{11} \ll \frac{S_2}{Q^2}, \quad r_{12} \ll \frac{S_2}{Q}, \quad r_{22} \ll S_2.$$

Diagonalize

$$G_0 = U \Lambda U^\top, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2),$$

with U orthogonal. Since

$$a \asymp Q, \quad c \asymp Q^3, \quad b = O(1),$$

we have

$$\lambda_1 \asymp Q, \quad \lambda_2 \asymp Q^3.$$

Moreover, if U is a rotation by angle θ , then

$$\tan(2\theta) = \frac{2b}{c-a} = O(Q^{-3}),$$

hence

$$\theta = O(Q^{-3}), \quad U = I + O(Q^{-3}).$$

Set

$$H := U^\top R U.$$

A direct computation gives

$$\begin{aligned} H_{11} &= r_{11} + 2\theta r_{12} + O(\theta^2 r_{22}), \\ H_{12} &= r_{12} + \theta(r_{22} - r_{11}) + O(\theta^2 r_{12}), \\ H_{22} &= r_{22} - 2\theta r_{12} + O(\theta^2 r_{11}). \end{aligned}$$

Using

$$\theta = O(Q^{-3}), \quad r_{11} \ll \frac{S_2}{Q^2}, \quad r_{12} \ll \frac{S_2}{Q}, \quad r_{22} \ll S_2,$$

we obtain

$$H_{11} \ll \frac{S_2}{Q^2}, \quad H_{12} \ll \frac{S_2}{Q}, \quad H_{22} \ll S_2.$$

Now let $f(x) = x^{-1/2}$. For a diagonal matrix Λ , the Fréchet derivative is given entrywise by the Löwner formula

$$Df_\Lambda(H)_{ij} = f^{[1]}(\lambda_i, \lambda_j) H_{ij},$$

where

$$f^{[1]}(u, v) = \begin{cases} \frac{u^{-1/2} - v^{-1/2}}{u - v}, & u \neq v, \\ -\frac{1}{2u^{3/2}}, & u = v. \end{cases}$$

Since

$$\lambda_1 \asymp Q, \quad \lambda_2 \asymp Q^3,$$

we have

$$f^{[1]}(\lambda_1, \lambda_1) \asymp Q^{-3/2}, \quad f^{[1]}(\lambda_2, \lambda_2) \asymp Q^{-9/2}, \quad f^{[1]}(\lambda_1, \lambda_2) \asymp Q^{-7/2}.$$

Therefore, if

$$Y := Df_\Lambda(H),$$

then

$$\begin{aligned} Y_{11} &\ll Q^{-3/2} \cdot \frac{S_2}{Q^2} = \frac{S_2}{Q^{7/2}}, \\ Y_{12} &\ll Q^{-7/2} \cdot \frac{S_2}{Q} = \frac{S_2}{Q^{9/2}}, \\ Y_{22} &\ll Q^{-9/2} \cdot S_2 = \frac{S_2}{Q^{9/2}}. \end{aligned}$$

Since

$$X = U Y U^\top$$

and $U = I + O(Q^{-3})$, these scales persist in the original basis:

$$X_{11} \ll \frac{S_2}{Q^{7/2}}, \quad X_{12}, X_{21} \ll \frac{S_2}{Q^{9/2}}, \quad X_{22} \ll \frac{S_2}{Q^{9/2}}.$$

Step 3: the left same-point term $\mathcal{T}_-$. The baseline mixed-block scales are

$$N^{(0)} = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix}, \quad G_+^{(0)-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix}.$$

We estimate only the off-diagonal entries, since

$$\Phi_K(Y) = Y_{12} + Y_{21}.$$

For $(\mathcal{T}_-)_{12}$, every summand is $O(S_2/Q^3)$. For example,

$$X_{11} N_{12} (G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{7/2}} \cdot Q^2 \cdot Q^{-3/2} = \frac{S_2}{Q^3},$$

and

$$X_{12} N_{22} (G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{9/2}} \cdot Q^3 \cdot Q^{-3/2} = \frac{S_2}{Q^3}.$$

All other terms are smaller. Thus

$$(\mathcal{T}_-)_{12} \ll \frac{S_2}{Q^3}.$$

The same argument gives

$$(\mathcal{T}_-)_{21} \ll \frac{S_2}{Q^3},$$

and hence

$$\Phi_K(\mathcal{T}_-) \ll \frac{S_2}{Q^3}.$$

By symmetry, the same estimate holds for the right same-point term:

$$\Phi_K(\mathcal{T}_+) \ll \frac{S_2}{Q^3}.$$

Step 4: the quadratic remainder. Every term in $\mathcal{E}_2$ contains at least two perturbation factors from $(R_-, \tilde{R}, R_+)$, and whitening contributes only polynomial Q -weights. Therefore

$$\Phi_K(\mathcal{E}_2) \ll \frac{S_2^2}{Q^3}.$$

By Theorem 6.1, $S_2 = o(Q)$, so this is lower order than S_2/Q^3 .

Combining the estimates for $\mathcal{T}_-$, $\mathcal{T}_{\text{mix}}$, $\mathcal{T}_+$, and $\mathcal{E}_2$ gives

$$\Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m) - \hat{\Omega}_\zeta^{(0)}(s; m)) \ll \frac{S_2}{Q^3},$$

as claimed. □

Corollary 9.10 (Sharpened calibration remainder estimate). *Fix a compact interval*

$$D \subset (0, \infty),$$

and work in the fixed-scaled regime

$$0 < x \leq x_0(D), \quad d \in D, \quad x = B_\zeta(m)\sigma, \quad |\sigma| \asymp Q^{-1}.$$

Write the corrected local deformation in the form

$$\Delta_\zeta(m, \sigma) = S(m)A_{\text{val}}(x) + R_\zeta(m, \sigma).$$

Then

$$|\langle A_{\text{val}}(x), R_\zeta(m, \sigma) \rangle_F| \ll_D \frac{x}{B_\zeta(m)} \frac{S_2}{Q^3},$$

and hence

$$\boxed{|\Psi_{x,d}(R_\zeta(m, \sigma))| \ll_D \frac{S_2}{Q^3}.$$

In particular, if

$$S_2 = o(xQ^2S(m)),$$

then

$$\Psi_{x,d}(R_\zeta(m, \sigma)) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

Proof. By Proposition 9.6, after subtraction of the explicit leading value-channel term $S(m)A_{\text{val}}(x)$, the remaining contribution $R_\zeta(m, \sigma)$ is internal to the same corrected transferred package controlled by the whitening perturbation analysis.

We therefore use the entry scales already established in the proof of Proposition 9.9. First, from Proposition 9.3 and the baseline inverse-square-root scales,

$$G_\pm^{(0)}(s)^{-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix}, \quad \tilde{R}(s) = \begin{pmatrix} O(S_2/Q^2) & O(S_2/Q) \\ O(S_2/Q) & O(S_2) \end{pmatrix},$$

one gets

$$\mathcal{T}_{\text{mix}}(s; m) = G_-^{(0)}(s)^{-1/2} \tilde{R}(s) G_+^{(0)}(s)^{-1/2} = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}.$$

Next, from the same proof,

$$X_{\pm}(s) = \begin{pmatrix} O(S_2/Q^{7/2}) & O(S_2/Q^{9/2}) \\ O(S_2/Q^{9/2}) & O(S_2/Q^{9/2}) \end{pmatrix}, \quad N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix},$$

and multiplying once more by $G_{\pm}^{(0)}(s)^{-1/2}$ gives

$$\mathcal{T}_{-}(s; m) = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}, \quad \mathcal{T}_{+}(s; m) = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}.$$

Finally, every entry of the quadratic remainder $\mathcal{E}_2(s; m)$ is

$$O(S_2^2/Q^3),$$

which is lower order because $S_2 = o(Q)$ by Theorem 6.1.

Thus the part of the corrected deformation seen after removal of the leading value-channel term satisfies

$$R_{\zeta}(m, \sigma) = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}$$

up to lower-order $O(S_2^2/Q^3)$ terms.

Now use the explicit small- x structure of the value-channel derivative: by Lemma 8.3,

$$A_{\text{val}}(x) = \frac{1}{B_{\zeta}(m)} \begin{pmatrix} O(x^2) & O(x) \\ O(x) & O(x^2) \end{pmatrix}.$$

Therefore

$$|\langle A_{\text{val}}(x), R_{\zeta}(m, \sigma) \rangle_{\text{F}}| \ll \frac{x}{B_{\zeta}(m)} \frac{S_2}{Q^3}.$$

By Proposition 8.4,

$$|\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}| \asymp_D \frac{x}{B_{\zeta}(m)},$$

so dividing gives

$$|\Psi_{x,d}(R_{\zeta}(m, \sigma))| \ll_D \frac{S_2}{Q^3}.$$

Finally, since $B_{\zeta}(m) \asymp Q$,

$$\frac{S_2/Q^3}{(x/B_{\zeta}(m))S(m)} \asymp \frac{S_2}{xQ^2S(m)} \rightarrow 0$$

under the hypothesis $S_2 = o(xQ^2S(m))$. This proves the final claim. $\square$

10. ODD HOLOMORPHIC TRANSVERSE CHANNEL

Define the corrected transverse scalar

$$(20) \quad H_m(s) := \Phi_K(\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m)).$$

By symmetry under $s \mapsto -s$, this is odd. By Corollary 8.2, the leading value channel is annihilated by Φ_K , so only the curvature, tail, and corrected finite- s remainder channels contribute.

Proposition 10.1 (Boundary estimate). *On the microscopic boundary*

$$|s| = \rho_0/Q,$$

one has

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3}.$$

Proof. By Proposition 9.6, the corrected local deformation may be written in the form

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

where $R_\zeta(m, \sigma)$ is internal to the full corrected cutoff-dependent scalar

$$H_{m,R}(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m)).$$

By Proposition 9.7, all cutoff-dependent tail, jet-correction, and estimation contributions are already absorbed into this full corrected scalar, so no external remainder term is carried separately at the boundary stage.

Applying Φ_K to the corrected local decomposition and using Corollary 8.2,

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0,$$

gives

$$H_{m,R}(s) = \Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m)) = \Phi_K(R_\zeta(m, \sigma)).$$

Now compare the full corrected whitened block with its baseline counterpart. By Proposition 9.9,

$$\Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m) - \widehat{\Omega}_\zeta^{(0),R}(s; m)) \ll \frac{S_2}{Q^3} \quad (|s| < \rho_0/Q).$$

On the microscopic boundary $|s| = \rho_0/Q$, the tail-curvature theorem and the corrected transfer bounds imply

$$S_2 \ll L(m)S(m)^2$$

in the corrected regime. Therefore

$$|H_{m,R}(s)| \ll \frac{L(m)S(m)^2}{Q^3} \quad (|s| = \rho_0/Q).$$

Since the cutoff bookkeeping has already been absorbed into the full corrected scalar $H_{m,R}(s)$, this is the boundary estimate for the complete corrected object, not merely for a leading part plus an external tail/error remainder. This proves the claim. $\square$

Proposition 10.2 (Odd microscopic expansion). *The function $H_m(s)$ is odd and holomorphic on $|s| < \rho_0/Q$, and therefore admits an expansion*

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}}, \quad |z| < \rho_0 Q,$$

with

$$|c_{2r+1}(m)| \ll_{\rho_0} L(m)S(m)^2 \rho_0^{-(2r+1)}.$$

Proof. Oddness follows from symmetric placement and the jet-basis swap symmetry:

$$\widehat{\Omega}_\zeta^{\text{corr}}(-s; m) = J_0 \widehat{\Omega}_\zeta^{\text{corr}}(s; m) J_0,$$

with

$$\Phi_K(J_0 X J_0) = -\Phi_K(X).$$

Holomorphy follows from Proposition 9.5, and the coefficient bounds are immediate from Cauchy's estimate applied to Proposition 10.1. $\square$

10.1. Coefficient-level corrected-whitening transfer and exterior odd-jet localization.

Lemma 10.3 (Baseline-scaled matrix gain for corrected whitening). *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Assume the baseline same-point and mixed blocks satisfy

$$G_\pm^{(0)}(s) = \begin{pmatrix} O(Q) & O(1) \\ O(1) & O(Q^3) \end{pmatrix}, \quad N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix},$$

uniformly on $|s| < \rho_0/Q$, and that $G_\pm^{(0)}(s)$ remain uniformly positive definite there.

Set

$$S_Q := \begin{pmatrix} Q^{-1/2} & 0 \\ 0 & Q^{-3/2} \end{pmatrix}, \quad \widehat{G}_\pm := S_Q^{-1} G_\pm S_Q^{-1}, \quad \widehat{N} := S_Q^{-1} N S_Q^{-1}.$$

Then

$$\widehat{G}_\pm^{(0)}(s) = \begin{pmatrix} O(1) & O(Q^{-1}) \\ O(Q^{-1}) & O(1) \end{pmatrix}, \quad \widehat{N}^{(0)}(s) = \begin{pmatrix} O(1) & O(1) \\ O(1) & O(1) \end{pmatrix},$$

and

$$\mathcal{W}(G_-, N, G_+) = S_Q \mathcal{W}(\widehat{G}_-, \widehat{N}, \widehat{G}_+) S_Q.$$

In particular, every off-diagonal entry of $\mathcal{W}(G_-, N, G_+)$ carries an explicit matrix-level factor Q^{-2} relative to the corresponding rescaled whitened block. Any additional Q -power appearing in the corrected transverse scalar must therefore come from the scalar normalization used in the definition of that scalar, not from the matrix identity itself.

Proof. The displayed bounds for $\widehat{G}_\pm^{(0)}$ and $\widehat{N}^{(0)}$ are immediate from the definitions and the baseline scales. Since

$$G_\pm = S_Q \widehat{G}_\pm S_Q,$$

functional calculus gives

$$G_\pm^{-1/2} = S_Q^{-1} \widehat{G}_\pm^{-1/2} S_Q^{-1}.$$

Hence

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2} = S_Q \widehat{G}_-^{-1/2} \widehat{N} \widehat{G}_+^{-1/2} S_Q = S_Q \mathcal{W}(\widehat{G}_-, \widehat{N}, \widehat{G}_+) S_Q.$$

The off-diagonal entries therefore acquire one factor $Q^{-1/2}$ from the upper-left entry of S_Q and one factor $Q^{-3/2}$ from the lower-right entry of S_Q , hence a total matrix-level factor Q^{-2} . $\square$

Proposition 10.4 (Uniform scaled corrected-whitening transfer). *Fix $0 < c_* < \rho_0/8$. For every Q and every radius R satisfying*

$$0 < R \leq c_* Q,$$

set

$$w := \frac{z}{R}, \quad s = \frac{z}{Q^2} = \frac{Rw}{Q^2}.$$

Let

$$\mathfrak{D}_Q(M) := \begin{pmatrix} Q^2 m_{11} & Q m_{12} \\ Q m_{21} & m_{22} \end{pmatrix}, \quad \mathfrak{D}_Q^{-1}(X) = \begin{pmatrix} Q^{-2} x_{11} & Q^{-1} x_{12} \\ Q^{-1} x_{21} & x_{22} \end{pmatrix}.$$

For $R' > 0$, let

$$A_{R'}(M_2) := \left\{ F(z) = \sum_{n \geq 0} F_n z^n : \|F\|_{A_{R'}} := \sum_{n \geq 0} \|F_n\| R'^n < \infty \right\},$$

and write $A_1(M_2)$ for the case $R' = 1$. For scalar holomorphic germs, $A_{R'}(\mathbb{C})$ is defined similarly.

Write the baseline corrected same-point and mixed blocks on the w -disk as

$$G_{\pm;Q,R}^{(0)}(w) := G_\pm^{(0)}(Rw/Q^2), \quad N_{Q,R}^{(0)}(w) := N^{(0)}(Rw/Q^2).$$

For a weighted perturbation triple

$$X = (X_-, Y, X_+) \in A_1(M_2)^3,$$

define

$$\delta G_\pm := \mathfrak{D}_Q^{-1} X_\pm, \quad \delta N := \mathfrak{D}_Q^{-1} Y,$$

and, whenever the corrected scalar perturbation

$$\Phi_K \left(\mathcal{W}(G_{-;Q,R}^{(0)} + \delta G_-, N_{Q,R}^{(0)} + \delta N, G_{+;Q,R}^{(0)} + \delta G_+) - \mathcal{W}(G_{-;Q,R}^{(0)}, N_{Q,R}^{(0)}, G_{+;Q,R}^{(0)}) \right)$$

vanishes at $w = 0$, define

$$\mathcal{T}_{Q,R}(X)(w) := \frac{Q}{Rw} \Phi_K \left(\mathcal{W}(G_{-;Q,R}^{(0)} + \delta G_-, N_{Q,R}^{(0)} + \delta N, G_{+;Q,R}^{(0)} + \delta G_+) - \mathcal{W}(G_{-;Q,R}^{(0)}, N_{Q,R}^{(0)}, G_{+;Q,R}^{(0)}) \right).$$

Then there exist absolute constants

$$\eta > 0, \quad C > 0, \quad B > 0$$

such that for every Q , every $0 < R \leq c_* Q$, and every

$$X \in A_1(M_2)^3 \quad \text{with} \quad \|X_-\|_{A_1} + \|Y\|_{A_1} + \|X_+\|_{A_1} \leq \eta,$$

the quotient $\mathcal{T}_{Q,R}(X)$ is holomorphic on $|w| < 1$ and admits an absolutely convergent homogeneous expansion

$$\mathcal{T}_{Q,R}(X) = \sum_{p \geq 1} \mathcal{T}_p(X),$$

with

$$\|\mathcal{T}_p(X)\|_{A_1} \leq CB^p (\|X_-\|_{A_1} + \|Y\|_{A_1} + \|X_+\|_{A_1})^p.$$

The constants C, B, η are independent of Q and R .

Moreover, if

$$X_{\pm}(w) = \sum_{m \geq 0} X_{\pm,m} w^m, \quad Y(w) = \sum_{m \geq 0} Y_m w^m,$$

and

$$\mathcal{T}_p(X)(w) = \sum_{n \geq 0} T_{p,n} w^n,$$

then with

$$\mathbf{X}_m := \max(\|X_{-,m}\|, \|Y_m\|, \|X_{+,m}\|),$$

one has the coefficient bound

$$|T_{p,n}| \leq CB^p \sum_{\substack{M \leq n \\ m_1 + \dots + m_p = M \\ m_i \geq 0}} \sum_{m_i \geq 0} \prod_{i=1}^p \mathbf{X}_{m_i}.$$

Proof. On $|w| \leq 2$,

$$\left| \frac{Rw}{Q^2} \right| \leq \frac{2R}{Q^2} \leq \frac{2c_*}{Q} < \frac{\rho_0}{Q},$$

so the baseline blocks

$$G_{\pm;Q,R}^{(0)}(w), \quad N_{Q,R}^{(0)}(w)$$

are holomorphic and uniformly bounded on $|w| \leq 2$, with bounds independent of Q and R . By Lemma 10.3, the same-point baseline blocks remain uniformly positive definite there.

Therefore the corrected-whitening map

$$(\delta G_-, \delta N, \delta G_+) \mapsto \mathcal{W}(G_{-;Q,R}^{(0)} + \delta G_-, N_{Q,R}^{(0)} + \delta N, G_{+;Q,R}^{(0)} + \delta G_+)$$

is analytic on a fixed Banach neighborhood in $A_1(M_2)^3$, uniformly in Q, R . Since Φ_K is linear, the corrected scalar perturbation depends analytically on the input triple. For those structured perturbations for which the corrected scalar perturbation vanishes at $w = 0$, the quotient by w in the definition of $\mathcal{T}_{Q,R}$ is holomorphic. This gives the homogeneous expansion with uniform constants C, B, η .

The coefficient bound follows by testing the homogeneous maps on monomials in $A_1(M_2)$ and using the Wiener-algebra estimate

$$|[w^n]F| \leq \|F\|_{A_1}$$

for scalar coefficients. □

Proposition 10.5 (Pair-radius kernel-class decomposition for one-pair perturbations). *Fix $0 < c_* < \rho_0/8$ and $0 < c \ll 1$. For one exterior zero pair ρ , set*

$$\Delta_\rho := |2m - \gamma_\rho|, \quad R_\rho := c \min(\Delta_\rho, c_* Q).$$

Let $f_\rho(t)$ denote the one-pair phase-derivative contribution, and remove its midpoint affine jet:

$$r_\rho(t) := f_\rho(t) - f_\rho(m) - f'_\rho(m)(t - m).$$

With

$$t_{\pm} := m \pm \frac{z}{2Q^2},$$

define

$$d_{\pm}(z) := r_{\rho}(t_{\pm}), \quad e_{\pm}(z) := r'_{\rho}(t_{\pm}), \quad g_{\pm}(z) := r''_{\rho}(t_{\pm}),$$

and

$$\eta(z) := \delta\Delta_{\rho}(z/Q^2) := \int_{t_+}^{t_-} r_{\rho}(u) du.$$

Then there exist holomorphic kernels $U_{\pm}, E_{\pm}, G_{\pm}, V$ such that

$$d_{\pm}(z) = \frac{z^2}{Q^4} U_{\pm}(z), \quad e_{\pm}(z) = \frac{z}{Q^2} E_{\pm}(z), \quad g_{\pm}(z) = G_{\pm}(z), \quad \eta(z) = \frac{z^3}{Q^6} V(z).$$

For every fixed $H_* > 1$, if $\Delta_{\rho} \geq H_*$, then

$$U_{\pm}, E_{\pm}, G_{\pm}, V$$

are holomorphic on $|z| < 2R_{\rho}$ and satisfy

$$\|U_{\pm}\|_{A_{R_{\rho}}} + \|E_{\pm}\|_{A_{R_{\rho}}} + \|G_{\pm}\|_{A_{R_{\rho}}} + \|V\|_{A_{R_{\rho}}} \ll \Delta_{\rho}^{-4}.$$

Let

$$X_{\rho} := (X_{\rho,-}, Y_{\rho}, X_{\rho,+})$$

be the weighted one-pair perturbation triple

$$X_{\rho,\pm} := \mathfrak{D}_Q(\delta G_{\rho,\pm}), \quad Y_{\rho} := \mathfrak{D}_Q(\delta N_{\rho}),$$

where $\delta G_{\rho,\pm}$ and δN_{ρ} are the actual one-pair perturbations of the corrected same-point and mixed blocks in the draft.

Then, on $|z| < 2R_{\rho}$, every entry of every block of $X_{\rho}(z)$ is linear in

$$U_{-}, U_{+}, E_{-}, E_{+}, G_{-}, G_{+}, V$$

modulo terms containing at least two such pair kernels. In particular,

$$\|X_{\rho}(R_{\rho})\|_{A_1} \ll \Delta_{\rho}^{-4}.$$

Proof. The one-pair poles lie at distance $\asymp \Delta_{\rho}$ in the s -plane, hence at distance $\asymp Q^2 \Delta_{\rho}$ in the z -plane. Since

$$R_{\rho} \ll \min(\Delta_{\rho}, Q),$$

the kernels $U_{\pm}, E_{\pm}, G_{\pm}, V$ are holomorphic on $|z| < 2R_{\rho}$, and Cauchy gives the displayed $A_{R_{\rho}}$ -bounds.

Now check the actual corrected formulas.

For the corrected same-point block,

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}.$$

Subtracting the baseline and using

$$q = q_0 + d_{\pm}, \quad q' = q'_0 + e_{\pm}, \quad q'' = q''_0 + g_{\pm},$$

the weighted perturbation entries are:

$$(1, 1) \sim (z/Q)^2 U_{\pm}, \quad (1, 2), (2, 1) \sim (z/Q) E_{\pm},$$

and

$$(2, 2) \sim G_{\pm} + \left(\frac{q_0}{Q}\right)^2 (z/Q)^2 U_{\pm} + \left(\frac{q_0}{Q}\right) (z/Q)^4 U_{\pm}^2 + (z/Q)^6 U_{\pm}^3.$$

On $|z| \leq 2R_{\rho}$, the normalized baseline factors

$$\frac{q_0 z}{Q^2}, \quad \frac{q_0^2 z^2}{Q^4}$$

are $O(1)$, so the same-point block is linear in the pair kernels modulo terms with at least two pair kernels.

For the corrected mixed block,

$$N_m^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q(t_+)s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q(t_-)s \cos \Delta}{s^2} & \frac{(q(t_-) + q(t_+))s \cos \Delta + (2 - q(t_-)q(t_+)s^2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Write

$$q_{\pm} = q_{0,\pm} + d_{\pm}, \quad \Delta = \Delta_0 + \eta.$$

For the $(1, 1)$ -entry, since

$$\eta(z) = \frac{z^3}{Q^6} V(z) = s^3 V(z),$$

$$Q^2 \frac{\sin(\Delta_0 + \eta) - \sin \Delta_0}{s}$$

is linear in V modulo higher pair order.

For the off-diagonal entries, the numerator difference is

$$(\sin(\Delta_0 + \eta) - \sin \Delta_0) + d_{\pm} s \cos(\Delta_0 + \eta) + q_{0,\pm} s (\cos(\Delta_0 + \eta) - \cos \Delta_0),$$

and after multiplying by Q/s^2 the linear terms are bounded-baseline times $U_{\pm}$ and V .

For the $(2, 2)$ -entry, define

$$F(q_-, q_+, \Delta; s) := (q_- + q_+)s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta.$$

Then the perturbation is

$$\frac{F(q_{0,-} + d_-, q_{0,+} + d_+, \Delta_0 + \eta; s) - F(q_{0,-}, q_{0,+}, \Delta_0; s)}{s^3}.$$

Taylor expansion in (d_-, d_+, η) around the baseline gives

$$\partial_{q_-} F_0 d_- + \partial_{q_+} F_0 d_+ + \partial_{\Delta} F_0 \eta$$

plus a remainder quadratic in (d_-, d_+, η) . After dividing by s^3 , the linear part is

$$\tilde{B}_-(z) U_-(z) + \tilde{B}_+(z) U_+(z) + \tilde{B}_{\Delta}(z) V(z),$$

with bounded baseline coefficients $\tilde{B}_-, \tilde{B}_+, \tilde{B}_{\Delta}$, and the remainder carries at least two pair kernels.

So every weighted mixed-block entry is also linear in the pair kernels modulo higher pair order.

Finally, on $|z| \leq 2R_{\rho}$, every factor z/Q is $O(1)$, every normalized baseline factor is $O(1)$, and each pair kernel has $A_{R_{\rho}}$ norm $O(\Delta_{\rho}^{-4})$. Therefore

$$\|X_{\rho}(R_{\rho} \cdot)\|_{A_1} \ll \Delta_{\rho}^{-4}.$$

□

Definition 10.6 (Tagged two-atom pre-whitening source block). Introduce formal source tags τ_1, τ_2 . For a normalized two-atom core $(a_1, h_1; a_2, h_2)$, define the tagged corrected source perturbations by additive substitution into the corrected phase variables:

$$d_{\pm}^{(2)} := \tau_1 a_1 d_{h_1, \pm} + \tau_2 a_2 d_{h_2, \pm}, \quad e_{\pm}^{(2)} := \tau_1 a_1 e_{h_1, \pm} + \tau_2 a_2 e_{h_2, \pm},$$

$$g_{\pm}^{(2)} := \tau_1 a_1 g_{h_1, \pm} + \tau_2 a_2 g_{h_2, \pm}, \quad \eta^{(2)} := \tau_1 a_1 \eta_{h_1} + \tau_2 a_2 \eta_{h_2}.$$

Equivalently,

$$q_{2,\pm} = q_{0,\pm} + d_{\pm}^{(2)}, \quad q'_{2,\pm} = q'_{0,\pm} + e_{\pm}^{(2)}, \quad q''_{2,\pm} = q''_{0,\pm} + g_{\pm}^{(2)}, \quad \Delta_2 = \Delta_0 + \eta^{(2)}.$$

Define

$$\mathcal{B}_2(a_1, h_1; a_2, h_2) := (G_{2,-}^{\text{corr}}, N_2^{\text{corr}}, G_{2,+}^{\text{corr}})$$

by the same corrected same-point and mixed block formulas used above for the one-pair corrected block, with $q_{\pm}, \Delta$ replaced by $q_{2,\pm}, \Delta_2$. Set

$$\mathcal{B}_0 := (G_-^{(0)}, N^{(0)}, G_+^{(0)}).$$

Definition 10.7 (Pair-kernel linear part). Let $\mathcal{K}$ be the filtration on entries of $\mathfrak{D}_Q(\mathcal{B}_2 - \mathcal{B}_0)$ in which each tagged source kernel

$$\tau_i U_{i,\pm}, \quad \tau_i E_{i,\pm}, \quad \tau_i G_{i,\pm}, \quad \tau_i V_i$$

has $\mathcal{K}$ -degree 1, while baseline factors, the amplitudes a_i , Q -scales, and bounded analytic coefficients have degree 0. Let $\text{Lin}_{\mathcal{K}}$ denote projection to $\mathcal{K}$ -degree 1.

Define the one-atom source-linear triple by

$$L_h := \text{Lin}_{\mathcal{K}} \mathfrak{D}_Q(\mathcal{B}_1(h) - \mathcal{B}_0),$$

where $\mathcal{B}_1(h)$ is the one-atom corrected block obtained from the same corrected block formulas.

Lemma 10.8 (Tagged pair-kernel linearity of the source block). *With source tags retained,*

$$\text{Lin}_{\mathcal{K}} \mathfrak{D}_Q(\mathcal{B}_2(a_1, h_1; a_2, h_2) - \mathcal{B}_0) = \tau_1 a_1 L_{h_1} + \tau_2 a_2 L_{h_2}.$$

After this tagged $\mathcal{K}$ -linear projection is taken, applying the augmentation $\tau_1, \tau_2 \mapsto 1$ gives

$$\text{Lin}_{\mathcal{K}} \mathfrak{D}_Q(\mathcal{B}_2 - \mathcal{B}_0) = a_1 L_{h_1} + a_2 L_{h_2}.$$

Proof. For the same-point blocks $G_{2,\pm}^{\text{corr}}$, the $(1, 1)$ and off-diagonal entries are linear in $d_{\pm}^{(2)}$ and $e_{\pm}^{(2)}$. The $(2, 2)$ entry has the form

$$g_{\pm}^{(2)} + \text{baseline} \cdot d_{\pm}^{(2)} + \text{terms of } \mathcal{K}\text{-degree at least 2},$$

by the same expansion as in the one-pair corrected block calculation.

For the mixed block N_2^{corr} , Taylor expansion in $(d_{-}^{(2)}, d_{+}^{(2)}, \eta^{(2)})$ gives a bounded-baseline linear combination of those variables and a remainder of $\mathcal{K}$ -degree at least 2. Since the tagged variables are additive sums of the two source contributions, the $\mathcal{K}$ -linear part is the sum of the two one-atom $\mathcal{K}$ -linear parts, with tags still attached. $\square$

Remark 10.9 (Scope of the tagged source block). The definitions and lemma above are pre-whitening statements. They prove tagged $\mathcal{K}$ -linear additivity of the corrected source block before the nonlinear corrected-whitening transfer

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

They do not by themselves prove transfer-domain vanishing, source-bidegree edge divisibility, quotient-output package coincidence, diagonal merger, the coefficient $B_7^{\mathfrak{f}}$, the quotient $Q_{7,m}^q$, or the projector $\Pi_{1,1}$.

In particular, the tagged pre-whitening block does not automatically induce the homogeneous scalar-grade source tower used in Corollary B.21. That identification is a separate source-model theorem for the actual $\mathfrak{D}_Q$ -scaled corrected-whitening package.

Remark 10.10 (Corrected package map for the two-point and mixed four-point frontiers). The current local frontier is best viewed as a single corrected package problem, not as two independent mixed-geometry problems. The two-point mixed branch and the mixed four-point branch both require the same source-provenance package:

$$\mathcal{B}_2 \longrightarrow C_{\mathcal{W}} \longrightarrow B_7^{\mathfrak{f}} \longrightarrow Q_{7,m}^q \longrightarrow \mathcal{H}_7^q.$$

Here $\mathcal{B}_2$ is the tagged two-atom pre-whitening source block of Definition 10.6, $C_{\mathcal{W}}$ is the corrected-whitening matrix cross-effect before Φ_K of Definition C.4, $B_7^{\mathfrak{f}}$ is the pre-determinant fixed-sector order-7 coefficient of Definition C.9, and

$$Q_{7,m}^q(C) := [B_7^{\mathfrak{f}}(C)] \in \mathfrak{f}/\text{CA}_5^{\mathfrak{f}}(m)$$

is the fixed midpoint quotient class. The normalized corrected package representative $C_{\mathcal{W}}^{\text{corr}}$ and its moving fixed-sector representative $B_{\text{pkg}}^{\mathfrak{f}}$ are defined in Section C.

For the two-point branch, the remaining source input is finite: prove swap symmetry, one-amplitude collapse, and diagonal merger for the actual corrected two-atom package. Once these hold, the quadratic collision factorization

$$\mathcal{I}_2 = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2$$

is formal. The role of the order-7 quotient package is to make the corresponding septic quotient defect an actual fixed-target object rather than an uncontrolled two-atom mixed representative.

For the mixed four-point branch, the current reduction says that a genuinely mixed clean branch cannot persist independently in the region

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}.$$

Thus every clean mixed branch must exit through one of the scalar walls

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

The package theorem is needed to prevent the corrected source cross-effect from creating a new non-scalar fixed-sector direction in the quotient

$$\mathfrak{f}/\mathbf{CA}_5^\mathfrak{f}(m).$$

Consequently, the local target is not a scalarized identity after Φ_K . It is the pre- Φ_K , pre-determinant quotient-gauge statement

$$B_7^\mathfrak{f}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{CA}_5^\mathfrak{f}(m).$$

If one keeps a full corrected package before source-bidegree projection, the equivalent full-package formulation is obtained by applying $(1 - \Pi_{1,1})_{\text{mix}}$ before fixed-sector extraction and quotienting. This implication does not by itself prove that the scalar exits

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0$$

are impossible. Its role is narrower: it removes the possibility that non- $(1,1)$ corrected source terms create an independent fixed-sector quotient direction. After that removal, the existing mixed four-point capture theorem reduces every clean mixed branch to the scalar exits, and the two-point residual is pushed into the midpoint quotient edge package.

If this package theorem is proved, the residual two-point mixed obstruction is absorbed into the midpoint quotient edge package, and the mixed four-point source-coupled walls have no independent clean local carrier beyond the scalar exits above.

Remark 10.11 (Conditional role of the corrected package in the mixed branches). The corrected package theorem does not by itself prove RH. Its role is to remove the residual local mixed carrier. If the package theorem closes, then the two-point mixed obstruction reduces to the fixed midpoint quotient edge package, and the mixed four-point branch has no independent clean local carrier outside the scalar exits

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

The remaining global proof still requires the finite-core lower-model transfer or a separate treatment of the near-zero-anomaly branch $\mathcal{N}_k(m, Q)$, hidden-state/ T -horizontal odd-block extraction, and the downstream endgame reductions, including the source-level interaction branch controlled by $J_1 = [s]\Phi_K(\mathcal{I}_{\geq 2})$.

Remark 10.12 (Sensitivity of the mixed four-point branch to Gate B). The mixed four-point branch is sensitive only to the fixed-sector quotient class of a Gate B residual. Let

$$E = eI + fS \in \mathfrak{f}_h.$$

If

$$E \in \mathbf{CA}_5^\mathfrak{f}(h),$$

then E is invisible in the quotient package. On a good patch, this is equivalently

$$ev_5 - u_5f = 0, \quad A_5^\mathfrak{f}(h) = u_5I + v_5S.$$

If instead

$$ev_5 - u_5f \neq 0,$$

then E shifts the order-7 quotient scalar:

$$\Delta_7 \longmapsto \Delta_7 + (ev_5 - u_5f).$$

Consequently $\lambda = \Delta_7/v_5$ and the downstream scalar package $\mathcal{Z}, \mathcal{R}, T_v$ generally change. Thus a non-gauge Gate B residual would create precisely the independent fixed-sector quotient carrier that the corrected package theorem is designed to remove.

The mixed four-point scalar-exit theorem should therefore be read after Gate B has removed non-(1,1) quotient directions. It is not a substitute for Gate B.

Theorem 10.13 (Mixed four-point dependence descends to the quotient determinant). *Work on a good fixed-sector patch on which*

$$c(h) \neq 0, \quad A_5^f(h) = u_5(h)I + v_5(h)S \neq 0.$$

Let

$$E = eI + fS \in \mathfrak{f}_h$$

be an order-7 fixed-sector residual added to

$$A_7^f = u_7I + v_7S,$$

with c, u_5, v_5 fixed. Then the mixed four-point scalar-exit package sees E only through the quotient determinant pairing

$$\omega_h(E, A_5^f(h)) = ev_5 - u_5f.$$

Equivalently,

$$\Delta_7 = u_7v_5 - u_5v_7$$

is transformed by

$$\Delta_7 \mapsto \Delta_7 + \omega_h(E, A_5^f(h)),$$

and every lifted v_5 -patch scalar

$$Q_v, \quad T_v, \quad \mathcal{Z}, \quad \mathcal{R}$$

changes only through this induced change of Δ_7 .

In particular, if

$$E \in \mathbf{CA}_5^f(h),$$

then E is invisible to the mixed four-point geometry. If

$$[E] \neq 0 \quad \text{in} \quad \mathfrak{f}_h/\mathbf{CA}_5^f(h),$$

then the resulting mixed four-point effect is exactly the already-defined quotient-septic obstruction, expressed through Δ_7 . No independent mixed four-point carrier is produced.

Proof. The lifted mixed four-point branch is built, on a v_5 -patch, from

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5},$$

and hence from

$$Q_v = (Y, x, S).$$

The scalar exits $T_v, \mathcal{Z}, \mathcal{R}$ are functions of these data and their x -derivatives. Thus the only possible order-7 input from A_7^f is through Δ_7 .

Replacing A_7^f by $A_7^f + E$, with $E = eI + fS$, changes

$$\Delta_7 = \omega_h(A_7^f, A_5^f)$$

by

$$\omega_h(E, A_5^f) = ev_5 - u_5f.$$

The same argument on a u_5 -patch uses the parallel quotient coordinate

$$-\frac{\Delta_7}{c u_5}.$$

Hence no raw representative data (u_7, v_7) or $\Phi_K(A_7^f)$ enter as independent mixed four-point inputs. $\square$

Corollary 10.14 (Connection residuals are quotient determinant failures). *For $M \in \{M_A, M_C\}$, set*

$$q_{\text{tot},7}[M] := R_7^{(M)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

If

$$q_{\text{tot},7}[M_A] = q_{\text{tot},7}[M_C] = 0,$$

then the residual monomials M_A, M_C create no mixed four-point carrier beyond the scalar exits already present in the lifted branch.

If one of these two scalars is nonzero, the obstruction is exactly a failure of quotient determinant closure:

$$\Delta_7 \mapsto \Delta_7 + q_{\text{tot},7}[M].$$

It is not a separate four-point mechanism.

Remark 10.15 (Three data layers in the mixed four-point audit). The mixed four-point carrier audit separates three logically different layers:

Layer	Data	Role
pre- Φ_K quotient package	$[B_7^f] \in \mathfrak{f}_m / \mathbf{C}A_5^f(m)$	Gate B / UV-026 obstruction ob
Φ_K -visible hidden state	$A_7^f \mapsto A_7^f + \chi A_5^f, \quad \Phi_K(\chi A_5^f) = 2\chi v_5$	visible to transform extraction, quotient
mixed four-point scalar exits	$Q_v, T_v, \mathcal{Z}, \mathcal{R}$ built from c, u_5, v_5, Δ_7	downstream four-point branch geo

Thus quotient invisibility is enough to remove an order-7 corrected source residual from the mixed four-point branch, but it is not by itself enough to solve the package-to-transform hidden-state bridge.

Theorem 10.16 (Mixed four-point factorization through quotient determinant). *Let*

$$A_5^f = u_5 I + v_5 S, \quad A_7^f = u_7 I + v_7 S, \quad c \neq 0,$$

and define

$$\Delta_7 = \omega_h(A_7^f, A_5^f) = u_7 v_5 - u_5 v_7, \quad T = \frac{v_7}{c}.$$

On every good mixed four-point patch, the mixed four-point scalar exits depend only on

$$c, \quad u_5, \quad v_5, \quad \Delta_7.$$

They do not depend on T independently.

Proof. On a $v_5 \neq 0$ patch, the mixed four-point branch is written through the normalized variables

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad \sigma_v = \frac{\Delta_7}{c v_5}.$$

All scalar exits are functions of these quantities and their local derivatives. Thus they are determined by

$$c, \quad u_5, \quad v_5, \quad \Delta_7,$$

and do not require u_7 or v_7 separately.

On a $u_5 \neq 0$ patch, one uses the parallel quotient coordinate

$$\sigma_u = -\frac{\Delta_7}{c u_5},$$

and the same factorization holds.

For an order-seven fixed-sector residual

$$E = eI + fS,$$

adding E to A_7^f changes the determinant by

$$\Delta_7 \mapsto \Delta_7 + \omega_h(E, A_5^f) = \Delta_7 + e v_5 - u_5 f.$$

The hidden coordinate changes by

$$T \mapsto T + \frac{f}{c}.$$

Thus the mixed four-point branch sees only the determinant projection $ev_5 - u_5 f$, while the T -coordinate records a separate Φ_K -visible projection. $\square$

Remark 10.17 (No independent T -dependency in mixed four-point). A pure quotient-line shift

$$E = \chi A_5^f$$

has

$$\omega_h(E, A_5^f) = 0,$$

and is therefore invisible to mixed four-point geometry. It changes

$$T$$

by

$$T \mapsto T + \chi \frac{v_5}{c},$$

and is generally Φ_K -visible on $v_5 \neq 0$. Hence mixed four-point closure is a quotient-determinant problem, whereas two-point transform closure also requires T -horizontal compatibility.

Corollary 10.18 (Single-pair exterior odd-coefficient bound). *Fix $0 < c_* < \rho_0/8$. Let $h_\rho(s)$ be the corrected transverse scalar contributed by one exterior zero pair at absolute distance*

$$\Delta_\rho := |2m - \gamma_\rho|,$$

and write

$$h_\rho(z/Q^2) = \sum_{r \geq 0} a_{2r+1}(\rho) \frac{z^{2r+1}}{Q^{2r+4}}.$$

Then for every fixed $H_* > 1$, if $\Delta_\rho \geq H_*$, one has

$$|a_{2r+1}(\rho)| \ll \frac{1}{Q} \begin{cases} \Delta_\rho^{-(2r+4)}, & \Delta_\rho \leq c_* Q, \\ (c_* Q)^{-2r} \Delta_\rho^{-4}, & \Delta_\rho > c_* Q. \end{cases}$$

Proof. Let

$$R_\rho := c \min(\Delta_\rho, c_* Q).$$

By Proposition 10.5,

$$\|X_\rho(R_\rho \cdot)\|_{A_1} \ll \Delta_\rho^{-4}.$$

For H_* large enough, this lies inside the uniform small ball from Proposition 10.4. Since the one-pair corrected scalar $h_\rho(s)$ is odd in s , the corresponding corrected scalar perturbation vanishes at $w = 0$, so the quotient map $\mathcal{T}_{Q,R_\rho}$ is holomorphic on the one-pair input. Therefore the normalized even germ

$$g_\rho(z) := \frac{Q^3}{z} h_\rho(z/Q^2)$$

satisfies

$$\|g_\rho(R_\rho \cdot)\|_{A_1} \ll \Delta_\rho^{-4}.$$

Hence

$$[z^{2r}] g_\rho(z) \ll R_\rho^{-2r} \Delta_\rho^{-4}.$$

Since

$$g_\rho(z) = \sum_{r \geq 0} \frac{a_{2r+1}(\rho)}{Q^{2r+1}} z^{2r},$$

this gives

$$|a_{2r+1}(\rho)| \ll Q^{2r+1} R_\rho^{-2r} \Delta_\rho^{-4}.$$

Using

$$R_\rho \asymp \min(\Delta_\rho, c_* Q)$$

yields the displayed hybrid bound. $\square$

Lemma 10.19 (Local zero-counting in dyadic shells). *Let*

$$\mathcal{S}_k := \{\rho : 2^k H < \Delta_\rho \leq 2^{k+1} H\}, \quad \Delta_\rho := |2m - \gamma_\rho|, \quad k \geq 0.$$

Then

$$\#\mathcal{S}_k \ll Q(2^k H + 1),$$

where $Q \asymp \log |m|$.

Proof. This is the local zero-counting estimate already used elsewhere in the draft, applied to the interval of ordinates

$$[2m - 2^{k+1} H, 2m + 2^{k+1} H].$$

Its length is $O(2^k H + 1)$, and the local zero density at height m is $O(Q)$. $\square$

Corollary 10.20 (Exterior higher odd-coefficient bound). *Let*

$$H_{\text{out}}(z/Q^2) = \sum_{r \geq 0} a_{2r+1}^{\text{out}}(m, H) \frac{z^{2r+1}}{Q^{2r+4}}$$

be the corrected transverse scalar contributed by all zeros outside absolute radius H from the local center. Then for every $r \geq 0$ and every $H \geq H_$,*

$$|a_{2r+1}^{\text{out}}(m, H)| \ll H^{-(2r+3)} + Q^{-(2r+2)}.$$

Proof. Decompose the exterior zeros into dyadic shells

$$\mathcal{S}_k := \{\rho : 2^k H < \Delta_\rho \leq 2^{k+1} H\}, \quad k \geq 0.$$

By Lemma 10.19,

$$\#\mathcal{S}_k \ll Q(2^k H + 1).$$

For shells with $2^{k+1} H \leq c_* Q$, Corollary 10.18 gives

$$\sum_{\rho \in \mathcal{S}_k} |a_{2r+1}(\rho)| \ll Q(2^k H + 1) \cdot \frac{1}{Q(2^k H)^{2r+4}} \ll (2^k H)^{-(2r+3)}.$$

Summing these shells yields

$$\ll H^{-(2r+3)}.$$

For shells with $2^k H > c_* Q$, Corollary 10.18 gives

$$\sum_{\rho \in \mathcal{S}_k} |a_{2r+1}(\rho)| \ll Q(2^k H + 1) \cdot \frac{1}{Q(c_* Q)^{2r}(2^k H)^4} \ll (c_* Q)^{-2r} (2^k H)^{-3}.$$

Summing the far shells gives

$$\ll Q^{-2r-2}.$$

Adding the near and far ranges proves the claim. $\square$

Corollary 10.21 (Core localization of the extracted odd jet). *Let $N = \lfloor \alpha Q \rfloor$ with $\alpha > 0$, and let $a_{2N-1}^{\leq H}(m)$ denote the coefficient contributed by the zeros with $\Delta_\rho \leq H$. Then for every fixed $H \geq H_*$,*

$$a_{2N-1}(m) = a_{2N-1}^{\leq H}(m) + O(H^{-(2N+1)} + Q^{-2N}).$$

In particular, the first surviving corrected odd coefficient is exponentially localized to any fixed finite core.

Proof. This is exactly Corollary 10.20 with $r = N - 1$. $\square$

11. N -POINT ODD-MOMENT CANCELLATION

For $N \leq \rho_0 Q$, set

$$s_j = \frac{j}{Q^2}, \quad j = 1, \dots, N,$$

and define

$$a_j^{(N)} := (-1)^{j+1} \frac{j \binom{2N}{N+j}}{\binom{2N-2}{N-1}}, \quad 1 \leq j \leq N.$$

Define the corrected N -point transverse scalar

$$(21) \quad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j).$$

Lemma 11.1 (Explicit odd-moment-canceling coefficients and ℓ^1 control). *The coefficients $a_j^{(N)}$ satisfy*

$$\sum_{j=1}^N a_j^{(N)} = 1, \quad \sum_{j=1}^N a_j^{(N)} j^{2r+1} = 0 \quad (0 \leq r \leq N-2).$$

Moreover,

$$\sum_{j=1}^N |a_j^{(N)}| = 2N-1.$$

Consequently, for every integer $k \geq 0$,

$$\left| \sum_{j=1}^N a_j^{(N)} j^k \right| \leq (2N-1)N^k.$$

Proof. Set

$$D_N := \binom{2N-2}{N-1}.$$

We first prove the normalization. Writing

$$j \binom{2N}{N+j} = N \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right),$$

one gets

$$\sum_{j=1}^N (-1)^{j+1} j \binom{2N}{N+j} = N \sum_{j=1}^N (-1)^{j+1} \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right).$$

Using the partial alternating-binomial identity

$$\sum_{k=0}^m (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m},$$

one finds

$$\sum_{k=N}^{2N-1} (-1)^k \binom{2N-1}{k} = (-1)^N \binom{2N-2}{N-1}.$$

Substituting this into the preceding display yields

$$\sum_{j=1}^N (-1)^{j+1} j \binom{2N}{N+j} = \binom{2N-2}{N-1} = D_N,$$

hence

$$\sum_{j=1}^N a_j^{(N)} = 1.$$

Next, for $0 \leq r \leq N - 2$, apply the standard finite-difference identity

$$\sum_{k=0}^{2N} (-1)^k \binom{2N}{k} P(k) = 0$$

to the polynomial

$$P(k) = (k - N)^{2r+2}, \quad \deg P = 2r + 2 < 2N.$$

After the change of variables $j = k - N$, this gives

$$\sum_{j=-N}^N (-1)^j \binom{2N}{N+j} j^{2r+2} = 0.$$

Since $2r + 2$ is even and the $j = 0$ term vanishes,

$$\sum_{j=1}^N (-1)^j \binom{2N}{N+j} j^{2r+2} = 0.$$

Dividing by D_N yields

$$\sum_{j=1}^N a_j^{(N)} j^{2r+1} = 0 \quad (0 \leq r \leq N - 2).$$

Finally, since the signs alternate,

$$\sum_{j=1}^N |a_j^{(N)}| = \frac{1}{D_N} \sum_{j=1}^N j \binom{2N}{N+j}.$$

Using again

$$j \binom{2N}{N+j} = N \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right),$$

this telescopes to

$$\sum_{j=1}^N j \binom{2N}{N+j} = N \binom{2N-1}{N}.$$

Therefore

$$\sum_{j=1}^N |a_j^{(N)}| = N \frac{\binom{2N-1}{N}}{\binom{2N-2}{N-1}} = 2N - 1.$$

The final moment bound is immediate from

$$\left| \sum_{j=1}^N a_j^{(N)} j^k \right| \leq \sum_{j=1}^N |a_j^{(N)}| j^k \leq (2N - 1) N^k. \quad \square$$

Lemma 11.2 (Closed-form Mellin symbol of the discrete N -point projector). *Define*

$$A_N(t) := \sum_{j=1}^N a_j^{(N)} t^j, \quad \hat{\mu}_{N,Q}(z) := A_N(e^{z/Q^2}).$$

Then

$$A_N(t) = \frac{2(2N-1)}{N+1} {}_2F_1(1-N, 2; N+2; t),$$

so in particular

$$\hat{\mu}_{N,Q}(z) = \frac{2(2N-1)}{N+1} e^{z/Q^2} {}_2F_1(1-N, 2; N+2; e^{z/Q^2}).$$

Moreover,

$$A_N(t) - A_N(t^{-1}) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} t^{-N} (1-t)^{2N-1} (1+t),$$

and hence

$$\widehat{\mu}_{N,Q}(z) - \widehat{\mu}_{N,Q}(-z) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} e^{-Nz/Q^2} (1 - e^{z/Q^2})^{2N-1} (1 + e^{z/Q^2}).$$

In particular,

$$\widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0 \quad (0 \leq r \leq N-2),$$

and the odd part of $\widehat{\mu}_{N,Q}$ starts only at order $2N-1$.

Proof. Let

$$B_N(t) := \frac{2(2N-1)}{N+1} t {}_2F_1(1-N, 2; N+2; t) = \sum_{j=1}^N b_j^{(N)} t^j.$$

Its coefficients satisfy

$$\frac{b_{j+1}^{(N)}}{b_j^{(N)}} = \frac{(1-N+j-1)(2+j-1)}{(N+2+j-1)j} = \frac{(j-N)(j+1)}{j(N+j+1)}.$$

On the other hand, the coefficients $a_j^{(N)}$ satisfy

$$\frac{a_{j+1}^{(N)}}{a_j^{(N)}} = \frac{(-1)^{j+2}(j+1)\binom{2N}{N+j+1}}{(-1)^{j+1}j\binom{2N}{N+j}} = \frac{(j-N)(j+1)}{j(N+j+1)}.$$

Thus $a_j^{(N)}$ and $b_j^{(N)}$ agree up to an overall constant. Matching the t^1 -coefficient gives

$$b_1^{(N)} = \frac{2(2N-1)}{N+1} = \frac{\binom{2N}{N+1}}{\binom{2N-2}{N-1}} = a_1^{(N)},$$

hence $A_N(t) = B_N(t)$.

For the second identity, define

$$F_N(t) := \sum_{j=-N}^N (-1)^j \binom{2N}{N+j} t^j.$$

By the binomial theorem,

$$F_N(t) = (-1)^N t^{-N} (1-t)^{2N}.$$

Differentiating gives

$$tF'_N(t) = \sum_{j=-N}^N j(-1)^j \binom{2N}{N+j} t^j = -N(-1)^N t^{-N} (1-t)^{2N-1} (1+t).$$

Now

$$A_N(t) - A_N(t^{-1}) = -\frac{1}{\binom{2N-2}{N-1}} \sum_{j=1}^N j(-1)^j \binom{2N}{N+j} (t^j - t^{-j}),$$

while the symmetry $j \mapsto -j$ in the preceding display shows

$$tF'_N(t) = \sum_{j=1}^N j(-1)^j \binom{2N}{N+j} (t^j - t^{-j}).$$

Substituting the closed form for $tF'_N(t)$ yields

$$A_N(t) - A_N(t^{-1}) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} t^{-N} (1-t)^{2N-1} (1+t).$$

Replacing t by e^{z/Q^2} gives the displayed formula for $\widehat{\mu}_{N,Q}(z) - \widehat{\mu}_{N,Q}(-z)$. The normalization $\widehat{\mu}_{N,Q}(0) = 1$ and the low odd-jet vanishing are equivalent to the moment identities in Lemma 11.1. $\square$

Lemma 11.3 (Exact odd-moment generating function). *Define*

$$\mu_k^{(N)} := \sum_{j=1}^N a_j^{(N)} j^k,$$

and, for $|u| < 1/N$,

$$\mathcal{M}_N(u) := \sum_{r \geq 0} \mu_{2r+1}^{(N)} u^{2r+1} = \sum_{j=1}^N \frac{a_j^{(N)} j u}{1 - j^2 u^2}.$$

Then

$$\mathcal{M}_N(u) = (-1)^{N+1} (2N-1)N!(N-1)! \frac{u^{2N-1}}{\prod_{j=1}^N (1 - j^2 u^2)}.$$

In particular, for every real $x > N$,

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| x^{-(2r+1)} = (2N-1)N!(N-1)! x^2 \frac{\Gamma(x-N)}{\Gamma(x+N+1)}.$$

Proof. The identity

$$\mathcal{M}_N(u) = \sum_{j=1}^N \frac{a_j^{(N)} j u}{1 - j^2 u^2}$$

is immediate from summing the geometric series

$$\frac{j u}{1 - j^2 u^2} = \sum_{r \geq 0} j^{2r+1} u^{2r+1} \quad (|u| < 1/j).$$

Set

$$D_N(u) := \prod_{j=1}^N (1 - j^2 u^2) \mathcal{M}_N(u).$$

Since $\mathcal{M}_N$ is a sum of simple fractions, $D_N(u)$ is an odd polynomial of degree at most $2N-1$. By Lemma 11.1,

$$\mu_{2r+1}^{(N)} = 0 \quad (0 \leq r \leq N-2),$$

so $\mathcal{M}_N(u) = O(u^{2N-1})$ as $u \rightarrow 0$. Hence

$$D_N(u) = C_N u^{2N-1}$$

for some constant C_N .

To determine C_N , compare residues at $u = 1/N$. The residue of $\mathcal{M}_N(u)$ there comes only from the $j = N$ term and equals

$$\operatorname{Res}_{u=1/N} \mathcal{M}_N(u) = -\frac{a_N^{(N)}}{2N}.$$

On the other hand,

$$\operatorname{Res}_{u=1/N} \frac{C_N u^{2N-1}}{\prod_{j=1}^N (1 - j^2 u^2)} = -\frac{C_N}{2N^2 \prod_{k=1}^{N-1} (1 - k^2/N^2)}.$$

Equating these residues gives

$$C_N = a_N^{(N)} N \prod_{k=1}^{N-1} (N^2 - k^2).$$

Now

$$a_N^{(N)} = (-1)^{N+1} \frac{N}{\binom{2N-2}{N-1}},$$

and

$$\prod_{k=1}^{N-1} (N^2 - k^2) = \prod_{k=1}^{N-1} (N - k) \prod_{k=1}^{N-1} (N + k) = (N - 1)! \frac{(2N - 1)!}{N!} = \frac{(2N - 1)!}{N}.$$

Therefore

$$C_N = (-1)^{N+1} \frac{N(2N - 1)!}{\binom{2N-2}{N-1}} = (-1)^{N+1} (2N - 1)N!(N - 1)!.$$

This proves the first formula.

For the tail identity, note that

$$(-1)^{N+1} \mathcal{M}_N(u) = (2N - 1)N!(N - 1)! \frac{u^{2N-1}}{\prod_{j=1}^N (1 - j^2 u^2)}$$

has nonnegative Taylor coefficients for $0 < u < 1/N$. Hence

$$(-1)^{N+1} \mu_{2r+1}^{(N)} \geq 0 \quad (r \geq N - 1),$$

and therefore, for $x > N$,

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| x^{-(2r+1)} = \left| \mathcal{M}_N \left(\frac{1}{x} \right) \right| = (2N - 1)N!(N - 1)! \frac{x^2}{\prod_{j=1}^N (x^2 - j^2)}.$$

Finally,

$$\prod_{j=1}^N (x^2 - j^2) = \prod_{j=1}^N (x - j) \prod_{j=1}^N (x + j) = \frac{\Gamma(x)}{\Gamma(x - N)} \cdot \frac{\Gamma(x + N + 1)}{\Gamma(x + 1)} = \frac{1}{x} \frac{\Gamma(x + N + 1)}{\Gamma(x - N)}.$$

Substituting this identity yields the claimed formula. $\square$

Remark 11.4 (Structure of the surviving odd tail). Expanding

$$\frac{1}{\prod_{j=1}^N (1 - j^2 u^2)} = \sum_{k \geq 0} h_k(1^2, \dots, N^2) u^{2k},$$

where h_k denotes the complete homogeneous symmetric polynomial, the exact formula of Lemma 11.3 yields

$$\mu_{2N-1+2k}^{(N)} = (-1)^{N+1} (2N - 1)N!(N - 1)! h_k(1^2, \dots, N^2), \quad k \geq 0.$$

Thus the surviving odd moments are not merely bounded: they are explicit and have a fixed sign,

$$(-1)^{N+1} \mu_{2r+1}^{(N)} \geq 0 \quad (r \geq N - 1).$$

This makes the post-cancellation odd tail completely structured rather than mysterious.

Corollary 11.5 (Exact action of the N -point operator on odd germs). *Let*

$$F(s) = \sum_{r \geq 0} b_{2r+1} s^{2r+1}$$

be an odd analytic germ at $s = 0$. Then

$$(\mathcal{P}_{N,Q} F) := \sum_{j=1}^N a_j^{(N)} F\left(\frac{j}{Q^2}\right) = \sum_{r \geq 0} b_{2r+1} \frac{\mu_{2r+1}^{(N)}}{Q^{4r+2}}.$$

Hence, by Remark 11.4,

$$(\mathcal{P}_{N,Q} F) = (-1)^{N+1} (2N - 1)N!(N - 1)! \sum_{k \geq 0} b_{2N-1+2k} h_k(1^2, \dots, N^2) Q^{-(4N+4k-2)}.$$

Thus the N -point operator does not merely annihilate the first $N - 1$ odd jets: on odd germs it acts as an explicit shifted complete-homogeneous transform of the surviving odd Taylor coefficients.

Proposition 11.6 (Exact finite-difference factorization on odd germs). *Let*

$$T_Q := e^{Q^{-2}D}, \quad D := \frac{d}{ds},$$

so that $T_Q F(s) = F(s + Q^{-2})$. Then for every odd analytic germ F at $s = 0$,

$$(\mathcal{P}_{N,Q}F) = \frac{N(-1)^N}{2\binom{2N-2}{N-1}} \left[T_Q^{-N} (1 - T_Q)^{2N-1} (1 + T_Q) F \right]_{s=0}.$$

Equivalently,

$$(\mathcal{P}_{N,Q}F) = \frac{(-1)^{N+1}N}{2\binom{2N-2}{N-1}} \left[T_Q^{-N} (T_Q - 1)^{2N-1} (1 + T_Q) F \right]_{s=0}.$$

In particular, on odd germs the N -point operator factors exactly through the $(2N - 1)$ -st forward difference.

Proof. Let

$$P_{N,Q}(D) := \sum_{j=1}^N a_j^{(N)} e^{(j/Q^2)D} = \sum_{j=1}^N a_j^{(N)} T_Q^j.$$

Then

$$(\mathcal{P}_{N,Q}F) = (P_{N,Q}(D)F)|_{s=0}.$$

By Lemma 11.2, with $t = T_Q$,

$$P_{N,Q}(D) - P_{N,Q}(-D) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} T_Q^{-N} (1 - T_Q)^{2N-1} (1 + T_Q).$$

If F is odd, then

$$(P_{N,Q}(-D)F)|_{s=0} = \sum_{j=1}^N a_j^{(N)} F(-j/Q^2) = - \sum_{j=1}^N a_j^{(N)} F(j/Q^2) = -(\mathcal{P}_{N,Q}F).$$

Hence

$$2(\mathcal{P}_{N,Q}F) = ((P_{N,Q}(D) - P_{N,Q}(-D))F)|_{s=0},$$

which gives the claimed formula. The second form follows from $(1 - T_Q)^{2N-1} = -(T_Q - 1)^{2N-1}$. $\square$

Remark 11.7 (Why the odd tail begins at order $2N - 1$). Proposition 11.6 shows that the disappearance of the first $N - 1$ odd jets is not merely a moment identity. It is built into the operator itself: on odd germs, the N -point projector contains the factor $(T_Q - 1)^{2N-1}$. The surviving odd tail therefore begins at order $2N - 1$ for operator-theoretic reasons, not only by coefficient cancellation.

Lemma 11.8 (Integral form of a forward difference). *For $m \geq 1$, $h > 0$, and an analytic germ f , one has*

$$(e^{hD} - 1)^m f(x) = h^m \int_0^m f^{(m)}(x + hu) \omega_m(u) du,$$

where

$$\omega_m(u) := \frac{1}{(m-1)!} \sum_{r=0}^m (-1)^r \binom{m}{r} (u-r)_+^{m-1}, \quad (u-r)_+ := \max\{u-r, 0\}.$$

The function ω_m is nonnegative, supported on $[0, m]$, and satisfies the symmetry

$$\omega_m(u) = \omega_m(m-u).$$

Proof. The identity

$$(e^{hD} - 1)f(x) = f(x + h) - f(x) = \int_0^h f'(x + t) dt$$

iterates to

$$(e^{hD} - 1)^m f(x) = \int_{[0, h]^m} f^{(m)}(x + t_1 + \cdots + t_m) dt_1 \cdots dt_m.$$

Rescaling $t_i = hu_i$ gives

$$(e^{hD} - 1)^m f(x) = h^m \int_{[0, 1]^m} f^{(m)}(x + h(u_1 + \cdots + u_m)) du_1 \cdots du_m.$$

Pushing forward Lebesgue measure on $[0, 1]^m$ under

$$(u_1, \dots, u_m) \mapsto u_1 + \cdots + u_m$$

yields the density ω_m , namely the classical Irwin–Hall density, which has the stated explicit formula, support, and symmetry. This proves the claim. $\square$

Proposition 11.9 (Positive-kernel representation of the odd projector). *Set*

$$m := 2N - 1, \quad h := Q^{-2},$$

and define, for $0 \leq u \leq N$,

$$\kappa_N(u) := \omega_{2N-1}(N - u) + \omega_{2N-1}(N - 1 - u),$$

where ω_{2N-1} is the density from Lemma 11.8, extended by 0 outside $[0, 2N - 1]$. Then $\kappa_N(u) \geq 0$ on $[0, N]$, and for every odd analytic germ F ,

$$(\mathcal{P}_{N, Q} F) = \frac{N(-1)^{N+1}}{\binom{2N-2}{N-1}} Q^{-(4N-2)} \int_0^N \kappa_N(u) F^{(2N-1)}\left(\frac{u}{Q^2}\right) du.$$

Moreover, $\kappa_N(u) > 0$ for $0 < u < N$.

Proof. Apply Proposition 11.6 and Lemma 11.8 with $m = 2N - 1$ and $h = Q^{-2}$. This gives

$$(\mathcal{P}_{N, Q} F) = \frac{N(-1)^{N+1}}{2\binom{2N-2}{N-1}} h^{2N-1} \int_0^{2N-1} \omega_{2N-1}(v) \left[F^{(2N-1)}((v-N)h) + F^{(2N-1)}((v-N+1)h) \right] dv.$$

Since F is odd, $F^{(2N-1)}$ is even. Using this evenness together with the symmetry

$$\omega_{2N-1}(v) = \omega_{2N-1}(2N - 1 - v),$$

the two terms may be folded onto $[0, N]$, yielding

$$(\mathcal{P}_{N, Q} F) = \frac{N(-1)^{N+1}}{\binom{2N-2}{N-1}} h^{2N-1} \int_0^N [\omega_{2N-1}(N - u) + \omega_{2N-1}(N - 1 - u)] F^{(2N-1)}(uh) du.$$

This is exactly the stated formula. The strict positivity on $0 < u < N$ follows because $N - u \in (0, 2N - 1)$ there, and ω_{2N-1} is strictly positive on the interior of its support. $\square$

Corollary 11.10 (Sign transfer from the first surviving derivative). *Let F be an odd analytic germ. If for some $\sigma \in \{\pm 1\}$,*

$$\sigma F^{(2N-1)}(s) \geq 0 \quad \text{for all } 0 \leq s \leq \frac{N}{Q^2},$$

then

$$\sigma(-1)^{N+1}(\mathcal{P}_{N, Q} F) \geq 0.$$

If moreover $F^{(2N-1)} \not\equiv 0$ on $[0, N/Q^2]$, then the inequality is strict.

Proof. This is immediate from Proposition 11.9, since the kernel κ_N is nonnegative and strictly positive on $(0, N)$. $\square$

Remark 11.11 (Why this is the right downstream target). Proposition 11.9 shows that the N -point layer is not merely an odd-moment cancellation device: on odd germs it is a positive average of the first surviving derivative $F^{(2N-1)}$ over the microscopic interval $[0, N/Q^2]$. Accordingly, any source-level argument that gives a fixed sign, one-sided lower bound, or variation control for

$$H_m^{(2N-1)}(s) \quad \left(0 \leq s \leq \frac{N}{Q^2}\right)$$

would transfer directly to the sign or size of $\Xi_\zeta^{(N)}(m)$. This isolates the next natural analytic target in the N -point route.

Remark 11.12 (Finite-order recurrence of the normalized surviving tail). Define

$$\nu_k := \frac{(-1)^{N+1} \mu_{2N-1+2k}^{(N)}}{(2N-1)N!(N-1)!} = h_k(1^2, \dots, N^2), \quad k \geq 0.$$

Then

$$\sum_{k \geq 0} \nu_k t^k = \frac{1}{\prod_{j=1}^N (1 - j^2 t)} = \frac{1}{1 - e_1 t + e_2 t^2 - \dots + (-1)^N e_N t^N},$$

where $e_r = e_r(1^2, \dots, N^2)$ is the r -th elementary symmetric polynomial in $1^2, \dots, N^2$. Consequently,

$$\nu_k = e_1 \nu_{k-1} - e_2 \nu_{k-2} + \dots + (-1)^{N+1} e_N \nu_{k-N} \quad (k \geq 1),$$

with the conventions $\nu_0 = 1$ and $\nu_k = 0$ for $k < 0$. Thus the surviving odd tail is governed by an exact order- N linear recurrence.

Proposition 11.13 (Partial-fraction form of the surviving odd tail). *Let*

$$G_N(t) := \frac{1}{\prod_{m=1}^N (1 - m^2 t)}.$$

Then G_N has the exact partial-fraction decomposition

$$G_N(t) = \sum_{j=1}^N \frac{B_{N,j}}{1 - j^2 t},$$

where

$$B_{N,j} = \frac{2(-1)^{N-j} j^{2N}}{(N-j)!(N+j)!}.$$

Equivalently,

$$\frac{1}{\prod_{m=1}^N (1 - m^2 u^2)} = \sum_{j=1}^N \frac{2(-1)^{N-j} j^{2N}}{(N-j)!(N+j)!} \frac{1}{1 - j^2 u^2}.$$

Consequently, for every $k \geq 0$,

$$h_k(1^2, \dots, N^2) = \sum_{j=1}^N \frac{2(-1)^{N-j} j^{2N+2k}}{(N-j)!(N+j)!},$$

and hence

$$\mu_{2N-1+2k}^{(N)} = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{j=1}^N \frac{2(-1)^{N-j} j^{2N+2k}}{(N-j)!(N+j)!}.$$

Proof. Since the poles of $G_N(t)$ are simple and located at $t = j^{-2}$ ($1 \leq j \leq N$), one has

$$G_N(t) = \sum_{j=1}^N \frac{B_{N,j}}{1 - j^2 t}$$

for uniquely determined coefficients $B_{N,j}$. Multiplying by $(1 - j^2 t)$ and evaluating at $t = j^{-2}$ gives

$$B_{N,j} = \frac{1}{\prod_{m \neq j} (1 - m^2 t)} \Big|_{t=j^{-2}} = \frac{1}{\prod_{m \neq j} (1 - m^2 / j^2)}.$$

Now

$$\prod_{m \neq j} \left(1 - \frac{m^2}{j^2}\right) = j^{-2N+2} \prod_{m \neq j} (j^2 - m^2),$$

and

$$\prod_{m \neq j} (j^2 - m^2) = \prod_{m \neq j} (j - m) \prod_{m \neq j} (j + m).$$

Here

$$\prod_{m \neq j} (j - m) = (j - 1)! (-1)^{N-j} (N - j)!,$$

while

$$\prod_{m \neq j} (j + m) = \frac{(N + j)!}{2j j!}.$$

Therefore

$$\prod_{m \neq j} (j^2 - m^2) = (-1)^{N-j} \frac{(N - j)! (N + j)!}{2j^2},$$

and hence

$$B_{N,j} = \frac{2(-1)^{N-j} j^{2N}}{(N - j)! (N + j)!}.$$

Expanding each geometric series

$$\frac{1}{1 - j^2 u^2} = \sum_{k \geq 0} j^{2k} u^{2k}$$

and comparing coefficients of u^{2k} yields the formula for $h_k(1^2, \dots, N^2)$. The identity for $\mu_{2N-1+2k}^{(N)}$ then follows from Remark 11.4. $\square$

Remark 11.14 (Dominance of the outermost pole for fixed N). For fixed N , the explicit decomposition in Proposition 11.13 shows that the surviving tail is asymptotically dominated by the outermost node $j = N$. Indeed,

$$h_k(1^2, \dots, N^2) = \frac{2N^{2N+2k}}{(2N)!} + O_N((N - 1)^{2k}) \quad (k \rightarrow \infty),$$

and therefore

$$\mu_{2N-1+2k}^{(N)} = (-1)^{N+1} \frac{N^{2N+2k}}{\binom{2N-2}{N-1}} + O_N((N - 1)^{2k}) \quad (k \rightarrow \infty).$$

So for fixed N , the far odd tail is eventually controlled by the single outermost pole $u = \pm 1/N$.

Proposition 11.15 (Kernel form and explicit corrected N -point transverse estimate). *Let*

$$H_m(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m))$$

be the full corrected transverse scalar. For $N = \lfloor \alpha Q \rfloor$ with $0 < \alpha < \rho_0$, define

$$s_j = \frac{j}{Q^2}, \quad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j).$$

Then

$$\Xi_\zeta^{(N)}(m) = \frac{1}{2\pi i} \oint_{|\xi|=\rho_0/Q} H_m(\xi) K_{N,Q}(\xi) d\xi,$$

where

$$K_{N,Q}(\xi) := 2 \sum_{j=1}^N a_j^{(N)} \frac{s_j}{\xi^2 - s_j^2} = 2(-1)^{N+1} (2N-1)N!(N-1)! Q^4 \xi \frac{\Gamma(Q^2 \xi - N)}{\Gamma(Q^2 \xi + N + 1)}.$$

Moreover, writing $x := \rho_0 Q$, one has the explicit bound

$$|\Xi_\zeta^{(N)}(m)| \ll_{\rho_0} \rho_0^2 \frac{(2N-1)N!(N-1)!}{Q} \frac{\Gamma(x-N)}{\Gamma(x+N+1)} L(m)S(m)^2.$$

Proof. By Proposition 9.5, the full corrected scalar

$$H_m(s) = \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m))$$

is holomorphic on $|s| < \rho_0/Q$. By Proposition 9.7, all cutoff-dependent tail, jet-correction, and estimation contributions are already internal to the full corrected scalar $H_m(s)$; hence no external remainder is carried into $\Xi_\zeta^{(N)}(m)$.

Since H_m is odd, the odd Cauchy formula gives

$$H_m(s) = \frac{1}{2\pi i} \oint_{|\xi|=\rho_0/Q} H_m(\xi) \frac{2s}{\xi^2 - s^2} d\xi.$$

Summing this identity against the coefficients $a_j^{(N)}$ at $s = s_j = j/Q^2$ yields

$$\Xi_\zeta^{(N)}(m) = \frac{1}{2\pi i} \oint_{|\xi|=\rho_0/Q} H_m(\xi) K_{N,Q}(\xi) d\xi$$

with

$$K_{N,Q}(\xi) = 2 \sum_{j=1}^N a_j^{(N)} \frac{s_j}{\xi^2 - s_j^2}.$$

Now

$$\frac{s_j}{\xi^2 - s_j^2} = \frac{1}{\xi} \frac{j/(Q^2 \xi)}{1 - j^2/(Q^4 \xi^2)},$$

so, using Lemma 11.3,

$$K_{N,Q}(\xi) = \frac{2}{\xi} \mathcal{M}_N\left(\frac{1}{Q^2 \xi}\right) = 2(-1)^{N+1} (2N-1)N!(N-1)! \frac{Q^2}{\prod_{j=1}^N (Q^4 \xi^2 - j^2)}.$$

Finally,

$$\prod_{j=1}^N (Q^4 \xi^2 - j^2) = \prod_{j=1}^N (Q^2 \xi - j) \prod_{j=1}^N (Q^2 \xi + j) = \frac{1}{Q^2 \xi} \frac{\Gamma(Q^2 \xi + N + 1)}{\Gamma(Q^2 \xi - N)},$$

which gives

$$K_{N,Q}(\xi) = 2(-1)^{N+1} (2N-1)N!(N-1)! Q^4 \xi \frac{\Gamma(Q^2 \xi - N)}{\Gamma(Q^2 \xi + N + 1)}.$$

For the size bound, Proposition 10.1 gives

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3} \quad (|s| = \rho_0/Q).$$

By Proposition 10.2, one may write

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}}, \quad |c_{2r+1}(m)| \ll L(m)S(m)^2 \rho_0^{-(2r+1)}.$$

Therefore

$$\Xi_\zeta^{(N)}(m) = \sum_{r \geq 0} c_{2r+1}(m) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}}.$$

By Lemma 11.1 and Remark 11.4, this becomes the exact surviving expansion

$$\Xi_{\zeta}^{(N)}(m) = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{k \geq 0} c_{2N-1+2k}(m) h_k(1^2, \dots, N^2) Q^{-(2N+2k+2)}.$$

In particular, the N -point operation isolates the first surviving corrected odd coefficient $c_{2N-1}(m)$ and propagates the higher odd coefficients through an explicit universal positive tail.

For the bound, the coefficient estimate yields

$$|\Xi_{\zeta}^{(N)}(m)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3} \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_0 Q)^{-(2r+1)}.$$

Applying Lemma 11.3 with $x = \rho_0 Q$ gives

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_0 Q)^{-(2r+1)} = (2N-1)N!(N-1)! (\rho_0 Q)^2 \frac{\Gamma(\rho_0 Q - N)}{\Gamma(\rho_0 Q + N + 1)}.$$

Substituting this proves the stated bound. $\square$

Remark 11.16 (Exact surviving-coefficient expansion on the zeta side). With the notation of the proof of Proposition 11.15,

$$\Xi_{\zeta}^{(N)}(m) = \sum_{r \geq 0} c_{2r+1}(m) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}} = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{k \geq 0} c_{2N-1+2k}(m) h_k(1^2, \dots, N^2) Q^{-(2N+2k+2)}.$$

Thus the N -point step does not merely remove the first $N-1$ odd jets: it converts the corrected odd Taylor expansion into an explicit universal tail whose leading term is the first surviving corrected odd coefficient $c_{2N-1}(m)$.

Corollary 11.17 (Finite-core localization of the transformed zeta-side scalar). *With the notation of Proposition 11.15 and Corollary 10.20, let*

$$H_{\text{out}}(z/Q^2) = \sum_{r \geq 0} a_{2r+1}^{\text{out}}(m, H) \frac{z^{2r+1}}{Q^{2r+4}}$$

be the corrected transverse scalar contributed by all zero pairs with $\Delta_{\rho} > H$, and define

$$\Xi_{\zeta, \text{out}}^{(N)}(m; H) := \sum_{j=1}^N a_j^{(N)} H_{\text{out}}(s_j), \quad \Xi_{\zeta, \leq H}^{(N)}(m) := \Xi_{\zeta}^{(N)}(m) - \Xi_{\zeta, \text{out}}^{(N)}(m; H).$$

If $HQ > N$, then

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \frac{(2N-1)N!(N-1)!}{Q} \frac{\Gamma(HQ - N)}{\Gamma(HQ + N + 1)} + (2N-1)N!(N-1)! \frac{\Gamma(Q^2 - N)}{\Gamma(Q^2 + N + 1)}.$$

Consequently,

$$\Xi_{\zeta}^{(N)}(m) = \Xi_{\zeta, \leq H}^{(N)}(m) + O\left(\frac{(2N-1)N!(N-1)!}{Q} \frac{\Gamma(HQ - N)}{\Gamma(HQ + N + 1)}\right) + O\left((2N-1)N!(N-1)! \frac{\Gamma(Q^2 - N)}{\Gamma(Q^2 + N + 1)}\right).$$

Proof. By the exact N -point action on odd germs used in the proof of Proposition 11.15,

$$\Xi_{\zeta, \text{out}}^{(N)}(m; H) = \sum_{r \geq 0} a_{2r+1}^{\text{out}}(m, H) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}}.$$

Since $\mu_{2r+1}^{(N)} = 0$ for $0 \leq r \leq N-2$,

$$\Xi_{\zeta, \text{out}}^{(N)}(m; H) = \sum_{r \geq N-1} a_{2r+1}^{\text{out}}(m, H) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}}.$$

Taking absolute values and applying Corollary 10.20,

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| \left(\frac{H^{-(2r+3)}}{Q^{2r+4}} + \frac{Q^{-(2r+2)}}{Q^{2r+4}} \right).$$

Thus

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \frac{1}{H^2 Q^3} \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (HQ)^{-(2r+1)} + \frac{1}{Q^4} \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (Q^2)^{-(2r+1)}.$$

Now apply Lemma 11.3 first with $x = HQ$ and then with $x = Q^2$. Since $HQ > N$, both evaluations are valid for all sufficiently large Q , and one obtains

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (HQ)^{-(2r+1)} = (2N-1)N!(N-1)!(HQ)^2 \frac{\Gamma(HQ-N)}{\Gamma(HQ+N+1)},$$

and

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (Q^2)^{-(2r+1)} = (2N-1)N!(N-1)!Q^4 \frac{\Gamma(Q^2-N)}{\Gamma(Q^2+N+1)}.$$

Substituting these identities proves the displayed bound for $\Xi_{\zeta, \text{out}}^{(N)}(m; H)$, and the decomposition

$$\Xi_{\zeta}^{(N)}(m) = \Xi_{\zeta, \leq H}^{(N)}(m) + \Xi_{\zeta, \text{out}}^{(N)}(m; H)$$

gives the second display. $\square$

Remark 11.18 (Asymptotic finite-core localization at linear depth). Let

$$N = \lfloor \alpha Q \rfloor, \quad 0 < \alpha < \rho_0,$$

and fix $H > \alpha$. Then Corollary 11.17 and Stirling's formula give

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \exp(-Hf(\alpha/H)Q + O(\log Q)) + \exp(-2\alpha Q \log Q + O(Q)),$$

where

$$f(\beta) = (1+\beta) \log(1+\beta) - 2\beta \log \beta - (1-\beta) \log(1-\beta).$$

In particular,

$$\Xi_{\zeta}^{(N)}(m) = \Xi_{\zeta, \leq H}^{(N)}(m) + O(e^{-Hf(\alpha/H)Q + O(\log Q)}) + O(e^{-2\alpha Q \log Q + O(Q)}).$$

Thus the full transformed zeta-side scalar is determined, up to exponentially small error, by a fixed finite core.

Definition 11.19 (Finite-core N -point scalar). Fix a finite same-scale core $\mathcal{C}$ of zero pairs around the midpoint m , and let

$$H_{\text{core}, m}(s) := \Phi_K(\widehat{\Omega}_{\text{core}}^{\text{corr}}(s; m))$$

denote the corrected transverse scalar contributed by this core, in the same finite- s normalization as $H_m(s)$. For $N \geq 1$, define

$$\Xi_{\text{core}}^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_{\text{core}, m}(j/Q^2).$$

If

$$H_{\text{core}, m}(z/Q^2) = \sum_{r \geq 0} c_{\text{core}, 2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}},$$

then the N -point moment cancellations give

$$\Xi_{\text{core}}^{(N)}(m) = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{k \geq 0} c_{\text{core}, 2N-1+2k}(m) h_k(1^2, \dots, N^2) Q^{-(2N+2k+2)}.$$

Thus the first finite-core coefficient seen by the N -point projector at depth N is

$$c_{\text{core}, 2N-1}(m).$$

Remark 11.20 (Finite-core localization is not a lower law). Corollary 11.17 is an upper/localization statement. It does not supply a lower bound for

$$\Xi_{\text{core}}^{(N)}(m).$$

In particular, it does not rule out cancellation among the finitely many core constituents:

$$c_{\text{core}, 2N-1}(m) = 0$$

or cancellation of several consecutive N -visible odd coefficients. A separate finite-core lower-transfer theorem is required before the N -point contradiction can be applied to genuinely non-pair-like cores.

Remark 11.21 (Exact Laurent tail of the contour kernel). The kernel in Proposition 11.15 has an exact large- ξ Laurent expansion. Indeed, using Lemma 11.3 and Remark 11.4,

$$K_{N,Q}(\xi) = \frac{2}{\xi} \mathcal{M}_N\left(\frac{1}{Q^2\xi}\right) = 2(-1)^{N+1}(2N-1)N!(N-1)!Q^{-4N+2}\xi^{-2N} \sum_{k \geq 0} h_k(1^2, \dots, N^2) Q^{-4k} \xi^{-2k}.$$

So the contour kernel is not merely small: its entire post-cancellation odd tail is explicit, universal, and sign-rigid term-by-term.

Corollary 11.22 (Optimized zeta-side cancellation depth). *Let*

$$N = \lfloor \alpha Q \rfloor, \quad 0 < \alpha < \rho_0, \quad \beta := \frac{\alpha}{\rho_0} \in (0, 1).$$

Then

$$|\Xi_{\zeta}^{(N)}(m)| \ll_{\rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha, \rho_0)Q},$$

where

$$\delta_*(\alpha, \rho_0) = \rho_0 \left((1 + \beta) \log(1 + \beta) - 2\beta \log \beta - (1 - \beta) \log(1 - \beta) \right).$$

Moreover, $\delta_(\alpha, \rho_0)$ is maximized uniquely at*

$$\beta = \frac{1}{\sqrt{2}}, \quad \text{i.e.} \quad \alpha_{\text{opt}} = \frac{\rho_0}{\sqrt{2}},$$

so the optimal zeta-side cancellation depth is

$$N_{\text{opt}} = \left\lfloor \frac{\rho_0}{\sqrt{2}} Q \right\rfloor,$$

and the corresponding exponential rate is

$$\delta_{\text{opt}} = 2\rho_0 \log(1 + \sqrt{2}).$$

Proof. Set $x := \rho_0 Q$, so that $N = \beta x + O(1)$ with $\beta \in (0, 1)$ fixed. By Proposition 11.15, it suffices to estimate

$$(2N-1)N!(N-1)! \frac{\Gamma(x-N)}{\Gamma(x+N+1)}.$$

A Stirling expansion gives

$$\log \left((2N-1)N!(N-1)! \frac{\Gamma(x-N)}{\Gamma(x+N+1)} \right) = -x f(\beta) + O(\log x),$$

where

$$f(\beta) = (1 + \beta) \log(1 + \beta) - 2\beta \log \beta - (1 - \beta) \log(1 - \beta).$$

Hence

$$(2N-1)N!(N-1)! \frac{\Gamma(x-N)}{\Gamma(x+N+1)} \ll_{\rho_0, \alpha} e^{-x f(\beta)},$$

and therefore

$$|\Xi_{\zeta}^{(N)}(m)| \ll_{\rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\rho_0 f(\beta)Q}.$$

This proves the first claim.

For the optimization, differentiate:

$$f'(\beta) = \log\left(\frac{1 - \beta^2}{\beta^2}\right).$$

Thus $f'(\beta) = 0$ exactly when $\beta = 1/\sqrt{2}$. Since

$$f''(\beta) = -\frac{2\beta}{1 - \beta^2} - \frac{2}{\beta} < 0 \quad (0 < \beta < 1),$$

this critical point is the unique maximizer. Finally,

$$f\left(\frac{1}{\sqrt{2}}\right) = 2 \log(1 + \sqrt{2}),$$

which gives the stated value of δ_{opt} . $\square$

Remark 11.23. Corollary 11.22 optimizes the *zeta-side* cancellation depth. After the exact odd-tail evaluation is imported into Proposition 11.37, the toy side imposes no asymptotic upper restriction on the linear choice $N = \lfloor \alpha Q \rfloor$: for every fixed $\alpha > 0$, the toy microscopic nodes $s_j = j/Q^2$ remain inside the toy analyticity disk for all sufficiently large Q , and the residual odd toy tail is in fact super-exponentially small on the Q -scale. Accordingly one may take the exact zeta-side optimum

$$\alpha = \frac{\rho_0}{\sqrt{2}}.$$

Remark 11.24 (Sharper prefactor in the optimized zeta-side bound). With the notation of Corollary 11.22, set

$$x := \rho_0 Q, \quad N = \beta x + O(1), \quad \beta = \frac{\alpha}{\rho_0} \in (0, 1).$$

A more precise Stirling expansion of the gamma-ratio factor in Proposition 11.15 gives

$$(2N - 1)N!(N - 1)! \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} = \frac{4\pi\beta}{\sqrt{1 - \beta^2}} e^{-xf(\beta)} (1 + o(1)),$$

where

$$f(\beta) = (1 + \beta) \log(1 + \beta) - 2\beta \log \beta - (1 - \beta) \log(1 - \beta).$$

Consequently,

$$|\Xi_{\zeta}^{(N)}(m)| \ll 2C_{\partial}(\rho_0)\rho_0 \frac{4\pi\beta}{\sqrt{1 - \beta^2}} \frac{L(m)S(m)^2}{Q^2} e^{-\delta_*(\alpha, \rho_0)Q} (1 + o(1)).$$

At the zeta-side optimum $\beta = 1/\sqrt{2}$, the prefactor simplifies to 4π , so

$$|\Xi_{\zeta}^{(N_{\text{opt}})}(m)| \ll 8\pi C_{\partial}(\rho_0)\rho_0 \frac{L(m)S(m)^2}{Q^2} e^{-2\rho_0 \log(1 + \sqrt{2})Q} (1 + o(1)).$$

This sharper prefactor is not needed for the qualitative contradiction, but it records that, once the boundary constant $C_{\partial}(\rho_0)$ is made explicit, the zeta-side N -point estimate becomes effective with an explicit leading constant.

Proposition 11.25 (Real profile of the exact N -point kernel). *Let*

$$C_N := (2N - 1)N!(N - 1)!, \quad \kappa_N(x) := x \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} \quad (x > N).$$

Then

$$\kappa_N(x) = \frac{1}{\prod_{j=1}^N (x^2 - j^2)},$$

and for real $x > N$,

$$K_{N,Q}\left(\frac{x}{Q^2}\right) = 2(-1)^{N+1} C_N Q^2 \kappa_N(x).$$

Moreover:

$$(1) \quad \kappa_N(x) > 0 \text{ for all } x > N;$$

- (2) $\kappa'_N(x) < 0$ for all $x > N$;
 (3) for every $x > N$,

$$\sup_{|\xi|=x/Q^2} |K_{N,Q}(\xi)| = 2C_N Q^2 \kappa_N(x) = \left| K_{N,Q}\left(\frac{x}{Q^2}\right) \right|.$$

Proof. From Proposition 11.15,

$$K_{N,Q}(\xi) = 2(-1)^{N+1} C_N Q^4 \xi \frac{\Gamma(Q^2 \xi - N)}{\Gamma(Q^2 \xi + N + 1)}.$$

Substituting $\xi = x/Q^2$ gives

$$K_{N,Q}\left(\frac{x}{Q^2}\right) = 2(-1)^{N+1} C_N Q^2 x \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} = 2(-1)^{N+1} C_N Q^2 \kappa_N(x).$$

Now

$$\Gamma(x + N + 1) = (x + N)(x + N - 1) \cdots x \Gamma(x),$$

and

$$\Gamma(x) = (x - 1)(x - 2) \cdots (x - N) \Gamma(x - N).$$

Hence

$$x \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} = \frac{1}{(x - N) \cdots (x - 1)(x + 1) \cdots (x + N)} = \frac{1}{\prod_{j=1}^N (x^2 - j^2)},$$

which proves the product formula. Positivity is immediate for $x > N$.

For monotonicity, differentiate the logarithm:

$$\frac{d}{dx} \log \kappa_N(x) = - \sum_{j=1}^N \frac{2x}{x^2 - j^2}.$$

Since $x > N$, each denominator is positive, so

$$\frac{d}{dx} \log \kappa_N(x) < 0,$$

hence $\kappa'_N(x) < 0$.

Finally, using the product form of the kernel,

$$K_{N,Q}(\xi) = 2(-1)^{N+1} C_N Q^2 \frac{1}{\prod_{j=1}^N (Q^4 \xi^2 - j^2)},$$

if $|\xi| = x/Q^2$ then

$$|Q^4 \xi^2 - j^2| \geq ||Q^2 \xi|^2 - j^2| = x^2 - j^2 \quad (1 \leq j \leq N),$$

so

$$|K_{N,Q}(\xi)| \leq 2C_N Q^2 \frac{1}{\prod_{j=1}^N (x^2 - j^2)} = 2C_N Q^2 \kappa_N(x).$$

Equality is attained at $\xi = x/Q^2$, proving the final claim. $\square$

Remark 11.26 (No contour loss in the kernel factor). Proposition 11.25 shows that the kernel factor in Proposition 11.15 is controlled exactly by its positive real value on the boundary circle. Thus the boundary estimate used in the zeta-side N -point bound carries no hidden contour-loss from the kernel itself: the worst point is the positive real boundary point

$$\xi = \frac{\rho_0}{Q}.$$

Moreover, since κ_N is strictly decreasing on (N, ∞) , the kernel factor is largest at the smallest admissible boundary radius.

Proposition 11.27 (Positive smoothing preserves the real kernel profile). *Let $\psi \in C_c^\infty(\mathbf{R})$ be even, nonnegative, and normalized by*

$$\int_{\mathbf{R}} \psi(u) du = 1.$$

Set

$$U_\psi := \sup\{|u| : u \in \text{supp } \psi\}.$$

For $x > N + \eta Q^2 U_\psi$, define the smoothed real profile

$$\kappa_{N,\eta}^\psi(x) := \int_{\mathbf{R}} \psi(u) \kappa_N(x + \eta Q^2 u) du,$$

where

$$\kappa_N(x) := x \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} = \frac{1}{\prod_{j=1}^N (x^2 - j^2)}.$$

Then

$$\kappa_{N,\eta}^\psi(x) > 0 \quad \text{and} \quad \frac{d}{dx} \kappa_{N,\eta}^\psi(x) < 0$$

for all $x > N + \eta Q^2 U_\psi$.

Proof. For $x > N + \eta Q^2 U_\psi$ and $u \in \text{supp } \psi$, one has

$$x + \eta Q^2 u > N.$$

By Proposition 11.25,

$$\kappa_N(x + \eta Q^2 u) > 0 \quad \text{and} \quad \kappa'_N(x + \eta Q^2 u) < 0.$$

Since $\psi \geq 0$ and $\int \psi = 1$, averaging preserves positivity:

$$\kappa_{N,\eta}^\psi(x) > 0.$$

Differentiating under the integral sign gives

$$\frac{d}{dx} \kappa_{N,\eta}^\psi(x) = \int_{\mathbf{R}} \psi(u) \kappa'_N(x + \eta Q^2 u) du < 0.$$

□

Remark 11.28 (Smoothed real profile). Proposition 11.27 shows that nonnegative smoothing preserves the qualitative real-axis behavior of the exact kernel: positivity and strict monotonic decay survive averaging. Thus the Mellin-side smoothing introduced later does not destroy the basic real-profile structure of the N -point kernel.

Remark 11.29 (Discrete Mellin projector and test-function viewpoint). Define the additive operator

$$(\mathcal{P}_{N,Q} F) := \sum_{j=1}^N a_j^{(N)} F\left(\frac{j}{Q^2}\right),$$

and the associated multiplicative distribution on $\mathbb{R}_+^\times$,

$$\mu_{N,Q} := \sum_{j=1}^N a_j^{(N)} \delta_{e^{j/Q^2}}.$$

Its Mellin symbol is precisely

$$\widehat{\mu}_{N,Q}(z) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2}.$$

Equivalently, if $D := \frac{d}{ds}$, then

$$\mathcal{P}_{N,Q} = P_{N,Q}(D)|_{s=0}, \quad P_{N,Q}(D) := \sum_{j=1}^N a_j^{(N)} e^{(j/Q^2)D}.$$

Thus $\mathcal{P}_{N,Q}$ is a polynomial in the translation semigroup, evaluated at the origin. If

$$F(s) = \sum_{n \geq 0} c_n s^n,$$

then

$$(\mathcal{P}_{N,Q}F) = \sum_{n \geq 0} c_n \frac{\mu_n^{(N)}}{Q^{2n}}, \quad \mu_n^{(N)} := \sum_{j=1}^N a_j^{(N)} j^n.$$

Lemma 11.2 shows that $\widehat{\mu}_{N,Q}(0) = 1$ and that the odd part of $\widehat{\mu}_{N,Q}$ starts only at order $2N - 1$; meanwhile, Corollary 11.5 identifies the exact action of $\mathcal{P}_{N,Q}$ on odd germs as a shifted complete-homogeneous transform of the surviving odd Taylor coefficients.

In this sense, the N -point operator may be viewed as a local discrete Mellin-side projector. This is formally reminiscent of the Weil/Connes test-function viewpoint, where one probes the zero set through Mellin/Fourier transforms of multiplicative test functions and explicit- or trace-formula pairings [3, 10, 11]. The present paper does not use any global trace formula; the comparison is only structural.

A natural smoothed version is obtained by fixing an even $\phi \in C_c^\infty(\mathbb{R})$ with $\widehat{\phi}(0) = 1$ and setting

$$f_{N,Q,\eta}(x) := \sum_{j=1}^N a_j^{(N)} \eta^{-1} \phi\left(\frac{\log x - j/Q^2}{\eta}\right).$$

Its Mellin transform is

$$\widehat{f}_{N,Q,\eta}(z) = \widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z).$$

Hence the same low odd-jet annihilation persists after smoothing, while the finite-support projector is replaced by an honest smooth multiplicative test function. This observation is not used elsewhere in the proof, but it places the N -point cancellation layer in a more canonical operator-theoretic setting.

Lemma 11.30 (Smoothed Mellin projector). *Let $\phi \in C_c^\infty(\mathbb{R})$ be even and normalized by*

$$\int_{\mathbb{R}} \phi(u) du = 1.$$

Define its bilateral Laplace transform by

$$\widehat{\phi}(w) := \int_{\mathbb{R}} \phi(u) e^{wu} du.$$

For $\eta > 0$, define the smoothed additive projector

$$(\mathcal{P}_{N,Q,\eta}^\phi F) := \sum_{j=1}^N a_j^{(N)} \int_{\mathbb{R}} \phi(u) F\left(\frac{j}{Q^2} + \eta u\right) du,$$

whenever F is holomorphic on a neighborhood of the integration region.

Then:

(1) *for every exponential mode $F(s) = e^{zs}$,*

$$\mathcal{P}_{N,Q,\eta}^\phi(e^{zs}) = \widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z), \quad \widehat{\mu}_{N,Q}(z) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2};$$

(2) *the smoothed symbol preserves the exact low odd-jet annihilation:*

$$\widehat{\phi}(0) \widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1}(\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z)) \Big|_{z=0} = 0 \quad (0 \leq r \leq N-2);$$

(3) *if F is holomorphic on a neighborhood Ω containing all points*

$$\frac{j}{Q^2} + \eta u \quad (1 \leq j \leq N, u \in \text{supp } \phi),$$

then

$$|\mathcal{P}_{N,Q,\eta}^\phi F - \mathcal{P}_{N,Q} F| \leq \frac{\eta^2}{2} (2N-1) m_2(\phi) \sup_{\Omega} |F''|,$$

where

$$m_2(\phi) := \int_{\mathbf{R}} |u|^2 |\phi(u)| du.$$

Proof. For $F(s) = e^{zs}$,

$$\mathcal{P}_{N,Q,\eta}^\phi(e^{zs}) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2} \int_{\mathbf{R}} \phi(u) e^{\eta zu} du = \widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z),$$

which proves the first statement.

Since ϕ is even, $\widehat{\phi}$ is an even entire function. In particular,

$$\widehat{\phi}(0) = \int_{\mathbf{R}} \phi(u) du = 1, \quad \widehat{\phi}^{(2r+1)}(0) = 0 \quad (r \geq 0).$$

Lemma 11.2 gives

$$\widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0 \quad (0 \leq r \leq N-2).$$

Applying Leibniz' rule to the product $\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z)$ shows that all odd derivatives up to order $2N-3$ vanish at $z=0$, proving the second statement.

For the quantitative comparison, expand F at j/Q^2 :

$$F\left(\frac{j}{Q^2} + \eta u\right) = F\left(\frac{j}{Q^2}\right) + \eta u F'\left(\frac{j}{Q^2}\right) + \frac{\eta^2 u^2}{2} F''(\xi_{j,u})$$

for some $\xi_{j,u} \in \Omega$. Since ϕ is even,

$$\int_{\mathbf{R}} \phi(u) u du = 0.$$

Also $\int \phi(u) du = 1$. Therefore

$$\int_{\mathbf{R}} \phi(u) F\left(\frac{j}{Q^2} + \eta u\right) du - F\left(\frac{j}{Q^2}\right) = \frac{\eta^2}{2} \int_{\mathbf{R}} \phi(u) u^2 F''(\xi_{j,u}) du.$$

Summing against $a_j^{(N)}$ and using Lemma 11.1,

$$\sum_{j=1}^N |a_j^{(N)}| = 2N-1,$$

gives

$$|\mathcal{P}_{N,Q,\eta}^\phi F - \mathcal{P}_{N,Q} F| \leq \frac{\eta^2}{2} (2N-1) m_2(\phi) \sup_{\Omega} |F''|. \quad \square$$

Remark 11.31 (Smoothing scale). Since $N \asymp Q$, Lemma 11.30 gives

$$\mathcal{P}_{N,Q,\eta}^\phi F = \mathcal{P}_{N,Q} F + O(Q\eta^2 \sup |F''|).$$

Thus any choice $\eta = o(Q^{-1/2})$ makes the smoothing error $o(1)$ while preserving the exact low odd-jet annihilation. This does not enter the present proof, but it shows that the discrete N -point projector admits a genuinely smooth Mellin-side avatar with the same exact cancellation pattern.

Proposition 11.32 (Smoothed comparison for the corrected N -point zeta scalar). *Let $\phi \in C_c^\infty(\mathbf{R})$ be even, normalized by*

$$\int_{\mathbf{R}} \phi(u) du = 1,$$

and set

$$U_\phi := \sup\{|u| : u \in \text{supp } \phi\}, \quad m_2(\phi) := \int_{\mathbf{R}} |u|^2 |\phi(u)| du.$$

Fix $0 < \alpha < \rho_0$, let

$$N = \lfloor \alpha Q \rfloor,$$

and assume

$$0 < \eta \leq \frac{\rho_0 - \alpha}{2U_\phi Q}.$$

Define the smoothed corrected scalar

$$H_{m,\eta}(s) := \int_{\mathbf{R}} \phi(u) H_m(s + \eta u) du,$$

and the corresponding smoothed N -point scalar

$$\Xi_{\zeta,\eta}^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_{m,\eta}\left(\frac{j}{Q^2}\right).$$

Then:

(1) $H_{m,\eta}$ is odd and holomorphic on

$$|s| < \frac{\rho_\eta}{Q}, \quad \rho_\eta := \rho_0 - U_\phi \eta Q > \alpha.$$

(2) One has the zeta-side exponential estimate

$$|\Xi_{\zeta,\eta}^{(N)}(m)| \ll_{\phi,\rho_0,\alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha,\rho_\eta)Q},$$

where

$$\delta_*(\alpha, \rho_\eta) = \rho_\eta \left((1 + \beta_\eta) \log(1 + \beta_\eta) - 2\beta_\eta \log \beta_\eta - (1 - \beta_\eta) \log(1 - \beta_\eta) \right), \quad \beta_\eta := \frac{\alpha}{\rho_\eta}.$$

(3) The smoothed and discrete N -point scalars satisfy

$$|\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m)| \leq \frac{4C_\partial(\rho_0) m_2(\phi)}{(\rho_0 - \alpha)^2} \frac{2N - 1}{Q} \eta^2 L(m) S(m)^2.$$

In particular,

$$|\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m)| \ll_{\phi,\rho_0,\alpha} \eta^2 L(m) S(m)^2.$$

Proof. Since H_m is odd and ϕ is even,

$$H_{m,\eta}(-s) = \int_{\mathbf{R}} \phi(u) H_m(-s + \eta u) du = \int_{\mathbf{R}} \phi(u) H_m(-(s + \eta u)) du = -H_{m,\eta}(s),$$

so $H_{m,\eta}$ is odd. If $|s| < \rho_\eta/Q$ and $u \in \text{supp } \phi$, then

$$|s + \eta u| \leq |s| + \eta U_\phi < \frac{\rho_0 - U_\phi \eta Q}{Q} + \eta U_\phi = \frac{\rho_0}{Q}.$$

Since H_m is holomorphic on $|s| < \rho_0/Q$, it follows that $H_{m,\eta}$ is holomorphic on $|s| < \rho_\eta/Q$. This proves (1).

For $|s| = \rho_\eta/Q$ and $u \in \text{supp } \phi$, the same estimate gives

$$|s + \eta u| \leq \frac{\rho_0}{Q}.$$

Hence Proposition 10.1 implies

$$|H_m(s + \eta u)| \leq C_\partial(\rho_0) \frac{L(m)S(m)^2}{Q^3}.$$

Therefore

$$|H_{m,\eta}(s)| \leq \|\phi\|_{L^1} C_\partial(\rho_0) \frac{L(m)S(m)^2}{Q^3} \quad \left(|s| = \frac{\rho_\eta}{Q} \right).$$

Since $H_{m,\eta}$ is odd and holomorphic on $|s| < \rho_\eta/Q$, the proof of Corollary 11.22 applies verbatim with ρ_0 replaced by ρ_η . This gives

$$|\Xi_{\zeta,\eta}^{(N)}(m)| \ll_{\phi,\rho_0,\alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha,\rho_\eta)Q},$$

proving (2).

For (3), note that

$$\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m) = (\mathcal{P}_{N,Q,\eta}^\phi - \mathcal{P}_{N,Q})H_m.$$

Apply Lemma 11.30 with $F = H_m$. It remains to bound H_m'' on the relevant region. Since

$$\frac{j}{Q^2} \leq \frac{N}{Q^2} \leq \frac{\alpha}{Q}, \quad \eta U_\phi \leq \frac{\rho_0 - \alpha}{2Q},$$

all points

$$\frac{j}{Q^2} + \eta u \quad (1 \leq j \leq N, u \in \text{supp } \phi)$$

lie in the disk

$$|s| \leq \frac{\alpha}{Q} + \eta U_\phi \leq \frac{\rho_0 + \alpha}{2Q}.$$

Thus their distance to the boundary circle $|s| = \rho_0/Q$ is at least

$$\frac{\rho_0 - \alpha}{2Q}.$$

By Cauchy's derivative estimate and Proposition 10.1,

$$\sup_{\Omega} |H_m''| \leq \frac{2 C_{\partial}(\rho_0) L(m)S(m)^2/Q^3}{((\rho_0 - \alpha)/(2Q))^2} = \frac{8 C_{\partial}(\rho_0)}{(\rho_0 - \alpha)^2} \frac{L(m)S(m)^2}{Q}.$$

Lemma 11.30 therefore yields

$$|\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m)| \leq \frac{\eta^2}{2} (2N - 1) m_2(\phi) \frac{8 C_{\partial}(\rho_0)}{(\rho_0 - \alpha)^2} \frac{L(m)S(m)^2}{Q},$$

which is exactly the stated bound. □

Remark 11.33 (Microscopic smoothing regime). If

$$\eta \ll Q^{-2},$$

then Proposition 11.32 gives

$$\Xi_{\zeta,\eta}^{(N)}(m) = \Xi_\zeta^{(N)}(m) + O\left(\frac{L(m)S(m)^2}{Q^4}\right).$$

At the same time,

$$\rho_\eta = \rho_0 + O(Q^{-1}),$$

so

$$\delta_*(\alpha, \rho_\eta) = \delta_*(\alpha, \rho_0) + O(Q^{-1}).$$

Thus microscopic smoothing preserves the same zeta-side exponential scale while replacing the discrete projector by a genuinely smooth Mellin-side test function.

Lemma 11.34 (Higher-order smoothed Mellin projector). *Fix an integer $M \geq 1$. Let $\phi_M \in C_c^\infty(\mathbf{R})$ be even and satisfy*

$$\int_{\mathbf{R}} \phi_M(u) du = 1, \quad \int_{\mathbf{R}} u^{2\ell} \phi_M(u) du = 0 \quad (1 \leq \ell \leq M - 1).$$

Define its bilateral Laplace transform by

$$\widehat{\phi}_M(w) := \int_{\mathbf{R}} \phi_M(u) e^{wu} du.$$

For $\eta > 0$, define

$$(\mathcal{P}_{N,Q,\eta}^{(M)} F) := \sum_{j=1}^N a_j^{(N)} \int_{\mathbf{R}} \phi_M(u) F\left(\frac{j}{Q^2} + \eta u\right) du,$$

whenever F is holomorphic on a neighborhood of the integration region.

Then:

(1) for every exponential mode $F(s) = e^{zs}$,

$$\mathcal{P}_{N,Q,\eta}^{(M)}(e^{zs}) = \widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z), \quad \widehat{\mu}_{N,Q}(z) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2};$$

(2) one has

$$\widehat{\phi}_M(0) = 1, \quad \widehat{\phi}_M^{(k)}(0) = 0 \quad (1 \leq k \leq 2M-1);$$

(3) the smoothed symbol preserves the exact low odd-jet annihilation:

$$\partial_z^{2r+1}(\widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z)) \Big|_{z=0} = 0 \quad (0 \leq r \leq N-2);$$

(4) if F is holomorphic on a neighborhood Ω containing all points

$$\frac{j}{Q^2} + \eta u \quad (1 \leq j \leq N, u \in \text{supp } \phi_M),$$

then

$$|\mathcal{P}_{N,Q,\eta}^{(M)} F - \mathcal{P}_{N,Q} F| \leq \frac{\eta^{2M}}{(2M)!} (2N-1) m_{2M}(\phi_M) \sup_{\Omega} |F^{(2M)}|,$$

where

$$m_{2M}(\phi_M) := \int_{\mathbf{R}} |u|^{2M} |\phi_M(u)| du.$$

Proof. For $F(s) = e^{zs}$,

$$\mathcal{P}_{N,Q,\eta}^{(M)}(e^{zs}) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2} \int_{\mathbf{R}} \phi_M(u) e^{\eta zu} du = \widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z),$$

which proves the first statement.

Since ϕ_M is even, all odd moments vanish. By the imposed even-moment conditions, one also has

$$\int_{\mathbf{R}} u^k \phi_M(u) du = 0 \quad (1 \leq k \leq 2M-1).$$

Differentiating $\widehat{\phi}_M$ at 0 gives

$$\widehat{\phi}_M^{(k)}(0) = \int_{\mathbf{R}} u^k \phi_M(u) du,$$

so

$$\widehat{\phi}_M(0) = 1, \quad \widehat{\phi}_M^{(k)}(0) = 0 \quad (1 \leq k \leq 2M-1).$$

This proves the second statement.

By Lemma 11.2,

$$\widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0 \quad (0 \leq r \leq N-2).$$

Applying Leibniz' rule to $\widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z)$ shows that all odd derivatives up to order $2N-3$ vanish at $z=0$, proving the third statement.

For the quantitative comparison, expand F to order $2M-1$ at j/Q^2 :

$$F\left(\frac{j}{Q^2} + \eta u\right) = \sum_{k=0}^{2M-1} \frac{(\eta u)^k}{k!} F^{(k)}\left(\frac{j}{Q^2}\right) + \frac{(\eta u)^{2M}}{(2M)!} F^{(2M)}(\xi_{j,u})$$

for some $\xi_{j,u} \in \Omega$. Integrating against ϕ_M , all terms of order 1 through $2M - 1$ vanish by the moment conditions, while the constant term survives because $\int \phi_M = 1$. Hence

$$\int_{\mathbf{R}} \phi_M(u) F\left(\frac{j}{Q^2} + \eta u\right) du - F\left(\frac{j}{Q^2}\right) = \frac{\eta^{2M}}{(2M)!} \int_{\mathbf{R}} \phi_M(u) u^{2M} F^{(2M)}(\xi_{j,u}) du.$$

Summing against $a_j^{(N)}$ and using Lemma 11.1,

$$\sum_{j=1}^N |a_j^{(N)}| = 2N - 1,$$

gives the claimed bound. $\square$

Proposition 11.35 (Higher-order smoothed comparison for the corrected N -point zeta scalar). *Fix an integer $M \geq 1$, and let $\phi_M \in C_c^\infty(\mathbf{R})$ be even, normalized, and moment-corrected as in Lemma 11.34. Set*

$$U_M := \sup\{|u| : u \in \text{supp } \phi_M\}, \quad m_{2M}(\phi_M) := \int_{\mathbf{R}} |u|^{2M} |\phi_M(u)| du.$$

Fix $0 < \alpha < \rho_0$, let

$$N = \lfloor \alpha Q \rfloor,$$

and assume

$$0 < \eta \leq \frac{\rho_0 - \alpha}{2U_M Q}.$$

Define

$$H_{m,\eta}^{(M)}(s) := \int_{\mathbf{R}} \phi_M(u) H_m(s + \eta u) du,$$

and

$$\Xi_{\zeta,\eta}^{(N,M)}(m) := \sum_{j=1}^N a_j^{(N)} H_{m,\eta}^{(M)}\left(\frac{j}{Q^2}\right).$$

Then:

(1) $H_{m,\eta}^{(M)}$ is odd and holomorphic on

$$|s| < \frac{\rho_{M,\eta}}{Q}, \quad \rho_{M,\eta} := \rho_0 - U_M \eta Q > \alpha;$$

(2) one has

$$|\Xi_{\zeta,\eta}^{(N,M)}(m)| \ll_{\phi_M, \rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha, \rho_{M,\eta})Q};$$

(3) the smoothed and discrete N -point scalars satisfy

$$|\Xi_{\zeta,\eta}^{(N,M)}(m) - \Xi_{\zeta}^{(N)}(m)| \leq C_{\partial}(\rho_0) (2N - 1) m_{2M}(\phi_M) \left(\frac{2}{\rho_0 - \alpha}\right)^{2M} \eta^{2M} Q^{2M-3} L(m)S(m)^2,$$

and hence

$$|\Xi_{\zeta,\eta}^{(N,M)}(m) - \Xi_{\zeta}^{(N)}(m)| \ll_{\phi_M, \rho_0, \alpha} (\eta Q)^{2M} \frac{L(m)S(m)^2}{Q^2}.$$

Proof. The proof of oddness and holomorphy is identical to that of Proposition 11.32, with ϕ replaced by ϕ_M . Likewise, on $|s| = \rho_{M,\eta}/Q$,

$$|H_{m,\eta}^{(M)}(s)| \leq \|\phi_M\|_{L^1} C_{\partial}(\rho_0) \frac{L(m)S(m)^2}{Q^3}.$$

Therefore the proof of Corollary 11.22 applies verbatim with ρ_0 replaced by $\rho_{M,\eta}$, giving the stated exponential bound.

For the comparison estimate,

$$\Xi_{\zeta,\eta}^{(N,M)}(m) - \Xi_{\zeta}^{(N)}(m) = (\mathcal{P}_{N,Q,\eta}^{(M)} - \mathcal{P}_{N,Q})H_m.$$

Apply Lemma 11.34 with $F = H_m$. Exactly as in the proof of Proposition 11.32, all relevant points lie at distance at least

$$\frac{\rho_0 - \alpha}{2Q}$$

from the boundary circle $|s| = \rho_0/Q$. Hence Cauchy's derivative estimate and Proposition 10.1 give

$$\sup_{\Omega} |H_m^{(2M)}| \leq (2M)! C_{\partial}(\rho_0) \left(\frac{2Q}{\rho_0 - \alpha} \right)^{2M} \frac{L(m)S(m)^2}{Q^3}.$$

Substituting this into Lemma 11.34 yields

$$|\Xi_{\zeta, \eta}^{(N, M)}(m) - \Xi_{\zeta}^{(N)}(m)| \leq C_{\partial}(\rho_0) (2N - 1) m_{2M}(\phi_M) \left(\frac{2}{\rho_0 - \alpha} \right)^{2M} \eta^{2M} Q^{2M-3} L(m)S(m)^2,$$

as claimed. Since $N \asymp Q$, the final big-Oh form follows. $\square$

Remark 11.36 (Saturation at the natural Q^{-1} smoothing scale). Proposition 11.35 shows that higher-order moment correction improves the comparison error to order $2M$, but under the largest admissible smoothing scale

$$\eta \asymp Q^{-1},$$

one still has

$$|\Xi_{\zeta, \eta}^{(N, M)}(m) - \Xi_{\zeta}^{(N)}(m)| \ll \frac{L(m)S(m)^2}{Q^2}.$$

Thus higher-order smoothing does not improve the Q -power at the natural admissible scale; its gain is in canonicity and in the ability to make the smoothed projector arbitrarily close to the discrete one once one works at a smaller scale. For example, if

$$\eta = Q^{-1-\sigma} \quad (\sigma > 0),$$

then

$$|\Xi_{\zeta, \eta}^{(N, M)}(m) - \Xi_{\zeta}^{(N)}(m)| \ll Q^{-2-2M\sigma} L(m)S(m)^2.$$

We isolate the toy microscopic expansion in Appendix A; the only fact used in the main argument is the following N -point preservation statement.

Proposition 11.37 (Toy N -point transverse preservation). *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

let ρ_{toy} and $u_0(D)$ be as in Lemma A.1, and fix

$$\alpha > 0.$$

For

$$N = \lfloor \alpha Q \rfloor, \quad s_j = \frac{j}{Q^2},$$

define

$$\Xi_{\text{toy}}^{(N)}(u, d) := \sum_{j=1}^N a_j^{(N)} \Phi_K(\Delta_{\text{toy}}(u, d; s_j)).$$

Then, for all sufficiently large Q (depending on D and α), for $d \in D$ and $|u| < u_0(D)$,

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

uniformly in $d \in D$. In particular,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2$$

uniformly on compact d -ranges.

Proof. By Lemma A.1,

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

where $E(u, d; s)$ is analytic in s on $|s| < \rho_{\text{toy}}$ and uniformly bounded there for $d \in D$ and $|u| < u_0(D)$. Applying the linear functional Φ_K , define

$$R_{u,d}(s) := \Phi_K(E(u, d; s)).$$

Then $R_{u,d}(s)$ is analytic on $|s| < \rho_{\text{toy}}$, and by Cauchy's estimate its Taylor coefficients satisfy

$$R_{u,d}(s) = \sum_{n \geq 0} b_n(u, d) s^n, \quad |b_n(u, d)| \ll \rho_{\text{toy}}^{-n},$$

uniformly for $d \in D$ and $|u| < u_0(D)$.

Therefore

$$\Phi_K(\Delta_{\text{toy}}(u, d; s_j)) = u^2 \Phi_K(M(d)) + u^4 R_{u,d}(s_j),$$

so

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + u^4 \sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j).$$

Expand $R_{u,d}$ at the microscopic nodes $s_j = j/Q^2$:

$$\sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j) = \sum_{n \geq 0} b_n(u, d) \frac{\mu_n^{(N)}}{Q^{2n}}, \quad \mu_n^{(N)} := \sum_{j=1}^N a_j^{(N)} j^n.$$

The constant term contributes

$$b_0(u, d) \mu_0^{(N)} = b_0(u, d),$$

since $\mu_0^{(N)} = \sum_j a_j^{(N)} = 1$ by Lemma 11.1. For the remaining terms, split into even and odd parts.

For the even terms, Lemma 11.1 gives

$$|\mu_{2\ell}^{(N)}| \leq (2N-1)N^{2\ell},$$

so

$$\sum_{\ell \geq 1} |b_{2\ell}(u, d)| \frac{|\mu_{2\ell}^{(N)}|}{Q^{4\ell}} \ll (2N-1) \sum_{\ell \geq 1} \rho_{\text{toy}}^{-2\ell} \frac{N^{2\ell}}{Q^{4\ell}} \ll \frac{1}{Q},$$

because $N \asymp Q$.

For the odd terms, Lemma 11.1 gives

$$\mu_{2r+1}^{(N)} = 0 \quad (0 \leq r \leq N-2),$$

so only the tail $r \geq N-1$ remains. Using the coefficient bound $|b_{2r+1}(u, d)| \ll \rho_{\text{toy}}^{-(2r+1)}$, we obtain

$$\sum_{r \geq N-1} |b_{2r+1}(u, d)| \frac{|\mu_{2r+1}^{(N)}|}{Q^{4r+2}} \ll \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_{\text{toy}} Q^2)^{-(2r+1)}.$$

Now apply Lemma 11.3 with

$$x = \rho_{\text{toy}} Q^2.$$

For all sufficiently large Q , one has $x > N$, and therefore

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_{\text{toy}} Q^2)^{-(2r+1)} = (2N-1)N!(N-1)!(\rho_{\text{toy}} Q^2)^2 \frac{\Gamma(\rho_{\text{toy}} Q^2 - N)}{\Gamma(\rho_{\text{toy}} Q^2 + N + 1)}.$$

Since $N = \lfloor \alpha Q \rfloor$ and $x = \rho_{\text{toy}} Q^2$, Stirling's formula gives

$$(2N-1)N!(N-1)!(\rho_{\text{toy}} Q^2)^2 \frac{\Gamma(\rho_{\text{toy}} Q^2 - N)}{\Gamma(\rho_{\text{toy}} Q^2 + N + 1)} = \exp(-2\alpha Q \log Q + O(Q)).$$

In particular, the residual odd toy tail is super-exponentially small on the Q -scale:

$$\sum_{r \geq N-1} |b_{2r+1}(u, d)| \frac{|\mu_{2r+1}^{(N)}|}{Q^{4r+2}} \ll \exp(-2\alpha Q \log Q + O(Q)).$$

Combining the constant, even, and odd contributions,

$$\sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j) = b_0(u, d) + O(Q^{-1}) + O(e^{-cQ}),$$

and in particular

$$\sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j) = O(1),$$

uniformly for $d \in D$ and $|u| < u_0(D)$. Therefore

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

uniformly in $d \in D$.

Because $\Phi_K(M(d)) \neq 0$ for $d > 0$, and is bounded away from 0 on every compact interval $D \subset (0, \infty)$, the final estimate

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2$$

follows. □

Remark 11.38 (The toy odd tail is negligible beyond all e^{-cQ} -scales). The proof of Proposition 11.37 shows more than the $O(u^4)$ preservation statement used later: after N -point cancellation, the residual odd toy tail is of size

$$\exp(-2\alpha Q \log Q + O(Q)).$$

Thus on the toy side the N -point odd leakage is not merely exponentially small in Q ; it is negligible beyond every e^{-cQ} -scale.

On the toy side, Proposition 11.37 gives

$$(22) \quad \Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

so on compact d -ranges,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2.$$

Using (19),

$$(23) \quad |\Xi_{\text{toy}}^{(N)}(u, d)| \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

Remark 11.39 (Work in progress: calibrated size versus microscopic toy regime). As presently written, the final calibration freezes the comparison scale at a fixed value $x = x_0(D)$, which yields

$$u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

With only the crude global bounds used later in the contradiction,

$$S(m) \ll Q^2, \quad B_\zeta(m) \asymp Q,$$

this gives at best

$$u^2 \ll x_0 Q,$$

and therefore does not by itself force the calibrated toy parameter u to lie in the microscopic toy regime $|u| < u_0(D)$ required by Proposition 11.37.

A plausible repair is to avoid freezing the scale at $x_0(D)$, and instead to choose the comparison scale adaptively:

$$x = x(m) \in (0, x_0(D)].$$

Since the pairing and calibration statements are naturally formulated for $0 < x \leq x_0(D)$, one expects to choose $x(m)$ so that

$$u^2 \asymp \frac{x(m)}{B_\zeta(m)} S(m)$$

is forced to remain below the toy threshold $u_0(D)^2$. For example, if the implicit upper bound

$$u^2 \leq C \frac{x}{B_\zeta(m)} S(m)$$

is available uniformly, then one may try to take

$$x(m) := \min \left\{ x_0(D), \frac{u_0(D)^2}{2C} \frac{B_\zeta(m)}{S(m)} \right\},$$

which would guarantee

$$u^2 \leq \frac{u_0(D)^2}{2}.$$

To make this repair rigorous, however, the draft must still verify that the entire calibration chain remains valid uniformly in such a variable x -regime. In particular, one still needs to check all of the following:

- (1) The pairing and denominator estimates remain uniform for all $0 < x \leq x_0(D)$, including the regime $x \rightarrow 0$, so that the calibration functional is still well-defined and the main term continues to scale linearly in x .
- (2) The fixed-scaled statements currently written in terms of

$$x = B_\zeta(m)\sigma, \quad |\sigma| \asymp Q^{-1},$$

can be upgraded to a variable/shrinking regime of the form

$$x = B_\zeta(m)\sigma, \quad |\sigma| \leq c Q^{-1},$$

or an equivalent $0 < x \leq x_0(D)$ formulation, without loss of the transfer, whitening, and calibration estimates used later.

- (3) The remainder bound after applying the calibrated functional remains uniform in the variable- x regime; in particular, one needs the normalized error estimate to stay independent of x , or at least sufficiently mild as $x \rightarrow 0$, so that shrinking x to enforce $|u| < u_0(D)$ does not destroy the zeta-side control.
- (4) The cutoff choice required in the calibration remainder estimate can still be made uniformly when $x = x(m)$ is allowed to shrink. Concretely, one must check that the condition

$$R^3 \gg \frac{1}{xQ S(m)}$$

remains compatible with the admissible cutoff regime under the adaptive choice of $x(m)$.

- (5) The final contradiction theorem must be reformulated so that the toy-side main term is compared at the variable scale $x(m)$, rather than at the fixed scale $x_0(D)$, and one must verify that the resulting lower bound still dominates the exponentially small zeta-side bound.

Accordingly, the small- u issue does not presently appear to be a fundamental obstruction, but the current argument should be read as conditional on a variable- x calibration rewrite that has not yet been carried out in the draft.

The following remark records the current status of the toy model in the overall proof strategy.

Remark 11.40 (Work in progress: pair-like versus finite-core endgame). At the current stage of the draft, the toy anomaly model should be understood as a *pair-core first-odd-jet normal form*, not yet as a universal normal form for an arbitrary finite same-scale off-line core.

More precisely, in the pure pair model one has first-odd-jet nondegeneracy:

$$a_1^{\text{toy}}(u, d) = \kappa(d)u^2 + O(u^4), \quad \kappa(d) \neq 0,$$

so the first odd jet already detects the off-line anomaly.

However, the fixed-core analysis developed later in the draft shows that for a general finite same-scale core, cancellation may occur at the first odd order even when a genuine corrected anomaly survives at a higher odd order. Thus the toy model is presently justified as the correct first-odd-jet simplification only under a *pair-like core* hypothesis (or a perturbative enlargement thereof), not yet for an arbitrary finite core.

Accordingly, the argument now has a genuine fork:

- (1) **Pair-like branch.** If the actual same-scale core is perturbatively pair-like in the corrected microscopic sense, then the present toy-model first-jet obstruction should still apply, and the contradiction can be driven by the first odd coefficient.
- (2) **Finite-core branch.** If the actual same-scale core is not perturbatively pair-like, then the correct endgame must be reformulated in terms of a genuine finite-core anomaly model, and the obstruction should be attached not specifically to the first odd jet, but to the *first nonzero odd jet* of the corrected transverse scalar.

Thus the toy model is best viewed, at present, as the first-odd-jet simplification of a pair-like core, rather than as a universal model for all finite cores. This is one of the main places where the present manuscript should still be read as a work in progress.

12. LOCAL PROJECTIVE RIGIDITY OF THE ONE-PAIR AND MULTI-PAIR FAMILIES

Throughout this section, $\mathcal{Q}^{\text{abs}}$ denotes the abstract odd-germ quotient by the value/gauge family (not the height scale $Q \asymp \log |m|$ used elsewhere in the paper). For the first residual step on the cubic-generic region, we will also introduce an explicit finite-order model $\mathcal{R}_h^{(5)}$ through order 5. The latter supplements, but does not globally replace, the ambient abstract quotient $\mathcal{Q}^{\text{abs}}$.

12.1. Cubic leading line and the first residual class for the one-pair family. Let

$$F_h(z) \in \mathcal{Q}^{\text{abs}}[[z^2]]$$

denote the corrected quotient odd germ of the normalized one-pair family, where $\mathcal{Q}^{\text{abs}}$ is the abstract odd-germ quotient by the value/gauge family.

Lemma 12.1 (First-order value-family generator). *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Let G_0 and N_0 denote the coincident background same-point and mixed blocks, and write

$$X_0 := G_0^{-1/2}.$$

Let L be the Fréchet derivative of the inverse-square-root map $G \mapsto G^{-1/2}$ at G_0 , so that

$$(G_0 + \varepsilon H)^{-1/2} = X_0 + \varepsilon L(H) + O(\varepsilon^2).$$

If $\dot{G}_{-,1}$, $\dot{N}_1$, and $\dot{G}_{+,1}$ are the first-order odd pure-value variations of the left same-point block, the mixed block, and the right same-point block, respectively, then the induced first-order odd coefficient in the corrected local block is

$$B_1 = L(\dot{G}_{-,1}) N_0 X_0 + X_0 \dot{N}_1 X_0 + X_0 N_0 L(\dot{G}_{+,1}).$$

Equivalently, the first-order family $V_{\text{val}}^{(1)}$ is the image of the linear map

$$(\dot{G}_{-,1}, \dot{N}_1, \dot{G}_{+,1}) \mapsto L(\dot{G}_{-,1}) N_0 X_0 + X_0 \dot{N}_1 X_0 + X_0 N_0 L(\dot{G}_{+,1}).$$

Proof. The corrected local block is obtained by the analytic whitening map

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

Expand the three inputs to first odd order under a pure value deformation:

$$G_-(s) = G_0 + \varepsilon \dot{G}_{-,1} + O(\varepsilon^2), \quad N(s) = N_0 + \varepsilon \dot{N}_1 + O(\varepsilon^2), \quad G_+(s) = G_0 + \varepsilon \dot{G}_{+,1} + O(\varepsilon^2).$$

Then

$$G_{\pm}(s)^{-1/2} = X_0 + \varepsilon L(\dot{G}_{\pm,1}) + O(\varepsilon^2),$$

so multiplying

$$G_-(s)^{-1/2} N(s) G_+(s)^{-1/2}$$

to first order gives

$$X_0 N_0 X_0 + \varepsilon \left(L(\dot{G}_{-,1}) N_0 X_0 + X_0 \dot{N}_1 X_0 + X_0 N_0 L(\dot{G}_{+,1}) \right) + O(\varepsilon^2).$$

The coefficient of ε is exactly the first-order generator B_1 . □

Lemma 12.2 (Whitened first-order reduction). *At the coincident background one has*

$$N_0 = G_0.$$

If

$$\tilde{H} := X_0 H X_0$$

denotes whitening by $X_0 = G_0^{-1/2}$, then the first-order generator of Lemma 12.1 becomes

$$\tilde{B}_1 = \tilde{N}_1 - \frac{1}{2} \tilde{\dot{G}}_{-,1} - \frac{1}{2} \tilde{\dot{G}}_{+,1}.$$

Proof. Since $X_0 G_0 X_0 = I$, whitening conjugates the background to the identity. The Fréchet derivative of $G \mapsto G^{-1/2}$ at I is

$$L_I(H) = -\frac{1}{2}H.$$

Therefore, after whitening the formula of Lemma 12.1 becomes

$$\tilde{B}_1 = L_I(\tilde{\dot{G}}_{-,1}) I I + I \tilde{N}_1 I + I I L_I(\tilde{\dot{G}}_{+,1}) = \tilde{N}_1 - \frac{1}{2} \tilde{\dot{G}}_{-,1} - \frac{1}{2} \tilde{\dot{G}}_{+,1}. \quad \square$$

Lemma 12.3 (First-order same-point span). *The first-order pure-value same-point variations span the full symmetric subspace:*

$$\text{span}\{\tilde{\dot{G}}_{-,1}\} = \text{span}\{\tilde{\dot{G}}_{+,1}\} = \text{Sym}_2(\mathbf{C}).$$

Proof. The corrected same-point block has the form

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}.$$

Therefore an infinitesimal pure-value variation at one side gives

$$\delta G_{\pm} = \frac{1}{\pi} \begin{pmatrix} 2\delta q_{\pm} & \frac{1}{2}\delta q'_{\pm} \\ \frac{1}{2}\delta q'_{\pm} & \frac{1}{12}(\delta q''_{\pm} + 6q_{\pm,0}^2 \delta q_{\pm}) \end{pmatrix}.$$

As $\delta q_{\pm}$, $\delta q'_{\pm}$, and $\delta q''_{\pm}$ vary independently, the off-diagonal entry, the (1,1)-entry, and the pure (2,2)-direction may be varied independently. Hence the span is the full space $\text{Sym}_2(\mathbf{C})$. Whitening by X_0 preserves the symmetric subspace and is invertible, so the same conclusion holds for $\tilde{\dot{G}}_{\pm,1}$. □

Lemma 12.4 (First-order mixed skew direction). *Let*

$$\mathfrak{so}_2(\mathbf{C}) := \left\{ \begin{pmatrix} 0 & \xi \\ -\xi & 0 \end{pmatrix} : \xi \in \mathbf{C} \right\}.$$

Then the first-order pure-value mixed variations contain the full skew line:

$$\mathfrak{so}_2(\mathbf{C}) \subset \text{span}\{\tilde{N}_1\}.$$

Equivalently, the first-order mixed family contains a nonzero skew direction.

Proof. Freeze the same-point data and vary only the phase gap:

$$\Delta_m(s) \mapsto \Delta_m(s) + \varepsilon \delta \Delta(s), \quad \delta \Delta(s) = \theta s^3 + O(s^5), \quad \theta \neq 0.$$

Linearizing the mixed block formula in ε , the two off-diagonal entries vary with opposite sign, while the diagonal entries do not contribute at first odd order. More precisely,

$$\delta N_{12}(s) = \frac{\theta}{\pi} s + O(s^3), \quad \delta N_{21}(s) = -\frac{\theta}{\pi} s + O(s^3),$$

up to the overall sign convention for θ . Hence the first odd mixed coefficient of this phase-gap variation is a nonzero element of

$$\mathfrak{so}_2(\mathbf{C}).$$

Whitening by X_0 is invertible, so the whitened first-order mixed family also contains a nonzero skew element. Since $\mathfrak{so}_2(\mathbf{C})$ is one-dimensional, this is equivalent to

$$\mathfrak{so}_2(\mathbf{C}) \subset \text{span}\{\tilde{N}_1\}.$$

□

Proposition 12.5 (Generic triviality of the first quotient order). *The first-order pure-value family fills the full matrix space:*

$$V_{\text{val}}^{(1)} = M_2(\mathbf{C}).$$

Consequently

$$\mathcal{J}_1/V_{\text{val}}^{(1)} = 0,$$

so the generic leading quotient order is at least 3.

Proof. By Lemma 12.2, the first-order generator is

$$\tilde{B}_1 = \tilde{N}_1 - \frac{1}{2}\tilde{G}_{-,1} - \frac{1}{2}\tilde{G}_{+,1}.$$

By Lemma 12.3, the same-point directions $\tilde{G}_{-,1}$ and $\tilde{G}_{+,1}$ span the full symmetric subspace $\text{Sym}_2(\mathbf{C})$. By Lemma 12.4, the mixed directions contain the skew line

$$\mathfrak{so}_2(\mathbf{C}).$$

Therefore the image of the generator contains

$$\text{Sym}_2(\mathbf{C}) \oplus \mathfrak{so}_2(\mathbf{C}) = M_2(\mathbf{C}).$$

Hence

$$V_{\text{val}}^{(1)} = M_2(\mathbf{C}),$$

and so

$$\mathcal{J}_1/V_{\text{val}}^{(1)} = 0.$$

Thus the generic leading quotient order is at least 3. □

Remark 12.6 (Raw versus full value family). The first-order surjectivity statement concerns the full value family as it appears in the three-input derivative

$$D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+].$$

This is larger than the raw swap-odd one-pair slice obtained by tying the left and right same-point variations together. In particular, at first order the combined family $V_{\text{val}}^{(1)}$ fills $M_2(\mathbf{C})$, while the odd coefficients of the raw corrected local block are constrained by additional swap/parity structure. Thus the present notation

$$V_{\text{val}}^{(2j+1)}$$

bundles together at least two logically distinct sources of ambiguity: raw pure-value odd jets and additional normalization/basis-gauge directions. The explicit finite-order model introduced later makes this split rigorous through order 5; outside that local model, the ambient notation $V_{\text{val}}^{(2j+1)}$ remains schematic.

Lemma 12.7 (Rigidity of the cubic scalar B). *Let I be a connected parameter interval, and define*

$$B(h) := \frac{q(h)^2 q'(h)}{4} - \frac{q'''(h)}{48}.$$

Then the following are equivalent:

- (1) $B \equiv 0$ on I ;
- (2) $q''' = 12q^2 q'$ on I ;

(3) there exists a constant $C_0 \in \mathbf{C}$ such that

$$q'' = 4q^3 + C_0 \quad \text{on } I.$$

In particular, the hypothesis $B \not\equiv 0$ is equivalent to excluding the special cubic ODE $q'' = 4q^3 + C_0$.

Proof. By definition,

$$48 B = 12q^2 q' - q''.$$

Hence $B \equiv 0$ is equivalent to

$$q''' = (4q^3)'.$$

Integrating on the connected interval I gives

$$q'' = 4q^3 + C_0$$

for some constant $C_0 \in \mathbf{C}$. The converse is immediate by differentiation. $\square$

Proposition 12.8 (Generic cubic leading quotient order). *Let I be a connected one-pair parameter interval, and let $J_3^{\text{pair}}(h)$ denote the cubic jet of the normalized one-pair family.*

Assume:

(1) the cubic value-family satisfies

$$V_{\text{val}}^{(3)} \subset \left\{ \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \right\};$$

(2) the background same-point block is nondegenerate, so that the diagonal entries of $X_0 = G_0^{-1/2}$ satisfy $x_1 \neq x_2$;

(3) the scalar $v(h)$ is not identically zero on I ;

(4) $B(h) \not\equiv 0$ on I , equivalently, by Lemma 12.7, q does not satisfy

$$q'' = 4q^3 + C_0$$

identically on I .

Define the cubic-generic region

$$\mathcal{U}_{\text{cub}} := \{ h \in I : v(h)B(h) \neq 0 \}.$$

Then $\mathcal{U}_{\text{cub}}$ is open and dense in I , and for every $h \in \mathcal{U}_{\text{cub}}$,

$$[J_3^{\text{pair}}(h)] \neq 0 \quad \text{in} \quad \mathcal{J}_3/V_{\text{val}}^{(3)}.$$

More precisely, the image of $J_3^{\text{pair}}(h)$ in the cubic quotient has nonzero diagonal-traceless component

$$v(h) B(h) (x_1 - x_2) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Consequently,

$$F_h(z) = v_0(h)z^3 + v_1(h)z^5 + \cdots, \quad v_0(h) := [J_3^{\text{pair}}(h)] \neq 0$$

for $h \in \mathcal{U}_{\text{cub}}$. Since the first quotient order is generically trivial, the generic leading quotient order is therefore exactly 3 on $\mathcal{U}_{\text{cub}}$.

Proof. By assumption,

$$\text{diag } J_3^{\text{pair}}(h) = v(h) B(h) (x_1 - x_2) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Because $V_{\text{val}}^{(3)}$ is confined to the off-diagonal sector, it cannot cancel any nonzero diagonal component. Hence for every h with $v(h)B(h)(x_1 - x_2) \neq 0$, the cubic quotient class $[J_3^{\text{pair}}(h)]$ is nonzero.

Since $x_1 \neq x_2$ is fixed and both v and B are not identically zero, the set $\mathcal{U}_{\text{cub}}$ is nonempty; if the dependence on h is real-analytic, its complement is a proper analytic subset, hence $\mathcal{U}_{\text{cub}}$ is open and dense. Finally, Proposition 12.5 shows that the leading quotient order is at least 3, while the present argument shows that it is equal to 3 on $\mathcal{U}_{\text{cub}}$. $\square$

Remark 12.9 (What is genuinely generic here). The cubic-leading statement depends on three logically distinct inputs:

- (1) $x_1 \neq x_2$, a background same-point nondegeneracy condition;
- (2) $v \neq 0$, a one-pair nonfreezing condition;
- (3) exclusion of the special ODE $q'' = 4q^3 + C_0$, equivalently $B \neq 0$.

This separates the structural cubic mechanism from the residual quintic bookkeeping needed later.

Definition 12.10 (Cubic leading line and first residual class). For $h \in \mathcal{U}_{\text{cub}}$, the cubic quotient coefficient

$$v_0(h) = [J_3^{\text{pair}}(h)]$$

is nonzero by Proposition 12.8. Define the cubic leading line by

$$\ell_h := \mathbf{C} v_0(h) \subset \mathcal{Q}^{\text{abs}}.$$

Let $v_1(h) := [J_5^{\text{pair}}(h)]$ denote the quintic quotient coefficient. Its image in the residual quotient

$$\mathcal{Q}^{\text{abs}}/\ell_h$$

is denoted by

$$\bar{v}_1(h) \in \mathcal{Q}^{\text{abs}}/\ell_h.$$

12.2. Normalization gauge at the first residual step. The phrase “value/gauge family” bundles together two logically distinct sources of ambiguity:

- (1) the *pure value family*, recorded coefficientwise by the spaces $V_{\text{val}}^{(2j+1)} \subset \mathcal{J}_{2j+1}$, and
- (2) the *normalization gauge*, consisting of even scalar renormalizations and even jet-basis changes.

For the first residual step it is useful to separate these two.

Definition 12.11 (Normalization gauge). Let $F(z)$ be a matrix-valued odd germ. The normalization gauge acts by

$$F(z) \mapsto a(z) P(z) F(z) P(z)^{-1},$$

where

$$a(z) = 1 + c_2 z^2 + c_4 z^4 + \dots$$

is an even scalar germ with $a(0) = 1$, and

$$P(z) = I + B_2 z^2 + B_4 z^4 + \dots$$

is an even 2×2 matrix germ with $P(0) = I$.

Proposition 12.12 (First-residual transformation law). *Let*

$$F(z) = A_3 z^3 + A_5 z^5 + O(z^7)$$

be an odd matrix germ, and let

$$F^{\text{new}}(z) := a(z) P(z) F(z) P(z)^{-1}$$

with a, P as in Definition 12.11. Then

$$A_3^{\text{new}} = A_3, \quad A_5^{\text{new}} = A_5 + c_2 A_3 + [B_2, A_3].$$

Proof. Expand

$$P(z)^{-1} = I - B_2 z^2 + O(z^4).$$

Then

$$P(z) F(z) P(z)^{-1} = (I + B_2 z^2) (A_3 z^3 + A_5 z^5) (I - B_2 z^2) + O(z^7),$$

so

$$P(z) F(z) P(z)^{-1} = A_3 z^3 + (A_5 + [B_2, A_3]) z^5 + O(z^7).$$

Multiplying by

$$a(z) = 1 + c_2 z^2 + O(z^4)$$

gives

$$F^{\text{new}}(z) = A_3 z^3 + (A_5 + c_2 A_3 + [B_2, A_3]) z^5 + O(z^7),$$

as claimed. $\square$

Corollary 12.13 (Residual effect of normalization gauge). *Let $\ell := \mathbf{C}A_3$. Under the normalization gauge, the quintic coefficient changes by*

$$A_5 \mapsto A_5 + c_2 A_3 + [B_2, A_3].$$

Hence its class in the residual quotient modulo the leading line satisfies

$$[A_5] \mapsto [A_5 + [B_2, A_3]] \quad \text{in} \quad \mathcal{I}_5/\ell.$$

In particular, scalar renormalization is invisible in the first residual quotient.

Proof. This is immediate from Proposition 12.12, since $c_2 A_3 \in \ell$. $\square$

Corollary 12.14 (Trace invariance under normalization gauge). *If A_3 is diagonal-traceless, then every commutator $[B_2, A_3]$ is traceless. Consequently*

$$\text{tr}(A_5^{\text{new}}) = \text{tr}(A_5)$$

for every normalization-gauge transform F^{new} of F .

Proof. The scalar renormalization term $c_2 A_3$ is traceless because A_3 is. The commutator term is traceless because

$$\text{tr}[B_2, A_3] = 0.$$

Hence Proposition 12.12 gives

$$\text{tr}(A_5^{\text{new}}) = \text{tr}(A_5).$$

$\square$

Proposition 12.15 (Second-residual transformation law). *Let*

$$F(z) = A_3 z^3 + A_5 z^5 + A_7 z^7 + O(z^9)$$

be an odd matrix germ, and let

$$F^{\text{new}}(z) := a(z) P(z) F(z) P(z)^{-1}$$

with a, P as in Definition 12.11. Then

$$A_3^{\text{new}} = A_3, \quad A_5^{\text{new}} = A_5 + c_2 A_3 + [B_2, A_3],$$

and the septic coefficient transforms by

$$A_7^{\text{new}} = A_7 + c_2 A_5 + c_4 A_3 + [B_2, A_5] + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Proof. By Proposition 12.12, the formulas for A_3^{new} and A_5^{new} are already known. It remains to compute the z^7 -coefficient.

Expand

$$P(z)^{-1} = I - B_2 z^2 + (B_2^2 - B_4) z^4 + O(z^6).$$

Then

$$P(z) F(z) P(z)^{-1} = (I + B_2 z^2 + B_4 z^4) (A_3 z^3 + A_5 z^5 + A_7 z^7) (I - B_2 z^2 + (B_2^2 - B_4) z^4) + O(z^9).$$

Collecting the z^7 -terms gives

$$A_7 + [B_2, A_5] + [B_4, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Multiplying by

$$a(z) = 1 + c_2 z^2 + c_4 z^4 + O(z^6)$$

adds the scalar contributions

$$c_2 A_5 + c_4 A_3 + c_2 [B_2, A_3].$$

Hence

$$A_7^{\text{new}} = A_7 + c_2 A_5 + c_4 A_3 + [B_2, A_5] + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2,$$

as claimed. $\square$

Lemma 12.16 (Quintic trace witness). *Let $J_{5,K_1}(h)$ denote the K_1 -transport contribution to the quintic jet. At the trace level, the relevant contribution is*

$$\mathrm{tr} J_{5,K_1}(h) = \kappa(h) \mathrm{tr} \left(X_0 J X_4(h) - X_4(h) J X_0 + X_3(h) J X_1(h) - X_1(h) J X_3(h) \right),$$

since

$$\mathrm{tr} (X_2(h) J X_2(h)) = 0$$

for symmetric $X_2(h)$. Therefore

$$\mathrm{tr} J_{5,K_1}(h) = -2\kappa(h) \left((x_1 - x_2)r(h) + \beta(h)(u(h) - w(h)) + v(h)(\gamma(h) - \alpha(h)) \right),$$

where $K_1 = \kappa(h)J$, $v(h)$ is the off-diagonal part of $X_1(h)$, and $\gamma(h) - \alpha(h)$ is the diagonal-traceless part of $X_3(h)$.

Moreover, the linear G_3 -contribution to $\gamma(h) - \alpha(h)$ is

$$(\gamma - \alpha)^{\mathrm{lin}} = \frac{1}{2\pi} \left[x_1^3 \frac{q'''}{24} - x_2^3 \left(\frac{q''''}{576} + \frac{q^2 q'''}{96} + \frac{qq'q''}{16} + \frac{(q')^3}{48} \right) \right].$$

Since

$$v(h) = -\frac{1}{4\pi(s_1 + s_2)} \eta_2(h),$$

the product $v(h)(\gamma(h) - \alpha(h))^{\mathrm{lin}}$ contains the monomial

$$\kappa(h) \eta_2(h) q''''(h)$$

with nonzero coefficient.

Lemma 12.17 (Parity of the whitening expansion at a diagonal base). *Let*

$$G_0 = \mathrm{diag}(\lambda_1, \lambda_2), \quad \lambda_1, \lambda_2 > 0,$$

and let

$$X(H) := (G_0 + H)^{-1/2}$$

for H sufficiently small. Write

$$E := \left\{ \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix} \right\}, \quad O := \left\{ \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \right\}.$$

Then every multilinear Taylor coefficient of $X(H)$ has definite matrix parity: a coefficient depending on perturbation factors $H_1, \dots, H_n$ lies in E if an even number of the H_j lie in O , and lies in O if an odd number of the H_j lie in O .

In particular, diagonal input gives diagonal output at linear order, and a bilinear coefficient with one diagonal and one off-diagonal input is off-diagonal.

Proof. Use the holomorphic functional calculus

$$(G_0 + H)^{-1/2} = \frac{1}{2\pi i} \oint_{\Gamma} z^{-1/2} (zI - G_0 - H)^{-1} dz,$$

where Γ encloses the spectrum of $G_0 + H$. Expanding the resolvent,

$$(zI - G_0 - H)^{-1} = R(z) + R(z)HR(z) + R(z)HR(z)HR(z) + \dots,$$

with

$$R(z) := (zI - G_0)^{-1},$$

every factor $R(z)$ is diagonal because G_0 is diagonal. Hence the matrix parity of each word

$$R(z)H_{i_1}R(z)H_{i_2} \cdots R(z)H_{i_n}R(z)$$

is exactly the parity of the product $H_{i_1} \cdots H_{i_n}$. Since

$$E \cdot E \subset E, \quad E \cdot O \subset O, \quad O \cdot E \subset O, \quad O \cdot O \subset E,$$

the parity rule follows. Integrating over z preserves the subspaces E and O . $\square$

Lemma 12.18 (No-cancellation of the quintic trace witness). *The monomial*

$$\kappa(h) \eta_2(h) q''''(h)$$

appearing in Lemma 12.16 is not produced by any other quintic contribution to $\text{tr } J_5^{\text{pair}}(h)$. More precisely, on the graded slice corresponding to

$$(K_1, \eta_2, q^{(5)}),$$

the only contributing channels in $\text{tr } J_5^{\text{pair}}(h)$ are

$$-2\kappa(h) (x_1 - x_2)r(h) \quad \text{and} \quad -2\kappa(h) v(h)(\gamma(h) - \alpha(h)),$$

and their total coefficient of $\kappa(h)\eta_2(h)q^{(5)}(h)$ is nonzero. Consequently

$$\text{tr } J_5^{\text{pair}}(h) \not\equiv 0$$

on the generic one-pair region.

Proof. Fix h and isolate the graded slice

$$\kappa(h) \eta_2(h) q^{(5)}(h).$$

We first exclude all non- K_1 contributions. Same-point quintic terms carry no κ -factor, so they cannot contribute. A K_5 -transport term leaves no same-point degree available, and a K_3 -transport term leaves only same-point degree 2, whereas the factor $q^{(5)}(h)$ first appears in same-point degree 3. Hence the target monomial can occur only in the K_1 -transport sector.

Now consider the same-point expansion. By the explicit formula

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix},$$

the factor $q^{(5)}(h)$ first appears only in the cubic same-point coefficient G_3 , and there only in the diagonal sector, namely through the cubic Taylor term of the $(2, 2)$ -entry coming from $q''(t_{\pm})$. Therefore, by Lemma 12.17, the linear $q^{(5)}$ -contribution to X_3 is diagonal. In particular it contributes to $\gamma - \alpha$, but not to β .

Likewise, the $\eta_2(h)$ -slice of $X_1(h)$ is off-diagonal, because

$$v(h) = -\frac{1}{4\pi(s_1 + s_2)} \eta_2(h).$$

Applying Lemma 12.17 again, the mixed $(\eta_2, q^{(5)})$ -slice of X_4 is off-diagonal. Hence it contributes to r , but not to the diagonal difference $u - w$.

It follows that on the graded slice $(K_1, \eta_2, q^{(5)})$, the term

$$-2\kappa(h) \beta(h)(u(h) - w(h))$$

does not contribute at all. Thus the only surviving K_1 -channels are

$$-2\kappa(h) (x_1 - x_2)r(h) \quad \text{and} \quad -2\kappa(h) v(h)(\gamma(h) - \alpha(h)).$$

To compute their combined coefficient, it suffices by multilinearity to restrict to the two perturbation directions carrying the relevant tags: an off-diagonal first-order perturbation A for the η_2 -slice and a diagonal third-order perturbation D for the $q^{(5)}$ -slice. Accordingly, write

$$G_0 = \text{diag}(a^2, b^2), \quad A = \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix},$$

and expand

$$X(\varepsilon, \delta) := (G_0 + \varepsilon A + \delta D)^{-1/2} = X_0 + \varepsilon U + \delta V + \varepsilon \delta W + O(\varepsilon^2, \delta^2).$$

Solving the identity

$$X(\varepsilon, \delta) (G_0 + \varepsilon A + \delta D) X(\varepsilon, \delta) = I$$

order by order gives

$$U_{12} = -\frac{u}{ab(a+b)}, \quad V_{22} = -\frac{d}{2b^3}, \quad W_{12} = \frac{du(a+2b)}{2ab^3(a+b)^2}.$$

Here U_{12} is the η_2 -slice of v , V_{22} is the $q^{(5)}$ -slice of $\gamma - \alpha$, and W_{12} is the mixed $(\eta_2, q^{(5)})$ -slice of r . Therefore the total coefficient of the target slice inside

$$(x_1 - x_2)r + v(\gamma - \alpha)$$

is

$$\left(\frac{1}{a} - \frac{1}{b}\right) W_{12} + U_{12}V_{22} = \frac{du}{a^2b^2(a+b)^2},$$

which is nonzero.

Thus the monomial $\kappa(h)\eta_2(h)q^{(5)}(h)$ occurs in $\text{tr } J_5^{\text{pair}}(h)$ with nonzero total coefficient and is not canceled by any other quintic contribution. Hence

$$\text{tr } J_5^{\text{pair}}(h) \neq 0.$$

□

12.3. Explicit finite-order model of the first residual quotient. For the first residual step it is enough to work with an explicit finite-order model through order 5 on the cubic-generic region. In this range the explicit model is not merely a proxy: Proposition 12.24 canonically identifies it with the order- ≤ 5 truncation of the ambient abstract quotient $\mathcal{Q}^{\text{abs}}$ after passage to the cubic line.

Accordingly, the first residual quintic detector is now rigorous at the finite-order ambient level. What is not addressed here is only a stronger global identification for later uses of $\mathcal{Q}^{\text{abs}}$ beyond order 5 and beyond the first residual layer.

Definition 12.19 (Finite-order truncation of the ambient abstract quotient). Fix $h \in \mathcal{U}_{\text{cub}}$, and let $A_3(h)$ represent the cubic leading line $\ell_h \subset \mathcal{Q}^{\text{abs}}$.

Define the order- ≤ 5 truncation of the ambient abstract quotient at h to be the quotient

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} := \frac{\mathbf{C} A_3(h)z^3 \oplus \mathcal{J}_5 z^5}{(V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)})z^5}.$$

Equivalently, $\mathcal{Q}_h^{\text{abs}(\leq 5)}$ is the cubic–quintic odd-germ quotient in which the quintic ambiguity is generated exactly by the raw pure-value quintic slice and the realized normalization-gauge shadows through order 5.

Its first residual quotient is

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} / \ell_h \cong \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}).$$

Definition 12.20 (Swap involution on cubic–quintic germs). Let

$$J_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For a truncated odd germ

$$F(z) = A_3 z^3 + A_5 z^5 + O(z^7)$$

define

$$\Sigma(F)(z) := J_0 F(-z) J_0 = -J_0 A_3 J_0 z^3 - J_0 A_5 J_0 z^5 + O(z^7).$$

Thus the induced involution on quintic coefficients is

$$\Sigma_5(A) := -J_0 A J_0.$$

Definition 12.21 (Cubic-normalized swap-symmetric raw corrected germ through order 5). Fix $h \in \mathcal{U}_{\text{cub}}$, and choose a diagonal-traceless matrix

$$A_3(h)$$

representing the cubic leading line ℓ_h .

A cubic-normalized swap-symmetric raw corrected germ through order 5 at h is a truncated odd germ of the form

$$R_h(z) = A_3(h)z^3 + A_5 z^5 + O(z^7)$$

obtained from the corrected one-pair local block by a cubic normalization that preserves the swap symmetry inherited from symmetric placement.

Its raw pure-value quintic slice is the coefficient space

$$V_{\text{raw}}^{(5)}(h)$$

consisting of all quintic coefficients A_5 arising in this way under pure value deformations at the fixed parameter h .

Lemma 12.22 (Raw quintic slice is anti-fixed). *For every $h \in \mathcal{U}_{\text{cub}}$,*

$$V_{\text{raw}}^{(5)}(h) \subset \{A \in M_2(\mathbf{C}) : J_0 A J_0 = -A\}.$$

In particular,

$$\text{tr}(V_{\text{raw}}^{(5)}(h)) = 0.$$

Proof. By Definition 12.21, every cubic-normalized raw corrected representative satisfies

$$R_h(-z) = J_0 R_h(z) J_0.$$

Writing

$$R_h(z) = A_3(h)z^3 + A_5z^5 + O(z^7),$$

coefficient comparison at order z^5 gives

$$J_0 A_5 J_0 = -A_5.$$

Hence every $A_5 \in V_{\text{raw}}^{(5)}(h)$ is anti-fixed. If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

satisfies $J_0 A J_0 = -A$, then

$$\begin{pmatrix} d & c \\ b & a \end{pmatrix} = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix},$$

so $d = -a$, and therefore $\text{tr}(A) = 0$. □

Definition 12.23 (Quintic ambiguity space and explicit first residual quotient). For $h \in \mathcal{U}_{\text{cub}}$, let $V_{\text{raw}}^{(5)}(h)$ be the raw pure-value quintic slice from Definition 12.21.

Define the normalization-gauge quintic shadow space $\mathcal{N}_h^{(5)}$ to be the set of all differences

$$\tilde{A}_5 - A_5$$

arising from two cubic-normalized swap-symmetric raw corrected germs

$$R_h(z) = A_3(h)z^3 + A_5z^5 + O(z^7), \quad \tilde{R}_h(z) = A_3(h)z^3 + \tilde{A}_5z^5 + O(z^7),$$

of the same raw corrected local object through order 5, when those two representatives are related by the allowed normalization gauge.

By Corollary 12.13, every such shadow has the form

$$\tilde{A}_5 - A_5 = c_2 A_3(h) + [B_2, A_3(h)],$$

but $\mathcal{N}_h^{(5)}$ is defined to consist only of the actually realized gauge shadows.

Set

$$V_{\text{amb}}^{(5)}(h) := V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}.$$

Define the explicit first residual quotient by

$$\mathcal{R}_h^{(5)} := \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h)).$$

Proposition 12.24 (Finite-order comparison theorem). *For each $h \in \mathcal{U}_{\text{cub}}$, the order- ≤ 5 truncation of the ambient abstract quotient and the explicit first residual model agree canonically:*

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} / \ell_h \cong \mathcal{R}_h^{(5)}.$$

Equivalently, after passing to the cubic-generic region and truncating through order 5, the first residual ambiguity in the ambient abstract quotient is exactly the ambiguity built into Definition 12.23.

Proof. By Definition 12.19,

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} = \frac{\mathbf{C} A_3(h) z^3 \oplus \mathcal{J}_5 z^5}{(V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}) z^5}.$$

Passing to the first residual quotient by the cubic line $\ell_h = \mathbf{C} A_3(h)$ gives

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} / \ell_h \cong \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}).$$

By Definition 12.23,

$$V_{\text{amb}}^{(5)}(h) = V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)},$$

so the right-hand side is exactly

$$\mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h)) = \mathcal{R}_h^{(5)}.$$

This is the claimed canonical identification. $\square$

Lemma 12.25 (Gauge shadows are anti-fixed). *For every $h \in \mathcal{U}_{\text{cub}}$,*

$$\mathcal{N}_h^{(5)} \subset \{A \in M_2(\mathbf{C}) : J_0 A J_0 = -A\}.$$

In particular,

$$\text{tr}(\mathcal{N}_h^{(5)}) = 0.$$

Proof. Let

$$R_h(z) = A_3(h) z^3 + A_5 z^5 + O(z^7), \quad \tilde{R}_h(z) = A_3(h) z^3 + \tilde{A}_5 z^5 + O(z^7)$$

be two cubic-normalized swap-symmetric representatives of the same raw corrected local object through order 5, related by the allowed normalization gauge. By Definition 12.21, both satisfy

$$R_h(-z) = J_0 R_h(z) J_0, \quad \tilde{R}_h(-z) = J_0 \tilde{R}_h(z) J_0.$$

Comparing coefficients at order z^5 gives

$$J_0 A_5 J_0 = -A_5, \quad J_0 \tilde{A}_5 J_0 = -\tilde{A}_5.$$

Subtracting,

$$J_0 (\tilde{A}_5 - A_5) J_0 = -(\tilde{A}_5 - A_5).$$

Since $\tilde{A}_5 - A_5 \in \mathcal{N}_h^{(5)}$ by Definition 12.23, every gauge shadow is anti-fixed. Tracelessness follows as in Lemma 12.22. $\square$

Proposition 12.26 (Trace-free quintic ambiguity in the explicit finite-order model). *For every $h \in \mathcal{U}_{\text{cub}}$,*

$$V_{\text{amb}}^{(5)}(h) \subset \{A \in M_2(\mathbf{C}) : J_0 A J_0 = -A\}.$$

Equivalently, the involution Σ_5 preserves the quintic ambiguity space and acts trivially on it.

In particular,

$$\text{tr}(V_{\text{amb}}^{(5)}(h)) = 0.$$

Proof. Combine Lemma 12.22 and Lemma 12.25; by Definition 12.23,

$$V_{\text{amb}}^{(5)}(h) = V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}.$$

$\square$

Corollary 12.27 (Trace descends on the explicit first residual quotient). *For every $h \in \mathcal{U}_{\text{cub}}$, trace annihilates*

$$\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h).$$

Hence trace descends to a well-defined linear functional

$$\tau_h : \mathcal{R}_h^{(5)} \rightarrow \mathbf{C}.$$

Proof. By Proposition 12.26, trace vanishes on $V_{\text{amb}}^{(5)}(h)$. It also vanishes on $\mathbf{C} A_3(h)$ because $A_3(h)$ is diagonal-traceless. $\square$

Proposition 12.28 (Generic first residual nonvanishing in the explicit finite-order model). *There exists an open dense subset*

$$\mathcal{U}_{\text{res}} \subset \mathcal{U}_{\text{cub}}$$

such that

$$\bar{v}_1(h) \neq 0 \quad \text{in } \mathcal{R}_h^{(5)}$$

for every $h \in \mathcal{U}_{\text{res}}$.

Hence the one-pair quotient odd germ is not projectively constant on $\mathcal{U}_{\text{res}}$ at the explicit first residual level.

Proof. By Corollary 12.27, trace defines a linear functional

$$\tau_h : \mathcal{R}_h^{(5)} \rightarrow \mathbf{C}.$$

By Lemma 12.18,

$$\text{tr } J_5^{\text{pair}}(h) \neq 0$$

on the cubic-generic region. Hence there is an open dense subset

$$\mathcal{U}_{\text{res}} \subset \mathcal{U}_{\text{cub}}$$

on which

$$\text{tr } J_5^{\text{pair}}(h) \neq 0.$$

For such h ,

$$\tau_h(\bar{v}_1(h)) = \text{tr } J_5^{\text{pair}}(h) \neq 0,$$

so $\bar{v}_1(h) \neq 0$ in $\mathcal{R}_h^{(5)}$. □

Remark 12.29 (Status of the explicit first residual model). The first residual step is now closed at the finite-order ambient level. Indeed, Proposition 12.24 identifies the explicit quotient $\mathcal{R}_h^{(5)}$ canonically with the order- ≤ 5 truncation of the ambient abstract quotient $\mathcal{Q}^{\text{abs}}$ modulo the cubic line on the cubic-generic region.

Thus the remaining issue is no longer a finite-order comparison theorem for the first residual layer. What remains, if one wants a stronger global ambient identification, is a separate theorem showing that later uses of $\mathcal{Q}^{\text{abs}}$ beyond order 5 do not introduce any new ambiguity relevant to the downstream multi-pair and global arguments.

Remark 12.30 (Beyond the first residual trace). The quintic trace is the first scalar residual detector, but it is not the end of the one-pair local structure. The later fixed-sector refinement $A_5^{\mathfrak{f}}, A_7^{\mathfrak{f}}, \Delta_7$ indicates that the residual germ carries further geometric information beyond the first trace channel.

12.4. Fixed-sector refinement at quintic and septic order. The trace witness at quintic order suggests a more structured refinement of the one-pair residual germ than the trace alone records. For bookkeeping purposes, it is convenient to decompose the quintic and septic coefficients into fixed and anti-fixed parts with respect to the involution

$$\iota(A) := J_0 A J_0, \quad J_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Write

$$\pi_{\mathfrak{f}}(A) := \frac{1}{2}(A + J_0 A J_0), \quad \pi_{\mathfrak{a}}(A) := \frac{1}{2}(A - J_0 A J_0).$$

Then

$$\mathfrak{f} = \text{span}\{I, S\}, \quad \mathfrak{a} = \text{span}\{D, J\},$$

where

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Definition 12.31 (Fixed-sector quintic and septic coefficients). Let

$$F_h(z) = A_3(h)z^3 + A_5(h)z^5 + A_7(h)z^7 + \dots$$

denote the normalized one-pair quotient odd germ on the cubic-generic region. Its fixed-sector quintic and septic coefficients are defined by

$$A_5^{\mathfrak{f}}(h) := \pi_{\mathfrak{f}}(A_5(h)), \quad A_7^{\mathfrak{f}}(h) := \pi_{\mathfrak{f}}(A_7(h)).$$

Write

$$A_5^{\mathfrak{f}}(h) = u_5(h)I + v_5(h)S, \quad A_7^{\mathfrak{f}}(h) = u_7(h)I + v_7(h)S.$$

Definition 12.32 (Second residual determinant). On the cubic-generic region, define

$$\Delta_7(h) := u_7(h)v_5(h) - u_5(h)v_7(h).$$

Remark 12.33 (Interpretation). The quintic fixed-sector vector $A_5^{\mathfrak{f}}(h)$ refines the first residual data beyond the trace detector alone, and the determinant $\Delta_7(h)$ records the next residual quotient at septic order. In the working notes, these form the first two nontrivial layers of a one-pair residual package:

$$c(h), \quad A_5^{\mathfrak{f}}(h), \quad \Delta_7(h).$$

The current draft now records explicit coefficient formulas for the quintic commutator contribution and the septic K_1 -transport contribution in the existing notation $(u, v, w, \alpha, \beta, \gamma, r, \dots)$. The canonical order-7 fixed-sector content is the quotient class

$$[A_7^{\mathfrak{f}}] \in \mathfrak{f}/\mathbf{C}A_5^{\mathfrak{f}},$$

equivalently the second residual determinant

$$\Delta_7 = u_7v_5 - u_5v_7.$$

What remains non-unique is only the raw representative pair (u_7, v_7) , which is defined only up to the one-dimensional septic gauge ambiguity along the quintic fixed-sector line $\mathbf{C}A_5^{\mathfrak{f}}$. Thus the later geometry should be formulated in terms of the canonical quotient-level order-7 data $[A_7^{\mathfrak{f}}]$ and Δ_7 , rather than a uniquely chosen raw pair (u_7, v_7) . At the canonical level, these are the natural variables for the geometric reformulation of the multi-pair problem; raw representatives (u_7, v_7) may be chosen locally by gauge-fixing when convenient, but are not themselves canonical.

Proposition 12.34 (Projected septic gauge law). *Assume the normalization gauge is symmetry-preserving in the sense that*

$$B_2, B_4 \in \mathfrak{f}, \quad A_3 \in \mathfrak{a},$$

with

$$\mathfrak{f} = \text{span}\{I, S\}, \quad \mathfrak{a} = \text{span}\{D, J\}.$$

Then the fixed-sector septic coefficient satisfies

$$(A_7^{\mathfrak{f}})^{\text{new}} = A_7^{\mathfrak{f}} + c_2A_5^{\mathfrak{f}}.$$

Equivalently, the septic fixed-sector ambiguity is one-dimensional and lies along the quintic fixed-sector line.

Proof. By Proposition 12.15,

$$A_7^{\text{new}} = A_7 + c_2A_5 + c_4A_3 + [B_2, A_5] + [B_4, A_3] + c_2[B_2, A_3] + A_3B_2^2 - B_2A_3B_2.$$

Apply the fixed-sector projection $\pi_{\mathfrak{f}}$ term by term.

Since $A_3 \in \mathfrak{a}$, one has

$$\pi_{\mathfrak{f}}(c_4A_3) = 0.$$

Because $B_4 \in \mathfrak{f}$ and $A_3 \in \mathfrak{a}$,

$$[B_4, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

so

$$\pi_{\mathfrak{f}}[B_4, A_3] = 0.$$

Likewise,

$$[B_2, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

hence

$$\pi_{\mathfrak{f}}(c_2[B_2, A_3]) = 0.$$

Now decompose

$$A_5 = A_5^{\mathfrak{f}} + A_5^{\mathfrak{a}}.$$

Since $\mathfrak{f} = \text{span}\{I, S\}$ is abelian,

$$[B_2, A_5^{\mathfrak{f}}] = 0.$$

And since $B_2 \in \mathfrak{f}$ and $A_5^{\mathfrak{a}} \in \mathfrak{a}$,

$$[B_2, A_5^{\mathfrak{a}}] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

so

$$\pi_{\mathfrak{f}}[B_2, A_5] = 0.$$

Finally, $B_2 \in \mathfrak{f}$ implies $B_2^2 \in \mathfrak{f}$. Therefore

$$A_3 B_2^2 \in \mathfrak{af} \subset \mathfrak{a}, \quad B_2 A_3 B_2 \in \mathfrak{faf} \subset \mathfrak{a},$$

and hence

$$\pi_{\mathfrak{f}}(A_3 B_2^2 - B_2 A_3 B_2) = 0.$$

Thus the only term surviving fixed projection is

$$c_2 A_5^{\mathfrak{f}},$$

and we conclude

$$(A_7^{\mathfrak{f}})^{\text{new}} = A_7^{\mathfrak{f}} + c_2 A_5^{\mathfrak{f}}. \quad \square$$

Corollary 12.35 (Gauge invariance of the second residual determinant). *Under a symmetry-preserving normalization gauge, the scalar*

$$\Delta_7 = u_7 v_5 - u_5 v_7$$

is invariant.

Proof. By Proposition 12.34,

$$(A_7^{\mathfrak{f}})^{\text{new}} = A_7^{\mathfrak{f}} + c_2 A_5^{\mathfrak{f}}.$$

Writing

$$A_5^{\mathfrak{f}} = u_5 I + v_5 S, \quad A_7^{\mathfrak{f}} = u_7 I + v_7 S,$$

the new septic pair is

$$(u_7^{\text{new}}, v_7^{\text{new}}) = (u_7, v_7) + c_2(u_5, v_5).$$

Therefore

$$\Delta_7^{\text{new}} = (u_7 + c_2 u_5) v_5 - u_5 (v_7 + c_2 v_5) = u_7 v_5 - u_5 v_7 = \Delta_7. \quad \square$$

Definition 12.36 (Septic normalization-shadow sectors). In the notation of Proposition 12.15, define the order-2 and order-4 septic normalization-shadow sectors by

$$A_{7,K_5} := c_2 A_5 + [B_2, A_5],$$

and

$$A_{7,K_3} := c_4 A_3 + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Then the exact septic transformation law may be written

$$A_7^{\text{new}} = A_7 + A_{7,K_5} + A_{7,K_3}.$$

When the normalization gauge is symmetry-preserving, these are the septic K_5 - and K_3 -transport shadow sectors.

Proposition 12.37 (Projected septic K_5 -shadow law). *Assume the normalization gauge is symmetry-preserving in the sense of Proposition 12.34. Then*

$$A_{7,K_5}^{\mathfrak{f}} = c_2 A_5^{\mathfrak{f}}.$$

In particular,

$$A_{7,K_5}^{\mathfrak{f}} \in \mathbf{C} A_5^{\mathfrak{f}}.$$

Proof. By Definition 12.36,

$$A_{7,K_5} = c_2 A_5 + [B_2, A_5].$$

Apply the fixed-sector projection $\pi_{\mathfrak{f}}$. Decompose

$$A_5 = A_5^{\mathfrak{f}} + A_5^{\mathfrak{a}}.$$

Since $\mathfrak{f} = \text{span}\{I, S\}$ is abelian,

$$[B_2, A_5^{\mathfrak{f}}] = 0.$$

And since $B_2 \in \mathfrak{f}$ and $A_5^{\mathfrak{a}} \in \mathfrak{a}$,

$$[B_2, A_5^{\mathfrak{a}}] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

so its fixed-sector projection vanishes. Therefore

$$A_{7,K_5}^{\mathfrak{f}} = \pi_{\mathfrak{f}}(c_2 A_5 + [B_2, A_5]) = c_2 A_5^{\mathfrak{f}}. \quad \square$$

Proposition 12.38 (Projected septic K_3 -shadow vanishing). *Assume the normalization gauge is symmetry-preserving in the sense of Proposition 12.34. Then*

$$A_{7,K_3}^{\mathfrak{f}} = 0.$$

Proof. By Definition 12.36,

$$A_{7,K_3} = c_4 A_3 + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Since $A_3 \in \mathfrak{a}$, one has

$$c_4 A_3 \in \mathfrak{a}.$$

Because $B_4 \in \mathfrak{f}$ and $A_3 \in \mathfrak{a}$,

$$[B_4, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a}.$$

Likewise,

$$[B_2, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

hence

$$c_2 [B_2, A_3] \in \mathfrak{a}.$$

Finally, $B_2 \in \mathfrak{f}$ implies $B_2^2 \in \mathfrak{f}$, so

$$A_3 B_2^2 \in \mathfrak{a}\mathfrak{f} \subset \mathfrak{a}, \quad B_2 A_3 B_2 \in \mathfrak{f}\mathfrak{a}\mathfrak{f} \subset \mathfrak{a}.$$

Thus every term of A_{7,K_3} is anti-fixed, and therefore

$$A_{7,K_3}^{\mathfrak{f}} = \pi_{\mathfrak{f}}(A_{7,K_3}) = 0. \quad \square$$

Remark 12.39 (Interpretation of the septic shadow sectors). Propositions 12.37 and 12.38 show that the lower-order normalization shadows at septic order contribute no new canonical fixed-sector information beyond the already-known quintic line $\mathbf{C}A_5^{\mathfrak{f}}$.

Proposition 12.40 (Explicit fixed-sector formulas at quintic order). *Write*

$$X_1 = \begin{pmatrix} u & v \\ v & w \end{pmatrix}, \quad X_3 = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}.$$

Then the fixed-sector part of the basic quintic commutator channel

$$X_3 J X_1 - X_1 J X_3$$

is

$$\pi_{\mathfrak{f}}(X_3 J X_1 - X_1 J X_3) = (v(\gamma - \alpha) + \beta(u - w))I + (\gamma u - \alpha w)S.$$

Consequently, for the quintic fixed-sector coefficient

$$A_5^{\mathfrak{f}}(h) = u_5(h)I + v_5(h)S,$$

the commutator-type contribution is explicitly

$$u_5^{(\text{comm})} = v(\gamma - \alpha) + \beta(u - w), \quad v_5^{(\text{comm})} = \gamma u - \alpha w.$$

Proof. A direct computation gives

$$X_3 J X_1 - X_1 J X_3 = \begin{pmatrix} -2\alpha v + 2\beta u & -\alpha w + \gamma u \\ -\alpha w + \gamma u & -2\beta w + 2\gamma v \end{pmatrix}.$$

The fixed-sector projection of a symmetric matrix

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix}$$

is

$$\pi_{\mathfrak{f}} \begin{pmatrix} A & B \\ B & C \end{pmatrix} = \frac{A+C}{2} I + B S.$$

Applying this to the displayed matrix gives

$$\pi_{\mathfrak{f}}(X_3 J X_1 - X_1 J X_3) = \frac{(-2\alpha v + 2\beta u) + (-2\beta w + 2\gamma v)}{2} I + (\gamma u - \alpha w) S,$$

which simplifies to

$$(v(\gamma - \alpha) + \beta(u - w)) I + (\gamma u - \alpha w) S.$$

□

Proposition 12.41 (Explicit septic K_1 -transport fixed-sector formulas). *Write*

$$\begin{aligned} X_0 &= \text{diag}(x_1, x_2), & X_1 &= \begin{pmatrix} u & v \\ v & w \end{pmatrix}, & X_2 &= \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \\ X_4 &= \begin{pmatrix} p & r \\ r & s \end{pmatrix}, & X_5 &= \begin{pmatrix} \alpha_5 & \beta_5 \\ \beta_5 & \gamma_5 \end{pmatrix}, & X_6 &= \begin{pmatrix} \mu_6 & \nu_6 \\ \nu_6 & \lambda_6 \end{pmatrix}. \end{aligned}$$

Then the fixed-sector parts of the three septic K_1 -transport channels are

$$\begin{aligned} \pi_{\mathfrak{f}}(X_5 J X_1 - X_1 J X_5) &= (\beta_5(u - w) + v(\gamma_5 - \alpha_5)) I + (\gamma_5 u - \alpha_5 w) S, \\ \pi_{\mathfrak{f}}(X_2 J X_4 - X_4 J X_2) &= (b(p - s) + r(c - a)) I + (cp - as) S, \end{aligned}$$

and

$$\pi_{\mathfrak{f}}(X_0 J X_6 - X_6 J X_0) = (x_2 - x_1) \nu_6 I + (x_2 \mu_6 - x_1 \lambda_6) S.$$

Consequently, if the septic K_1 -transport fixed-sector coefficient is written

$$A_{7,K_1}^{\mathfrak{f}}(h) = u_{7,K_1}(h) I + v_{7,K_1}(h) S,$$

then

$$\begin{aligned} u_{7,K_1} &= \kappa \left((x_2 - x_1) \nu_6 + \beta_5(u - w) + v(\gamma_5 - \alpha_5) + b(p - s) + r(c - a) \right), \\ v_{7,K_1} &= \kappa \left(x_2 \mu_6 - x_1 \lambda_6 + \gamma_5 u - \alpha_5 w + cp - as \right). \end{aligned}$$

Proof. For two symmetric matrices

$$X = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad Y = \begin{pmatrix} p & q \\ q & r \end{pmatrix},$$

a direct calculation gives

$$X J Y - Y J X = \begin{pmatrix} -2aq + 2bp & -ar + cp \\ -ar + cp & -2br + 2cq \end{pmatrix}.$$

Taking fixed-sector projection yields

$$\pi_{\mathfrak{f}}(X J Y - Y J X) = (b(p - r) + q(c - a)) I + (cp - ar) S.$$

Applying this formula to $(X, Y) = (X_5, X_1)$ and $(X, Y) = (X_2, X_4)$ gives the first two displayed identities.

For the X_0/X_6 channel, a direct multiplication gives

$$X_0 J X_6 - X_6 J X_0 = \begin{pmatrix} -2x_1 \nu_6 & -x_1 \lambda_6 + x_2 \mu_6 \\ -x_1 \lambda_6 + x_2 \mu_6 & 2x_2 \nu_6 \end{pmatrix},$$

whose fixed-sector part is

$$(x_2 - x_1)\nu_6 I + (x_2\mu_6 - x_1\lambda_6)S.$$

Summing the three channels gives the stated formulas for u_{7,K_1} and v_{7,K_1} . $\square$

Definition 12.42 (Septic K_1 -residual determinant). If

$$A_{7,K_1}^f(h) = u_{7,K_1}(h) I + v_{7,K_1}(h) S,$$

define the corresponding septic K_1 -residual determinant by

$$\Delta_{7,K_1}(h) := u_{7,K_1}(h)v_5(h) - u_5(h)v_{7,K_1}(h).$$

Corollary 12.43 (Direct $q^{(7)}$ -driven septic slice). *On the direct linear $q^{(7)}$ -driven slice, the septic K_1 -transport fixed-sector contribution is I -directional. More precisely,*

$$v_{7,K_1}^{(q^{(7)})} = 0,$$

and

$$u_{7,K_1}^{(q^{(7)})} = \kappa\left((x_2 - x_1)\nu_6^{(q^{(7)})} + v(\gamma_5 - \alpha_5)^{(q^{(7)})}\right),$$

so in particular

$$\Delta_7^{(q^{(7)})} = u_7^{(q^{(7)})}v_5.$$

Proof. On the direct $q^{(7)}$ -driven slice, the linear X_5 -input is diagonal, hence

$$\beta_5^{(q^{(7)})} = 0.$$

The mixed X_6 -slice is off-diagonal on this grading, so

$$\mu_6^{(q^{(7)})} = \lambda_6^{(q^{(7)})} = 0.$$

Moreover the X_2/X_4 -channel does not lie on the pure $(K_1, \eta_2, q^{(7)})$ -slice. Therefore the S -component vanishes and the displayed formula for $u_{7,K_1}^{(q^{(7)})}$ follows from Proposition 12.41. Since

$$\Delta_7 = u_7v_5 - u_5v_7,$$

the direct $q^{(7)}$ -slice satisfies

$$\Delta_7^{(q^{(7)})} = u_7^{(q^{(7)})}v_5. \quad \square$$

Proposition 12.44 (Noncancellation of the direct septic K_1 witness). *On the graded slice*

$$(K_1, \eta_2, q^{(7)}),$$

the only contributing channels to

$$u_{7,K_1}^{(q^{(7)})}$$

are

$$\kappa(x_2 - x_1)\nu_6^{(q^{(7)})} \quad \text{and} \quad \kappa v(\gamma_5 - \alpha_5)^{(q^{(7)})},$$

and their total coefficient of the monomial

$$\kappa(h)\eta_2(h)q^{(7)}(h)$$

is nonzero. Consequently,

$$u_{7,K_1}^{(q^{(7)})}(h) \neq 0$$

on the generic one-pair region.

Proof. Fix h and isolate the graded slice

$$\kappa(h)\eta_2(h)q^{(7)}(h).$$

By Corollary 12.43, the direct $q^{(7)}$ -driven slice is I -directional and is given by

$$u_{7,K_1}^{(q^{(7)})} = \kappa\left((x_2 - x_1)\nu_6^{(q^{(7)})} + v(\gamma_5 - \alpha_5)^{(q^{(7)})}\right).$$

Thus it remains only to show that the two displayed channels do not cancel.

Now consider the same-point expansion. By the explicit formula

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix},$$

the factor $q^{(7)}(h)$ first appears only in the quintic same-point coefficient G_5 , and there only in the diagonal sector, namely through the quintic Taylor term of the $(2, 2)$ -entry coming from $q''(t_{\pm})$. Therefore, by Lemma 12.17, the linear $q^{(7)}$ -contribution to X_5 is diagonal. In particular it contributes to $\gamma_5 - \alpha_5$, but not to β_5 .

Likewise, the $\eta_2(h)$ -slice of $X_1(h)$ is off-diagonal, because

$$v(h) = -\frac{1}{4\pi(s_1 + s_2)} \eta_2(h).$$

Applying Lemma 12.17 again, the mixed $(\eta_2, q^{(7)})$ -slice of X_6 is off-diagonal. Hence it contributes to ν_6 , but not to μ_6 or λ_6 .

Thus the only contributing channels on the graded slice

$$(K_1, \eta_2, q^{(7)})$$

are exactly

$$\kappa(x_2 - x_1)\nu_6^{(q^{(7)})} \quad \text{and} \quad \kappa v(\gamma_5 - \alpha_5)^{(q^{(7)})}.$$

To compute their combined coefficient, it suffices by multilinearity to restrict to the two perturbation directions carrying the relevant tags: an off-diagonal first-order perturbation A for the η_2 -slice and a diagonal fifth-order perturbation D for the $q^{(7)}$ -slice. Accordingly, write

$$G_0 = \text{diag}(a^2, b^2), \quad A = \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix},$$

and expand

$$X(\varepsilon, \delta) := (G_0 + \varepsilon A + \delta D)^{-1/2} = X_0 + \varepsilon U + \delta V + \varepsilon \delta W + O(\varepsilon^2, \delta^2).$$

Solving the identity

$$X(\varepsilon, \delta) (G_0 + \varepsilon A + \delta D) X(\varepsilon, \delta) = I$$

order by order gives

$$U_{12} = -\frac{u}{ab(a+b)}, \quad V_{22} = -\frac{d}{2b^3}, \quad W_{12} = \frac{du(a+2b)}{2ab^3(a+b)^2}.$$

Here U_{12} is the η_2 -slice of v , V_{22} is the $q^{(7)}$ -slice of $\gamma_5 - \alpha_5$, and W_{12} is the mixed $(\eta_2, q^{(7)})$ -slice of ν_6 . Therefore the total coefficient of the target slice inside

$$(x_2 - x_1)\nu_6 + v(\gamma_5 - \alpha_5)$$

is

$$\left(\frac{1}{a} - \frac{1}{b}\right) W_{12} + U_{12}V_{22} = \frac{du}{a^2b^2(a+b)^2},$$

which is nonzero.

Thus the monomial

$$\kappa(h) \eta_2(h) q^{(7)}(h)$$

occurs in $u_{7,K_1}^{(q^{(7)})}(h)$ with nonzero total coefficient, so

$$u_{7,K_1}^{(q^{(7)})}(h) \neq 0.$$

□

Remark 12.45 (Septic K_1 trace target). At the graded level, the septic K_1 -transport channel is the first odd transport sector capable of coupling the new same-point $q^{(7)}$ -slice into the fixed-sector septic residual. In particular, the expected highest new contribution to the septic determinant is carried by the K_1 -channel, in parallel with the earlier quintic trace witness at order 5.

Lemma 12.46 (Parity elimination of the pure same-point septic sector). *Let $t_{\pm} = h \pm s/2$, and let the odd two-point germ be defined as the odd part under the involution*

$$s \mapsto -s \quad (\text{equivalently } t_+ \leftrightarrow t_-).$$

Let $A_{7,\text{pure}}$ denote the pure same-point septic sector, i.e. the septic contribution built only from the same-point blocks and their whitening, with no transport factor coming from the ordered pair (t_+, t_-) .

Then $A_{7,\text{pure}}$ is even in s , hence vanishes in the odd two-point germ. In particular,

$$A_{7,\text{pure}} = 0 \quad \text{in the odd germ}, \quad A_{7,\text{pure}}^f = 0.$$

Consequently, pure same-point septic terms do not contribute to Δ_7 , and in particular do not contribute to the weight-one part $[\Delta_7]^{[1]}$.

Proof. A pure same-point septic term depends only on midpoint jet data at h , i.e. on the same-point expansion at the midpoint, and carries no transport factor coming from the ordered pair (t_+, t_-) . Therefore it is invariant under exchanging the two points:

$$t_+ \leftrightarrow t_-.$$

Since this exchange is exactly the involution $s \mapsto -s$, every pure same-point septic term is an even function of s .

By definition, the odd two-point germ is obtained by projecting to the odd part under $s \mapsto -s$. Hence every even contribution vanishes in that projection. Therefore

$$A_{7,\text{pure}} = 0 \quad \text{in the odd germ.}$$

Applying fixed-sector projection gives

$$A_{7,\text{pure}}^f = 0.$$

Since Δ_7 is extracted from the odd pair germ after fixed-sector projection and quotient normalization, pure same-point septic terms contribute nothing to Δ_7 , and therefore nothing to $[\Delta_7]^{[1]}$. $\square$

Corollary 12.47 (Weight-one septic reduction to the direct K_1 slice). *Let $[\cdot]^{[1]}$ denote the part linear in the first genuinely new same-point $q^{(7)}$ -slice. Then*

$$[\Delta_7]^{[1]} = [\Delta_{7,K_1}]^{[1]}.$$

Equivalently, the highest genuinely new $q^{(7)}$ -coefficient of the septic residual determinant comes only from the direct septic K_1 -transport slice.

Proof. The first genuinely new $q^{(7)}$ -slice enters the same-point expansion at order 5, so

$$X_j^{[1]} = 0 \quad (j < 5).$$

Hence every septic transport sector built only from lower same-point orders is weight zero; in particular, the K_3 - and K_5 -transport sectors do not contribute to $[\Delta_7]^{[1]}$.

By Lemma 12.46, pure same-point septic terms also do not contribute to $[\Delta_7]^{[1]}$, because they vanish already in the odd two-point germ.

Thus the only septic sector that can see the weight-one $q^{(7)}$ -slice is the direct K_1 -transport sector, and therefore

$$[\Delta_7]^{[1]} = [\Delta_{7,K_1}]^{[1]}. \quad \square$$

Proposition 12.48 (Generic nonvanishing of the direct septic slice under nontrivial quintic fixed-sector coefficient). *Let J be a connected open interval contained in the quintic-generic region. Assume*

$$v_5 \not\equiv 0 \quad \text{on } J.$$

Then the direct $q^{(7)}$ -driven slice of the full total septic residual determinant is generically nonzero on J . Equivalently,

$$\Delta_7^{(q^{(7)})} \not\equiv 0 \quad \text{on } J.$$

Proof. By Corollary 12.47,

$$[\Delta_7]^{[1]} = [\Delta_{7,K_1}]^{[1]}.$$

Thus the highest genuinely new direct $q^{(7)}$ -coefficient of the total septic residual determinant is the same as that of the direct K_1 -transport slice.

By Proposition 12.44, the direct septic K_1 witness is generically nonzero on the quintic-generic region. In particular,

$$u_{7,K_1}^{(q^{(7)})} \neq 0 \quad \text{on } J.$$

Moreover, by Corollary 12.43, on the direct $q^{(7)}$ -driven slice one has

$$v_{7,K_1}^{(q^{(7)})} = 0, \quad \Delta_{7,K_1}^{(q^{(7)})} = u_{7,K_1}^{(q^{(7)})} v_5.$$

Since $v_5 \neq 0$ on J , it follows that

$$\Delta_{7,K_1}^{(q^{(7)})} \neq 0 \quad \text{on } J.$$

Therefore

$$\Delta_7^{(q^{(7)})} \neq 0 \quad \text{on } J,$$

as claimed. $\square$

Theorem 12.49 (Quotient-septic closure). *Let the septic odd-germ coefficient be decomposed as*

$$A_7 = A_{7,K_1} + A_{7,K_5} + A_{7,K_3} + A_{7,\text{pure}},$$

where A_{7,K_1} is the direct K_1 -transport sector, A_{7,K_5} and A_{7,K_3} are the septic normalization-shadow sectors from Definition 12.36, and $A_{7,\text{pure}}$ is the pure same-point septic sector from Lemma 12.46.

Then

$$A_7^{\mathfrak{f}} = A_{7,K_1}^{\mathfrak{f}} + A_{7,K_5}^{\mathfrak{f}},$$

with

$$A_{7,K_5}^{\mathfrak{f}} \in \mathbf{C} A_5^{\mathfrak{f}}.$$

Consequently,

$$[A_7^{\mathfrak{f}}] = [A_{7,K_1}^{\mathfrak{f}}] \quad \text{in } \mathfrak{f}/\mathbf{C}A_5^{\mathfrak{f}},$$

and

$$\Delta_7 = \Delta_{7,K_1}.$$

Proof. By Proposition 12.38,

$$A_{7,K_3}^{\mathfrak{f}} = 0.$$

By Lemma 12.46,

$$A_{7,\text{pure}}^{\mathfrak{f}} = 0.$$

Therefore

$$A_7^{\mathfrak{f}} = A_{7,K_1}^{\mathfrak{f}} + A_{7,K_5}^{\mathfrak{f}}.$$

By Proposition 12.37,

$$A_{7,K_5}^{\mathfrak{f}} \in \mathbf{C} A_5^{\mathfrak{f}}.$$

Hence $A_7^{\mathfrak{f}}$ and $A_{7,K_1}^{\mathfrak{f}}$ define the same class in the quotient

$$\mathfrak{f}/\mathbf{C}A_5^{\mathfrak{f}}.$$

Writing

$$A_5^{\mathfrak{f}} = u_5 I + v_5 S, \quad A_{7,K_1}^{\mathfrak{f}} = u_{7,K_1} I + v_{7,K_1} S,$$

this quotient equality means precisely that

$$A_7^{\mathfrak{f}} = A_{7,K_1}^{\mathfrak{f}} + \chi A_5^{\mathfrak{f}}$$

for some scalar χ . Therefore

$$\Delta_7 = (u_{7,K_1} + \chi u_5) v_5 - u_5 (v_{7,K_1} + \chi v_5) = u_{7,K_1} v_5 - u_5 v_{7,K_1} = \Delta_{7,K_1},$$

where Δ_{7,K_1} is as in Definition 12.42. $\square$

Remark 12.50 (Status of the order-7 fixed-sector package). The order-7 fixed-sector analysis is now closed at the canonical quotient level. By Theorem 12.49, the full total septic fixed-sector class satisfies

$$[A_7^f] = [A_{7,K_1}^f] \quad \text{in } \mathfrak{f}/\mathbf{C}A_5^f,$$

and the canonical second residual determinant is

$$\Delta_7 = \Delta_{7,K_1}.$$

What remains non-unique is only the raw representative of A_7^f inside the affine line

$$A_{7,K_1}^f + \mathbf{C}A_5^f,$$

as recorded by Proposition 12.34. Equivalently, the full lower/mixed septic completion of the raw total pair (u_7, v_7) is not canonically determined. This remaining gauge-dependent choice does not affect the quotient class $[A_7^f]$ or the residual determinant Δ_7 used in the later geometric arguments.

Remark 12.51 (Local gauge-fixing for raw septic representatives). Although the raw representative pair (u_7, v_7) is not canonical, local choices may be made by fixing a gauge section along the quintic fixed-sector line. Indeed, Proposition 12.34 gives

$$(u_7^{\text{new}}, v_7^{\text{new}}) = (u_7, v_7) + c_2(u_5, v_5).$$

Hence on any region where $v_5 \neq 0$, one may choose

$$c_2 = -\frac{v_7}{v_5},$$

so that

$$v_7^{\text{new}} = 0, \quad u_7^{\text{new}} = \frac{\Delta_7}{v_5}.$$

Similarly, on any region where $u_5 \neq 0$, one may choose

$$c_2 = -\frac{u_7}{u_5},$$

so that

$$u_7^{\text{new}} = 0, \quad v_7^{\text{new}} = -\frac{\Delta_7}{u_5}.$$

Thus raw septic representatives can be chosen locally by gauge-fixing, but such choices are auxiliary and are not canonical.

12.5. Normalized planar curve and curvature bridge. On the region where $c(h) \neq 0$, define the normalized scalars

$$\rho_1(h) := \frac{v_5(h)}{c(h)}, \quad \rho_2(h) := \frac{\Delta_7(h)}{c(h)},$$

and the associated planar invariant curve

$$\Gamma(h) := (\rho_1(h), \rho_2(h)).$$

Proposition 12.52 (Non-affinity of the second normalized residual scalar). *Let J be a connected open interval contained in the quintic-generic region and in $\{h : c(h) \neq 0\}$. Assume*

$$v_5 \not\equiv 0 \quad \text{on } J.$$

Then there do not exist constants $A, B \in \mathbf{C}$ such that

$$\rho_2(h) \equiv A\rho_1(h) + B$$

on J .

Equivalently,

$$\Delta_7(h) \not\equiv A v_5(h) + B c(h)$$

on J .

Proof. Suppose

$$\Delta_7(h) \equiv A v_5(h) + B c(h)$$

on J . Then the direct $q^{(7)}$ -driven slice of the right-hand side vanishes, because v_5 and c are governed by the cubic/quintic layers and carry no direct septic $q^{(7)}$ -slice.

On the other hand, by Proposition 12.48, the direct $q^{(7)}$ -driven slice of Δ_7 is generically nonzero on J . This is a contradiction. Dividing by $c(h)$ yields

$$\rho_2(h) \neq A \rho_1(h) + B. \quad \square$$

Definition 12.53 (Normalized derivative numerators). Define

$$M(h) := c(h)v_5'(h) - c'(h)v_5(h), \quad N(h) := c(h)\Delta_7'(h) - c'(h)\Delta_7(h).$$

Then

$$\rho_1'(h) = \frac{M(h)}{c(h)^2}, \quad \rho_2'(h) = \frac{N(h)}{c(h)^2}.$$

Proposition 12.54 (Curvature bridge for the normalized planar curve). *Let*

$$K_\Gamma(h) := \rho_1'(h)\rho_2''(h) - \rho_2'(h)\rho_1''(h)$$

denote the affine curvature numerator of the planar curve $\Gamma(h)$. Then

$$K_\Gamma(h) = \frac{M(h)N'(h) - N(h)M'(h)}{c(h)^4}.$$

In particular, if $K_\Gamma \equiv 0$ on a connected open interval J on which

$$c \neq 0, \quad M \neq 0, \quad v_5 \neq 0,$$

then ρ_2 is affine in ρ_1 on J , contradicting Proposition 12.52. Hence on every such interval one has

$$K_\Gamma \neq 0.$$

Proof. By Definition 12.53,

$$\rho_1' = \frac{M}{c^2}, \quad \rho_2' = \frac{N}{c^2}.$$

Differentiating gives

$$\rho_1'' = \frac{M'c - 2Mc'}{c^3}, \quad \rho_2'' = \frac{N'c - 2Nc'}{c^3}.$$

Therefore

$$K_\Gamma = \frac{M}{c^2} \frac{N'c - 2Nc'}{c^3} - \frac{N}{c^2} \frac{M'c - 2Mc'}{c^3} = \frac{MN' - NM'}{c^4}.$$

If $K_\Gamma \equiv 0$ on a connected open set with $M \neq 0$, then

$$0 = MN' - NM' = M^2 \left(\frac{N}{M} \right)',$$

so N/M is constant there:

$$\frac{N}{M} = A.$$

Since

$$\frac{N}{M} = \frac{\rho_2'}{\rho_1'},$$

it follows that

$$\rho_2' = A \rho_1',$$

hence

$$\rho_2 = A \rho_1 + B$$

on that connected interval. If in addition

$$c \neq 0, \quad v_5 \neq 0,$$

there, this contradicts Proposition 12.52. The final claim follows. $\square$

Remark 12.55 (Geometric consequence). Proposition 12.54 is the local-to-global bridge from the order-7 fixed-sector analysis back to the secant/tangent program. On the open set where

$$c(h) \neq 0, \quad M(h) \neq 0, \quad M(h)N'(h) - N(h)M'(h) \neq 0,$$

the normalized planar curve $\Gamma(h)$ decomposes into C^2 branches in the affine chart (ρ_1, ρ_2) , each of which is strictly convex or strictly concave. Thus the remaining global obstruction is not same-branch secancy, but the possibility of a common four-secant line meeting two separated ECT branches twice each.

Definition 12.56 (Projective Wronskian of the strengthened one-pair triple). Define

$$W(h) := \det \begin{pmatrix} c(h) & v_5(h) & \Delta_7(h) \\ c'(h) & v_5'(h) & \Delta_7'(h) \\ c''(h) & v_5''(h) & \Delta_7''(h) \end{pmatrix}.$$

Proposition 12.57 (Projective-Wronskian bridge). *With $M(h)$ and $N(h)$ as in Definition 12.53, one has*

$$M(h)N'(h) - N(h)M'(h) = c(h) W(h).$$

Consequently,

$$K_\Gamma(h) = \frac{W(h)}{c(h)^3}$$

on the region where $c(h) \neq 0$.

Proof. Using

$$M = c v_5' - c' v_5, \quad N = c \Delta_7' - c' \Delta_7,$$

a direct expansion gives

$$MN' - NM' = (c v_5' - c' v_5)(c \Delta_7'' - c'' \Delta_7) - (c \Delta_7' - c' \Delta_7)(c v_5'' - c'' v_5).$$

Expanding and regrouping yields

$$MN' - NM' = c \det \begin{pmatrix} c & v_5 & \Delta_7 \\ c' & v_5' & \Delta_7' \\ c'' & v_5'' & \Delta_7'' \end{pmatrix} = c W.$$

The displayed formula for K_Γ now follows from Proposition 12.54. □

Lemma 12.58 (Exact identities for W and W'). *Let*

$$F(h) := (c(h), v_5(h), \Delta_7(h)), \quad M(h) := c(h)v_5'(h) - c'(h)v_5(h).$$

Then

$$W(h) = \det(F(h), F'(h), F''(h))$$

admits the exact expansion

$$W = M \Delta_7'' + (v_5 c'' - c v_5'') \Delta_7' + (c' v_5'' - v_5' c'') \Delta_7,$$

and

$$W'(h) = \det(F(h), F'(h), F'''(h))$$

admits the exact expansion

$$W' = M \Delta_7''' + (v_5 c''' - c v_5''') \Delta_7' + (c' v_5''' - v_5' c''') \Delta_7.$$

Proof. Expand the determinant $\det(F, F', F'')$ along the third column. This gives

$$W = \det \begin{pmatrix} c & v_5 \\ c' & v_5' \end{pmatrix} \Delta_7'' - \det \begin{pmatrix} c & v_5 \\ c'' & v_5'' \end{pmatrix} \Delta_7' + \det \begin{pmatrix} c' & v_5' \\ c'' & v_5'' \end{pmatrix} \Delta_7,$$

which is exactly the first identity.

For W' , differentiate $W = \det(F, F', F'')$. The terms $\det(F', F', F'')$ and $\det(F, F'', F'')$ vanish by repeated columns, so

$$W' = \det(F, F', F''').$$

Expanding along the third column gives the second formula. □

Proposition 12.59 (Exclusion of W -flat intervals on the $c \neq 0$, $M \neq 0$ region). *Let J be a connected open interval on which*

$$c(h) \neq 0, \quad M(h) \neq 0.$$

Then

$$W \not\equiv 0 \quad \text{on } J.$$

Proof. Suppose for contradiction that

$$W \equiv 0 \quad \text{on } J.$$

By Proposition 12.57,

$$K_\Gamma(h) = \frac{W(h)}{c(h)^3} \equiv 0 \quad \text{on } J.$$

Since

$$M = c v'_5 - c' v_5$$

and $c \neq 0$ on J , the hypothesis $M \neq 0$ on J rules out $v_5 \equiv 0$ on J . Hence $v_5 \not\equiv 0$ on J .

By Proposition 12.54, it follows that

$$\rho_2(h) = A \rho_1(h) + B$$

for some constants $A, B \in \mathbf{C}$, i.e.

$$\Delta_7(h) = A v_5(h) + B c(h) \quad \text{on } J.$$

This contradicts Proposition 12.52. Therefore

$$W \not\equiv 0 \quad \text{on } J. \quad \square$$

Remark 12.60 (Interpretation of the λ -reduction). On the direct highest-new slice one has

$$\Delta_7^{[1]} = \lambda^{[1]} v_5$$

by Corollary 12.43 together with Corollary 12.47. Thus if $W \equiv 0$ on a connected open interval with $c \neq 0$ and $M \neq 0$, then

$$v_5 \not\equiv 0$$

on that interval, because $M = c v'_5 - c' v_5$. The affine dependence

$$\Delta_7 = A v_5 + B c$$

forced by Proposition 12.59 implies

$$\Delta_7^{[1]} = A v_5,$$

hence

$$\lambda^{[1]} \equiv A$$

on that interval. In this sense, the present argument excludes W -flatness on an $M \neq 0$ region without requiring a separate differential equation for $\lambda^{[1]}$; what remains open is not the canonical quotient-level order-7 invariant, but only the lower/mixed septic completion of a raw representative pair (u_7, v_7) , which is irrelevant to the quotient class $[A_7^f]$ and to Δ_7 .

Definition 12.61 (Projective graph and dual tangent-line variables). On the region where $v_5 \neq 0$, define

$$\mu := \frac{c}{v_5}, \quad \lambda := \frac{\Delta_7}{v_5}.$$

On the subregion where $M \neq 0$, define

$$L := v_5 \Delta'_7 - v'_5 \Delta_7, \quad \beta := \frac{d\lambda}{d\mu}, \quad m := \lambda - \mu \beta.$$

Proposition 12.62 (Closed minor system and dual tangent-line curve). *Let*

$$F := (c, v_5, \Delta_7), \quad U := (L, -N, M).$$

Then

$$F \times F' = U, \quad U \times U' = W F.$$

Equivalently,

$$MN' - NM' = cW, \quad ML' - LM' = v_5W, \quad NL' - LN' = \Delta_7W.$$

On the region where $v_5M \neq 0$,

$$\mu' = -\frac{M}{v_5^2}, \quad \beta = -\frac{L}{M}, \quad m = \frac{N}{M},$$

and

$$\beta' = -\frac{v_5W}{M^2}, \quad m' = \frac{cW}{M^2}.$$

Hence the dual tangent-line curve

$$T(h) := (\beta(h), m(h))$$

satisfies

$$T'(h) = \frac{W(h)}{M(h)^2} (-v_5(h), c(h)).$$

Moreover,

$$\det(T'(h), T''(h)) = \frac{W(h)^2}{M(h)^3}.$$

Proof. The identity

$$F \times F' = (L, -N, M)$$

is the direct expansion of the cross product. Differentiating gives

$$U' = F \times F''.$$

Applying the vector identity

$$(A \times B) \times (A \times C) = \det(A, B, C) A$$

with $A = F$, $B = F'$, $C = F''$ yields

$$U \times U' = (F \times F') \times (F \times F'') = \det(F, F', F'') F = W F,$$

which is exactly the three displayed scalar identities.

Next,

$$\mu' = \left(\frac{c}{v_5} \right)' = \frac{c'v_5 - cv_5'}{v_5^2} = -\frac{M}{v_5^2},$$

and

$$\lambda' = \left(\frac{\Delta_7}{v_5} \right)' = \frac{v_5\Delta_7' - v_5'\Delta_7}{v_5^2} = \frac{L}{v_5^2},$$

so

$$\beta = \frac{\lambda'}{\mu'} = -\frac{L}{M}.$$

Also

$$\mu L + M\lambda = \frac{c}{v_5}(v_5\Delta_7' - v_5'\Delta_7) + (cv_5' - c'v_5)\frac{\Delta_7}{v_5} = c\Delta_7' - c'\Delta_7 = N,$$

hence

$$m = \lambda - \mu\beta = \lambda + \mu\frac{L}{M} = \frac{N}{M}.$$

The formulas for β' and m' follow immediately from

$$ML' - LM' = v_5W, \quad MN' - NM' = cW.$$

Finally, write

$$T' = \alpha V, \quad \alpha := \frac{W}{M^2}, \quad V := (-v_5, c).$$

Then

$$\det(T', T'') = \alpha^2 \det(V, V') = \frac{W^2}{M^4} \det \begin{pmatrix} -v_5 & -v'_5 \\ c & c' \end{pmatrix} = \frac{W^2}{M^3},$$

because

$$\det(V, V') = cv'_5 - c'v_5 = M.$$

□

Corollary 12.63 (Graph curvature identities in (μ, λ) -coordinates). *On the region where $v_5 M \neq 0$,*

$$\lambda_{\mu\mu} = \frac{v_5^3}{M^3} W,$$

and

$$\lambda_{\mu\mu\mu} = -\frac{v_5^5}{M^4} \left(W' + 3 \left(\frac{v'_5}{v_5} - \frac{M'}{M} \right) W \right).$$

In particular, at a point where $W = 0$, $M \neq 0$, and $v_5 \neq 0$,

$$\lambda_{\mu\mu\mu} = -\frac{v_5^5}{M^4} W'.$$

Proof. Since

$$\frac{d}{d\mu} = \frac{1}{\mu'} \frac{d}{dh} = -\frac{v_5^2}{M} \frac{d}{dh},$$

the identity

$$\lambda_{\mu\mu} = \frac{v_5^3}{M^3} W$$

follows from

$$\lambda_\mu = \beta, \quad \beta' = -\frac{v_5 W}{M^2}.$$

Differentiating once more gives

$$\lambda_{\mu\mu\mu} = -\frac{v_5^2}{M} \frac{d}{dh} \left(\frac{v_5^3}{M^3} W \right),$$

which expands to the displayed formula. The last statement is the specialisation $W = 0$. □

Proposition 12.64 (Projectivized tangent-angle flow and contact identity). *On any simply connected subinterval of the region*

$$v_5 \neq 0, \quad M \neq 0,$$

choose a local lift Θ of the projectivized tangent direction by

$$(\cos \Theta, \sin \Theta) = \frac{1}{\sqrt{M^2 + L^2}} (M, -L).$$

Then

$$\Theta' = -\frac{v_5 W}{M^2 + L^2}.$$

Moreover, with

$$\mu = \frac{c}{v_5}, \quad \beta = -\frac{L}{M}, \quad m = \frac{N}{M},$$

one has the exact contact identity

$$dm + \mu d\beta = 0.$$

Equivalently, on every interval where β' does not vanish,

$$\frac{dm}{d\beta} = -\mu.$$

In particular, at a generic turning boundary with

$$M(h_b) = 0, \quad N(h_b) \neq 0, \quad v_5(h_b) \neq 0,$$

the projectivized slope direction is regular: one has

$$L(h_b) \neq 0,$$

so the compactified tangent direction extends smoothly across the boundary. Thus the only critical points of the compactified tangent-direction map are the zeros of W .

Proof. Differentiate the angle of the vector $(M, -L)$:

$$\Theta' = \frac{M(-L)' - (-L)M'}{M^2 + L^2} = -\frac{ML' - LM'}{M^2 + L^2}.$$

By Proposition 12.62,

$$ML' - LM' = v_5 W,$$

hence

$$\Theta' = -\frac{v_5 W}{M^2 + L^2}.$$

Next, by Definition 12.61,

$$m = \lambda - \mu\beta.$$

Since $\beta = d\lambda/d\mu$, one has

$$d\lambda = \beta d\mu,$$

and therefore

$$dm = d\lambda - \beta d\mu - \mu d\beta = -\mu d\beta.$$

This is exactly

$$dm + \mu d\beta = 0.$$

Finally, at a generic turning boundary with $v_5(h_b) \neq 0$, one has

$$\mu(h_b) = \frac{c(h_b)}{v_5(h_b)} \neq 0.$$

Using

$$N = \mu L + M\lambda$$

from Proposition 12.62, and evaluating at h_b where $M(h_b) = 0$, gives

$$N(h_b) = \mu(h_b)L(h_b).$$

Since $N(h_b) \neq 0$ and $\mu(h_b) \neq 0$, it follows that

$$L(h_b) \neq 0.$$

Hence

$$M(h_b)^2 + L(h_b)^2 \neq 0,$$

so the projectivized tangent direction is regular there. $\square$

Remark 12.65 (Determinant interpretation on a fixed-sign W -interval). On any connected interval where c , M , and W are nonzero and W keeps one sign, the graph $\lambda = \lambda(\mu)$ is strictly convex or strictly concave. Consequently, for any ordered triple of distinct points on that interval, the determinant

$$D(s, t, u) = \det(F(s), F(t), F(u))$$

has the corresponding fixed Vandermonde sign. Thus the unresolved global issue is not interval-wise sign regularity, but propagation across turning seams and possible nonadjacent recurrence of tangent-slope / tangent-line data.

Proposition 12.66 (Local determinant sign transport across a generic turning boundary). *Let h_b be a generic turning boundary:*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Then there exists a neighborhood U of h_b and a sign $\sigma_{\text{turn}} \in \{\pm 1\}$ such that for any three distinct points $h_1, h_2, h_3 \in U$,

$$\text{sgn } D(h_1, h_2, h_3) = \sigma_{\text{turn}} \text{sgn}((h_2 - h_1)(h_3 - h_1)(h_3 - h_2)),$$

where

$$D(h_1, h_2, h_3) := \det(F(h_1), F(h_2), F(h_3)), \quad F = (c, v_5, \Delta_7).$$

Equivalently, the triple (c, v_5, Δ_7) is locally discretely sign-regular of order 3 across a generic turning boundary.

Proof. Work in the normalized affine chart

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

At a generic turning boundary one has

$$x'(h_b) = 0, \quad x''(h_b) \neq 0, \quad y'(h_b) \neq 0.$$

Hence y is a valid local coordinate near h_b , and the normalized curve may be written as

$$x = \chi(y)$$

with

$$\chi'(y_b) = 0, \quad \chi''(y_b) \neq 0.$$

After shrinking U , the sign of χ'' is constant on U .

Now for $h_i \in U$, write

$$x_i := x(h_i), \quad y_i := y(h_i).$$

Since

$$F(h_i) = c(h_i) (1, x_i, y_i),$$

we have

$$D(h_1, h_2, h_3) = c(h_1)c(h_2)c(h_3) \det \begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{pmatrix}.$$

Using $x_i = \chi(y_i)$, the 3×3 determinant factors as

$$\det \begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{pmatrix} = (y_2 - y_1)(y_3 - y_1)(y_3 - y_2) \chi[y_1, y_2, y_3],$$

where $\chi[y_1, y_2, y_3]$ is the second divided difference of χ . By the scalar divided-difference mean-value theorem,

$$\chi[y_1, y_2, y_3] = \frac{1}{2} \chi''(\eta)$$

for some η between the y_i . Thus $\chi[y_1, y_2, y_3]$ has fixed sign on U .

Also c has fixed sign on a sufficiently small neighborhood of h_b , and since $y'(h_b) \neq 0$, the map $h \mapsto y(h)$ is monotone on U . Hence

$$\operatorname{sgn}((y_2 - y_1)(y_3 - y_1)(y_3 - y_2)) = \operatorname{sgn}(y')^3 \operatorname{sgn}((h_2 - h_1)(h_3 - h_1)(h_3 - h_2)),$$

with $\operatorname{sgn}(y')$ constant on U .

Collecting the fixed signs of c , y' , and χ'' gives the desired constant σ_{turn} . $\square$

Proposition 12.67 (Dual asymptote at a generic turning boundary). *Let h_b be a generic turning boundary:*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Assume $v_5(h_b) \neq 0$, and write

$$\mu_b := \frac{c(h_b)}{v_5(h_b)}, \quad \lambda_b := \frac{\Delta_7(h_b)}{v_5(h_b)}.$$

Let

$$T(h) = (\beta(h), m(h)) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right)$$

be the dual tangent-line curve on the neighboring $M \neq 0$ branches. Then both one-sided dual arcs have the common asymptote

$$m = -\mu_b \beta + \lambda_b,$$

and, more precisely,

$$m + \mu_b \beta - \lambda_b = \frac{1}{4a\beta} + o\left(\frac{1}{\beta}\right) \quad \text{as } |\beta| \rightarrow \infty,$$

where $a \neq 0$ is the quadratic coefficient in the local turning-point normal form

$$\mu - \mu_b = a(\lambda - \lambda_b)^2 + O((\lambda - \lambda_b)^3).$$

In particular, the two adjacent dual arcs approach the same asymptote from opposite sides. This asymptotic separation is the dual-curve explanation of the local adjacent turning-boundary exclusion theorem proved later.

Proof. At a generic turning boundary one has $\mu'(h_b) = 0$ and $\lambda'(h_b) \neq 0$, so λ is a valid local coordinate and

$$\mu - \mu_b = a(\lambda - \lambda_b)^2 + O((\lambda - \lambda_b)^3), \quad a \neq 0.$$

Differentiating gives

$$\frac{d\mu}{d\lambda} = 2a(\lambda - \lambda_b) + O((\lambda - \lambda_b)^2),$$

hence

$$\beta = \frac{d\lambda}{d\mu} = \frac{1}{2a(\lambda - \lambda_b)} + O(1).$$

Thus $|\beta| \rightarrow \infty$ as $h \rightarrow h_b$ from either side.

Now

$$m = \lambda - \mu\beta,$$

so

$$m + \mu_b \beta - \lambda_b = (\lambda - \lambda_b) - (\mu - \mu_b)\beta.$$

Using

$$\mu - \mu_b = a(\lambda - \lambda_b)^2 + O((\lambda - \lambda_b)^3)$$

and

$$\beta = \frac{1}{2a(\lambda - \lambda_b)} + O(1),$$

we obtain

$$(\mu - \mu_b)\beta = \frac{1}{2}(\lambda - \lambda_b) + O((\lambda - \lambda_b)^2).$$

Therefore

$$m + \mu_b \beta - \lambda_b = \frac{1}{2}(\lambda - \lambda_b) + O((\lambda - \lambda_b)^2).$$

Substituting

$$\lambda - \lambda_b = \frac{1}{2a\beta} + o\left(\frac{1}{\beta}\right)$$

gives

$$m + \mu_b \beta - \lambda_b = \frac{1}{4a\beta} + o\left(\frac{1}{\beta}\right).$$

Since $\beta \rightarrow +\infty$ on one side and $\beta \rightarrow -\infty$ on the other, the correction term changes sign, so the two one-sided dual arcs approach the same asymptote from opposite sides. $\square$

Proposition 12.68 (ECT intervals for the strengthened one-pair triple). *Let I be a connected interval on which*

$$c(h) \neq 0, \quad M(h) \neq 0, \quad W(h) \neq 0.$$

Then the triple

$$(c, v_5, \Delta_7)$$

is an extended complete Chebyshev system on I . Equivalently, every nontrivial linear combination

$$Ac(h) + Bv_5(h) + C\Delta_7(h)$$

has at most two zeros on I , counting multiplicity.

In particular, in the affine chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

every affine line meets $\Gamma(I)$ in at most two points, counting tangency with multiplicity.

Proof. Define

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Since the one-pair family is real-analytic on the generic region, the functions x and y are C^2 on I .

By Definition 12.53,

$$x'(h) = \frac{M(h)}{c(h)^2}, \quad y'(h) = \frac{N(h)}{c(h)^2}.$$

Because $M(h) \neq 0$ and $c(h) \neq 0$ on I , one has

$$x'(h) \neq 0 \quad (h \in I),$$

so x is strictly monotone on I and may be used as a parameter on $\Gamma(I)$.

Moreover,

$$\frac{dy}{dx} = \frac{y'(h)}{x'(h)} = \frac{N(h)}{M(h)}.$$

Differentiating with respect to x gives

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dh}(N/M)}{x'(h)} = \frac{(MN' - NM')/M^2}{M/c^2}.$$

Using Proposition 12.57,

$$MN' - NM' = cW,$$

hence

$$\frac{d^2y}{dx^2} = \frac{c(h)^3 W(h)}{M(h)^3}.$$

Since $c(h) \neq 0$, $M(h) \neq 0$, and $W(h) \neq 0$ on I , it follows that

$$\frac{d^2y}{dx^2} \neq 0 \quad (h \in I).$$

Therefore $\Gamma(I)$ is the graph of a strictly convex or strictly concave C^2 function $y = y(x)$.

Now let

$$L(h) := A c(h) + B v_5(h) + C \Delta_7(h)$$

be a nontrivial linear combination. Since $c(h) \neq 0$ on I ,

$$L(h) = 0 \iff A + B x(h) + C y(h) = 0.$$

Thus zeros of L correspond exactly to intersections of $\Gamma(I)$ with the affine line

$$A + B x + C y = 0.$$

If $C = 0$, then the equation is

$$A + B x = 0,$$

which has at most one solution because x is strictly monotone on I . If $C \neq 0$, the equation may be written as

$$y = -\frac{A}{C} - \frac{B}{C}x,$$

and a strictly convex or strictly concave graph meets an affine line in at most two points, counting tangency with multiplicity.

Hence every nontrivial linear combination

$$A c + B v_5 + C \Delta_7$$

has at most two zeros on I , counting multiplicity. This is exactly the ECT property for the triple (c, v_5, Δ_7) . The affine-line statement is the same conclusion in the chart (x, y) . $\square$

Definition 12.69 (ECT interval). An ECT interval is a connected interval

$$I \subset \{h : c(h) \neq 0, M(h) \neq 0, W(h) \neq 0\}.$$

Equivalently, by Proposition 12.68, it is a connected interval on which the triple

$$(c, v_5, \Delta_7)$$

is an extended complete Chebyshev system.

Corollary 12.70 (At most two points from one ECT interval). *Let I be an ECT interval. Then every affine line in the chart*

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

meets $\Gamma(I)$ in at most two points, counting tangency with multiplicity.

In particular, if a finite configuration is cut out by a single affine relation in the normalized chart, then its contribution from any fixed ECT interval has cardinality at most two.

Proof. This is exactly the affine-line formulation of Proposition 12.68. $\square$

Remark 12.71 (Branch geometry on an ECT interval). On an ECT interval I , the quantities c , M , and W have fixed nonzero sign. Hence

$$x = \frac{v_5}{c}$$

is a monotone parameter on I , and in the affine chart

$$\Gamma(I) = \left\{ \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right) : h \in I \right\}$$

is a single C^2 branch which is strictly convex or strictly concave according to the sign of

$$\frac{c(h)^3 W(h)}{M(h)^3}.$$

The later tangent-separator analysis is therefore an adjacent-branch problem, not a same-branch secancy problem.

Definition 12.72 (Tangent-separator kernel). Let

$$F(h) := (c(h), v_5(h), \Delta_7(h)).$$

For s, t in the generic region, define

$$K(s, t) := \det(F(s), F'(s), F(t)).$$

For fixed s , the zeros of $t \mapsto K(s, t)$ are exactly the points $\Gamma(t)$ lying on the affine tangent line to Γ at $\Gamma(s)$. Equivalently, $K(s, t) = 0$ is the tangent-separator incidence condition for the tangent at s and the point at t .

Lemma 12.73 (Affine form of the tangent-separator kernel). *On the affine chart $c \neq 0$, write*

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Then for every s, t with $c(s)c(t) \neq 0$,

$$K(s, t) = c(s)^2 c(t) x'(s) \left((y(t) - m(s)x(t)) - (y(s) - m(s)x(s)) \right),$$

where

$$m(s) = \frac{dy}{dx}(s) = \frac{N(s)}{M(s)}.$$

Equivalently,

$$K(s, t) = c(s)^2 c(t) x'(s) (\phi_{m(s)}(t) - \tau_s),$$

with

$$\phi_{m(s)}(t) := y(t) - m(s)x(t), \quad \tau_s := y(s) - m(s)x(s).$$

Proof. Write

$$F(h) = (c(h), c(h)x(h), c(h)y(h)).$$

Then

$$F'(s) = (c'(s), c'(s)x(s) + c(s)x'(s), c'(s)y(s) + c(s)y'(s)).$$

Subtracting $x(s)$ times the first column from the second and $y(s)$ times the first column from the third gives

$$K(s, t) = \det \begin{pmatrix} c(s) & 0 & 0 \\ c'(s) & c(s)x'(s) & c(s)y'(s) \\ c(t) & c(t)(x(t) - x(s)) & c(t)(y(t) - y(s)) \end{pmatrix}.$$

Expanding along the first row yields

$$K(s, t) = c(s)^2 c(t) (x'(s)(y(t) - y(s)) - y'(s)(x(t) - x(s))).$$

Since

$$m(s) = \frac{y'(s)}{x'(s)},$$

this becomes

$$K(s, t) = c(s)^2 c(t) x'(s) ((y(t) - m(s)x(t)) - (y(s) - m(s)x(s))),$$

as claimed. $\square$

Lemma 12.74 (Basic properties of the tangent-separator kernel). *For every s in the generic region,*

$$K(s, s) = 0, \quad \partial_t K(s, s) = 0, \quad \partial_t^2 K(s, s) = W(s).$$

In particular, on any interval where W has fixed sign, the tangent line at $\Gamma(s)$ is locally supporting.

Proof. The first two identities follow because, at $t = s$, the third column of the determinant equals $F(s)$, and after differentiating once in t it equals $F'(s)$. Differentiating a second time gives

$$\partial_t^2 K(s, s) = \det(F(s), F'(s), F''(s)) = W(s). \quad \square$$

Lemma 12.75 (Compact first-zero reduction on adjacent intervals). *Let $I_1 < I_2$ be adjacent ECT intervals in the affine $x = v_5/c$ ordering. Assume there are compact subintervals*

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

such that every zero of the tangent-separator kernel

$$K(s, t) = \det(F(s), F'(s), F(t)), \quad F(h) = (c(h), v_5(h), \Delta_7(h)),$$

with $s \in I_2$, $t \in I_1$, lies in $J_2 \times J_1$.

If K has a zero on $J_2 \times J_1$, then there exists a zero

$$(s_*, t_*) \in J_2 \times J_1$$

such that s_ is minimal in the $x = v_5/c$ ordering among all s -coordinates of zeros of K on $J_2 \times J_1$.*

Proof. Because K is continuous, its zero set in the compact set $J_2 \times J_1$ is closed, hence compact. The projection of that zero set to the s -axis is therefore compact as well. Since $x = v_5/c$ is monotone on I_2 , the induced x -ordering is equivalent to the interval ordering on J_2 . Thus the projected zero set has a minimal element s_* , and there exists $t_* \in J_1$ such that (s_*, t_*) is a zero of K . $\square$

Lemma 12.76 (First adjacent zero implies tangency). *In the setting of Lemma 12.75, let*

$$(s_*, t_*) \in J_2 \times J_1$$

be a zero with minimal s -coordinate in the $x = v_5/c$ ordering. Then

$$\partial_t K(s_*, t_*) = 0.$$

Consequently,

$$K(s_*, t_*) = 0, \quad \partial_t K(s_*, t_*) = 0$$

force the tangent line at $\Gamma(s_*)$ to be tangent also at $\Gamma(t_*)$.

Proof. Assume for contradiction that

$$\partial_t K(s_*, t_*) \neq 0.$$

Since

$$(s_*, t_*) \in J_2 \times J_1 \subseteq I_2 \times I_1,$$

the implicit function theorem applies in the open set $I_2 \times I_1$ and gives a local C^1 zero curve

$$t = t(s)$$

through (s_*, t_*) , defined for s in a neighborhood of s_* , such that

$$K(s, t(s)) = 0.$$

For s sufficiently close to s_* with $s < s_*$, one still has

$$(s, t(s)) \in I_2 \times I_1$$

and

$$K(s, t(s)) = 0.$$

By the zero-capture hypothesis in Lemma 12.75, every zero of K with $s \in I_2$ and $t \in I_1$ lies in $J_2 \times J_1$. Hence these nearby zeros also lie in $J_2 \times J_1$, contradicting the minimality of s_* among all zeros of K on $J_2 \times J_1$.

Therefore

$$\partial_t K(s_*, t_*) = 0.$$

Finally, the identities

$$K(s, t) = \det(F(s), F'(s), F(t)), \quad \partial_t K(s, t) = \det(F(s), F'(s), F'(t))$$

show that $K(s_*, t_*) = 0$ places $\Gamma(t_*)$ on the tangent line at $\Gamma(s_*)$, while $\partial_t K(s_*, t_*) = 0$ places the tangent direction at t_* on that same line. Hence the line is tangent to Γ at both $\Gamma(s_*)$ and $\Gamma(t_*)$; that is, it is a common tangent line. $\square$

Proposition 12.77 (Adjacent slope-range separation). *Let $I_1 < I_2$ be adjacent ECT intervals, meeting at a boundary point h_b .*

(1) *If*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0,$$

then the interior tangent-slope ranges of I_1 and I_2 are disjoint.

(2) *If*

$$W(h_b) = 0, \quad M(h_b) \neq 0,$$

then $x = v_5/c$ remains a valid local parameter, but no adjacent tangent separation statement is asserted here. In that case the local affine jet is governed by

$$\left. \frac{d^2 y}{dx^2} \right|_{h_b} = 0, \quad \left. \frac{d^3 y}{dx^3} \right|_{h_b} = \frac{c(h_b)^5}{M(h_b)^4} W'(h_b),$$

so the generic behavior is an ordinary affine inflection unless further special structure forces $W'(h_b) = 0$.

Proof. In case (1), $M(h_b) = 0$ is a simple zero and $N(h_b) \neq 0$. Therefore

$$m(h) = \frac{N(h)}{M(h)}$$

has opposite one-sided blow-up:

$$m(h) \rightarrow \pm\infty$$

as $h \rightarrow h_b^\pm$. Hence the interior tangent-slope ranges on the two sides are disjoint.

Case (2) is the local affine-jet computation recorded in Remark 12.85; it identifies the cubic-inflection coefficient but does not produce a tangent-separation theorem. $\square$

Lemma 12.78 (Endpoint tangent exclusion at a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a boundary point h_b , and assume*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Then the boundary tangent line at $\Gamma(h_b)$ is vertical and meets neither adjacent interval in its interior.

Proof. The boundary tangent line is

$$x = \rho_1(h_b), \quad \rho_1(h) = \frac{v_5(h)}{c(h)}.$$

Since $M(h_b) = 0$ is a simple zero,

$$\rho_1'(h_b) = 0, \quad \rho_1''(h_b) \neq 0,$$

so ρ_1 has a strict local extremum at h_b . Therefore

$$\rho_1(h) - \rho_1(h_b)$$

has fixed sign on each adjacent interval and vanishes only at the boundary point. Hence the vertical tangent line meets neither adjacent interval in its interior. $\square$

Theorem 12.79 (Conditional adjacent tangent-separator sign regularity across turning boundaries). *Let $I_1 < I_2$ be adjacent ECT intervals, and assume their common boundary point h_b is a generic turning boundary:*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Assume in addition the zero-capture hypothesis of Lemma 12.75 on the adjacent pair $I_2 \times I_1$. Then

$$K(s, t) \neq 0 \quad (s \in I_2, t \in I_1),$$

where

$$K(s, t) = \det(F(s), F'(s), F(t)), \quad F(h) = (c(h), v_5(h), \Delta_7(h)).$$

Equivalently, tangent lines from the right interval do not meet the adjacent left interval across a generic turning boundary.

Proof. By Lemma 12.78, for a generic turning boundary there is a neighborhood of the boundary pair on which $K(s, t) \neq 0$ for $s \in I_2, t \in I_1$.

Assume for contradiction that some zero $(s_0, t_0) \in I_2 \times I_1$ exists. By the zero-capture hypothesis, there are compact subintervals

$$J_1 \Subset I_1, \quad J_2 \Subset I_2$$

such that every zero of K on $I_2 \times I_1$ lies in $J_2 \times J_1$. Replacing them by smaller compact subintervals still containing the zero set, using the boundary-near zero-free region furnished by Lemma 12.78, we may also assume that $J_2 \times J_1$ is disjoint from that boundary-near region. Then the zero set of K in the compact set $J_2 \times J_1$ is nonempty and closed, so by Lemma 12.75 there exists a zero $(s_*, t_*) \in J_2 \times J_1$ whose s -coordinate is minimal in the $x = v_5/c$ ordering among all zeros in $J_2 \times J_1$.

By Lemma 12.76,

$$K(s_*, t_*) = 0, \quad \partial_t K(s_*, t_*) = 0,$$

so the tangent line at $\Gamma(s_*)$ is also tangent at $\Gamma(t_*)$.

Across a generic turning boundary, Proposition 12.77 shows that the interior tangent-slope ranges of the two adjacent intervals are disjoint. Hence no common interior tangent slope can occur. This contradicts the existence of (s_*, t_*) . Therefore no such zero exists, and

$$K(s, t) \neq 0 \quad (s \in I_2, t \in I_1). \quad \square$$

Remark 12.80 (Status of the turning-boundary adjacent theorem). Theorem 12.79 remains a conditional turning-boundary theorem: the local compact first-zero and first-zero-implies-tangency mechanism is in place, but it still depends on a separate zero-capture statement for the directional tangent-separator kernel $K_{21}(s, t)$ on the full adjacent pair $I_2 \times I_1$.

However, by Theorem 12.96, full zero-capture is no longer needed to exclude actual adjacent pair-pair lines. What remains open here is therefore only the stronger support-ordering / directional-kernel route, not the local adjacent $2 + 2$ exclusion itself.

Lemma 12.81 (Boundary-near exclusion for both directional kernels at a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Define

$$K_{21}(s, t) := \det(F(s), F'(s), F(t)), \quad K_{12}(t, s) := \det(F(t), F'(t), F(s)),$$

with

$$F(h) = (c(h), v_5(h), \Delta_7(h)).$$

Then there is a neighborhood of the boundary pair on which both directional kernels are nonzero:

$$K_{21}(s, t) \neq 0, \quad K_{12}(t, s) \neq 0$$

for $s \in I_2, t \in I_1$ sufficiently close to h_b .

Proof. Write

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Since

$$y'(h) = \frac{N(h)}{c(h)^2},$$

the hypothesis $N(h_b) \neq 0$ implies that y is a valid local coordinate near h_b . Thus the normalized planar curve may be written locally as a graph

$$x = \chi(y).$$

At h_b ,

$$\frac{dx}{dy} = \frac{M(h)}{N(h)},$$

so

$$\left. \frac{dx}{dy} \right|_{h_b} = 0.$$

Differentiating once more,

$$\frac{d^2x}{dy^2} = \frac{c(h)^2}{N(h)} \cdot \frac{M'(h)N(h) - M(h)N'(h)}{N(h)^2},$$

and since $M(h_b) = 0$,

$$\left. \frac{d^2x}{dy^2} \right|_{h_b} = \frac{c(h_b)^2 M'(h_b)}{N(h_b)^2} \neq 0.$$

Hence, after shrinking to a small neighborhood of h_b , the graph $x = \chi(y)$ is strictly convex or strictly concave.

For such a graph, the tangent line at any point meets the graph only at its point of tangency. Therefore no tangent line from one side can meet the opposite side inside a sufficiently small neighborhood of the boundary. Equivalently, both directional tangent-separator kernels are nonzero near the boundary pair. $\square$

Lemma 12.82 (Mirrored first adjacent zero implies tangency). *Let $J_1 \Subset I_1$, $J_2 \Subset I_2$ be compact interior subintervals across a generic turning boundary, and define*

$$K_{12}(t, s) := \det(F(t), F'(t), F(s)), \quad t \in I_1, s \in I_2.$$

Assume every zero of K_{12} on $I_1 \times I_2$ lies in $J_1 \times J_2$. Let $(t_, s_*) \in J_1 \times J_2$ be a zero whose t -coordinate is minimal in the $x = v_5/c$ ordering among all zeros of K_{12} on $I_1 \times I_2$. Then*

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Proof. Because $M \neq 0$ on the open interval I_1 , the function

$$x(h) = \frac{v_5(h)}{c(h)}$$

is a valid local coordinate on I_1 . Since every zero of K_{12} lies in the compact interior rectangle $J_1 \times J_2$, the chosen point (t_*, s_*) is interior in the ambient domain $I_1 \times I_2$.

Suppose

$$\partial_s K_{12}(t_*, s_*) \neq 0.$$

Then by the implicit function theorem there is a C^1 local zero curve

$$s = s(t)$$

through (t_*, s_*) , defined for t near t_* , such that

$$K_{12}(t, s(t)) = 0.$$

Because t_* is interior in I_1 , there are nearby points with smaller $x = v_5/c$ ordering still in I_1 . For such t , the point $(t, s(t))$ is also a zero of K_{12} , contradicting the minimality of t_* . Therefore

$$\partial_s K_{12}(t_*, s_*) = 0.$$

□

Theorem 12.83 (Conditional symmetric directional-kernel exclusion across turning boundaries). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Assume:

- (1) *the zero-capture hypothesis required in Theorem 12.79 for the kernel K_{21} on $I_2 \times I_1$;*
- (2) *the symmetric zero-capture hypothesis for K_{12} , namely that every zero of*

$$K_{12}(t, s) = \det(F(t), F'(t), F(s))$$

on $I_1 \times I_2$ lies in some compact interior rectangle $J_1 \times J_2 \Subset I_1 \times I_2$.

Then both directional kernels are zero-free:

$$K_{21}(s, t) \neq 0 \quad (s \in I_2, t \in I_1),$$

and

$$K_{12}(t, s) \neq 0 \quad (t \in I_1, s \in I_2).$$

Proof. The first assertion is exactly Theorem 12.79.

For the second assertion, boundary-near exclusion for K_{12} is provided by Lemma 12.81. Assume for contradiction that K_{12} has a zero. By the symmetric zero-capture hypothesis, every zero lies in a compact interior rectangle $J_1 \times J_2$. Choose a zero $(t_*, s_*) \in J_1 \times J_2$ whose t -coordinate is minimal in the $x = v_5/c$ ordering among all zeros of K_{12} on $I_1 \times I_2$.

By Lemma 12.82,

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Together with

$$K_{12}(t_*, s_*) = 0,$$

this implies that the tangent line at $\Gamma(t_*)$ is also tangent at $\Gamma(s_*)$. Hence the two adjacent intervals share a common interior tangent slope, contradicting Proposition 12.77. Therefore K_{12} has no zero. □

Remark 12.84 (Status of the directional-kernel reduction). Lemma 12.91 reduces an adjacent pair-pair line across a generic turning boundary to a compact interior zero of one of the two directional tangent-separator kernels. Theorem 12.83 shows that the remaining turning-boundary issue is no longer orientation selection between K_{21} and K_{12} , but the symmetric zero-capture problem for these two kernels.

Remark 12.85 (The $W = 0$ obstruction and the cubic-inflection coefficient). Let

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

At a boundary point h_b with

$$W(h_b) = 0, \quad M(h_b) \neq 0,$$

the coordinate x remains a valid local parameter. Since

$$\frac{d^2y}{dx^2} = \frac{c(h)^3 W(h)}{M(h)^3},$$

one has

$$\left. \frac{d^2y}{dx^2} \right|_{h_b} = 0.$$

Differentiating once more with respect to x gives

$$\left. \frac{d^3y}{dx^3} \right|_{h_b} = \frac{c(h_b)^5}{M(h_b)^4} W'(h_b).$$

Hence, after translating the point to the origin and subtracting the tangent line, the normalized curve has local expansion

$$y = \frac{c(h_b)^5 W'(h_b)}{6M(h_b)^4} x^3 + O(x^4).$$

Therefore a generic $W = 0$, $M \neq 0$ boundary is an ordinary affine inflection unless additional special structure forces

$$W'(h_b) = 0.$$

In particular, the tangent-separator route is not presently justified across such boundaries, and the real remaining geometric question is whether the actual triple (c, v_5, Δ_7) imposes any non-generic vanishing of $W'(h_b)$.

Lemma 12.86 (Fixed-slope intercept sets for a strictly convex ECT branch). *Let I be an ECT interval in the affine chart*

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

and suppose $\Gamma(I)$ is a strictly convex C^2 graph

$$\Gamma(I) = \{(x, \psi_I(x)) : x \in J_I\}$$

over an open interval $J_I \subset \mathbf{R}$.

For each slope $m \in \mathbf{R}$, define

$$\mathcal{B}_I(m) := \{\psi_I(x) - mx : x \in J_I\}.$$

Then:

- (1) $\mathcal{B}_I(m)$ is an interval.
- (2) A line $y = mx + b$ meets $\Gamma(I)$ if and only if

$$b \in \mathcal{B}_I(m).$$

- (3) If m lies in the interior tangent-slope range of I , there is a unique $x_I(m) \in J_I$ such that

$$\psi'_I(x_I(m)) = m,$$

and the tangent line of slope m has intercept

$$\tau_I(m) := \psi_I(x_I(m)) - m x_I(m).$$

Moreover,

$$\tau_I(m) = \inf \mathcal{B}_I(m).$$

(4) For such m , a line $y = mx + b$ is 2-secant to $\Gamma(I)$ if and only if

$$b \in \text{int } \mathcal{B}_I(m).$$

Equivalently, the 2-secant intercepts of slope m are exactly the interior points of the intercept interval.

Proof. For fixed m , put

$$\phi_m(x) := \psi_I(x) - mx.$$

Since ϕ_m is continuous on the interval J_I , its image

$$\phi_m(J_I) = \mathcal{B}_I(m)$$

is an interval. This proves (1), and (2) is immediate from

$$(x, \psi_I(x)) \in \{y = mx + b\} \iff \psi_I(x) - mx = b.$$

If m lies in the interior tangent-slope range, strict convexity implies that ψ'_I is strictly monotone, hence there is a unique $x_I(m)$ with $\psi'_I(x_I(m)) = m$. At that point,

$$\phi'_m(x_I(m)) = \psi'_I(x_I(m)) - m = 0,$$

and since

$$\phi''_m(x) = \psi''_I(x) > 0,$$

the point $x_I(m)$ is the unique minimizer of ϕ_m . Thus

$$\tau_I(m) = \phi_m(x_I(m)) = \inf \mathcal{B}_I(m),$$

which proves (3).

Finally, because ϕ_m is strictly convex with a unique minimum, a level b is attained at two points exactly when it lies in the interior of the interval $\phi_m(J_I) = \mathcal{B}_I(m)$. This is precisely the 2-secant condition, proving (4). $\square$

Proposition 12.87 (Support-curve calculus on an ECT branch). *Let I be as in Lemma 12.86, and assume m lies in the interior tangent-slope range of I . Then*

$$x'_I(m) = \frac{1}{\psi''_I(x_I(m))} > 0,$$

$$\tau'_I(m) = -x_I(m),$$

and

$$\tau''_I(m) = -\frac{1}{\psi''_I(x_I(m))} < 0.$$

In particular, the tangent-intercept function $m \mapsto \tau_I(m)$ is strictly concave on the interior tangent-slope range.

Proof. The defining relation

$$\psi'_I(x_I(m)) = m$$

implies, by differentiation in m ,

$$\psi''_I(x_I(m)) x'_I(m) = 1,$$

hence

$$x'_I(m) = \frac{1}{\psi''_I(x_I(m))} > 0.$$

Now

$$\tau_I(m) = \psi_I(x_I(m)) - m x_I(m).$$

Differentiate:

$$\tau'_I(m) = \psi'_I(x_I(m)) x'_I(m) - x_I(m) - m x'_I(m) = m x'_I(m) - x_I(m) - m x'_I(m) = -x_I(m).$$

Differentiating once more gives

$$\tau_I''(m) = -x_I'(m) = -\frac{1}{\psi_I''(x_I(m))} < 0.$$

□

Corollary 12.88 (Single crossing and betweenness of support curves). *Let $I_i < I_j$ be two ordered strict convex ECT branches in the affine chart. On every common interior tangent-slope interval,*

$$(\tau_{I_j} - \tau_{I_i})'(m) = -(x_{I_j}(m) - x_{I_i}(m)) < 0.$$

Hence τ_{I_i} and τ_{I_j} cross at most once; equivalently, the two ordered branches admit at most one common interior tangent slope.

More generally, let

$$I_i < I_j < I_k$$

be three ordered strict convex ECT branches, active on a common interior slope interval. If τ_{I_i} and τ_{I_j} cross at m_{ij} and τ_{I_j} and τ_{I_k} cross at m_{jk} with

$$m_{ij} < m_{jk},$$

then there is a unique

$$m_{ik} \in (m_{ij}, m_{jk})$$

such that

$$\tau_{I_i}(m_{ik}) = \tau_{I_k}(m_{ik}).$$

So the support curves

$$b = \tau_I(m)$$

form a pseudoline arrangement on every common slope window.

Proof. Since the branches are ordered in the $x = v_5/c$ coordinate, one has

$$x_{I_i}(m) < x_{I_j}(m)$$

whenever both tangencies exist. Therefore

$$(\tau_{I_j} - \tau_{I_i})'(m) = -(x_{I_j}(m) - x_{I_i}(m)) < 0,$$

so $\tau_{I_j} - \tau_{I_i}$ is strictly decreasing and can vanish at most once.

For the betweenness statement, set

$$\delta_{ab}(m) := \tau_{I_b}(m) - \tau_{I_a}(m).$$

Each δ_{ab} is strictly decreasing. If

$$\delta_{ij}(m_{ij}) = 0, \quad \delta_{jk}(m_{jk}) = 0, \quad m_{ij} < m_{jk},$$

then

$$\delta_{ik}(m_{ij}) = \delta_{ij}(m_{ij}) + \delta_{jk}(m_{ij}) = \delta_{jk}(m_{ij}) > 0,$$

because δ_{jk} vanishes later at m_{jk} , and similarly

$$\delta_{ik}(m_{jk}) = \delta_{ij}(m_{jk}) + \delta_{jk}(m_{jk}) = \delta_{ij}(m_{jk}) < 0.$$

By strict monotonicity, δ_{ik} has a unique zero in (m_{ij}, m_{jk}) . □

Proposition 12.89 (Ordered-pair overlap windows are one-shot). *Let $I_1 < I_2$ be two ordered strict convex ECT branches in the affine chart, with*

$$\Gamma(I_j) = \{(x, \psi_j(x)) : x \in J_j\}, \quad J_1 < J_2.$$

For $j = 1, 2$, let

$$\mathcal{B}_{I_j}(m) = \{\psi_j(x) - mx : x \in J_j\}, \quad u_j(m) := \sup \mathcal{B}_{I_j}(m),$$

and let $\tau_j(m)$ be the tangent intercept on the interior tangent-slope range.

Then the directional overlap window

$$\Sigma_{2 \rightarrow 1} := \{m : \tau_1(m) \leq \tau_2(m) \leq u_1(m)\}$$

is an interval, possibly empty.

Likewise the pair-pair overlap window

$$\Sigma_{12}^{\text{pp}} := \{m : \text{int } \mathcal{B}_{I_1}(m) \cap \text{int } \mathcal{B}_{I_2}(m) \neq \emptyset\}$$

is an interval, possibly empty.

More precisely:

(1) on every slope subinterval where one fixed endpoint support line

$$m \mapsto y_\xi - m\xi$$

realizes $u_1(m)$, one has

$$\frac{d}{dm}(u_1(m) - \tau_2(m)) = x_{I_2}(m) - \xi > 0;$$

(2) on every slope subinterval where one fixed endpoint support line

$$m \mapsto y_\eta - m\eta$$

realizes $u_2(m)$, one has

$$\frac{d}{dm}(u_2(m) - \tau_1(m)) = x_{I_1}(m) - \eta < 0.$$

In particular, for a fixed ordered pair, directional or pair-pair overlap cannot disappear and later reappear.

Proof. By Corollary 12.88,

$$\tau_2(m) - \tau_1(m)$$

is strictly decreasing on the common interior slope range. Therefore the set

$$\{m : \tau_1(m) \leq \tau_2(m)\}$$

is an interval.

Now fix a slope subinterval on which one endpoint support line

$$m \mapsto y_\xi - m\xi$$

realizes $u_1(m)$. Then

$$u_1(m) = y_\xi - m\xi,$$

so

$$\frac{d}{dm}(u_1(m) - \tau_2(m)) = -\xi - (-x_{I_2}(m)) = x_{I_2}(m) - \xi.$$

Because every point of J_2 lies strictly to the right of every endpoint of J_1 , one has

$$x_{I_2}(m) - \xi > 0.$$

Hence

$$u_1(m) - \tau_2(m)$$

is strictly increasing on that piece. Therefore the set

$$\{m : \tau_2(m) \leq u_1(m)\}$$

is an interval after patching across the finitely many endpoint-switch slopes. Thus

$$\Sigma_{2 \rightarrow 1} = \{m : \tau_1(m) \leq \tau_2(m)\} \cap \{m : \tau_2(m) \leq u_1(m)\}$$

is an interval.

Similarly, on a slope subinterval where one endpoint support line

$$m \mapsto y_\eta - m\eta$$

realizes $u_2(m)$,

$$\frac{d}{dm}(u_2(m) - \tau_1(m)) = -\eta - (-x_{I_1}(m)) = x_{I_1}(m) - \eta < 0,$$

because every point of J_1 lies strictly to the left of every endpoint of J_2 .

Finally, by Lemma 12.86,

$$\text{int } \mathcal{B}_{I_1}(m) \cap \text{int } \mathcal{B}_{I_2}(m) \neq \emptyset$$

is equivalent to

$$\tau_2(m) < u_1(m) \quad \text{and} \quad \tau_1(m) < u_2(m).$$

The first inequality holds on an interval because $u_1 - \tau_2$ is piecewise strictly increasing; the second holds on an interval because $u_2 - \tau_1$ is piecewise strictly decreasing. Their intersection is therefore an interval. $\square$

Lemma 12.90 (Support-overlap truncation across a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$c(h_b) \neq 0, \quad M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

For a fixed finite slope m , define

$$\phi_m(h) := \frac{\Delta_7(h) - mv_5(h)}{c(h)}, \quad \mathcal{B}_I(m) := \phi_m(I).$$

Assume

$$b \in \mathcal{B}_{I_1}(m) \cap \mathcal{B}_{I_2}(m).$$

Then there exists a neighborhood U of h_b and compact subintervals

$$J_1 \subseteq I_1 \setminus U, \quad J_2 \subseteq I_2 \setminus U$$

such that

$$b \in \mathcal{B}_{J_1}(m) \cap \mathcal{B}_{J_2}(m).$$

Proof. Differentiate:

$$\phi'_m(h) = \frac{N(h) - mM(h)}{c(h)^2}.$$

At $h = h_b$, since $M(h_b) = 0$,

$$\phi'_m(h_b) = \frac{N(h_b)}{c(h_b)^2} \neq 0.$$

Hence for each fixed finite m , after shrinking to a small neighborhood U of h_b , the function ϕ_m is strictly monotone on each one-sided piece

$$I_1 \cap U, \quad I_2 \cap U.$$

Therefore, for each $i = 1, 2$, the equation

$$\phi_m(h) = b$$

has at most one solution in $I_i \cap U$.

But the assumption

$$b \in \mathcal{B}_{I_i}(m)$$

means exactly that there is at least one solution of $\phi_m(h) = b$ somewhere on I_i . Therefore, after removing the boundary neighborhood U , at least one such solution

$$h_i^\sharp \in I_i \setminus U$$

remains on each side. Choose small compact subintervals

$$J_i \subseteq I_i \setminus U$$

with $h_i^\sharp \in J_i$. Then by continuity,

$$b = \phi_m(h_i^\sharp) \in \phi_m(J_i) = \mathcal{B}_{J_i}(m),$$

for $i = 1, 2$. This proves the claim. $\square$

Lemma 12.91 (Compact adjacent pair-pair implies a directional kernel zero). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary, and let*

$$L : y = mx + b$$

be a finite-slope line that is 2-secant to both $\Gamma(I_1)$ and $\Gamma(I_2)$. Then, after removing a sufficiently small boundary neighborhood, there exist compact subintervals

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

and points $s \in J_2, t \in J_1$ such that one of the two directional kernels vanishes:

$$K_{21}(s, t) := \det(F(s), F'(s), F(t)) = 0,$$

or

$$K_{12}(t, s) := \det(F(t), F'(t), F(s)) = 0,$$

where

$$F(h) = (c(h), v_5(h), \Delta_7(h)).$$

Proof. Because L is 2-secant to each branch, Lemma 12.86 shows that m lies in the interior tangent-slope range of each interval. Hence each branch has a unique interior tangent line of slope m , with tangent intercept

$$\tau_{I_i}(m) = \inf \mathcal{B}_{I_i}(m).$$

At a generic turning boundary,

$$\frac{dy/dh}{dx/dh} = \frac{N(h)}{M(h)}$$

has one-sided blow-up, so no finite-slope tangency can occur arbitrarily close to h_b . Therefore, after shrinking away a boundary neighborhood and applying Lemma 12.90, we may choose compact subintervals

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

such that:

- (1) $b \in \mathcal{B}_{J_1}(m) \cap \mathcal{B}_{J_2}(m)$;
- (2) the unique slope- m tangency point of each branch lies in the corresponding J_i .

Hence each $\mathcal{B}_{J_i}(m)$ is a compact interval whose lower endpoint is the tangent intercept

$$\tau_i := \tau_{J_i}(m) = \inf \mathcal{B}_{J_i}(m).$$

Now set

$$b_* := \inf(\mathcal{B}_{J_1}(m) \cap \mathcal{B}_{J_2}(m)).$$

Since the intersection of two compact intervals is again a compact interval,

$$b_* = \max(\tau_1, \tau_2).$$

So b_* is equal to one of the two tangent intercepts, and it still lies in the support set of the other branch.

If $b_* = \tau_2$, then the slope- m tangent line to $\Gamma(J_2)$ meets $\Gamma(J_1)$. Thus there are points $s \in J_2, t \in J_1$ such that

$$\det(F(s), F'(s), F(t)) = 0,$$

i.e.

$$K_{21}(s, t) = 0.$$

If $b_* = \tau_1$, then the slope- m tangent line to $\Gamma(J_1)$ meets $\Gamma(J_2)$. Thus there are points $t \in J_1, s \in J_2$ such that

$$\det(F(t), F'(t), F(s)) = 0,$$

i.e.

$$K_{12}(t, s) = 0.$$

This proves the lemma. □

Lemma 12.92 (Strip exclusion for K_{21} against a compact opposite branch). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Let $J_1 \Subset I_1$ be a compact subinterval. Then there exists a one-sided neighborhood $U_2 \subset I_2$ of the shared boundary point h_b such that

$$K_{21}(s, t) := \det(F(s), F'(s), F(t)) \neq 0 \quad (s \in U_2, t \in J_1),$$

where $F(h) = (c(h), v_5(h), \Delta_7(h))$.

Proof. Work in the affine chart

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

At a generic turning boundary one has

$$x'(h_b) = 0, \quad x''(h_b) \neq 0, \quad y'(h_b) \neq 0.$$

Hence y is a valid local coordinate and the normalized curve may be written near h_b as a graph

$$x = \chi(y)$$

with

$$\chi'(y_b) = 0, \quad \chi''(y_b) \neq 0.$$

Thus x has a strict local extremum at the boundary, while the tangent slope

$$m(h) = \frac{dy/dh}{dx/dh} = \frac{N(h)}{M(h)}$$

blows up one-sidedly as $h \rightarrow h_b$ from either adjacent interval.

Because $J_1 \Subset I_1$ is compact away from h_b , there exists $\delta > 0$ such that

$$|x(t) - x(h_b)| \geq 2\delta \quad (t \in J_1).$$

After shrinking to a small one-sided neighborhood $U_2 \subset I_2$ of h_b , we may also assume

$$|x(s) - x(h_b)| \leq \delta \quad (s \in U_2).$$

Hence

$$|x(t) - x(s)| \geq \delta \quad (s \in U_2, t \in J_1).$$

Fix a preliminary compact neighborhood $U_2^0 \Subset I_2$ of h_b and set

$$B := \sup\{|y(t) - y(s)| : t \in J_1, s \in U_2^0\} < \infty.$$

Since $m(s) \rightarrow \pm\infty$ as $s \rightarrow h_b$ in I_2 , we may shrink further to $U_2 \subset U_2^0$ so that

$$|m(s)| > \frac{2B}{\delta} \quad (s \in U_2).$$

If $K_{21}(s, t) = 0$, then the tangent line at $\Gamma(s)$ passes through $\Gamma(t)$, equivalently

$$y(t) - y(s) = m(s)(x(t) - x(s)).$$

Taking absolute values gives

$$B \geq |y(t) - y(s)| = |m(s)| |x(t) - x(s)| > \frac{2B}{\delta} \cdot \delta = 2B,$$

a contradiction. Therefore $K_{21}(s, t) \neq 0$ on $U_2 \times J_1$. □

Lemma 12.93 (Strip exclusion for K_{12} against a compact opposite branch). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Let $J_2 \Subset I_2$ be a compact subinterval. Then there exists a one-sided neighborhood $U_1 \subset I_1$ of the shared boundary point h_b such that

$$K_{12}(t, s) := \det(F(t), F'(t), F(s)) \neq 0 \quad (t \in U_1, s \in J_2),$$

where $F(h) = (c(h), v_5(h), \Delta_7(h))$.

Proof. This is the left-right mirror of Lemma 12.92. $\square$

Lemma 12.94 (Compact first-contact for K_{21}). *Let $J_1 \in I_1$, $J_2 \in I_2$ be compact subintervals across a generic turning boundary. Assume there is a relatively open collar $V_2 \subset J_2$ at the boundary-facing end of J_2 such that*

$$K_{21}(s, t) \neq 0 \quad (s \in V_2, t \in J_1).$$

Assume also that the zero set of K_{21} in $J_2 \times J_1$ is nonempty. Then there exists a zero $(s_, t_*) \in J_2 \times J_1$ such that:*

- (1) s_* is boundary-nearest in the $x = v_5/c$ ordering among all $s \in J_2$ for which $K_{21}(s, t) = 0$ for some $t \in J_1$;
- (2)

$$\partial_t K_{21}(s_*, t_*) = 0.$$

Proof. Define

$$A_{21} := \{s \in J_2 : \exists t \in J_1 \text{ with } K_{21}(s, t) = 0\}.$$

Since the zero set of K_{21} in $J_2 \times J_1$ is compact and nonempty, the set A_{21} is compact and nonempty. By hypothesis,

$$A_{21} \cap V_2 = \emptyset.$$

Hence A_{21} has a boundary-nearest element s_* in the $x = v_5/c$ ordering on J_2 . Choose $t_* \in J_1$ with

$$K_{21}(s_*, t_*) = 0.$$

Suppose

$$\partial_t K_{21}(s_*, t_*) \neq 0.$$

Then by the implicit function theorem there is a C^1 local zero curve

$$t = t(s)$$

through (s_*, t_*) , defined for s near s_* , such that

$$K_{21}(s, t(s)) = 0.$$

Because $s_* \notin V_2$ and V_2 is an open collar at the boundary-facing end of J_2 , there exist nearby points $s \in J_2$ that are boundary-nearer than s_* in the $x = v_5/c$ ordering. For such s , the point $(s, t(s))$ is also a zero of K_{21} , contradicting the minimality of s_* . Therefore

$$\partial_t K_{21}(s_*, t_*) = 0.$$

$\square$

Lemma 12.95 (Compact first-contact for K_{12}). *Let $J_1 \in I_1$, $J_2 \in I_2$ be compact subintervals across a generic turning boundary. Assume there is a relatively open collar $V_1 \subset J_1$ at the boundary-facing end of J_1 such that*

$$K_{12}(t, s) \neq 0 \quad (t \in V_1, s \in J_2).$$

Assume also that the zero set of K_{12} in $J_1 \times J_2$ is nonempty. Then there exists a zero $(t_, s_*) \in J_1 \times J_2$ such that:*

- (1) t_* is boundary-nearest in the $x = v_5/c$ ordering among all $t \in J_1$ for which $K_{12}(t, s) = 0$ for some $s \in J_2$;
- (2)

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Proof. Define

$$A_{12} := \{t \in J_1 : \exists s \in J_2 \text{ with } K_{12}(t, s) = 0\}.$$

Since the zero set of K_{12} in $J_1 \times J_2$ is compact and nonempty, the set A_{12} is compact and nonempty. By hypothesis,

$$A_{12} \cap V_1 = \emptyset.$$

Hence A_{12} has a boundary-nearest element t_* in the $x = v_5/c$ ordering on J_1 . Choose $s_* \in J_2$ with

$$K_{12}(t_*, s_*) = 0.$$

Suppose

$$\partial_s K_{12}(t_*, s_*) \neq 0.$$

Then by the implicit function theorem there is a C^1 local zero curve

$$s = s(t)$$

through (t_*, s_*) , defined for t near t_* , such that

$$K_{12}(t, s(t)) = 0.$$

Because $t_* \notin V_1$ and V_1 is an open collar at the boundary-facing end of J_1 , there exist nearby points $t \in J_1$ that are boundary-nearer than t_* in the $x = v_5/c$ ordering. For such t , the point $(t, s(t))$ is also a zero of K_{12} , contradicting the minimality of t_* . Therefore

$$\partial_s K_{12}(t_*, s_*) = 0.$$

□

Theorem 12.96 (Adjacent pair-pair exclusion across a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Then no finite-slope affine line in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is 2-secant to both $\Gamma(I_1)$ and $\Gamma(I_2)$.

Proof. Assume for contradiction that a finite-slope line

$$L : y = mx + b$$

is 2-secant to both adjacent branches. By Lemma 12.91, after removing a small boundary neighborhood there exist compact subintervals

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

and a zero of one of the two directional kernels.

First suppose there exist $s_0 \in J_2$, $t_0 \in J_1$ such that

$$K_{21}(s_0, t_0) = 0.$$

By Lemma 12.92, there exists a one-sided neighborhood $U_2 \subset I_2$ of h_b such that

$$K_{21}(s, t) \neq 0 \quad (s \in U_2, t \in J_1).$$

Choose a compact interval $\tilde{J}_2 \subseteq I_2$ containing s_0 and whose boundary-facing collar V_2 lies in U_2 . Then the zero set of K_{21} in $\tilde{J}_2 \times J_1$ is nonempty, and Lemma 12.94 yields a zero

$$(s_*, t_*) \in \tilde{J}_2 \times J_1$$

such that

$$\partial_t K_{21}(s_*, t_*) = 0.$$

Together with $K_{21}(s_*, t_*) = 0$, this means that the tangent line at $\Gamma(s_*)$ is also tangent at $\Gamma(t_*)$. Since $s_* \in I_2$ and $t_* \in I_1$ are interior points, this gives a common interior tangent slope across the adjacent branches, contradicting Proposition 12.77.

Now suppose instead there exist $t_0 \in J_1$, $s_0 \in J_2$ such that

$$K_{12}(t_0, s_0) = 0.$$

By Lemma 12.93, there exists a one-sided neighborhood $U_1 \subset I_1$ of h_b such that

$$K_{12}(t, s) \neq 0 \quad (t \in U_1, s \in J_2).$$

Choose a compact interval $\tilde{J}_1 \subseteq I_1$ containing t_0 and whose boundary-facing collar V_1 lies in U_1 . Then the zero set of K_{12} in $\tilde{J}_1 \times J_2$ is nonempty, and Lemma 12.95 yields a zero

$$(t_*, s_*) \in \tilde{J}_1 \times J_2$$

such that

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Together with $K_{12}(t_*, s_*) = 0$, this means that the tangent line at $\Gamma(t_*)$ is also tangent at $\Gamma(s_*)$. Again this gives a common interior tangent slope across the adjacent branches, contradicting Proposition 12.77.

Both cases are impossible. Therefore no finite-slope line is 2-secant to both adjacent branches across a generic turning boundary. $\square$

Remark 12.97 (What the unconditional adjacent turning theorem does and does not yet replace). Theorem 12.96 gives an unconditional exclusion of finite-slope pair-pair lines across a single generic turning boundary. It is stronger than the older local directional-kernel reduction in that it bypasses full symmetric zero-capture.

However, it does not by itself replace Theorem 12.100. To obtain a full global no-four-secant theorem from it, one would still need a separate propagation statement reducing nonadjacent pair-pair lines to the adjacent turning-boundary case.

Remark 12.98 (Status of the directional-kernel reduction). Lemma 12.91 reduces an adjacent pair-pair line across a generic turning boundary to a compact interior zero of one of the two directional tangent-separator kernels.

There are now two closures of that reduction in the draft:

- (1) Theorem 12.96 gives an unconditional closure for actual adjacent pair-pair lines, via strip exclusion and compact first-contact.
- (2) Theorem 12.83 gives a stronger zero-freeness statement for both directional kernels, but only conditionally on symmetric zero-capture.

Thus orientation selection between K_{21} and K_{12} is no longer the main issue. What remains open only for the stronger tangent-separator / support-ordering route is the symmetric zero-capture problem on the full adjacent rectangle.

Lemma 12.99 (Adjacent support-ordering). *Let $I_1 < I_2$ be adjacent ECT intervals under the hypotheses of Theorem 12.79. Then, for every slope $m \in \mathbf{R}$ such that both fixed-slope intercept sets are nonempty,*

$$\mathcal{B}_{I_1}(m) \cap \mathcal{B}_{I_2}(m) = \emptyset.$$

More precisely, one has one of the two strict alternatives

$$\sup \mathcal{B}_{I_1}(m) < \inf \mathcal{B}_{I_2}(m) \quad \text{or} \quad \sup \mathcal{B}_{I_2}(m) < \inf \mathcal{B}_{I_1}(m),$$

and the same alternative holds for all such m on the adjacent pair.

Proof. Fix m such that both $\mathcal{B}_{I_1}(m)$ and $\mathcal{B}_{I_2}(m)$ are nonempty.

First assume m lies in the interior tangent-slope range of I_2 . Let $s \in I_2$ be the unique point with tangent slope m . Then by Lemma 12.86,

$$\tau_{I_2}(m) = \inf \mathcal{B}_{I_2}(m).$$

By Theorem 12.79,

$$K(s, t) \neq 0 \quad (t \in I_1).$$

By Lemma 12.73,

$$K(s, t) = c(s)^2 c(t) x'(s) (\phi_m(t) - \tau_{I_2}(m)), \quad \phi_m(t) := y(t) - mx(t).$$

Since $c(s)^2 c(t) x'(s)$ has fixed sign on I_1 and $K(s, t)$ is continuous and nowhere zero there, the quantity

$$\phi_m(t) - \tau_{I_2}(m)$$

has constant sign on I_1 . To determine which sign occurs, move s toward the common boundary point from inside I_2 . By Lemma 12.78, the limiting boundary tangent line does not meet I_1 , so

I_1 lies entirely on one side of that tangent line. By continuity, the same side persists for nearby interior tangents. Indeed, if the side changed as the tangency point s moved along I_2 , then by the connectedness of I_2 and continuity of the tangent-line data there would exist an intermediate point $s_* \in I_2$ for which the tangent line at $\Gamma(s_*)$ meets $\Gamma(I_1)$. Equivalently, there would exist $t_* \in I_1$ such that

$$K(s_*, t_*) = 0,$$

contradicting Theorem 12.79. Hence the same side holds for all $s \in I_2$. Therefore either

$$\phi_m(t) < \tau_{I_2}(m) \quad \text{for all } t \in I_1,$$

or

$$\phi_m(t) > \tau_{I_2}(m) \quad \text{for all } t \in I_1.$$

In the first case,

$$\sup \mathcal{B}_{I_1}(m) \leq \tau_{I_2}(m) = \inf \mathcal{B}_{I_2}(m),$$

and since equality would force an intersection with the tangent line, the inequality is strict. The second case is symmetric.

Now suppose m is not in the interior tangent-slope range of I_2 , but both intercept sets are nonempty. Then m lies on one side of the tangent slope range of I_2 , so ϕ_m is strictly monotone on I_2 , and $\mathcal{B}_{I_2}(m)$ is an interval with endpoint attained in the one-sided limit at the adjacent boundary. Passing to that boundary-support line and using Lemma 12.78 again shows that I_1 lies strictly on one side of that boundary-support line. Hence the same strict ordering of intercept sets holds in this case as well.

Thus $\mathcal{B}_{I_1}(m)$ and $\mathcal{B}_{I_2}(m)$ are disjoint for every slope m for which both are nonempty, and the side ordering is constant on the adjacent pair. $\square$

Theorem 12.100 (Partial global no-secant theorem on turning-boundary generic blocks). *Assume every adjacent pair of ECT intervals is separated by a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0,$$

and assume moreover that the zero-capture hypothesis required in Theorem 12.79 holds for every adjacent turning-boundary pair. Then no affine line in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is 2-secant to two distinct ECT intervals. In particular, there is no global four-secant line for the normalized planar curve $\Gamma(h)$ on such a turning-boundary generic block.

Proof. Under the stated hypotheses, every adjacent boundary is of turning type and Theorem 12.79 applies to each adjacent pair. For such adjacent pairs, Lemma 12.99 applies. Hence adjacent fixed-slope intercept sets are strictly ordered. Iterating this ordering over the $x = v_5/c$ ordering of ECT intervals shows that no affine line can be 2-secant to two distinct ECT intervals. Therefore no global four-secant line exists on the block. $\square$

Corollary 12.101 (Partial exclusion of pair-pair branch configurations). *Under the hypotheses of Theorem 12.100, a bad finite core cannot contain two distinct ECT intervals each contributing a pair of points on a turning-boundary generic block.*

Proof. Each ECT interval contributes at most two points to the zero set of an affine line in the normalized chart, and by Theorem 12.100 no affine line can be 2-secant to two different ECT intervals on such a block. Therefore a pair-pair configuration across two distinct ECT intervals is impossible there. $\square$

Remark 12.102 (Strip picture and clean-block compression). For each ECT interval I in the normalized affine chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

it is convenient to package the fixed-slope 2-secant data by the open secant strip

$$\mathcal{S}_I := \{(m, b) : b \in \text{int } \mathcal{B}_I(m)\},$$

where $\mathcal{B}_I(m)$ is the intercept interval from Lemma 12.86. Thus

$$(m, b) \in \mathcal{S}_I$$

if and only if the affine line $y = mx + b$ is 2-secant to $\Gamma(I)$.

Under the hypotheses of Theorem 12.100, these strips are pairwise disjoint across distinct ECT intervals on the block:

$$\mathcal{S}_{I_i} \cap \mathcal{S}_{I_j} = \emptyset \quad (i \neq j).$$

Equivalently, a pair-pair state can use at most one ECT interval from such a turning-boundary generic block. In the later finite-core language, every such block behaves as a single transport chamber rather than as an independent source of $2 + 2$ combinatorics.

Remark 12.103 (Canonical quotient-level closure versus the honest order-7 lane). The direct $q^{(7)}$ -slice of total Δ_7 is now closed at the weight-one level by Proposition 12.48. Together with Theorem 12.49, this closes the canonical quotient-level order-7 package.

However, this does *not* mean that order 7 has disappeared from the later endgame. The honest transverse order-7 datum is not raw septic equality of representatives, but the quotient-septic class

$$[A_7^{\mathfrak{f}}] \in \mathfrak{f}/\mathbf{CA}_5^{\mathfrak{f}}.$$

Once two corrected germs agree through cubic and quintic order and differ only by odd reparametrization plus the already-quotiented lower scalar/value gauge, the remaining septic mismatch is automatically a multiple of $A_5^{\mathfrak{f}}$, so the quotient-septic defect vanishes. What is not yet proved is that cubic/quintic merger alone forces that odd-reparametrization comparability.

Accordingly, the honest order-7 theorem target is not “prove raw septic equality.” It is a provenance-sensitive package statement, for example:

- same reduced image germ at coincidence; or
- collision-functoriality of the corrected block triple on the collision/cancellation locus.

If such a statement holds, the quotient-septic collapse becomes formal afterward. So order 7 is canonically closed, but not yet globally exhausted.

Remark 12.104 (Canonical quotient-septic closure is still locally free). The canonical quotient-level order-7 package is now closed, but that does not yet make the quotient-septic geometry rigid.

Indeed, once the lower one-pair data (c, u_5, v_5) are fixed, the canonical septic residual

$$\Delta_7 = u_7 v_5 - u_5 v_7$$

is still locally free. On a region where $v_5 \neq 0$ one may gauge-fix

$$v_7^{\text{new}} = 0, \quad u_7^{\text{new}} = \frac{\Delta_7}{v_5},$$

while on a region where $u_5 \neq 0$ one may gauge-fix

$$u_7^{\text{new}} = 0, \quad v_7^{\text{new}} = -\frac{\Delta_7}{u_5}.$$

So after the lower one-pair data are fixed, the canonical quotient-septic curve

$$\Gamma(h) = \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right)$$

is still locally determined by an arbitrary analytic choice of Δ_7 .

Accordingly, no theorem using only arbitrary canonical quotient-septic geometry of (c, v_5, Δ_7) can by itself close the later repeated-source / discriminant branch. Any successful higher-order hybrid must therefore be provenance-sensitive, not merely quotient-septic.

Remark 12.105 (Status and ceiling of the highest-new direct provenance slice). On the region $v_5 \neq 0$, set

$$\mu := \frac{c}{v_5}, \quad \lambda := \frac{\Delta_7}{v_5}, \quad \phi := \lambda^{[1]}.$$

By Corollary 12.43, Corollary 12.47, and Theorem 12.49, the highest-new quotient-level septic scalar is already reduced to the direct $q^{(7)}$ -driven K_1 -slice:

$$\Delta_7^{[1]} = v_5 \phi, \quad \phi = \kappa \left((x_2 - x_1) \nu_6^{(q^{(7)})} + v(\gamma_5 - \alpha_5)^{(q^{(7)})} \right).$$

On the subregion where $M \neq 0$, the direct highest-new dual variables are

$$\beta^{[1]} = \phi_\mu, \quad m^{[1]} = \phi - \mu \phi_\mu,$$

and by Corollary 12.63,

$$\phi_{\mu\mu} = \frac{v_5^3}{M^3} W^{[1]}.$$

Thus the direct highest-new quotient-septic geometry is exactly the ordinary tangent-line geometry of the scalar graph

$$y = \phi(\mu).$$

In particular, any bitangent line or four-secant line for the graph $y = \phi(\mu)$, and any nonadjacent slope re-entry for

$$\beta^{[1]} = \phi_\mu,$$

forces at least two zeros of

$$\phi_{\mu\mu},$$

equivalently of

$$W^{[1]}.$$

So the global direct-slice propagation problem may be viewed as a no-recurrence problem for the scalar graph ϕ .

However, the explicit direct provenance formula only shows that

$$\phi = A q^{(7)} \quad (A := \kappa \eta_2),$$

with A generically nonvanishing on a generic one-pair block. By Proposition 13.8 below, this still leaves ϕ fully universal on each such block. So the direct graph picture is now a precise dictionary, not by itself a live rigidity theorem.

Remark 12.106 (Status of the $W = 0$, $M \neq 0$ side). On the region $c \neq 0$, $M \neq 0$, Proposition 12.59 excludes open-interval flatness $W \equiv 0$. At an isolated boundary

$$W(h_b) = 0, \quad M(h_b) \neq 0,$$

the normalized planar curve in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is an ordinary odd-order affine inflection whenever W changes sign, so adjacent tangent-separator sign regularity fails there in general.

However, Theorem 12.107 shows that every finite-order isolated boundary of type $W = 0$, $M \neq 0$ is locally sterile for pair-pair birth. Thus the remaining issue on this side is not local creation of a bad $2 + 2$ line, but global continuation / recurrence of determinant-zero or tangent-slope data across multiple intervals.

The exact identities for W and W' continue to show that this residual obstruction is still encoded through the canonical order-7 package Δ_7 , not by a genuinely new primitive order-9 invariant.

Theorem 12.107 (Finite-order $W = 0$ boundaries are locally sterile for pair-pair birth). *Let h_b satisfy*

$$c(h_b) \neq 0, \quad M(h_b) \neq 0, \quad W(h_b) = 0,$$

and assume that W has finite vanishing order $r \geq 1$ at h_b :

$$W(h) = (h - h_b)^r \widetilde{W}(h), \quad \widetilde{W}(h_b) \neq 0.$$

Define

$$F(h) := (c(h), v_5(h), \Delta_7(h)), \quad D(s, t, u) := \det(F(s), F(t), F(u)).$$

Then, after shrinking to a sufficiently small neighborhood U of h_b , the following holds.

If $s < t < h_b$ lie in U , then:

- (1) if r is even, the function $u \mapsto D(s, t, u)$ has no zero on the opposite side $h_b < u \in U$;
- (2) if r is odd, the function $u \mapsto D(s, t, u)$ has at most one zero on the opposite side $h_b < u \in U$.

The same conclusion holds with left and right exchanged. In particular, no finite-order boundary of type $W = 0$, $M \neq 0$ can support a local pair-pair birth.

Proof. Work in the affine chart

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Since

$$x'(h) = \frac{M(h)}{c(h)^2},$$

the hypothesis $M(h_b) \neq 0$ implies that x is a valid local coordinate near h_b . After translating so that $x(h_b) = 0$ and subtracting the tangent line, we may write the normalized curve as

$$y = f(x), \quad f(0) = f'(0) = 0.$$

By Proposition 12.57,

$$\frac{d^2 y}{dx^2} = \frac{c(h)^3 W(h)}{M(h)^3}.$$

Because $x'(h_b) \neq 0$, the finite vanishing order of W in h is the same as its finite vanishing order in the local coordinate x . Therefore

$$f''(x) = \lambda x^r + O(x^{r+1}) \quad (\lambda \neq 0).$$

Fix two same-side points

$$x_1 < x_2 < 0$$

sufficiently close to 0, and define

$$\widehat{D}(x_1, x_2, u) := \det \begin{pmatrix} 1 & x_1 & f(x_1) \\ 1 & x_2 & f(x_2) \\ 1 & u & f(u) \end{pmatrix}.$$

Then

$$\widehat{D}(x_1, x_2, u) = (x_2 - x_1)(u - x_1)(u - x_2) H_{x_1, x_2}(u),$$

where

$$H_{x_1, x_2}(u) := f[x_1, x_2, u]$$

is the second divided difference of f .

If r is even, then f'' has fixed sign near 0. By the Hermite–Genocchi formula for second divided differences, $H_{x_1, x_2}(u)$ has that same fixed sign for $u > 0$ sufficiently small. Hence $\widehat{D}(x_1, x_2, u) \neq 0$ on the opposite side.

If r is odd, then

$$f'''(x) = \lambda r x^{r-1} + O(x^r)$$

has fixed nonzero sign near 0, because $r - 1$ is even. Differentiating with respect to the repeated node gives

$$\partial_u H_{x_1, x_2}(u) = f[x_1, x_2, u, u].$$

By the Hermite–Genocchi formula for repeated-node divided differences, $f[x_1, x_2, u, u]$ has the same fixed nonzero sign for $u > 0$ sufficiently small. Thus $u \mapsto H_{x_1, x_2}(u)$ is strictly monotone on the opposite side, and therefore has at most one zero there.

Finally,

$$D(s, t, u) = c(s)c(t)c(u) \widehat{D}(x(s), x(t), x(u)),$$

and c is nonzero on a small neighborhood of h_b , so the zero set of D is exactly the zero set of $\widehat{D}$. The theorem follows. $\square$

Remark 12.108 (Consequence for the $W = 0$, $M \neq 0$ side). Theorem 12.107 shows that finite-order $W = 0$, $M \neq 0$ boundaries are not local birthplaces of pair-pair configurations. Combined with Lemma 12.109 and Corollary 12.110, this yields a sharper reduction of the one-pair propagation problem.

Namely, after removing local births at generic turning boundaries and at finite-order isolated $W = 0$, $M \neq 0$ boundaries, any surviving nonadjacent pair-pair obstruction must come from a *remote* interior zero of one of the endpoint-gap functions

$$G_{\xi,J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h)),$$

where (ξ, y_{ξ}) is an endpoint support point and J is an ordered target branch.

Since each $G_{\xi,J}$ is strictly monotone on J , each endpoint/branch pair can contribute at most one such tangency. Thus the remaining obstruction on the $W = 0$ side is no longer local birth at an affine inflection seam, but a finite remote-incidence problem.

Lemma 12.109 (Endpoint-gap monotonicity on an ordered target branch). *Let J be an ECT interval in the affine chart*

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

and let (ξ, y_{ξ}) be a fixed support point with

$$\xi < x(h) \quad (h \in J).$$

Define

$$G_{\xi,J}(h) := y_{\xi} - y(h) - m(h)(\xi - x(h)), \quad m(h) = \frac{N(h)}{M(h)}.$$

Then

$$G'_{\xi,J}(h) = \frac{c(h)W(h)}{M(h)^2} (x(h) - \xi).$$

In particular, $G_{\xi,J}$ is strictly monotone on J .

Equivalently, a fixed support point strictly to the left of J lies on at most one tangent line to $\Gamma(J)$. The symmetric statement holds for a fixed support point strictly to the right of J .

Proof. Using

$$y'(h) = m(h)x'(h),$$

differentiate:

$$G'_{\xi,J}(h) = -y'(h) - m'(h)(\xi - x(h)) + m(h)x'(h) = m'(h)(x(h) - \xi).$$

By Proposition 12.62,

$$m'(h) = \frac{c(h)W(h)}{M(h)^2},$$

so

$$G'_{\xi,J}(h) = \frac{c(h)W(h)}{M(h)^2} (x(h) - \xi).$$

On an ECT interval, c , W , and M have fixed nonzero sign, and by assumption $x(h) - \xi > 0$ on J , so $G'_{\xi,J}$ has fixed nonzero sign. Therefore $G_{\xi,J}$ is strictly monotone. $\square$

Corollary 12.110 (Turning-boundary inertness for endpoint contacts). *Let J be an ECT interval ending at a generic turning boundary h_b with*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0, \quad c(h_b) \neq 0.$$

Assume $v_5(h_b) \neq 0$, and let

$$x_b := \frac{v_5(h_b)}{c(h_b)}.$$

If (ξ, y_{ξ}) is a fixed support point with $\xi < x_b$, then

$$G_{\xi,J}(h) \rightarrow \pm\infty \quad \text{as } h \rightarrow h_b \text{ from within } J.$$

In particular, endpoint-support contact cannot be created or destroyed at a generic turning-boundary end of an ordered branch.

Proof. By Proposition 12.62,

$$m(h) = \frac{N(h)}{M(h)} \rightarrow \pm\infty \quad (h \rightarrow h_b),$$

while

$$x(h) \rightarrow x_b, \quad y(h) \rightarrow y_b := \frac{\Delta_7(h_b)}{c(h_b)}.$$

Hence

$$G_{\xi,J}(h) = (y_\xi - y(h)) + m(h)(x(h) - \xi).$$

Since $x(h) - \xi \rightarrow x_b - \xi > 0$, the second term dominates and has fixed sign for h sufficiently close to h_b . Therefore

$$G_{\xi,J}(h) \rightarrow \pm\infty.$$

□

Remark 12.111 (Endpoint purification for bounded strip-overlap components). Let

$$\mathcal{P} := \bigcup_{i < j-1} (\mathcal{S}_{I_i} \cap \mathcal{S}_{I_j})$$

denote the nonadjacent pair-pair strip-overlap set on a compact patch, where the union runs over the ECT intervals in the normalized affine chart.

By Theorem 12.96 and Theorem 12.107, a bounded connected component

$$U \subset \mathcal{P}$$

cannot be created or destroyed at a generic turning boundary, and cannot have a first entrance or last exit at a finite-order boundary of type

$$W = 0, \quad M \neq 0.$$

Moreover, by Lemma 12.109, each fixed endpoint support point can meet a given ordered target branch at most once. Thus any surviving bounded strip-overlap component is controlled by a finite collection of isolated remote endpoint/branch tangencies, rather than by diffuse local birth along a seam.

So the remaining nonadjacent pair-pair problem is best viewed simply as a finite remote-incidence problem.

Remark 12.112 (Audit: residual $W = 0$ pathology). By Proposition 12.59, the $W = 0$, $M \neq 0$ side is no longer obstructed by open-interval flatness $W \equiv 0$ on a component with $c \neq 0$ and $M \neq 0$. By Theorem 12.107, every finite-order isolated $W = 0$ boundary is locally sterile for pair-pair birth. Thus any remaining $W = 0$ issue would have to come either from non-analytic / infinite-order flatness or from a genuinely global continuation phenomenon, not from a generic local inflection birth mechanism.

12.6. From the one-pair residual germ to two-pair and minimal-core rigidity.

Remark 12.113 (Strengthened organizing principle). The original version of this subsection was organized around the cubic leading line and the first residual trace witness. The later working analysis suggests a stronger one-pair package:

$$c(h), \quad A_5^f(h), \quad \Delta_7(h),$$

or equivalently the normalized affine curve

$$h \mapsto \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right).$$

The intended two-pair and minimal-core rigidity statements should ultimately be reformulated in terms of this strengthened one-pair invariant geometry.

Remark 12.114 (Subsection guide). This subsection is best read as a rigidity program built on the strengthened one-pair local package

$$c(h), \quad A_5^f(h), \quad \Delta_7(h),$$

or equivalently the normalized planar curve

$$h \mapsto \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right).$$

The projective coincidence conditions remain the organizing principle for the two-pair and minimal-core analysis. The finite-order discussion below now isolates the exact source-level criterion that would force the two-point quadratic interaction factorization through order 7, but the current attack also exhibits two backup lanes: a weaker quotient-diagonal exact exclusion route and, on good overlap patches, a mixed λ -shear / projective-shadow route. Beyond that, the remaining steps are the verification of one of these two-point lanes, the genuine multi-pair interaction bookkeeping, and the extraction into the finite-core endgame.

Remark 12.115 (Interpretation of residual notation in the multi-pair architecture). In the remainder of this section, the symbols

$$F_h(z), \quad \ell_h, \quad \bar{v}_1(h), \quad \mathcal{Q}^{\text{abs}}/\ell_h$$

are used schematically to describe the intended cubic-leading / first-residual architecture of multi-pair and minimal-core configurations.

The fully explicit rigorous first-residual object established earlier is

$$\mathcal{R}_h^{(5)} = \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h)).$$

Accordingly, whenever a later statement invokes the first residual class in a theorem-strength way, it should be interpreted through $\mathcal{R}_h^{(5)}$. The present two-pair subsection now isolates the exact finite-order source criterion through order 7 together with two weaker backup routes. What remains architectural is the verification of one of these two-point routes for the actual corrected two-atom package, together with its extension to genuine multi-pair and minimal-bad-core interaction terms.

Let $F_h(z)$ denote the corrected quotient odd germ of the normalized one-pair family. By Proposition 12.8, on the cubic-generic region we have

$$F_h(z) = v_0(h)z^3 + v_1(h)z^5 + \cdots, \quad v_0(h) \neq 0,$$

with cubic leading line

$$\ell_h := \mathbf{C} v_0(h) \subset \mathcal{Q}^{\text{abs}}.$$

Under the additional hypotheses of Proposition 12.28, there is an open dense subset $\mathcal{U}_{\text{res}} \subset \mathcal{U}_{\text{cub}}$ on which the explicit first residual class

$$\bar{v}_1(h) \in \mathcal{R}_h^{(5)}$$

is nonzero.

In the strengthened fixed-sector refinement, the one-pair local package is recorded by

$$c(h), \quad A_5^f(h) = u_5(h)I + v_5(h)S, \quad \Delta_7(h),$$

and hence by the normalized planar curve

$$\Gamma(h) := \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right) = (\rho_1(h), \rho_2(h)).$$

By Proposition 12.54, the affine curvature numerator of Γ is

$$K_\Gamma(h) = \rho_1'(h)\rho_2''(h) - \rho_2'(h)\rho_1''(h) = \frac{M(h)N'(h) - N(h)M'(h)}{c(h)^4},$$

where

$$M(h) = c(h)v_5'(h) - c'(h)v_5(h), \quad N(h) = c(h)\Delta_7'(h) - c'(h)\Delta_7(h).$$

Thus the strengthened one-pair geometry is governed by the curve $\Gamma(h)$ and its secant/tangent behavior on the region where

$$c(h) \neq 0, \quad M(h) \neq 0, \quad M(h)N'(h) - N(h)M'(h) \neq 0.$$

Definition 12.116 (Amplitude-invariant strengthened datum). On the region where

$$c(h) \neq 0,$$

define

$$\widehat{\Psi}(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right).$$

Lemma 12.117 (Scaling law for the strengthened package). *If the normalized one-pair odd germ is multiplied by a scalar λ , then*

$$c \mapsto \lambda c, \quad A_5^f \mapsto \lambda A_5^f, \quad A_7^f \mapsto \lambda A_7^f,$$

and hence

$$\Delta_7 \mapsto \lambda^2 \Delta_7.$$

Consequently the datum

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right)$$

is invariant under scalar rescaling of the full one-pair germ.

Proof. The cubic, quintic, and septic coefficients of the odd germ all scale linearly under multiplication of the germ by λ . Since

$$A_5^f = u_5 I + v_5 S, \quad A_7^f = u_7 I + v_7 S,$$

it follows that

$$u_5 \mapsto \lambda u_5, \quad v_5 \mapsto \lambda v_5, \quad u_7 \mapsto \lambda u_7, \quad v_7 \mapsto \lambda v_7.$$

Therefore

$$\Delta_7 = u_7 v_5 - u_5 v_7$$

scales as

$$\Delta_7 \mapsto \lambda^2 \Delta_7,$$

which gives the claim. □

Remark 12.118 (Two normalizations: geometric versus projective). The affine curve

$$\Gamma(h) = \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right)$$

is the correct object for the one-pair differential geometry, because its derivatives are governed by the projective Wronskians M , N , and W .

For the multi-pair coincidence problem, however, the natural scalar-invariant datum is $\widehat{\Psi}(h)$, not $\Gamma(h)$. The reason is exactly the scaling law in Lemma 12.117: under scalar rescaling of the full one-pair germ, Δ_7/c is not invariant, whereas Δ_7/c^2 is.

In this formulation, the geometric secant/tangent obstruction is well controlled on ECT intervals and across generic turning-point boundaries. The remaining global issue is no longer a purely local $W = 0$, $M \neq 0$ obstruction, but the global propagation / recurrence problem for the determinant-zero and tangent configurations that survive the local sterility results proved later in the paper.

Definition 12.119 (Two-pair decomposition). For a normalized two-pair core with amplitudes a_1, a_2 and offsets h_1, h_2 , write its quotient odd germ schematically as

$$F_{(a_1, h_1; a_2, h_2)}(z) = a_1 F_{h_1}(z) + a_2 F_{h_2}(z) + \mathcal{I}_{12}(z),$$

where $\mathcal{I}_{12}(z)$ denotes the pair-interaction germ.

In the explicit finite-order corrected source-level model through order 7 constructed below, the cubic scalar, quintic fixed-sector, and septic fixed-sector projections of $\mathcal{I}_{12}$ are quadratic in the amplitudes and divisible by $(h_1 - h_2)^2$.

Proposition 12.120 (Exact strengthened two-pair coincidence through order 7). *Let $(a_1, h_1; a_2, h_2)$ be a normalized two-pair core with*

$$F_{(a_1, h_1; a_2, h_2)}(z) \equiv 0.$$

Assume moreover that, after passage to the explicit corrected finite-order model through order 7, the interaction contribution vanishes at the three relevant levels, namely:

$$a_1 c(h_1) + a_2 c(h_2) = 0,$$

$$a_1 A_5^{\mathfrak{f}}(h_1) + a_2 A_5^{\mathfrak{f}}(h_2) = 0,$$

and, after the second identity identifies the common quotient line

$$\mathbf{C}A_5^{\mathfrak{f}}(h_1) = \mathbf{C}A_5^{\mathfrak{f}}(h_2),$$

the second residual classes satisfy

$$a_1 [A_7^{\mathfrak{f}}](h_1) + a_2 [A_7^{\mathfrak{f}}](h_2) = 0 \quad \text{in} \quad \mathfrak{f}/\mathbf{C}A_5^{\mathfrak{f}}(h_1).$$

Then

$$\widehat{\Psi}(h_1) = \widehat{\Psi}(h_2).$$

Equivalently,

$$\frac{u_5(h_1)}{c(h_1)} = \frac{u_5(h_2)}{c(h_2)}, \quad \frac{v_5(h_1)}{c(h_1)} = \frac{v_5(h_2)}{c(h_2)}, \quad \frac{\Delta_7(h_1)}{c(h_1)^2} = \frac{\Delta_7(h_2)}{c(h_2)^2}.$$

Proof. Set

$$\lambda := -\frac{a_2}{a_1}.$$

The cubic identity gives

$$c(h_1) = \lambda c(h_2).$$

The quintic fixed-sector identity gives

$$A_5^{\mathfrak{f}}(h_1) = \lambda A_5^{\mathfrak{f}}(h_2).$$

Writing

$$A_5^{\mathfrak{f}}(h_i) = u_5(h_i)I + v_5(h_i)S,$$

we obtain

$$u_5(h_1) = \lambda u_5(h_2), \quad v_5(h_1) = \lambda v_5(h_2).$$

Hence

$$\frac{u_5(h_1)}{c(h_1)} = \frac{u_5(h_2)}{c(h_2)}, \quad \frac{v_5(h_1)}{c(h_1)} = \frac{v_5(h_2)}{c(h_2)}.$$

Now the second identity implies

$$\mathbf{C}A_5^{\mathfrak{f}}(h_1) = \mathbf{C}A_5^{\mathfrak{f}}(h_2),$$

so the residual quotient spaces are canonically identified. The order-7 residual identity therefore says

$$[A_7^{\mathfrak{f}}](h_1) = \lambda [A_7^{\mathfrak{f}}](h_2)$$

in the common quotient. Equivalently, there exists $\mu \in \mathbf{C}$ such that

$$A_7^{\mathfrak{f}}(h_1) = \lambda A_7^{\mathfrak{f}}(h_2) + \mu A_5^{\mathfrak{f}}(h_2).$$

Writing

$$A_7^{\mathfrak{f}}(h_i) = u_7(h_i)I + v_7(h_i)S,$$

and using

$$\Delta_7(h) = u_7(h)v_5(h) - u_5(h)v_7(h),$$

we obtain

$$\Delta_7(h_1) = \det(A_7^{\mathfrak{f}}(h_1), A_5^{\mathfrak{f}}(h_1)) = \det(\lambda A_7^{\mathfrak{f}}(h_2) + \mu A_5^{\mathfrak{f}}(h_2), \lambda A_5^{\mathfrak{f}}(h_2)) = \lambda^2 \Delta_7(h_2).$$

Since also $c(h_1) = \lambda c(h_2)$, it follows that

$$\frac{\Delta_7(h_1)}{c(h_1)^2} = \frac{\Delta_7(h_2)}{c(h_2)^2}.$$

Thus $\widehat{\Psi}(h_1) = \widehat{\Psi}(h_2)$. □

Theorem 12.121 (Quantitative defect-thickened strengthened coincidence). *Let $(a_1, h_1; a_2, h_2)$ be a normalized bad two-pair core, and write*

$$c_i := c(h_i), \quad A_{5,i}^{\mathfrak{f}} := A_5^{\mathfrak{f}}(h_i) = u_{5,i}I + v_{5,i}S, \quad \Delta_i := \Delta_7(h_i).$$

Assume

$$a_1 \neq 0, \quad c_1 c_2 \neq 0.$$

Suppose the cubic, quintic, and septic quotient defects are written as

$$a_1 c_1 + a_2 c_2 + E_{12}^{(3)} = 0,$$

$$a_1 A_{5,1}^{\mathfrak{f}} + a_2 A_{5,2}^{\mathfrak{f}} + E_{12}^{(5)} = 0,$$

and

$$a_1 [A_{7,1}^{\mathfrak{f}}] + a_2 [A_{7,2}^{\mathfrak{f}}] + \overline{E}_{12}^{(7;1)} = 0 \quad \text{in} \quad \mathfrak{f}/\mathbf{C}A_{5,1}^{\mathfrak{f}}.$$

Set

$$\lambda := -\frac{a_2}{a_1}, \quad e := \frac{E_{12}^{(3)}}{a_1}, \quad E := \frac{E_{12}^{(5)}}{a_1} = e_u I + e_v S.$$

Choose any representative $R \in \mathfrak{f}$ of the quotient class

$$-\frac{1}{a_1} \overline{E}_{12}^{(7;1)},$$

so that

$$A_{7,1}^{\mathfrak{f}} = \lambda A_{7,2}^{\mathfrak{f}} + R + \eta A_{5,1}^{\mathfrak{f}}$$

for some $\eta \in \mathbf{C}$. Then the strengthened invariant

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right)$$

satisfies the exact identities

$$\begin{aligned} \frac{u_{5,1}}{c_1} - \frac{u_{5,2}}{c_2} &= \frac{u_{5,2}e - c_2 e_u}{c_1 c_2}, \\ \frac{v_{5,1}}{c_1} - \frac{v_{5,2}}{c_2} &= \frac{v_{5,2}e - c_2 e_v}{c_1 c_2}, \end{aligned}$$

and

$$\frac{\Delta_1}{c_1^2} - \frac{\Delta_2}{c_2^2} = \frac{-\lambda c_2^2 \det(A_{7,2}^{\mathfrak{f}}, E) + c_2^2 \det(R, A_{5,1}^{\mathfrak{f}}) + \Delta_2(2\lambda c_2 e - e^2)}{c_1^2 c_2^2}.$$

Proof. The cubic defect identity gives

$$c_1 = \lambda c_2 - e.$$

The quintic defect identity gives

$$A_{5,1}^{\mathfrak{f}} = \lambda A_{5,2}^{\mathfrak{f}} - E,$$

hence

$$u_{5,1} = \lambda u_{5,2} - e_u, \quad v_{5,1} = \lambda v_{5,2} - e_v.$$

Therefore

$$\frac{u_{5,1}}{c_1} - \frac{u_{5,2}}{c_2} = \frac{u_{5,1}c_2 - u_{5,2}c_1}{c_1 c_2} = \frac{(\lambda u_{5,2} - e_u)c_2 - u_{5,2}(\lambda c_2 - e)}{c_1 c_2} = \frac{u_{5,2}e - c_2 e_u}{c_1 c_2},$$

and similarly

$$\frac{v_{5,1}}{c_1} - \frac{v_{5,2}}{c_2} = \frac{v_{5,2}e - c_2 e_v}{c_1 c_2}.$$

For the third coordinate, by the choice of R we have

$$A_{7,1}^f = \lambda A_{7,2}^f + R + \eta A_{5,1}^f$$

for some $\eta \in \mathbf{C}$. Using

$$\Delta_i = \det(A_{7,i}^f, A_{5,i}^f),$$

we obtain

$$\Delta_1 = \det(A_{7,1}^f, A_{5,1}^f) = \lambda \det(A_{7,2}^f, A_{5,1}^f) + \det(R, A_{5,1}^f),$$

since the $\eta A_{5,1}^f$ term drops out. Substituting

$$A_{5,1}^f = \lambda A_{5,2}^f - E$$

gives

$$\det(A_{7,2}^f, A_{5,1}^f) = \lambda \Delta_2 - \det(A_{7,2}^f, E),$$

hence

$$\Delta_1 - \lambda^2 \Delta_2 = -\lambda \det(A_{7,2}^f, E) + \det(R, A_{5,1}^f).$$

Also,

$$c_1 = \lambda c_2 - e$$

implies

$$\lambda^2 c_2^2 - c_1^2 = 2\lambda c_2 e - e^2.$$

Therefore

$$\frac{\Delta_1}{c_1^2} - \frac{\Delta_2}{c_2^2} = \frac{c_2^2 \Delta_1 - c_1^2 \Delta_2}{c_1^2 c_2^2} = \frac{c_2^2 (\Delta_1 - \lambda^2 \Delta_2) + \Delta_2 (\lambda^2 c_2^2 - c_1^2)}{c_1^2 c_2^2},$$

which is exactly the claimed formula.

Finally, the expression

$$\det(R, A_{5,1}^f)$$

is independent of the chosen representative of the quotient defect, since replacing R by $R + \xi A_{5,1}^f$ does not change the determinant. $\square$

Corollary 12.122 (Bad two-pair cores are defect-scale close on good $M \neq 0$ patches). *Let $J \subseteq \{c \neq 0, M \neq 0\}$ be a compact interval. Assume*

$$|a_1| \geq a_* > 0$$

for the normalized bad two-pair core under consideration. Then there exists a constant $C_J > 0$ such that whenever $h_1, h_2 \in J$,

$$|h_1 - h_2| \leq C_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right).$$

In particular, on good $M \neq 0$ patches, bad two-pair cores force the two parameters to be quantitatively close once the cubic/quintic interaction defects are small.

Proof. The second coordinate of $\widehat{\Psi}$ is

$$x(h) := \frac{v_5(h)}{c(h)}, \quad x'(h) = \frac{M(h)}{c(h)^2}.$$

Since $J \subseteq \{c \neq 0, M \neq 0\}$, there exists $\eta_J > 0$ such that

$$|x(h_1) - x(h_2)| \geq \eta_J |h_1 - h_2| \quad (h_1, h_2 \in J).$$

On the other hand, Theorem 12.121 gives the exact identity

$$x(h_1) - x(h_2) = \frac{v_{5,2}e - c_2 e_v}{c_1 c_2},$$

with

$$e = \frac{E_{12}^{(3)}}{a_1}, \quad e_v = S\text{-coefficient of } \frac{E_{12}^{(5)}}{a_1}.$$

Since J is compact and a_1 is bounded away from 0, the right-hand side is bounded by

$$C'_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right)$$

for some constant C'_J . Combining the two estimates yields the claim. $\square$

Proposition 12.123 (No sublinear two-pair defects on good patches). *Let*

$$J \Subset \{c \neq 0, M \neq 0\}$$

be a compact interval. Then there exists a constant $C_J > 0$ such that every normalized bad two-pair core with $h_1, h_2 \in J$ and $h_1 \neq h_2$ satisfies

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| \geq C_J^{-1} |h_1 - h_2|.$$

In particular, there is no sequence of nontrivial normalized bad two-pair cores in J for which

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| = o(|h_1 - h_2|).$$

Proof. By Corollary 12.122, there exists a constant $C'_J > 0$ such that

$$|h_1 - h_2| \leq C'_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right)$$

for every normalized bad two-pair core with $h_1, h_2 \in J$. Since

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \leq |E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}|,$$

the same estimate yields

$$|h_1 - h_2| \leq C'_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| \right).$$

Renaming C'_J as C_J proves the first claim. The final statement is immediate: if a sequence satisfied

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| = o(|h_1 - h_2|),$$

then the displayed lower bound would force

$$|h_1 - h_2| = o(|h_1 - h_2|),$$

which is impossible for a nontrivial two-point core with $h_1 \neq h_2$. $\square$

Lemma 12.124 (Minimal finite-order source criterion for quadratic two-point factorization). *Let V be a finite-dimensional complex vector space, and let*

$$\mathfrak{P}_2(a_1, h_1; a_2, h_2) \in V$$

be analytic on a neighborhood of the small two-atom data. Assume that there is a one-atom package

$$F_h \in V$$

such that the following three identities hold:

$$\mathfrak{P}_2(a_1, h_1; a_2, h_2) = \mathfrak{P}_2(a_2, h_2; a_1, h_1),$$

$$\mathfrak{P}_2(0, h_1; a_2, h_2) = a_2 F_{h_2}, \quad \mathfrak{P}_2(a_1, h_1; 0, h_2) = a_1 F_{h_1},$$

and

$$\mathfrak{P}_2(a_1, h; a_2, h) = (a_1 + a_2) F_h.$$

Define the interaction remainder

$$\mathcal{I}_2(a_1, h_1; a_2, h_2) := \mathfrak{P}_2(a_1, h_1; a_2, h_2) - a_1 F_{h_1} - a_2 F_{h_2}.$$

Then there exists an analytic map

$$\mathcal{J}_2 : (\text{small two-atom data}) \rightarrow V$$

such that

$$\boxed{\mathcal{I}_2(a_1, h_1; a_2, h_2) = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2(a_1, h_1; a_2, h_2).}$$

Proof. By the one-amplitude collapse identities,

$$\mathcal{I}_2(0, h_1; a_2, h_2) = 0, \quad \mathcal{I}_2(a_1, h_1; 0, h_2) = 0.$$

Hence analyticity implies divisibility by $a_1 a_2$:

$$\mathcal{I}_2(a_1, h_1; a_2, h_2) = a_1 a_2 G(h_1, h_2)$$

for an analytic

$$G : (h_1, h_2) \mapsto V.$$

Swap symmetry gives

$$G(h_1, h_2) = G(h_2, h_1).$$

On the diagonal,

$$\mathcal{I}_2(a_1, h; a_2, h) = \mathfrak{P}_2(a_1, h; a_2, h) - a_1 F_h - a_2 F_h = (a_1 + a_2) F_h - a_1 F_h - a_2 F_h = 0,$$

so

$$G(h, h) = 0.$$

Now set

$$m = \frac{h_1 + h_2}{2}, \quad s = h_2 - h_1, \quad \tilde{G}(m, s) := G\left(m - \frac{s}{2}, m + \frac{s}{2}\right).$$

Since G is symmetric, $\tilde{G}(m, s)$ is even in s , and since $G(h, h) = 0$, one has

$$\tilde{G}(m, 0) = 0.$$

Therefore $\tilde{G}(m, s)$ is divisible by s^2 :

$$\tilde{G}(m, s) = s^2 H(m, s)$$

for analytic H . Returning to (h_1, h_2) gives

$$\mathcal{I}_2(a_1, h_1; a_2, h_2) = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2(a_1, h_1; a_2, h_2),$$

as claimed. $\square$

Corollary 12.125 (Conditional quadratic defect estimate and two-point exclusion). *Assume that the actual corrected two-pair finite-order package through order 7 admits a source-level realization*

$$\mathfrak{P}_2(a_1, h_1; a_2, h_2) \in \mathbf{C} \oplus \mathfrak{f} \oplus (\mathfrak{f}/\mathbf{C}A_5^\mathfrak{f})$$

satisfying the three hypotheses of Lemma 12.124, with

$$F_h = (c(h), A_5^\mathfrak{f}(h), [A_7^\mathfrak{f}](h)).$$

Then on compacta,

$$\mathcal{E}^{(3)} = a_1 a_2 O(\delta^2), \quad \mathcal{E}^{(5)} = a_1 a_2 O(\delta^2), \quad \overline{\mathcal{E}}^{(7)} = a_1 a_2 O(\delta^2), \quad \delta := |h_1 - h_2|.$$

If the normalized defects are obtained by dividing by a_1 , then on normalized families with $|a_2|$ bounded one has

$$E_{12}^{(3)} = O(\delta^2), \quad E_{12}^{(5)} = O(\delta^2), \quad \overline{E}_{12}^{(7;1)} = O(\delta^2).$$

Consequently, the two-point bad-core branch is excluded on compact good patches

$$J \subseteq \{c \neq 0, M \neq 0\}.$$

Proof. Apply Lemma 12.124 to the cubic, quintic fixed-sector, and septic quotient outputs, then argue exactly as in the present proof using Theorem 12.121. $\square$

Proposition 12.126 (Conditional exact two-point exclusion from quotient diagonal collapse). *Let*

$$J \subseteq \{c \neq 0\}$$

be a compact interval. Assume there exists an actual corrected two-atom quotient map

$$\mathcal{F}_2(a_1, h_1; a_2, h_2) \in \mathcal{Q}^{\text{abs}}[[z^2]]$$

defined on normalized two-pair data, with the following properties:

(1) **Continuity.** The map $\mathcal{F}_2$ is continuous in the normalized two-pair parameters.

(2) **Diagonal collapse.** For every $h \in J$ and every normalized amplitude pair (a_1, a_2) ,

$$\mathcal{F}_2(a_1, h; a_2, h) = F_h,$$

where F_h is the normalized one-pair quotient odd germ.

(3) **Precompactness of normalized amplitudes.** Every sequence of normalized two-pair data with

$$h_{1,n}, h_{2,n} \in J, \quad |h_{1,n} - h_{2,n}| \rightarrow 0$$

admits a subsequence along which the full normalized parameter tuple converges.

Then there is no sequence of nontrivial normalized bad two-pair cores

$$(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n})$$

with

$$h_{1,n}, h_{2,n} \in J, \quad 0 < |h_{1,n} - h_{2,n}| \rightarrow 0.$$

Equivalently, under the three assumptions above, the exact close two-point bad-core branch is excluded on compact subsets of $\{c \neq 0\}$.

Proof. Assume for contradiction that there exists a sequence of nontrivial normalized bad two-pair cores

$$(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n})$$

with

$$h_{1,n}, h_{2,n} \in J, \quad 0 < |h_{1,n} - h_{2,n}| \rightarrow 0.$$

Since each core is bad, its quotient odd germ vanishes identically, so

$$\mathcal{F}_2(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n}) = 0 \quad \text{for all } n.$$

By the precompactness assumption, after passing to a subsequence we may assume that the full normalized parameter tuple converges:

$$a_{1,n} \rightarrow a_1^*, \quad a_{2,n} \rightarrow a_2^*, \quad h_{1,n} \rightarrow m, \quad h_{2,n} \rightarrow m$$

for some $m \in J$. By continuity of $\mathcal{F}_2$,

$$0 = \lim_{n \rightarrow \infty} \mathcal{F}_2(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n}) = \mathcal{F}_2(a_1^*, m; a_2^*, m).$$

Applying diagonal collapse gives

$$0 = \mathcal{F}_2(a_1^*, m; a_2^*, m) = F_m.$$

But $m \in J \subseteq \{c \neq 0\}$, so the normalized one-pair quotient germ F_m is nonzero: its cubic leading coefficient is nonzero on $\{c \neq 0\}$. This contradiction proves the claim. $\square$

Remark 12.127 (A weaker exact two-point route). Proposition 12.126 is strictly weaker than the quadratic defect-factorization route of Lemma 12.124. It does not produce

$$\mathcal{I}_2 = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2$$

and does not yield the quantitative defect estimates used in Theorem 12.121. Instead, it isolates a smaller alternative input: a genuine corrected two-atom quotient map, continuity in normalized parameters, and exact diagonal collapse to the one-pair quotient class.

Thus there are now two distinct two-point routes in the draft:

- (1) the stronger finite-order source-level route, which would force quadratic defect bounds and hence quantitative defect-scale closeness;
- (2) the weaker quotient-diagonal route, which would already exclude exact close two-point bad cores on compact subsets of $\{c \neq 0\}$.

At present, the draft does not prove either route outright. The first remains a finite-order source-level merger problem; the second remains a continuity and diagonal-collapse problem for the actual corrected two-atom quotient germ.

Remark 12.128 (Exact remaining source-level input on the two-point branch). Lemma 12.124 isolates the exact finite-order source identities needed for the two-point branch. It is enough to verify, for the actual corrected two-pair cubic/quintic/septic package, only:

- (1) swap symmetry;
- (2) one-amplitude collapse;
- (3) diagonal merger

$$\mathfrak{P}_2(a_1, h; a_2, h) = (a_1 + a_2)F_h.$$

Once these identities are known, the quadratic factorization

$$\mathcal{I}_2 = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2$$

is formal. So the remaining issue on the two-point branch is not another asymptotic estimate, but verification of this finite-order source-level merger statement for the corrected two-atom package.

Remark 12.129 (Audit: even-amplitude obstruction for the naive source-summed model). The most obvious attempt to prove the source-level merger theorem is to insert a weighted one-atom datum

$$q_{a,h} = a q_h, \quad \Phi_{a,h} = a \Phi_h$$

directly into the corrected same-point and mixed block formulas, whiten, and then extract the cubic/quintic/septic outputs. This route cannot by itself prove the required one-atom linearity

$$\mathfrak{P}_2(a\delta_h) = a F_h.$$

Indeed the explicit corrected block formulas have the form

$$G_{a,h,\pm}^{\text{corr}}(s) = a \mathcal{G}_{h,\pm}(a^2; s), \quad N_{a,h}^{\text{corr}}(s) = a \mathcal{N}_h(a^2; s),$$

so after whitening the overall factor of a cancels:

$$\widehat{G}_{a,h}^{\text{corr}}(s) = \mathcal{G}_{h,-}(a^2; s)^{-1/2} \mathcal{N}_h(a^2; s) \mathcal{G}_{h,+}(a^2; s)^{-1/2}.$$

Hence every fixed finite-order coefficient extracted from the whitened block is an even analytic function of the source weight a . In particular, the naive source-summed corrected block package cannot satisfy analytic one-atom linearity or coincident-atom additivity unless it vanishes identically. So the remaining two-point theorem must be proved using a different lift/transport object, not the raw source-summed whitened coefficients themselves.

Remark 12.130 (Audit: quotient and single-determinant routes are shear-blind). On the mixed two-point branch, any route based only on cubic/quintic source minors, the septic quotient class, or the pointwise patch gauges $v_7 = 0$ and $u_7 = 0$ is exactly blind to the repeated-node shear parameter.

For an exact bad two-pair core, the cubic/quintic defect vector has the form

$$d = \Lambda s_2 - s_1, \quad \Lambda := -\frac{a_2}{a_1}, \quad s_i = (c_i, u_{5,i}, v_{5,i}),$$

so

$$s_2 \wedge d = s_1 \wedge s_2$$

is independent of Λ . Likewise, any pointwise representative extracted only from the septic quotient class and $A_{5,1}^{\mathfrak{f}}$ carries no more information than the same determinant scalar already seen by the quotient. Thus the remaining mixed two-point burden is genuinely source-coupled rather than quotient-determinantal.

Proposition 12.131 (Projective compactness of the moving λ -shear defects). *Assume a close exact mixed two-point bad branch on a compact good overlap patch*

$$U \subseteq \{c \neq 0, v_5 \neq 0, M \neq 0\},$$

and write

$$\Lambda_n := -\frac{a_{2,n}}{a_{1,n}}, \quad \delta_n f := f(h_{1,n}) - \Lambda_n f(h_{2,n}), \quad \lambda = \frac{\Delta_7}{v_5}.$$

Then the mixed branch equations are equivalent to the existence of branch-generated moving shears $(\eta_n, \xi_n, \gamma_n)$ such that

$$\delta_n(c) = \eta_n \delta_n(\lambda), \quad \delta_n(v_5) = \xi_n \delta_n(\lambda), \quad \delta_n(u_5) = \gamma_n \delta_n(\lambda).$$

Set

$$d_n := (\delta_n(c), \delta_n(v_5), \delta_n(u_5), \delta_n(\lambda)) \in \mathbf{R}^4.$$

Then, after passing to a subsequence, exactly one of the following holds.

- (1) $d_n = 0$ for all n in the subsequence. Equivalently, along that subsequence one already has exact repeated-source transport simultaneously for

$$c, \quad v_5, \quad u_5, \quad \lambda.$$

- (2) $d_n \neq 0$ for all n in the subsequence, and the normalized defect directions

$$\hat{d}_n := \frac{d_n}{\|d_n\|} \in S^3$$

converge. Equivalently, the projective classes

$$[d_n] = [\delta_n(c) : \delta_n(v_5) : \delta_n(u_5) : \delta_n(\lambda)] \in \mathbf{P}^3$$

admit a convergent subsequence.

Moreover, on any subsequence for which $\delta_n(\lambda) \neq 0$ and the projective limit lies in the affine chart with nonzero λ -coordinate, the affine λ -shears converge:

$$\eta_n = \frac{\delta_n(c)}{\delta_n(\lambda)} \rightarrow \eta_*, \quad \xi_n = \frac{\delta_n(v_5)}{\delta_n(\lambda)} \rightarrow \xi_*, \quad \gamma_n = \frac{\delta_n(u_5)}{\delta_n(\lambda)} \rightarrow \gamma_*.$$

For the limiting affine shear one has only the asymptotic fixed-shear identities

$$\begin{aligned} \delta_n(c - \eta_* \lambda) &= (\eta_n - \eta_*) \delta_n(\lambda), \\ \delta_n(v_5 - \xi_* \lambda) &= (\xi_n - \xi_*) \delta_n(\lambda), \\ \delta_n(u_5 - \gamma_* \lambda) &= (\gamma_n - \gamma_*) \delta_n(\lambda). \end{aligned}$$

Hence a close exact mixed branch automatically reduces only to a projectively frozen secant-shadow problem. In general it does not reduce to exact repeated-source transport for a single fixed triple

$$(\eta_*, \xi_*, \gamma_*).$$

Proof. The identities

$$\delta_n(c) = \eta_n \delta_n(\lambda), \quad \delta_n(v_5) = \xi_n \delta_n(\lambda), \quad \delta_n(u_5) = \gamma_n \delta_n(\lambda)$$

are just a rewriting of the branch equations. If $d_n = 0$ infinitely often, pass to that subsequence. Otherwise $d_n \neq 0$ for all large n , and compactness of the unit sphere gives a convergent subsequence of $\hat{d}_n$, equivalently of the associated projective classes in $\mathbf{P}^3$.

On the affine chart $\delta_n(\lambda) \neq 0$, the projective coordinates are exactly

$$\eta_n = \frac{\delta_n(c)}{\delta_n(\lambda)}, \quad \xi_n = \frac{\delta_n(v_5)}{\delta_n(\lambda)}, \quad \gamma_n = \frac{\delta_n(u_5)}{\delta_n(\lambda)},$$

so convergence in that chart is precisely convergence of the affine λ -shears. The last three identities are immediate from the definitions. $\square$

Remark 12.132 (Affine freezing is stronger than projective freezing on the mixed two-point branch). Proposition 12.131 shows that the automatic compactness on the mixed two-point branch is projective, not affine. Even on a good overlap patch, the affine λ -shears can diverge if $\delta_n(\lambda)$ vanishes to higher order than $\delta_n(c), \delta_n(v_5), \delta_n(u_5)$. And even when

$$(\eta_n, \xi_n, \gamma_n) \rightarrow (\eta_*, \xi_*, \gamma_*),$$

one gets only asymptotic transport for

$$c - \eta_* \lambda, \quad v_5 - \xi_* \lambda, \quad u_5 - \gamma_* \lambda,$$

not exact transport at each stage of the branch. Therefore the downstream derivative-geometric reduction should be formulated in terms of the projective defect direction

$$[\delta_n(c) : \delta_n(v_5) : \delta_n(u_5) : \delta_n(\lambda)],$$

or equivalently in any affine chart on which the projective limit has nonzero coordinate, rather than as an exact frozen- λ repeated-source branch.

Remark 12.133 (Collision-cancellation chart for close two-point cores). On a compact good patch

$$J \subseteq \{c \neq 0, M \neq 0\},$$

a close exact two-point bad core is naturally studied not in the raw diagonal chart $(h_1, h_2) \rightarrow (h, h)$, but in the cancellation chart

$$m := \frac{h_1 + h_2}{2}, \quad \delta := h_2 - h_1, \quad \lambda := -\frac{a_2}{a_1}, \quad \omega := \frac{\lambda - 1}{\lambda + 1}.$$

Indeed, the cubic equation

$$a_1 c(h_1) + a_2 c(h_2) = 0$$

forces $\lambda \rightarrow 1$ whenever $\delta \rightarrow 0$ on J .

The variable ω is the natural swap-compatible amplitude coordinate: under interchange of the two atoms one has

$$\lambda \mapsto \lambda^{-1}, \quad (m, \omega, \delta) \mapsto (m, -\omega, -\delta).$$

Writing

$$\mathcal{D}_h := (c(h), u_5(h), v_5(h)),$$

and using

$$\lambda = \frac{1 + \omega}{1 - \omega} = 1 + 2\omega + O(\omega^2),$$

one gets the first-order expansion

$$\mathcal{D}_{h_1} - \lambda \mathcal{D}_{h_2} = -2\omega \mathcal{D}_m - \delta \mathcal{D}'_m + O((|\omega| + |\delta|)^2).$$

After polar blow-up

$$(2\omega, \delta) = \rho(\cos \theta, \sin \theta),$$

the leading cancellation datum is therefore the projective source direction

$$\Theta_{m,\theta} := \cos \theta \mathcal{D}_m + \sin \theta \mathcal{D}'_m.$$

So any future cubic/quintic limit theorem for close cancelling two-point cores should be phrased in this blown-up collision-cancellation chart rather than as a Taylor expansion through the singular zero-source point.

Remark 12.134 (Work in progress: parity/projective factorization and the local good-patch frontier). The collision-cancellation chart suggests a substantial weakening of the hypotheses needed for local exclusion on the residual exact fixed-shear lane.

Suppose that an actual corrected defect $E(m, \omega, \delta)$ is analytic in this swap-compatible chart, is swap-even,

$$E(m, -\omega, -\delta) = E(m, \omega, \delta),$$

and vanishes on the collision locus

$$E(m, 0, 0) = 0.$$

Then its Taylor expansion has no linear terms, and with

$$\kappa := \frac{2\omega}{\delta}$$

one can write

$$E(m, \omega, \delta) = \delta^2 \mathcal{H}(m, \kappa, \delta^2)$$

for an analytic germ $\mathcal{H}$.

So the entire first blown-up interaction layer is parity-sterile: the first surviving layer is already quadratic and one-projective-parameter. Along an involutive exact fixed-shear branch this becomes quotient-visible, and on the quartic–sextic rung the quadratic coefficient is expected to be controlled by the single quotient transport scalar Q .

A sharper downstream target is the following conditional edge law for the actual corrected cubic/quintic defects:

$$E_{12}^{(3)} = \delta^2 \mathcal{H}_3(m, \kappa, \delta^2), \quad E_{12}^{(5)} = \delta^2 \mathcal{H}_5(m, \kappa, \delta^2),$$

with edge values

$$\mathcal{H}_3(m, \kappa, 0) = c'(m) - \frac{\kappa}{2}c(m), \quad \mathcal{H}_5(m, \kappa, 0) = (A_5^{\mathfrak{f}})'(m) - \frac{\kappa}{2}A_5^{\mathfrak{f}}(m).$$

On the cubic-killing slope

$$\kappa_*(m) = 2 \frac{c'(m)}{c(m)},$$

this would give the reduced quintic obstruction

$$\mathfrak{R}(m, 0) = \frac{M(m)}{c(m)}, \quad M := c(A_5^{\mathfrak{f}})' - c' A_5^{\mathfrak{f}}.$$

Hence on every compact good patch

$$J \in \{c \neq 0, M \neq 0\},$$

sufficiently close cancelling bad cores would already be excluded.

Beyond that good-patch region, the same edge-law package would force any quartic–sextic survivor into the exceptional locus $M = 0$, where the residual problem should reduce, after Weierstrass preparation, to finitely many prepared branches and then to a one-variable compatibility / resultant / norm test. This is a reduction of the remaining burden, not a closed theorem at present.

Gate A and Gate B.. The corrected-package audit uses two separate gates. We call them Gate A and Gate B to prevent the leading homogeneous cubic audit from being confused with the full raw edge-divisibility theorem.

Remark 12.135 (Roadmap of the Gate A/Gate B correction). The corrected-package audit separates into two layers.

Gate A is the leading homogeneous cubic audit. It is a finite, order-7, leading-source calculation. Under the homogeneous scalar-grade convention, Gate A closes at the aggregate fixed-sector level:

$$F_1^{\text{lead cubic}} = 0.$$

Gate B is the raw edge-divisibility theorem. It concerns the full non-(1, 1) projected raw corrected-whitening cross-effect

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

on the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta).$$

Gate B is the assertion

$$[z^7][\delta^1] \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0.$$

The proof of Gate B below uses four inputs:

- (1) Gate A aggregate closure;
- (2) odd total-grade subset-complement cancellation for amplitude terms;
- (3) moving Taylor-frame telescoping for the source-motion terms;
- (4) 2-jet source-frame normalization, which removes the affine lower boundary.

Only after Gate B is proved may one form the normalized edge package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

Definition 12.136 (Gate A: leading homogeneous cubic audit). Work under the homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(h)}{(a+2)!} (t-h)^{a+2}.$$

Gate A is the audit of the leading homogeneous cubic non- $(1,1)$ source sector at ordinary order 7. In the active convention, this is the leading $(1,1,5)$ -type cubic sector. The Gate A families are

$$\begin{aligned} L_1 Y R_1, & \quad L_2 Y R_0, & \quad L_0 Y R_2, \\ L_2 N_0 R_1, & \quad L_1 N_0 R_2, \end{aligned}$$

and

$$L_3 N_0 R_0, \quad L_0 N_0 R_3.$$

Gate A asks whether these leading homogeneous non- $(1,1)$ cubic terms produce a fixed-sector order-7 obstruction.

Definition 12.137 (Gate B: raw edge-divisibility). Gate B is the raw edge-divisibility theorem for the full projected corrected-whitening cross-effect. Let

$$C_{\mathcal{W}}^{\text{raw}} = \mathcal{W}(\mathcal{B}_2) - \mathcal{W}(\mathcal{B}_1(a_1, h_1)) - \mathcal{W}(\mathcal{B}_1(a_2, h_2)) + \mathcal{W}(\mathcal{B}_0),$$

where

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

Apply the source-bidegree projection before fixed-sector extraction, determinant scalarization, or quotienting:

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

On the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

Gate B is the assertion

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1 \delta^2)$$

through fixed-sector ordinary order 7. Equivalently, the first-collision coefficient

$$F_1 := [z^7][\delta^1] \pi_{f_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

vanishes.

Remark 12.138 (Why Gate B must be raw). Gate B must be proved before forming the normalized package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

If $F_1 \neq 0$, this normalized expression has a δ^{-1} -pole and the post-edge package is not holomorphic. Therefore one must not use the post-edge quotient statement

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

as the first proof of edge divisibility. Quotient compatibility is a post-edge statement unless it is reformulated at the raw edge level.

Lemma 12.139 (Order exhaustion for Gate A). *Under the homogeneous scalar-grade convention, every leading homogeneous cubic non- $(1,1)$ contribution to $B_7^{\dagger}$ is one of the seven families*

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2, \quad L_2 N_0 R_1, \quad L_1 N_0 R_2, \quad L_3 N_0 R_0, \quad L_0 N_0 R_3.$$

The three Y-slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

are absent at $B_7^{\dagger}$. The non-Y families reduce to the leading placements

$$L_2^{\{1,1\}} N_0[0] R_1^{[5]}, \quad L_2^{\{1,5\}} N_0[0] R_1^{[1]},$$

and their right-side mirrors, together with the all-left/all-right placements

$$L_3^{\{1,1,5\}} N_0[0] R_0, \quad L_0 N_0[0] R_3^{\{1,1,5\}}.$$

Proof sketch. The Y -slot absence is the homogeneous scalar-grade order-count statement: under this convention the relevant Y -middle entries do not reach ordinary order 7. The remaining non- Y placements are the only ones whose Fréchet degree and homogeneous source grades can sum to 7 with the baseline middle $N_0[0]$. This is scoped to the homogeneous scalar-grade convention and does not apply to an ordinary matrix-output $M^{[5]}$ convention. $\square$

Lemma 12.140 (Aggregate closure of the leading $L_2N_0R_1/L_1N_0R_2$ gate). *Work under the homogeneous scalar-grade convention. Let*

$$N := N_0[0], \quad W := W_{-,0} = G_0^{-1/2},$$

and let Σ_- denote the Sylvester inverse

$$G_0^{1/2}Z + ZG_0^{1/2} = C.$$

The leading same-point source inputs satisfy

$$X_{-,1}^{[1]} = -\frac{1}{48\pi Q^2}(I - D),$$

and

$$X_{-,5}^{[5]} = -\frac{1}{92160\pi Q^{10}}(I - D) = \lambda X_{-,1}^{[1]}, \quad \lambda := \frac{1}{1920Q^8}.$$

Set

$$A := S_{-,1}^{[1]} = \Sigma_-(X_{-,1}^{[1]}), \quad M := WAW,$$

and

$$T_2^{1,1} := \Sigma_-(-2AA), \quad H := 2WAWAW - WT_2^{1,1}W.$$

Then

$$S_{-,5}^{[5]} = \lambda A, \quad T_6^{1,5} = \lambda T_2^{1,1}, \quad H_{-,6}^{1,5} = \lambda H.$$

Consequently the leading fixed-sector tag vectors satisfy

$$C_{112}^{L_2N_0R_1} = C_{122}^{L_2N_0R_1} = \lambda \pi_{\mathfrak{f}}(HNM),$$

and

$$C_{112}^{L_1N_0R_2} = C_{122}^{L_1N_0R_2} = -\lambda \pi_{\mathfrak{f}}(MNH).$$

Since H, N, M are symmetric and

$$(HNM)^T = MNH,$$

while $\pi_{\mathfrak{f}}$ is transpose-invariant, one has

$$\pi_{\mathfrak{f}}(HNM) = \pi_{\mathfrak{f}}(MNH).$$

Therefore

$$C_{112}^{L_2N_0R_1} + C_{112}^{L_1N_0R_2} = 0,$$

and

$$C_{122}^{L_2N_0R_1} + C_{122}^{L_1N_0R_2} = 0.$$

Thus the leading homogeneous non- Y $L_2N_0R_1/L_1N_0R_2$ gate closes at the aggregate fixed-sector level before quotienting.

Lemma 12.141 (Aggregate closure of the leading all-left/all-right cubic gate). *Assume the midpoint parity convention extends to the third-Fréchet coefficient table:*

$$G_{n;a,b,c}^+ = (-1)^n G_{n;a,b,c}^-.$$

Under the homogeneous scalar-grade convention, the first all-left/all-right source multiset is $(1, 1, 5)$, and it contributes at ordinary order 7. Let

$$G := G_{7;1,1,5}^-, \quad W := W_{-,0} = W_{+,0}, \quad N := N_0[0].$$

Then

$$G_{7;1,1,5}^+ = -G.$$

The leading all-left and all-right fixed-sector contributions are

$$C_{1,1,5}^{L_3 N_0 R_0} = \pi_{\mathfrak{f}}(GNW),$$

and

$$C_{1,1,5}^{L_0 N_0 R_3} = -\pi_{\mathfrak{f}}(WNG).$$

Since G, N, W are symmetric,

$$(GNW)^T = WNG.$$

Because $\pi_{\mathfrak{f}}$ is transpose-invariant,

$$\pi_{\mathfrak{f}}(GNW) = \pi_{\mathfrak{f}}(WNG).$$

Hence

$$C_{1,1,5}^{L_3 N_0 R_0} + C_{1,1,5}^{L_0 N_0 R_3} = 0 \quad \text{in } \mathfrak{f}.$$

Thus the leading all-left/all-right homogeneous third-Fréchet gate closes at the aggregate fixed-sector level before quotienting.

Corollary 12.142 (Gate A closure). *Under the homogeneous scalar-grade convention and the stated midpoint-parity conventions,*

$$F_1^{\text{lead cubic}} = 0.$$

Equivalently, the leading homogeneous cubic non-(1,1) sector contributes no fixed-sector $B_7^{\mathfrak{f}}$ obstruction and no first-collision Gate B obstruction.

Remark 12.143 (Aggregate closure versus family-by-family gauge). The preceding Gate A closures are aggregate fixed-sector cancellations. They do not assert that the individual families

$$L_2 N_0 R_1, \quad L_1 N_0 R_2, \quad L_3 N_0 R_0, \quad L_0 N_0 R_3$$

are separately proportional to $A_5^{\mathfrak{f}}(h)$. The package theorem requires the aggregate corrected-whitening cross-effect, not a family-by-family gauge statement. No determinant scalarization, no Φ_K -scalar invisibility, and no downstream diagonal merger is used in these closures.

Remark 12.144 (Scope of Gate A). Gate A is scoped to the active homogeneous scalar-grade convention. It does not apply to an ordinary matrix-output $M^{[5]}$ convention, nor to a bookkeeping system that treats lower-shadow grades 0, 2, 4 as independent ordinary matrix-output inputs. Such conventions require a separate no-double-counting theorem.

Remark 12.145 (Corrected package theorem after the Gate A/Gate B split). The current corrected-package frontier is no longer the raw seven-family Fréchet shortcut. The proof architecture is layered as follows.

First, Gate A closes the leading homogeneous cubic non-(1,1) audit at the aggregate fixed-sector level. This uses the homogeneous scalar-grade convention, the absence of the Y -slot families at B_7 , the aggregate $L_2 N_0 R_1 / L_1 N_0 R_2$ cancellation, and the aggregate $L_3 N_0 R_0 / L_0 N_0 R_3$ cancellation, the last under the stated third-Fréchet midpoint parity convention.

Second, Gate B proves raw edge-divisibility for the projected raw corrected-whitening cross-effect:

$$[z^7][\delta^1] \pi_{\mathfrak{f}_h}(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0.$$

This is the raw theorem needed before the amplitude-normalized package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

can be formed.

Only after Gate B is proved does the corrected fixed-target package become available for the quotient-gauge theorem

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{C}A_5^{\mathfrak{f}}(m),$$

equivalently

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) = 0.$$

Thus any statement of the form

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

is post-edge bookkeeping unless it is explicitly reformulated as a raw edge-divisibility statement.

The older raw seven-family finite-coefficient tests remain useful only as diagnostic coefficient tools. They are not the theorem object and should not be used in place of the corrected package representative.

Definition 12.146 (Homogeneous scalar grade). For a residual scalar source r , define

$$\text{Gr}_a r := \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}.$$

Thus grade 1 corresponds to the cubic scalar jet $r^{(3)}(m)$, and grade 5 corresponds to the septic scalar jet $r^{(7)}(m)$.

Lemma 12.147 (Transpose parity of the source-linear mixed input). *For the full source-linear scaled mixed input $M(z)$ before Φ_K , one has*

$$M(-z) = M(z)^T.$$

Consequently the diagonal entries of $M(z)$ are even in z , while the odd ordinary powers occur in the off-diagonal antisymmetric channel. Through ordinary order 7, the coefficient $[z^5]M$ is off-diagonal antisymmetric and is supported by the even source derivatives

$$r^{(2)}(m), \quad r^{(4)}(m), \quad r^{(6)}(m),$$

not by the homogeneous grade-five source $r^{(7)}(m)$.

Remark 12.148 (Verification sketch for the transpose-parity mixed input). Changing z to $-z$ interchanges the two endpoints in the scaled mixed block. The mixed block is therefore transformed by endpoint swap, which is matrix transpose in the left/right basis. Hence the source-linear mixed input satisfies

$$M(-z) = M(z)^T.$$

The diagonal and symmetric off-diagonal channels are even, while the antisymmetric off-diagonal channel is odd. The audited order-7 expansion places $[z^5]M$ in that antisymmetric channel, supported by

$$r^{(2)}(m), \quad r^{(4)}(m), \quad r^{(6)}(m),$$

not by the homogeneous grade-five source $r^{(7)}(m)$.

Lemma 12.149 (Y -slot families are absent at B_7 under homogeneous grade). *Under Definition 12.146, the Y -slot cubic families*

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

do not contribute to $B_7^\dagger$.

More precisely, the homogeneous grade-five mixed input associated to $\text{Gr}_5 r$ has no ordinary z^5 middle coefficient in the mixed block; its first possible mixed contribution occurs too late to combine with the corresponding first- or second-Fréchet same-point factors at total ordinary order 7. Consequently the three Y -slot families above start at ordinary order at least 8.

Remark 12.150 (Scope of Lemma 12.149). Lemma 12.149 is an order-count absence theorem, not a determinant-proportionality theorem. It is scoped to the homogeneous scalar grade convention. If a future theorem defines $M^{[5]}$ instead as an ordinary matrix-output order-5 projection, or as a parity-corrected non-homogeneous source projection, then the Y -slot families must be reopened and audited separately. Such a convention also requires a no-double-counting theorem against the natural grade 0, 2, 4 same-point shadows.

Remark 12.151 (Status of the homogeneous-grade lemmas after Gate A/B). Lemmas 12.147 and 12.149 are now inputs to the Gate A/Gate B closure mechanism.

Under the homogeneous scalar-grade convention, the Y -slot families are absent at B_7 , the leading non- Y Gate A aggregate closes in f_h , and Gate B closes through ordinary order 7 after all-subset midpoint parity, moving source-frame telescoping, and affine boundary removal are imposed.

This does not mean that lower-shadow grades 0, 2, 4 are discarded. They may appear in the moving Taylor tower. Interior lower-shadow terms are handled by the telescoping identity of Theorem B.17; only the affine grade -1 boundary is removed by the 2-jet source-frame normalization of Lemma B.18.

Remark 12.152 (Grade-zero sector after the Gate B source-frame audit). The homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}$$

does not discard the grade-zero component

$$\text{Gr}_0 r = \frac{r^{(2)}(m)}{2!} (t-m)^2.$$

Affine removal eliminates only the constant and affine modes. Thus $r^{(2)}(m)$ is a genuine source-frame direction, not baseline data.

The Gate B result treats this direction differently from the old open-status notes. Grade 0 is allowed in the moving homogeneous Taylor tower. Its amplitude contributions are covered by the odd total-grade subset-complement cancellation, including zero-padded labelled multisets. Its source-motion lower boundary is

$$(\partial_h \text{Gr}_0 r)_{\text{lower}} = -r^{(2)}(h)(t-h),$$

which is affine and is removed by the 2-jet source-frame projection $\Pi_{\geq 2, h}$.

Consequently grade 0 is neither an independent leftover Gate B residual nor something silently absorbed into q_0 . It is a live source-frame direction whose first-collision contribution is killed by the Gate B source-frame mechanism.

Remark 12.153 (Residual-source tower before highest-new reduction). The homogeneous scalar-grade convention organizes the local residual source tower, but it does not by itself identify all residual-source coefficients with the highest-new tower $q^{(7)}$. Finite quotient-gauge identities must first be tested in the residual-source variables supplied by the corrected source block. Only a residual that survives this package-level test should be passed to the later same-tower / highest-new reduction.

Thus the statement that the eventual local bridge should run through the single highest-new source $q^{(7)}$ is a downstream package-to-jet bridge claim. It is not a substitute for the pre- Φ_K , pre-determinant fixed-sector quotient-gauge calculation.

Remark 12.154 (Quarantined matrix-output $M^{[5]}$ convention). A nonzero ordinary matrix-output order-5 convention for $M^{[5]}$ is not the active homogeneous scalar-grade convention. If such a convention is later introduced, then the Y -slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

must be reopened. That route requires a no-double-counting theorem separating the ordinary matrix-output order-5 contribution from the natural homogeneous scalar grades 0, 2, 4, whose source derivatives also have same-point shadows. The earlier raw-seven-family numerical failure recorded in Remark B.42 rejects the raw matrix-output shortcut; it does not reject the homogeneous scalar-grade gate of Lemma 12.149 or the corrected package representative of Section C.

Remark 12.155 (Historical role of the first non- Y cubic gate). Before the aggregate Gate A closure was isolated, the first non- Y homogeneous cubic gate was the pair

$$L_2 N_0 R_1 \quad \text{and} \quad L_1 N_0 R_2.$$

The finite coefficient inputs for this diagnostic gate were

$$\begin{aligned} [z^2] \mathcal{L}_-^{\{1,1\}}, & \quad [z^6] \mathcal{L}_-^{\{1,5\}}, \\ [z^2] \mathcal{R}_+^{\{1,1\}}, & \quad [z^6] \mathcal{R}_+^{\{1,5\}}, \end{aligned}$$

together with

$$N_0[0], \quad [z^1]\Lambda_{\pm}^{[1]}, \quad [z^5]\Lambda_{\pm}^{[5]}.$$

These data are still useful for auditing the finite coefficient tables.

However, in the current proof they are no longer an open Gate A target. Lemma 12.140 proves that the $L_2N_0R_1/L_1N_0R_2$ leading homogeneous gate closes at the aggregate fixed-sector level. Therefore this remark records historical coefficient-tool provenance, not a remaining theorem obligation.

Remark 12.156 (Superseded coefficient test for the first non- Y gate). The earlier coefficient-level audit asked for the four fixed-sector tag vectors

$$C_{112}^{L_2N_0R_1}, \quad C_{122}^{L_2N_0R_1}, \quad C_{112}^{L_1N_0R_2}, \quad C_{122}^{L_1N_0R_2}.$$

That was the correct finite test before the aggregate closure mechanism was identified.

In the current draft this test is superseded by Lemma 12.140, which proves the cancellation in $\mathfrak{f}$ directly:

$$C_{112}^{L_2N_0R_1} + C_{112}^{L_1N_0R_2} = 0, \quad C_{122}^{L_2N_0R_1} + C_{122}^{L_1N_0R_2} = 0.$$

Thus the old determinant checks against $A_5^{\mathfrak{f}}$ should not be read as remaining obligations for Gate A. They remain only as a useful coordinate-level audit if the homogeneous scalar-grade convention is later changed or the aggregate proof is challenged.

Remark 12.157 (All-left/all-right cubic families: aggregate status). The all-left/all-right families

$$L_3N_0R_0, \quad L_0N_0R_3$$

are not closed family-by-family. However, under the homogeneous scalar-grade convention and assuming third-Fréchet midpoint parity, their leading $(1, 1, 5)$ aggregate satisfies

$$L_3N_0R_0 + L_0N_0R_3 = 0 \quad \text{in } \mathfrak{f}.$$

This is stronger than quotient-gauge for the package-level Gate A target. It does not assert that either family is separately proportional to $A_5^{\mathfrak{f}}$.

If a coefficient table uses ordered triples rather than symmetric multilinear coefficients, the aggregate must include all ordered permutations with the same convention on both left and right sides before this cancellation is invoked.

Remark 12.158 (Current UV chain and status). The current package chain is as follows.

- (1) **UV-022.** Define an invariant source-weight-linear input for the corrected block and obtain exact one-amplitude collapse at the package level.
- (2) **UV-023.** Establish the raw first-collision edge law for the non- $(1, 1)$ projected corrected-whitening cross-effect through ordinary order 7:

$$[z^7][\delta^1]\pi_{\mathfrak{f}_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = 0.$$

In the present draft this is supplied, conditionally, by Corollary B.21. The mechanism is not a $\mathcal{T}_2$ -only truncation theorem, not source-weight linearization, and not post-edge quotient compatibility. It uses Gate A aggregate closure, odd total-grade subset-complement cancellation, grade-preserving source-motion cancellation, moving source-frame telescoping, affine boundary removal, and full-tower differentiation before ordinary-order truncation.

- (3) **UV-024.** Define the pre-determinant fixed-sector coefficient

$$B_7^{\mathfrak{f}}(C)$$

for the actual corrected-whitening matrix cross-effect before Φ_K , and define

$$Q_{7,m}^q(C) := [B_7^{\mathfrak{f}}(C)] \in \mathfrak{f}/\mathbf{CA}_5^{\mathfrak{f}}(m).$$

- (4) **UV-025.** Supply the tagged pre-whitening source block of Definition 10.6. This is now a guarded source-level input, not a quotient theorem.

- (5) **Gate A / UV-026 leading cubic audit.** Under the homogeneous scalar-grade convention, the leading homogeneous cubic non- $(1, 1)$ sector is closed at the aggregate fixed-sector level. The Y -slot families are absent at B_7 , and the non- Y aggregate pairs

$$L_2 N_0 R_1 + L_1 N_0 R_2, \quad L_3 N_0 R_0 + L_0 N_0 R_3$$

vanish in $\mathfrak{f}$, the latter assuming third-Fréchet midpoint parity.

- (6) **Gate B raw edge-divisibility.** The projected raw corrected-whitening cross-effect satisfies

$$[z^7][\delta^1] \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0$$

through ordinary order 7, provided the all-subset midpoint parity, leading Y -middle parity, moving source-frame telescoping, and affine boundary-removal lemmas are imposed. This supplies the raw edge divisibility needed before amplitude normalization and before the lower package table $Z_j[M]$ is formed.

- (7) **Connection correction subtarget.** After Gate B, compute the transported moving line coefficients

$$A_i = [(h - m)^i] \tau_{h \rightarrow m} A_5^{\mathfrak{f}}(h), \quad 1 \leq i \leq 7,$$

and the pre-trivialization lower edge coefficients

$$Z_j[M_A], \quad Z_j[M_C], \quad 0 \leq j \leq 6,$$

for

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The required cancellation identities are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

These are package-level pre-trivialization connection identities, not raw seven-family identities and not post-trivialization $B_j[M]$ identities.

Shortcut proofs using Φ_K -scalar hiding, determinant scalarization, swap symmetry, one-pair gauge, or the UV-025 source block alone do not prove UV-026. The theorem must be proved at the pre- Φ_K , pre-determinant fixed-sector quotient level of Section C.

In this formulation, UV-026 is a theorem about the normalized corrected package representative. The raw seven-family Fréchet sum is only a diagnostic object; its failure under a matrix-output $M^{[5]}$ convention is recorded in Remark B.42 and does not replace the corrected package theorem.

Remark 12.159 (Superseded Gate B status notes). Earlier WIP notes treated $p \geq 4$ transfer filtration and lower-shadow absorption as open obstructions. That was the correct status before the homogeneous transfer audit. The current proof replaces those open obligations as follows:

- (1) positive-grade amplitude terms of total grade 7 vanish by odd total-grade subset-complement cancellation;
- (2) the grade-preserving part of X_h vanishes by the same cancellation;
- (3) the grade-lowering part of X_h telescopes across adjacent moving Taylor grades;
- (4) the sole lower boundary produced by Gr_0 is affine and is removed by 2-jet source-frame normalization.

Thus the remaining role of the old notes is as guardrails: do not prove Gate B by post-edge quotient compatibility, do not demand family-by-family vanishing, and do not truncate the source tower before differentiating.

Remark 12.160 (The old $d \geq 4$ residual partition list is superseded). Earlier audits listed positive-grade residual partitions such as

$$(1, 1, 1, 4), \quad (1, 1, 2, 3), \quad (1, 2, 2, 2), \\ (1, 1, 1, 1, 3), \quad (1, 1, 1, 2, 2), \quad (1, 1, 1, 1, 1, 2), \quad (1, 1, 1, 1, 1, 1, 1)$$

as live $d \geq 4$ transfer-filtration candidates. Those lists are now superseded by the odd total-grade subset-complement cancellation and the moving source-frame telescoping theorem. They should be used only as historical diagnostics for checking that the aggregate proof has included all labelled placements, not as remaining theorem obligations.

Remark 12.161 (Lower-shadow sectors and zero-padding after Gate B). The old lower-shadow obstruction was not a finite missing table. If grade 0 is treated as an independent source factor, then the source-grade partition inventory is non-finite:

$$(0, 1, 1, 5), \quad (0, 0, 1, 1, 5), \quad (\underbrace{0, \dots, 0}_N, 1, 1, 5), \quad \dots$$

and similarly for

$$(\underbrace{0, \dots, 0}_N, 7), \quad (\underbrace{0, \dots, 0}_N, 1, 2, 4), \quad (\underbrace{0, \dots, 0}_N, 2, 2, 3).$$

The Gate B proof does not close this tower by finite enumeration. It closes it uniformly.

More precisely, lower-shadow grades 0, 2, 4 are not reclassified as baseline data and are not silently discarded. They remain part of the moving homogeneous Taylor source tower. Their amplitude contributions of total homogeneous grade 7 are killed by odd total-grade subset-complement cancellation for arbitrary labelled transfer degree. Their source-motion contributions are handled by the moving source-frame telescoping identity, and the only lower boundary generated by differentiating Gr_0 is the affine grade -1 mode, which is removed by the 2-jet source-frame normalization.

Thus the correct post-Gate B status is: lower shadows are live source-frame directions, but not live Gate B first-collision residuals. They should not be listed as an infinite family of remaining residual partitions after Corollary B.21.

Remark 12.162 (Counterterm route if Gate B dependencies fail). The present draft proves Gate B under the stated midpoint-parity, source-frame telescoping, and 2-jet normalization hypotheses. If any of those hypotheses fails in the final source model, then one cannot simply subtract the residual after determinant scalarization.

A legitimate counterterm would have to be a canonical pre- Φ_K , pre-determinant, source-bidegree-compatible edge counterterm $C_{\text{diag/edge}}^{\text{raw}}$ determined by the raw edge jets

$$[z^7][\delta^0]\pi_{\text{fh}}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}, \quad [z^7][\delta^1]\pi_{\text{fh}}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}.$$

It must be natural under atom swap, must preserve the $(1 - \Pi_{1,1})_{\text{mix}}$ source-bidegree sector, and must be defined before Φ_K , determinant pairing, and quotienting. Otherwise it is merely subtracting the obstruction by hand.

Remark 12.163 (Relation to the package-to-jet bridge). The corrected package program above is not an alternative to the zeta-side odd-jet extractor. The extractor side is already available; the missing bridge is the package-to-jet theorem identifying the first surviving corrected odd coefficient from finite-core package data. In the fixed-shear corner, the finite quotient-visible transport has already been compressed to the remaining package-level data. Thus any surviving obstruction after the corrected package quotient theorem must be treated as a package-to-transform or hidden-state problem, not as another planar or scalarized local detector.

In particular, Gate A and Gate B are package-construction results. They do not by themselves prove the final finite-core contradiction or the package-to- $\Xi_{\zeta}^{(N)}$ hidden-state theorem. Any residual Φ_K -visible affine-lift or higher odd-block state remains a separate package-to-transform issue.

Remark 12.164 (Open theorem target: finite-core lower transfer). The finite-core branch would be closed by a theorem of the following form.

Finite-core lower-transfer theorem. Let $\mathcal{C}$ be a normalized genuinely finite same-scale core whose corrected package anomaly is nonzero after all quotient and hidden-state identifications relevant to Φ_K . Then there exist an admissible $N = N(m)$, a sign $\sigma_{\mathcal{C}} \in \{\pm 1\}$, and a constant $c_{\mathcal{C}} > 0$, stable on the compact core-parameter patch, such that

$$|\Xi_{\text{core}}^{(N)}(m)| \geq c_{\mathcal{C}} \frac{x}{B_{\zeta}(m)} S(m).$$

The present draft does not prove this theorem. The corrected order-7 package controls the local quotient construction and removes the known two-point/package quotient residuals, but it does not by itself determine the N -visible coefficient

$$c_{\text{core}, 2N-1}(m)$$

for a genuinely non-pair-like finite core, nor does it rule out cancellation among finite-core constituents.

Definition 12.165 (Nonsingular actual-source k -core stratum). A compact parameter patch Θ_k is called a nonsingular actual-source k -core stratum if the following conditions hold.

- (1) Θ_k is contained in a finite-dimensional real-analytic parameter manifold with fixed core cardinality k , fixed source combinatorics, and fixed collision/blow-up chart.
- (2) The corrected finite-core transverse scalar

$$H_{\text{core}, \theta}(z) = \Phi_K \left(\hat{\Omega}_{\text{core}, \theta}^{\text{corr}}(z/Q^2; m) \right)$$

is real-analytic jointly in (θ, z) on a neighborhood of $\Theta_k \times \{0\}$.

- (3) The whitening, inverse-square-root, and midpoint-trivialization maps used to define $\hat{\Omega}_{\text{core}, \theta}^{\text{corr}}$ are nonsingular and analytic on the patch.
- (4) The chosen coordinate chart does not divide by a vanishing quantity. In particular, if the chart uses c^{-1} , u_5^{-1} , v_5^{-1} , or a determinant coordinate relative to A_5^f , the corresponding quantity is nonzero on the patch.

Theorem 12.166 (Finite odd-germ determinacy on compact actual-source strata). *Let Θ_k be a compact nonsingular actual-source k -core stratum. Write*

$$H_{\text{core}, \theta}(z) = \sum_{r \geq 0} c_{\text{core}, 2r+1}(\theta) z^{2r+1}.$$

Then there exists an integer $R_{\Theta_k} \geq 0$ such that

$$c_{\text{core}, 1}(\theta) = c_{\text{core}, 3}(\theta) = \cdots = c_{\text{core}, 2R_{\Theta_k}+1}(\theta) = 0$$

implies

$$H_{\text{core}, \theta} \equiv 0$$

as an odd germ.

Equivalently, the zero-anomaly locus

$$\mathcal{Z}_k = \{\theta \in \Theta_k : H_{\text{core}, \theta} \equiv 0\}$$

is cut out on Θ_k by finitely many odd coefficient equations.

Proof. By joint analyticity of $H_{\text{core}, \theta}(z)$, each coefficient

$$c_{\text{core}, 2r+1}(\theta)$$

is real-analytic on a neighborhood of Θ_k .

Fix $\theta_0 \in \Theta_k$. Let

$$\mathcal{I}_{\theta_0} = (c_{\text{core}, 1}, c_{\text{core}, 3}, c_{\text{core}, 5}, \dots) \subset \mathcal{O}_{\Theta_k, \theta_0}$$

be the ideal generated by the odd coefficient germs in the local analytic ring at θ_0 . The local analytic ring is Noetherian. Hence $\mathcal{I}_{\theta_0}$ is finitely generated. Therefore there are indices $r_1, \dots, r_s$ such that

$$\mathcal{I}_{\theta_0} = (c_{\text{core}, 2r_1+1}, \dots, c_{\text{core}, 2r_s+1})$$

as germs at θ_0 .

Set

$$R(\theta_0) = \max(r_1, \dots, r_s).$$

After shrinking to a neighborhood U_{θ_0} , every coefficient $c_{\text{core}, 2r+1}$ is an analytic linear combination of

$$c_{\text{core}, 1}, c_{\text{core}, 3}, \dots, c_{\text{core}, 2R(\theta_0)+1}.$$

Thus, on U_{θ_0} , vanishing of all odd coefficients up to order $2R(\theta_0) + 1$ forces every odd coefficient to vanish. Hence $H_{\text{core}, \theta} \equiv 0$ as an odd germ.

The compactness of Θ_k gives a finite subcover

$$U_{\theta_1}, \dots, U_{\theta_M}.$$

Taking

$$R_{\Theta_k} = \max_{1 \leq j \leq M} R(\theta_j)$$

proves the claim uniformly on Θ_k . □

Remark 12.167 (Strata excluded from one finite-determinacy patch). Theorem 12.166 is stratum-local. The following loci must be separated before applying it.

(1) *Amplitude-drop strata:*

$$a_j = 0,$$

where a k -core degenerates to a smaller core.

(2) *Collision strata:*

$$h_i = h_j,$$

unless a collision/blow-up coordinate chart has been chosen in which the corrected source construction remains analytic.

(3) *Whitening singularities:* loci where the relevant Gram blocks lose invertibility or the inverse-square-root branches cease to be analytic.

(4) *$c = 0$ strata:* these must be excluded from charts using

$$T = \frac{v_7}{c}$$

or any other c^{-1} -normalization.

(5) $A_5^f = 0$ strata: these are rank-jump loci for the quotient line

$$\mathbb{C}A_5^f.$$

(6) $v_5 = 0$ and $u_5 = 0$ patch boundaries: these must be separated when using the corresponding v_5 -patch or u_5 -patch coordinates.

(7) *Hidden-state rank jumps:* loci where the augmented package coordinate

$$T = \frac{v_7}{c}$$

or the Φ_K -visible lift ceases to be represented by the chosen analytic chart.

After separating these loci, finite odd-germ determinacy applies on each compact nonsingular analytic stratum.

Corollary 12.168 (Coefficient floor away from the zero-anomaly locus). *Let Θ_k be a compact nonsingular actual-source k -core stratum, and let R_{Θ_k} be the finite-determinacy depth from Theorem 12.166. Define*

$$\mathcal{Z}_k = \{\theta \in \Theta_k : H_{\text{core}, \theta} \equiv 0\}.$$

For $\eta > 0$, set

$$\Theta_{k, \eta} = \{\theta \in \Theta_k : \text{dist}(\theta, \mathcal{Z}_k) \geq \eta\}.$$

Then there exists $\gamma_\eta > 0$ such that every $\theta \in \Theta_{k,\eta}$ satisfies

$$\max_{0 \leq r \leq R_{\Theta_k}} |c_{\text{core},2r+1}(\theta)| \geq \gamma_\eta.$$

Proof. The function

$$D(\theta) = \max_{0 \leq r \leq R_{\Theta_k}} |c_{\text{core},2r+1}(\theta)|$$

is continuous on Θ_k . By Theorem 12.166,

$$D(\theta) = 0 \iff \theta \in \mathcal{Z}_k.$$

Thus $D > 0$ on the compact set $\Theta_{k,\eta}$, and hence D has a positive minimum there. $\square$

Remark 12.169 (Finite determinacy is not zeta-scale transfer). Theorem 12.166 is qualitative. It proves only

$$H_{\text{core},\theta} \neq 0 \implies \exists N \leq R_{\Theta_k} + 1 : c_{\text{core},2N-1}(\theta) \neq 0.$$

It does not prove

$$|\Xi_{\text{core}}^{(N)}(m)| \gtrsim \frac{x}{B_\zeta(m)} S(m).$$

Nor does it provide a sign law, a tail-domination estimate for the filtered N -point coefficient, or an explicit bound for R_{Θ_k} depending only on k .

The remaining quantitative problem is to compare the actual zeta core's distance from

$$\mathcal{Z}_k$$

with the calibration scale

$$\frac{x}{B_\zeta(m)} S(m).$$

That is the calibrated separation problem, not a consequence of Noetherian finite determinacy.

Remark 12.170 (Later refinement of calibrated separation). The distance formulation below is an earlier geometric form of the finite-core quantitative gap. The current proof-state uses the sharper sampled-vector formulation in Subsection 14.24. In that formulation the finite-determinacy vector is decomposed as

$$C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}},$$

and the live hypothesis is the sampled vector lower bound

$$\|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_\infty \geq \Lambda(m, Q).$$

Thus the branch

$$\mathcal{N}_k(m, Q)$$

defined below should be read as a geometric precursor to the sharper interaction-aligned branch

$$\mathcal{N}_k^{\text{align}}(m, Q).$$

Hypothesis 12.171 (Calibrated finite-core separation). Let Θ_k be a compact nonsingular actual-source k -core stratum, with zero-anomaly locus

$$\mathcal{Z}_k = \{\theta : H_{\text{core},\theta} \equiv 0\}.$$

For actual zeta finite cores $\theta(m)$, either

$$\theta(m) \in \mathcal{Z}_k,$$

or $\theta(m)$ lies in a controlled incidence stratum, or else

$$\text{dist}(\theta(m), \mathcal{Z}_k) \geq D_{\Theta_k}(m, Q),$$

where $D_{\Theta_k}(m, Q)$ is large enough to force

$$\left| c_{\text{core},2N-1} + \sum_{j \geq 1} c_{\text{core},2N-1+2j} h_j(1^2, \dots, N^2) Q^{-2j} \right| \gtrsim \frac{Q^{2N+2}}{(2N-1)N!(N-1)!} \frac{x}{B_\zeta(m)} S(m)$$

for some

$$N \leq R_{\Theta_k} + 1.$$

Definition 12.172 (Near-zero-anomaly finite-core branch). On a compact nonsingular actual-source k -core stratum Θ_k , define

$$\mathcal{N}_k(m, Q) := \{\theta(m) \in \Theta_k : 0 < \text{dist}(\theta(m), \mathcal{Z}_k) < D_{\Theta_k}(m, Q), \quad \theta(m) \notin \mathcal{E}_{\text{geom}}\}.$$

This is the branch where the corrected finite-core anomaly is nonzero but too close to the zero-anomaly locus for the finite-determinacy–Łojasiewicz argument to imply

$$|\Xi_{\text{core}}^{(N)}(m)| \gtrsim \frac{x}{B_\zeta(m)} S(m).$$

Remark 12.173 (Status of the near-zero-anomaly branch). The present draft does not prove

$$\mathcal{N}_k(m, Q) = \emptyset.$$

In particular, proximity to $\mathcal{Z}_k$ is not currently shown to imply membership in a controlled incidence stratum such as

$$M = 0, \quad W = 0, \quad K_v = 0, \quad T_v = 0,$$

nor collision, patch-boundary, endpoint-gap incidence, amplitude drop, whitening singularity, $c = 0$, $A_5^f = 0$, or hidden-state rank jump. Thus finite-core closure requires either calibrated anti-tangency from $\mathcal{Z}_k$ or a separate analysis of $\mathcal{N}_k(m, Q)$.

Definition 12.174 (First interaction branch). For a normalized same-scale finite core, write

$$H_{\text{core}}(s) = \Phi_K(F_{\text{core}})(s) = \sum_{r \geq 0} c_{\text{core}, 2r+1} s^{2r+1},$$

and decompose

$$F_{\text{core}} = \sum_{\nu} a_{\nu} F_{h_{\nu}} + \mathcal{I}_{\geq 2}.$$

Define

$$J_1 := [s] \Phi_K(\mathcal{I}_{\geq 2}).$$

The first interaction branch is

$$\mathfrak{Z}_{\text{int}} = \left\{ \sum_{\nu} a_{\nu} K(d_{\nu}) + J_1 = 0 \right\} \setminus \mathcal{E}_{\text{geom}},$$

where $\mathcal{E}_{\text{geom}}$ denotes the union of the controlled geometric, collision, patch-boundary, and endpoint-gap incidence strata.

Remark 12.175 (Interaction rigidity remains a source-level obligation). The current ECT and lifted-curve geometry controls defect-free affine incidence. It does not by itself control the genuinely interacting scalar

$$J_1 = [s] \Phi_K(\mathcal{I}_{\geq 2}).$$

The known structural properties of $\mathcal{I}_{\geq 2}$, such as one-amplitude collapse and collision divisibility, force J_1 to vanish on some boundary strata, but they do not force

$$J_1 \equiv 0,$$

nor do they impose a sign condition or place J_1 in the radical of the controlled incidence ideal.

Thus the finite-core proof needs an additional interaction-rigidity theorem: J_1 must be shown to be zero, sign-controlled, or governed by the controlled incidence equations. Without such a theorem,

$$\mathfrak{Z}_{\text{int}}$$

remains a separate finite-core branch.

Remark 12.176 (Source-level first-interaction rigidity remains open). At two-source order, the first interaction coefficient has the form

$$\mathfrak{j}_{ij}^{(2)} = [s] \Phi_K \left(D\mathcal{W}_0[\mathcal{B}_{ij}^{(2)}] + D^2\mathcal{W}_0[\mathcal{B}_i^{(1)}, \mathcal{B}_j^{(1)}] \right).$$

The direct Hessian contribution and the bilinear whitening contribution split into the independent channels

$$dd, \quad d\eta + \eta d, \quad \eta\eta,$$

together with bilinear whitening products. No current parity, endpoint-symmetry, ECT, or controlled-incidence theorem forces their sum to vanish, gives it a calibrated sign/cone condition, or places it in the radical of the controlled incidence ideal.

Thus

$$J_1 = [s]\Phi_K(\mathcal{I}_{\geq 2})$$

remains an independent finite-core obstruction unless a separate source-level first-interaction rigidity theorem is proved.

Remark 12.177 (Scope after Gate B). Gate B supplies the raw δ^2 -edge divisibility needed to construct the order-7 edge-normalized corrected package. It does not by itself prove the final quotient-gauge theorem, diagonal merger, κ -independence, the connection cancellations for M_A, M_C , or the package-to- $\Xi_\zeta^{(N)}$ bridge.

The M_A, M_C connection branch is now closed under the source-grade reflection convention by Theorem C.21. Thus the remaining post-Gate-B local obstruction is not the $Z_j[M_A], Z_j[M_C]$ table. The remaining local actual-source/Bottleneck-C obstruction is the promoted quadratic mixed-block coefficient

$$E_{A,Y}^{\text{BC}}$$

of Definition 12.199. Its quotient test is

$$e_{A,Y}v_5 - u_5f_{A,Y} = 0,$$

and its transform-safe test is

$$f_{A,Y} = 0.$$

Remark 12.178 (Reduced package equality does not imply hidden-state equality). The reduced datum $\widehat{\Psi}$ forgets the raw septic representative in the affine line

$$A_7^{\mathfrak{f}} \mapsto A_7^{\mathfrak{f}} + \chi A_5^{\mathfrak{f}}.$$

This forgotten direction is not automatically invisible to the transverse functional. Writing

$$A_5^{\mathfrak{f}} = u_5I + v_5S,$$

one has

$$\Phi_K(A_5^{\mathfrak{f}}) = 2v_5.$$

Thus reduced-package equality alone cannot imply agreement modulo $\ker \Phi_K$ through the first surviving odd order, except on the special locus $v_5 = 0$ or after an additional theorem showing cancellation in the actual corrected block. The missing hidden-state theorem is therefore a transform-level state-locality statement modulo $\ker \Phi_K$, not a formal consequence of quotient-septic closure.

Definition 12.179 (Φ_K -visible septic state). On the fixed target fiber

$$\mathfrak{f}_m = \mathbf{C}I \oplus \mathbf{C}S,$$

write

$$A_7^{\mathfrak{f}} = u_7I + v_7S.$$

The Φ_K -visible septic state is the class

$$[A_7^{\mathfrak{f}}]_{\Phi} := A_7^{\mathfrak{f}} \bmod \ker(\Phi_K|_{\mathfrak{f}_m}),$$

equivalently the scalar

$$\phi_7 := \Phi_K(A_7^{\mathfrak{f}}) = 2v_7.$$

If the c -normalized package is being used, the normalized visible septic state is

$$\theta_7 := \frac{\phi_7}{c} = \frac{2v_7}{c}.$$

Proposition 12.180 (Formal non-descent of the Φ_K -visible septic state). *On the generic locus $v_5 \neq 0$, the quotient class*

$$[A_7^f] \in \mathfrak{f}_m / \mathbf{C}A_5^f(m)$$

does not determine the Φ_K -visible septic state

$$[A_7^f]_\Phi \in \mathfrak{f}_m / \ker(\Phi_K|_{\mathfrak{f}_m}).$$

In particular, quotient package equality does not imply equality of the first Φ_K -visible corrected odd coefficient.

Proof. Write

$$A_5^f = u_5 I + v_5 S, \quad A_7^f = u_7 I + v_7 S.$$

The quotient identifies

$$A_7^f \sim A_7^f + \chi A_5^f.$$

However

$$\Phi_K(A_7^f + \chi A_5^f) - \Phi_K(A_7^f) = \chi \Phi_K(A_5^f) = 2\chi v_5.$$

Thus, if $v_5 \neq 0$, the forgotten quotient direction is visible to Φ_K .

Equivalently, take the formal model

$$A_5^f = S, \quad A_7^{f,\chi} = \chi S.$$

Then

$$[A_7^{f,\chi}] = 0 \quad \text{in} \quad \mathfrak{f}_m / \mathbf{C}S$$

for every χ , but

$$\Phi_K(A_7^{f,\chi}) = 2\chi.$$

Hence the reduced quotient package does not determine the Φ_K -visible septic state. $\square$

Remark 12.181 (Package-to-transform bridge after augmentation). The quotient-only package-to-transform bridge is impossible on the generic patch $v_5 \neq 0$, since

$$A_7^f \sim A_7^f + \chi A_5^f$$

preserves the quotient class but changes

$$\Phi_K(A_7^f)$$

by

$$2\chi v_5.$$

The correct theorem object is therefore the transform-safe augmented package

$$\widehat{\Psi}_{\text{red}}^+ = (\widehat{\Psi}_{\text{red}}, T), \quad T = \frac{v_7}{c}.$$

With this augmentation, the first Φ_K -visible septic coefficient is read off as

$$\Phi_K(A_7^f) = 2cT.$$

The remaining compatibility problem is not quotient-only state-locality; it is the hidden-horizontal diagonal-merger condition

$$\partial_\kappa T(m, \kappa, 0) = 0.$$

Definition 12.182 (Transform-safe augmented reduced package). Work on a good patch with

$$c \neq 0, \quad A_5^f = u_5 I + v_5 S.$$

Write

$$A_7^f = u_7 I + v_7 S, \quad \Delta_7 := u_7 v_5 - u_5 v_7.$$

The transform-safe augmented reduced package is

$$\widehat{\Psi}_{\text{red}}^+ := (\widehat{\Psi}_{\text{red}}, T), \quad T := \frac{v_7}{c}.$$

Equivalently, on the v_5 -patch one may write

$$\widehat{\Psi}_{\text{red}}^+ = (c, u_5, v_5, \Delta_7, T).$$

The normalized Φ_K -visible coordinate is

$$\theta_{\Phi,7} := \frac{\Phi_K(A_7^f)}{c} = 2T.$$

Theorem 12.183 (Augmented package-to-transform readout). *Let*

$$\widehat{\Psi}_{\text{red}}^+ = (c, u_5, v_5, \Delta_7, T)$$

be the transform-safe augmented package on a good patch with $c \neq 0$. Then the first Φ_K -visible septic coefficient of the corrected finite-s two-point transform is determined by the augmented package:

$$\Phi_K(A_7^f) = 2v_7 = 2cT.$$

Equivalently,

$$\frac{1}{c}[s^7]\Phi_K(\widehat{\Omega}^{\text{corr}}(s; m)) = 2T.$$

Thus the two-point transform-visible septic state factors through $\widehat{\Psi}_{\text{red}}^+$, not through $\widehat{\Psi}_{\text{red}}$ alone.

Remark 12.184 (Why the old quotient-only bridge must be abandoned). The reduced quotient package remembers only

$$[A_7^f] \in \mathfrak{f}_m / \mathbf{C}A_5^f(m).$$

It therefore identifies

$$A_7^f \sim A_7^f + \chi A_5^f.$$

But

$$\Phi_K(A_5^f) = 2v_5.$$

Hence, on the generic patch $v_5 \neq 0$, the quotient-forgotten line direction is Φ_K -visible:

$$\Phi_K(A_7^f + \chi A_5^f) - \Phi_K(A_7^f) = 2\chi v_5.$$

Therefore no functional of $\widehat{\Psi}_{\text{red}}$ alone can determine $\Phi_K(A_7^f)$ on the generic patch. The missing coordinate is exactly

$$T = \frac{v_7}{c}.$$

Theorem 12.185 (Sharp no-descent criterion for the Φ_K -visible septic state). *Let*

$$\mathfrak{f}_m = \mathbf{C}I \oplus \mathbf{C}S, \quad A_5^f = u_5I + v_5S.$$

Let

$$q : \mathfrak{f}_m \rightarrow \mathfrak{f}_m / \mathbf{C}A_5^f$$

be the quotient map. There exists a functional

$$\Theta_{\Phi,7}$$

on the quotient such that

$$\Phi_K(A_7^f) = \Theta_{\Phi,7}(q(A_7^f))$$

for all $A_7^f \in \mathfrak{f}_m$ if and only if

$$\Phi_K(A_5^f) = 0.$$

Equivalently, since

$$\Phi_K(A_5^f) = 2v_5,$$

such descent is possible if and only if

$$v_5 = 0.$$

Proof. The quotient identifies

$$A_7^f \sim A_7^f + \chi A_5^f.$$

Therefore Φ_K descends to the quotient if and only if it is constant on these affine fibers, i.e.

$$\Phi_K(A_7^f + \chi A_5^f) = \Phi_K(A_7^f)$$

for all A_7^f and all χ . By linearity this is equivalent to

$$\chi \Phi_K(A_5^f) = 0$$

for all χ , hence to

$$\Phi_K(A_5^f) = 0.$$

Writing

$$A_5^f = u_5 I + v_5 S$$

gives

$$\Phi_K(A_5^f) = 2v_5.$$

Thus descent holds exactly on the locus $v_5 = 0$. $\square$

Theorem 12.186 (Augmented Bottleneck C condition). *On the good patch $c \neq 0$, diagonal merger for the transform-safe package*

$$\widehat{\Psi}_{\text{red}}^+ = (c, u_5, v_5, \Delta_7, T)$$

requires both

$$\partial_\kappa Q_{7,m}^q(m, \kappa, 0) = 0$$

and

$$\partial_\kappa T(m, \kappa, 0) = 0.$$

The first condition is quotient-horizontal merger. The second is hidden-horizontal merger. On the generic patch $v_5 \neq 0$, the first does not imply the second.

Remark 12.187 (Scalar Bottleneck C condition). Let

$$P_7(a_1, a_2) = [z^7][\delta^1]\tau_{h \rightarrow m}\pi_{f_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}.$$

The quotient Bottleneck C condition is

$$\partial_{a_2} P_7(a_1, -a_1) \in \mathbf{C}A_5^f(m).$$

The transform-safe condition is instead

$$\Phi_K(\partial_{a_2} P_7(a_1, -a_1)) = 0.$$

Equivalently, if

$$P_{7,\Phi}(a_1, a_2) := \frac{1}{c} \Phi_K(P_7(a_1, a_2)),$$

then the scalar condition is

$$P_{7,\Phi}(a_1, a_2) \in (a_1 + a_2)^2 \mathbf{C}[a_1, a_2].$$

This scalar condition is independent of quotient-valued double-balanced divisibility on the generic patch $v_5 \neq 0$.

Lemma 12.188 (Hidden-horizontal exceptional divisor condition). *Work on a good patch with*

$$c(m) \neq 0, \quad A_5^f(m) = u_5 I + v_5 S, \quad v_5 \neq 0.$$

Let

$$F_7(a_1, a_2, \delta) = [z^7]\tau_{h \rightarrow m}\pi_{f_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = \delta P_7(a_1, a_2) + O(\delta^2).$$

Then the augmented hidden coordinate $T = v_7/c$ is horizontal on the exceptional divisor if and only if

$$\Phi_K(\partial_{a_2} P_7(a_1, -a_1)) = 0.$$

Equivalently,

$$\partial_\kappa T(m, \kappa, 0) = 0 \quad \Longleftrightarrow \quad \frac{1}{c} \Phi_K(\partial_{a_2} P_7(a_1, -a_1)) = 0.$$

Proof. Along the collision chart

$$a_2 = -a_1(1 + \kappa\delta),$$

one has

$$P_7(a_1, a_2) = P_7(a_1, -a_1) - a_1\kappa\delta \partial_{a_2} P_7(a_1, -a_1) + O(\delta^2).$$

After the standard edge normalization, the exceptional κ -slope is represented by

$$\partial_{a_2} P_7(a_1, -a_1).$$

Applying Φ_K/c gives the slope of the normalized transform-visible coordinate. Since $T = \Phi_K(A_7^f)/(2c)$, the displayed criterion follows. $\square$

Proposition 12.189 (Formal transform-visible line-gauge leakage). *On a good patch with*

$$c \neq 0, \quad A_5^f = u_5 I + v_5 S, \quad v_5 \neq 0,$$

quotient-horizontal Bottleneck C does not imply T-horizontal merger.

Proof. Consider an order-7 fixed-sector edge carrier of the form

$$[\delta^1] A_7^{f\chi} = \lambda(m) a_1 a_2 (a_1 + a_2) A_5^f(m).$$

Since it is pointwise parallel to $A_5^f(m)$, its quotient class vanishes:

$$[A_7^{f\chi}] = 0 \quad \text{in} \quad \mathfrak{f}_m / \mathbf{C} A_5^f(m).$$

Thus it is invisible to quotient Bottleneck C.

However,

$$a_1 a_2 (a_1 + a_2) \big|_{a_2 = -a_1} = 0,$$

but

$$\partial_{a_2} (a_1 a_2 (a_1 + a_2)) \big|_{a_2 = -a_1} = -a_1^2.$$

Moreover,

$$\Phi_K(A_5^f) = 2v_5.$$

Therefore the induced hidden-state slope is

$$[\delta^1] \partial_{a_2} T_\chi(a_1, -a_1, \delta) = -a_1^2 \lambda(m) \frac{v_5(m)}{c(m)}.$$

On the generic patch $c \neq 0$, $v_5 \neq 0$, this is nonzero unless

$$\lambda(m) = 0.$$

Thus a quotient-invisible line-gauge edge carrier can be transform-visible. $\square$

Corollary 12.190 (First live T -slope carrier). *Let*

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S$$

be the frozen Bottleneck-C coefficient of the promoted nonlinear mixed-block carrier. The amplitude factor of this carrier is

$$a_1 a_2 (a_1 + a_2),$$

and

$$\partial_{a_2} (a_1 a_2 (a_1 + a_2)) \big|_{a_2 = -a_1} = -a_1^2.$$

Consequently its contribution to the hidden-horizontal slope is

$$\partial_\kappa T \big|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \frac{\Phi_K(E_{A,Y}^{\text{BC}})}{2c} = -a_1^2 \frac{f_{A,Y}}{c}.$$

Thus the augmented transform-safe package requires

$$f_{A,Y} = 0.$$

In particular, if

$$E_{A,Y}^{\text{BC}} = \lambda A_5^f,$$

then

$$\partial_\kappa T|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \lambda \frac{v_5}{c}.$$

On the generic $v_5 \neq 0$ patch, a nonzero line-gauge $E_{A,Y}^{\text{BC}}$ closes the quotient condition but still leaks through T .

Remark 12.191 (Local diagnostic for the hidden septic state). On the quintic-good patch

$$\{c \neq 0, v_5 \neq 0\},$$

one useful local diagnostic enlargement of the reduced package is

$$\tilde{\Psi}_{\min} = (x, Y, S, T) = \left(\frac{v_5}{c}, \frac{u_5}{c}, \frac{\Delta_7}{cv_5}, \frac{v_7}{c} \right).$$

This tuple is useful only as normalized fixed-sector septic data. It is not a canonical global theorem object and it does not give finite determination of the first surviving odd jet. The global target remains the transform-level first Φ_K -visible corrected odd coefficient, modulo $\ker \Phi_K$.

Remark 12.192 (Bottleneck C is exceptional-divisor fiber selection). For Bottleneck C, the obstruction is not the choice of a local septic representative such as $v_7 = 0$, $u_7 = 0$, or $T = v_7/c$. Those are diagnostic data for the hidden-state problem. The C-side object has already quotiented raw septic representatives away. Its remaining obstruction is the exceptional-divisor trace

$$B(m, \kappa) := \tilde{\Psi}_{\text{red}}^{\text{corr}}(m, \kappa, 0).$$

Analyticity, swap-evenness, fixed codomain, scalar normalization, and the central value $B(m, 0) = \hat{\Psi}(m)$ do not force

$$\partial_\kappa B = 0.$$

A valid C proof must supply actual diagonal merger, same-reduced-image, or collision-functoriality for the corrected two-atom package.

Remark 12.193 (Formal C properties do not rule out slope leakage). Formal properties such as analyticity, swap-evenness, fixed codomain, scalar normalization, and the central value at $\kappa = 0$ do not by themselves force the C-side exceptional-divisor trace to be independent of κ . A formal model may have

$$B_3 = Z + \varepsilon \kappa,$$

or, at the quotient-defect level,

$$R \mapsto R + R_\kappa, \quad \det(R_\kappa, A_5^f) = c^2 \varepsilon \kappa.$$

Thus representative-gauge invariance only kills additions parallel to A_5^f . It does not kill determinant-changing quotient-defect data. A C proof must control the actual corrected exceptional-divisor quotient class, not merely the formal shape of the reduced package.

Remark 12.194 (Atom swap does not impose $\kappa \mapsto -\kappa$ on the exceptional divisor). In the collision chart

$$h_1 = m - \frac{\delta}{2}, \quad h_2 = m + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

interchanging the two labelled atoms gives

$$\delta \mapsto -\delta, \quad a_1 \mapsto -a_1(1 + \kappa\delta), \quad \kappa \mapsto \frac{\kappa}{1 + \kappa\delta}.$$

Hence on the exceptional divisor $\delta = 0$, the slope coordinate is fixed:

$$\kappa \mapsto \kappa.$$

Therefore atom-swap symmetry does not force the exceptional trace

$$B(m, \kappa)$$

to be even in κ , and in particular it does not force

$$\partial_\kappa B(m, \kappa) = 0.$$

Swap symmetry removes the wrong-parity terms in the transverse collision variable; it does not remove quotient-visible dependence on the exceptional slope.

Remark 12.195 (Missing theorem: quotient-valued diagonal merger). Bottleneck C requires a quotient-valued diagonal merger theorem. On every good patch on which

$$c(m) \neq 0, \quad A_5^{\mathfrak{f}}(m) \neq 0,$$

the corrected two-atom collision package must factor through the blow-down of the exceptional divisor:

$$(m, \kappa, 0) \longmapsto m.$$

Equivalently,

$$B(m, \kappa) = B(m, 0) = \widehat{\Psi}(m) \quad \text{for all sufficiently small } \kappa.$$

At the order-7 quotient level this means

$$B_7^{\mathfrak{f}}(m, \kappa, 0) - B_7^{\mathfrak{f}}(m, 0, 0) \in \mathbf{C}A_5^{\mathfrak{f}}(m).$$

This is stronger than choosing a local septic gauge such as $v_7 = 0$ or $u_7 = 0$. Such a gauge selects a representative inside a fixed quotient class; it does not prove that the quotient class itself is independent of the exceptional slope κ .

Definition 12.196 (Quotient-valued two-atom collision functor). On a good patch with

$$A_5^{\mathfrak{f}}(m) \neq 0,$$

define the order-7 quotient-valued two-atom collision functor by the ordered composite

$$\begin{aligned} \mathcal{B}_2 &\longmapsto C_{\mathcal{W}}^{\text{raw}} \longmapsto (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} \longmapsto -a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} \\ &\longmapsto \pi_{\mathfrak{f}_h} [z^{\leq 7}] \longmapsto \tau_{h \rightarrow m} \longmapsto [z^7] \longmapsto \mathfrak{f}_m / \mathbf{C}A_5^{\mathfrak{f}}(m). \end{aligned}$$

In formulas,

$$\mathcal{C}_7^q(m, \kappa, \delta) := [[z^7] \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}})].$$

The exceptional-divisor trace is

$$B(m, \kappa) = \mathcal{C}_7^q(m, \kappa, 0).$$

Thus quotient-valued diagonal merger is precisely the statement that

$$\mathcal{C}_7^q(m, \kappa, 0) = \mathcal{C}_7^q(m, 0, 0),$$

or equivalently that $\mathcal{C}_7^q$ factors on the exceptional divisor through the blow-down map

$$(m, \kappa, 0) \longmapsto m.$$

Proposition 12.197 (Formal quotient-visible exceptional-slope leakage). *The formal properties*

analyticity, atom-swap compatibility, fixed codomain, scalar normalization, $B(m, 0) = \widehat{\Psi}(m)$ do not imply quotient-valued diagonal merger.

Proof. Work on a good patch and choose

$$E_m \in \mathfrak{f}_m$$

with

$$\omega_m(E_m, A_5^{\mathfrak{f}}(m)) = 1.$$

Thus E_m is nonzero in

$$\mathfrak{f}_m / \mathbf{C}A_5^{\mathfrak{f}}(m).$$

In the collision chart

$$h_1 = m - \frac{\delta}{2}, \quad h_2 = m + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

atom swap acts by

$$\delta \longmapsto -\delta, \quad \kappa \longmapsto \frac{\kappa}{1 + \kappa\delta}.$$

Define

$$H(\kappa, \delta) := \frac{1}{2} \left(\kappa + \frac{\kappa}{1 + \kappa\delta} \right).$$

Then H is analytic near $(\kappa, \delta) = (0, 0)$, satisfies

$$H\left(\frac{\kappa}{1 + \kappa\delta}, -\delta\right) = H(\kappa, \delta),$$

and restricts to

$$H(\kappa, 0) = \kappa$$

on the exceptional divisor.

Let $Z_0(m) \in \mathfrak{f}_m$ be any representative of $\widehat{\Psi}(m)$, and define the formal normalized septic edge representative

$$\mathcal{R}_\varepsilon(m, \kappa, \delta; z) = z^7 (Z_0(m) + \varepsilon H(\kappa, \delta) E_m).$$

Equivalently, before edge normalization one may write the raw model as

$$\mathcal{N}_\varepsilon = -a_1 \delta^2 z^7 (Z_0(m) + \varepsilon H(\kappa, \delta) E_m).$$

This raw model is $O(a_1 \delta^2)$, so it is compatible with raw edge-divisibility. It is analytic and atom-swap compatible by construction.

At $\kappa = 0$,

$$B_\varepsilon(m, 0) = [Z_0(m)] = \widehat{\Psi}(m).$$

But on the exceptional divisor,

$$B_\varepsilon(m, \kappa) = [Z_0(m) + \varepsilon \kappa E_m],$$

and hence

$$\partial_\kappa B_\varepsilon(m, \kappa) = \varepsilon [E_m] \neq 0$$

in

$$\mathfrak{f}_m / \mathbf{C} A_5^{\mathfrak{f}}(m).$$

Therefore the listed formal properties do not force quotient-valued diagonal merger. $\square$

Remark 12.198 (Balancing does not impose hidden κ -horizontality). The balancing condition

$$a_2 = -a_1(1 + \kappa\delta)$$

gives

$$\partial_\kappa a_2 = -a_1 \delta.$$

This single factor of δ does not force the normalized exceptional slope to vanish after the δ^{-2} edge normalization. A raw term of the form

$$-a_1 \delta^2 \kappa E_m$$

is still of the allowed Gate-B edge order and survives normalization as

$$\kappa E_m.$$

Equivalently, with $\kappa = 2\omega/\delta$, the leakage term

$$2\varepsilon\omega\delta E_m = \varepsilon\kappa\delta^2 E_m$$

is invisible before edge normalization but visible on the exceptional divisor. Thus balancing kills the zeroth source moment; it does not by itself kill the projective exceptional slope.

Definition 12.199 (Promoted quadratic mixed-block Bottleneck-C coefficient). Let

$$Y_{12}^{(2)} := D^2 N_0[\xi_1, \xi_2]$$

denote the $\mathcal{K}^2$, source-bidegree $(1, 1)$ nonlinear Taylor remainder of the actual corrected mixed block $N^{\text{corr}}(d_-, d_+, \eta)$.

The promoted channel has amplitude factor

$$a_1 a_2 (a_1 + a_2).$$

On the collision chart

$$a_2 = -a_1(1 + \kappa\delta),$$

this amplitude already contributes the first δ -edge:

$$a_1 a_2 (a_1 + a_2) = a_1^3 \kappa \delta + O(\delta^2).$$

Therefore the Bottleneck-C coefficient for this promoted channel is the frozen internal coefficient

$$E_{A,Y}^{\text{BC}} := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0.$$

Equivalently,

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{g=4}^7 \sum_{r+b=7-g} H_{r,0}^A Y_{g,0}^\chi W_{+,b}.$$

Write

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S.$$

The quotient Bottleneck-C obstruction is

$$Q_{A,Y} := e_{A,Y} v_5 - u_5 f_{A,Y}.$$

The transform-safe obstruction is

$$\Phi_K(E_{A,Y}^{\text{BC}}) = 2f_{A,Y}.$$

For comparison, the literal source-position edge

$$E_{A,Y}^{\text{pos}} := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7 \delta^1] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0$$

belongs to the next source-motion layer and should not be confused with the first Bottleneck-C amplitude-derivative coefficient.

Definition 12.200 (Active Bottleneck-C source-linear index set). Let

$$\mathcal{A}_{\text{BC}} \subseteq \{2, 3, 4, 5, 6\}$$

denote the set of same-point source-linear Fréchet primitive indices that can enter the promoted mixed-block Bottleneck-C coefficient

$$E_{A,Y}^{\text{BC}}.$$

In the finite Fréchet convention of Section B, the lead top-row obstruction uses

$$a = 2, \quad \Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22},$$

so $2 \in \mathcal{A}_{\text{BC}}$. Older shorthand using $a \in \{1, 5\}$ should be read only as a deprecated table convention and not as the active finite Fréchet index set.

Remark 12.201 (Executable table for $E_{A,Y}^{\text{BC}}$). The frozen Bottleneck-C coefficient is computed from the rows

$$\mathcal{E}_{\chi;g,r,b}^a := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \left(\Lambda_{a,r}^- Y_{g,0}^\chi W_{+,b} \right),$$

where

$$\begin{aligned} a &\in \mathcal{A}_{\text{BC}}, & \chi &\in \{dd, d\eta + \eta d, \eta\eta\}, \\ 4 &\leq g \leq 7, & r + b &= 7 - g. \end{aligned}$$

Thus

$$E_{A,Y}^{\text{BC}} = \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{a \in \mathcal{A}_{\text{BC}}} \sum_{\substack{4 \leq g \leq 7 \\ r+b=7-g}} \mathcal{E}_{\chi;g,r,b}^a.$$

The skew row

$$Y_{d\eta - \eta d}^{(2)}$$

is excluded because it is killed by fixed-sector projection.

Remark 12.202 (Status of $E_{A,Y}^{\text{BC}}$). The recurrences for

$$W_{+,0}, W_{+,1}, W_{+,2}, W_{+,3}$$

and

$$\Lambda_{a,0}^-, \Lambda_{a,1}^-, \Lambda_{a,2}^-, \Lambda_{a,3}^-$$

reduce $E_{A,Y}^{\text{BC}}$ to a finite symbolic coefficient table. However, the present draft does not yet supply the scalar midpoint source jets needed to evaluate

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S.$$

The missing scalar jets are

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

and

$$(r_i^{[a]})^{(k)}(m), \quad i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9.$$

Moreover, the active top row

$$(a, r, g, b) = (2, 0, 7, 0), \quad \chi = \eta\eta_{\text{odd}},$$

recorded in Subsection 14.3, gives a concrete Φ_K -visible survivor unless

$$\nu_{21,7} = 0$$

or unless it is cancelled by another physical top-row contribution. Thus the current local obstruction is sharper than an unevaluated table: the package must prove cancellation of the odd- $\eta\eta$ top row, or add a physical correction cancelling it.

Remark 12.203 (First live promoted nonlinear mixed-block obstruction). The same-point cubic q^3 defect is fixed-sector invisible in same-point-only Fréchet responses by endpoint-swap/transpose symmetry. The first remaining actual-source/Bottleneck-C carrier is the promoted nonlinear mixed-block quadratic remainder

$$Y_{12}^{(2)} = D^2 N_0[\xi_1, \xi_2],$$

coupled to one same-point linear whitening slot.

The amplitude factor of this channel is

$$a_1 a_2 (a_1 + a_2).$$

Since

$$a_1 a_2 (a_1 + a_2) = a_1^3 \kappa \delta + O(\delta^2) \quad \text{on } a_2 = -a_1(1 + \kappa \delta),$$

the first Bottleneck-C coefficient is

$$E_{A,Y}^{\text{BC}},$$

not the internal source-position coefficient $E_{A,Y}^{\text{pos}}$.

Thus quotient-valued Bottleneck C for this channel is exactly

$$Q_{A,Y} = e_{A,Y} v_5 - u_5 f_{A,Y} = 0.$$

The augmented transform-safe package requires the stronger scalar condition

$$f_{A,Y} = 0, \quad \text{equivalently} \quad \Phi_K(E_{A,Y}^{\text{BC}}) = 0.$$

On the generic $v_5 \neq 0$ patch, quotient closure together with T -horizontal closure forces

$$E_{A,Y}^{\text{BC}} = 0.$$

Lemma 12.204 (Promoted mixed-block coefficient table). *In the notation of Definition 12.199, the frozen Bottleneck-C coefficient of the promoted quadratic mixed-block channel is*

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{g=4}^7 \sum_{r+b=7-g} H_{r,0}^A Y_{g,0}^\chi W_{+,b}.$$

Here

$$H_{r,0}^A = [z^r] D(G_-^{-1/2})[A_-], \quad W_{+,b} = [z^b] G_+^{-1/2},$$

and

$$Y_{g,0}^\chi = [z^g]Y_\chi^{(2)}$$

for

$$\chi \in \{dd, d\eta + \eta d, \eta\eta\}.$$

The skew combination

$$Y_{d\eta-\eta d}^{(2)}$$

is not included, since it is killed by fixed-sector projection.

If

$$E_{A,Y}^{\text{BC}} = e_{A,Y}I + f_{A,Y}S,$$

then the quotient obstruction is

$$e_{A,Y}v_5 - u_5f_{A,Y},$$

and the transform-safe obstruction is

$$f_{A,Y}.$$

Remark 12.205 (Current status of the promoted mixed-block coefficient table). The finite Fréchet/Sylvester machinery determines the shape of $E_{A,Y}^{\text{BC}}$, but the present draft does not yet tabulate the scalar midpoint jet data needed to decide

$$E_{A,Y}^{\text{BC}} = 0, \quad E_{A,Y}^{\text{BC}} \in \mathbb{C}A_5^{\dagger}(m), \quad \text{or} \quad (E_{A,Y}^{\text{BC}})^q \neq 0.$$

The exact missing finite table is

$$\left\{ W_{+,0}, W_{+,1}, W_{+,2}, W_{+,3}, \quad \Lambda_{a,0}^-, \Lambda_{a,1}^-, \Lambda_{a,2}^-, \Lambda_{a,3}^-, \right. \\ \left. Y_{g,0}^{dd}, \quad Y_{g,0}^{d\eta+\eta d}, \quad Y_{g,0}^{\eta\eta} \quad (4 \leq g \leq 7) \right\}.$$

Until that table is evaluated, $E_{A,Y}^{\text{BC}}$ remains the first unresolved local actual-source/Bottleneck-C obstruction.

Remark 12.206 (Frozen Bottleneck-C coefficient versus source-position edge). The coefficient $E_{A,Y}^{\text{BC}}$ is the first Bottleneck-C amplitude-ratio carrier. It uses the frozen internal coefficient because the amplitude factor

$$a_1a_2(a_1 + a_2)$$

already supplies the first δ -edge on the collision chart

$$a_2 = -a_1(1 + \kappa\delta).$$

The literal source-position edge

$$E_{A,Y}^{\text{pos}} := \tau_{h \rightarrow m} \pi_{\text{fh}} \text{Sym}[z^7 \delta^1] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0$$

belongs to the next source-motion layer and should not be used as the first Bottleneck-C amplitude-derivative coefficient.

Remark 12.207 (Shortcut routes excluded for the corrected package theorem). The following routes do not prove the corrected package quotient theorem:

- (1) Φ_K -scalar invisibility or scalar hiding after Φ_K .
- (2) Determinant scalarization used as the definition of the quotient class.
- (3) Swap symmetry or one-pair K_5/K_3 gauge laws without the actual two-atom package object.
- (4) The tagged pre-whitening source block alone.
- (5) The raw seven-family Fréchet sum; tested source-constrained nonconstant baselines leave nonzero determinant residuals.
- (6) Cubic/quintic edge laws plus the abstract source criterion alone.
- (7) Existing one-pair order-7 or centered D_2 formulas promoted directly to a two-atom order-7 Hessian.
- (8) Signed source-weight lift alone; it restores oddness but leaves higher $a^3, a^5, \dots$ amplitude terms.

Every positive route must work before Φ_K , before determinant scalarization, and inside the fixed-target quotient package.

Remark 12.208 (UV-010 boundary: edge law before descent). The midpoint quotient trivialization is available on the good patch

$$A_5^f(m) \neq 0,$$

but this does not construct the order-7 two-atom edge package. The UV-010 target is the normalized fixed-midpoint quotient edge law

$$\overline{E}_{12}^{(7;1)} = \delta^2 \mathcal{H}_7^q(m, \kappa, \delta^2),$$

with

$$[R]_{\text{edge}} = -\mathcal{H}_7^q(m, \kappa, 0).$$

Descent, κ -independence, and diagonal merger are downstream of this edge-law construction. The Gate B edge-divisibility theorem constructs the holomorphic environment needed for the corrected package representative; it should not be conflated with the further UV-010 task of identifying the exceptional-divisor value $[R]_{\text{edge}}$ with the C-side reduced-package trace. Determinant scalarization is good-patch bookkeeping after the quotient-valued edge object exists; it is not a substitute for the edge package.

Remark 12.209 (What remains in the two-pair rewrite). Theorem 12.121 still provides the correct algebraic bridge from the two-pair core to the strengthened invariant $\hat{\Psi}$, and Lemma 12.124 still isolates the finite-order source criterion that would close the close two-point branch on compact good patches.

What remains, however, is no longer raw bookkeeping through orders 3, 5, 7 and no longer a search for another local mixed branch taxonomy. The overlap-side geometry is now sharply split:

$$(K_v = 0, S \neq 0)$$

is already mixed low-order,

$$(S = 0, K_v \neq 0)$$

is already absorbed by nonmixed 4-point geometry, and the only genuinely residual mixed corner is

$$(S = 0, K_v = 0).$$

There the branch is quotient-diagonal / involutive, and the remaining burden is best read through the three-front compression of Remark 12.210.

Accordingly, the live downstream two-point programs are now the following.

First, a good-patch collision-cancellation theorem for the *actual* corrected cubic/quintic defects: enough regularity and swap-evenness in the blown-up cancellation chart to justify the quadratic/projective edge-law package.

Second, a quotient-diagonal closure theorem on the exact fixed-shear corner: for example state-locality, transport closure, or a diagonal-collapse result for the actual corrected defect package after quotient-visible descent.

Third, a provenance-sensitive package theorem — for example same reduced image germ at coincidence or collision-functoriality on the collision/cancellation locus — strong enough to force the later quotient-septic collapse formally.

These are conditional downstream programs, not yet closure theorems. What the present draft already shows is that the residual burden is concentrated there, rather than in a missing local branch classification.

Remark 12.210 (Current three-front compression of the remaining burden). The two-point endgame is now substantially narrower than a generic “one more local sign argument” picture.

On the nonmixed/global affine side, the remaining burden is an RH-specific exclusion problem for the isolated remote endpoint/branch incidences encoded by the monotone gap functions $G_{\xi, J}$.

On the mixed side, the only genuinely residual overlap contribution is the exact fixed-shear corner

$$\{S = 0\} \cap \{K_v = 0\},$$

already reduced to a quotient-diagonal / involutive transport problem. On the honest higher-order side, the relevant target is no longer raw order-7 equality of representatives, but a provenance-sensitive package statement such as same reduced image germ at coincidence or collision-functoriality on the collision/cancellation locus.

Accordingly, what remains is not a diffuse interaction heuristic. It is a three-front problem: remote endpoint-gap rigidity, exact fixed-shear quotient transport, and honest package-level coincidence/functoriality.

Remark 12.211 (Minimal-core reformulation). The proposition above shows that the correct projective datum for the downstream multi-pair and minimal-core extraction is the amplitude-invariant strengthened map

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right),$$

not merely the pair $(\ell_h, \bar{v}_1(h))$.

By contrast, the affine curve

$$\Gamma(h) = \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right)$$

remains the correct object for the one-pair secant/tangent geometry. The finite-core endgame should therefore be viewed as a two-stage problem: first extract defect-corrected coincidence information for $\widehat{\Psi}$, then convert that information into affine-line incidence constraints for Γ .

Remark 12.212 (Finite-core compression to a remote endpoint/branch incidence problem). The two-stage reformulation above suggests a useful downstream combinatorial compression.

Fix a compact relevant patch J . By Theorem 12.100, every maximal turning-boundary-generic block contributes no pair-pair geometry, so such a block behaves as a single transport chamber. By Theorem 12.107, the isolated finite-order $W = 0$, $M \neq 0$ boundaries are locally sterile for pair-pair birth. Moreover, on the repeated-source side, Proposition 12.298, Corollary 12.314, Proposition 12.316, and Corollary 12.318 show that odd-order $\mathcal{R}$ -boundaries are relay-type rather than arbitrary first-root events.

After these absorptions, any surviving minimal bad core contribution from one-pair incidence must come through the finite remote endpoint/branch tangencies of Lemma 12.109 and Remark 13.5, equivalently through isolated interior zeros of the strictly monotone endpoint-gap functions

$$G_{\xi,J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h)).$$

On the mixed side, the only residual contribution is the exact fixed-shear corner $\{S = 0\} \cap \{K_v = 0\}$, already treated separately as a quotient-diagonal transport problem.

Thus the remaining global nonadjacent recurrence problem is cleanly a finite remote-incidence problem for the $G_{\xi,J}$, not a diffuse local-birth problem on a graph of generic gates.

Remark 12.213 (Ceiling of the remaining affine/global lane). The derivative-geometric reductions on the nonmixed/global side now have a clean positive content and a clean ceiling.

The positive content is that fixed-slope tangency/incidence has been reduced to a finite remote endpoint/branch incidence problem: support curves are strictly concave, have single-crossing and betweenness, and the endpoint-gap functions

$$G_{\xi,J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h))$$

are strictly monotone on ordered target branches. Thus each endpoint/branch pair contributes at most one remote tangency.

The ceiling is that this affine/pseudoline structure is universal for analytic plane curves. So no further purely planar, interval-graph, pseudoline, or local convexity argument can by itself

finish this lane. Any final contradiction here must use RH-specific source structure constraining the possible zeros of the functions $G_{\xi,J}$.

Definition 12.214 (Minimal bad core). A normalized finite core is called bad if its quotient odd germ vanishes identically. A minimal bad core is a bad normalized finite core with minimal number of constituent pairs.

Theorem 12.215 (Defect-free quintic circuit reduction). *Let*

$$F_{\text{core}}(z) = \sum_{j=1}^k a_j F_{h_j}(z) + \mathcal{I}_{\geq 2}(z)$$

be a normalized finite core, and assume that the multi-pair interaction germ $\mathcal{I}_{\geq 2}(z)$ contributes nothing to the cubic and quintic coefficients. Write

$$A_5^{\mathfrak{f}}(h) = u_5(h)I + v_5(h)S, \quad V(h) := (c(h), u_5(h), v_5(h)) \in \mathbf{C}^3,$$

and on $\{c \neq 0\}$ define

$$\Xi(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)} \right) \in \mathbf{C}^2.$$

If

$$F_{\text{core}}(z) \equiv 0,$$

then

$$\sum_{j=1}^k a_j V(h_j) = 0.$$

Equivalently, with

$$b_j := a_j c(h_j),$$

one has

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j \Xi(h_j) = 0.$$

Thus the points $\Xi(h_j)$ are affinely dependent.

Proof. Taking the cubic coefficient in the identity

$$F_{\text{core}}(z) \equiv 0$$

and using the hypothesis that $\mathcal{I}_{\geq 2}$ contributes nothing at cubic order gives

$$\sum_{j=1}^k a_j c(h_j) = 0.$$

Taking the quintic fixed-sector coefficient and using the corresponding defect-free hypothesis at order 5 gives

$$\sum_{j=1}^k a_j A_5^{\mathfrak{f}}(h_j) = 0.$$

Since

$$A_5^{\mathfrak{f}}(h_j) = u_5(h_j)I + v_5(h_j)S$$

and I, S are linearly independent, this is equivalent to

$$\sum_{j=1}^k a_j u_5(h_j) = 0, \quad \sum_{j=1}^k a_j v_5(h_j) = 0.$$

Together these three identities are exactly

$$\sum_{j=1}^k a_j V(h_j) = 0.$$

If $c(h_j) \neq 0$ for all j , dividing the last two identities by $c(h_j)$ after setting $b_j = a_j c(h_j)$ yields

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j \Xi(h_j) = 0,$$

which is the affine dependence statement. $\square$

Corollary 12.216 (Defect-free minimal quintic circuits have size at most four). *In the defect-free order- ≤ 5 model of Theorem 12.215, every minimal bad core has size at most 4. More precisely:*

- (1) if $k = 2$, then $\Xi(h_1) = \Xi(h_2)$;
- (2) if $k = 3$, then the three points $\Xi(h_1), \Xi(h_2), \Xi(h_3)$ are collinear;
- (3) if $k = 4$, then the four points $\Xi(h_1), \dots, \Xi(h_4)$ form an affine circuit in $\mathbf{C}^2$.

Proof. By Theorem 12.215, a minimal defect-free bad core produces a minimal affine dependence among finitely many points of the affine plane $\mathbf{C}^2$. Such a minimal affine dependent set has size at most 4. The cases $k = 2, 3, 4$ are the standard affine configurations in dimension 2: coincidence, collinear triple, or four-point affine circuit. $\square$

Definition 12.217 (Lifted v_5 -patch coordinates). On a connected interval $U \subset \{c \neq 0, v_5 \neq 0\}$, define

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{u_5(h)}{c(h)}, \quad \sigma_v(h) := \frac{\Delta_7(h)}{c(h) v_5(h)}.$$

The associated lifted planar and space curves are

$$\Lambda_v(h) := (x(h), \sigma_v(h)) \in \mathbf{C}^2, \quad Q_v(h) := (y(h), x(h), \sigma_v(h)) \in \mathbf{C}^3.$$

Proposition 12.218 (Defect-free lifted affine dependence on a v_5 -patch). *Let*

$$F_{\text{core}}(z) = \sum_{j=1}^k a_j F_{h_j}(z) + \mathcal{I}_{\geq 2}(z)$$

be a normalized finite core, and assume:

- (1) all parameters h_j lie in one connected interval $U \subset \{c \neq 0, v_5 \neq 0\}$;
- (2) the multi-pair interaction germ $\mathcal{I}_{\geq 2}(z)$ contributes nothing to the cubic, quintic, or septic coefficients.

Then, after choosing on U the local septic gauge $v_7 = 0$, the vanishing identity

$$F_{\text{core}}(z) \equiv 0$$

implies

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j y(h_j) = 0, \quad \sum_{j=1}^k b_j x(h_j) = 0, \quad \sum_{j=1}^k b_j \sigma_v(h_j) = 0,$$

where $b_j := a_j c(h_j)$. Equivalently, the points $Q_v(h_j) \in \mathbf{C}^3$ are affinely dependent.

Proof. The cubic and quintic identities are exactly those in Theorem 12.215, so after setting $b_j = a_j c(h_j)$ we obtain

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j y(h_j) = 0, \quad \sum_{j=1}^k b_j x(h_j) = 0.$$

On the interval U we may choose the local septic gauge $v_7 = 0$. Then

$$u_7(h) = \frac{\Delta_7(h)}{v_5(h)} = c(h) \sigma_v(h).$$

Since the septic interaction is assumed absent, taking the septic fixed-sector coefficient of $F_{\text{core}} \equiv 0$ yields

$$\sum_{j=1}^k a_j u_7(h_j) = 0.$$

Substituting $u_7(h_j) = c(h_j)\sigma_v(h_j)$ gives

$$\sum_{j=1}^k b_j \sigma_v(h_j) = 0.$$

Hence the points $Q_v(h_j)$ are affinely dependent. $\square$

Corollary 12.219 (Geometric meaning of defect-free lifted affine dependence). *Assume the hypotheses of Proposition 12.218. Then:*

- (1) if $k = 2$, the two points $Q_v(h_1)$ and $Q_v(h_2)$ coincide;
- (2) if $k = 3$, the three points $Q_v(h_1), Q_v(h_2), Q_v(h_3)$ are collinear in $\mathbf{C}^3$;
- (3) if $k = 4$, the four points $Q_v(h_1), \dots, Q_v(h_4)$ are coplanar in $\mathbf{C}^3$.

Proof. By Proposition 12.218, the points $Q_v(h_j)$ are affinely dependent. For $k = 2$, affine dependence of two points is exactly coincidence. For $k = 3$, affine dependence of three points is exactly collinearity. For $k = 4$, affine dependence of four points is exactly coplanarity. $\square$

Corollary 12.220 (Explicit four-point determinant form on a v_5 -patch). *Assume the hypotheses of Proposition 12.218 with $k = 4$. Then the coplanarity of the four lifted points is equivalent to*

$$\det \begin{pmatrix} 1 & y(h_1) & x(h_1) & \sigma_v(h_1) \\ 1 & y(h_2) & x(h_2) & \sigma_v(h_2) \\ 1 & y(h_3) & x(h_3) & \sigma_v(h_3) \\ 1 & y(h_4) & x(h_4) & \sigma_v(h_4) \end{pmatrix} = 0.$$

Equivalently,

$$\det \begin{pmatrix} 1 & \frac{u_5(h_1)}{c(h_1)} & \frac{v_5(h_1)}{c(h_1)} & \frac{\Delta_7(h_1)}{c(h_1)v_5(h_1)} \\ 1 & \frac{u_5(h_2)}{c(h_2)} & \frac{v_5(h_2)}{c(h_2)} & \frac{\Delta_7(h_2)}{c(h_2)v_5(h_2)} \\ 1 & \frac{u_5(h_3)}{c(h_3)} & \frac{v_5(h_3)}{c(h_3)} & \frac{\Delta_7(h_3)}{c(h_3)v_5(h_3)} \\ 1 & \frac{u_5(h_4)}{c(h_4)} & \frac{v_5(h_4)}{c(h_4)} & \frac{\Delta_7(h_4)}{c(h_4)v_5(h_4)} \end{pmatrix} = 0.$$

Multiplying row j by $v_5(h_j)$ gives the canonical determinant identity

$$\det \begin{pmatrix} v_5(h_1) & u_5(h_1) & \frac{v_5(h_1)^2}{c(h_1)} & \frac{\Delta_7(h_1)}{c(h_1)} \\ v_5(h_2) & u_5(h_2) & \frac{v_5(h_2)^2}{c(h_2)} & \frac{\Delta_7(h_2)}{c(h_2)} \\ v_5(h_3) & u_5(h_3) & \frac{v_5(h_3)^2}{c(h_3)} & \frac{\Delta_7(h_3)}{c(h_3)} \\ v_5(h_4) & u_5(h_4) & \frac{v_5(h_4)^2}{c(h_4)} & \frac{\Delta_7(h_4)}{c(h_4)} \end{pmatrix} = 0.$$

Equivalently, multiplying instead row j by $c(h_j)v_5(h_j)$ gives the fully polynomial version

$$\det \begin{pmatrix} c(h_1)v_5(h_1) & u_5(h_1)v_5(h_1) & v_5(h_1)^2 & \Delta_7(h_1) \\ c(h_2)v_5(h_2) & u_5(h_2)v_5(h_2) & v_5(h_2)^2 & \Delta_7(h_2) \\ c(h_3)v_5(h_3) & u_5(h_3)v_5(h_3) & v_5(h_3)^2 & \Delta_7(h_3) \\ c(h_4)v_5(h_4) & u_5(h_4)v_5(h_4) & v_5(h_4)^2 & \Delta_7(h_4) \end{pmatrix} = 0.$$

Proof. For $k = 4$, Corollary 12.219 shows that the four lifted points

$$Q_v(h_j) = (y(h_j), x(h_j), \sigma_v(h_j))$$

are coplanar. This is equivalent to the vanishing of the affine 4×4 determinant with first column 1. The displayed reformulations follow by substituting the definitions of x, y, σ_v and multiplying rows by the indicated nonzero factors. $\square$

Remark 12.221 (Finite-core meaning of the lifted v_5 -patch geometry). The defect-free finite-core reduction now places the lifted v_5 -patch geometry into a sharp global context.

Indeed, by Theorem 12.215, Corollary 12.216, and Proposition 12.218, defect-free bad cores on a good v_5 -patch reduce to affine dependence of the lifted points

$$Q_v(h) = (y(h), x(h), \sigma_v(h)),$$

with effective ceiling

$$k \leq 4.$$

Thus the three low core sizes have the following exact lifted meanings:

$$\begin{aligned} k = 2 &\iff Q_v(h_1) = Q_v(h_2), \\ k = 3 &\iff Q_v(h_1), Q_v(h_2), Q_v(h_3) \text{ are collinear}, \\ k = 4 &\iff Q_v(h_1), \dots, Q_v(h_4) \text{ are coplanar}. \end{aligned}$$

So the lifted 3-point and 4-point analyses are not auxiliary side computations: they are exactly the $k = 3$ and $k = 4$ geometric layers in the finite-core program. In particular, once the lifted 3-point branch is globally rigid on good v_5 -patches, the remaining finite-core burden shifts upward to the two-point source-level merger problem and to the mixed/nonmixed lifted 4-point mechanisms.

Remark 12.222 (Parallel u_5 -patch version). There is a completely parallel local story on any connected interval in $\{c \neq 0, u_5 \neq 0\}$, obtained by choosing the local septic gauge $u_7 = 0$ instead of $v_7 = 0$. Then one defines

$$\tau_u(h) := -\frac{\Delta_7(h)}{c(h)u_5(h)}, \quad Q_u(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \tau_u(h) \right),$$

and the same defect-free affine-dependence conclusions hold with Q_v replaced by Q_u . Since the present one-pair geometry already uses

$$x(h) = \frac{v_5(h)}{c(h)}$$

as the distinguished ECT coordinate, the $v_5 \neq 0$ patch is the natural first setting and will be used below.

Proposition 12.223 (Centered quintic-patch dichotomy). *Assume the centered diagonal family is a genuine analytic branch in the underlying h -geometry, and that the centered scalar package from Remark 12.378 holds on that branch. Then, after shrinking, there exist one-sided punctured subarcs on which at least one of the following holds:*

(1)

$$c \neq 0, \quad M \neq 0, \quad v_5 \neq 0,$$

so the branch enters a good v_5 -patch;

(2)

$$c \neq 0, \quad u_5 \neq 0,$$

so the parallel u_5 -patch of Remark 12.222 applies.

In particular, the centered diagonal branch enters some quintic patch on one-sided punctured arcs.

Proof. By Corollary 12.382,

$$A_5^f(s) \neq 0,$$

so at least one of u_5 or v_5 is a nonzero analytic germ on the centered branch.

If $v_5 \neq 0$, then its zeros are isolated. Since

$$c(s) = C_0 + C_2 s^2 + \dots, \quad C_0 \neq 0,$$

we also have $c \neq 0$ near the basepoint. On any one-sided punctured arc where $v_5 \neq 0$, define

$$x = \frac{v_5}{c}.$$

This is a nonconstant analytic germ. If $M \equiv 0$ on such an arc, then

$$x' = \left(\frac{v_5}{c}\right)' = \frac{M}{c^2} \equiv 0,$$

so x would be constant, contradiction. Hence M is not identically zero on that arc, and by analyticity its zeros are isolated. After shrinking again, one obtains

$$c \neq 0, \quad M \neq 0, \quad v_5 \neq 0,$$

i.e. a good v_5 -patch.

If instead $v_5 \equiv 0$, then $u_5 \neq 0$. Its zeros are isolated, so after shrinking to a one-sided punctured arc one has

$$c \neq 0, \quad u_5 \neq 0,$$

and the u_5 -patch of Remark 12.222 applies. $\square$

Corollary 12.224 (Conditional punctured v_5 -patch promotion). *Under the hypotheses of Proposition 12.223, if*

$$v_5 \neq 0$$

on the centered diagonal branch, then after shrinking there exist one-sided punctured subarcs contained in

$$\{c \neq 0, M \neq 0, v_5 \neq 0\}.$$

Hence on such arcs

$$x = \frac{v_5}{c}$$

is automatically a valid local coordinate.

Proof. This is exactly the first branch of the proof of Proposition 12.223. $\square$

Proposition 12.225 (Linear independence of $1, u_5/c, v_5/c$). *Let J be a connected open interval on which*

$$c \neq 0, \quad M \neq 0.$$

Then the three functions

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c}$$

are linearly independent on J .

Proof. Assume there exist constants $A, B, C \in \mathbf{C}$, not all zero, such that

$$A + B \frac{u_5}{c} + C \frac{v_5}{c} \equiv 0 \quad \text{on } J.$$

Multiplying by c gives

$$A c + B u_5 + C v_5 \equiv 0.$$

If $B = 0$, then

$$A c + C v_5 \equiv 0,$$

so v_5/c is constant on J . But

$$\left(\frac{v_5}{c}\right)' = \frac{M}{c^2},$$

and $M \neq 0$ on J . This is impossible.

Hence $B \neq 0$, and after renormalization we may write

$$u_5 = \alpha v_5 + \beta c$$

for constants α, β .

Now isolate the graded slice corresponding to

$$(K_1, \eta_2, q^{(5)}).$$

By Lemmas 12.16 and 12.18, the tagged monomial

$$\kappa(h) \eta_2(h) q^{(5)}(h)$$

occurs with nonzero coefficient in $\text{tr } J_5^{\text{pair}}(h)$. Since

$$A_5^f(h) = u_5(h)I + v_5(h)S$$

and S is traceless, this means exactly that the same tagged monomial occurs in the I -coefficient $u_5(h)$.

It cannot occur in $c(h)$, since c is cubic and carries no quintic $q^{(5)}$ -slice. It also cannot occur in $v_5(h)$. Indeed, on the tagged slice, the only surviving quintic K_1 -transport channels are those already isolated in Lemma 12.18. For the channel

$$X_0 J X_4 - X_4 J X_0,$$

Lemma 12.17 shows that the relevant $(\eta_2, q^{(5)})$ -slice is diagonal, hence contributes only to the I -component. For the commutator channel

$$X_3 J X_1 - X_1 J X_3,$$

Proposition 12.40 gives

$$v_5^{(\text{comm})} = \gamma u - \alpha w.$$

On the tagged slice the η_2 -input in X_1 is purely off-diagonal, so $u = w = 0$, while the $q^{(5)}$ -input in X_3 is diagonal. Hence

$$v_5^{(\text{comm})} = 0$$

on that tagged slice.

Thus the monomial $\kappa \eta_2 q^{(5)}$ occurs in u_5 but in neither v_5 nor c , contradicting

$$u_5 = \alpha v_5 + \beta c.$$

This proves the claim. $\square$

Corollary 12.226 (Automatic u_5 -patch on a good $M \neq 0$ interval). *Let J be a connected open interval on which*

$$c \neq 0, \quad M \neq 0.$$

Then $u_5 \neq 0$ on J . Consequently, every connected component of

$$J \setminus u_5^{-1}(0)$$

is a valid local u_5 -patch in the sense of Remark 12.222.

Proof. If $u_5 \equiv 0$ on J , then the three functions

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c}$$

would be linearly dependent, contradicting Proposition 12.225. Thus $u_5 \neq 0$ on J , so its zeros are isolated and each connected component of $J \setminus u_5^{-1}(0)$ lies in $\{c \neq 0, u_5 \neq 0\}$. The parallel u_5 -patch geometry then follows from Remark 12.222. $\square$

Proposition 12.227 (Non-affinity of the lifted septic scalar). *Let J be a connected open interval contained in the quintic-generic region and in*

$$\{h : c(h) \neq 0, v_5(h) \neq 0\}.$$

Then there do not exist constants $A, B \in \mathbf{C}$ such that

$$\sigma_v(h) \equiv A x(h) + B \quad \text{on } J.$$

Equivalently,

$$\Delta_7(h) \not\equiv A v_5(h)^2 + B c(h) v_5(h) \quad \text{on } J.$$

Proof. If

$$\sigma_v(h) \equiv A x(h) + B,$$

then multiplying by $c(h)v_5(h)$ gives

$$\Delta_7(h) \equiv A v_5(h)^2 + B c(h) v_5(h).$$

The right-hand side is built only from the cubic and quintic layers and therefore has no direct septic $q^{(7)}$ -slice. But by Proposition 12.48, the direct $q^{(7)}$ -slice of Δ_7 is generically nonzero on J . This is a contradiction. $\square$

Definition 12.228 (Lifted derivative numerator and curvature numerator). On $\{c \neq 0, v_5 \neq 0\}$, define

$$P_v(h) := c(h)v_5(h)\Delta'_7(h) - \Delta_7(h)(c(h)v_5(h))'.$$

Then

$$\sigma'_v(h) = \frac{P_v(h)}{c(h)^2 v_5(h)^2}.$$

Define also the affine curvature numerator of the lifted planar curve

$$\Lambda_v(h) = (x(h), \sigma_v(h))$$

by

$$K_v(h) := x'(h)\sigma''_v(h) - \sigma'_v(h)x''(h).$$

Proposition 12.229 (Curvature bridge for the lifted v_5 -patch curve). On $\{c \neq 0, M \neq 0, v_5 \neq 0\}$,

$$\frac{d\sigma_v}{dx} = \frac{P_v}{M v_5^2},$$

and

$$K_v(h) = \frac{M(h)^2}{c(h)^4} \frac{d}{dh} \left(\frac{P_v(h)}{M(h) v_5(h)^2} \right).$$

In particular, if $K_v \equiv 0$ on a connected open interval $J \subset \{c \neq 0, M \neq 0, v_5 \neq 0\}$, then σ_v is affine in x on J , contradicting Proposition 12.227. Hence

$$K_v \neq 0$$

on every such interval.

Proof. Since

$$x = \frac{v_5}{c}, \quad x' = \frac{M}{c^2}, \quad \sigma'_v = \frac{P_v}{c^2 v_5^2},$$

we obtain

$$\frac{d\sigma_v}{dx} = \frac{\sigma'_v}{x'} = \frac{P_v}{M v_5^2}.$$

Differentiating with respect to h and multiplying by

$$x' = \frac{M}{c^2}$$

gives

$$K_v = x'^2 \frac{d}{dx} \left(\frac{d\sigma_v}{dx} \right) = \frac{M^2}{c^4} \frac{d}{dh} \left(\frac{P_v}{M v_5^2} \right).$$

If $K_v \equiv 0$ on a connected interval with $M \neq 0$, then

$$\frac{P_v}{M v_5^2}$$

is constant there, so $d\sigma_v/dx$ is constant and σ_v is affine in x . This contradicts Proposition 12.227. $\square$

Corollary 12.230 (Local defect-free three-point exclusion on lifted ECT subintervals). Let $J \subset \{c \neq 0, M \neq 0, v_5 \neq 0\}$ be a connected interval on which K_v has fixed nonzero sign. Then the triple

$$1, \quad x, \quad \sigma_v$$

is an ECT system on J . In particular, no affine line meets the lifted planar curve

$$\Lambda_v(h) = (x(h), \sigma_v(h))$$

in three points of J . Hence the defect-free three-point branch is excluded on such a subinterval.

Proof. The Wronskian of $1, x, \sigma_v$ is exactly K_v . If K_v has fixed nonzero sign on J , the ECT criterion applies. $\square$

Corollary 12.231 (Lifted order-7 Wronskian is not identically zero). *Let J be a connected open interval contained in the quintic-generic region and in*

$$\{h : c(h) \neq 0, M(h) \neq 0, v_5(h) \neq 0\}.$$

Then

$$T_v(h) := W_h\left(1, \frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)v_5(h)}\right)$$

is not identically zero on J .

Proof. Write

$$y := \frac{u_5}{c}, \quad x := \frac{v_5}{c}, \quad \sigma_v := \frac{\Delta_7}{c v_5}.$$

If $T_v \equiv 0$ on J , then by analyticity the four functions

$$1, y, x, \sigma_v$$

are locally linearly dependent on J , hence linearly dependent on all of the connected interval J . By Proposition 12.225, the first three are linearly independent, so there exist constants A, B, C such that

$$\sigma_v = A + By + Cx.$$

Multiplying by $c v_5$ gives

$$\Delta_7 = A c v_5 + B u_5 v_5 + C v_5^2.$$

The right-hand side is built entirely from the cubic/quintic layers and therefore carries no direct $q^{(7)}$ -slice. This contradicts Proposition 12.48, which gives

$$\Delta_7^{(q^{(7)})} \not\equiv 0$$

on J . $\square$

Proposition 12.232 (Exact x -coordinate identities for the lifted 4-point invariants). *On a good v_5 -patch, write*

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

Then:

(1) *the lifted planar curvature numerator satisfies*

$$K_v = (x')^3 S''(x);$$

(2) *the lifted 4-point Wronskian satisfies*

$$T_v = (x')^6 W_x(1, Y, x, S) = (x')^6 (S''Y''' - S'''Y'');$$

(3) *equivalently,*

$$T_v = (x')^6 S''^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right) = K_v^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right).$$

In particular, K_v and S'' have the same zero set and the same vanishing order on every good v_5 -patch.

Proof. By definition,

$$K_v = x' \sigma_v'' - \sigma_v' x''.$$

Since $S = \sigma_v$ as a function of the local coordinate x , we have

$$\sigma_v' = S'x', \quad \sigma_v'' = S''(x')^2 + S'x'',$$

hence

$$K_v = x' (S''(x')^2 + S'x'') - S'x'x'' = (x')^3 S''.$$

Next,

$$T_v = W_h(1, Y, x, S).$$

Under reparametrization $h \mapsto x(h)$, a 4×4 Wronskian gains the factor $(x')^{0+1+2+3} = (x')^6$, so

$$T_v = (x')^6 W_x(1, Y, x, S).$$

Now

$$W_x(1, Y, x, S) = \det \begin{pmatrix} 1 & Y & x & S \\ 0 & Y' & 1 & S' \\ 0 & Y'' & 0 & S'' \\ 0 & Y''' & 0 & S''' \end{pmatrix} = S''Y''' - S'''Y'',$$

which proves the second identity.

Finally,

$$S''Y''' - S'''Y'' = S''^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right),$$

and using $K_v = (x')^3 S''$ gives

$$(x')^6 S''^2 = K_v^2.$$

This yields the third identity. □

Proposition 12.233 (Canonical scalar invariants $\mathcal{Z}$ and $\mathcal{R}$ on a nonmixed lifted patch). *On a good v_5 -patch, write*

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

On the nonmixed subregion where $K_v \neq 0$, define

$$\mathcal{Z} := \frac{Y''}{S''}, \quad \mathcal{R} := \frac{d\mathcal{Z}}{dx}.$$

Then:

(1)

$$\mathcal{R} = \frac{T_v}{K_v^2};$$

(2) *under the affine gauge*

$$Y \mapsto Y + \alpha + \beta x + \gamma S,$$

one has

$$\mathcal{Z} \mapsto \mathcal{Z} + \gamma, \quad \mathcal{R} \mapsto \mathcal{R}.$$

Hence $\mathcal{R}$ is the canonical scalar invariant of the lifted 4-point branch on every nonmixed patch.

Proof. By Proposition 12.232,

$$T_v = (x')^6 S''^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right) = K_v^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right),$$

so

$$\mathcal{R} = \frac{d}{dx} \left(\frac{Y''}{S''} \right) = \frac{T_v}{K_v^2}.$$

For the gauge transformation, one has

$$(Y + \alpha + \beta x + \gamma S)'' = Y'' + \gamma S'',$$

hence

$$\mathcal{Z} \mapsto \frac{Y'' + \gamma S''}{S''} = \mathcal{Z} + \gamma.$$

Differentiating with respect to x gives $\mathcal{R} \mapsto \mathcal{R}$. □

Remark 12.234 (Below-lifted Wronskian-ratio reformulation). Let

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \frac{\Delta_7}{c v_5}, \quad \lambda := \frac{\Delta_7}{v_5}, \quad M := c v_5' - c' v_5.$$

For any scalar germ f , define

$$J_f(h) := \det \begin{pmatrix} c(h) & v_5(h) & f(h) \\ c'(h) & v_5'(h) & f'(h) \\ c''(h) & v_5''(h) & f''(h) \end{pmatrix}.$$

Then

$$\frac{d^2}{dx^2} \left(\frac{f}{c} \right) = \frac{c^3}{M^3} J_f.$$

Applying this to $f = u_5$ and $f = \lambda$ gives

$$Y'' = \frac{c^3}{M^3} J_{u_5}, \quad S'' = \frac{c^3}{M^3} J_\lambda,$$

hence

$$\mathcal{Z} := \frac{Y''}{S''} = \frac{J_{u_5}}{J_\lambda}.$$

Differentiating in the x -coordinate yields

$$\mathcal{R} := \mathcal{Z}' = \frac{T_v}{K_v^2} = \frac{c^2}{M} \frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right).$$

Thus the lifted 4-point branch may be viewed equivalently as a parity/divisibility problem for the Wronskian ratio J_{u_5}/J_λ . On the nonmixed side, however, the later source-level no-go results show that the ratio/Wronskian package by itself is not yet restrictive enough: the remaining burden is a genuine source-coupling theorem, not a generic Wronskian-ratio argument.

Proposition 12.235 (Source-level Wronskian for the nonmixed 4-point branch). *Define*

$$\mathcal{W}_{\text{src}} := J_\lambda J'_{u_5} - J_{u_5} J'_\lambda.$$

Then

$$\mathcal{R} = \frac{c^2}{M} \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

In particular, on every nonmixed arc on which $J_\lambda \neq 0$, the analytic scalars $\mathcal{R}$ and $\mathcal{W}_{\text{src}}$ differ only by a nonvanishing analytic factor. Hence:

- (1) $\mathcal{R} = 0 \iff \mathcal{W}_{\text{src}} = 0$;
- (2) $\text{ord } \mathcal{R} = \text{ord } \mathcal{W}_{\text{src}}$;
- (3) *the local parity classification of isolated zeros of $\mathcal{R}$ transfers verbatim to $\mathcal{W}_{\text{src}}$.*

Moreover, under the affine gauge change

$$u_5 \mapsto u_5 + \alpha c + \beta v_5 + \gamma \lambda$$

one has

$$J_{u_5} \mapsto J_{u_5} + \gamma J_\lambda,$$

and therefore

$$\mathcal{W}_{\text{src}} \mapsto \mathcal{W}_{\text{src}}.$$

So $\mathcal{W}_{\text{src}}$ is the canonical source-level gauge-invariant nonmixed scalar.

Proof. Differentiate the ratio

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}$$

from Remark 12.234:

$$\frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right) = \frac{J_\lambda J'_{u_5} - J_{u_5} J'_\lambda}{J_\lambda^2} = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

Substituting this into

$$\mathcal{R} = \frac{c^2}{M} \frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right)$$

gives the displayed identity.

On a nonmixed arc, $J_\lambda \neq 0$ and $c^2/M \neq 0$, so $\mathcal{R}$ and $\mathcal{W}_{\text{src}}$ differ by a nonvanishing analytic factor. This proves the zero-set and parity statements.

For gauge invariance, the terms αc and βv_5 do not change J_{u_5} , while $\gamma \lambda$ adds γJ_λ . Therefore

$$(J_{u_5} + \gamma J_\lambda)' = J'_{u_5} + \gamma J'_\lambda,$$

and

$$J_\lambda(J'_{u_5} + \gamma J'_\lambda) - (J_{u_5} + \gamma J_\lambda)J'_\lambda = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda. \quad \square$$

Remark 12.236 (Determinant-level parity on a centered good patch). Assume the centered scalar package from Remark 12.378 and suppose that $x = v_5/c$ is being used as the induced odd collision-side coordinate. Then

$$u_5, \lambda := \frac{\Delta_7}{v_5} \text{ are odd in } s, \quad c \text{ is even in } s, \quad v_5 \text{ is odd in } s.$$

Hence the rows of the defining determinants for J_{u_5} and J_λ have parities

$$(e, o, o), \quad (o, e, e), \quad (e, o, o),$$

so each determinant monomial is odd. Therefore

$$J_{u_5}(s) \text{ and } J_\lambda(s) \text{ are odd in } s.$$

Consequently,

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}$$

is even and

$$\mathcal{R} = \mathcal{Z}'$$

is odd, in agreement with the parity package obtained from the direct Y''/S'' -analysis.

Corollary 12.237 (Centered parity of the source-level Wronskian). *Under the hypotheses of Remark 12.236, the source-level Wronskian*

$$\mathcal{W}_{\text{src}}(s) = J_\lambda(s) J'_{u_5}(s) - J_{u_5}(s) J'_\lambda(s)$$

is odd in the centered variable s .

Consequently, any isolated centered zero of $\mathcal{W}_{\text{src}}$ has odd order if and only if it is a potentially branch-seeding zero for the nonmixed 4-point geometry; even-order centered zeros are locally sterile.

Proof. By Remark 12.236,

$$J_{u_5}(s) \text{ and } J_\lambda(s) \text{ are odd in } s.$$

Hence

$$J'_{u_5}(s) \text{ and } J'_\lambda(s) \text{ are even in } s.$$

Therefore

$$\mathcal{W}_{\text{src}}(s) = J_\lambda(s) J'_{u_5}(s) - J_{u_5}(s) J'_\lambda(s)$$

is odd in s . The final statement follows from Proposition 12.235, which transfers the parity classification from $\mathcal{R}$ to $\mathcal{W}_{\text{src}}$. $\square$

Corollary 12.238 (Mixed low-order exclusion on a centered good v_5 -patch). *Under the hypotheses of Remark 12.236, one has*

$$\mathcal{Z}(x) \text{ even}, \quad \mathcal{R}(x) \text{ odd}, \quad K_v(x) \text{ odd}.$$

Consequently,

$$(r, \text{ord}_0 \mathcal{Z}) \notin \{(1, 1), (1, 3), (2, 2)\}.$$

Equivalently, none of the explicit mixed low-order models

$$(m, n) = (3, 4), (3, 6), (4, 6)$$

can occur on such a centered good patch.

Proof. Since $\mathcal{Z}$ is even, its finite vanishing order must be even, so the cases

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3)$$

are impossible. Since K_v is odd, its vanishing order $r = \text{ord}_0 K_v$ must be odd, so

$$(r, \text{ord}_0 \mathcal{Z}) = (2, 2)$$

is impossible as well. $\square$

Theorem 12.239 (Local exhaustion of the mixed 4-point branch near the centered base). *Assume the centered scalar package from Remark 12.378, the right-pincer local reduction from Corollary 12.387, and the finite-jet scalar descent from Corollary 12.376. Then every sufficiently small nearby genuinely new mixed 4-point obstruction near the centered diagonal base is excluded.*

Equivalently, near the centered base the mixed 4-point branch is locally exhausted: the clean branch $V_1 \neq 0$ is excluded by the centered good- v_5 -patch parity package, while every late- v_5 branch is either locally sterile on the u_5 -patch or already absorbed into the defect-free planar quintic-circuit regime.

Proof. By Corollary 12.387, every sufficiently small nearby genuinely new mixed 4-point obstruction must lie on the clean branch

$$V_1 \neq 0, \quad \text{equivalently} \quad \text{ord}_s v_5 = 1.$$

On that branch, Corollary 12.376 gives

$$c(s) = C_0 + C_2 s^2 + C_4 s^4 + \dots$$

Since $v_5(s)$ is odd with $V_1 \neq 0$, the induced patch coordinate

$$x(s) = \frac{v_5(s)}{c(s)}$$

is odd and has odd inverse. Since $u_5(s)$ is odd and $\Delta_7(s)$ is even, it follows that

$$Y(x) = \frac{u_5}{c}(x) \quad \text{and} \quad S(x) = \frac{\Delta_7}{cv_5}(x)$$

are odd in x . Hence

$$\mathcal{Z}(x) = \frac{Y''}{S''} \text{ is even,} \quad K_v = (x')^3 S''$$

has odd vanishing order, so by Corollary 12.238 none of the explicit mixed low-order obstruction patterns

$$(r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}$$

can occur on the clean branch.

Thus no sufficiently small nearby genuinely new mixed 4-point obstruction remains. The late- v_5 branches were already removed by Corollary 12.387, and the clean branch is excluded as above. $\square$

Remark 12.240 (What the mixed local exhaustion theorem now accomplishes). Theorem 12.239 closes the mixed 4-point route locally near the centered diagonal base in a self-contained way: the clean branch is eliminated by the canonical scalar descent together with the centered good- v_5 -patch parity package, while every late- v_5 branch is already removed by the u_5 -patch right-pincer package as either locally sterile or already-quintic.

Remark 12.241 (What remains after local mixed exhaustion). Theorem 12.239 exhausts the mixed 4-point branch in the centered-base local regime. Any remaining mixed 4-point escape must therefore be genuinely global/nonlocal: it cannot come from a new local mixed septic-jet pattern near the centered diagonal base.

Equivalently, the later mixed/global problem is no longer a local normal-form problem. It is a finite-core / global propagation problem for the normalized line-incidence geometry, with the residual mixed two-point derivative geometry concentrated on the overlap exceptional set and, after the off- $K_v = 0$ handoffs, sharply focused on the exact fixed-shear $K_v = 0$ slice.

Remark 12.242 (How mixed local exhaustion constrains later hybrid routes). Theorem 12.239 is useful beyond the mixed local lane itself. It shows that any later hybrid argument trying to constrain the nonmixed repeated-source / source-coupling branch cannot appeal to a new local mixed 4-point septic jet pattern near the centered diagonal base, because that possibility is already exhausted. Thus any remaining mixed input capable of affecting the nonmixed branch must be genuinely global/nonlocal, or must enter through external canonical/multi-pair structure rather than through a new local mixed obstruction.

Remark 12.243 (A second below-lifted ratio does not add a new obstruction). Let

$$\Theta := \frac{W_{u_5}}{W},$$

where W denotes the corresponding projective-chart Wronskian in the $(c, v_5, \cdot)$ -frame. In the gauge $c \equiv 1$, $v_5 = x$, one has the exact relation

$$\Theta = \mathcal{Z} \frac{S''}{xS'' + 2S'}, \quad (\mathcal{Z} - x\Theta)S'' = 2\Theta S'.$$

So Θ does not provide an independent local obstruction beyond $(\mathcal{Z}, \mathcal{R})$: whenever $2\Theta/(\mathcal{Z} - x\Theta)$ is analytic, one can solve locally for S and then recover Y from $Y'' = \mathcal{Z}S''$. In particular, attacks based only on a generic differential relation between $\mathcal{Z}$ and Θ are too weak to close the 4-point branch.

Corollary 12.244 (Local defect-free four-point exclusion on lifted ECT subintervals). *Let $J \subset \{c \neq 0, M \neq 0, v_5 \neq 0\}$ be a connected interval on which T_v has fixed nonzero sign. Then the quadruple*

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c}, \quad \frac{\Delta_7}{c v_5}$$

is an ECT system on J . Equivalently, no affine plane meets the lifted space curve

$$Q_v(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h) v_5(h)} \right)$$

in four points of J . Hence the defect-free four-point branch is excluded on such a subinterval.

Proof. The Wronskian of the displayed quadruple is exactly T_v . If T_v has fixed nonzero sign on J , the ECT criterion applies. $\square$

Lemma 12.245 (Osculating determinant expansion for shrinking affine circuits). *Let $Q : I \rightarrow \mathbb{C}^d$ be real-analytic on an interval I , and define*

$$\Delta_{d+1}(t_0, \dots, t_d) := \det \begin{pmatrix} 1 & Q(t_0) \\ \vdots & \vdots \\ 1 & Q(t_d) \end{pmatrix}.$$

Fix $h_ \in I$, and let*

$$t_j = h_* + \varepsilon \lambda_j \quad (j = 0, \dots, d),$$

where the λ_j are pairwise distinct and $\varepsilon \rightarrow 0$. Then

$$\Delta_{d+1}(t_0, \dots, t_d) = \frac{\varepsilon^{d(d+1)/2}}{\prod_{k=1}^d k!} V(\lambda_0, \dots, \lambda_d) \det(Q'(h_*), Q''(h_*), \dots, Q^{(d)}(h_*)) + O(\varepsilon^{d(d+1)/2+1}),$$

where

$$V(\lambda_0, \dots, \lambda_d) := \prod_{0 \leq i < j \leq d} (\lambda_j - \lambda_i)$$

is the Vandermonde determinant.

In particular, if there is a sequence of exact affine $(d+1)$ -circuits

$$\Delta_{d+1}(t_{0,n}, \dots, t_{d,n}) = 0$$

with pairwise distinct nodes $t_{j,n} \rightarrow h_$, then*

$$\det(Q'(h_*), Q''(h_*), \dots, Q^{(d)}(h_*)) = 0.$$

Proof. Write the Taylor expansion

$$Q(h_* + \varepsilon\lambda) = Q(h_*) + \sum_{m=1}^d \frac{\varepsilon^m \lambda^m}{m!} Q^{(m)}(h_*) + O(\varepsilon^{d+1}),$$

uniformly for λ in bounded sets. Subtract the first row $(1, Q(t_0))$ from each of the remaining rows. The first column then becomes $(1, 0, \dots, 0)^\top$, so Δ_{d+1} reduces to the determinant of the $d \times d$ matrix whose j -th row ($j = 1, \dots, d$) is

$$Q(t_j) - Q(t_0) = \sum_{m=1}^d \frac{\varepsilon^m (\lambda_j^m - \lambda_0^m)}{m!} Q^{(m)}(h_*) + O(\varepsilon^{d+1}).$$

Factor the column vectors $Q'(h_*), \dots, Q^{(d)}(h_*)$. This gives

$$\Delta_{d+1}(t_0, \dots, t_d) = \det(Q'(h_*), \dots, Q^{(d)}(h_*)) \det\left(\frac{\varepsilon^m (\lambda_j^m - \lambda_0^m)}{m!}\right)_{1 \leq j, m \leq d} + O(\varepsilon^{d(d+1)/2+1}).$$

Pulling $\varepsilon^m/m!$ from the m -th column yields the factor

$$\frac{\varepsilon^{1+\dots+d}}{\prod_{k=1}^d k!} = \frac{\varepsilon^{d(d+1)/2}}{\prod_{k=1}^d k!}.$$

The remaining determinant is the standard Vandermonde determinant $V(\lambda_0, \dots, \lambda_d)$. This proves the expansion.

If $\Delta_{d+1}(t_{0,n}, \dots, t_{d,n}) = 0$ for a sequence of shrinking distinct-node circuits, divide by the nonzero Vandermonde factor and let $\varepsilon \rightarrow 0$. The leading term must vanish, giving

$$\det(Q'(h_*), Q''(h_*), \dots, Q^{(d)}(h_*)) = 0. \quad \square$$

Remark 12.246 (Planar curvature and lifted torsion as the first derivative obstructions). Lemma 12.245 shows that shrinking exact affine circuits are controlled, at first nontrivial order, by derivative-side osculating determinants. For a planar curve q , shrinking exact collinear triples force

$$\det(q', q'') = 0,$$

and for a space curve Q , shrinking exact coplanar quadruples force

$$\det(Q', Q'', Q''') = 0.$$

Thus the defect-free lifted 3-point and 4-point branches are not merely incidence problems: they are rooted at curvature- and torsion-type derivative invariants of the corresponding lifted curves.

Proposition 12.247 (Shrinking exact lifted 3-point circuits concentrate into $K_v = 0$). *Let $P \in \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact, and use*

$$x = \frac{v_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

Suppose there is a sequence of distinct triples

$$x_{1,n}, x_{2,n}, x_{3,n} \in x(P), \quad x_{1,n}, x_{2,n}, x_{3,n} \rightarrow x_*,$$

such that

$$\det \begin{pmatrix} 1 & x_{1,n} & S(x_{1,n}) \\ 1 & x_{2,n} & S(x_{2,n}) \\ 1 & x_{3,n} & S(x_{3,n}) \end{pmatrix} = 0 \quad \text{for every } n.$$

Then

$$K_v(x_*) = 0.$$

Proof. Apply Lemma 12.245 with $d = 2$ to the planar lifted curve

$$q(x) := (x, S(x)).$$

The exact collinearity condition in the statement is precisely the vanishing of

$$\det \begin{pmatrix} 1 & x_1 & S(x_1) \\ 1 & x_2 & S(x_2) \\ 1 & x_3 & S(x_3) \end{pmatrix}.$$

Hence shrinking exact lifted 3-point circuits force

$$\det(q'(x_*), q''(x_*)) = 0.$$

Since

$$q'(x) = (1, S'(x)), \quad q''(x) = (0, S''(x)),$$

one gets

$$\det(q'(x_*), q''(x_*)) = S''(x_*).$$

On the x -patch, Proposition 12.232 gives

$$K_v = (x')^3 S'',$$

and $x' \neq 0$ on a good v_5 -patch. Therefore $S''(x_*) = 0$ implies

$$K_v(x_*) = 0. \quad \square$$

Corollary 12.248 (Uniform small-diameter exclusion away from $K_v = 0$). *Let $P_0 \subseteq \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact and assume*

$$P_0 \cap \{K_v = 0\} = \emptyset.$$

Then there exists $\rho_3 > 0$ such that no exact lifted 3-point circuit contained in P_0 has x -diameter $< \rho_3$.

Proof. Otherwise there would exist a sequence of exact lifted 3-point circuits in P_0 with diameters tending to 0. By compactness, after passing to a subsequence the three nodes would converge to some $x_* \in x(P_0)$. This contradicts Proposition 12.247. $\square$

Proposition 12.249 (Shrinking exact lifted 4-point circuits concentrate into $T_v = 0$). *Let $P \subseteq \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact, and use*

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

Suppose there is a sequence of distinct quadruples

$$x_{1,n}, x_{2,n}, x_{3,n}, x_{4,n} \in x(P), \quad x_{1,n}, x_{2,n}, x_{3,n}, x_{4,n} \rightarrow x_*,$$

such that

$$\det \begin{pmatrix} 1 & Y(x_{1,n}) & x_{1,n} & S(x_{1,n}) \\ 1 & Y(x_{2,n}) & x_{2,n} & S(x_{2,n}) \\ 1 & Y(x_{3,n}) & x_{3,n} & S(x_{3,n}) \\ 1 & Y(x_{4,n}) & x_{4,n} & S(x_{4,n}) \end{pmatrix} = 0 \quad \text{for every } n.$$

Then

$$T_v(x_*) = 0.$$

Proof. Apply Lemma 12.245 with $d = 3$ to the lifted space curve

$$Q(x) := (Y(x), x, S(x)).$$

The exact coplanarity condition in the statement is precisely the vanishing of

$$\det \begin{pmatrix} 1 & Y(x_1) & x_1 & S(x_1) \\ 1 & Y(x_2) & x_2 & S(x_2) \\ 1 & Y(x_3) & x_3 & S(x_3) \\ 1 & Y(x_4) & x_4 & S(x_4) \end{pmatrix}.$$

Hence shrinking exact lifted 4-point circuits force

$$\det(Q'(x_*), Q''(x_*), Q'''(x_*)) = 0.$$

But

$$\det(Q'(x), Q''(x), Q'''(x)) = W_x(1, Y, x, S)(x),$$

and Proposition 12.232 gives

$$T_v = (x')^6 W_x(1, Y, x, S).$$

Since $x' \neq 0$ on a good v_5 -patch, it follows that

$$T_v(x_*) = 0. \quad \square$$

Corollary 12.250 (Uniform small-diameter exclusion away from $T_v = 0$). *Let $P_0 \in \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact and assume*

$$P_0 \cap \{T_v = 0\} = \emptyset.$$

Then there exists $\rho_4 > 0$ such that no exact lifted 4-point circuit contained in P_0 has x -diameter $< \rho_4$.

Proof. Otherwise there would exist a sequence of exact lifted 4-point circuits in P_0 with diameters tending to 0. By compactness, after passing to a subsequence the four nodes would converge to some $x_* \in x(P_0)$. This contradicts Proposition 12.249. $\square$

Definition 12.251 (Repeated-node tangent-contact determinants on a v_5 -patch). On a $v_5 \neq 0$ patch with $x = v_5/c$, $Y = u_5/c$, and $S = \Delta_7/(c v_5)$, define

$$H_3(a, u) := \det \begin{pmatrix} 1 & a & S(a) \\ 0 & 1 & S'(a) \\ 1 & u & S(u) \end{pmatrix} = S(u) - S(a) - S'(a)(u - a),$$

and

$$H_4(a, u, v) := \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 0 & Y'(a) & 1 & S'(a) \\ 1 & Y(u) & u & S(u) \\ 1 & Y(v) & v & S(v) \end{pmatrix}.$$

Thus $H_3(a, u) = 0$ means that $(u, S(u))$ lies on the tangent line to Λ_v at a , while $H_4(a, u, v) = 0$ means that $Q_v(u)$ and $Q_v(v)$ lie in the tangent plane to Q_v at a .

Proposition 12.252 (Repeated-node limits of exact lifted determinants). *For the exact lifted determinants*

$$\Delta_3(a, b, u) := \det \begin{pmatrix} 1 & a & S(a) \\ 1 & b & S(b) \\ 1 & u & S(u) \end{pmatrix},$$

and

$$\Delta_4(a, b, u, v) := \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 1 & Y(b) & b & S(b) \\ 1 & Y(u) & u & S(u) \\ 1 & Y(v) & v & S(v) \end{pmatrix},$$

one has

$$\frac{\Delta_3(a, b, u)}{b - a} \longrightarrow H_3(a, u) \quad (b \rightarrow a),$$

and

$$\frac{\Delta_4(a, b, u, v)}{b - a} \longrightarrow H_4(a, u, v) \quad (b \rightarrow a).$$

Proof. Subtract the first row from the second row in each determinant and divide by $b - a$. The resulting second row is the first difference quotient of the corresponding row entries, and the limit $b \rightarrow a$ gives the derivative row. $\square$

Theorem 12.253 (Local classification of the lifted 3-point branch near a finite-order K_v -zero). *Let $x = 0$ be an isolated finite-order zero of K_v on a good v_5 -patch. After affine normalization of $\Lambda_v = (x, S)$, write*

$$S(x) = x^m + O(x^{m+1}), \quad m = r + 2 \geq 3,$$

where $r = \text{ord}_0 K_v$. Then the repeated-node tangent-contact equation

$$H_3(a, u) = 0$$

with $a < 0 < u$ has the following local classification:

- (1) *if r is even (equivalently m is even), there is no opposite-side tangent-contact branch near $(a, u) = (0, 0)$;*
- (2) *if r is odd (equivalently m is odd), there exists a unique real-analytic opposite-side branch*

$$u = \Upsilon(a), \quad a < 0, \quad |a| \ll 1,$$

with asymptotic expansion

$$\Upsilon(a) = \lambda_m(-a) + O(a^2),$$

where $\lambda_m > 1$ is the unique positive root of

$$P_m(\lambda) := \lambda^m - m\lambda - (m-1).$$

Proof. Set $a = -\alpha$ with $\alpha > 0$, and write $u = \lambda\alpha$ with $\lambda > 0$. For the model germ $S(x) = x^m$, the tangent-contact equation is

$$H_3(-\alpha, \lambda\alpha) = \alpha^m \begin{cases} \lambda^m + m\lambda + (m-1), & m \text{ even,} \\ \lambda^m - m\lambda - (m-1), & m \text{ odd.} \end{cases}$$

If m is even, the displayed factor is strictly positive for every $\lambda > 0$, so no opposite-side branch exists.

If m is odd, define

$$P_m(\lambda) := \lambda^m - m\lambda - (m-1).$$

Then

$$P_m(1) = -2(m-1) < 0, \quad P_m(\lambda) \rightarrow +\infty \text{ as } \lambda \rightarrow +\infty,$$

and

$$P'_m(\lambda) = m(\lambda^{m-1} - 1)$$

is negative on $(0, 1)$ and positive on $(1, \infty)$. Therefore P_m has a unique positive root $\lambda_m > 1$.

For the perturbed germ $S(x) = x^m + O(x^{m+1})$, divide $H_3(-\alpha, \lambda\alpha)$ by α^m . The resulting equation is

$$P_m(\lambda) + O(\alpha) = 0.$$

If m is even, positivity of the model factor persists for α small, so there is still no branch. If m is odd, the root λ_m is simple, so the implicit function theorem yields a unique analytic branch

$$\lambda = \lambda(\alpha) = \lambda_m + O(\alpha),$$

equivalently

$$u = \lambda_m(-a) + O(a^2).$$

This proves the classification. □

Corollary 12.254 (One-sided exclusion for repeated-node tangent contact). *Under the hypotheses of Theorem 12.253, there exists $\varepsilon > 0$ such that*

$$H_3(a, u) \neq 0$$

for every same-side pair with

$$-\varepsilon < a < u < 0 \quad \text{or} \quad 0 < a < u < \varepsilon.$$

Equivalently, every sufficiently small off-diagonal solution of

$$H_3(a, u) = 0$$

is necessarily opposite-side.

Proof. Using

$$H_3(a, u) = S(u) - S(a) - S'(a)(u - a),$$

one has the exact integral identity

$$H_3(a, u) = \int_a^u (u - t) S''(t) dt.$$

Since $x = 0$ is an isolated finite-order zero of $K_v = (x')^3 S''$ on a good v_5 -patch, the analytic function S'' has constant nonzero sign on each punctured one-sided interval

$$(-\varepsilon, 0), \quad (0, \varepsilon)$$

after shrinking. If $a < u$ lie on the same side of 0, then

$$u - t > 0$$

for all $t \in (a, u)$, and $S''(t)$ has fixed nonzero sign on the whole interval. So the integrand has fixed nonzero sign, and therefore

$$H_3(a, u) \neq 0. \quad \square$$

Corollary 12.255 (No infinite-order flatness for K_v on a good v_5 -patch). *On a good v_5 -patch, every zero of K_v has finite order. Equivalently, the lifted curvature $S''(x)$ has no infinite-order zero on such a patch.*

Proof. By Proposition 12.232,

$$K_v = (x')^3 S''(x),$$

and on a good v_5 -patch one has $x' \neq 0$. Thus K_v and S'' have the same zero set and the same vanishing order.

If K_v had an infinite-order zero at some interior point x_* , then so would S'' . Since S'' is analytic in the local coordinate x , this would force

$$S'' \equiv 0$$

on a neighborhood of x_* . Hence S would be affine in x there, contrary to Proposition 12.227. $\square$

Lemma 12.256 (Slope–intercept parametrization for the lifted 3-point branch). *Assume the affine normalization from Theorem 12.253, so that*

$$S(0) = S'(0) = 0, \quad S'' < 0 \text{ on the source side } I_- = (\alpha, 0).$$

Let

$$I_+ = (0, \beta)$$

be the maximal open interval to the right of 0 on which S' is strictly increasing. Equivalently, S'' does not change sign on I_+ , and any zeros of S'' there are isolated even-order zeros.

For

$$m \in J_- := S'(I_-),$$

define the inverse source point

$$a = a(m) := (S'|_{I_-})^{-1}(m),$$

and the source tangent intercept

$$\tau_-(m) := S(a(m)) - m a(m).$$

For

$$m \in J_+ := S'(I_+),$$

define the inverse target tangent point

$$u_+(m) := (S'|_{I_+})^{-1}(m),$$

and the target tangent intercept

$$\tau_+(m) := S(u_+(m)) - m u_+(m).$$

Then

$$\tau'_-(m) = -a(m), \quad \tau'_+(m) = -u_+(m).$$

Hence on

$$J := J_- \cap J_+$$

the intercept gap

$$D(m) := \tau_-(m) - \tau_+(m)$$

satisfies

$$D'(m) = u_+(m) - a(m) > 0.$$

Proof. Differentiate

$$\tau_-(m) = S(a(m)) - m a(m)$$

to obtain

$$\tau'_-(m) = S'(a(m))a'(m) - a(m) - m a'(m) = -a(m),$$

because $S'(a(m)) = m$. The same computation gives

$$\tau'_+(m) = -u_+(m).$$

Subtracting yields

$$D'(m) = u_+(m) - a(m).$$

Since $a(m) < 0 < u_+(m)$, the derivative is strictly positive. $\square$

Theorem 12.257 (Global continuation / nonrecurrence of the odd-order lifted 3-point branch). *Assume the hypotheses and affine normalization of Theorem 12.253, and suppose that $r = \text{ord}_0 K_v$ is odd, so the unique local opposite-side branch exists.*

Let

$$I_- = (\alpha, 0)$$

be the source interval on which $S'' < 0$, and let

$$I_+ = (0, \beta)$$

be the maximal open interval to the right of 0 on which S' is strictly increasing. Set

$$J := S'(I_-) \cap S'(I_+).$$

Then there exists a unique maximal open interval

$$J_{\text{br}} = (0, m_*) \subset J$$

and a unique real-analytic function

$$u : J_{\text{br}} \rightarrow I_+$$

such that

$$H_3(a(m), u(m)) = 0, \quad a(m) := (S'|_{I_-})^{-1}(m), \quad u(m) > 0,$$

and for small $m > 0$ this branch agrees with the local branch from Theorem 12.253.

Moreover:

(1) the intercept gap

$$D(m) := \tau_-(m) - \tau_+(m)$$

is strictly positive on all of J , and

$$D'(m) = u_+(m) - a(m) > 0;$$

(2) along the branch one has

$$\partial_u H_3(a(m), u(m)) = S'(u(m)) - m \neq 0,$$

so there is no interior common-tangent event;

(3) the branch is uniquely continued on J_{br} , with

$$u'(m) = \frac{u(m) - a(m)}{S'(u(m)) - m},$$

and the source coordinate is strictly monotone,

$$a'(m) = \frac{1}{S''(a(m))} < 0;$$

(4) if $m_* < \sup J$, then

$$u(m) \longrightarrow \beta \quad (m \uparrow m_*).$$

Hence the branch cannot end in the interior of I_+ ; it can stop only when it reaches the boundary of the source/target slope-overlap block.

In particular, the odd-order lifted 3-point branch has no interior recurrence, no self-return, no interior accumulation, and no infinite-order flatness. Its only remaining endpoint mechanism is boundary exit of the relevant good-patch / monotone-slope block.

Proof. For

$$m \in J,$$

set

$$\Phi(m, u) := S(u) - m u - \tau_-(m).$$

Then

$$\Phi(m, u) = H_3(a(m), u).$$

Step 1: positivity of the intercept gap. By the odd-order local classification theorem, for all sufficiently small $m > 0$ there exists a point $u(m) \in I_+$ such that

$$\Phi(m, u(m)) = 0.$$

Fix such a small m . Since S' is strictly increasing on I_+ , the function

$$u \longmapsto \Phi(m, u)$$

has strictly increasing derivative

$$\partial_u \Phi(m, u) = S'(u) - m,$$

hence a unique critical point, namely $u_+(m)$, and that critical point is its unique minimum on I_+ . Therefore

$$\Phi(m, u_+(m)) < 0.$$

But

$$\Phi(m, u_+(m)) = S(u_+(m)) - m u_+(m) - \tau_-(m) = \tau_+(m) - \tau_-(m) = -D(m),$$

so $D(m) > 0$ for all sufficiently small $m > 0$.

By Lemma 12.256,

$$D'(m) = u_+(m) - a(m) > 0$$

on all of J . Hence $D(m) > 0$ for every $m \in J$.

Step 2: no interior common tangency. Suppose

$$\Phi(m, u) = 0$$

for some $m \in J$ and $u \in I_+$. If also

$$\partial_u \Phi(m, u) = S'(u) - m = 0,$$

then by uniqueness of the inverse point on I_+ one has $u = u_+(m)$, and thus

$$0 = \Phi(m, u_+(m)) = -D(m),$$

contrary to $D(m) > 0$. Therefore every branch point satisfies

$$\partial_u \Phi(m, u) \neq 0.$$

Equivalently, there is no interior common-tangent point on the odd-order branch.

Step 3: uniqueness and analytic continuation. For fixed $m \in J$, the derivative

$$u \longmapsto \partial_u \Phi(m, u) = S'(u) - m$$

is strictly increasing on I_+ , and

$$\partial_u \Phi(m, 0) = S'(0) - m = -m < 0.$$

Hence $\Phi(m, \cdot)$ first decreases and then increases, so it has at most one zero on I_+ . Thus the local branch from Theorem 12.253 is unique on every connected continuation interval.

Let

$$E := \{m \in J : \exists u \in I_+ \text{ with } \Phi(m, u) = 0\}.$$

By Step 2, every zero is simple in the u -variable, so the implicit function theorem shows that E is open and that the zero depends real-analytically on m . Let

$$J_{\text{br}} = (0, m_*)$$

be the connected component of E containing all sufficiently small positive m . This gives the unique maximal analytic continuation

$$u = u(m), \quad m \in J_{\text{br}}.$$

Differentiating

$$\Phi(m, u(m)) = 0$$

gives

$$(S'(u(m)) - m)u'(m) - u(m) + a(m) = 0,$$

hence

$$u'(m) = \frac{u(m) - a(m)}{S'(u(m)) - m}.$$

Also

$$a'(m) = \frac{1}{S''(a(m))} < 0$$

because $S'' < 0$ on the source interval I_- . So the source coordinate is strictly monotone and cannot return to a previous value.

Step 4: endpoint control. Assume that

$$m_* < \sup J$$

but that $u(m)$ does not approach β as $m \uparrow m_*$. Then there exists a compact subinterval

$$K \subseteq I_+$$

and a sequence $m_n \uparrow m_*$ with

$$u(m_n) \in K.$$

After passing to a subsequence,

$$u(m_n) \rightarrow u_* \in K.$$

By continuity,

$$\Phi(m_*, u_*) = 0.$$

Since $m_* < \sup J$, one has $m_* \in J$, so by Step 2,

$$\partial_u \Phi(m_*, u_*) \neq 0.$$

The implicit function theorem therefore extends the branch analytically past m_* , contradicting maximality of J_{br} .

Hence

$$u(m) \rightarrow \beta \quad (m \uparrow m_*).$$

This proves that the branch cannot terminate by an interior fold, interior common tangency, or interior accumulation.

Finally, by Corollary 12.255, every zero of K_v has finite order. Since even-order zeros do not change the sign of S'' , they do not break the strict increase of S' on I_+ . Therefore the endpoint β can only be the ambient boundary of the good patch or the first odd-order zero of K_v to the right. $\square$

Remark 12.258 (Structural package for the global lifted 3-point branch). The correct global parameter on the odd-order lifted 3-point branch is the source tangent slope

$$m = S'(a).$$

In that parameter, the branch is governed by the source/target tangent intercepts

$$\tau_-(m) = S(a(m)) - m a(m), \quad \tau_+(m) = S(u_+(m)) - m u_+(m),$$

and their gap

$$D(m) = \tau_-(m) - \tau_+(m), \quad D'(m) = u_+(m) - a(m) > 0.$$

Thus interior common tangency is exactly the equation

$$D(m) = 0,$$

but this never occurs on the odd-order continuation branch because $D > 0$ there. So the global 3-point problem is not a hidden recurrence problem inside the patch: once born at an odd-order K_v -zero, the branch is a one-parameter analytic graph with monotone source coordinate, and even-order K_v -zeros are invisible to it except as isolated points where S'' vanishes without changing the monotone-slope block.

Remark 12.259 (Target-block form of the intercept-gap monotonicity). The slope–intercept monotonicity from Lemma 12.256 is not special to the first right-hand target block.

Let I be any open interval to the right of the source interval I_- on which S' is monotone and hence invertible. For

$$m \in S'(I_-) \cap S'(I),$$

define

$$u_I(m) := (S'|_I)^{-1}(m), \quad \tau_I(m) := S(u_I(m)) - m u_I(m),$$

and

$$D_I(m) := \tau_-(m) - \tau_I(m).$$

Then exactly the same envelope calculation gives

$$\tau'_I(m) = -u_I(m), \quad D'_I(m) = u_I(m) - a(m) > 0.$$

Thus for any monotone-slope target block, interior common tangency against the fixed source interval I_- is again the scalar equation

$$D_I(m) = 0.$$

So any future continuation across successive odd-order K_v -zeros must be organized blockwise through these monotone intercept-gap functions rather than through a hidden recurrence mechanism inside a block.

Remark 12.260 (Derivative-geometric meaning of the lifted 3-point branch). The lifted 3-point branch is now best viewed as a derivative-geometric propagation theorem for the planar lifted curve

$$\Lambda_v = (x, S).$$

By Lemma 12.245 and Proposition 12.247, shrinking exact lifted 3-point circuits are detected by the osculating determinant and therefore concentrate into the lifted curvature-zero set

$$K_v = (x')^3 S''(x) = 0.$$

Theorem 12.253 then identifies the local repeated-node branch structure at those curvature zeros, and Theorem 12.257 shows that the odd-order branch propagates globally on each monotone-slope target block by the slope–intercept envelope identities

$$\tau'_-(m) = -a(m), \quad \tau'_+(m) = -u_+(m), \quad D'(m) = u_+(m) - a(m) > 0.$$

So the lifted 3-point theory is no longer an ad hoc repeated-node analysis. It is the global propagation of an osculating-contact defect for the curve Λ_v , organized by the derivative geometry of its tangent lines and curvature.

Remark 12.261 (Exact nonmixed reduction in terms of $\mathcal{Z}$ and $\mathcal{R}$). At a nonmixed lifted point

$$K_v(0) \neq 0, \quad T_v(0) = 0,$$

Propositions 12.232 and 12.233 give

$$\mathcal{Z} = \frac{Y''}{S''}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}.$$

Since $K_v(0) \neq 0$, one has $S''(0) \neq 0$, so after analytic reparametrization and affine target normalization preserving coplanarity one may write the lifted germ as

$$Q_v(\xi) = (\tilde{Y}(\xi), \xi, \xi^2), \quad \tilde{Y}(0) = \tilde{Y}'(0) = \tilde{Y}''(0) = 0.$$

In these normalized coordinates,

$$\mathcal{Z}(\xi) = \frac{1}{2} \tilde{Y}''(\xi), \quad \mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi).$$

Thus the nonmixed lifted 4-point problem is exactly the jet problem of excluding odd-order zeros of $\mathcal{R}$. Equivalently, it is the problem of excluding even leading degrees of $\tilde{Y}$ in the normalized chart.

In particular, the low-order dangerous nonmixed cases are exactly:

$$\begin{aligned} \text{ord}_0 \mathcal{R} = 1 &\iff \tilde{Y}'''(0) = 0, \tilde{Y}^{(4)}(0) \neq 0, \\ \text{ord}_0 \mathcal{R} = 3 &\iff \tilde{Y}^{(3)}(0) = \tilde{Y}^{(4)}(0) = \tilde{Y}^{(5)}(0) = 0, \tilde{Y}^{(6)}(0) \neq 0, \\ \text{ord}_0 \mathcal{R} = 5 &\iff \tilde{Y}^{(3)}(0) = \dots = \tilde{Y}^{(7)}(0) = 0, \tilde{Y}^{(8)}(0) \neq 0. \end{aligned}$$

Equivalently, the dangerous nonmixed low-order list is

$$\text{ord}_0 \mathcal{Z} \in \{2, 4, 6\} \iff \text{ord}_0 \mathcal{R} \in \{1, 3, 5\}.$$

So the nonmixed 4-point endgame has been reduced to a finite jet problem for the canonical invariant $\mathcal{R}$.

Lemma 12.262 (Repeated-node tangent-plane determinant in the normalized nonmixed chart). *In the normalized nonmixed chart of Remark 12.261, define*

$$\tilde{H}_4(a, u, v) := \det \begin{pmatrix} 1 & \tilde{Y}(a) & a & a^2 \\ 0 & \tilde{Y}'(a) & 1 & 2a \\ 1 & \tilde{Y}(u) & u & u^2 \\ 1 & \tilde{Y}(v) & v & v^2 \end{pmatrix}.$$

Then

$$\tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2 \left(\tilde{Y}[a, a, v] - \tilde{Y}[a, a, u] \right),$$

where

$$\tilde{Y}[a, a, t] := \frac{\tilde{Y}(t) - \tilde{Y}(a) - \tilde{Y}'(a)(t - a)}{(t - a)^2}$$

is the repeated-node second divided difference. Equivalently, for $u \neq v$,

$$\tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2(v - u) \tilde{Y}[a, a, u, v].$$

Proof. Subtract the first row from the third and fourth rows. Since

$$u^2 - a^2 = (u - a)(u + a), \quad v^2 - a^2 = (v - a)(v + a),$$

this gives

$$\tilde{H}_4(a, u, v) = (u - a)(v - a) \det \begin{pmatrix} 1 & \tilde{Y}(a) & a & a^2 \\ 0 & \tilde{Y}'(a) & 1 & 2a \\ 0 & \frac{\tilde{Y}(u) - \tilde{Y}(a)}{u - a} & 1 & u + a \\ 0 & \frac{\tilde{Y}(v) - \tilde{Y}(a)}{v - a} & 1 & v + a \end{pmatrix}.$$

Expanding along the first column yields

$$\tilde{H}_4(a, u, v) = (u - a)(v - a) \det \begin{pmatrix} \tilde{Y}'(a) & 1 & 2a \\ \frac{\tilde{Y}(u) - \tilde{Y}(a)}{u - a} & 1 & u + a \\ \frac{\tilde{Y}(v) - \tilde{Y}(a)}{v - a} & 1 & v + a \end{pmatrix}.$$

Now subtract the first row from the second and third rows:

$$\tilde{H}_4(a, u, v) = (u - a)(v - a) \det \begin{pmatrix} \tilde{Y}'(a) & 1 & 2a \\ (u - a)\tilde{Y}[a, a, u] & 0 & u - a \\ (v - a)\tilde{Y}[a, a, v] & 0 & v - a \end{pmatrix}.$$

Expanding along the middle column gives

$$\tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2 \left(\tilde{Y}[a, a, v] - \tilde{Y}[a, a, u] \right),$$

which is the first formula. Since

$$\tilde{Y}[a, a, u, v] = \frac{\tilde{Y}[a, a, v] - \tilde{Y}[a, a, u]}{v - u},$$

the second formula follows for $u \neq v$. □

Proposition 12.263 (Exact nonmixed tangent-plane contacts force nearby zeros of $\mathcal{R}$). *In the normalized nonmixed chart of Remark 12.261, assume $a, u, v \in \mathbf{R}$ are distinct and satisfy*

$$\tilde{H}_4(a, u, v) = 0.$$

Then there exists

$$w \in \text{conv}\{a, u, v\}$$

such that

$$\mathcal{R}(w) = 0.$$

Equivalently,

$$\tilde{Y}'''(w) = 0.$$

Proof. By Lemma 12.262, since a, u, v are distinct,

$$0 = \tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2(v - u) \tilde{Y}[a, a, u, v].$$

Hence

$$\tilde{Y}[a, a, u, v] = 0.$$

By the Hermite–Genocchi formula for repeated-node divided differences, there exists

$$w \in \text{conv}\{a, u, v\}$$

such that

$$\tilde{Y}[a, a, u, v] = \frac{\tilde{Y}'''(w)}{3!}.$$

Therefore

$$\tilde{Y}'''(w) = 0.$$

In the normalized nonmixed chart one has

$$\mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi),$$

so $\mathcal{R}(w) = 0$. □

Corollary 12.264 (Sign-definite nonmixed intervals are exactly-contact sterile). *Let J be an open interval in the normalized nonmixed chart such that*

$$\mathcal{R} \text{ has fixed nonzero sign on } J.$$

Then there do not exist distinct points $a, u, v \in J$ such that

$$\tilde{H}_4(a, u, v) = 0.$$

Equivalently, there are no exact nonmixed tangent-plane contacts with all three nodes in J .

Proof. If such a, u, v existed, Proposition 12.263 would produce some

$$w \in \text{conv}\{a, u, v\} \subset J$$

with $\mathcal{R}(w) = 0$, contradicting the sign assumption. $\square$

Remark 12.265 (Derivative-side interpretation of exact nonmixed contacts). Proposition 12.263 strengthens the shrinking-circuit statement Proposition 12.249. It is not merely a limit theorem: every exact nonmixed tangent-plane contact already forces a zero of the canonical derivative invariant inside the convex hull of the three nodes,

$$\mathcal{R} = \frac{1}{2} \tilde{Y}'''.$$

Thus the nonmixed 4-point branch is controlled not only asymptotically by $T_v = 0$, but pointwise by the zero set of a single analytic scalar.

Proposition 12.266 (Interval-persistent nonmixed 4-point degeneracy forces affine planarity). *Let J be a connected open interval in a good v_5 -patch such that*

$$K_v \neq 0 \quad \text{on } J.$$

Assume

$$T_v \equiv 0 \quad \text{on } J.$$

Then on J one has

$$\mathcal{R} \equiv 0, \quad \mathcal{Z} \equiv \gamma$$

for some constant $\gamma \in \mathbf{C}$, and hence

$$Y = \gamma S + ax + b$$

for some constants $a, b \in \mathbf{C}$. Equivalently, the lifted curve

$$x \mapsto (Y(x), x, S(x))$$

lies in an affine plane on J .

In particular, any interval-persistent nonmixed exact lifted 4-point branch collapses to an affine-planar residual rather than remaining genuinely three-dimensional.

Proof. On the nonmixed interval J , Proposition 12.233 gives

$$\mathcal{R} = \frac{T_v}{K_v^2}.$$

Since $T_v \equiv 0$ and $K_v \neq 0$ on J , it follows that

$$\mathcal{R} \equiv 0 \quad \text{on } J.$$

By definition,

$$\mathcal{R} = \frac{d\mathcal{Z}}{dx},$$

so $\mathcal{Z}$ is constant on J :

$$\mathcal{Z} \equiv \gamma.$$

Using

$$\mathcal{Z} = \frac{Y''}{S''},$$

we obtain

$$Y'' = \gamma S''.$$

Integrating twice with respect to x yields

$$Y = \gamma S + ax + b$$

for constants $a, b \in \mathbf{C}$. Therefore the lifted curve satisfies the affine linear equation

$$Y - \gamma S - ax - b = 0$$

throughout J , so it lies in an affine plane. $\square$

Corollary 12.267 (Recurrent exact nonmixed contacts force affine-planar collapse). *Let J be a connected open interval in a good v_5 -patch such that*

$$K_v \neq 0 \quad \text{on } J.$$

Assume that every nonempty open subinterval $I \subset J$ contains an exact nonmixed tangent-plane contact triple. Equivalently, after local nonmixed normalization on I , there exist distinct points $a, u, v \in I$ such that

$$\tilde{H}_4(a, u, v) = 0.$$

Then

$$\mathcal{R} \equiv 0 \quad \text{on } J,$$

hence

$$T_v \equiv 0 \quad \text{on } J,$$

and therefore the lifted curve is affine-planar on J :

$$Y = \gamma S + ax + b$$

for some constants $\gamma, a, b \in \mathbf{C}$.

Proof. Let $I \subset J$ be any nonempty open subinterval. By assumption, I contains an exact nonmixed tangent-plane contact triple. After nonmixed normalization on I , Proposition 12.263 produces a point $w_I \in I$ such that

$$\mathcal{R}(w_I) = 0.$$

Thus every nonempty open subinterval of J contains a zero of $\mathcal{R}$. Since $\mathcal{R}$ is analytic, its zero set cannot have an accumulation point in J unless

$$\mathcal{R} \equiv 0 \quad \text{on } J.$$

Because $K_v \neq 0$ on J and

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

it follows that

$$T_v \equiv 0 \quad \text{on } J.$$

Now apply Proposition 12.266 to conclude that

$$Y = \gamma S + ax + b$$

on J . $\square$

Lemma 12.268 (Derivative formula for the repeated-node nonmixed profile). *In the normalized nonmixed chart, define*

$$D_a(t) := \tilde{Y}[a, a, t] = \frac{\tilde{Y}(t) - \tilde{Y}(a) - \tilde{Y}'(a)(t - a)}{(t - a)^2}, \quad D_a(a) := \frac{1}{2} \tilde{Y}''(a).$$

Then $D_a(t)$ extends real-analytically across $t = a$, and

$$D_a(t) = \int_0^1 (1 - s) \tilde{Y}''(a + s(t - a)) ds.$$

Moreover,

$$D'_a(t) = \int_0^1 s(1 - s) \tilde{Y}'''(a + s(t - a)) ds = 2 \int_0^1 s(1 - s) \mathcal{R}(a + s(t - a)) ds.$$

Proof. Taylor's integral formula gives

$$\tilde{Y}(t) = \tilde{Y}(a) + \tilde{Y}'(a)(t-a) + (t-a)^2 \int_0^1 (1-s) \tilde{Y}''(a+s(t-a)) ds.$$

Dividing by $(t-a)^2$ proves the first formula for $t \neq a$, and the analytic extension across $t = a$ is immediate. Differentiating under the integral sign gives

$$D'_a(t) = \int_0^1 s(1-s) \tilde{Y}'''(a+s(t-a)) ds.$$

Since $\mathcal{R} = \frac{1}{2} \tilde{Y}'''$ in the normalized nonmixed chart, the last formula follows. $\square$

Proposition 12.269 (Parity classification of isolated nonmixed zeros of $\mathcal{R}$). *Let $\xi_0 \in \mathbf{R}$ be an isolated zero of the nonmixed invariant*

$$\mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi),$$

and write

$$\mathcal{R}(\xi) = (\xi - \xi_0)^m \rho(\xi), \quad m \geq 1, \quad \rho(\xi_0) \neq 0.$$

(i) Even-order sterility. *If m is even, then there exists a neighborhood U of ξ_0 such that for every $a \in U$, the function*

$$D_a(t) := \tilde{Y}[a, a, t]$$

is strictly monotone on U . In particular, there do not exist distinct $a, u, v \in U$ such that

$$\tilde{H}_4(a, u, v) = 0.$$

(ii) Odd-order birth. *If m is odd, then there exists a neighborhood U of ξ_0 and a real-analytic function $\tau : U \rightarrow U$ such that*

$$\tau(\xi_0) = \xi_0, \quad \tau'(\xi_0) = -1, \quad D'_a(\tau(a)) = 0 \quad (a \in U).$$

Moreover,

$$D''_a(\tau(a)) \neq 0 \quad (a \in U \setminus \{\xi_0\}),$$

so for each $a \in U \setminus \{\xi_0\}$, the function D_a has a strict local extremum at $t = \tau(a)$. Consequently, for each such a , there exist distinct points $u_-(a), u_+(a) \in U$ with

$$u_-(a) < \tau(a) < u_+(a), \quad D_a(u_-(a)) = D_a(u_+(a)),$$

and hence

$$\tilde{H}_4(a, u_-(a), u_+(a)) = 0.$$

Proof. By translation in the normalized chart, we may assume $\xi_0 = 0$. Thus

$$\mathcal{R}(\xi) = \xi^m \rho(\xi), \quad \rho(0) \neq 0.$$

For the even-order case, shrink to a neighborhood U on which ρ has fixed nonzero sign. Since m is even, $\xi^m \geq 0$, so $\mathcal{R}$ has fixed sign on U , vanishing only at 0. By Lemma 12.268,

$$D'_a(t) = 2 \int_0^1 s(1-s) \mathcal{R}(a+s(t-a)) ds.$$

For $a \neq t$, the affine segment $a+s(t-a)$ is not identically zero, so the integrand has fixed sign and is not identically zero. Hence $D'_a(t)$ has fixed nonzero sign for all $a \neq t$ in U , and therefore each D_a is strictly monotone on U . Thus D_a is injective on U , and by Lemma 12.262 there are no exact contacts $\tilde{H}_4(a, u, v) = 0$ with distinct $a, u, v \in U$.

Now assume m is odd. Write $a = h$ and $t = \lambda h$. Then Lemma 12.268 gives

$$D'_h(\lambda h) = 2h^m G(h, \lambda),$$

where

$$G(h, \lambda) := \int_0^1 s(1-s)(1+s(\lambda-1))^m \rho(h(1+s(\lambda-1))) ds.$$

This is real-analytic in (h, λ) , and

$$G(0, \lambda) = \rho(0) J_m(\lambda), \quad J_m(\lambda) := \int_0^1 s(1-s)(1+s(\lambda-1))^m ds.$$

Since m is odd,

$$J_m(-1) = \int_0^1 s(1-s)(1-2s)^m ds = 0$$

by the symmetry $s \mapsto 1-s$, while

$$J'_m(-1) = m \int_0^1 s^2(1-s)(1-2s)^{m-1} ds > 0$$

because $m-1$ is even. Thus

$$\partial_\lambda G(0, -1) = \rho(0) J'_m(-1) \neq 0.$$

By the implicit function theorem, there exists a unique real-analytic function $\lambda(h)$ near 0 such that

$$\lambda(0) = -1, \quad G(h, \lambda(h)) = 0.$$

Define

$$\tau(h) := \lambda(h)h.$$

Then

$$\tau(0) = 0, \quad \tau'(0) = \lambda(0) = -1,$$

and

$$D'_h(\tau(h)) = 0.$$

It remains to show that this critical point is a strict local extremum. By Lemma 12.268,

$$D''_a(t) = 2 \int_0^1 s^2(1-s) \mathcal{R}'(a+s(t-a)) ds.$$

Substituting $a = h$, $t = \lambda(h)h$, and using

$$\mathcal{R}'(\xi) = m\xi^{m-1}\rho(\xi) + \xi^m\rho'(\xi),$$

one obtains

$$D''_h(\tau(h)) = 2h^{m-1}\rho(0) J'_m(-1) + O(h^m).$$

Since $\rho(0)J'_m(-1) \neq 0$, it follows that

$$D''_h(\tau(h)) \neq 0$$

for $h \neq 0$ sufficiently small. Therefore $t = \tau(h)$ is a strict local extremum of D_h . Hence there exist distinct points $u_-(h)$, $u_+(h)$ arbitrarily close to $\tau(h)$ such that

$$D_h(u_-(h)) = D_h(u_+(h)).$$

By Lemma 12.262,

$$\tilde{H}_4(h, u_-(h), u_+(h)) = (h - u_-(h))^2(h - u_+(h))^2(D_h(u_+(h)) - D_h(u_-(h))) = 0.$$

This proves the odd-order birth statement. $\square$

Remark 12.270 (Reflection law for odd-order birth). In the odd-order case of Proposition 12.269, the critical-point branch satisfies

$$\tau(a) = 2\xi_0 - a + O((a - \xi_0)^2).$$

Thus the local exact nonmixed contact family born at any odd-order zero of $\mathcal{R}$ has the same first-order reflection geometry across that zero as in the simple-zero case.

Proposition 12.271 (Odd-zero critical branch is uniformly unrooted). *Let ξ_0 be an isolated odd-order zero of*

$$\mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi),$$

and, after translation, write

$$\xi_0 = 0, \quad \mathcal{R}(\xi) = \xi^m \rho(\xi), \quad m \geq 1 \text{ odd}, \quad \rho(0) \neq 0.$$

Let

$$D_a(t) := \tilde{Y}[a, a, t], \quad D_a(a) = \frac{1}{2} \tilde{Y}''(a) = \mathcal{Z}(a).$$

Then there exists $\varepsilon > 0$ and a unique real-analytic function

$$\mu : (-\varepsilon, \varepsilon) \rightarrow \mathbf{R}, \quad \mu(0) = 0,$$

such that for every sufficiently small $h > 0$ the unique nontrivial critical pair

$$a(h) := h(\mu(h) - 1), \quad x(h) := h(\mu(h) + 1)$$

satisfies

$$D'_{a(h)}(x(h)) = 0,$$

and every critical pair

$$a < x, \quad |a| + |x| < \varepsilon, \quad D'_a(x) = 0$$

is of this form.

Moreover,

$$D_{a(h)}(x(h)) - D_{a(h)}(a(h)) = -\frac{\rho(0)}{m+2} h^{m+1} + O(h^{m+2}).$$

Equivalently,

$$D_{a(h)}(x(h)) - D_{a(h)}(a(h)) = -\frac{\rho(0)}{2^{m+1}(m+2)} (x(h) - a(h))^{m+1} + o((x(h) - a(h))^{m+1}).$$

In particular, after shrinking ε if needed, every nontrivial nearby critical pair satisfies

$$D_a(x) \neq D_a(a),$$

indeed

$$\operatorname{sgn}(D_a(x) - D_a(a)) = -\operatorname{sgn} \rho(0).$$

So no sheared repeated-node tangent-contact event can occur in a sufficiently small neighborhood of an isolated odd zero of $\mathcal{R}$.

Proof. Write every nearby pair as

$$a = c - h, \quad x = c + h, \quad h = \frac{x - a}{2} > 0, \quad \mu = \frac{c}{h}.$$

By Lemma 12.268,

$$D'_a(x) = 2 \int_0^1 s(1-s) \mathcal{R}(a + s(x-a)) ds.$$

With the change of variables $u = 2s - 1$, this becomes

$$D'_{c-h}(c+h) = \frac{1}{4} \int_{-1}^1 (1-u^2) \mathcal{R}(c+hu) du = h^m F(h, \mu),$$

where

$$F(h, \mu) := \frac{1}{4} \int_{-1}^1 (1-u^2)(\mu+u)^m \rho(h(\mu+u)) du.$$

This is real-analytic in (h, μ) near $(0, 0)$. At $h = 0$,

$$F(0, \mu) = \frac{\rho(0)}{4} K_m(\mu), \quad K_m(\mu) := \int_{-1}^1 (1-u^2)(\mu+u)^m du.$$

Since m is odd,

$$K_m(0) = 0,$$

and

$$K'_m(\mu) = m \int_{-1}^1 (1-u^2)(\mu+u)^{m-1} du > 0$$

for all real μ , because $m-1$ is even. Hence $\mu = 0$ is the unique real root of K_m , and it is simple. Therefore

$$\partial_\mu F(0,0) \neq 0,$$

so the implicit function theorem gives a unique real-analytic branch

$$\mu = \mu(h), \quad \mu(0) = 0,$$

parameterizing all nearby critical pairs.

Now write

$$\mathcal{Z}(\xi) = \mathcal{Z}(0) + \xi^{m+1}\eta(\xi), \quad \eta(0) = \frac{\rho(0)}{m+1},$$

since $\mathcal{Z}' = \mathcal{R} = \xi^m \rho(\xi)$. Using

$$D_a(x) - D_a(a) = \frac{1}{2} \int_{-1}^1 (1-u)(\mathcal{Z}(c+hu) - \mathcal{Z}(c-h)) du,$$

we obtain

$$D_{c-h}(c+h) - D_{c-h}(c-h) = h^{m+1}G(h, \mu),$$

where

$$G(h, \mu) := \frac{1}{2} \int_{-1}^1 (1-u) \left((\mu+u)^{m+1} \eta(h(\mu+u)) - (\mu-1)^{m+1} \eta(h(\mu-1)) \right) du.$$

Again G is real-analytic. Evaluating at $(0,0)$ gives

$$G(0,0) = \frac{\eta(0)}{2} \int_{-1}^1 (1-u)(u^{m+1} - 1) du = -\frac{\rho(0)}{m+2} \neq 0.$$

Therefore along the critical branch,

$$D_{a(h)}(x(h)) - D_{a(h)}(a(h)) = h^{m+1}G(h, \mu(h)) = -\frac{\rho(0)}{m+2}h^{m+1} + O(h^{m+2}),$$

which has fixed nonzero sign for h sufficiently small. This proves the claim. $\square$

Corollary 12.272 (Sheared repeated-node tangent-contact is impossible near an isolated odd $\mathcal{R}$ -zero). *Let b be an isolated odd-order zero of $\mathcal{R}$ on a genuine opened nonmixed arc. Then there exists a neighborhood U of b such that no pair*

$$a < x, \quad a, x \in U,$$

can satisfy

$$\tilde{Y}''_{\mathcal{Z}(a)}(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Equivalently, no rooted sheared repeated-node tangent-contact event occurs in a sufficiently small neighborhood of an isolated odd zero of $\mathcal{R}$.

Proof. In the normalized nonmixed chart centered at b , the rooted sheared event is exactly the conjunction

$$D'_a(x) = 0, \quad D_a(x) = D_a(a) = \mathcal{Z}(a).$$

Proposition 12.271 shows that for every sufficiently small nontrivial critical pair one has

$$D_a(x) - D_a(a) \neq 0.$$

So the rooted sheared event is impossible near b . $\square$

Proposition 12.273 (Quadratic-envelope factorization on a nonmixed interval). *Let J be a connected open interval in a good v_5 -patch such that*

$$K_v \neq 0 \quad \text{on } J.$$

Write

$$Q_v(h) = (Y(h), x(h), S(h)), \quad \mathcal{Z} = \frac{Y''}{S''}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}.$$

For a fixed source point $a \in J$, define

$$A_a(x) := Y(x) - Y(a) - Y'(a)(x - a),$$

$$B_a(x) := S(x) - S(a) - S'(a)(x - a),$$

and

$$\Gamma_a(x) := \frac{A_a(x)}{B_a(x)} \quad (x \neq a).$$

Then:

(i) Tangent-contact denominator. *For $x \neq a$,*

$$B_a(x) = \int_a^x (x - t) S''(t) dt,$$

so $B_a(x) \neq 0$ on $J \setminus \{a\}$.

(ii) Exact repeated-source factorization. *For distinct $a, u, v \in J$,*

$$H_4(a, u, v) = -B_a(u) B_a(v) (\Gamma_a(u) - \Gamma_a(v)).$$

(iii) Integral representation. *For $x \neq a$,*

$$\Gamma_a(x) = \frac{\int_a^x (x - t) \mathcal{Z}(t) S''(t) dt}{\int_a^x (x - t) S''(t) dt}.$$

(iv) Positive-kernel derivative formula. *For $x \neq a$,*

$$\Gamma'_a(x) = \frac{1}{B_a(x)^2} \int_a^x \Omega_{a,x}(r) \mathcal{R}(r) dr,$$

where

$$\Omega_{a,x}(r) := \int_a^r \int_r^x (t - s) S''(s) S''(t) dt ds.$$

Moreover, after orienting x so that S'' has fixed sign on J ,

$$\Omega_{a,x}(r) > 0 \quad (a < r < x).$$

Proof. Since $K_v = (x')^3 S''$ and $x' \neq 0$ on a good v_5 -patch, the function S'' has fixed nonzero sign on J . Therefore

$$B_a(x) = \int_a^x (x - t) S''(t) dt$$

has fixed nonzero sign for $x \neq a$, proving (i).

For (ii), use the repeated-node determinant

$$H_4(a, u, v) = \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 0 & Y'(a) & 1 & S'(a) \\ 1 & Y(u) & u & S(u) \\ 1 & Y(v) & v & S(v) \end{pmatrix}.$$

Subtract from the u -row and v -row the corresponding affine tangent data at a . This gives

$$H_4(a, u, v) = \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 0 & Y'(a) & 1 & S'(a) \\ 0 & A_a(u) & 0 & B_a(u) \\ 0 & A_a(v) & 0 & B_a(v) \end{pmatrix} = A_a(u) B_a(v) - A_a(v) B_a(u),$$

hence

$$H_4(a, u, v) = -B_a(u)B_a(v)(\Gamma_a(u) - \Gamma_a(v)).$$

For (iii), Taylor's integral formula gives

$$A_a(x) = \int_a^x (x-t) Y''(t) dt = \int_a^x (x-t) \mathcal{Z}(t) S''(t) dt,$$

while

$$B_a(x) = \int_a^x (x-t) S''(t) dt.$$

Dividing yields the formula for Γ_a .

For (iv), set

$$N_a(x) := \int_a^x (x-t) \mathcal{Z}(t) S''(t) dt, \quad D_a(x) := \int_a^x (x-t) S''(t) dt = B_a(x).$$

Then $\Gamma_a = N_a/D_a$, so

$$\Gamma'_a(x) = \frac{N'_a(x)D_a(x) - N_a(x)D'_a(x)}{D_a(x)^2}.$$

A symmetrization gives

$$N'_a D_a - N_a D'_a = \frac{1}{2} \int_a^x \int_a^x (\mathcal{Z}(t) - \mathcal{Z}(s))(t-s) S''(s) S''(t) ds dt.$$

Since $\mathcal{Z}(t) - \mathcal{Z}(s) = \int_s^t \mathcal{R}(r) dr$, Fubini yields

$$N'_a D_a - N_a D'_a = \int_a^x \Omega_{a,x}(r) \mathcal{R}(r) dr$$

with

$$\Omega_{a,x}(r) = \int_a^r \int_r^x (t-s) S''(s) S''(t) dt ds.$$

Because $t-s > 0$ on the region of integration and S'' has fixed sign on J , one has $\Omega_{a,x}(r) > 0$ after orienting the x -coordinate. This proves (iv). $\square$

Corollary 12.274 (Repeated-source monotonicity on one-sign $\mathcal{R}$ -blocks). *Let $I \subset J$ be a subinterval on which*

$\mathcal{R}$ has fixed nonzero sign.

Then for every $a \in I$, the function $x \mapsto \Gamma_a(x)$ is strictly monotone on each component of $I \setminus \{a\}$. In particular, there do not exist points

$$a < u < v \quad \text{in } I$$

such that

$$H_4(a, u, v) = 0.$$

Proof. This is immediate from Proposition 12.273: the derivative $\Gamma'_a(x)$ is a positive-kernel average of $\mathcal{R}$, so it has the same sign as $\mathcal{R}$ on each one-sign block. The factorization of H_4 then excludes repeated-source exact contacts there. $\square$

Remark 12.275 (Below-lifted source-determinant reformulation of Γ_a). Let

$$\lambda := \frac{\Delta_7}{v_5}, \quad x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \frac{\lambda}{c}.$$

For any scalar germ f , define

$$D_f(a, h) := \det \begin{pmatrix} c(a) & v_5(a) & f(a) \\ c'(a) & v'_5(a) & f'(a) \\ c(h) & v_5(h) & f(h) \end{pmatrix}.$$

A direct row reduction gives

$$D_f(a, h) = c(a)^2 c(h) \left(x'(a) (F(h) - F(a)) - F'(a) (x(h) - x(a)) \right), \quad F = \frac{f}{c}.$$

In particular,

$$\Psi_a(h) := \frac{D_{u_5}(a, h)}{D_\lambda(a, h)} = \Gamma_{x(a)}(x(h)).$$

Thus the repeated-source nonmixed 4-point geometry is already encoded by a finite ratio of source determinants.

Likewise, if one sets

$$\mathfrak{D}(a, h) := \det \begin{pmatrix} c(a) & v_5(a) & \lambda(a) & u_5(a) \\ c'(a) & v_5'(a) & \lambda'(a) & u_5'(a) \\ c(h) & v_5(h) & \lambda(h) & u_5(h) \\ c'(h) & v_5'(h) & \lambda'(h) & u_5'(h) \end{pmatrix},$$

then a Plücker identity yields

$$\frac{d}{dh} \Psi_a(h) = \frac{W_h(c, v_5)(a) \mathfrak{D}(a, h)}{D_\lambda(a, h)^2}.$$

So the sign of the quadratic-envelope derivative is governed by one explicit 4×4 source determinant.

Moreover, combining the identity

$$\Psi_a(h) = \Gamma_{x(a)}(x(h))$$

with Proposition 12.273 gives the exact source-level kernel formula

$$\mathfrak{D}(a, h) = c(a)^2 c(h)^2 x'(a) x'(h) \int_{x(a)}^{x(h)} \Omega_{x(a), x(h)}(r) \mathcal{R}(r) dr.$$

After orienting the x -coordinate so that $x' > 0$ on the interval, the sign of $\mathfrak{D}(a, h)$ is therefore exactly the sign of the corresponding positive-kernel average of $\mathcal{R}$. In particular, on every one-sign $\mathcal{R}$ -block the quadruple

$$(c, v_5, \lambda, u_5)$$

is osculatory-ECT in the sense that $\mathfrak{D}(a, h)$ has fixed nonzero sign for $a < h$.

Corollary 12.276 (Source-level Wronskian consequences for the nonmixed branch). *On a good nonmixed v_5 -patch, the following hold.*

- (1) *If there is an exact nonmixed tangent-plane contact, then there exists a source point h_* in the source convex hull of the contact nodes such that*

$$\mathcal{W}_{\text{src}}(h_*) = 0.$$

- (2) *If $\mathcal{W}_{\text{src}}$ has fixed nonzero sign on an open nonmixed interval, then there are no exact nonmixed tangent-plane contacts on that interval.*
- (3) *If every nonempty open subinterval of a nonmixed interval contains an exact nonmixed tangent-plane contact, then*

$$\mathcal{W}_{\text{src}} \equiv 0$$

on that interval, hence

$$\mathcal{R} \equiv 0,$$

and therefore the lifted curve collapses to affine-planar geometry.

Proof. For (1), Proposition 12.263 produces a point in the convex hull of the three normalized nodes where $\mathcal{R} = 0$. Pulling back through the monotone source-to- x coordinate on the nonmixed patch and using Proposition 12.235 gives $\mathcal{W}_{\text{src}} = 0$ at the corresponding source point.

For (2), by Proposition 12.235, $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ have the same sign on a nonmixed arc. So Corollary 12.274 applies.

For (3), every nonempty open subinterval contains a zero of $\mathcal{W}_{\text{src}}$ by (1), so analyticity forces $\mathcal{W}_{\text{src}} \equiv 0$. Proposition 12.235 then gives $\mathcal{R} \equiv 0$, and Corollary 12.267 gives affine-planar collapse. $\square$

Remark 12.277 (Chebyshev formulation of the nonmixed burden). Corollary 12.276 shows that the nonmixed 4-point branch is controlled source-level by the pair

$$(J_\lambda, J_{u_5}).$$

If this pair were a Chebyshev pair on an opened nonmixed arc, i.e.

$$\mathcal{W}_{\text{src}} \neq 0$$

throughout that arc, then the entire nonmixed 4-point branch would be sterile there. Thus a one-sign theorem for $\mathcal{W}_{\text{src}}$ would still be sufficient. However, the later source-level no-go theorem shows that generic Wronskian algebra alone is not enough to force such a statement. The actual remaining burden is therefore a source-coupling theorem that excludes the analytically possible odd zeros of $\mathcal{W}_{\text{src}}$ on genuine opened nonmixed arcs.

Proposition 12.278 (Lift of the source-level Wronskian to a 4×4 Wronskian). *With*

$$J_\lambda = W(c, v_5, \lambda), \quad J_{u_5} = W(c, v_5, u_5), \quad M = W(c, v_5) = cv'_5 - c'v_5,$$

one has the exact identity

$$\boxed{\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda = W(J_\lambda, J_{u_5}) = M W(c, v_5, \lambda, u_5).}$$

Thus, on a nonmixed arc where $M \neq 0$, the odd-zero problem for $\mathcal{W}_{\text{src}}$ is equivalent to the odd-zero problem for the source-level 4×4 Wronskian

$$\mathcal{U}_{\text{src}} := W(c, v_5, \lambda, u_5).$$

Proof. By definition,

$$\mathcal{W}_{\text{src}} = W(J_\lambda, J_{u_5}).$$

Now apply the Sylvester–Wronski identity

$$W(W(f_1, f_2, f_3), W(f_1, f_2, f_4)) = W(f_1, f_2) W(f_1, f_2, f_3, f_4)$$

with

$$(f_1, f_2, f_3, f_4) = (c, v_5, \lambda, u_5).$$

This gives

$$W(J_\lambda, J_{u_5}) = W(c, v_5) W(c, v_5, \lambda, u_5) = M W(c, v_5, \lambda, u_5),$$

which is exactly the claimed identity. $\square$

Proposition 12.279 (The current source-level Wronskian package is not yet a contradiction engine for nonmixed odd zeros). *Let I be an interval on which*

$$c \neq 0, \quad M = W(c, v_5) \neq 0, \quad J_\lambda = W(c, v_5, \lambda) \neq 0.$$

Then:

(1) *for every analytic forcing $F : I \rightarrow \mathbf{C}$, there exists an analytic function u_5 on I such that*

$$W(c, v_5, \lambda, u_5) = F;$$

(2) *consequently, for every analytic forcing $G : I \rightarrow \mathbf{C}$, there exists an analytic function u_5 on I such that*

$$\mathcal{W}_{\text{src}} = G;$$

(3) *equivalently, for every analytic forcing $R : I \rightarrow \mathbf{C}$, there exists an analytic function u_5 on I such that*

$$\mathcal{R} = R.$$

In particular, at the current source-level ratio/Wronskian/ 4×4 -determinant level, the zero set, parity, and odd/even vanishing pattern of $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ are formally unconstrained.

Proof. Expand the 4×4 Wronskian along the u_5 -column. Since the highest derivative of u_5 appearing in $W(c, v_5, \lambda, u_5)$ is u_5''' , with coefficient exactly

$$W(c, v_5, \lambda) = J_\lambda,$$

there exist analytic coefficients a_0, a_1, a_2 on I such that

$$W(c, v_5, \lambda, u_5) = J_\lambda u_5''' + a_2 u_5'' + a_1 u_5' + a_0 u_5.$$

Because $J_\lambda \neq 0$ on I , this is a third-order linear analytic ODE for u_5 with nonvanishing leading coefficient. Standard analytic ODE existence therefore gives an analytic solution u_5 on I for every analytic forcing F . This proves (1).

For (2), given analytic G , set

$$F := \frac{G}{M}.$$

Since $M \neq 0$ on I , F is analytic. By (1), choose u_5 such that

$$W(c, v_5, \lambda, u_5) = F.$$

Then Proposition 12.278 gives

$$\mathcal{W}_{\text{src}} = M W(c, v_5, \lambda, u_5) = G.$$

For (3), given analytic R , set

$$G := \frac{M J_\lambda^2}{c^2} R.$$

Since $c \neq 0$, $M \neq 0$, and $J_\lambda \neq 0$ on I , this is analytic. By (2), choose u_5 such that $\mathcal{W}_{\text{src}} = G$. Then Proposition 12.235 gives

$$\mathcal{R} = \frac{c^2}{M} \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2} = \frac{c^2}{M} \frac{G}{J_\lambda^2} = R. \quad \square$$

Remark 12.280 (Ceiling of the current source-level reduction). Propositions 12.278 and 12.279 show that the current source-level reduction has been pushed to its natural ceiling. The nonmixed 4-point branch is indeed equivalent to an odd-zero problem for the gauge-invariant source-level Wronskian

$$\mathcal{W}_{\text{src}},$$

and even to the 4×4 Wronskian $W(c, v_5, \lambda, u_5)$. But at that level the odd-zero pattern is still formally free.

Accordingly, the missing ingredient can no longer be a generic Wronskian or determinant argument. It must be a genuine *source-coupling theorem*: one that uses additional identities coming from the actual RH source structure to show that the analytically possible u_5 -freedom exhibited above is not realizable on a genuine opened nonmixed arc.

Remark 12.281 (First-loss rigidification of the quadratic-envelope scalar). Set

$$M(a, x) := B_a(x)^2 \Gamma'_a(x) = A'_a(x) B_a(x) - A_a(x) B'_a(x).$$

Then

$$\partial_x M(a, x) = S''(x) B_a(x) (\mathcal{Z}(x) - \Gamma_a(x)).$$

Moreover, at any point where $M(a, x) = 0$,

$$\partial_a M(a, x) = S''(a) C_a(x) (\mathcal{Z}(a) - \Gamma_a(x)),$$

where

$$C_a(x) := S(a) - S(x) + S'(x)(x - a) = \int_a^x (t - a) S''(t) dt.$$

Thus any formally attained first loss of monotonicity for Γ_a forces

$$\Gamma_a(x) = \mathcal{Z}(a).$$

After the affine shear

$$\tilde{Y}_\gamma := Y - \gamma S, \quad \gamma = \mathcal{Z}(a),$$

this yields

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

So a first loss of quadratic-envelope monotonicity can only occur through a sheared repeated-node tangent-contact configuration rooted at a curvature-zero point. In particular, the remaining nonmixed obstruction is diagonal in nature: it is governed by odd zeros of $\mathcal{R}$, not by an independent separated source-target mechanism.

Proposition 12.282 (The current 3-point package is not by itself a contradiction engine for nonmixed first loss). *Fix an analytic function S on an interval I with*

$$S'' \neq 0 \quad \text{on } I.$$

Let $a < x$ be two points of I . Then there exist an analytic function Y on I and a constant $\gamma \in \mathbf{R}$ such that, with

$$\tilde{Y}_\gamma := Y - \gamma S,$$

one has

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0,$$

where

$$\tilde{H}_3(a, u) := \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}_\gamma'(a)(u - a).$$

In particular, the rigidified repeated-node tangent-contact event from Remark 12.281 is formally compatible with an arbitrary fixed nonmixed planar curve

$$\Lambda_v = (x, S),$$

so the established 3-point package for Λ_v alone cannot exclude that event.

Proof. Define

$$T_{a,x}(t) := (t - a)^3(t - x)^2.$$

Then

$$T_{a,x}(a) = 0, \quad T'_{a,x}(a) = 0, \quad T''_{a,x}(a) = 0, \quad T_{a,x}(x) = 0, \quad T'_{a,x}(x) = 0.$$

Hence

$$T_{a,x}(x) - T_{a,x}(a) - T'_{a,x}(a)(x - a) = 0, \quad T'_{a,x}(x) - T'_{a,x}(a) = 0.$$

Now fix any $\gamma \in \mathbf{R}$ and set

$$Y := \gamma S + T_{a,x}.$$

Then

$$\tilde{Y}_\gamma = Y - \gamma S = T_{a,x},$$

so

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Since S itself has not changed and $S'' \neq 0$ on I , the original 3-point geometry of $\Lambda_v = (x, S)$ on I is exactly the same as before. Thus the sheared repeated-node tangent-contact event is not ruled out by the planar derivative geometry of Λ_v alone. $\square$

Proposition 12.283 (A sheared repeated-node event forces two interior $\mathcal{R}$ -zeros). *Let J be a connected opened nonmixed arc, and let*

$$a < x \quad \text{lie in } J.$$

Set

$$\gamma := \mathcal{Z}(a), \quad \tilde{Y}_\gamma := Y - \gamma S,$$

and let

$$\tilde{H}_3(a, u) := \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}_\gamma'(a)(u - a).$$

Assume

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Then one of the following holds:

$$(1) \mathcal{R} \equiv 0 \text{ on } [a, x];$$

(2) there exist points

$$a < r_1 < r_2 < x$$

such that

$$\mathcal{Z}(r_1) = \mathcal{Z}(r_2) = \mathcal{Z}(a).$$

Consequently there exist

$$\xi_1 \in (a, r_1), \quad \xi_2 \in (r_1, r_2)$$

with

$$\mathcal{R}(\xi_1) = \mathcal{R}(\xi_2) = 0.$$

In particular, every nontrivial sheared repeated-node event contains at least two interior zeros of $\mathcal{R}$.

Proof. Since

$$\tilde{Y}_\gamma'' = Y'' - \gamma S'' = (\mathcal{Z} - \gamma)S'',$$

the choice $\gamma = \mathcal{Z}(a)$ gives

$$\tilde{Y}_\gamma'' = (\mathcal{Z} - \mathcal{Z}(a))S''.$$

Also,

$$\partial_x \tilde{H}_3(a, x) = \tilde{Y}_\gamma'(x) - \tilde{Y}_\gamma'(a) = \int_a^x \tilde{Y}_\gamma''(t) dt,$$

and

$$\tilde{H}_3(a, x) = \int_a^x (x - t) \tilde{Y}_\gamma''(t) dt.$$

Therefore the hypotheses imply

$$\int_a^x (\mathcal{Z}(t) - \mathcal{Z}(a)) S''(t) dt = 0,$$

and

$$\int_a^x (x - t)(\mathcal{Z}(t) - \mathcal{Z}(a)) S''(t) dt = 0.$$

Choose $\varepsilon \in \{\pm 1\}$ so that

$$\varepsilon S'' > 0 \quad \text{on } J,$$

and define the positive measure

$$d\mu(t) := \varepsilon S''(t) dt, \quad \phi(t) := \mathcal{Z}(t) - \mathcal{Z}(a).$$

Then

$$\int_a^x \phi d\mu = 0, \quad \int_a^x (x - t)\phi d\mu = 0.$$

Hence also

$$\int_a^x t \phi(t) d\mu(t) = 0.$$

If $\phi \equiv 0$ on $[a, x]$, then

$$\mathcal{R} = \phi' \equiv 0 \quad \text{on } [a, x],$$

which is alternative (1).

Assume now that $\phi \not\equiv 0$. Then ϕ cannot have fixed sign on $[a, x]$, because μ is positive and

$$\int_a^x \phi d\mu = 0.$$

It also cannot have exactly one sign change. Indeed, if $c \in (a, x)$ were the unique sign-change point, then

$$(t - c)\phi(t) \geq 0 \quad \text{on } [a, x],$$

with strict inequality on a set of positive μ -measure. Therefore

$$\int_a^x (t - c)\phi(t) d\mu(t) > 0.$$

But

$$\int_a^x (t - c)\phi(t) d\mu(t) = \int_a^x t\phi(t) d\mu(t) - c \int_a^x \phi(t) d\mu(t) = 0,$$

a contradiction.

So ϕ has at least two sign changes on (a, x) . Since $\phi(a) = 0$ and ϕ is continuous, there exist

$$a < r_1 < r_2 < x$$

such that

$$\phi(r_1) = \phi(r_2) = 0,$$

i.e.

$$\mathcal{Z}(r_1) = \mathcal{Z}(r_2) = \mathcal{Z}(a).$$

Applying Rolle's theorem to $\phi = \mathcal{Z} - \mathcal{Z}(a)$ on $[a, r_1]$ and on $[r_1, r_2]$ gives

$$\mathcal{R}(\xi_1) = 0 \quad \text{for some } \xi_1 \in (a, r_1), \quad \mathcal{R}(\xi_2) = 0 \quad \text{for some } \xi_2 \in (r_1, r_2).$$

This proves (2). $\square$

Corollary 12.284 (Isolated $\mathcal{R}$ -zeros are shear-sterile). *Let $b \in J$ be an isolated zero of $\mathcal{R}$. Then there exists a neighborhood $U \subset J$ of b such that no pair*

$$a < x, \quad a, x \in U,$$

can satisfy

$$\gamma = \mathcal{Z}(a), \quad \tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Proof. Choose U so small that b is the only zero of $\mathcal{R}$ in U . If such a pair $a < x$ existed in U , then Proposition 12.283 would produce two interior zeros of $\mathcal{R}$ in $(a, x) \subset U$, a contradiction. $\square$

Corollary 12.285 (Diagonal sheared recurrence forces affine-planar collapse). *Let J be a connected opened nonmixed arc. Suppose there exists a sequence*

$$a_n < x_n, \quad a_n, x_n \in J, \quad a_n, x_n \rightarrow b, \quad x_n - a_n \rightarrow 0,$$

such that, with

$$\gamma_n := \mathcal{Z}(a_n), \quad \tilde{Y}_{\gamma_n} := Y - \gamma_n S,$$

one has

$$\tilde{Y}_{\gamma_n}''(a_n) = 0, \quad \tilde{H}_{3,n}(a_n, x_n) = 0, \quad \partial_x \tilde{H}_{3,n}(a_n, x_n) = 0,$$

where

$$\tilde{H}_{3,n}(a, u) := \tilde{Y}_{\gamma_n}(u) - \tilde{Y}_{\gamma_n}(a) - \tilde{Y}_{\gamma_n}'(a)(u - a).$$

Then

$$\mathcal{R} \equiv 0$$

on a neighborhood of b . Hence on some nontrivial interval containing b one has

$$Y = \gamma S + \alpha x + \beta,$$

so the lifted curve is locally affine-planar there.

Proof. If for some n the first alternative in Proposition 12.283 holds, then $\mathcal{R} \equiv 0$ on $[a_n, x_n]$ and we are done.

Otherwise, for every n the interval (a_n, x_n) contains at least two zeros of $\mathcal{R}$. Since

$$a_n, x_n \rightarrow b, \quad x_n - a_n \rightarrow 0,$$

these zeros accumulate at b . Because $\mathcal{R}$ is analytic, this implies

$$\mathcal{R} \equiv 0$$

on a neighborhood of b . Therefore $\mathcal{Z}$ is locally constant, so

$$Y'' = \gamma S''$$

there, and integrating twice gives

$$Y = \gamma S + \alpha x + \beta$$

on a nontrivial interval around b . $\square$

Proposition 12.286 (Uniform fixed-level nonrecurrence for the sheared 3-point branch). *Let J be a connected opened nonmixed arc, fix a real scalar γ , and set*

$$\tilde{Y}_\gamma := Y - \gamma S = \frac{u_5 - \gamma\lambda}{c}, \quad f_\gamma := u_5 - \gamma\lambda, \quad J_\gamma := J_{f_\gamma} = J_{u_5} - \gamma J_\lambda.$$

Then

$$\tilde{Y}_\gamma'' = \frac{c^3}{M^3} J_\gamma.$$

Assume that $b \in J$ is an isolated finite-order odd zero of J_γ . After affine normalization of the planar curve

$$(x, \tilde{Y}_\gamma)$$

at b , let

$$I_- = (\alpha, 0)$$

be the source interval on which

$$\tilde{Y}_\gamma'' < 0,$$

and let

$$I_+ = (0, \beta)$$

be the maximal open interval to the right of 0 on which

$$\tilde{Y}_\gamma'$$

is strictly increasing. Set

$$\mathcal{J}_\gamma := \tilde{Y}_\gamma'(I_-) \cap \tilde{Y}_\gamma'(I_+).$$

For

$$m \in \mathcal{J}_\gamma,$$

write

$$a_\gamma(m) := (\tilde{Y}_\gamma'|_{I_-})^{-1}(m),$$

and define

$$\tilde{H}_{3,\gamma}(a, u) := \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}_\gamma'(a)(u - a).$$

Then there exists a unique maximal open interval

$$\mathcal{J}_{\gamma, \text{br}} = (0, m_{\gamma,*}) \subset \mathcal{J}_\gamma$$

and a unique real-analytic function

$$u_\gamma : \mathcal{J}_{\gamma, \text{br}} \rightarrow I_+$$

such that

$$\tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) = 0 \quad (m \in \mathcal{J}_{\gamma, \text{br}}),$$

and this branch agrees near $m = 0$ with the unique local opposite-side branch born at the odd zero b .

Moreover, along the whole branch one has

$$\partial_u \tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) = \tilde{Y}_\gamma'(u_\gamma(m)) - m \neq 0,$$

so there is no interior common-tangent event, no interior recurrence, and no interior accumulation. If

$$m_{\gamma,*} < \sup \mathcal{J}_\gamma,$$

then

$$u_\gamma(m) \longrightarrow \beta \quad (m \uparrow m_{\gamma,*}),$$

so the branch can stop only by boundary exit of the relevant slope-overlap block.

Equivalently, using

$$\tilde{H}_{3,\gamma}(a, u) = A_a(u) - \gamma B_a(u) = B_a(u)(\Gamma_a(u) - \gamma),$$

one has along the branch

$$\Gamma_{a_\gamma(m)}(u_\gamma(m)) = \gamma, \quad \Gamma'_{a_\gamma(m)}(u_\gamma(m)) \neq 0.$$

In particular, for every fixed γ , the distinguished level γ of the repeated-source scalar Γ_a has no interior critical point on the sheared odd-zero branch.

Proof. By Remark 12.234, for any scalar germ f one has

$$\frac{d^2}{dx^2} \left(\frac{f}{c} \right) = \frac{c^3}{M^3} J_f.$$

Applying this with

$$f = f_\gamma = u_5 - \gamma\lambda$$

gives

$$\tilde{Y}_\gamma'' = \frac{c^3}{M^3} J_\gamma.$$

Hence b is an isolated finite-order odd zero of J_γ if and only if it is an isolated finite-order odd zero of the planar curvature

$$\tilde{Y}_\gamma''.$$

Now apply Theorem 12.253 and Theorem 12.257 with the affine-normalized analytic function

$$\tilde{Y}_\gamma$$

playing the role of the planar graphing function S . The proofs of those results use only the planar repeated-node tangent-contact equation for the chosen analytic function, its curvature parity at the base point, and the maximal monotone-slope block on the opposite side. Therefore they apply verbatim to

$$\tilde{H}_{3,\gamma}(a, u) = \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}_\gamma'(a)(u - a),$$

and yield the unique local opposite-side branch, its unique continuation on the maximal slope-overlap block, the nonvanishing of

$$\partial_u \tilde{H}_{3,\gamma},$$

and the boundary-exit alternative. This proves the first part.

For the reformulation in terms of Γ_a , note from the definitions of A_a , B_a , and Γ_a that

$$\tilde{H}_{3,\gamma}(a, u) = (Y(u) - Y(a) - Y'(a)(u - a)) - \gamma(S(u) - S(a) - S'(a)(u - a)) = A_a(u) - \gamma B_a(u).$$

Since

$$A_a(u) = \Gamma_a(u) B_a(u),$$

this becomes

$$\tilde{H}_{3,\gamma}(a, u) = B_a(u)(\Gamma_a(u) - \gamma).$$

By Proposition 12.273,

$$B_a(u) \neq 0 \quad (u \neq a),$$

so along the branch

$$\tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) = 0 \iff \Gamma_{a_\gamma(m)}(u_\gamma(m)) = \gamma.$$

Differentiating

$$\tilde{H}_{3,\gamma}(a, u) = B_a(u)(\Gamma_a(u) - \gamma)$$

with respect to u gives

$$\partial_u \tilde{H}_{3,\gamma}(a, u) = B_a'(u)(\Gamma_a(u) - \gamma) + B_a(u)\Gamma_a'(u).$$

On the branch, where

$$\Gamma_a(u) = \gamma,$$

this simplifies to

$$\partial_u \tilde{H}_{3,\gamma}(a, u) = B_a(u)\Gamma_a'(u).$$

Since the first part already gives

$$\partial_u \tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) \neq 0$$

and

$$B_{a_\gamma(m)}(u_\gamma(m)) \neq 0,$$

it follows that

$$\Gamma'_{a_\gamma(m)}(u_\gamma(m)) \neq 0$$

along the whole branch. This proves the final statement. $\square$

Remark 12.287 (No-go for the fixed-level chamber-exit route). One might hope to strengthen Proposition 12.286 as follows: if a fixed level γ is chosen near a strict local extremum of $\mathcal{Z}$, so that

$$J_\gamma = (\mathcal{Z} - \gamma)J_\lambda$$

has two nearby simple zeros

$$r_- < r_+,$$

then the corresponding sheared opposite-side 3-point branch should be unable to reach the chamber boundary $u = r_+$ without first encountering an interior common-tangent point

$$\partial_u \tilde{H}_{3,\gamma} = 0.$$

This is false already at the level of the local planar/sheared equations.

Indeed, consider the model functions

$$S_{\text{mod}}(x) := x^2, \quad \mathcal{Z}_{\text{mod}}(x) := 1 - x^2, \quad \mathcal{R}_{\text{mod}}(x) := \mathcal{Z}'_{\text{mod}}(x) = -2x,$$

and fix the level

$$\gamma := 0.$$

Set

$$F_{\text{mod}}(x) := x^2 - \frac{x^4}{6}.$$

Then

$$F''_{\text{mod}}(x) = 2(1 - x^2) = (\mathcal{Z}_{\text{mod}}(x) - \gamma) S''_{\text{mod}}(x),$$

so the two simple zeros of the sheared curvature are

$$r_- = -1, \quad r_+ = 1.$$

Let

$$H_{\text{mod}}(a, u) := F_{\text{mod}}(u) - F_{\text{mod}}(a) - F'_{\text{mod}}(a)(u - a).$$

A direct computation gives

$$H_{\text{mod}}(a, u) = -\frac{(u - a)^2}{6} (3a^2 + 2au + u^2 - 6).$$

Hence the nontrivial opposite-side branch issued from the odd zero $r_- = -1$ is

$$u(a) = -a - \sqrt{6 - 2a^2}, \quad -\frac{5}{3} < a < -1.$$

Along this branch one has

$$-1 < u(a) < 1, \quad u(a) \uparrow 1 \quad \text{as} \quad a \downarrow -\frac{5}{3},$$

so the branch runs inside the chamber and stops by boundary exit at the second curvature zero.

Moreover,

$$\partial_u H_{\text{mod}}(a, u) = F'_{\text{mod}}(u) - F'_{\text{mod}}(a) = -\frac{u - a}{3} (a^2 + au + u^2 - 3).$$

Substituting $u = u(a)$ yields

$$a^2 + a u(a) + u(a)^2 - 3 = -a^2 + a\sqrt{6 - 2a^2} + 3 < 0 \quad \left(-\frac{5}{3} < a < -1 \right),$$

and therefore

$$\partial_u H_{\text{mod}}(a, u(a)) > 0 \quad \left(-\frac{5}{3} < a < -1 \right).$$

So there is *no* interior common-tangent point anywhere on the branch before boundary exit. In fact, if one continues the same algebraic branch past the chamber boundary, the first common tangent occurs only later at

$$(a, u) = (-\sqrt{3}, \sqrt{3}),$$

strictly after the chamber has already been crossed.

Thus fixed-level sheared nonrecurrence cannot be sharpened to a chamber-exit contradiction by planar arguments alone. The remaining obstruction is therefore genuinely source-coupled: any successful odd-zero exclusion must use the source-pinned moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

or some equivalent RH-specific coupling, rather than only a frozen level γ and the planar branch geometry of

$$\tilde{Y}_\gamma = Y - \gamma S.$$

Remark 12.288 (Source-level form of the shear transfer). For fixed $a \in J$, set

$$f_a := u_5 - \mathcal{Z}(a)\lambda.$$

Then

$$\tilde{Y}_{\mathcal{Z}(a)} = \frac{f_a}{c}, \quad J_{f_a} = J_{u_5} - \mathcal{Z}(a)J_\lambda = (\mathcal{Z} - \mathcal{Z}(a))J_\lambda,$$

and

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(a) D_\lambda(a, h).$$

Thus the sheared repeated-node conditions are exactly a repeated-node tangent-contact event for the source triple (c, v_5, f_a) :

$$J_{f_a}(a) = 0, \quad D_{f_a}(a, h) = 0, \quad \partial_h D_{f_a}(a, h) = 0.$$

Proposition 12.283 shows that such a source-level event forces either $J_{f_a} \equiv 0$ on the interval or at least two interior zeros of J_{f_a} , equivalently at least two interior zeros of $\mathcal{R}$ and hence of $\mathcal{W}_{\text{src}}$.

Proposition 12.289 (Source-determinant form of the odd-zero unrooted defect). *Let J be a good nonmixed v_5 -patch, and let $b \in J$ be an isolated odd-order zero of $\mathcal{R}$. Use the local x -coordinate on J centered at b , so that*

$$x(b) = 0, \quad \mathcal{R}(x) = x^m \rho(x), \quad m \geq 1 \text{ odd}, \quad \rho(0) \neq 0.$$

For each source point $a \in J$, set

$$\Psi_a(h) := \frac{D_{u_5}(a, h)}{D_\lambda(a, h)}, \quad f_a := u_5 - \mathcal{Z}(x(a))\lambda.$$

Then, after shrinking J around b if necessary, the following hold.

(1) The critical locus

$$\partial_h \Psi_a(h) = 0 \quad \text{equivalently} \quad \mathfrak{D}(a, h) = 0$$

is the graph of a unique real-analytic branch

$$h = \sigma(a), \quad \sigma(b) = b, \quad \sigma(a) \neq a \quad (a \neq b).$$

(2) One has the exact identity

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h) = D_\lambda(a, h) (\Psi_a(h) - \mathcal{Z}(x(a))).$$

(3) Along the critical branch,

$$\Psi_a(\sigma(a)) - \mathcal{Z}(x(a)) = -\frac{\rho(0)}{2^{m+1}(m+2)} (x(\sigma(a)) - x(a))^{m+1} + o(|x(\sigma(a)) - x(a)|^{m+1}).$$

(4) Consequently,

$$D_{f_a}(a, \sigma(a)) = -\frac{c(b)^4 J_\lambda(b) \rho(0)}{2^{m+2}(m+2)M(b)^2} (x(\sigma(a)) - x(a))^{m+3} + o(|x(\sigma(a)) - x(a)|^{m+3}).$$

In particular, for all sufficiently small $a \neq b$ one has

$$D_{f_a}(a, \sigma(a)) \neq 0,$$

and indeed

$$\operatorname{sgn} D_{f_a}(a, \sigma(a)) = -\operatorname{sgn}(J_\lambda(b)\rho(0)).$$

So near an isolated odd zero of $\mathcal{R}$, the born repeated-source critical branch

$$\mathfrak{D}(a, h) = 0$$

is uniformly unrooted in source-determinant language:

$$D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h) \neq 0.$$

Proof. By Remark 12.275,

$$\Psi_a(h) = \Gamma_{x(a)}(x(h)).$$

Therefore the critical locus

$$\partial_h \Psi_a(h) = 0$$

is exactly the pullback through the source-to- x coordinate of the local critical branch

$$\partial_x \Gamma_{x(a)}(x) = 0.$$

So item (1) is the source-coordinate form of Proposition 12.271.

For item (2), linearity of the determinant in the third column gives

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h).$$

Dividing by $D_\lambda(a, h) \neq 0$ on the nonmixed patch yields

$$\frac{D_{f_a}(a, h)}{D_\lambda(a, h)} = \Psi_a(h) - \mathcal{Z}(x(a)).$$

For item (3), apply Proposition 12.271 to the x -coordinate pair

$$(x(a), x(\sigma(a))).$$

Since

$$\Psi_a(\sigma(a)) = \Gamma_{x(a)}(x(\sigma(a))),$$

that proposition gives exactly

$$\Psi_a(\sigma(a)) - \mathcal{Z}(x(a)) = -\frac{\rho(0)}{2^{m+1}(m+2)}(x(\sigma(a)) - x(a))^{m+1} + o(|x(\sigma(a)) - x(a)|^{m+1}).$$

For item (4), apply the row-reduction identity from Remark 12.275 to

$$f = f_a, \quad F = \frac{f_a}{c} = Y - \mathcal{Z}(x(a))S.$$

This gives

$$D_{f_a}(a, h) = c(a)^2 c(h) x'(a) \tilde{H}_{3, \mathcal{Z}(x(a))}(x(a), x(h)),$$

and since

$$\tilde{H}_{3, \mathcal{Z}(x(a))}(x(a), x(h)) = B_{x(a)}(x(h))(\Psi_a(h) - \mathcal{Z}(x(a))),$$

we obtain

$$D_{f_a}(a, h) = c(a)^2 c(h) x'(a) B_{x(a)}(x(h))(\Psi_a(h) - \mathcal{Z}(x(a))).$$

Along the branch $h = \sigma(a)$, write

$$\delta(a) := x(\sigma(a)) - x(a).$$

Then

$$B_{x(a)}(x(\sigma(a))) = \frac{1}{2} S''(b) \delta(a)^2 + o(|\delta(a)|^2),$$

while

$$x'(b) = \frac{M(b)}{c(b)^2}, \quad S''(b) = \frac{c(b)^3}{M(b)^3} J_\lambda(b)$$

by Remark 12.234. Therefore

$$c(a)^2 c(\sigma(a)) x'(a) B_{x(a)}(x(\sigma(a))) = \frac{c(b)^4 J_\lambda(b)}{2M(b)^2} \delta(a)^2 + o(|\delta(a)|^2).$$

Multiplying by item (3) gives

$$D_{f_a}(a, \sigma(a)) = -\frac{c(b)^4 J_\lambda(b) \rho(0)}{2^{m+2}(m+2)M(b)^2} \delta(a)^{m+3} + o(|\delta(a)|^{m+3}),$$

which is the claimed asymptotic. Since $m+3$ is even, the leading term has fixed nonzero sign for all sufficiently small $a \neq b$. $\square$

Remark 12.290 (Ceiling of the local shear-transfer route). Proposition 12.289 shows that near an isolated odd zero of $\mathcal{R}$, the genuinely born local repeated-source branch is the critical branch

$$\mathfrak{D}(a, h) = 0 \quad \text{equivalently} \quad \partial_h \Psi_a(h) = 0,$$

but its rooted numerator

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h)$$

has a fixed nonzero leading term there.

So the rooted sheared repeated-node tangent-contact condition is not the local source avatar of diagonal odd-zero birth. In particular, no argument using only the local odd-zero model and the repeated-source critical-branch equations can prove an implication of the form

$$\mathfrak{D}(a, h) = 0 \implies D_{f_a}(a, h) = 0$$

near odd zeros.

This is a different obstruction from Remark 12.280: the latter says the generic source-level Wronskian/ 4×4 package is formally too free, while the present remark says that even along the *actual local odd-zero branch*, the rooted numerator fails to vanish with a universal signed defect. Any successful shear-transfer theorem must therefore use genuinely additional RH source coupling beyond the local odd-zero critical-branch analysis.

Proposition 12.291 (Repeated-source first loss is exactly diagonal odd-zero birth). *Let J be a connected opened nonmixed arc, so*

$$K_v \neq 0 \quad \text{on } J.$$

For $a < x$ in J , define

$$A_a(x) := Y(x) - Y(a) - Y'(a)(x - a), \quad B_a(x) := S(x) - S(a) - S'(a)(x - a),$$

$$\Gamma_a(x) := \frac{A_a(x)}{B_a(x)}, \quad M(a, x) := B_a(x)^2 \Gamma'_a(x).$$

Equivalently,

$$M(a, x) = A'_a(x) B_a(x) - A_a(x) B'_a(x).$$

Let

$$\Sigma_M := \{(a, x) \in J^2 : a < x, M(a, x) = 0\}.$$

Then the following hold.

(1) No attained separated first loss. *There is no pair $(a_*, x_*) \in \Sigma_M$ such that*

$$x_* = \inf\{x : \exists a < x \text{ with } (a, x) \in \Sigma_M\}.$$

(2) Any first-loss sequence is diagonal. *If $(a_n, x_n) \in \Sigma_M$ and*

$$x_n \downarrow x_* := \inf\{x : \exists a < x \text{ with } (a, x) \in \Sigma_M\},$$

then

$$x_n - a_n \rightarrow 0.$$

(3) Diagonal accumulation forces a zero of $\mathcal{R}$. If $(a_n, x_n) \in \Sigma_M$ and

$$a_n, x_n \rightarrow b, \quad x_n - a_n \rightarrow 0,$$

then

$$\mathcal{R}(b) = 0.$$

(4) Even-order zeros are diagonally sterile. If b is an isolated even-order zero of $\mathcal{R}$, then there exists a neighborhood U of b such that

$$\Sigma_M \cap (U \times U) = \emptyset.$$

(5) Odd-order zeros seed diagonal first loss. If b is an isolated odd-order zero of $\mathcal{R}$, then every neighborhood U of b contains points

$$(a, x) \in \Sigma_M \quad \text{with} \quad a < x, \quad a, x \in U.$$

Accordingly, repeated-source first loss on a nonmixed arc is not an independent obstruction mechanism: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$.

Proof. For (1), suppose $(a_*, x_*) \in \Sigma_M$ were an attained separated first-loss pair. Then for fixed a_* one has

$$M(a_*, x) > 0 \quad (a_* < x < x_*),$$

by the definition of x_* . Since $B_{a_*}(x) \neq 0$ on a nonmixed arc, this implies

$$\Gamma'_{a_*}(x) > 0 \quad (a_* < x < x_*),$$

so Γ_{a_*} is strictly increasing on (a_*, x_*) . Therefore

$$\Gamma_{a_*}(x_*) > \lim_{x \downarrow a_*} \Gamma_{a_*}(x) = \mathcal{Z}(a_*).$$

But Remark 12.281 shows that every formally attained first loss satisfies

$$\Gamma_{a_*}(x_*) = \mathcal{Z}(a_*),$$

a contradiction. Thus no attained separated first-loss pair exists.

For (2), suppose $(a_n, x_n) \in \Sigma_M$ and $x_n \downarrow x_*$, but that $x_n - a_n \not\rightarrow 0$. Passing to a subsequence, we may assume

$$a_n \rightarrow a_* < x_*.$$

By continuity of M ,

$$M(a_*, x_*) = 0,$$

so $(a_*, x_*) \in \Sigma_M$, contradicting (1). Hence $x_n - a_n \rightarrow 0$.

For (3), since $B_a(x) \neq 0$ on a nonmixed arc, the condition $(a_n, x_n) \in \Sigma_M$ is equivalent to

$$\Gamma'_{a_n}(x_n) = 0.$$

In the quadratic-envelope notation from Proposition 12.273,

$$\Gamma'_a(x) = \frac{1}{B_a(x)^2} \int_a^x \Omega_{a,x}(r) \mathcal{R}(r) dr.$$

Equivalently, after local nonmixed normalization at b , this becomes the repeated-node profile derivative formula

$$D'_a(x) = 2 \int_0^1 s(1-s) \mathcal{R}(a + s(x-a)) ds.$$

Letting $a_n, x_n \rightarrow b$ with $x_n - a_n \rightarrow 0$, continuity yields

$$0 = \lim_{n \rightarrow \infty} D'_{a_n}(x_n) = 2 \left(\int_0^1 s(1-s) ds \right) \mathcal{R}(b) = \frac{1}{3} \mathcal{R}(b).$$

Hence $\mathcal{R}(b) = 0$.

For (4), if b is an isolated even-order zero of $\mathcal{R}$, then Proposition 12.269 gives a neighborhood U of b on which every repeated-node profile D_a is strictly monotone. Therefore $D'_a(x) \neq 0$ for $a \neq x$ in U , hence $\Gamma'_a(x) \neq 0$ and

$$\Sigma_M \cap (U \times U) = \emptyset.$$

For (5), if b is an isolated odd-order zero of $\mathcal{R}$, then Proposition 12.269 gives, after local normalization, a local repeated-node extremum branch for the profiles D_a . In particular, every neighborhood U of b contains points $a < x$ with

$$D'_a(x) = 0.$$

Since $B_a(x) \neq 0$ on the nonmixed arc, this is equivalent to

$$\Gamma'_a(x) = 0,$$

hence $(a, x) \in \Sigma_M \cap (U \times U)$.

The final statement is the combination of (2), (4), and (5). $\square$

Remark 12.292 (Effect on the rigidified first-loss lane). Proposition 12.291 shows that the repeated-source / first-loss mechanism is not an independent obstruction: no attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at isolated odd-order zeros of $\mathcal{R}$.

At the same time, Corollary 12.272 shows that near an isolated odd zero of $\mathcal{R}$, the rooted sheared repeated-node tangent-contact event itself is locally absent; and Corollary 12.285 shows that any genuine diagonal realization of the sheared event already forces the affine-planar collapse case

$$\mathcal{R} \equiv 0.$$

So the sheared bottleneck closes in the collapse direction, not by a purely planar contradiction for (x, S) .

Proposition 12.286 shows moreover that for each fixed shear level γ , the corresponding sheared opposite-side 3-point branch is already globally rigid on its maximal slope-overlap block: it has no interior common-tangent event, no interior recurrence, and no interior accumulation. Thus the remaining issue is not sheared 3-point recurrence at fixed γ , but the source-coupling / level-selection step that would force the distinguished level

$$\gamma = \mathcal{Z}(b)$$

arising from a hypothetical odd zero of $\mathcal{W}_{\text{src}}$ to be hit by the relevant critical-value configuration of Γ_a .

Remark 12.287 shows that this fixed-level route cannot be pushed further by planar chamber-exit arguments alone: even with two nearby roots of $\mathcal{Z} = \gamma$, the sheared opposite-side branch may run to the second curvature zero by boundary exit without any interior common-tangent event. Thus the remaining step must be genuinely source-coupled, involving the source-pinned moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

or an equivalent RH-specific coupling, rather than only the frozen-level geometry of

$$\tilde{Y}_\gamma = Y - \gamma S.$$

Remark 12.293 (The repeated-source lane is now closed). Proposition 12.291 shows that the repeated-source / first-loss mechanism does not define an additional nonmixed obstruction beyond the odd-zero problem for $\mathcal{R}$. No attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at isolated odd-order zeros of $\mathcal{R}$. Thus the genuinely remaining nonmixed burden is purely the global/source-level control of odd zeros of $\mathcal{R}$, equivalently of the source-level Wronskian $\mathcal{W}_{\text{src}}$.

Proposition 12.294 (Single-sign-change source segment cannot realize rigidified first loss). *Let J be an opened nonmixed arc, and let $a < x$ in J . After local nonmixed normalization on $[a, x]$, write*

$$R_{a,x}(s) := \mathcal{R}(a + s(x - a)), \quad 0 \leq s \leq 1.$$

Then

$$\Gamma'_a(x) = 2 \int_0^1 s(1-s) R_{a,x}(s) ds,$$

and

$$\Gamma_a(x) - \mathcal{Z}(a) = (x - a) \int_0^1 (1 - s)^2 R_{a,x}(s) ds.$$

Assume that $R_{a,x}$ has exactly one sign change on $(0, 1)$. If

$$\Gamma'_a(x) = 0,$$

then

$$\operatorname{sgn}(\Gamma_a(x) - \mathcal{Z}(a))$$

is the sign of $R_{a,x}$ on the initial side of the sign change. In particular,

$$\Gamma_a(x) \neq \mathcal{Z}(a).$$

Equivalently, no pair (a, x) for which $\mathcal{R}$ changes sign exactly once on (a, x) can satisfy simultaneously

$$\Gamma'_a(x) = 0, \quad \Gamma_a(x) = \mathcal{Z}(a).$$

Hence any rigidified first-loss segment must cross at least two sign changes of $\mathcal{R}$.

Proof. In the normalized nonmixed chart, $S(t) = t^2$, so $\Gamma_a = D_a$ with

$$D_a(t) = \int_0^1 (1 - s) \tilde{Y}''(a + s(t - a)) ds, \quad D_a(a) = \mathcal{Z}(a).$$

Differentiating in t gives

$$D'_a(t) = \int_0^1 s(1 - s) \tilde{Y}'''(a + s(t - a)) ds = 2 \int_0^1 s(1 - s) \mathcal{R}(a + s(t - a)) ds,$$

which is the first displayed formula at $t = x$. Integrating $\tilde{Y}''(a + s(t - a))$ by parts in s yields

$$D_a(t) - D_a(a) = (t - a) \int_0^1 (1 - s)^2 \mathcal{R}(a + s(t - a)) ds,$$

which is the second displayed formula at $t = x$.

Let

$$k_1(s) := s(1 - s), \quad k_0(s) := (1 - s)^2,$$

so that

$$\frac{k_0(s)}{k_1(s)} = \frac{1 - s}{s}$$

is strictly decreasing on $(0, 1)$.

Write $R_{a,x} = \sigma\rho$ with $\rho \geq 0$ and σ changing sign exactly once at some $s_0 \in (0, 1)$. By hypothesis,

$$0 = \int_0^1 k_1(s) \sigma(s) \rho(s) ds.$$

Split

$$I := \int_0^1 k_0(s) \sigma(s) \rho(s) ds = \int_0^{s_0} \frac{k_0}{k_1}(s) k_1(s) \sigma(s) \rho(s) ds + \int_{s_0}^1 \frac{k_0}{k_1}(s) k_1(s) \sigma(s) \rho(s) ds.$$

Let $\epsilon := \operatorname{sgn}(\sigma|_{(0, s_0)})$. Then $k_1\sigma\rho \geq 0$ with sign ϵ on $(0, s_0)$ and sign $-\epsilon$ on $(s_0, 1)$. Because k_0/k_1 is strictly decreasing, it is bounded below by its value at s_0 on $(0, s_0)$ and bounded above by its value at s_0 on $(s_0, 1)$. Using

$$\int_0^{s_0} k_1\sigma\rho = - \int_{s_0}^1 k_1\sigma\rho$$

and strict monotonicity on the support of ρ gives

$$\epsilon I > 0.$$

Hence

$$\Gamma_a(x) - \mathcal{Z}(a) = (x - a) I \neq 0,$$

with sign ϵ (after orienting $x > a$). This proves the claim. $\square$

Corollary 12.295 (Source-level two-sign-change obstruction). *Let $a < h$ lie on an opened nonmixed arc in source coordinates. Assume*

$$\Psi'_a(h) = 0.$$

If the source-level Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda$$

changes sign exactly once on the source interval $[a, h]$, then

$$\Psi_a(h) \neq \mathcal{Z}(a).$$

Equivalently, the simultaneous source-level first-loss conditions

$$\Psi'_a(h) = 0, \quad \Psi_a(h) = \mathcal{Z}(a)$$

force at least two sign changes of $\mathcal{W}_{\text{src}}$ on $[a, h]$.

Proof. By Remark 12.275, $\Psi_a(h) = \Gamma_{x(a)}(x(h))$, and on the nonmixed arc $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ differ only by a nonvanishing analytic factor (Proposition 12.235). So a single sign change of $\mathcal{W}_{\text{src}}$ on $[a, h]$ pulls back to a single sign change of $\mathcal{R}$ on $[x(a), x(h)]$. Apply Proposition 12.294. $\square$

Proposition 12.296 (Transport equations for a repeated-source critical branch). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let $U \subset J$ be an interval, and let

$$\tau : U \rightarrow J$$

be a real-analytic function such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in U).$$

Set

$$V(a) := D_a(\tau(a)).$$

Then:

(1)

$$V'(a) = \frac{2}{\tau(a) - a} (V(a) - \mathcal{Z}(a)).$$

(2)

$$D''_a(\tau(a)) = \frac{2}{(\tau(a) - a)^2} (\mathcal{Z}(\tau(a)) - V(a)).$$

(3) *At any point $a \in U$ where $D''_a(\tau(a)) \neq 0$,*

$$\tau'(a) = \frac{V(a) - \mathcal{Z}(a)}{\mathcal{Z}(\tau(a)) - V(a)}.$$

Proof. Write

$$D_a(t) = \frac{\tilde{Y}(t) - \tilde{Y}(a) - \tilde{Y}'(a)(t - a)}{(t - a)^2}.$$

Differentiating in a gives

$$\partial_a D_a(t) = \frac{2}{t - a} (D_a(t) - \mathcal{Z}(a)).$$

Evaluating at $t = \tau(a)$ and using $D'_a(\tau(a)) = 0$ yields

$$V'(a) = \frac{d}{da} D_a(\tau(a)) = \partial_a D_a(\tau(a)) = \frac{2}{\tau(a) - a} (V(a) - \mathcal{Z}(a)),$$

which proves (1).

Now use the rigidification scalar

$$M(a, t) := B_a(t)^2 \Gamma'_a(t).$$

In the normalized chart,

$$B_a(t) = (t - a)^2, \quad \Gamma_a = D_a, \quad M(a, t) = (t - a)^4 D'_a(t).$$

By Remark 12.281,

$$\partial_t M(a, t) = 2(t - a)^2 (\mathcal{Z}(t) - D_a(t)).$$

At $t = \tau(a)$ we have $D'_a(\tau(a)) = 0$, so

$$\partial_t M(a, \tau(a)) = (\tau(a) - a)^4 D''_a(\tau(a)).$$

Comparing the two formulas gives

$$(\tau(a) - a)^4 D''_a(\tau(a)) = 2(\tau(a) - a)^2 (\mathcal{Z}(\tau(a)) - V(a)),$$

hence (2).

Finally, still by Remark 12.281, at points where $M(a, t) = 0$ one has

$$\partial_a M(a, t) = 2(t - a)^2 (\mathcal{Z}(a) - D_a(t)),$$

in the normalized chart. Since

$$M(a, \tau(a)) = 0,$$

differentiating in a gives

$$0 = \frac{d}{da} M(a, \tau(a)) = \partial_a M(a, \tau(a)) + \partial_t M(a, \tau(a)) \tau'(a).$$

Substituting the formulas above yields

$$0 = 2(\tau(a) - a)^2 (\mathcal{Z}(a) - V(a)) + 2(\tau(a) - a)^2 (\mathcal{Z}(\tau(a)) - V(a)) \tau'(a).$$

If $D''_a(\tau(a)) \neq 0$, then by (2),

$$\mathcal{Z}(\tau(a)) - V(a) \neq 0,$$

so solving for $\tau'(a)$ gives (3). □

Corollary 12.297 (One-sign-change chamber is no-return for the born critical branch). *In the setting of Proposition 12.296, assume moreover that for every $a \in U$ the segment*

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$.

Then for every $a \in U$:

(1)

$$\text{sgn}(V(a) - \mathcal{Z}(a))$$

is the sign of the initial lobe of $\mathcal{R}$ on $[a, \tau(a)]$;

(2)

$$\text{sgn}(\mathcal{Z}(\tau(a)) - V(a))$$

is the sign of the terminal lobe of $\mathcal{R}$ on $[a, \tau(a)]$;

(3)

$$(V(a) - \mathcal{Z}(a))(\mathcal{Z}(\tau(a)) - V(a)) < 0;$$

(4)

$$D''_a(\tau(a)) \neq 0, \quad \tau'(a) < 0;$$

(5)

$$V(a) \notin [\mathcal{Z}(a), \mathcal{Z}(\tau(a))].$$

In particular,

$$V(a) \neq \mathcal{Z}(a),$$

so the branch cannot realize the rigidified first-loss condition anywhere on U .

Proof. The first sign statement is exactly Proposition 12.294 applied to the critical pair $(a, \tau(a))$.

For the second sign statement, write

$$R_{a, \tau(a)}(s) := \mathcal{R}(a + s(\tau(a) - a)).$$

Since

$$\mathcal{Z}(\tau(a)) - \mathcal{Z}(a) = (\tau(a) - a) \int_0^1 R_{a, \tau(a)}(s) ds$$

and

$$V(a) - \mathcal{Z}(a) = (\tau(a) - a) \int_0^1 (1-s)^2 R_{a,\tau(a)}(s) ds,$$

one gets

$$\mathcal{Z}(\tau(a)) - V(a) = (\tau(a) - a) \int_0^1 (2s - s^2) R_{a,\tau(a)}(s) ds.$$

At the critical pair,

$$0 = D'_a(\tau(a)) = 2 \int_0^1 s(1-s) R_{a,\tau(a)}(s) ds,$$

so

$$\int_0^1 (2s - s^2) R_{a,\tau(a)}(s) ds = \int_0^1 s^2 R_{a,\tau(a)}(s) ds.$$

Hence

$$\mathcal{Z}(\tau(a)) - V(a) = (\tau(a) - a) \int_0^1 s^2 R_{a,\tau(a)}(s) ds.$$

Now set

$$k_1(s) := s(1-s), \quad k_2(s) := s^2.$$

Then

$$\frac{k_2(s)}{k_1(s)} = \frac{s}{1-s}$$

is strictly increasing on $(0, 1)$. Repeating the weighted signed-mass argument from the proof of Proposition 12.294 shows that

$$\operatorname{sgn}(\mathcal{Z}(\tau(a)) - V(a))$$

is the sign of the terminal lobe of $\mathcal{R}$.

The two signs are therefore opposite, which proves (3). Then Proposition 12.296(2) gives

$$D''_a(\tau(a)) \neq 0,$$

and Proposition 12.296(3) gives

$$\tau'(a) < 0.$$

Finally, opposite signs of

$$V(a) - \mathcal{Z}(a) \quad \text{and} \quad \mathcal{Z}(\tau(a)) - V(a)$$

mean exactly that $V(a)$ lies strictly outside the endpoint-curvature interval

$$[\mathcal{Z}(a), \mathcal{Z}(\tau(a))].$$

□

Proposition 12.298 (First rooting requires a second odd zero along the born repeated-source branch). *Let b be an isolated odd-order zero of $\mathcal{R}$ on an opened nonmixed arc, and let*

$$t = \tau(a)$$

denote an analytic continuation of the local born repeated-source critical branch, so that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0.$$

Assume that U is a connected continuation interval on which, for every $a \in U$, the segment

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$.

Then for every $a \in U$ one has

$$D_a(\tau(a)) \neq \mathcal{Z}(a).$$

Equivalently, on the corresponding source branch

$$h = \sigma(a), \quad \mathfrak{D}(a, \sigma(a)) = 0,$$

one has

$$\Psi_a(\sigma(a)) \neq \mathcal{Z}(x(a)),$$

or in determinant form,

$$D_{u_5}(a, \sigma(a)) - \mathcal{Z}(x(a)) D_\lambda(a, \sigma(a)) \neq 0.$$

Consequently, any first parameter at which the analytic continuation of the born repeated-source branch realizes a genuine repeated-source obstruction inside the opened nonmixed arc must occur only after the corresponding segment ceases to be one-sign-change. Internally, away from affine-planar collapse and without leaving the arc, this can happen only when the segment acquires a second odd-order zero of $\mathcal{R}$, equivalently of

$$\mathcal{W}_{\text{src}}.$$

Proof. The first claim is exactly Corollary 12.297(5): on any continuation interval on which each segment

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$, the born critical branch cannot realize the rigidified first-loss condition

$$D_a(\tau(a)) = \mathcal{Z}(a).$$

Via Remark 12.275,

$$\Psi_a(h) = \Gamma_{x(a)}(x(h)),$$

so the same statement in source coordinates is

$$\Psi_a(\sigma(a)) \neq \mathcal{Z}(x(a)).$$

Using Proposition 12.289(2), this is equivalent to

$$D_{u_5}(a, \sigma(a)) - \mathcal{Z}(x(a)) D_\lambda(a, \sigma(a)) \neq 0.$$

For the final statement, as long as the continued branch remains inside the opened nonmixed arc and the corresponding segment remains one-sign-change, the first part shows that rooting cannot occur. Thus any first rooted point must be preceded by failure of the one-sign-change condition.

Now even-order zeros of $\mathcal{R}$ do not change sign, so internally the sign-change count on the segment can change only when an endpoint of the segment meets another odd-order zero of $\mathcal{R}$ (unless the branch exits the arc or enters the affine-planar collapse case). Finally, on a nonmixed arc the functions $\mathcal{R}$ and

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda$$

differ only by a nonvanishing analytic factor (Proposition 12.235), so odd zeros of the two objects coincide. $\square$

Proposition 12.299 (Discriminant entry into the adjacent-two-odd-zero chamber is not a first rooted point). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let

$$\tau : (a_-, a_0] \rightarrow J$$

be a real-analytic continuation of the born repeated-source critical branch, so that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in (a_-, a_0]).$$

Assume:

(1) for every $a \in (a_-, a_0)$, the segment

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$;

(2) the target endpoint reaches the next adjacent odd zero at $a = a_0$, i.e.

$$q := \tau(a_0)$$

is an odd-order zero of $\mathcal{R}$, there are no odd zeros of $\mathcal{R}$ in $(\tau(a), q)$ for $a < a_0$ close to a_0 , and the segment

$$[a_0, q]$$

still contains exactly one sign change of $\mathcal{R}$.

Set

$$V(a) := D_a(\tau(a)).$$

Define the sheared discriminant and the RH-selected slice by

$$\begin{aligned} \mathcal{D} &:= \{(a, \gamma, h) : a < h, D_{f_\gamma}(a, h) = 0, \partial_h D_{f_\gamma}(a, h) = 0\}, \\ \mathcal{S}_{\text{RH}} &:= \{(a, \gamma, h) : \gamma = \mathcal{Z}(a)\}. \end{aligned}$$

Then

$$P(a) := (a, V(a), \tau(a)) \in \mathcal{D} \quad (a \in (a_-, a_0]),$$

but at the adjacent-odd-zero entrance point one has

$$V(a_0) - \mathcal{Z}(a_0) \neq 0.$$

More precisely,

$$V(a_0) - \mathcal{Z}(a_0) = (q - a_0) \int_0^1 (1 - s)^2 \mathcal{R}(a_0 + s(q - a_0)) ds,$$

and this quantity has the sign of the initial lobe of $\mathcal{R}$ on $[a_0, q]$.

Consequently,

$$P(a_0) \notin \mathcal{S}_{\text{RH}},$$

and there exists a neighborhood N of $P(a_0)$ in

$$J \times \mathbf{R} \times J$$

such that

$$N \cap \mathcal{D} \cap \mathcal{S}_{\text{RH}} = \emptyset.$$

In particular, a first rooted point cannot be created at the moment when the next adjacent odd zero first enters the active segment. If rooting occurs at all, it must occur only after the branch has moved into a genuine two-sign-change chamber of positive width.

Proof. By Remark 12.275,

$$\Psi_a(h) = \Gamma_{x(a)}(x(h)),$$

and on a nonmixed patch

$$D_{f_\gamma}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \gamma).$$

Since

$$D_\lambda(a, h) \neq 0 \quad (a < h),$$

we have

$$D_{f_\gamma}(a, h) = 0, \partial_h D_{f_\gamma}(a, h) = 0 \iff \gamma = \Psi_a(h), \partial_h \Psi_a(h) = 0.$$

Along the continued born branch,

$$\partial_h \Psi_a(\tau(a)) = 0, \quad \Psi_a(\tau(a)) = V(a),$$

so indeed

$$P(a) = (a, V(a), \tau(a)) \in \mathcal{D}.$$

At $a = a_0$, the integral formula for D_a gives

$$V(a_0) - \mathcal{Z}(a_0) = (q - a_0) \int_0^1 (1 - s)^2 \mathcal{R}(a_0 + s(q - a_0)) ds.$$

Also, since $D'_{a_0}(q) = 0$,

$$0 = D'_{a_0}(q) = 2 \int_0^1 s(1 - s) \mathcal{R}(a_0 + s(q - a_0)) ds.$$

Set

$$R_0(s) := \mathcal{R}(a_0 + s(q - a_0)), \quad k_0(s) := (1 - s)^2, \quad k_1(s) := s(1 - s).$$

Then

$$\frac{k_0(s)}{k_1(s)} = \frac{1 - s}{s}$$

is strictly decreasing on $(0, 1)$. Because R_0 has exactly one sign change on $(0, 1)$, the same weighted signed-mass argument used in Proposition 12.294 implies that

$$\int_0^1 k_0(s) R_0(s) ds$$

has the sign of the initial lobe of R_0 , hence is nonzero. Therefore

$$V(a_0) - \mathcal{Z}(a_0) \neq 0.$$

Thus

$$P(a_0) = (a_0, V(a_0), q) \in \mathcal{D} \quad \text{but} \quad P(a_0) \notin \mathcal{S}_{\text{RH}}.$$

Since both $\mathcal{D}$ along the continued branch and the function

$$(a, \gamma, h) \mapsto \gamma - \mathcal{Z}(a)$$

are continuous, there is a neighborhood N of $P(a_0)$ on which

$$\gamma - \mathcal{Z}(a)$$

keeps the sign of

$$V(a_0) - \mathcal{Z}(a_0).$$

Hence

$$N \cap \mathcal{D} \cap \mathcal{S}_{\text{RH}} = \emptyset.$$

The final statement is immediate: a rooted point is exactly a point of

$$\mathcal{D} \cap \mathcal{S}_{\text{RH}},$$

so no such point can be born at the adjacent-odd-zero entrance. $\square$

Remark 12.300 (What the discriminant analysis now leaves open). Proposition 12.299 shows that a rooted point is not created *at the entry* into the adjacent-two-odd-zero chamber. Corollary 12.314 shows more: on any continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

does not cross the continued discriminant curve at all.

Thus the adjacent-two-odd-zero chamber is not itself the residual obstruction. What remains open is narrower and genuinely global: whether a rooted crossing can occur only after the anchor passes through an odd-order zero of $\mathcal{R}$ (equivalently of $\mathcal{W}_{\text{src}}$), or after branch exit, or after affine-planar collapse. In other words, the remaining discriminant-theoretic bottleneck is a *delayed post-anchor no-crossing problem*: not an entry-point singularity problem, and not crossing at the anchor odd zero itself, but a later crossing on a dangerous post-anchor chamber.

Proposition 12.301 (Rooted-defect transport and stationary first root). *In the setting of Proposition 12.296, define*

$$L(a) := \tau(a) - a, \quad G(a) := V(a) - \mathcal{Z}(a) = D_a(\tau(a)) - \mathcal{Z}(a).$$

Then

$$G'(a) = \frac{2}{L(a)} G(a) - \mathcal{R}(a).$$

Hence, if $a_* \in U$ satisfies

$$G(a_*) = 0,$$

then

$$G'(a_*) = -\mathcal{R}(a_*).$$

If in addition

$$\mathcal{Z}(\tau(a_*)) \neq \mathcal{Z}(a_*),$$

then

$$D''_{a_*}(\tau(a_*)) \neq 0, \quad \tau'(a_*) = 0, \quad \tau''(a_*) = -\frac{\mathcal{R}(a_*)}{\mathcal{Z}(\tau(a_*)) - \mathcal{Z}(a_*)}.$$

Proof. Since

$$G(a) = V(a) - \mathcal{Z}(a),$$

Proposition 12.296(1) gives

$$G'(a) = V'(a) - \mathcal{R}(a) = \frac{2}{\tau(a) - a} (V(a) - \mathcal{Z}(a)) - \mathcal{R}(a) = \frac{2}{L(a)} G(a) - \mathcal{R}(a).$$

Evaluating at a root a_* of G gives

$$G'(a_*) = -\mathcal{R}(a_*).$$

Now assume also that

$$\mathcal{Z}(\tau(a_*)) \neq \mathcal{Z}(a_*).$$

Since $G(a_*) = 0$, this means

$$\mathcal{Z}(\tau(a_*)) \neq V(a_*).$$

So Proposition 12.296(2) yields

$$D''_{a_*}(\tau(a_*)) = \frac{2}{(\tau(a_*) - a_*)^2} (\mathcal{Z}(\tau(a_*)) - V(a_*)) \neq 0.$$

Then Proposition 12.296(3) applies near a_* and gives

$$\tau'(a) = \frac{G(a)}{\mathcal{Z}(\tau(a)) - V(a)}.$$

Evaluating at a_* gives

$$\tau'(a_*) = 0.$$

Differentiate this identity once more. Since $G(a_*) = 0$ and

$$\mathcal{Z}(\tau(a_*)) - V(a_*) = \mathcal{Z}(\tau(a_*)) - \mathcal{Z}(a_*) \neq 0,$$

one gets

$$\tau''(a_*) = \frac{G'(a_*)}{\mathcal{Z}(\tau(a_*)) - V(a_*)} = -\frac{\mathcal{R}(a_*)}{\mathcal{Z}(\tau(a_*)) - \mathcal{Z}(a_*)},$$

as claimed. □

Proposition 12.302 (Nondegenerate rooted point is a fold of the moving-shear equation).
Work in a local normalized nonmixed chart on which

$$S(t) = t^2.$$

Define

$$\mathcal{F}(a, t) := A_a(t) - \mathcal{Z}(a) B_a(t) = (t - a)^2 (D_a(t) - \mathcal{Z}(a)).$$

Let

$$a_* < t_*$$

satisfy

$$\mathcal{F}(a_*, t_*) = 0, \quad \partial_t \mathcal{F}(a_*, t_*) = 0,$$

and assume

$$\mathcal{R}(a_*) \neq 0, \quad \mathcal{Z}(t_*) \neq \mathcal{Z}(a_*).$$

Then

$$\partial_a \mathcal{F}(a_*, t_*) = -(t_* - a_*)^2 \mathcal{R}(a_*),$$

and

$$\partial_{tt} \mathcal{F}(a_*, t_*) = 2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*)).$$

In particular,

$$\partial_a \mathcal{F}(a_*, t_*) \neq 0, \quad \partial_{tt} \mathcal{F}(a_*, t_*) \neq 0.$$

Hence near (a_*, t_*) the zero set

$$\mathcal{F}(a, t) = 0$$

is the graph of a unique real-analytic function

$$a = \alpha(t)$$

with

$$\alpha(t_*) = a_*, \quad \alpha'(t_*) = 0, \quad \alpha''(t_*) = \frac{2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*))}{(t_* - a_*)^2 \mathcal{R}(a_*)}.$$

Equivalently,

$$a - a_* = \frac{\mathcal{Z}(t_*) - \mathcal{Z}(a_*)}{(t_* - a_*)^2 \mathcal{R}(a_*)} (t - t_*)^2 + o((t - t_*)^2).$$

Proof. Since

$$\mathcal{F}(a, t) = (t - a)^2 (D_a(t) - \mathcal{Z}(a)),$$

the conditions

$$\mathcal{F}(a_*, t_*) = 0, \quad \partial_t \mathcal{F}(a_*, t_*) = 0$$

imply

$$D_{a_*}(t_*) = \mathcal{Z}(a_*), \quad D'_{a_*}(t_*) = 0.$$

So (a_*, t_*) is a rooted critical point.

Differentiate $\mathcal{F}$ with respect to a :

$$\partial_a \mathcal{F}(a, t) = -2(t - a)(D_a(t) - \mathcal{Z}(a)) + (t - a)^2 (\partial_a D_a(t) - \mathcal{R}(a)).$$

At the rooted point,

$$D_{a_*}(t_*) - \mathcal{Z}(a_*) = 0,$$

and by Proposition 12.296,

$$\partial_a D_a(t) = \frac{2}{t - a} (D_a(t) - \mathcal{Z}(a)),$$

so

$$\partial_a \mathcal{F}(a_*, t_*) = -(t_* - a_*)^2 \mathcal{R}(a_*).$$

Next,

$$\partial_t \mathcal{F}(a, t) = 2(t - a)(D_a(t) - \mathcal{Z}(a)) + (t - a)^2 D'_a(t),$$

hence

$$\partial_{tt} \mathcal{F}(a, t) = 2(D_a(t) - \mathcal{Z}(a)) + 4(t - a)D'_a(t) + (t - a)^2 D''_a(t).$$

At the rooted critical point this becomes

$$\partial_{tt} \mathcal{F}(a_*, t_*) = (t_* - a_*)^2 D''_{a_*}(t_*).$$

Applying Proposition 12.296(2) and using

$$D_{a_*}(t_*) = \mathcal{Z}(a_*),$$

one gets

$$\partial_{tt} \mathcal{F}(a_*, t_*) = 2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*)).$$

By the two nonvanishing assumptions,

$$\partial_a \mathcal{F}(a_*, t_*) \neq 0.$$

So the implicit function theorem gives a unique real-analytic graph

$$a = \alpha(t)$$

for the zero set of $\mathcal{F}$ near (a_*, t_*) . Since

$$\partial_t \mathcal{F}(a_*, t_*) = 0,$$

one has

$$\alpha'(t_*) = 0.$$

Differentiating

$$\mathcal{F}(\alpha(t), t) = 0$$

twice at $t = t_*$ gives

$$\partial_a \mathcal{F}(a_*, t_*) \alpha''(t_*) + \partial_{tt} \mathcal{F}(a_*, t_*) = 0,$$

hence

$$\alpha''(t_*) = -\frac{\partial_{tt} \mathcal{F}(a_*, t_*)}{\partial_a \mathcal{F}(a_*, t_*)} = \frac{2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*))}{(t_* - a_*)^2 \mathcal{R}(a_*)}.$$

The Taylor expansion of α then yields the stated quadratic form. $\square$

Remark 12.303 (First rooted point: local fold data versus the actual remaining burden). Let b be an isolated odd-order zero of $\mathcal{R}$, and let

$$t = \tau(a)$$

be an analytic continuation of the born repeated-source critical branch. By Proposition 12.298, any first rooted point along this continuation can occur only after the segment has acquired a second odd-order zero of $\mathcal{R}$.

At such a rooted point, if

$$\mathcal{R}(a_*) \neq 0, \quad \mathcal{Z}(\tau(a_*)) \neq \mathcal{Z}(a_*),$$

then Proposition 12.301 shows that the rooted defect

$$G(a) = D_a(\tau(a)) - \mathcal{Z}(a) \quad \text{equivalently} \quad G(a) = \Psi_a(\sigma(a)) - \mathcal{Z}(a)$$

crosses transversely and that the target map is stationary there:

$$\tau'(a_*) = 0.$$

Moreover, Proposition 12.302 shows that the moving-shear equation

$$A_a(x) - \mathcal{Z}(a) B_a(x) = 0 \quad \text{equivalently} \quad D_{f_a}(a, h) = 0, \quad f_a = u_5 - \mathcal{Z}(a)\lambda,$$

has a local fold at such a nondegenerate rooted point.

Thus the transport/fold package determines the *local shape* of any prospective rooted crossing. However, the later route-2 results show that this is not the final obstruction theorem. Proposition 12.299 rules out birth of the first rooted point at entry into the adjacent-two-odd-zero chamber; Proposition 12.313 and Corollary 12.314 exclude rooted crossing on continuation intervals where the anchor-side value keeps fixed sign; Proposition 12.316 rules out rooted crossing at an odd anchor zero itself or immediately after passage; and Proposition 12.321 together with Corollary 12.322 show that even delayed rooted crossing on dangerous post-anchor chambers cannot be excluded from the current planar/route-2 package alone.

Accordingly, this remark should be read only as local fold data. The remaining route-2 burden is no longer a “first-fold exclusion” theorem, but the search for the first genuinely non-universal source-coupled input beyond the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$.

Proposition 12.304 (At most one odd zero implies no genuine repeated-source obstruction). *Let J be an opened nonmixed arc. Assume that $\mathcal{R}$ has at most one odd-order zero on J . Then there is no pair $a < x$ in J such that*

$$\Gamma'_a(x) = 0, \quad \Gamma_a(x) = \mathcal{Z}(a).$$

Equivalently, there is no source-level pair $a < h$ on the corresponding source arc such that

$$\Psi'_a(h) = 0, \quad \Psi_a(h) = \mathcal{Z}(a).$$

Proof. On a nonmixed arc, $\mathcal{W}_{\text{src}}$ differs from $\mathcal{R}$ only by a nonvanishing analytic factor (Proposition 12.235), so odd zeros and sign changes of the two objects coincide.

If $\mathcal{R}$ has at most one odd-order zero on J , then every segment contained in J contains at most one sign change of $\mathcal{R}$ (equivalently of $\mathcal{W}_{\text{src}}$), since even-order zeros do not change sign. Hence Corollary 12.295 excludes any simultaneous source-level solution of

$$\Psi'_a(h) = 0, \quad \Psi_a(h) = \mathcal{Z}(a).$$

Via Remark 12.275, this is equivalent to the planar formulation

$$\Gamma'_a(x) = 0, \quad \Gamma_a(x) = \mathcal{Z}(a).$$

□

Remark 12.305 (Minimal unresolved configuration on this route). Proposition 12.304 rules out every opened nonmixed arc carrying at most one odd zero of $\mathcal{R}$, and Corollary 12.314 rules out rooted crossing on the standard adjacent-two-odd-zero chamber as long as the anchor-side value $\mathcal{R}(a)$ keeps a fixed sign.

The new odd-anchor passage theorem (Proposition 12.316) shows that no rooted crossing can occur at the anchor odd zero itself or immediately after it is crossed, while Corollary 12.318 shows that the subsequent post-anchor chambers alternate dangerous and safe, beginning with a dangerous first post-anchor chamber.

Remark 12.319 shows that the first dangerous chamber is already formally compatible with the local scalar rooted equations and fold data. Accordingly, the first genuinely unresolved configuration is no longer the adjacent-two-odd-zero chamber itself, nor the anchor odd zero passage, but a *delayed* crossing on a dangerous post-anchor chamber. Thus the sharpened source-coupling target is a delayed no-crossing / profile-realizability exclusion theorem for the moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

on genuine opened nonmixed arcs.

Remark 12.306 (No-go for a pure level-set / barrier theorem for $\mathcal{Z}$). A natural next guess would be that the remaining adjacent-odd-zero case could be excluded by a theorem depending only on the level geometry of

$$\mathcal{Z}, \quad \mathcal{R} = \mathcal{Z}',$$

together with the first-root transport identities

$$G(a_*) := V(a_*) - \mathcal{Z}(a_*) = 0, \quad G'(a_*) = -\mathcal{R}(a_*), \quad \tau'(a_*) = 0,$$

and the fact that a rooted point forces two interior recrossings of the level

$$\mathcal{Z}(r) = \mathcal{Z}(a_*).$$

At the current level of abstraction this is false: the level-set geometry of $\mathcal{Z}$ alone is still too free.

Indeed, already in the normalized planar determinant model one can realize all of the above with exactly two simple odd zeros of $\mathcal{R}$ and with a nondegenerate rooted critical pair. Take

$$S(t) = t^2, \quad \mathcal{R}(t) = 5t^3 + 5t^2 - 7t + 1, \quad \mathcal{Z}(t) = \int_0^t \mathcal{R}(u) du = t - \frac{7}{2}t^2 + \frac{5}{3}t^3 + \frac{5}{4}t^4.$$

Then $\mathcal{R}$ has exactly two simple zeros in $(0, 1)$, numerically

$$t \approx 0.165726, \quad t \approx 0.660736,$$

so on $[0, 1]$ the sign pattern is $+, -, +$ with exactly two odd sign changes. Now choose an analytic function $\tilde{Y}$ such that

$$\frac{1}{2} \tilde{Y}'' = \mathcal{Z}.$$

For the normalized determinant function $D_a(x)$ associated to $(x, \tilde{Y})$, the rooted critical conditions at $(a, x) = (0, 1)$ become

$$D'_0(1) = 2 \int_0^1 s(1-s) \mathcal{R}(s) ds, \quad D_0(1) - \mathcal{Z}(0) = \int_0^1 (1-s)^2 \mathcal{R}(s) ds.$$

For this $\mathcal{R}$, both moments vanish:

$$\int_0^1 s(1-s) \mathcal{R}(s) ds = 0, \quad \int_0^1 (1-s)^2 \mathcal{R}(s) ds = 0.$$

Hence

$$D'_0(1) = 0, \quad D_0(1) = \mathcal{Z}(0),$$

so $(0, 1)$ is an exact rooted repeated-source critical pair.

Moreover,

$$D_0''(1) = 2 \int_0^1 s^2 \mathcal{R}(s) ds = \frac{5}{6} > 0,$$

so the critical branch is nondegenerate and the implicit-function theorem gives a real-analytic continuation

$$x = \tau(a), \quad \tau(0) = 1, \quad \tau'(0) = 0.$$

Also, with

$$V(a) := D_a(\tau(a)), \quad G(a) := V(a) - \mathcal{Z}(a),$$

one has at the rooted point

$$G(0) = 0, \quad G'(0) = -\mathcal{R}(0) = -1.$$

Finally, the level equation

$$\mathcal{Z}(t) = \mathcal{Z}(0) = 0$$

has two interior solutions,

$$t \approx 0.368001, \quad t \approx 0.851550,$$

so the rooted point comes with exactly the two interior recrossings of the moving level demanded by the source-pinned shear picture.

Thus a theorem using only the scalar level-set topology of $\mathcal{Z}$ together with the first-root identities above cannot suffice to exclude the remaining adjacent-odd-zero configuration. Any successful next step must use additional structure: either the provenance of $(a, \tau(a))$ as the *continued born branch*, or a genuinely source-coupled constraint such as the moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda,$$

rather than only the abstract level geometry of $\mathcal{Z}$.

Remark 12.307 (No-go for branch provenance alone). One might try to strengthen Remark 12.306 by using not just the scalar level geometry of $\mathcal{Z}$, but also the *provenance* of the rooted point: namely, that $(a, \tau(a))$ lies on the continued branch born at an earlier odd zero of $\mathcal{R}$. At the planar normalized D_a -level, this is still not enough. There are explicit analytic models in which the born branch from one odd zero continues to a genuine rooted point before leaving the interval bounded by the adjacent odd zeros.

A concrete example is obtained by taking

$$S(t) = t^2, \quad \mathcal{R}(t) := t(t-1)(t^2 - 4t - 2) = t^4 - 5t^3 + 2t^2 + 2t,$$

and

$$\mathcal{Z}(t) := \int_0^t \mathcal{R}(u) du = \frac{1}{5}t^5 - \frac{5}{4}t^4 + \frac{2}{3}t^3 + t^2.$$

Then the odd zeros of $\mathcal{R}$ are

$$2 - \sqrt{6}, \quad 0, \quad 1, \quad 2 + \sqrt{6},$$

so 0 is an odd zero and its adjacent odd zero on the left is $2 - \sqrt{6}$.

Now choose an analytic function $\tilde{Y}$ such that

$$\frac{1}{2} \tilde{Y}'' = \mathcal{Z},$$

and let $D_a(x)$ denote the normalized determinant/divided-difference function for the planar graph $(x, \tilde{Y})$. Near the odd zero 0 there is the usual born critical branch

$$x = \tau(a), \quad a < 0 < x, \quad D_a'(\tau(a)) = 0,$$

with

$$\tau(a) = -a + O(a^2),$$

and the local odd-zero analysis gives

$$G(a) := D_a(\tau(a)) - \mathcal{Z}(a) < 0$$

for $a < 0$ sufficiently close to 0.

However, along this same connected born branch one finds

$$G(-0.5801) < 0, \quad G(-0.5901) > 0,$$

so by continuity there exists

$$a_* \in (-0.5901, -0.5801)$$

such that

$$G(a_*) = 0.$$

Thus the *continued born branch* reaches a rooted point

$$D'_{a_*}(\tau(a_*)) = 0, \quad D_{a_*}(\tau(a_*)) = \mathcal{Z}(a_*).$$

Numerically,

$$a_* \approx -0.5805673462, \quad \tau(a_*) \approx 0.3888336023,$$

and

$$D''_{a_*}(\tau(a_*)) > 0,$$

so this rooted point is nondegenerate; in particular,

$$\tau'(a_*) = 0$$

as predicted by the transport system.

Moreover, the segment

$$[a_*, \tau(a_*)]$$

contains exactly the born odd zero 0 together with the adjacent odd zero $2 - \sqrt{6}$, and no others. So even the strengthened statement

“continued born-branch provenance plus adjacent-odd-zero geometry already excludes rooting” is false in the planar normalized model.

Therefore the remaining exclusion cannot depend only on abstract branch continuation data for

$$D'_a(\tau(a)) = 0$$

together with the scalar geometry of $\mathcal{Z}$. Any successful next step must use genuinely source-coupled structure, such as the actual quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

the moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda,$$

or some equivalent RH-specific relation invisible in the planar D_a model.

Remark 12.308 (No-go for disconjugacy/ECT alone). A further natural idea is to recast the moving-shear family through the third-order operator

$$L[y] := \frac{W(c, v_5, \lambda, y)}{J_\lambda},$$

and try to exclude rooted moving-shear events by proving that L is disconjugate, Chebyshev, or sign-regular on opened nonmixed arcs. At the present level of abstraction this is still not enough.

Indeed, already in the normalized source model

$$c(t) = 1, \quad v_5(t) = t, \quad \lambda(t) = t^2,$$

one has

$$J_f = W(c, v_5, f) = f'', \quad J_\lambda = 2, \quad L[y] = \frac{W(c, v_5, \lambda, y)}{J_\lambda} = \frac{1}{2} y''''.$$

So the homogeneous equation $L[y] = 0$ has basis

$$1, \ t, \ t^2,$$

hence is maximally disconjugate / ECT on every interval. Nevertheless, the planar normalized counterexample from Remark 12.307 fits exactly into this framework: for a suitable analytic u_5 and the source-pinned family

$$f_a := u_5 - \mathcal{Z}(a)\lambda,$$

one has

$$L[f_a] = L[u_5], \quad J_{f_a}(a) = 0,$$

and along the continued born branch there is still a rooted point

$$D_{f_a}(a, h) = 0, \quad \partial_h D_{f_a}(a, h) = 0.$$

Thus disconjugacy, ECT, or sign-regularity of the operator L by itself cannot exclude the remaining rooted moving-shear configuration. Any successful next step must use additional source-coupled structure beyond the abstract third-order ODE viewpoint, such as special positivity or monotonicity for the distinguished quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)}$$

or for the particular slice

$$f_a = u_5 - \mathcal{Z}(a)\lambda.$$

Remark 12.309 (Ceiling of the present scalar/discriminant analysis). The preceding results sharply delimit what can and cannot be obtained from the current planar/scalar reformulation.

On the positive-kernel side, one-sign-change chambers are sterile for genuine repeated-source first loss: the born critical branch is simple, has strictly negative slope, and cannot satisfy

$$D_a(\tau(a)) = \mathcal{Z}(a)$$

before the active segment acquires another odd zero of $\mathcal{R}$ (Proposition 12.298). On the discriminant side, the adjacent-odd-zero entrance itself cannot create the first rooted point (Proposition 12.299).

The new anchor-sign and odd-anchor-passage arguments remove a substantial part of the remaining freedom. On any continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps fixed sign, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

cannot cross the continued discriminant curve. When the anchor itself passes through an odd-order zero of $\mathcal{R}$, the integrating-factor defect keeps its incoming sign across that odd anchor zero, so a rooted crossing cannot occur there or immediately after it. The ensuing post-anchor chambers then alternate dangerous and safe, beginning with a dangerous first post-anchor chamber.

However, two no-go facts now show that the present scalar/discriminant package has reached its ceiling. First, Remark 12.319 shows that already on the first dangerous post-anchor chamber, the local normalized rooted equations and fold/transversality data are formally realizable in the planar scalar model. Second, Proposition 12.321 and Corollary 12.322 show that the entire current route-2 source package

$$\Psi_a, \quad \mathfrak{D}, \quad \mathcal{W}_{\text{src}}, \quad D_{f_a}$$

is universal under normalization: every analytic normalized planar model lifts to a genuine analytic source model for this package.

Accordingly, after the fixed-sign and odd-anchor exclusions, the remaining problem is *not* a delayed no-crossing / profile-realizability problem inside the present scalar/discriminant package. It is to find the first genuinely non-universal source-coupled input, i.e. structure not already captured by the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$.

So any further progress on the nonmixed repeated-source branch must use genuinely source-coupled structure beyond the present universal package, possibly through stronger provenance

laws, higher canonical/mixed data, or another RH-specific input invisible in the planar normalized model.

Proposition 12.310 (Discriminant reduction for the moving-shear family). *Let J be a good nonmixed v_5 -patch. For $\gamma \in \mathbf{R}$, set*

$$f_\gamma := u_5 - \gamma\lambda.$$

Define the sheared discriminant set

$$\mathcal{D} := \{(a, \gamma, h) \in J \times \mathbf{R} \times J : a < h, D_{f_\gamma}(a, h) = 0, \partial_h D_{f_\gamma}(a, h) = 0\}.$$

Then on the region $a < h$ one has the exact equivalence

$$(a, \gamma, h) \in \mathcal{D} \iff \gamma = \Psi_a(h), \quad \mathfrak{D}(a, h) = 0.$$

Equivalently,

$$\mathcal{D} = \{(a, \Psi_a(h), h) : a < h, \mathfrak{D}(a, h) = 0\}.$$

In particular, on a nonmixed patch the sheared discriminant is not an independent two-dimensional catastrophe surface in (a, γ, h) -space: it is the graph of the repeated-source critical locus

$$\mathfrak{D}(a, h) = 0$$

in the extra γ -direction.

Proof. By Proposition 12.289(2), for every (a, h) with $a < h$ one has

$$D_{f_\gamma}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \gamma).$$

Differentiating in h gives

$$\partial_h D_{f_\gamma}(a, h) = \partial_h D_\lambda(a, h)(\Psi_a(h) - \gamma) + D_\lambda(a, h) \partial_h \Psi_a(h).$$

On a good nonmixed patch,

$$D_\lambda(a, h) \neq 0 \quad (a < h).$$

Therefore

$$D_{f_\gamma}(a, h) = 0 \iff \gamma = \Psi_a(h).$$

At such a point the first term in the formula for $\partial_h D_{f_\gamma}$ vanishes, so

$$\partial_h D_{f_\gamma}(a, h) = 0 \iff D_\lambda(a, h) \partial_h \Psi_a(h) = 0 \iff \partial_h \Psi_a(h) = 0.$$

Finally, by Remark 12.275,

$$\partial_h \Psi_a(h) = \frac{W_h(c, v_5)(a) \mathfrak{D}(a, h)}{D_\lambda(a, h)^2},$$

and the prefactor is nonvanishing on the nonmixed patch. Hence

$$\partial_h \Psi_a(h) = 0 \iff \mathfrak{D}(a, h) = 0.$$

Combining the two equivalences proves the result. $\square$

Proposition 12.311 (Generic rooted discriminant point is a transverse crossing). *In the setting of Proposition 12.310, define the RH-selected slice*

$$\mathcal{S}_{\text{RH}} := \{(a, \gamma, h) \in J \times \mathbf{R} \times J : \gamma = \mathcal{Z}(x(a))\}.$$

Let

$$(a_*, \gamma_*, h_*) \in \mathcal{D} \cap \mathcal{S}_{\text{RH}}$$

and assume

$$\partial_h^2 D_{f_{\gamma_*}}(a_*, h_*) \neq 0, \quad \mathcal{R}(x(a_*)) \neq 0.$$

Then there is a neighborhood of (a_, γ_*, h_*) in which $\mathcal{D}$ is a unique real-analytic curve*

$$a \mapsto (a, \gamma(a), h(a)),$$

with

$$\gamma(a) = \Psi_a(h(a)).$$

Moreover one has

$$\gamma'(a_*) = 0, \quad h'(a_*) = 0,$$

and

$$\left. \frac{d}{da} (\gamma(a) - \mathcal{Z}(x(a))) \right|_{a=a_*} = -\mathcal{R}(x(a_*)) x'(a_*) \neq 0.$$

Hence the discriminant curve $\mathcal{D}$ meets the RH-selected slice $\mathcal{S}_{\text{RH}}$ transversely at (a_*, γ_*, h_*) .

In fact the γ -projection of the discriminant curve has a nondegenerate local extremum there:

$$\gamma''(a_*) = -\frac{2x'(a_*)^2}{x(h_*) - x(a_*)} \mathcal{R}(x(a_*)).$$

Proof. By Proposition 12.310, the discriminant is cut out by

$$\partial_h \Psi_a(h) = 0, \quad \gamma = \Psi_a(h).$$

At the given point,

$$\partial_h^2 D_{f_{\gamma_*}}(a_*, h_*) \neq 0.$$

Since

$$D_{f_\gamma}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \gamma),$$

differentiating twice in h and evaluating at a discriminant point gives

$$\partial_h^2 D_{f_{\gamma_*}}(a_*, h_*) = D_\lambda(a_*, h_*) \partial_h^2 \Psi_{a_*}(h_*).$$

Because $D_\lambda(a_*, h_*) \neq 0$, this implies

$$\partial_h^2 \Psi_{a_*}(h_*) \neq 0.$$

So by the implicit function theorem there is a unique real-analytic function

$$h = h(a)$$

with

$$h(a_*) = h_*, \quad \partial_h \Psi_a(h(a)) = 0.$$

Setting

$$\gamma(a) := \Psi_a(h(a))$$

gives the asserted local parametrization of $\mathcal{D}$.

Now pass to the local normalized nonmixed x -coordinate and write

$$\alpha := x(a), \quad \tau(\alpha) := x(h(a)), \quad V(\alpha) := D_\alpha(\tau(\alpha)).$$

By Remark 12.275,

$$\gamma(a) = \Psi_a(h(a)) = V(\alpha).$$

Since $(a_*, \gamma_*, h_*) \in \mathcal{S}_{\text{RH}}$, one has

$$V(\alpha_*) = \mathcal{Z}(\alpha_*), \quad \alpha_* := x(a_*).$$

By Proposition 12.296, at a rooted noncollapse point one has

$$V'(\alpha_*) = 0, \quad \tau'(\alpha_*) = 0.$$

Therefore

$$\gamma'(a_*) = V'(\alpha_*) x'(a_*) = 0,$$

and

$$x'(h_*) h'(a_*) = \tau'(\alpha_*) x'(a_*) = 0,$$

hence

$$h'(a_*) = 0.$$

For the transverse-crossing derivative, differentiate

$$\gamma(a) - \mathcal{Z}(x(a)) = V(\alpha) - \mathcal{Z}(\alpha)$$

to get

$$\frac{d}{da} (\gamma(a) - \mathcal{Z}(x(a))) = x'(a)(V'(\alpha) - \mathcal{R}(\alpha)).$$

At $a = a_*$, since $V'(\alpha_*) = 0$, this becomes

$$\left. \frac{d}{da} (\gamma(a) - \mathcal{Z}(x(a))) \right|_{a=a_*} = -\mathcal{R}(\alpha_*) x'(a_*),$$

which is nonzero by hypothesis. This proves transversality.

Finally, differentiate

$$V'(\alpha) = \frac{2}{\tau(\alpha) - \alpha} (V(\alpha) - \mathcal{Z}(\alpha))$$

from Proposition 12.296(1). At the rooted point,

$$V(\alpha_*) - \mathcal{Z}(\alpha_*) = 0, \quad \tau'(\alpha_*) = 0,$$

so

$$V''(\alpha_*) = \frac{2}{\tau(\alpha_*) - \alpha_*} (V'(\alpha_*) - \mathcal{R}(\alpha_*)) = -\frac{2}{\tau(\alpha_*) - \alpha_*} \mathcal{R}(\alpha_*).$$

Hence

$$\gamma''(a_*) = V''(\alpha_*) x'(a_*)^2 = -\frac{2 x'(a_*)^2}{x(h_*) - x(a_*)} \mathcal{R}(x(a_*)),$$

as claimed. $\square$

Remark 12.312 (What the discriminant/catastrophe route actually reduces to). The discriminant reformulation sharpens the geometry, but it also shows the limit of purely local catastrophe arguments on this route.

Indeed, by Proposition 12.310, the sheared discriminant on a nonmixed patch is already one-dimensional: it is the graph of the repeated-source critical locus

$$\mathfrak{D}(a, h) = 0$$

in the γ -direction. And by Proposition 12.311, a generic rooted point is not a tangential meeting of sheets, nor a local fold/cusp obstruction for the RH-selected slice

$$\gamma = \mathcal{Z}(x(a)).$$

It is a *transverse crossing* of a smooth discriminant curve by that moving slice, with oriented crossing sign controlled by

$$-\mathcal{R}(x(a_*)).$$

Accordingly, the discriminant/catastrophe route does not reduce the adjacent odd-zero problem to a local singularity classification in (a, γ, h) -space. The remaining issue is global: to exclude the existence of such a transverse crossing along the continued born branch before the segment exits the arc, collapses affine-planarly, or acquires the next odd zero of $\mathcal{W}_{\text{src}}$.

Proposition 12.313 (Anchor-sign preservation excludes first rooting). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let

$$I = (a_1, a_0] \subset J$$

and let

$$\tau : I \rightarrow J$$

be a real-analytic continuation of the repeated-source critical branch such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in I).$$

Set

$$V(a) := D_a(\tau(a)), \quad G(a) := V(a) - \mathcal{Z}(a).$$

Assume that there exists $\epsilon \in \{\pm 1\}$ such that

$$\epsilon \mathcal{R}(a) > 0 \quad \text{for every } a \in I,$$

and

$$\epsilon G(a_0) > 0.$$

Then

$$\epsilon G(a) > 0 \quad \text{for every } a \in I.$$

In particular,

$$G(a) \neq 0 \quad (a \in I),$$

so the branch cannot realize the rigidified first-loss condition anywhere on I .

Proof. Assume for contradiction that G vanishes somewhere on I . Since G is continuous, the set

$$Z := \{a \in I : G(a) = 0\}$$

is closed in I and nonempty. Let

$$a_* := \sup Z.$$

Then $a_* < a_0$, and by the choice of a_* one has

$$\epsilon G(a) > 0 \quad \text{for all } a \in (a_*, a_0].$$

Hence the one-sided derivative at a_* satisfies

$$\epsilon G'(a_*) \geq 0.$$

On the other hand, by Proposition 12.301,

$$G'(a) = \frac{2}{\tau(a) - a} G(a) - \mathcal{R}(a).$$

Evaluating at the root $a = a_*$ gives

$$G'(a_*) = -\mathcal{R}(a_*).$$

Since $\epsilon \mathcal{R}(a_*) > 0$ by hypothesis, this yields

$$\epsilon G'(a_*) = -\epsilon \mathcal{R}(a_*) < 0,$$

contradicting

$$\epsilon G'(a_*) \geq 0.$$

Therefore G has no zero on I . Since

$$\epsilon G(a_0) > 0,$$

the sign of G on all of I is ϵ . □

Corollary 12.314 (No crossing on the adjacent-two-odd-zero chamber under anchor-sign preservation). *Let $b < q$ be adjacent odd-order zeros of $\mathcal{R}$ on an opened nonmixed arc J . Work in a local normalized nonmixed chart, and let*

$$I = (a_1, a_0] \subset J, \quad \tau : I \rightarrow J$$

be a real-analytic continuation of the born repeated-source critical branch such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in I),$$

with

$$\tau(a_0) = q.$$

Assume that for every $a \in I$ with $a < a_0$, the segment

$$[a, \tau(a)]$$

contains exactly two sign changes of $\mathcal{R}$, occurring at the odd zeros b and q , and that

$$\mathcal{R}(a)$$

has fixed sign on I .

Then

$$\operatorname{sgn}(G(a)) = \operatorname{sgn}(\mathcal{R}(a)) \quad (a \in I),$$

hence

$$G(a) \neq 0 \quad (a \in I).$$

Equivalently, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

does not cross the continued discriminant curve

$$(a, \gamma, h) = (a, V(a), \tau(a))$$

anywhere on I .

Proof. At the entry point $a = a_0$, one has

$$\tau(a_0) = q,$$

and the segment

$$[a_0, q]$$

contains exactly one sign change of $\mathcal{R}$, namely at b . Therefore Proposition 12.294 applied to the critical pair (a_0, q) gives

$$\operatorname{sgn}(G(a_0)) = \operatorname{sgn}(\mathcal{R}(a_0)).$$

Now apply Proposition 12.313 with

$$\epsilon := \operatorname{sgn}(\mathcal{R}|_I).$$

□

Remark 12.315 (What this really removes from the previous caveat). The preceding corollary shows that the geometric hypothesis “the anchor stays on the original birth-side outer lobe” was stronger than necessary. The proof uses only the intrinsic condition that the anchor-side value

$$\mathcal{R}(a)$$

keep a fixed sign along the continuation interval.

Accordingly, the remaining unresolved possibility on this route is narrower, but it is not yet exhausted. A rooted point can occur only after the anchor itself passes through an odd-order zero of $\mathcal{R}$, or after branch exit, or after affine-planar collapse. Moreover, Proposition 12.316 shows that no rooted crossing can occur at that anchor odd zero itself or immediately after passage, and Corollary 12.318 shows that the subsequent post-anchor chambers alternate dangerous and safe. Thus the true residual obstruction is no longer the raw adjacent-two-odd-zero chamber itself. But by the universality threshold for the current route-2 package, by the local freedom of arbitrary canonical quotient-septic geometry, and by the full-block universality of the direct provenance slice

$$\phi = Aq^{(7)},$$

even the delayed post-anchor scenario cannot be excluded inside those packages alone.

Proposition 12.316 (Odd-anchor passage has no sign reset). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let

$$q \in J$$

be an isolated odd-order zero of $\mathcal{R}$, and let

$$\tau : (q - \delta, q + \delta) \rightarrow J$$

be a real-analytic continuation of the repeated-source critical branch such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in (q - \delta, q + \delta)).$$

Set

$$G(a) := D_a(\tau(a)) - \mathcal{Z}(a), \quad L(a) := \tau(a) - a,$$

and define

$$\mathcal{A}'(a) := \frac{2}{L(a)}, \quad \mathcal{G}(a) := e^{-\mathcal{A}(a)} G(a).$$

Assume that there exists $\epsilon \in \{\pm 1\}$ such that

$$\epsilon \mathcal{R}(a) > 0, \quad \epsilon G(a) > 0 \quad \text{for every } a \in (q, q + \delta).$$

Then

$$\epsilon G(q) > 0.$$

In particular, $G(q) \neq 0$. Hence, after possibly shrinking δ ,

$$\epsilon G(a) > 0 \quad \text{for every } a \in (q - \delta, q + \delta).$$

Since q is an odd-order zero of $\mathcal{R}$, the sign of $\mathcal{R}$ flips across q ; therefore on the outgoing side

$$\operatorname{sgn} G(a) = -\operatorname{sgn} \mathcal{R}(a) \quad (a \in (q - \delta, q)).$$

So a rooted crossing cannot occur at the anchor odd zero itself, nor immediately after passage across it.

Proof. By Proposition 12.301,

$$G'(a) = \frac{2}{L(a)} G(a) - \mathcal{R}(a),$$

hence

$$\mathcal{G}'(a) = -e^{-\mathcal{A}(a)} \mathcal{R}(a).$$

Assume for contradiction that $G(q) = 0$. Then $\mathcal{G}(q) = 0$, and for $a \in (q, q + \delta)$,

$$\mathcal{G}(a) = \mathcal{G}(q) + \int_q^a \mathcal{G}'(s) ds = - \int_q^a e^{-\mathcal{A}(s)} \mathcal{R}(s) ds.$$

Since $e^{-\mathcal{A}(s)} > 0$ and $\epsilon \mathcal{R}(s) > 0$ on $(q, q + \delta)$, this gives

$$\epsilon \mathcal{G}(a) < 0 \quad (a \in (q, q + \delta)),$$

hence

$$\epsilon G(a) < 0 \quad (a \in (q, q + \delta)),$$

contradicting the hypothesis. Therefore $G(q) \neq 0$. Since

$$\epsilon G(a) > 0 \quad (a \in (q, q + \delta)),$$

continuity yields

$$\epsilon G(q) \geq 0,$$

and the nonvanishing just proved forces

$$\epsilon G(q) > 0.$$

A second continuity argument then gives

$$\epsilon G(a) > 0 \quad (a \in (q - \delta, q + \delta))$$

after shrinking δ if necessary. Finally, because q is an odd-order zero of $\mathcal{R}$, its sign flips across q , so on $(q - \delta, q)$ one has

$$\operatorname{sgn} G(a) = -\operatorname{sgn} \mathcal{R}(a).$$

In particular G cannot vanish there. □

Proposition 12.317 (Weighted-defect recursion on post-anchor chambers). *Assume the setting of Proposition 12.316. Let*

$$q_1 > q_2 > \cdots > q_N$$

be successive odd-order zeros of $\mathcal{R}$ crossed by the anchor along the continued branch after the adjacent-two-odd-zero chamber, and set

$$I_j := (q_{j+1}, q_j) \quad (1 \leq j \leq N-1).$$

Assume that the continued branch exists real-analytically on

$$\bigcup_{j=1}^{N-1} I_j.$$

For each j , let

$$\epsilon_j := \operatorname{sgn}(\mathcal{R}|_{I_j}) \in \{\pm 1\}, \quad m_j(a) := \int_a^{q_j} e^{-\mathcal{A}(s)} |\mathcal{R}(s)| ds \quad (a \in I_j),$$

and

$$M_j := m_j(q_{j+1}) > 0.$$

Then on each I_j ,

$$\mathcal{G}(a) = \mathcal{G}(q_j) + \epsilon_j m_j(a).$$

Consequently:

(1) if

$$\operatorname{sgn} \mathcal{G}(q_j) = \epsilon_j,$$

then $\mathcal{G}$ has no zero on I_j ;

(2) if

$$\operatorname{sgn} \mathcal{G}(q_j) = -\epsilon_j,$$

then $\mathcal{G}$ has a zero on I_j if and only if

$$M_j > |\mathcal{G}(q_j)|,$$

and in that case the zero is unique;

(3) if $\mathcal{G}$ has no zero on I_j , then

$$\operatorname{sgn} \mathcal{G}(q_{j+1}) = \operatorname{sgn} \mathcal{G}(q_j).$$

Since $\epsilon_{j+1} = -\epsilon_j$, the chamber type

$$\operatorname{sgn} \mathcal{G}(q_j) \text{ versus } \epsilon_j$$

alternates from one post-anchor chamber to the next as long as no rooted crossing has yet occurred.

Proof. On I_j one has

$$\mathcal{G}'(a) = -e^{-\mathcal{A}(a)} \mathcal{R}(a) = -\epsilon_j e^{-\mathcal{A}(a)} |\mathcal{R}(a)|.$$

Hence for $a \in I_j$,

$$\mathcal{G}(a) = \mathcal{G}(q_j) + \int_{q_j}^a \mathcal{G}'(s) ds = \mathcal{G}(q_j) + \epsilon_j \int_a^{q_j} e^{-\mathcal{A}(s)} |\mathcal{R}(s)| ds,$$

which is the stated formula.

Now $m_j(a)$ is strictly increasing as a decreases from q_j to q_{j+1} , with

$$m_j(q_j) = 0, \quad m_j(q_{j+1}) = M_j.$$

So if

$$\operatorname{sgn} \mathcal{G}(q_j) = \epsilon_j,$$

then

$$\mathcal{G}(a) = \epsilon_j (|\mathcal{G}(q_j)| + m_j(a))$$

has fixed sign on I_j and no zero. If instead

$$\operatorname{sgn} \mathcal{G}(q_j) = -\epsilon_j,$$

then

$$\mathcal{G}(a) = \epsilon_j (-|\mathcal{G}(q_j)| + m_j(a)),$$

so a zero occurs precisely when $m_j(a) = |\mathcal{G}(q_j)|$, which happens if and only if

$$M_j > |\mathcal{G}(q_j)|.$$

Because m_j is strictly monotone, that zero is unique.

Finally, if no zero occurs on I_j , then $\mathcal{G}$ keeps one sign throughout the interval, so in particular

$$\operatorname{sgn} \mathcal{G}(q_{j+1}) = \operatorname{sgn} \mathcal{G}(q_j).$$

Since the ϵ_j alternate at successive odd zeros, the relative chamber type alternates as claimed. $\square$

Corollary 12.318 (The first post-anchor chamber is dangerous). *Let q_1 be the first odd-order zero of $\mathcal{R}$ crossed by the anchor after the adjacent-two-odd-zero chamber, and let*

$$I_1 := (q_2, q_1)$$

be the first post-anchor chamber. Assume the continued born branch extends real-analytically across q_1 and on I_1 . Then:

- (1) *no rooted crossing occurs at q_1 or immediately after crossing it;*
- (2) *on I_1 one has*

$$\operatorname{sgn} \mathcal{G}(q_1) = -\epsilon_1,$$

so I_1 is the first dangerous post-anchor chamber;

- (3) *as long as no rooted crossing has yet occurred, the post-anchor chambers alternate dangerous/safe/dangerous/safe. . . .*

In particular, the first chamber in which a rooted crossing is even eligible after the anchor-sign-preservation interval is the first post-anchor chamber. Without an additional diagonal-capture theorem, however, this eligibility does not by itself determine the odd-zero count on the active segment.

Proof. The adjacent-two-odd-zero chamber is root-free by Corollary 12.314. At its left endpoint q_1 , the anchor crosses an odd-order zero of $\mathcal{R}$. Proposition 12.316 shows that $\mathcal{G}$ keeps its incoming sign across q_1 , while $\mathcal{R}$ changes sign there. Thus on the outgoing chamber I_1 the signs are opposite:

$$\operatorname{sgn} \mathcal{G}(q_1) = -\epsilon_1.$$

The no-root statement at and immediately after q_1 is also exactly Proposition 12.316. The alternation of chamber types then follows from Proposition 12.317(3). $\square$

Remark 12.319 (No-go for purely local scalar exclusion on the first dangerous post-anchor chamber). At a rooted critical point, after normalizing the active interval to

$$x = a + sL, \quad 0 \leq s \leq 1, \quad L = \tau(a) - a,$$

the profile

$$R_a(s) := \mathcal{R}(a + sL)$$

satisfies the two local scalar moment identities

$$\int_0^1 (1-s)^2 R_a(s) ds = 0, \quad \int_0^1 s(1-s) R_a(s) ds = 0.$$

Once nonlocal-target landing is admitted, the first dangerous post-anchor chamber need not correspond to a fixed odd-zero count or to a rigid four-lobe pattern on the active segment. Even in the simplest three-lobe sign pattern

$$+, -, +$$

(or its global negative), the local scalar rooted equations are already formally realizable.

Indeed, for any $A > 0$, define

$$R_A(s) := \begin{cases} A, & 0 < s < \frac{1}{3}, \\ -\frac{7}{2}A, & \frac{1}{3} < s < \frac{2}{3}, \\ \frac{11}{2}A, & \frac{2}{3} < s < 1. \end{cases}$$

A direct calculation gives

$$\int_0^1 (1-s)^2 R_A(s) ds = 0, \quad \int_0^1 s(1-s) R_A(s) ds = 0,$$

and also

$$\int_0^1 R_A(s) ds = \int_0^1 s R_A(s) ds = \int_0^1 s^2 R_A(s) ds = A.$$

So the local rooted-point equations and the nondegenerate fold data are already formally compatible with a three-lobe profile.

Thus no argument using only the local normalized rooted profile, the two vanishing scalar moments, and the local transversality/fold signs can exclude rooting on the first dangerous post-anchor chamber. Any successful next step must use genuinely source-coupled structure invisible in the planar normalized D_a model.

Remark 12.320 (What route 2 now reduces to). The discriminant analysis first showed that a rooted point is not born *at the entry* into the adjacent-two-odd-zero chamber. The stronger anchor-sign argument showed more: on any continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

cannot cross the discriminant curve at all.

The odd-anchor analysis sharpened this further: when the anchor itself passes through an odd-order zero of $\mathcal{R}$, the integrating-factor defect

$$\mathcal{G}(a) = e^{-\mathcal{A}(a)}(D_a(\tau(a)) - \mathcal{Z}(a))$$

keeps its incoming sign across that odd anchor zero, so a rooted crossing cannot occur there or immediately after passage. The subsequent post-anchor chambers then alternate dangerous and safe, beginning with a dangerous first post-anchor chamber (Corollary 12.318).

However, two no-go facts are now clear. First, Remark 12.319 shows that already on that first dangerous post-anchor chamber, the local normalized rooted equations and fold data are formally realizable in the planar scalar model. Second, the current route-2 source package itself is still universal under the normalized lift

$$c = 1, \quad v_5 = t, \quad \lambda = t^2, \quad u_5'' = 2\mathcal{Z}$$

(Proposition 12.321), so even the source-level objects

$$\Psi_a, \quad \mathfrak{D}, \quad \mathcal{W}_{\text{src}}, \quad D_{f_a}$$

do not yet distinguish the RH branch from an arbitrary analytic normalized model.

So route 2 no longer reduces merely to a delayed no-crossing / profile-realizability problem *within the current package*. It reduces to finding the first genuinely *non-universal* source-coupled input, i.e. an ingredient not already captured by the normalized planar D_a model.

Within the present draft, the first plausible places for such new input lie beyond the current universal quartet, for example in continuum-strength provenance constraints on q over whole one-pair blocks, or in external mixed/canonical structure not already absorbed by the normalized planar model.

Proposition 12.321 (Universality threshold for the current route-2 package). *Let $I \subset \mathbf{R}$ be an interval, let*

$$\mathcal{Z} : I \rightarrow \mathbf{R}$$

be real-analytic, and set

$$\mathcal{R} := \mathcal{Z}'.$$

Define analytic source functions on I by

$$c(t) := 1, \quad v_5(t) := t, \quad \lambda(t) := t^2,$$

and choose u_5 so that

$$u_5''(t) = 2\mathcal{Z}(t).$$

Then:

(1)

$$x = \frac{v_5}{c} = t, \quad S = \frac{\lambda}{c} = t^2, \quad Y = \frac{u_5}{c} = u_5;$$

(2) for every analytic f ,

$$J_f = W(c, v_5, f) = W(1, t, f) = f'',$$

hence

$$J_\lambda = 2, \quad J_{u_5} = 2\mathcal{Z}, \quad \mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda = 4\mathcal{R};$$

(3) the repeated-source ratio is exactly the normalized planar one:

$$\Psi_a(h) = \Gamma_a(h) = D_a(h);$$

(4) the critical determinant equation is exactly the normalized planar critical-point equation:

$$\mathfrak{D}(a, h) = 0 \iff \partial_h \Psi_a(h) = 0 \iff \partial_h D_a(h) = 0;$$

(5) for the source-pinned moving shear

$$f_a := u_5 - \mathcal{Z}(a)\lambda,$$

one has

$$J_{f_a} = 2(\mathcal{Z} - \mathcal{Z}(a)),$$

and

$$D_{f_a}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \mathcal{Z}(a)).$$

Thus every analytic normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$ lifts to a genuine analytic source model for the entire current route-2 package.

Proof. The identities

$$x = \frac{v_5}{c} = t, \quad S = \frac{\lambda}{c} = t^2, \quad Y = \frac{u_5}{c} = u_5$$

are immediate. Also,

$$J_f = W(c, v_5, f) = W(1, t, f) = f'',$$

so

$$J_\lambda = 2, \quad J_{u_5} = 2\mathcal{Z}.$$

Hence

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda = 2(2\mathcal{Z}') - 2\mathcal{Z} \cdot 0 = 4\mathcal{R}.$$

Because the source coordinate and the normalized planar coordinate coincide,

$$x(t) = t,$$

the source repeated-source quotient is exactly the planar normalized quadratic envelope:

$$\Psi_a(h) = \Gamma_a(h) = D_a(h).$$

Therefore the source critical determinant equation is exactly the planar critical-point equation

$$\mathfrak{D}(a, h) = 0 \iff \partial_h D_a(h) = 0.$$

Finally,

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

satisfies

$$J_{f_a} = u_5'' - 2\mathcal{Z}(a) = 2(\mathcal{Z} - \mathcal{Z}(a)),$$

and Proposition 12.289(2) gives

$$D_{f_a}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \mathcal{Z}(a)).$$

So all route-2 source objects reduce exactly to the normalized planar ones. $\square$

Corollary 12.322 (The current route-2 package is universal under normalization). *Any configuration formally realizable by an analytic normalized planar model*

$$(\mathcal{Z}, \mathcal{R}, D_a)$$

is also formally realizable by a genuine analytic source model for the current route-2 package

$$(c, v_5, \lambda, u_5), \quad \Psi_a, \quad \mathfrak{D}, \quad \mathcal{W}_{\text{src}}, \quad D_{f_a}.$$

Accordingly, no contradiction theorem phrased only in terms of the current route-2 package can distinguish the RH branch from an arbitrary analytic normalized planar model.

Proof. This is immediate from Proposition 12.321. □

Remark 12.323 (Explicit source-level countermodel to the current route-2 package). Take

$$\mathcal{R}(t) := t(t-1)(t^2 - 4t - 2)$$

and

$$\mathcal{Z}(t) := \int_0^t \mathcal{R}(u) du = \frac{1}{5}t^5 - \frac{5}{4}t^4 + \frac{2}{3}t^3 + t^2,$$

and realize this normalized planar model by

$$c(t) = 1, \quad v_5(t) = t, \quad \lambda(t) = t^2, \quad u_5''(t) = 2\mathcal{Z}(t).$$

Let

$$q := 2 - \sqrt{6}.$$

In this genuine analytic source model, the unique born repeated-source critical branch from the odd zero 0 continues analytically down to $a = q$, lands there with a nonlocal target

$$\tau(q) \approx 0.3540339256 \neq q,$$

satisfies the anchor-sign safe relation on $(q, 0)$, crosses the odd anchor zero q with no sign reset, and then undergoes a delayed rooted crossing at

$$a_* \approx -0.5805673462, \quad \tau(a_*) \approx 0.3888336023.$$

Thus diagonal capture fails, and the whole delayed-crossing scenario is realized inside a genuine analytic source model for the current route-2 package.

Remark 12.324 (The next live target must be non-universal). By Corollary 12.322, the present lower route-2 package is exhausted: any further progress must use a genuinely non-universal source-coupled input, i.e. an ingredient not already absorbed by the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$.

However, the thread analysis shows that this non-universal input cannot be identified merely with arbitrary canonical quotient-septic geometry, nor even with the explicit direct $q^{(7)}$ -slice by itself. Indeed, Remark 12.104 shows that arbitrary canonical Δ_7 is still locally free once the lower one-pair data are fixed, and Proposition 13.8 together with Corollary 13.9 show that the direct highest-new provenance slice

$$\phi = A q^{(7)}$$

is fully universal on each generic one-pair block, even after any finite family of linear provenance/matching conditions on q .

So any viable continuation must use something stronger: either a *continuum-strength provenance law* for q on a whole block, or an external mixed/canonical theorem that constrains q beyond finite jet data. Within the present draft, the first plausible places for such genuinely non-universal input lie in the mixed / canonical package above arbitrary quotient-septic geometry.

Remark 12.325 (What remains on the nonmixed 4-point side). The preceding results now give a full local derivative-side picture of the nonmixed exact-contact geometry. Exact nonmixed tangent-plane contacts force zeros of $\mathcal{R}$ (Proposition 12.263); intervals on which $\mathcal{R}$ has fixed nonzero sign are exactly-contact sterile (Corollary 12.264); recurrent exact contacts force affine-planar collapse (Corollary 12.267); and isolated zeros of $\mathcal{R}$ are classified by parity (Proposition 12.269).

At source level, the same structure is encoded by the gauge-invariant Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor on a nonmixed arc (Proposition 12.235). Thus exact nonmixed contacts force zeros of $\mathcal{W}_{\text{src}}$, one-sign $\mathcal{W}_{\text{src}}$ -intervals are sterile, and recurrent contacts force $\mathcal{W}_{\text{src}} \equiv 0$ (Corollary 12.276).

Moreover, Proposition 12.278 lifts $\mathcal{W}_{\text{src}}$ to the 4×4 Wronskian

$$W(c, v_5, \lambda, u_5),$$

while Proposition 12.279 shows that, at the current source-level ratio/Wronskian/ 4×4 -determinant level, the odd-zero pattern of $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ is still formally unconstrained. Thus generic determinant algebra alone is no longer enough to close the nonmixed branch.

The repeated-source / first-loss lane is now also closed. For each fixed source point a , repeated-source exact contacts factor through the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently through the source-determinant quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

with derivative controlled by the osculatory determinant $\mathfrak{D}(a, h)$. The formal first-loss event rigidifies to a sheared repeated-node tangent-contact configuration (Remark 12.281), but this is no longer a live independent bottleneck. Proposition 12.291 shows that no attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at isolated odd-order zeros of $\mathcal{R}$. Moreover, Proposition 12.286 shows that for each fixed shear level γ , the corresponding sheared opposite-side 3-point branch is globally rigid on its maximal slope-overlap block; and Proposition 12.271 together with Corollary 12.272 show that near an isolated odd zero of $\mathcal{R}$, the diagonal critical branch is uniformly unrooted, so the rooted sheared repeated-node tangent-contact event itself does not occur there. Thus the sheared lane is closed both in the fixed- γ nonrecurrence direction and in the local isolated-odd-zero direction.

Accordingly, the genuinely live nonmixed 4-point burden is no longer local birth, nor repeated-source first loss, nor generic determinant sign. The positive-kernel route now shows more: on any chamber where the repeated-source segment sees exactly one sign change of $\mathcal{R}$, the born critical branch is simple, has strictly negative slope, and cannot realize the rigidified first-loss condition

$$\Gamma_a(x) = \mathcal{Z}(a).$$

At source level, the same branch is characterized by

$$\mathfrak{D}(a, h) = 0,$$

but Proposition 12.289 shows that its rooted numerator

$$D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h)$$

has a fixed signed leading term near an isolated odd zero, so the local branch is uniformly unrooted there. Thus a local branch of

$$\mathfrak{D}(a, h) = 0$$

cannot become a genuine obstruction until the source segment acquires a second odd zero (equivalently a second sign-changing zero) of

$$\mathcal{W}_{\text{src}},$$

or exits the local one-sign-change chamber. By Proposition 12.304, the positive-kernel route already excludes every opened nonmixed arc carrying at most one odd zero of $\mathcal{W}_{\text{src}}$ (equivalently of $\mathcal{R}$).

Moreover, once one follows an analytic continuation of the born branch beyond the local birth chamber, the rooted defect

$$G(a) = D_a(\tau(a)) - \mathcal{Z}(a)$$

satisfies an exact first-order transport equation, and any nondegenerate rooted point is forced to be a stationary point of the target map and a local fold of the moving-shear equation

$$A_a(x) - \mathcal{Z}(a) B_a(x) = 0$$

(Propositions 12.301 and 12.302). The discriminant analysis rules out birth of such a rooted point at entry into the adjacent-two-odd-zero chamber (Proposition 12.299), and the anchor-sign argument rules out any rooted crossing on a continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign (Proposition 12.313 and Corollary 12.314).

Beyond that, Proposition 12.316 shows that even after the anchor crosses an odd-order zero of $\mathcal{R}$, no rooted crossing can occur at that odd anchor zero itself or immediately after passage, while Corollary 12.318 shows that the ensuing post-anchor chambers alternate dangerous and safe. However, Remark 12.319 shows that the local normalized rooted equations and fold data are already formally compatible with rooting on the first dangerous post-anchor chamber. More strongly, Proposition 12.321 and Corollary 12.322 show that the entire current route-2 source package is universal under normalization, and Remark 12.323 exhibits a genuine analytic source model realizing safe adjacent-two-odd-zero behavior, nonlocal-target landing at the next odd anchor zero, odd-anchor no-reset, and delayed rooted crossing.

So the sharpened remaining nonmixed burden is no longer a delayed no-crossing / profile-realizability theorem *within the current route-2 package*. The current lower package is already exhausted, and the thread analysis shows that arbitrary canonical quotient-septic geometry is still too free, while even the explicit direct provenance slice

$$\phi = A q^{(7)}$$

is fully universal on each generic one-pair block. More strongly, any finite family of linear provenance or matching conditions on q is still too weak to exclude direct-slice recurrence geometry.

Accordingly, any further progress must use a genuinely non-universal continuum-strength provenance law for q , or else an external mixed/canonical input that constrains q on a whole block and is not already absorbed by the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$. By Theorem 12.239, such an external mixed input cannot come from a new sufficiently small local mixed 4-point obstruction near the centered diagonal base; any useful mixed contribution must therefore be genuinely global/nonlocal, or arise from deeper canonical/multi-pair structure.

Remark 12.326 (Compact status summary for the nonmixed 4-point branch). The nonmixed 4-point branch is now reduced to a source-level odd-zero problem. Locally, exact nonmixed contacts force zeros of $\mathcal{R}$, and isolated zeros are classified by parity: even-order zeros are sterile, while odd-order zeros seed local exact contacts. Repeated-source first loss is not a separate mechanism: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$. The sheared repeated-node lane is also closed in the collapse direction: any genuine diagonal realization of the rigidified sheared event already forces $\mathcal{R} \equiv 0$ locally, hence affine-planar collapse.

On a nonmixed arc, the same structure is encoded by the gauge-invariant source-level Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor. Repeated-source first loss is not a separate mechanism: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$, and the corresponding rooted sheared repeated-node event is locally absent near an isolated odd zero of $\mathcal{R}$. Near such an isolated odd zero, the born repeated-source branch

$$\mathfrak{D}(a, h) = 0$$

is uniformly unrooted at source level:

$$D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h) \neq 0$$

with a fixed signed leading term (Proposition 12.289). For each fixed shear level γ , the associated sheared opposite-side 3-point branch is globally rigid on its maximal slope-overlap block. Moreover, the positive-kernel / source-determinant route now excludes every opened nonmixed arc carrying at most one odd zero of $\mathcal{W}_{\text{src}}$: on any chamber where the repeated-source segment sees exactly one sign change of $\mathcal{R}$, the born critical branch is simple, has strictly negative slope, and cannot realize the rigidified first-loss condition. Beyond that chamber, the discriminant analysis rules out birth of a rooted point at entry into the adjacent-two-odd-zero chamber, and the anchor-sign argument rules out any crossing on a continuation interval where the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign.

The odd-anchor passage theorem shows that once the anchor itself crosses an odd-order zero of $\mathcal{R}$, the integrating-factor defect keeps its incoming sign across that odd anchor zero, so no rooted crossing can occur there or immediately after it. The subsequent post-anchor chambers alternate dangerous and safe. However, the local normalized rooted equations and the local fold/transversality data are already formally realizable on the first dangerous post-anchor chamber, and more strongly the entire current route-2 source package is universal under normalization. Even arbitrary canonical quotient-septic geometry remains too free, and the explicit direct provenance slice

$$\phi = A q^{(7)}$$

is fully universal on each generic one-pair block; indeed, even after any finite family of linear provenance/matching conditions on q , direct-slice recurrence geometry is still realizable. Thus the genuinely remaining nonmixed burden is no longer a problem inside the current route-2 or arbitrary quotient-septic package at all. Any further progress must use a genuinely non-universal continuum-strength provenance law for q , or an external mixed/canonical theorem that constrains q on a whole block and is not already absorbed by the normalized planar model. By the centered-base mixed local exhaustion theorem, this external mixed input cannot be a new local mixed 4-point septic-jet phenomenon near the diagonal base; it must be genuinely global/nonlocal or come from deeper canonical/multi-pair structure.

Remark 12.327 (Fixed-slope interval graphs, support pseudolines, and the remaining nonmixed global burden). There is a useful combinatorial reformulation of the nonadjacent pair-pair geometry on a nonmixed patch.

For fixed slope m , Lemma 12.86 shows that each ECT interval I contributes an intercept interval

$$\mathcal{B}_I(m) = \{\psi_I(x) - mx : x \in J_I\},$$

and that a line $y = mx + b$ is 2-secant to $\Gamma(I)$ exactly when

$$b \in \text{int } \mathcal{B}_I(m).$$

So for fixed m , the same-slope pair-pair incidence graph is an interval graph on the b -axis. In particular, any genuinely minimal surviving nonadjacent recurrence cannot live at one fixed slope; it must be slope-changing.

There is also a sharper support-curve reformulation. By Proposition 12.87, Corollary 12.88, and Proposition 12.89, the support curves

$$b = \tau_I(m)$$

are strictly concave pseudolines on common slope windows: each ordered pair has at most one common interior tangent slope, triple crossings obey the usual betweenness law, and each ordered pair has at most one directional or pair-pair overlap window in slope.

Thus any genuinely minimal surviving nonadjacent recurrence cannot be fixed slope, cannot be pair-repeating, and cannot disappear and later reappear on the same ordered pair.

There is also a useful endpoint-gap reformulation. For every endpoint support point (ξ, y_ξ) and every ordered target branch J , the endpoint-gap function

$$G_{\xi,J}(h) := y_\xi - y(h) - m(h)(\xi - x(h))$$

is strictly monotone on J by Lemma 12.109. So each endpoint/branch pair can contribute at most one remote tangent incidence.

These observations give the sharp current picture of the remaining nonmixed global burden: any surviving bad core must be slope-changing, pair-nonrepeating, and supported by a finite table of remote endpoint/branch incidences.

It is equally important to record the ceiling of this reduction. By Remark 13.4, intrinsic one-pair affine geometry is universal for analytic plane curves. Thus pure planarity, parity, interval-graph combinatorics, fixed-slope support order, or pseudoline combinatorics by itself cannot close the remaining branch. Any final contradiction must therefore use RH-specific source/provenance/canonical structure, not only abstract strip/support geometry.

Accordingly, the natural next global target is not another local normal-form analysis, but an RH-specific sign theorem for the remote endpoint-gap functions $G_{\xi,J}$.

Remark 12.328 (Derivative-geometry meaning of the nonmixed invariant $\mathcal{R}$). On a nonmixed patch,

$$\mathcal{R} = \frac{T_v}{K_v^2} = \frac{d}{dx} \left(\frac{Y''}{S''} \right).$$

Thus $\mathcal{R}$ measures the failure of the lifted space curve

$$x \mapsto (Y(x), x, S(x))$$

to be affine-planar at the derivative level: $\mathcal{R} = 0$ means the ratio Y''/S'' is locally constant, equivalently that the curve has vanishing affine torsion and lies in an affine plane after integrating twice. In this sense, the nonmixed 4-point branch is controlled by a single scalar derivative invariant rather than by raw coplanarity alone.

Remark 12.329 (Exact mixed reduction in terms of $\mathcal{Z}$ and $\mathcal{R}$). At a mixed lifted point

$$K_v(0) = 0, \quad T_v(0) = 0,$$

set

$$r := \text{ord}_0 K_v, \quad s := \text{ord}_0 T_v.$$

By Proposition 12.232,

$$K_v = (x')^3 S'',$$

so on a good v_5 -patch one has

$$S''(x) = \kappa x^r + O(x^{r+1}), \quad \kappa \neq 0.$$

After removing affine terms and integrating twice,

$$S(x) = \frac{\kappa}{(r+1)(r+2)} x^{r+2} + O(x^{r+3}).$$

Thus the mixed exponent m of the monomial model

$$Q_v(x) \sim (x^n, x, x^m)$$

is exactly

$$m = r + 2.$$

Now let n be the first degree of $Y(x)$ not absorbed by $\text{span}\{1, x, S\}$, so that

$$Y(x) = \alpha x^n + O(x^{n+1}), \quad \alpha \neq 0, \quad n > r + 2.$$

Using

$$T_v = (x')^6 (S''Y''' - S'''Y''),$$

one finds

$$s = n + r - 3, \quad n = s - r + 3.$$

Equivalently, on the nonmixed complement of $K_v = 0$,

$$\mathcal{Z} = \frac{Y''}{S''}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}$$

satisfy the order bookkeeping

$$\text{ord}_0 \mathcal{Z} = n - r - 2 = s - 2r + 1, \quad \text{ord}_0 \mathcal{R} = s - 2r.$$

Thus the dangerous low-order mixed models are finite and explicit:

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2),$$

or

$$(r, \text{ord}_0 \mathcal{R}) = (1, 0), (1, 2), (2, 1).$$

So the mixed lifted 4-point endgame has also been reduced to a finite jet obstruction list, now phrased in terms of the invariant pair

$$(\mathcal{Z}, \mathcal{R}).$$

Remark 12.330 (Exact finite obstruction coefficients for the low-order 4-point branch). The explicit low-order 4-point obstruction list can be reduced to a finite jet package.

Nonmixed side. Suppose $K_v(0) \neq 0$, and write

$$s_k := S^{(k)}(0), \quad y_k := Y^{(k)}(0), \quad s_2 \neq 0.$$

Then the Taylor expansion

$$\mathcal{Z}(x) = z_0 + z_1x + z_2x^2 + z_3x^3 + z_4x^4 + z_5x^5 + z_6x^6 + O(x^7)$$

has canonical even coefficients

$$z_2, \quad z_4, \quad z_6,$$

and the dangerous nonmixed low-order cases are exactly:

$$n = 4 \iff z_2 \neq 0, \quad n = 6 \iff z_2 = 0, z_4 \neq 0, \quad n = 8 \iff z_2 = z_4 = 0, z_6 \neq 0.$$

Explicitly,

$$z_2 = \frac{s_2^2 y_4 - 2s_2 s_3 y_3 + (2s_3^2 - s_2 s_4) y_2}{2s_2^3},$$

$$z_4 = \frac{s_2^4 y_6 - s_2^3 (4s_3 y_5 + 6s_4 y_4 + 4s_5 y_3 + s_6 y_2) + 2s_2^2 (6s_3^2 y_4 + 12s_3 s_4 y_3 + 4s_3 s_5 y_2 + 3s_4^2 y_2) - 12s_2 s_3^2 (2s_3 y_3 + 3s_4 y_2)}{24s_2^5}$$

and

$$z_6 = \frac{\mathcal{P}_8(s_2, \dots, s_8; y_2, \dots, y_8)}{720s_2^7},$$

where $\mathcal{P}_8$ is the explicit degree-8 numerator polynomial obtained by Taylor division of $\mathcal{Z} = Y''/S''$.

Mixed side. Suppose $K_v(0) = T_v(0) = 0$, and let

$$r := \text{ord}_0 K_v.$$

For the two low-order mixed cases $r = 1, 2$, Taylor division of $\mathcal{Z} = Y''/S''$ gives the exact obstruction coefficients

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2},$$

defined by the expansions

$$\mathcal{Z}(x) = \frac{y_2}{s_3} \frac{1}{x} + \left(\frac{y_3}{s_3} - \frac{s_4 y_2}{2s_3^2} \right) + \kappa_{1,1}x + \kappa_{1,2}x^2 + \kappa_{1,3}x^3 + O(x^4) \quad (r = 1),$$

and

$$\mathcal{Z}(x) = \frac{2y_2}{s_4} \frac{1}{x^2} + \left(\frac{2y_3}{s_4} - \frac{2s_5y_2}{3s_4^2} \right) \frac{1}{x} + \left(\frac{y_4}{s_4} - \frac{2s_5y_3}{3s_4^2} - \frac{s_6y_2}{6s_4^2} + \frac{2s_5^2y_2}{9s_4^3} \right) + \kappa_{2,1}x + \kappa_{2,2}x^2 + O(x^3) \quad (r = 2).$$

The dangerous mixed low-order list is then exactly:

$$\begin{aligned} (m, n) = (3, 4) &\iff \kappa_{1,1} \neq 0, \\ (m, n) = (3, 6) &\iff \kappa_{1,3} \neq 0, \\ (m, n) = (4, 6) &\iff \kappa_{2,2} \neq 0. \end{aligned}$$

Thus the full currently explicit *low-order* 4-point obstruction list is equivalent to the finite set of coefficients

$$\boxed{z_2, z_4, z_6, \kappa_{1,1}, \kappa_{1,3}, \kappa_{2,2}}.$$

On the mixed side, the three coefficients

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2}$$

exhaust the presently explicit low-order obstruction list.

On the nonmixed side, however, the parity classification from Proposition 12.269 shows that local exact-contact birth occurs at every isolated odd-order zero of $\mathcal{R}$. Accordingly,

$$z_2, \quad z_4, \quad z_6$$

should now be viewed only as the first three explicit low-order nonmixed obstruction coefficients, not as a complete local nonmixed list.

Remark 12.331 (Wronskian-order reformulation of the mixed low-order list). On a good v_5 -patch one has

$$Y'' = \frac{c^3}{M^3} J_{u_5}, \quad S'' = \frac{c^3}{M^3} J_\lambda, \quad \mathcal{Z} = \frac{J_{u_5}}{J_\lambda}, \quad \lambda = \frac{\Delta_7}{v_5}.$$

Thus, if

$$r := \text{ord}_0 J_\lambda = \text{ord}_0 K_v, \quad q := \text{ord}_0 J_{u_5} = \text{ord}_0 Y'',$$

then

$$m = r + 2, \quad n = q + 2, \quad \text{ord}_0 \mathcal{Z} = q - r.$$

Hence the dangerous mixed models

$$(m, n) = (3, 4), (3, 6), (4, 6)$$

are exactly the Wronskian order-pairs

$$(\text{ord}_0 J_\lambda, \text{ord}_0 J_{u_5}) = (1, 2), (1, 4), (2, 4).$$

If

$$J_\lambda(x) = x^r A(x), \quad J_{u_5}(x) = x^r B(x), \quad A(0) \neq 0,$$

then

$$\mathcal{Z}(x) = \frac{B(x)}{A(x)}.$$

For $r = 1$, the coefficients $\kappa_{1,1}$ and $\kappa_{1,3}$ are exactly the x^1 - and x^3 -coefficients of the regularized ratio B/A ; for $r = 2$, $\kappa_{2,2}$ is exactly the x^2 -coefficient of B/A .

Moreover, since

$$J_c = J_{v_5} = 0,$$

replacing

$$u_5 \mapsto u_5 + a c + b v_5 + \gamma \lambda$$

changes J_{u_5} only by

$$J_{u_5} \mapsto J_{u_5} + \gamma J_\lambda.$$

Thus the nonconstant Taylor coefficients of $\mathcal{Z}$ are canonical. In particular, the currently explicit mixed low-order list is already a finite Wronskian-ratio obstruction list.

Remark 12.332 (Regularized source scalar for the mixed residual quotient). In the notation of Remark 12.331, write locally

$$J_\lambda(x) = x^r A(x), \quad J_{u_5}(x) = x^r B(x), \quad A(0) \neq 0,$$

and set

$$Q(x) := \frac{B(x)}{A(x)} = \frac{J_{u_5}(x)}{J_\lambda(x)}.$$

Then

$$Q'(x) = \frac{J_\lambda J'_{u_5} - J_{u_5} J'_\lambda}{J_\lambda^2} = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2},$$

where

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda$$

is the source-level Wronskian from Proposition 12.235.

Moreover, under the affine gauge change

$$u_5 \mapsto u_5 + \alpha c + \beta v_5 + \gamma \lambda$$

one has

$$J_{u_5} \mapsto J_{u_5} + \gamma J_\lambda, \quad Q \mapsto Q + \gamma,$$

so

$$Q'(x) = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}$$

is gauge-invariant. Thus, modulo the trivial additive constant in the regularized ratio Q , the mixed residual quotient germ is already controlled by the single scalar

$$\mathcal{D} := Q' = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

On the nonmixed complement of $K_v = 0$, Remark 12.234 and Proposition 12.235 give

$$\mathcal{D} = \frac{M}{c^2} \mathcal{R}.$$

So the exact fixed-shear residual lane is governed by the same regularized source scalar already present on the nonmixed side, rather than by a second primitive pointwise highest-new field.

Proposition 12.333 (Projective secant-shadow reduction on an overlap patch). *Let*

$$U \subseteq \{c \neq 0, u_5 \neq 0, v_5 \neq 0, M \neq 0\}$$

be a good overlap patch. Set

$$\lambda := \frac{\Delta_7}{v_5}, \quad x := \frac{v_5}{c}, \quad Y := \frac{u_5}{c}, \quad S := \frac{\lambda}{c} = \frac{\Delta_7}{c v_5}.$$

Assume there exists a close exact mixed two-point bad branch

$$(h_{1,n}, h_{2,n}, \Lambda_n), \quad h_{1,n} \neq h_{2,n}, \quad h_{1,n}, h_{2,n} \in U, \quad h_{1,n}, h_{2,n} \rightarrow m \in U,$$

and let $(\eta_n, \xi_n, \gamma_n)$ be the branch-generated moving λ -shears, so that

$$\begin{aligned} (c - \eta_n \lambda)(h_{1,n}) &= \Lambda_n(c - \eta_n \lambda)(h_{2,n}), \\ (v_5 - \xi_n \lambda)(h_{1,n}) &= \Lambda_n(v_5 - \xi_n \lambda)(h_{2,n}), \\ (u_5 - \gamma_n \lambda)(h_{1,n}) &= \Lambda_n(u_5 - \gamma_n \lambda)(h_{2,n}). \end{aligned}$$

For each n , set

$$A_n := c - \eta_n \lambda, \quad B_n := v_5 - \xi_n \lambda, \quad F_n := u_5 - \gamma_n \lambda.$$

Then, after discarding finitely many terms, the branch determines the exact projective secant shadow

$$\Sigma(h_{1,n}, h_{2,n}) := [x_1 S_2 - x_2 S_1 : Y_1 S_2 - Y_2 S_1 : S_2 - S_1],$$

where

$$x_i = x(h_{i,n}), \quad Y_i = Y(h_{i,n}), \quad S_i = S(h_{i,n}).$$

More precisely, for all large n one has the exact identity

$$[B_n(h_{1,n}) : F_n(h_{1,n}) : A_n(h_{1,n})] = [B_n(h_{2,n}) : F_n(h_{2,n}) : A_n(h_{2,n})] = \Sigma(h_{1,n}, h_{2,n}).$$

In particular, the secant shadow is independent of the moving shears $(\eta_n, \xi_n, \gamma_n)$ and of Λ_n .

Equivalently, in raw coordinates,

$$\Sigma(h_1, h_2) = [v_{5,1}\lambda_2 - v_{5,2}\lambda_1 : u_{5,1}\lambda_2 - u_{5,2}\lambda_1 : c_1\lambda_2 - c_2\lambda_1].$$

As $h_2 \rightarrow h_1 = a$, the secant shadow converges projectively to the tangent shadow

$$\Pi(a) := [xS_x - S : YS_x - SY_x : S_x]_{|x=x(a)}.$$

A direct calculation gives

$$\Pi(x) \wedge \partial_x \Pi(x) = S(x) (S_x Y_{xx} - S_{xx} Y_x, S_{xx}, -S_{xx}).$$

Hence Π is projectively immersive wherever

$$S \neq 0 \quad \text{and} \quad S_{xx} \neq 0.$$

Since on the v_5 -patch

$$K_v = (x')^3 S_{xx},$$

any such close exact mixed two-point bad branch can accumulate only on

$$\{S = 0\} \cup \{K_v = 0\}.$$

Proof. Since

$$x' = \left(\frac{v_5}{c}\right)' = \frac{M}{c^2},$$

the hypothesis $M \neq 0$ implies that $x = v_5/c$ is a local coordinate near m . After discarding finitely many terms we may therefore assume

$$x(h_{1,n}) \neq x(h_{2,n}).$$

Fix such an n , and abbreviate

$$\eta = \eta_n, \quad \xi = \xi_n, \quad \gamma = \gamma_n.$$

Write

$$p_i := (x_i, Y_i, 1) \in \mathbf{R}^3, \quad V := (\xi, \gamma, \eta) \in \mathbf{R}^3, \quad i = 1, 2.$$

Since

$$A_n = c(1 - \eta S), \quad B_n = c(x - \xi S), \quad F_n = c(Y - \gamma S),$$

the exact repeated-source equations imply

$$[B_n(h_{1,n}) : F_n(h_{1,n}) : A_n(h_{1,n})] = [p_1 - S_1 V]$$

and

$$[B_n(h_{2,n}) : F_n(h_{2,n}) : A_n(h_{2,n})] = [p_2 - S_2 V].$$

Because these two projective points are equal, there is a common projective value.

If $S_1 = S_2$, then

$$[p_1 - S_1 V] = [p_2 - S_1 V]$$

forces $p_1 = p_2$, hence $x_1 = x_2$, contradicting the choice of local coordinate. Therefore

$$S_1 \neq S_2.$$

Subtracting

$$p_1 - S_1 V = p_2 - S_2 V$$

gives

$$p_1 - p_2 = (S_1 - S_2)V,$$

hence

$$V = \frac{p_1 - p_2}{S_1 - S_2}.$$

Substituting back,

$$p_1 - S_1 V = p_1 - S_1 \frac{p_1 - p_2}{S_1 - S_2} = \frac{S_2 p_1 - S_1 p_2}{S_2 - S_1}.$$

Therefore the common projective value is exactly

$$[S_2 p_1 - S_1 p_2] = [x_1 S_2 - x_2 S_1 : Y_1 S_2 - Y_2 S_1 : S_2 - S_1],$$

which is the asserted secant shadow and is independent of $(\eta_n, \xi_n, \gamma_n)$ and of Λ_n . The raw-coordinate formula is the same identity multiplied through by $c_1 c_2$.

Now divide the homogeneous secant vector by $x_2 - x_1$ and let $h_2 \rightarrow h_1 = a$. The three components converge to

$$xS_x - S, \quad YS_x - SY_x, \quad S_x,$$

so the secant shadow converges projectively to

$$\mathbf{\Pi}(a) = [xS_x - S : YS_x - SY_x : S_x].$$

Differentiating $\mathbf{\Pi}$ in the local x -coordinate gives

$$\mathbf{\Pi}(x) \wedge \partial_x \mathbf{\Pi}(x) = S(x) (S_x Y_{xx} - S_{xx} Y_x, S_{xx}, -S_{xx}).$$

Therefore $\mathbf{\Pi}$ is projectively immersive wherever $S \neq 0$ and $S_{xx} \neq 0$. On a close bad branch the secant shadows converge to tangent shadows, so accumulation is possible only where immersion fails, i.e. on

$$\{S = 0\} \cup \{S_{xx} = 0\}.$$

Finally, on the v_5 -patch one has

$$K_v = (x')^3 S_{xx},$$

so $\{S_{xx} = 0\} = \{K_v = 0\}$ on U . □

Remark 12.334 (Useful overlap scalar). On the same overlap patch, set

$$\tau_u := -\frac{\Delta_7}{c u_5}, \quad \mathcal{G} := x' S_x + Y' \tau'_u.$$

Then $\mathcal{G}$ admits the exact factorization

$$\mathcal{G} = -\frac{W(u_5, v_5) W(u_5, \lambda)}{c^2 u_5^2}, \quad \lambda = \frac{\Delta_7}{v_5},$$

where $W(f, g) = fg' - f'g$. Equivalently,

$$\mathcal{G} = \frac{u_5 v'_5 - u'_5 v_5}{c^2} \left(\frac{\Delta_7}{u_5 v_5} \right)'.$$

In particular, $\mathcal{G} \neq 0$ on any connected overlap patch contained in $\{c \neq 0, M \neq 0\}$: the factor $W(u_5, v_5)$ cannot vanish identically there without forcing linear dependence of

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c},$$

and $W(u_5, \lambda)$ cannot vanish identically there without forcing

$$\Delta_7 = \kappa u_5 v_5,$$

which is incompatible with the direct septic slice carried by Δ_7 . This does not yet exclude the mixed two-point branch, but it records the first genuinely nontrivial Y - S overlap coupling scalar exposed by the current attack.

At highest-new weight, this scalar remains in the same differential tower as the direct septic source field:

$$\mathcal{G}^{[1]} = -\frac{W(u_5, v_5)}{c^2 u_5^2} W(u_5, A q^{(7)}), \quad A = \kappa \eta_2.$$

So $\mathcal{G}$ does not produce a second primitive pointwise highest-new generator. It does, however, provide the first useful external-looking rank-2 pointwise coupling on the overlap patch: together,

$$\lambda^{[1]} = A q^{(7)} \quad \text{and} \quad \mathcal{G}^{[1]}$$

recover $(q^{(7)}, (q^{(7)})')$ generically.

Proposition 12.335 (Same-tower closure for the current explicit highest-new overlap/canonical package). *Work on a good overlap patch*

$$U \subseteq \{c \neq 0, u_5 \neq 0, v_5 \neq 0, M \neq 0\}.$$

Let K_{low} denote the differential field generated by the lower controlled fields on U , in particular by

$$c, u_5, v_5, M, A = \kappa\eta_2$$

and their analytic derivatives, and write

$$r := q^{(7)}, \quad x := \frac{v_5}{c}, \quad Y := \frac{u_5}{c}, \quad S := \frac{\Delta_7}{c v_5} = \frac{\lambda}{c}.$$

Then every currently explicit pointwise highest-new field obtained from the quotient-septic package (Δ_7, λ) , the overlap fields (x, Y, S, τ_u) , the dual tangent-line fields $(\mu, \lambda, \beta, m, W)$, and the exact fixed-shear overlap constructions by finitely many algebraic operations, x -derivatives, Wronskians, finite jet extractions, and projectivizations lies in the differential algebra

$$K_{\text{low}}\langle r \rangle.$$

More precisely, the weight-one part of each such field is K_{low} -linear in finitely many x -jets

$$\partial_x^j S^{[1]}, \quad j = 0, \dots, N,$$

for some finite N , and

$$S^{[1]} = \frac{A}{c} r.$$

In particular, the currently explicit pointwise highest-new fields

$$\Delta_7^{[1]}, \lambda^{[1]}, \beta^{[1]}, m^{[1]}, W^{[1]}, S^{[1]}, \mathcal{G}^{[1]}, J_\lambda^{[1]}, \mathcal{W}_{\text{src}}^{[1]},$$

the projected fixed-shear germ at weight one, and the quartic–sextic coefficient $\kappa_{2,2}$ are all same-tower. Hence no second primitive pointwise highest-new generator appears in the current explicit overlap/canonical package. This is a statement about the present explicit pointwise package only, and does not address nonlocal, relational, or downstream transported-state obstructions.

Proof. On the overlap patch, the lower fields x, Y, μ, c, u_5, v_5 , and the coordinate-change factors are already fixed at weight zero. Thus all highest-new weight enters only through

$$\lambda^{[1]} = A r, \quad S^{[1]} = \frac{\lambda^{[1]}}{c} = \frac{A}{c} r.$$

Linearizing any allowed algebraic/differential expression at weight one kills quadratic interactions among highest-new pieces, so the weight-one part is a K_{low} -linear combination of finitely many x -jets of $S^{[1]}$. Since

$$\partial_x = \frac{c^2}{M} \partial_h$$

on U , each $\partial_x^j S^{[1]}$ lies in $K_{\text{low}}\langle r \rangle$. Projectivization only divides by lower nonvanishing factors, and finite jet coefficients such as $\kappa_{2,2}$ are extracted from finitely many such jets. The stated list therefore remains same-tower. $\square$

Corollary 12.336 (Best current external-looking rank-two coupling). *On the same overlap patch,*

$$\begin{pmatrix} \lambda^{[1]} \\ \mathcal{G}^{[1]} \end{pmatrix} = \begin{pmatrix} A & 0 \\ -\frac{W(u_5, v_5)}{c^2 u_5^2} (u_5 A' - u_5' A) & -\frac{W(u_5, v_5)}{c^2 u_5} A \end{pmatrix} \begin{pmatrix} r \\ r' \end{pmatrix}.$$

In particular,

$$\det \begin{pmatrix} A & 0 \\ -\frac{W(u_5, v_5)}{c^2 u_5^2} (u_5 A' - u_5' A) & -\frac{W(u_5, v_5)}{c^2 u_5} A \end{pmatrix} = -\frac{W(u_5, v_5)}{c^2 u_5} A^2.$$

So wherever

$$A \neq 0, \quad u_5 \neq 0, \quad W(u_5, v_5) \neq 0,$$

the pair $(\lambda^{[1]}, \mathcal{G}^{[1]})$ is a genuine rank-two pointwise coupling to (r, r') . It does not produce a new generator, but it is the best current external-looking rank-two coupling on the overlap patch.

Proof. This is the explicit highest-new expansion of Remark 12.334, using

$$\lambda^{[1]} = Ar, \quad \mathcal{G}^{[1]} = -\frac{W(u_5, v_5)}{c^2 u_5^2} (u_5 A r' + (u_5 A' - u_5' A) r).$$

□

Remark 12.337 (Work in progress: explicit-pointwise bridge to the local good-patch detector). Proposition 12.335 and Corollary 12.336 show that the present explicit highest-new package has only one currently visible pointwise source, namely the same tower generated by

$$r = q^{(7)},$$

even though the pair

$$(\lambda^{[1]}, \mathcal{G}^{[1]})$$

recovers (r, r') generically on the overlap patch.

This shifts the next bridge question. The issue is no longer whether another explicit pointwise highest-new field appears, but whether this single explicit pointwise direction is already seen by the local good-patch compensator. If it is, then the first same-order local compensator should already be quotient-septic-visible; if it is not, then the entire current explicit pointwise package is invisible to that local detector, and any surviving obstruction must already lie in nonlocal transport, collision-functoriality failure, or other provenance-sensitive two-atom structure.

At present this is a theorem target, not a proved equivalence.

Remark 12.338 (Ceiling of the current pointwise highest-new package). The present explicit highest-new audit has now reached a stable negative conclusion. All currently visible pointwise highest-new fields — including

$$\Delta_7^{[1]}, \lambda^{[1]}, \beta^{[1]}, m^{[1]}, W^{[1]}, S^{[1]}, \mathcal{G}^{[1]}$$

and the quartic–sextic transported coefficient package — remain in the differential algebra generated by the lower data together with

$$r = q^{(7)}.$$

So no second primitive explicit pointwise highest-new generator has emerged.

Combined with Remark 12.103 and Remark 12.129, this means that the honest remaining order-7/two-point burden is not “one more local field” and not the naive source-summed corrected-block route. What remains is package-level and provenance-sensitive: for example a reduced-image-germ theorem or collision-functoriality strong enough to force quotient-septic collapse.

Proposition 12.339 (Exact fixed-shear overlap branches off $K_v = 0$ force a fold at $S = 0$). Work on the overlap patch of Proposition 12.333, and write

$$\Phi(x) := \left(\frac{x - \xi_* S}{1 - \eta_* S}, \frac{Y - \gamma_* S}{1 - \eta_* S} \right), \quad S = \frac{\Delta_7}{c v_5}.$$

Assume there is a close exact fixed-shear mixed 2-point branch

$$x_{1,n} \neq x_{2,n}, \quad x_{1,n}, x_{2,n} \rightarrow a, \quad \Phi(x_{1,n}) = \Phi(x_{2,n}) \quad (n \geq 1).$$

If

$$K_v(a) \neq 0,$$

then

$$S(a) = 0, \quad \Phi'(a) = 0, \quad S_x(a) \neq 0.$$

More precisely,

$$0 = \Phi'_1(a) = 1 - (\xi_* - \eta_* x(a)) S_x(a),$$

$$0 = \Phi'_2(a) = Y_x(a) - (\gamma_* - \eta_* Y(a))S_x(a).$$

In particular, setting

$$\beta := \xi_* - \eta_* x(a) = \frac{1}{S_x(a)},$$

and centering the local x -coordinate at a , the first-coordinate fold model

$$U(x) := (x - x(a)) - \beta S(x)$$

satisfies

$$U(a) = 0, \quad U_x(a) = 0, \quad U_{xx}(a) = -\beta S_{xx}(a) \neq 0.$$

Hence, after analytic source reparametrization, the first coordinate of the fixed-shear projected germ is

$$t^2.$$

So any exact fixed-shear mixed branch away from $K_v = 0$ is already forced into the same fold regime that underlies the nonmixed odd-zero analysis. In particular, the remaining off- $K_v = 0$ burden is best understood as an absorption problem into the existing nonmixed odd-zero / affine-planar package, rather than as a search for a new mixed local carrier.

Proof. A fixed-shear branch is, in particular, a close exact mixed two-point bad branch on the same overlap patch. By Proposition 12.333, any accumulation point of such a branch lies in

$$\{S = 0\} \cup \{K_v = 0\}.$$

So if $K_v(a) \neq 0$, then necessarily

$$S(a) = 0.$$

Because

$$\Phi(x_{1,n}) = \Phi(x_{2,n}), \quad x_{1,n} \neq x_{2,n}, \quad x_{1,n}, x_{2,n} \rightarrow a,$$

the mean value theorem applied to each coordinate of Φ gives points

$$\zeta_{1,n}, \zeta_{2,n} \in (x_{1,n}, x_{2,n})$$

such that

$$\Phi'_1(\zeta_{1,n}) = 0, \quad \Phi'_2(\zeta_{2,n}) = 0.$$

Passing to the limit yields

$$\Phi'(a) = 0.$$

Differentiating the two coordinates of Φ gives

$$\begin{aligned} \Phi'_1 &= \frac{1 - (\xi_* - \eta_* x)S_x - \eta_* S}{(1 - \eta_* S)^2}, \\ \Phi'_2 &= \frac{Y_x - (\gamma_* - \eta_* Y)S_x - \eta_* S Y_x}{(1 - \eta_* S)^2}. \end{aligned}$$

Evaluating at $x = a$ and using $S(a) = 0$ yields the displayed identities

$$\begin{aligned} 0 &= \Phi'_1(a) = 1 - (\xi_* - \eta_* x(a))S_x(a), \\ 0 &= \Phi'_2(a) = Y_x(a) - (\gamma_* - \eta_* Y(a))S_x(a). \end{aligned}$$

The first identity implies

$$S_x(a) \neq 0 \quad \text{and} \quad \beta := \xi_* - \eta_* x(a) = \frac{1}{S_x(a)}.$$

Now center the local x -coordinate at a and set

$$U(x) := (x - x(a)) - \beta S(x).$$

Then

$$U(a) = 0, \quad U_x(a) = 1 - \beta S_x(a) = 0, \quad U_{xx}(a) = -\beta S_{xx}(a).$$

Since on the v_5 -patch

$$K_v = (x')^3 S_{xx}, \quad x' = \frac{M}{c^2} \neq 0,$$

the hypothesis $K_v(a) \neq 0$ implies

$$S_{xx}(a) \neq 0.$$

Hence

$$U_{xx}(a) \neq 0.$$

By the analytic Morse lemma in one variable, after analytic source reparametrization the first coordinate is t^2 . $\square$

Proposition 12.340 (Off-septic-zero K_v -accumulation is already mixed low-order). *Work on the overlap patch of Proposition 12.333. Assume there exists a close exact mixed two-point bad branch accumulating at a point a such that*

$$K_v(a) = 0, \quad S(a) \neq 0.$$

Then

$$Y_{xx}(a) = 0.$$

Equivalently, on the v_5 -patch one has

$$J_\lambda(a) = 0, \quad J_{u_5}(a) = 0.$$

So any such accumulation already lies in the existing mixed low-order obstruction regime.

Proof. By Proposition 12.333, any close exact mixed two-point bad branch determines the tangent shadow

$$\Pi(x) = [xS_x - S : YS_x - SY_x : S_x]$$

with projective immersion defect

$$\Pi(x) \wedge \partial_x \Pi(x) = S(x)(S_x Y_{xx} - S_{xx} Y_x, S_{xx}, -S_{xx}).$$

At a point a with $K_v(a) = 0$, one has $S_{xx}(a) = 0$ on the v_5 -patch, so this simplifies to

$$\Pi(a) \wedge \partial_x \Pi(a) = S(a) Y_{xx}(a) (S_x(a), 0, S(a) - x(a)S_x(a)).$$

If $S(a) \neq 0$ and $Y_{xx}(a) \neq 0$, then the vector on the right is nonzero, so Π is projectively immersive at a , contradicting accumulation of secant shadows there. Hence

$$Y_{xx}(a) = 0.$$

On the v_5 -patch,

$$J_\lambda = \frac{M^3}{c^3} S_{xx}, \quad J_{u_5} = \frac{M^3}{c^3} Y_{xx},$$

so $K_v(a) = 0$ implies $J_\lambda(a) = 0$, and the displayed vanishing of $Y_{xx}(a)$ gives $J_{u_5}(a) = 0$. $\square$

Corollary 12.341 (Simple septic-zero fixed-shear branches off $K_v = 0$ transfer to nonmixed four-point geometry). *Work on the overlap patch of Proposition 12.339. Assume there is a close exact fixed-shear mixed 2-point branch*

$$x_{1,n} \neq x_{2,n}, \quad x_{1,n}, x_{2,n} \rightarrow a, \quad \Phi(x_{1,n}) = \Phi(x_{2,n}),$$

and

$$K_v(a) \neq 0.$$

Then, after shrinking to a sufficiently small neighborhood of a , this lane produces a local interval-persistent nonmixed 4-point family for the lifted curve

$$Q(x) = (x, Y(x), S(x)).$$

Consequently it is absorbed by the existing nonmixed 4-point theory; in particular, it is not a surviving genuinely mixed 2-point derivative-geometric obstruction.

Proof. By Proposition 12.339,

$$S(a) = 0, \quad S_x(a) \neq 0,$$

and the first coordinate of the fixed-shear projected germ is in fold normal form. Thus, after analytic source reparametrization, there is a local parameter t with

$$\Phi(x(t)) = (t^2, \mathcal{V}(t))$$

and repeated nearby values force

$$\mathcal{V}(t) = \mathcal{V}(-t)$$

for all sufficiently small t . Equivalently, in the actual local septic coordinate $s := S$, there are even analytic germs E_x, E_Y such that

$$\frac{x(s) - \xi_* s}{1 - \eta_* s} = E_x(s), \quad \frac{Y(s) - \gamma_* s}{1 - \eta_* s} = E_Y(s), \quad E_x(-s) = E_x(s), \quad E_Y(-s) = E_Y(s).$$

Define

$$Q(s) := (x(s), Y(s), s).$$

If $\eta_* \neq 0$, set

$$P := \left(\frac{\xi_*}{\eta_*}, \frac{\gamma_*}{\eta_*}, \frac{1}{\eta_*} \right).$$

Then the parity formulas imply that $Q(s), Q(-s), P$ are collinear for all small s . If $\eta_* = 0$, then

$$Q(s) - Q(-s) = 2s(\xi_*, \gamma_*, 1),$$

so all symmetric secants are parallel to one fixed direction, which is the same projective statement with a point at infinity. Hence for every small $0 < s_1 < s_2$, the four points

$$Q(-s_2), Q(-s_1), Q(s_1), Q(s_2)$$

are coplanar.

Finally, by Proposition 12.339,

$$x_s(a) = \frac{1}{S_x(a)} \neq 0,$$

so x is monotone in s near a . Thus the four coplanar points above lie on one local x -arc in the nonmixed ordered pattern, yielding a local interval-persistent nonmixed 4-point family. The final claim follows from Proposition 12.266. $\square$

Remark 12.342 (Exact shadow-adapted freezing on the simple septic-zero slice). Work on a good overlap patch and suppose

$$S = \frac{\Delta_7}{c v_5}$$

is a local coordinate near a simple septic-zero point a , so

$$S(a) = 0, \quad S_x(a) \neq 0.$$

Write $s := S$ and let

$$C(s) := (x(s), Y(s)), \quad x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}.$$

For any distinct nearby secant pair $s_1 \neq s_2$, define

$$\xi(s_1, s_2) := \frac{x(s_1) - x(s_2)}{s_1 - s_2}, \quad \gamma(s_1, s_2) := \frac{Y(s_1) - Y(s_2)}{s_1 - s_2}.$$

Then one has the exact identities

$$x(s_1) - \xi(s_1, s_2) s_1 = x(s_2) - \xi(s_1, s_2) s_2,$$

$$Y(s_1) - \gamma(s_1, s_2) s_1 = Y(s_2) - \gamma(s_1, s_2) s_2.$$

Equivalently, on the simple septic-zero slice every exact secant-shadow pair already carries an exact fixed-shear normalization with

$$\eta = 0, \quad \xi = \xi(s_1, s_2), \quad \gamma = \gamma(s_1, s_2).$$

Thus the apparently projective, non-affinely-frozen septic-zero lane is not a separate carrier mechanism on the simple $S = 0$ slice: in the shadow-adapted chart it freezes exactly and therefore falls under the same exact fixed-shear analysis used in Corollary 12.341.

Remark 12.343 (Status of the mixed λ -shear route). Proposition 12.333 shows that the downstream secant-shadow reduction is already exact and fully projective: a close exact mixed two-point bad branch determines the projective secant shadow

$$\Sigma(h_1, h_2) = [x_1 S_2 - x_2 S_1 : Y_1 S_2 - Y_2 S_1 : S_2 - S_1], \quad S = \frac{\Delta_7}{c v_5},$$

independently of any affine freezing of the branch-generated λ -shears. On the simple septic-zero slice $S = 0$, $S_x \neq 0$, there is moreover a canonical shadow-adapted exact freezing with $\eta = 0$; see Remark 12.342.

The overlap geometry is now sharply split into three pieces.

First, Proposition 12.340 shows that if a close mixed branch accumulates at a point with

$$K_v = 0, \quad S \neq 0,$$

then

$$J_\lambda = J_{u_5} = 0,$$

so this part of the $K_v = 0$ locus is already absorbed by the existing mixed low-order obstruction package.

Second, Corollary 12.341 shows that in the conditional exact fixed-shear overlap lane, the slice

$$\{S = 0\} \cap \{K_v \neq 0\}$$

is not a surviving mixed 2-point carrier lane: it transfers into the established nonmixed 4-point geometry.

Thus the only genuinely residual exact fixed-shear mixed corner is

$$\{S = 0\} \cap \{K_v = 0\}.$$

On that corner, Proposition 12.353 shows that every finite parity-normalized transport jet descends analytically to the quotient variable $u = t^2$. In the quartic–sextic case, Corollary 12.354 shows that every finite quotient-visible transport state is locally an analytic function of Q alone, and Corollary 12.355 rules out any third local reset beyond the involution symmetry. So the residual branch is no longer merely quotient-diagonal / involutive; its finite transport state has local rank 1.

If the actual corrected cubic/quintic defects are state-local in this descended transport state, then Proposition 12.356 already excludes the corner. Under the stronger natural merger law, Corollary 12.357 shows that the interaction remainder dies identically along the involutive branch. Thus any still-live mixed two-point obstruction on

$$\{S = 0\} \cap \{K_v = 0\}$$

must come from genuinely relational or provenance-sensitive two-atom data beyond the descended transport state.

Remark 12.344 (Conditional parity reduction on the exact fixed-shear $K_v = 0$ slice). In the additional conditional exact fixed-shear overlap lane, the residual $K_v = 0$ slice should be read as quotient-diagonal / involutive rather than as a new mixed local branch-classification problem. Indeed, after centering and analytic source reparametrization, the projected germ has the form

$$\Phi(t) = (t^m, G(t)),$$

and exact repeated values force

$$m \text{ even}, \quad G(-t) = G(t).$$

So, after quotienting by analytic functions of the first coordinate, the genuinely mixed residue is an even/even quotient germ

$$(t^m, t^n + \cdots), \quad m, n \text{ even}, \quad n > m.$$

In particular, among the current explicit low-order mixed models

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

only the quartic–sextic case

$$(4, 6)$$

can survive on the exact fixed-shear overlap $K_v = 0$ slice. The subsequent quotient-transport reformulation will show that this is the first rung of an even-support residual ladder

$$(m, n) = (r + 2, r + 2e + 4), \quad r \in 2\mathbf{Z}_{\geq 0}, \quad e = \text{ord}_0 \widehat{\mathcal{D}}.$$

Lemma 12.345 (Local exact repeated-value locus on the involutive fixed-shear slice). *Assume the setup of Remark 12.344, so that in a centered analytic source coordinate*

$$\Phi(t) = (t^m, G(t)), \quad m \text{ even}, \quad G(-t) = G(t).$$

After shrinking to a sufficiently small neighborhood of $t = 0$, one has

$$\Phi(t_1) = \Phi(t_2), \quad t_1 \neq t_2 \iff t_2 = -t_1.$$

Equivalently, the local exact repeated-value locus is exactly the involution graph

$$\{(t, -t) : 0 < |t| < \varepsilon\}.$$

In particular, on the exact fixed-shear $K_v = 0$ slice there is no local hidden partner choice beyond the involution $t \leftrightarrow -t$.

Proof. If $\Phi(t_1) = \Phi(t_2)$, then comparing first coordinates gives

$$t_1^m = t_2^m.$$

Since m is even and the source coordinate is real, this implies

$$|t_1| = |t_2|.$$

If $t_1 \neq t_2$, then necessarily $t_2 = -t_1$. Conversely, because m is even and G is even,

$$\Phi(-t) = \Phi(t)$$

for all sufficiently small t . This proves the claim. $\square$

Proposition 12.346 (Quotient transport scalar and residual ladder on the exact fixed-shear $K_v = 0$ slice). *Assume the additional conditional exact fixed-shear overlap lane from Remark 12.344. Let*

$$r := \text{ord}_0 J_\lambda = \text{ord}_0 K_v, \quad q := \text{ord}_0 J_{u_5},$$

and write in the centered involutive source coordinate t

$$J_\lambda(t) = t^r A(t), \quad J_{u_5}(t) = t^q B(t), \quad A(0) \neq 0.$$

Set

$$Q(t) := \frac{B(t)}{A(t)} = \frac{J_{u_5}(t)}{J_\lambda(t)}.$$

Then the following hold.

(1) Q is even:

$$Q(-t) = Q(t).$$

Hence there exists a unique analytic germ $\widehat{Q}$ in

$$u := t^2$$

such that

$$Q(t) = \widehat{Q}(t^2).$$

(2) If

$$\widehat{\mathcal{D}}(u) := \widehat{Q}'(u),$$

then

$$\frac{\mathcal{W}_{\text{src}}(t)}{J_\lambda(t)^2} = Q'(t) = 2t \widehat{\mathcal{D}}(t^2),$$

and therefore

$$\mathcal{W}_{\text{src}}(t) = 2t^{2r+1} A(t)^2 \widehat{\mathcal{D}}(t^2).$$

(3) If

$$e := \text{ord}_0 \widehat{\mathcal{D}} \in \mathbf{Z}_{\geq 0},$$

then

$$q - r = 2e + 2, \quad m = r + 2, \quad n = r + 2e + 4,$$

and

$$\text{ord}_0 \mathcal{W}_{\text{src}} = 2r + 1 + 2e.$$

(4) In particular,

$$r \text{ is even, } \quad q - r \in 2\mathbf{Z}_{\geq 2}, \quad \text{ord}_0 \mathcal{W}_{\text{src}} \text{ is odd.}$$

So the exact fixed-shear residual lane lies on an even-support ladder over $K_v = 0$, while the associated source-level Wronskian vanishes to odd order.

(5) The currently surviving quartic–sextic residual model is exactly the first rung of this ladder:

$$(m, n) = (4, 6) \iff (r, e) = (2, 0).$$

Proof. By Remark 12.344, the projected germ has the form

$$\Phi(t) = (t^m, G(t)), \quad m \text{ even}, \quad G(-t) = G(t).$$

Thus the quotient-diagonal mixed residue is even/even, so the regularized ratio $Q = B/A$ is even, giving the factorization

$$Q(t) = \widehat{Q}(t^2).$$

Differentiating yields

$$Q'(t) = 2t \widehat{Q}'(t^2) = 2t \widehat{\mathcal{D}}(t^2).$$

On the other hand, Remark 12.332 gives

$$Q'(t) = \frac{\mathcal{W}_{\text{src}}(t)}{J_\lambda(t)^2}.$$

Since

$$J_\lambda(t)^2 = t^{2r} A(t)^2,$$

this gives

$$\mathcal{W}_{\text{src}}(t) = 2t^{2r+1} A(t)^2 \widehat{\mathcal{D}}(t^2).$$

If $e = \text{ord}_0 \widehat{\mathcal{D}}$, then

$$\widehat{Q}(u) - \widehat{Q}(0) \sim u^{e+1},$$

so

$$Q(t) - Q(0) \sim t^{2e+2}.$$

Hence

$$q - r = \text{ord}_0(Q(t) - Q(0)) = 2e + 2.$$

Now Remark 12.331 gives

$$m = r + 2, \quad n = q + 2,$$

so

$$n = r + 2e + 4.$$

Also

$$\text{ord}_0 \mathcal{W}_{\text{src}} = 2r + 1 + 2e.$$

Since m is even, $r = m - 2$ is even. The final statements follow immediately. $\square$

Proposition 12.347 (Sign-run compression and the three-run criterion on a quotient-visible exact fixed-shear branch). *Assume the quotient-visible exact fixed-shear reduction, and let*

$$[a, b] \subset \mathbf{R}_u$$

be a connected compact quotient branch on which the descended transport potential

$$\widehat{Q} : [a, b] \rightarrow \mathbf{R}$$

is analytic and nonconstant, with

$$\widehat{\mathcal{D}} = \widehat{Q}'.$$

Let

$$a = s_0 < s_1 < \cdots < s_\ell = b$$

be the sign-change partition of $\widehat{\mathcal{D}}$: on each run

$$R_j := (s_j, s_{j+1}), \quad j = 0, \dots, \ell - 1,$$

the derivative $\widehat{\mathcal{D}}$ has fixed nonzero sign, and the sign alternates from run to run. Define

$$d_j := \widehat{Q}(s_j), \quad \Omega_j := (\min\{d_j, d_{j+1}\}, \max\{d_j, d_{j+1}\}),$$

and the run amplitudes

$$B_j := |d_{j+1} - d_j| = \int_{R_j} |\widehat{\mathcal{D}}(u)| du.$$

Then the following hold.

(1) *For every regular value*

$$\gamma \notin \widehat{Q}(\{\widehat{\mathcal{D}} = 0\}),$$

one has the exact multiplicity formula

$$\#\widehat{Q}^{-1}(\gamma) = \#\{j : \gamma \in \Omega_j\}.$$

(2) *For every*

$$0 \leq j \leq \ell - 3,$$

one has the exact distance-two overlap criterion

$$\Omega_j \cap \Omega_{j+2} \neq \emptyset \iff B_j + B_{j+2} > B_{j+1}.$$

Equivalently,

$$\Omega_j \cap \Omega_{j+2} = \emptyset \iff B_{j+1} \geq B_j + B_{j+2}.$$

(3) *Two consecutive distance-two separations are impossible. More precisely, there is no j such that both*

$$\Omega_j \cap \Omega_{j+2} = \emptyset \quad \text{and} \quad \Omega_{j+1} \cap \Omega_{j+3} = \emptyset.$$

In particular, if the branch has no nonadjacent overlap of regular values, then

$$\ell \leq 3.$$

Proof. On each run R_j , the function $\widehat{Q}$ is strictly monotone, so for a regular value γ it has exactly one preimage on R_j if and only if γ lies in the open image interval Ω_j , and otherwise no preimage there. Summing over runs proves (1).

For (2), fix j and, after replacing $\widehat{Q}$ by $-\widehat{Q}$ if needed, assume that the signs on R_j, R_{j+1}, R_{j+2} are

$$+, -, +.$$

Then

$$d_j < d_{j+1} > d_{j+2} < d_{j+3},$$

with

$$B_j = d_{j+1} - d_j, \quad B_{j+1} = d_{j+1} - d_{j+2}, \quad B_{j+2} = d_{j+3} - d_{j+2}.$$

In this configuration,

$$\Omega_j = (d_j, d_{j+1}), \quad \Omega_{j+2} = (d_{j+2}, d_{j+3}),$$

and since $d_{j+2} < d_{j+1}$ automatically, these open intervals overlap if and only if

$$d_j < d_{j+3}.$$

But

$$d_{j+3} - d_j = (d_{j+1} - d_j) - (d_{j+1} - d_{j+2}) + (d_{j+3} - d_{j+2}) = B_j - B_{j+1} + B_{j+2},$$

so

$$d_j < d_{j+3} \iff B_j + B_{j+2} > B_{j+1}.$$

This proves (2).

For (3), if both consecutive separations held, then by (2) one would have

$$B_{j+1} \geq B_j + B_{j+2}, \quad B_{j+2} \geq B_{j+1} + B_{j+3},$$

which is impossible because all $B_k > 0$. Hence among any two consecutive distance-two comparisons, at least one overlap is forced. If $\ell \geq 4$, such a consecutive comparison exists, so no-nonadjacent-overlap forces $\ell \leq 3$. $\square$

Corollary 12.348 (Three-run classification on a quotient-visible exact fixed-shear branch). *In the setting of Proposition 12.347, assume that*

$$\ell = 3.$$

Then the branch has no nonadjacent overlap of regular values if and only if

$$B_1 \geq B_0 + B_2.$$

If

$$B_1 > B_0 + B_2,$$

the middle run separates the two outer windows by a genuine gap. If

$$B_1 = B_0 + B_2,$$

the outer windows touch only at a critical value, so there is still no regular value with nonadjacent re-entry.

Proof. When $\ell = 3$, the only nonadjacent comparison is (Ω_0, Ω_2) , so the claim is exactly Proposition 12.347(2). $\square$

Remark 12.349 (Touching is sterile in the three-run quotient model). In the equality case

$$B_1 = B_0 + B_2,$$

Corollary 12.348 says that the outer windows meet only at a boundary critical value. Thus the same-value relation between the two outer runs is zero-dimensional downstairs in the quotient variable $u = t^2$: there is no open value interval carrying a same-value branch between the outer runs.

Accordingly, the equality case should be viewed as critical-value touching, not as a genuine open transport overlap. The genuinely live open-overlap three-run regime is

$$B_0 + B_2 > B_1,$$

while

$$B_1 > B_0 + B_2$$

is the separation regime.

Remark 12.350 (Separation is not a transport lane in the three-run quotient model). In the strict separation regime

$$B_1 > B_0 + B_2,$$

the outer windows are disjoint, so there is no outer-overlap interval

$$\Omega_0 \cap \Omega_2$$

and hence no inverse-branch-gap transport problem between the two outer runs. Accordingly, the inverse-branch-gap / Δ / Ξ setup belongs only to the genuine outer-overlap regime

$$B_0 + B_2 > B_1,$$

while the strict separation regime is already a no-reentry regime downstairs in the quotient variable.

Corollary 12.351 (Quartic–sextic quotient injectivity and local no-reentry). *Assume the quartic–sextic residual case*

$$(m, n) = (4, 6).$$

Then

$$Q(t) = Q(0) + \kappa_{2,2}t^2 + O(t^4), \quad \widehat{Q}(u) = Q(0) + \kappa_{2,2}u + O(u^2),$$

so

$$\widehat{D}(0) = \kappa_{2,2} \neq 0.$$

Equivalently,

$$\kappa_{2,2} = \frac{1}{2} \left(\frac{\mathcal{W}_{\text{src}}}{J_\lambda^2} \right)'(0) = \frac{\mathcal{W}_{\text{src}}^{(5)}(0)}{60(J_\lambda''(0))^2}.$$

Hence $\widehat{Q}$ is a local analytic diffeomorphism in the quotient variable $u = t^2$. In particular, after quotienting the involution $t \leftrightarrow -t$, there is no local exact fixed-shear re-entry: every nearby fixed-shear level is attained by at most one quotient point.

Proof. In the quartic–sextic case, Proposition 12.346 gives

$$(r, e) = (2, 0).$$

So $\widehat{D}(0) \neq 0$, which is exactly the displayed coefficient $\kappa_{2,2}$. The explicit formulas follow from Remark 12.331 and the identity

$$Q'(t) = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

Since

$$\widehat{Q}'(0) = \widehat{D}(0) \neq 0,$$

the analytic inverse function theorem implies that $\widehat{Q}$ is a local diffeomorphism in $u = t^2$. $\square$

Remark 12.352 (The quartic–sextic rung is inert for the lobe-amplitude obstruction). On the first residual rung

$$(m, n) = (4, 6),$$

Proposition 12.346 gives

$$(r, e) = (2, 0),$$

and Corollary 12.351 gives

$$\widehat{D}(0) = \kappa_{2,2} \neq 0.$$

So the quartic–sextic basepoint is not a sign-change point of $\widehat{D}$, contributes no compressed node on the quotient branch, and does not participate in any three-run run-amplitude obstruction. The first possible quotient-visible bad core therefore begins only at a higher-rung critical point of $\widehat{D}$.

Proposition 12.353 (Finite transport-jet quotient descent on the exact fixed-shear corner). *Assume the additional conditional exact fixed-shear overlap lane from Remark 12.344, and work along a local involutive branch*

$$(h_1, h_2) = (h(t), h(-t)).$$

For each $k \geq 0$, define the parity-normalized transport jet

$$\Theta_k(t) := \begin{cases} Q^{(k)}(t), & k \text{ even}, \\ t^{-1}Q^{(k)}(t), & k \text{ odd}, \end{cases}$$

where

$$Q(t) = \frac{J_{u_5}(t)}{J_\lambda(t)} = \widehat{Q}(t^2).$$

Then every Θ_k is even analytic. Equivalently, for each $k \geq 0$ there exists a unique analytic quotient germ $\tilde{\Theta}_k(u)$ such that

$$\Theta_k(t) = \tilde{\Theta}_k(t^2).$$

In particular, the canonical descended scalar

$$X(t) := \frac{Q'(t)}{t}$$

is even analytic and satisfies

$$X(t) = 2\hat{Q}'(t^2) = 2\hat{D}(t^2).$$

Proof. Since $Q(t) = \hat{Q}(t^2)$ is even, $Q^{(k)}$ has the usual parity: $Q^{(k)}$ is even for k even and odd for k odd. Every odd analytic germ is divisible by t , so Θ_k is analytic for odd k . In both cases Θ_k is even, hence factors uniquely through $u = t^2$. The final identity is immediate from

$$Q'(t) = 2t\hat{Q}'(t^2) = 2t\hat{D}(t^2).$$

□

Corollary 12.354 (Rank-one quotient-visible transport state on the quartic–sextic rung). *Assume the quartic–sextic residual case*

$$(m, n) = (4, 6).$$

For every $N \geq 0$, set

$$\mathcal{T}_N(t) := (\Theta_0(t), \dots, \Theta_N(t)).$$

Then there exists an analytic map

$$\mathbf{F}_N$$

defined near $Q(0)$ such that

$$\mathcal{T}_N(t) = \mathbf{F}_N(Q(t)).$$

Equivalently, every finite quotient-visible transport state is locally an analytic function of the single scalar Q alone.

Proof. By Corollary 12.351, $\hat{Q}$ is a local analytic coordinate in $u = t^2$. Since each $\Theta_k(t) = \tilde{\Theta}_k(t^2)$, each descended jet $\tilde{\Theta}_k(u)$ can be re-expressed analytically as a function of $\hat{Q}(u)$, hence of $Q(t)$. Stacking these functions gives $\mathbf{F}_N$. □

Corollary 12.355 (No finite-order hidden reset on the quartic–sextic residual corner). *Assume the quartic–sextic residual case. Let*

$$\mathcal{T}_N(t) := (\Theta_0(t), \dots, \Theta_N(t))$$

be any finite quotient-visible transport state. Then

$$\mathcal{T}_N(t_1) = \mathcal{T}_N(t_2) \implies t_2 = \pm t_1.$$

In particular, no finite quotient-visible transport state produces a third local reset beyond the involution symmetry $t \leftrightarrow -t$.

Proof. By Corollary 12.354, equality of $\mathcal{T}_N$ implies equality of Q . But

$$Q(t) = \hat{Q}(t^2),$$

and Corollary 12.351 shows that $\hat{Q}$ is locally injective in $u = t^2$. Hence $t_1^2 = t_2^2$, so $t_2 = \pm t_1$. □

Proposition 12.356 (Conditional state-local defect closure on the residual exact fixed-shear corner). *Assume the setup of Proposition 12.369. Suppose there exists $N \geq 0$ and analytic functionals*

$$\mathfrak{e}_3, \mathfrak{e}_5$$

defined near $\mathcal{T}_N(0)$ such that along the involutive branch

$$E_{12}^{(3)}(t) = \mathfrak{e}_3(\mathcal{T}_N(t)), \quad E_{12}^{(5)}(t) = \mathfrak{e}_5(\mathcal{T}_N(t)),$$

and

$$\mathfrak{e}_3(\mathcal{T}_N(0)) = \mathfrak{e}_5(\mathcal{T}_N(0)) = 0.$$

Then

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2),$$

hence

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2),$$

and no such nontrivial exact residual fixed-shear involutive bad branch can persist on a compact good patch

$$J \in \{c \neq 0, M \neq 0\}.$$

Proof. By Proposition 12.353, each Θ_k is even analytic, so

$$\mathcal{T}_N(t) = \mathcal{T}_N(0) + O(t^2).$$

Analyticity of $\mathfrak{e}_3, \mathfrak{e}_5$ and vanishing at $\mathcal{T}_N(0)$ therefore give

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2).$$

Since

$$h_1 - h_2 = h(t) - h(-t) = 2h'(0)t + O(t^3),$$

this becomes

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2).$$

Now apply Corollary 12.122. □

Corollary 12.357 (Stronger conditional package collapse on the involutive branch). *Assume the setup of Proposition 12.369. Suppose that for some $N \geq 0$ the corrected two-pair package along the involutive branch is state-local in the sense that*

$$\mathcal{P}_2(t) = \mathfrak{P}_2(a_1(t), a_2(t); \mathcal{T}_N(t), \mathcal{T}_N(t)),$$

and the corresponding one-pair package is

$$F_{h_1(t)} = F_{h_2(t)} = \mathfrak{F}(\mathcal{T}_N(t)).$$

Assume moreover the natural merger law

$$\mathfrak{P}_2(a_1, a_2; Y, Y) = (a_1 + a_2)\mathfrak{F}(Y).$$

Then the interaction remainder

$$\mathcal{I}_2(t) := \mathcal{P}_2(t) - a_1(t)F_{h_1(t)} - a_2(t)F_{h_2(t)}$$

vanishes identically along the involutive branch:

$$\mathcal{I}_2(t) \equiv 0.$$

Proof. Substitute the displayed state-local forms into the definition of $\mathcal{I}_2(t)$, then apply the merger law with $Y = \mathcal{T}_N(t)$. □

Remark 12.358 (Any remaining residual obstruction is genuinely relational). The preceding results show that on the quartic–sextic residual corner the finite quotient-visible transport state has local rank 1 and no finite-order hidden reset. Therefore, if the exact fixed-shear residual corner is still genuinely live, the obstruction cannot come from:

- a second local transport variable on the quartic–sextic rung;
- a third local reset inside any finite quotient-visible transport state;
- the explicit quotient-visible three-run run-amplitude obstruction on the quartic–sextic rung, since Corollary 12.351 gives $\widehat{\mathcal{D}}(0) \neq 0$ there;
- a two-pair model built only from the descended one-atom transport state, assuming the natural merger law.

So any still-live obstruction on

$$\{S = 0\} \cap \{K_v = 0\}$$

must come from the first genuinely relational or provenance-sensitive two-atom input that survives after quotient-visible transport descent, or else from non-finite-order dependence.

Remark 12.359 (Quotient-cusp interpretation and the no-reset issue). The involutive quotient picture on the residual exact fixed-shear $K_v = 0$ slice is best read as a one-branched quotient singularity rather than as a new branching mixed carrier.

Indeed, in the notation of Remark 12.344, one has

$$\Phi(t) = (t^m, G(t)), \quad m \text{ even}, \quad G(-t) = G(t),$$

so after passing to the quotient parameter

$$u = t^2$$

the projected germ descends to a single analytic quotient branch

$$\bar{\Phi}(u) = (u^{m/2}, \bar{G}(u)).$$

In particular, in the quartic–sextic case

$$(m, n) = (4, 6),$$

the quotient branch is cusp-like:

$$\bar{\Phi}(u) \sim (u^2, u^3 + \dots).$$

Thus any local quantity that genuinely factors through the quotient-diagonal projected germ — equivalently through the quotient-visible fixed-shear transport data — is single-valued in u and has trivial local monodromy at the involutive point. In this sense, the residual exact fixed-shear $K_v = 0$ locus is best viewed as a quotient-cusp splice rather than as a new local source of cycle rank.

At the same time, this does *not* yet by itself prove global no-reentry on the $K_v = 0$ lane: one still has to show that the full transported overlap / critical-value state relevant to the later finite-core problem is indeed quotient-visible. So the live mixed residual question is now a *no-hidden-reset* question at $K_v = 0$, not another local branch classification problem.

Remark 12.360 (What the quotient-transport package does and does not prove). The quotient-transport package now does more than give a single pointwise scalar compression of the residual exact fixed-shear $K_v = 0$ slice.

First, Proposition 12.353 shows that every finite parity-normalized transport jet descends analytically to the quotient variable

$$u = t^2.$$

Second, on the quartic–sextic rung, Corollary 12.354 shows that every finite quotient-visible transport state is locally an analytic function of the single scalar

$$Q(t) = \hat{Q}(t^2)$$

alone, and Corollary 12.355 rules out any third local reset inside that descended state beyond the involution symmetry $t \leftrightarrow -t$.

What is *not* proved here is that the actual corrected two-pair defect package is already state-local in this descended transport state. If that state-locality held, then Proposition 12.356 would already exclude the residual branch, and under the stronger natural merger law Corollary 12.357 would make the interaction remainder vanish identically.

Accordingly, the present residual mixed burden should now be read as follows: the exact fixed-shear corner is no longer blocked by a missing local branch classification or by hidden finite-order transport dynamics. Any remaining obstruction must come from genuinely relational or provenance-sensitive two-atom structure, or from non-finite-order dependence, not from the descended transport state alone.

Remark 12.361 (No finite local rescue on the exact fixed-shear corner). The current quotient-transport package leaves very little residual finite local freedom on the exact fixed-shear corner.

First, every finite parity-normalized transport jet descends to the quotient variable $u = t^2$. Second, on the quartic–sextic rung the descended transport state has local rank one and no

third local reset beyond the involution $t \leftrightarrow -t$. Third, the explicit three-run run-amplitude obstruction is already inert there, because on that rung

$$\widehat{\mathcal{D}}(0) = \widehat{Q}'(0) \neq 0.$$

So if the residual exact fixed-shear corner still survives, it cannot be explained by a second local transport variable, by a hidden finite-order reset, or by the explicit lobe-amplitude obstruction on the quartic–sextic rung. Any remaining obstruction must therefore be genuinely nonlocal on the transport side, or package/provenance-sensitive on the two-atom side.

Proposition 12.362 (Local scalar repeated-value classification). *Let f be a real-analytic function on an open neighborhood of $a \in \mathbb{R}$ with $f'(a) = 0$, and let $m \geq 2$ be the order of vanishing of $f - f(a)$ at a , so*

$$f(u) - f(a) = c_m(u - a)^m + O((u - a)^{m+1}), \quad c_m \neq 0.$$

Then there exist a sign $\sigma \in \{+1, -1\}$ and a local real-analytic chart x near a with $x(a) = 0$ and $x'(a) \neq 0$ such that

$$f(u) = f(a) + \sigma x(u)^m.$$

The local repeated-value locus

$$\{(u_1, u_2) : u_1 \neq u_2, f(u_1) = f(u_2)\}$$

near (a, a) is:

- (Even m .) *If m is even, it is the involution graph*

$$u_2 = \iota(u_1) := x^{-1}(-x(u_1)),$$

together with $u_1 = u_2$. The involution ι fixes a and is analytic with $\iota'(a) = -1$; the repeated-value branch is a single local fold half-arc attached to a .

- (Odd m .) *If m is odd, then f is a local analytic homeomorphism near a and the nondiagonal repeated-value locus is empty near (a, a) ; the point a is sterile.*

Proof. By the Malgrange/Morse-type preparation for analytic functions of one variable with an isolated critical point, one may write $f(u) - f(a) = \sigma h(u)^m$ for a local analytic chart h with $h(a) = 0$ and $h'(a) \neq 0$, where $\sigma = \text{sgn}(c_m)$ when m is even and $\sigma = +1$ absorbing the sign into h when m is odd. Setting $x := h$ gives the stated normal form.

For even m , $f(u_1) = f(u_2)$ near (a, a) is equivalent to $x(u_1)^m = x(u_2)^m$, i.e. $x(u_2) = \pm x(u_1)$; the plus sign is the diagonal, the minus sign is $u_2 = x^{-1}(-x(u_1)) = \iota(u_1)$. That ι is involutive with $\iota(a) = a$ and $\iota'(a) = -1$ is immediate from $x'(a) \neq 0$ and $x(a) = 0$.

For odd m , $x(u_1)^m = x(u_2)^m$ forces $x(u_1) = x(u_2)$, hence $u_1 = u_2$ near a since x is a local homeomorphism. $\square$

Corollary 12.363 (Local scalar-bud decomposition on the close exact fixed-shear lane). *Apply Proposition 12.362 to $f = \widehat{Q}$ on the nonconstant quotient branch in the exact fixed-shear corner $\{S = 0\} \cap \{K_v = 0\}$. Let a be a zero of $\widehat{\mathcal{D}} = \widehat{Q}'$ of order $m - 1 \geq 1$. Then:*

- *If m is even (i.e. a is an odd-order zero of $\widehat{\mathcal{D}}$), the local close exact repeated-value carrier at a is exactly one fold half-arc, namely the involution graph $\{u_2 = \iota(u_1)\}$ of Proposition 12.362.*
- *If m is odd (i.e. a is an even-order zero of $\widehat{\mathcal{D}}$), the local close exact repeated-value carrier at a is empty; the basepoint is sterile.*

In particular, on a compact nonconstant quotient branch in the exact fixed-shear corner, the close exact repeated-value carrier is a finite disjoint union of local fold half-arcs, one per odd zero of $\widehat{\mathcal{D}}$, with even zeros sterile. No connected nonlocal close carrier remains on that branch.

Proof. By Lemma 12.345, away from zeros of $\widehat{Q}'$ the close exact repeated-value locus is empty, so any close carrier accumulates only at zeros of $\widehat{\mathcal{D}}$. At each such zero a the local structure is dictated by Proposition 12.362 applied to $\widehat{Q}$, with $m = \text{ord}_a(\widehat{Q} - \widehat{Q}(a)) = \text{ord}_a \widehat{\mathcal{D}} + 1$; the two cases transcribe to odd vs. even order of $\widehat{\mathcal{D}}$. Compactness then bounds the number of such basepoints, and the carrier is their finite disjoint union of local fold half-arcs. $\square$

Remark 12.364 (Constant-branch scalarization and the scalar ceiling). There is also a constant-quotient branch. If

$$\widehat{D} \equiv 0,$$

then the quotient identities give

$$Y_{xx} = \gamma_0 S_{xx}, \quad Y = \gamma_0 S + \beta_0 x + \alpha_0.$$

After affine gauge one may reduce to $Y \equiv 0$, and the exact fixed-shear overlap equations then force $\gamma_* = 0$. So the constant branch reduces to the repeated-value problem for the single planar scalar

$$V(t) := U_{\xi_*, \eta_*}(x(t), S(t)), \quad U_{\xi, \eta}(x, S) := \frac{x - \xi S}{1 - \eta S},$$

to which Proposition 12.362 applies pointwise in t . At basepoints with $V'(t_0) \neq 0$, V is a local homeomorphism and close exact repeated-value pairs are excluded; Corollary 12.351 rules out the basepoint case $\widehat{D}(0) \neq 0$, and at basepoints where V has a critical point the local carrier is again a single fold half-arc. Thus the constant exact fixed-shear branch also contributes only local fold half-arcs.

In this form the close exact fixed-shear lane contributes no connected nonlocal close carrier, only local scalar fold half-arcs. Corollary 12.363 is the theorem-level scalar ceiling on that lane: the remaining intrinsic obstruction on the close exact fixed-shear side is now equality of consecutive lobe amplitudes across these local fold half-arcs, not a hidden nonlocal carrier.

Remark 12.365 (Connected-carrier sterility versus scalar multiplicity on the close exact lane). Corollary 12.363 and Remark 12.364 do more than say that close exact repeated values are locally scalar. They also identify the global graph type of the close carrier: after shrinking the close threshold, each connected component is a single local fold bud attached at one odd scalar seed. In particular, each component is one-sided and pendant, never a cycle edge, and there is no connected nonlocal close carrier on the exact fixed-shear lane.

What this closes is the connected-carrier question, not abstract scalar multiplicity. Pure scalar analyticity can still realize multiplicity-4 repeated-value levels, equivalently equal consecutive lobe amplitudes, so no truthful purely scalar theorem can bound the close multiplicity by 2 in full generality. The point is instead that such multiplicity does not organize into a connected global close carrier on the exact fixed-shear lane.

Remark 12.366 (Work in progress: inverse-branch-gap endgame in the overlap regime on the strict three-run lane). Stay in the setting of Proposition 12.347 and Corollary 12.348, and assume the three-run overlap regime

$$B_0 + B_2 > B_1.$$

Then the outer windows overlap on an open value interval

$$\mathcal{I} := \Omega_0 \cap \Omega_2,$$

so there are inverse branches

$$\phi_0 : \mathcal{I} \rightarrow R_0, \quad \phi_2 : \mathcal{I} \rightarrow R_2$$

of $\widehat{Q}$. Define the inverse-branch gap

$$\Delta(q) := \phi_2(q) - \phi_0(q),$$

and the transport-curvature scalar

$$\Xi(u) := \frac{\widehat{Q}''(u)}{(\widehat{Q}'(u))^3}, \quad \Upsilon(q) := \Xi(\phi_0(q)) - \Xi(\phi_2(q)).$$

Then

$$\Delta'(q) = \frac{1}{\widehat{Q}'(\phi_2(q))} - \frac{1}{\widehat{Q}'(\phi_0(q))}, \quad \Delta''(q) = \Upsilon(q).$$

So the strict three-run overlap problem may be studied through the one-variable gap function Δ . Any interior critical point $q_* \in \mathcal{I}$ yields a same-value / same-slope coincidence: with

$$u_0 := \phi_0(q_*), \quad u_2 := \phi_2(q_*),$$

one has

$$\widehat{Q}(u_0) = \widehat{Q}(u_2), \quad \widehat{Q}'(u_0) = \widehat{Q}'(u_2).$$

Thus the overlap obstruction is already a concrete outer-pair witness, not merely an abstract transport mismatch.

It is natural to package the higher mismatch by

$$J_1(u) := \frac{1}{\widehat{Q}'(u)}, \quad J_{n+1}(u) := J_1(u) \partial_u J_n(u).$$

The sharp current expectation is that any genuine overlap witness should force a finite-order even jet mismatch: for some smallest $m \geq 1$,

$$J_2(u_0) = J_2(u_2), \dots, J_{2m-1}(u_0) = J_{2m-1}(u_2), \quad J_{2m}(u_0) \neq J_{2m}(u_2),$$

with sign corresponding to a local minimum of Δ .

There is also a sharper descent fork, now correctly understood as conditional on actual overlap. If Ξ descends to value space across the two outer runs, then $\Upsilon \equiv 0$, so Δ is affine. Any theorem excluding nonconstant affine same-value transport between the two outer runs would then rule out the overlap regime. A stronger rigid alternative, if one can justify the endpoint propagation, would be exact translated replication

$$\Omega_0 = \Omega_2, \quad B_0 = B_2, \quad \phi_2(q) - \phi_0(q) \equiv C > 0, \quad \widehat{Q}(u + C) = \widehat{Q}(u) \quad (u \in R_0).$$

So the genuine three-run *overlap* regime now has the following corrected endgame fork: either Ξ -descent fails on an actual outer-overlap interval, giving the first explicit nonlocal transport obstruction, or Ξ -descent holds and the problem rigidifies to affine, and potentially translated, outer transport. In particular, the inverse-branch-gap setup belongs only to the overlap regime

$$B_0 + B_2 > B_1,$$

whereas

$$B_1 > B_0 + B_2$$

is already the separation regime and carries no such outer-transport lane.

Equivalently, the quantitative overlap scalar is

$$B_0 + B_2 - B_1,$$

and one may write

$$\Delta'(q) = \Delta'(q_0) + \int_{q_0}^q \Upsilon(s) ds,$$

so a transport-mass bound too small to cancel the initial inverse-slope gap would exclude genuine overlap.

This is a sharpened endgame formulation, not yet a closure theorem.

Remark 12.367 (No-go: overlap geometry alone does not force a witness). The corrected strict-overlap transport lane should not be read as if open outer overlap by itself automatically forces an interior same-value / same-slope witness in every endpoint configuration. Outside the stricter endpoint-sign- forcing subcases, the endpoint behavior of

$$W(q) := \Delta'(q) = \frac{1}{\widehat{Q}'(\phi_2(q))} - \frac{1}{\widehat{Q}'(\phi_0(q))}$$

depends on which outer branch is singular or regular at each overlap endpoint. So overlap geometry alone does not determine witness existence in full generality.

Equivalently, outside the strict sign-forcing cases, one needs extra input: endpoint slope comparison, a branch-independent closure law, or genuinely source/package-level structure. Thus the corrected overlap lane has reached a real ceiling for pure overlap geometry.

Remark 12.368 (Conditional rigidity: H -descent forces affine replication). Still in the corrected overlap regime, let

$$H(u) := \frac{\widehat{Q}''(u)}{(\widehat{Q}'(u))^2}, \quad J_1(u) := \frac{1}{\widehat{Q}'(u)}.$$

If H descends to value space across the two outer inverse branches, in the sense that

$$H(\phi_0(q)) = H(\phi_2(q))$$

on the overlap interval, then the reciprocal-slope branches

$$y_i(q) := J_1(\phi_i(q))$$

satisfy the same linear first-order ODE

$$y' = -h(q) y.$$

Hence

$$\frac{y_2(q)}{y_0(q)} = \Lambda$$

is constant, so

$$J_1(\phi_2(q)) = \Lambda J_1(\phi_0(q)), \quad \phi_2'(q) = \Lambda \phi_0'(q),$$

and therefore

$$\phi_2(q) = \Lambda \phi_0(q) + B.$$

So H -descent does not merely give a soft transport coincidence: it forces the outer inverse branches into affine replication. In particular, the translated / affine-copy exceptional branch is a genuinely rigid symmetry branch, not a soft hidden-state loophole.

Proposition 12.369 (Residual involutive defect criterion on the exact fixed-shear corner). *Assume the additional conditional exact fixed-shear overlap lane near a residual accumulation point*

$$a \in \{S = 0\} \cap \{K_v = 0\}.$$

Let a local exact repeated-value branch be parametrized by

$$(h_1, h_2) = (h(t), h(-t)), \quad h(0) = a, \quad h'(0) \neq 0,$$

with h analytic. Let

$$E_{12}^{(3)}(t), \quad E_{12}^{(5)}(t)$$

denote the cubic and quintic normalized interaction defects along this branch.

Assume:

$$E_{12}^{(3)}(t) = E_{12}^{(3)}(-t), \quad E_{12}^{(5)}(t) = E_{12}^{(5)}(-t),$$

and

$$E_{12}^{(3)}(0) = 0, \quad E_{12}^{(5)}(0) = 0.$$

Then

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2).$$

Since

$$h_1 - h_2 = h(t) - h(-t) = 2h'(0)t + O(t^3),$$

it follows that

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2).$$

Consequently no such nontrivial exact residual fixed-shear involutive bad branch can persist on a compact good patch

$$J \in \{c \neq 0, M \neq 0\}.$$

Proof. By evenness and vanishing at $t = 0$, the Taylor expansions have no constant or odd-order terms:

$$E_{12}^{(3)}(t) = \alpha_2 t^2 + O(t^4), \quad E_{12}^{(5)}(t) = \beta_2 t^2 + O(t^4).$$

So

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2).$$

Since

$$h(t) - h(-t) = 2h'(0)t + O(t^3),$$

we have

$$t = O(h_1 - h_2),$$

and therefore

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2).$$

Now apply Corollary 12.122 on a compact good patch

$$J \in \{c \neq 0, M \neq 0\}.$$

It gives

$$|h_1 - h_2| \leq C_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right) = O((h_1 - h_2)^2).$$

For sufficiently small nonzero $|h_1 - h_2|$ this is impossible. Hence no such nontrivial exact residual fixed-shear involutive bad branch can persist. $\square$

Corollary 12.370 (Package-level reduction on the residual exact fixed-shear corner). *In the setting of Proposition 12.369, it is enough to verify for the actual corrected cubic and quintic two-pair defects that:*

- (1) *they are invariant under swapping the two atoms along the involutive parametrization $t \leftrightarrow -t$; and*
- (2) *they vanish on coincident atoms.*

Under these two package-level inputs, the exact residual fixed-shear corner

$$\{S = 0\} \cap \{K_v = 0\}$$

is excluded on compact good overlap patches.

Proof. The two hypotheses are exactly the evenness and diagonal-vanishing inputs of Proposition 12.369. $\square$

Remark 12.371 (The residual exact fixed-shear corner is now a package-level problem). The point of Proposition 12.369 is that it does not require construction of the full corrected two-atom quotient map. Once the residual exact fixed-shear branch is recognized as quotient-diagonal / involutive, the remaining burden is no longer a new local mixed branch classification. It is the package-level collapse of the actual finite-order two-pair defects on coincident atoms, or equivalently a no-hidden-reset statement for whatever transported overlap state survives on the quotient side.

Frame convention for the scalar block. On the cubic-generic one-pair locus we fix the cubic-normalized frame so that the distinguished cubic line is the diagonal-traceless line $\mathbf{C}D$, with

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J_0 D J_0 = -D.$$

Thus the scalar coefficient extracted below is not an auxiliary choice: it is the canonical coefficient of the cubic line in this fixed cubic-generic frame.

Definition 12.372 (Canonical scalar extraction from the raw one-pair stack). Let

$$R(z) = A_3 z^3 + A_5 z^5 + A_7 z^7 + O(z^9)$$

be a cubic-generic raw one-pair corrected germ in the cubic-normalized frame. Write

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

We use the Frobenius pairing

$$\langle X, Y \rangle_F := \text{tr}(X^* Y).$$

The scalar channel of the cubic coefficient is defined by projection to the diagonal-traceless line **CD**:

$$c(R) := \frac{\langle A_3, D \rangle_F}{\langle D, D \rangle_F} = \frac{1}{2} \text{tr}(D A_3).$$

Equivalently, the scalar normalization equations are

$$\pi_I(A_3) = 0, \quad \pi_S(A_3) = 0, \quad \pi_K(A_3) = 0, \quad A_3 = c(R) D,$$

where π_I, π_S, π_K denote the Frobenius projections to the lines **CI**, **CS**, **CK**.

Remark 12.373 (Compatibility with the existing gauge laws). By the already-proved gauge law

$$A_3^{\text{new}} = A_3,$$

the cubic coefficient is unchanged by the admissible one-pair normalization gauge. Hence the scalar extraction in Definition 12.372 is gauge independent on the raw one-pair stack.

Lemma 12.374 (Scalar normalization is Σ -invariant). *Let*

$$\Sigma(F)(z) := J_0 F(-z) J_0$$

be the centered reflection involution, and let

$$\Sigma_3(A) := -J_0 A J_0$$

be its induced action on cubic coefficients. Assume

$$J_0 D J_0 = -D.$$

Then the scalar extraction

$$c(R) = \frac{1}{2} \text{tr}(D A_3)$$

from Definition 12.372 is Σ -invariant:

$$c(\Sigma R) = c(R).$$

Proof. If $R(z) = A_3 z^3 + O(z^5)$, then

$$\Sigma(R)(z) = J_0 R(-z) J_0 = (-J_0 A_3 J_0) z^3 + O(z^5) = \Sigma_3(A_3) z^3 + O(z^5).$$

Therefore

$$c(\Sigma R) = \frac{1}{2} \text{tr}(D(-J_0 A_3 J_0)).$$

Using cyclicity of trace,

$$\text{tr}(D(-J_0 A_3 J_0)) = \text{tr}((-J_0 D J_0) A_3).$$

Since $J_0 D J_0 = -D$, we get

$$-J_0 D J_0 = D,$$

hence

$$c(\Sigma R) = \frac{1}{2} \text{tr}(D A_3) = c(R).$$

□

Lemma 12.375 (Uniqueness of the scalar output). *Let R be a cubic-generic raw one-pair corrected germ. Then the scalar $c(R)$ from Definition 12.372 is uniquely determined. In particular, if*

$$A_3 = cD = \tilde{c}D,$$

then $c = \tilde{c}$.

Proof. If

$$(c - \tilde{c})D = 0,$$

then $c - \tilde{c} = 0$ because $D \neq 0$. Equivalently,

$$c = \frac{1}{2} \operatorname{tr}(DA_3)$$

is the unique Frobenius coefficient of A_3 along the line $\mathbf{CD}$. $\square$

Corollary 12.376 (Finite-jet scalar descent on the centered branch). *Let R_s be the centered one-pair branch, and assume*

$$R_{-s} = \Sigma(R_s).$$

Then the scalar coefficient

$$c(s) := c(R_s)$$

is even:

$$c(-s) = c(s).$$

In particular,

$$C_1 = C_3 = C_5 = 0.$$

Proof. By Lemma 12.374,

$$c(R_{-s}) = c(\Sigma R_s) = c(R_s).$$

Thus $c(-s) = c(s)$. The vanishing of the odd Taylor coefficients follows immediately. $\square$

Remark 12.377 (Why the scalar block is the only remaining mixed-local bottleneck). Through quintic order, the quotient-level ambiguity is already confined to the J_0 -anti-fixed traceless sector, while at septic order the canonical datum is the gauge-invariant residual Δ_7 . Thus, on the clean branch $V_1 \neq 0$, the only remaining mixed-local theorem is the scalar descent

$$C_1 = C_3 = C_5 = 0.$$

Once this holds, the finite-jet bridge to the good v_5 -patch coordinate $x = v_5/c$ is formal and the explicit mixed low-order obstruction list

$$(r, \operatorname{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}$$

is excluded immediately.

Remark 12.378 (Centered symmetric-placement Taylor package and diagonal-base local exclusion). Let

$$t_{\pm} = h \pm \frac{s}{2},$$

and regard $s \mapsto -s$ as swapping the two microscopic points. For any symmetric 2×2 coefficient matrix, the diagonal half-sums are even in s , the diagonal half-differences are odd in s , and the off-diagonal entries are even in s . Applying this to the explicit formulas already in the draft gives

$$u_5(s), v_5(s) \text{ odd}, \quad u_{7,K_1}(s), v_{7,K_1}(s) \text{ odd},$$

Writing

$$u_5(s) = U_1 s + U_3 s^3 + U_5 s^5 + \cdots, \quad v_5(s) = V_1 s + V_3 s^3 + V_5 s^5 + \cdots,$$

the first coefficients are

$$U_1 = 2(v_0 \delta \alpha_1 + \beta_0 \delta u_1),$$

$$U_3 = 2(v_0 \delta \alpha_3 + v_2 \delta \alpha_1 + \beta_0 \delta u_3 + \beta_2 \delta u_1),$$

$$U_5 = 2(v_0 \delta \alpha_5 + v_2 \delta \alpha_3 + v_4 \delta \alpha_1 + \beta_0 \delta u_5 + \beta_2 \delta u_3 + \beta_4 \delta u_1),$$

and

$$V_1 = 2(\bar{\alpha}_0 \delta u_1 + \delta \alpha_1 \bar{u}_0),$$

$$V_3 = 2(\bar{\alpha}_0 \delta u_3 + \bar{\alpha}_2 \delta u_1 + \delta \alpha_1 \bar{u}_2 + \delta \alpha_3 \bar{u}_0),$$

$$V_5 = 2(\bar{\alpha}_0 \delta u_5 + \bar{\alpha}_2 \delta u_3 + \bar{\alpha}_4 \delta u_1 + \delta \alpha_1 \bar{u}_4 + \delta \alpha_3 \bar{u}_2 + \delta \alpha_5 \bar{u}_0).$$

Likewise, writing

$$u_{7,K_1}(s) = \widehat{U}_1 s + \widehat{U}_3 s^3 + \cdots, \quad v_{7,K_1}(s) = \widehat{V}_1 s + \widehat{V}_3 s^3 + \cdots,$$

one gets

$$\begin{aligned} \widehat{U}_1 &= 2\kappa(\delta x_1 \nu_{6,0} + \beta_{5,0} \delta u_1 + v_0 \delta \alpha_{5,1} + b_0 \delta p_1 + r_0 \delta a_1), \\ \widehat{V}_1 &= 2\kappa(\delta x_1 \bar{\mu}_{6,0} - \bar{x}_0 \delta \mu_{6,1} + \bar{\alpha}_{5,0} \delta u_1 + \delta \alpha_{5,1} \bar{u}_0 + \bar{a}_0 \delta p_1 + \delta a_1 \bar{p}_0), \end{aligned}$$

and

$$\begin{aligned} \widehat{U}_3 &= 2\kappa(\delta x_1 \nu_{6,2} + \delta x_3 \nu_{6,0} + \beta_{5,0} \delta u_3 + \beta_{5,2} \delta u_1 + v_0 \delta \alpha_{5,3} + v_2 \delta \alpha_{5,1} + b_0 \delta p_3 + b_2 \delta p_1 + r_0 \delta a_3 + r_2 \delta a_1), \\ \widehat{V}_3 &= 2\kappa(\delta x_1 \bar{\mu}_{6,2} + \delta x_3 \bar{\mu}_{6,0} - \bar{x}_0 \delta \mu_{6,3} - \bar{x}_2 \delta \mu_{6,1} + \bar{\alpha}_{5,0} \delta u_3 + \bar{\alpha}_{5,2} \delta u_1 + \delta \alpha_{5,1} \bar{u}_2 + \delta \alpha_{5,3} \bar{u}_0 + \bar{a}_0 \delta p_3 + \bar{a}_2 \delta p_1 + \delta a_1 \bar{p}_2 + \delta a_3 \bar{p}_0). \end{aligned}$$

and therefore

$$\Delta_7(s) = u_{7,K_1}(s) v_5(s) - u_5(s) v_{7,K_1}(s)$$

is even in s , with Taylor expansion

$$\Delta_7(s) = D_2 s^2 + D_4 s^4 + D_6 s^6 + \cdots,$$

where

$$\begin{aligned} D_2 &= \widehat{U}_1 V_1 - U_1 \widehat{V}_1, \\ D_4 &= \widehat{U}_1 V_3 + \widehat{U}_3 V_1 - U_1 \widehat{V}_3 - U_3 \widehat{V}_1, \\ D_6 &= \widehat{U}_1 V_5 + \widehat{U}_3 V_3 + \widehat{U}_5 V_1 - U_1 \widehat{V}_5 - U_3 \widehat{V}_3 - U_5 \widehat{V}_1. \end{aligned}$$

Assume now that

$$c(s) = C_0 + C_2 s^2 + C_4 s^4 + \cdots, \quad C_0 \neq 0.$$

Then

$$x(s) = \frac{v_5(s)}{c(s)}$$

is an odd series

$$x(s) = X_1 s + X_3 s^3 + X_5 s^5 + \cdots$$

with

$$\begin{aligned} X_1 &= \frac{V_1}{C_0}, \\ X_3 &= \frac{V_3 C_0 - V_1 C_2}{C_0^2}, \\ X_5 &= \frac{V_5 C_0^2 - V_3 C_0 C_2 + V_1 (C_2^2 - C_0 C_4)}{C_0^3}. \end{aligned}$$

In particular, if $V_1 \neq 0$, then x is an odd local coordinate with odd inverse

$$s(x) = \sigma_1 x + \sigma_3 x^3 + \sigma_5 x^5 + \cdots,$$

where

$$\sigma_1 = \frac{C_0}{V_1}, \quad \sigma_3 = -\frac{C_0(V_3 C_0 - V_1 C_2)}{V_1^4},$$

and so on.

Similarly,

$$\lambda(s) := \frac{\Delta_7(s)}{v_5(s)}$$

is an odd series

$$\lambda(s) = L_1 s + L_3 s^3 + L_5 s^5 + \cdots$$

with

$$\begin{aligned} L_1 &= \frac{D_2}{V_1}, \\ L_3 &= \frac{D_4 V_1 - D_2 V_3}{V_1^2}, \\ L_5 &= \frac{D_6 V_1^2 - D_4 V_1 V_3 + D_2 (V_3^2 - V_1 V_5)}{V_1^3}. \end{aligned}$$

So, whenever x is used as the induced odd collision-side coordinate, parity in s transfers to parity in x , and one obtains

$$Y(x) = \frac{u_5}{c}(x) \text{ odd}, \quad \lambda(x) = \frac{\Delta_7}{v_5}(x) \text{ odd}, \quad S(x) = \frac{\lambda}{c}(x) \text{ odd}.$$

Consequently,

$$Y''(x) \text{ odd}, \quad S''(x) \text{ odd},$$

hence

$$\mathcal{Z}(x) = \frac{Y''}{S''} \text{ even}, \quad \mathcal{R}(x) = \mathcal{Z}'(x) \text{ odd}, \quad K_v = (x')^3 S'' \text{ odd}.$$

This is the parity package underlying the provisional mixed-branch exclusion discussed below.

If $x = v_5/c$ is used as the induced odd collision-side coordinate, then the centered diagonal germ has the odd/odd form

$$Q(x) \sim (x^n, x, x^m), \quad m, n \text{ odd}, \quad n > m \geq 3.$$

In that regime one obtains the exact factorizations

$$S''(x) = x^{m-2} \Sigma_m[A](x^2), \quad \mathcal{Z}(x) = x^{n-m} H(x^2), \quad \mathcal{R}(x) = x^{n-m-1} J(x^2),$$

with $H(0) \neq 0$ and $J(0) \neq 0$. Hence there exists $\varepsilon > 0$ such that for

$$0 < |x| < \varepsilon$$

one has

$$K_v(x) \neq 0, \quad T_v(x) \neq 0.$$

So, in the diagonal-based local regime where this odd collision-side coordinate description is valid, the centered diagonal base has a punctured neighborhood free of $K_v = 0$ and $T_v = 0$ points. In particular, the currently explicit low-order dangerous patterns do not arise at the symmetric diagonal base itself.

Work in progress. The odd/even transfer from the centered microscopic variable s to a general good-patch coordinate $x = v_5/c$ should still be treated as a live theorem-level bridge outside the diagonal-based local regime. What is no longer open at the same level is total quintic collapse on the centered branch: by Lemma 12.381 and Corollary 12.382, one already has

$$u_5(s) \not\equiv 0, \quad A_5^\dagger(s) \not\equiv 0.$$

So the remaining mixed-side task is not to prove that some quintic channel survives, but to promote this centered quintic nontriviality into an actual punctured v_5 - or u_5 -patch.

Lemma 12.379 (Total centered quintic collapse forces source-level degeneration). *On the centered symmetric-placement branch one has*

$$u_5 = 2(v\delta\alpha + \beta\delta u), \quad v_5 = 2(\bar{\alpha}\delta u + \bar{u}\delta\alpha).$$

If

$$u_5 \equiv 0, \quad v_5 \equiv 0,$$

then

$$\begin{pmatrix} \beta(s) & v(s) \\ \bar{\alpha}(s) & \bar{u}(s) \end{pmatrix} \begin{pmatrix} \delta u(s) \\ \delta\alpha(s) \end{pmatrix} = 0$$

as an identity of analytic germs. Consequently, either

$$\Delta_5(s) := \beta(s)\bar{u}(s) - v(s)\bar{\alpha}(s) \equiv 0,$$

or

$$\delta u(s) \equiv 0, \quad \delta\alpha(s) \equiv 0.$$

Proof. This is immediate from the displayed formulas for u_5 and v_5 . If the 2×2 coefficient matrix has nonzero determinant, then the only solution is $\delta u = \delta\alpha \equiv 0$. If its determinant vanishes identically, then $\Delta_5 \equiv 0$. $\square$

Corollary 12.380 (Total centered quintic collapse forces total canonical transport collapse). *If*

$$u_5 \equiv 0, \quad v_5 \equiv 0$$

on the centered branch, then

$$A_5^f \equiv 0, \quad \Delta_7 \equiv 0.$$

In particular, total centered quintic collapse forces collapse of the canonical quintic/septic transport package on that branch.

Proof. The identity

$$A_5^f = u_5 I + v_5 S$$

gives $A_5^f \equiv 0$ immediately. Likewise

$$\Delta_7 = u_{7,K_1} v_5 - u_5 v_{7,K_1}$$

vanishes identically when $u_5 \equiv v_5 \equiv 0$. □

Lemma 12.381 (Centered tagged quintic witness survives specialization). *On the tagged slice*

$$(K_1, \eta_2, q^{(5)}),$$

the first centered odd coefficient

$$U_1$$

of

$$u_5(s) = U_1 s + U_3 s^3 + U_5 s^5 + \cdots$$

has a nonzero $\kappa \eta_2 q^{(5)}$ -coefficient. In particular

$$U_1 \neq 0$$

on that tagged slice, and hence

$$u_5(s) \not\equiv 0$$

as a centered germ.

Proof. The centered coefficient formula is

$$U_1 = 2(v_0 \delta \alpha_1 + \beta_0 \delta u_1).$$

By the tagged-slice analysis already used in the proof of Lemma 12.18, the η_2 -slice of X_1 is off-diagonal, so

$$\delta u_1 = 0,$$

and the $q^{(5)}$ -slice of X_3 is diagonal, so

$$\beta_0 = 0.$$

Hence

$$U_1 = 2v_0 \delta \alpha_1$$

on the tagged slice.

Now use the explicit two-direction computation from the proof of Lemma 12.18. There one writes

$$G_0 = \text{diag}(a^2, b^2), \quad A = \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix},$$

and expands

$$X(\varepsilon, \delta) = X_0 + \varepsilon U + \delta V + \varepsilon \delta W + O(\varepsilon^2, \delta^2).$$

The proof gives

$$U_{12} = -\frac{u}{ab(a+b)} \neq 0, \quad V_{22} = -\frac{d}{2b^3} \neq 0.$$

Here U_{12} gives the tagged η_2 -contribution to the off-diagonal channel v_0 , while V_{22} gives the tagged $q^{(5)}$ -contribution to the diagonal-difference channel $\gamma - \alpha = 2\delta \alpha_1$. Thus the tagged slice contributes nontrivially to the product $v_0 \delta \alpha_1$, and hence to $U_1 = 2v_0 \delta \alpha_1$. □

Corollary 12.382 (Centered quintic fixed-sector nontriviality). *On the centered diagonal branch,*

$$A_5^f(s) = u_5(s)I + v_5(s)S$$

is not identically zero. Equivalently,

$$(u_5(s), v_5(s)) \neq (0, 0).$$

Proof. By Lemma 12.381, one has $u_5(s) \neq 0$. Since I and S are linearly independent, the fixed-sector package $A_5^f(s) = u_5(s)I + v_5(s)S$ cannot vanish identically. $\square$

Lemma 12.383 (Valuation form of the septic transport constraint). *Assume on the centered branch that*

$$\lambda = \frac{\Delta_7}{v_5}$$

is analytic odd. Write

$$\text{ord}_s u_5 = 2m + 1, \quad \text{ord}_s v_5 = 2r + 1, \quad r \geq m.$$

Then

$$\text{ord}_s v_{7,K_1} \geq \text{ord}_s v_5 - \text{ord}_s u_5 + 1 = 2(r - m) + 1.$$

Equivalently, if

$$U_1 = \cdots = U_{2m-1} = 0, \quad U_{2m+1} \neq 0,$$

and

$$V_1 = \cdots = V_{2r-1} = 0, \quad V_{2r+1} \neq 0,$$

then

$$\widehat{V}_1 = \widehat{V}_3 = \cdots = \widehat{V}_{2(r-m)-1} = 0,$$

and the first potentially surviving even coefficient of Δ_7 is

$$D_{2r+2} = \widehat{U}_1 V_{2r+1} - U_{2m+1} \widehat{V}_{2(r-m)+1}.$$

Proof. Since λ is analytic odd and

$$\Delta_7 = \lambda v_5,$$

one has

$$\text{ord}_s \Delta_7 \geq \text{ord}_s v_5 + 1 = 2r + 2.$$

Suppose

$$\text{ord}_s v_{7,K_1} = 2d + 1 < 2(r - m) + 1.$$

Then

$$\text{ord}_s(u_5 v_{7,K_1}) = (2m + 1) + (2d + 1) \leq 2r,$$

whereas

$$\text{ord}_s(u_{7,K_1} v_5) \geq 1 + (2r + 1) = 2r + 2$$

because u_{7,K_1} is odd. Therefore the leading term of

$$\Delta_7 = u_{7,K_1} v_5 - u_5 v_{7,K_1}$$

would occur at order at most $2r$, contradicting $\text{ord}_s \Delta_7 \geq 2r + 2$. So

$$\text{ord}_s v_{7,K_1} \geq 2(r - m) + 1.$$

The coefficient statement is the corresponding triangular coefficient form: vanishing of

$$D_2, D_4, \dots, D_{2r}$$

forces

$$\widehat{V}_1, \widehat{V}_3, \dots, \widehat{V}_{2(r-m)-1}$$

to vanish successively, and the first potentially surviving coefficient is

$$D_{2r+2} = \widehat{U}_1 V_{2r+1} - U_{2m+1} \widehat{V}_{2(r-m)+1}.$$

$\square$

Lemma 12.384 (Late- v_5 u_5 -patch affine-normalization transplant). *Assume the centered scalar package from Remark 12.378, and assume*

$$U_1 \neq 0.$$

Then

$$y := \frac{u_5}{c}$$

is an odd local coordinate near the centered base. Suppose in addition that

$$\text{ord}_s v_5 = 2r + 1 \geq 3.$$

On the corresponding u_5 -patch write

$$x(y) := \frac{v_5}{c}, \quad \tau_u(y) := -\frac{\Delta_7}{c u_5}, \quad \tilde{\tau}_u(y) := \tau_u(y) - \tau'_u(0) y.$$

Then:

(1) $x(y)$ is odd and satisfies

$$x(y) = O(y^{2r+1});$$

in particular $\text{ord}_0 x \geq 3$;

(2) $\tilde{\tau}_u(y)$ is odd and either vanishes identically or has odd order at least 3;

(3) if $\tilde{\tau}_u \neq 0$, then after an ambient affine normalization consisting of a coordinate permutation and subtraction of a suitable odd multiple of one higher-order channel from the other, the germ Q_u is carried to the same odd/odd local normal form

$$(y^n, y, y^m), \quad n > m \geq 3, \quad m, n \text{ odd},$$

already analyzed in Remark 12.378.

Consequently, in the subcase $\tilde{\tau}_u \neq 0$, the late- v_5 corner is locally sterile on punctured u_5 -patch arcs: no genuinely new mixed low-order 4-point branch is born there.

Proof. By Lemma 12.381,

$$u_5(s) = U_1 s + O(s^3), \quad c(s) = C_0 + O(s^2), \quad C_0 \neq 0,$$

so

$$y(s) = \frac{u_5(s)}{c(s)}$$

is an odd local coordinate with odd inverse $s = s(y)$.

Since v_5 is odd and c is even in s ,

$$x(y) = \frac{v_5}{c}$$

is odd in y . Because $\text{ord}_s u_5 = 1$ and $\text{ord}_s v_5 = 2r + 1$, one has

$$x(y) = O(y^{2r+1}),$$

so $\text{ord}_0 x \geq 3$.

Likewise Δ_7 is even while c is even and u_5 is odd, hence

$$\tau_u(y) = -\frac{\Delta_7}{c u_5}$$

is odd in y . Therefore

$$\tilde{\tau}_u(y) := \tau_u(y) - \tau'_u(0) y$$

is odd and either identically zero or of odd order at least 3.

Assume now that $\tilde{\tau}_u \neq 0$. Then the normalized germ is

$$Q_u(y) = (y, x(y), \tilde{\tau}_u(y)),$$

with the last two channels odd and of order at least 3. Apply the ambient coordinate permutation

$$(Y, X, Z) \mapsto (X, Y, Z).$$

This preserves affine dependence/coplanarity geometry. If the two higher-order odd channels have the same leading odd order, subtract a suitable odd multiple of one from the other to separate those leading orders. After this affine normalization one reaches a germ of the form

$$(y^n B(y^2), y, y^m A(y^2)), \quad n > m \geq 3, \quad m, n \text{ odd},$$

with $A(0)B(0) \neq 0$, i.e. the same odd/odd local normal form analyzed in Remark 12.378. The same local calculation then gives a punctured neighborhood free of nearby mixed and nonmixed 4-point obstruction points. $\square$

Lemma 12.385 (Planar-lift absorption on the u_5 -patch). *Assume the hypotheses of Lemma 12.384, and suppose*

$$\tilde{\tau}_u \equiv 0.$$

Equivalently, there is a constant $a \in \mathbf{C}$ such that

$$\Delta_7 = -a u_5^2$$

locally on the u_5 -patch. Then after an ambient affine shear one has

$$Q_u \sim (y, x, 0), \quad y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}.$$

In particular, the lifted curve is affine-planar, and the remaining local content is only the planar affine dependence of

$$\Xi(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)} \right).$$

Moreover, in the defect-free order- ≤ 5 finite-core setting of Theorem 12.215, this residual corner is absorbed into the already-known quintic-circuit regime: it does not define a genuinely new mixed/septic 4-point branch.

Proof. The identity $\tilde{\tau}_u \equiv 0$ means exactly that

$$\tau_u(y) = a y$$

for some constant a . Since

$$\tau_u = -\frac{\Delta_7}{c u_5}, \quad y = \frac{u_5}{c},$$

this is equivalent to

$$-\frac{\Delta_7}{c u_5} = a \frac{u_5}{c},$$

hence

$$\Delta_7 = -a u_5^2.$$

Thus

$$Q_u = (y, x, ay).$$

Applying the ambient affine shear

$$(Y, X, Z) \mapsto (Y, X, Z - aY)$$

gives

$$Q_u \sim (y, x, 0).$$

So the lifted curve is affine-planar.

Now assume additionally the defect-free order- ≤ 5 finite-core hypotheses from Theorem 12.215. By Remark 12.222, the same defect-free affine-dependence conclusions hold with Q_v replaced by Q_u . Since the third coordinate of Q_u vanishes after the shear, affine dependence of the lifted points is equivalent to affine dependence of the planar points

$$\Xi(h_j) = \left(\frac{u_5(h_j)}{c(h_j)}, \frac{v_5(h_j)}{c(h_j)} \right) \in \mathbf{C}^2.$$

But this is exactly the planar quintic-circuit geometry organized by Theorem 12.215 and Corollary 12.216. Hence this residual corner is already-quintic rather than a genuinely new mixed septic branch. $\square$

Corollary 12.386 (Right-pincer trichotomy near the centered base). *Near the centered diagonal base, the mixed local picture is reduced to the following three cases:*

$$V_1 \neq 0 \implies \text{clean } v_5\text{-patch branch};$$

$$V_1 = 0, \tilde{\tau}_u \neq 0 \implies \text{late-}v_5 \text{ branch locally sterile on } Q_u;$$

$$V_1 = 0, \tilde{\tau}_u \equiv 0 \iff \Delta_7 = -a u_5^2 \implies \text{planar-lift corner, hence already-quintic.}$$

So every late- v_5 branch near the centered base is either locally sterile or already absorbed into the quintic-circuit regime. The only genuinely new mixed local branch left to study there is the clean branch

$$\text{ord}_s v_5 = 1.$$

Proof. Combine Lemma 12.384 and Lemma 12.385, together with the fact that v_5 is odd in the centered variable. $\square$

Corollary 12.387 (Right-pincer local reduction near the centered base). *Assume the centered scalar package from Remark 12.378, and assume also that the centered tagged quintic witness gives*

$$U_1 \neq 0.$$

Then every sufficiently small nearby genuinely new mixed 4-point obstruction point near the centered base must lie on the clean branch

$$\text{ord}_s v_5 = 1,$$

equivalently

$$V_1 \neq 0.$$

In particular, every late- v_5 branch near the centered base is either locally sterile on the u_5 -patch or already absorbed into the defect-free quintic-circuit regime.

Proof. Because v_5 is odd in the centered variable, either

$$\text{ord}_s v_5 = 1 \quad \text{or} \quad \text{ord}_s v_5 \geq 3 \text{ odd.}$$

If $\text{ord}_s v_5 \geq 3$, then Lemma 12.384 and Lemma 12.385 show that the branch is either locally sterile on the u_5 -patch or planar-lift and hence already-quintic. So a sufficiently small nearby genuinely new mixed 4-point obstruction cannot lie on such a late- v_5 branch. Therefore any such genuinely new mixed local branch must lie on the only remaining possibility,

$$\text{ord}_s v_5 = 1. \quad \square$$

Proposition 12.388 (Realization lemma for lifted germs). *Let $\tilde{Y}(\xi)$ and $\tilde{S}(\xi)$ be arbitrary analytic germs at $\xi = 0$. Define*

$$c(\xi) \equiv 1, \quad v_5(\xi) := 1 + \xi, \quad u_5(\xi) := \tilde{Y}(\xi), \quad \Delta_7(\xi) := (1 + \xi)\tilde{S}(\xi).$$

Then on a sufficiently small neighborhood of 0,

$$c \neq 0, \quad v_5 \neq 0, \quad M = c v_5' - c' v_5 = 1 \neq 0,$$

so this is a good v_5 -patch. The normalized lifted coordinates are

$$x = \frac{v_5}{c} = 1 + \xi, \quad Y = \frac{u_5}{c} = \tilde{Y}(\xi), \quad S = \frac{\Delta_7}{c v_5} = \tilde{S}(\xi).$$

After the affine shift $x \mapsto x - 1$, the lifted germ becomes

$$Q_v(\xi) = (\tilde{Y}(\xi), \xi, \tilde{S}(\xi)).$$

Thus every analytic pair of germs $(\tilde{Y}, \tilde{S})$ occurs, after affine normalization, as the lifted pair (Y, S) on some good v_5 -patch.

Proof. The displayed formulas are immediate from the definitions. The hypotheses $c \neq 0$, $v_5 \neq 0$, and $M = 1 \neq 0$ hold on a sufficiently small neighborhood of 0, and the formulas for x, Y, S are direct. $\square$

Corollary 12.389 (No new obstruction for $(\mathcal{Z}, \mathcal{R})$ is available at the current lifted level). *No identity involving only*

$$c, \quad v_5, \quad u_5, \quad \Delta_7$$

and their derivatives can force either of the two dream conclusions:

$$\mathcal{R} \text{ is even in the normalized nonmixed coordinate,}$$

or the mixed-branch exclusion

$$(r, \text{ord}_0 \mathcal{Z}) \notin \{(1, 1), (1, 3), (2, 2)\} \quad \text{at every mixed point.}$$

In particular, the nonmixed models

$$Q_v(\xi) = (\xi^n, \xi, \xi^2), \quad n \geq 4 \text{ even,}$$

are all realizable at the present lifted level; the cases

$$n = 4, 6, 8$$

are simply the first currently explicit low-order examples. The dangerous mixed models

$$Q_v(\xi) = (\xi^n, \xi, \xi^m), \quad (m, n) = (3, 4), (3, 6), (4, 6),$$

are all realizable at the present lifted level.

Consequently, any RH-specific obstruction to the low-order dangerous 4-point patterns must use information not encoded in the current lifted variables

$$(c, v_5, u_5, \Delta_7)$$

alone. It must come from deeper source-level corrected quotient-germ structure, matrix-level identities behind (u_5, Δ_7) , or an additional symmetry not yet visible in the present lifted formulation.

Proof. Apply Proposition 12.388 with

$$(\tilde{Y}, \tilde{S}) = (\xi^n, \xi^2)$$

for the nonmixed cases, and with

$$(\tilde{Y}, \tilde{S}) = (\xi^n, \xi^m)$$

for the mixed cases. These realize exactly the dangerous low-order patterns for $(\mathcal{Z}, \mathcal{R})$ listed above. $\square$

Remark 12.390 (Updated nonmixed status versus mixed local exhaustion). The local nonmixed lifted route is now closed at the derivative-geometric level. Exact nonmixed contacts force zeros of $\mathcal{R}$; one-sign $\mathcal{R}$ -intervals are sterile; recurrent exact contacts force affine-planar collapse; and isolated zeros of $\mathcal{R}$ are classified by parity. In addition, the repeated-source / first-loss mechanism is now fully reduced: no attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at odd zeros of $\mathcal{R}$. Thus the nonmixed burden is no longer a local parity theorem or a separate repeated-source obstruction. It is the global/source-level control of which odd zeros of

$$\mathcal{R} = \frac{T_v}{K_v^2}, \quad \text{equivalently} \quad \mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

can actually arise along opened nonmixed arcs.

On the mixed side, the decisive pair-level object remains

$$A_5^{\mathbf{f}}(s) = u_5(s)I + v_5(s)S.$$

By Corollary 12.382, this package is not identically zero, so after shrinking to one-sided punctured arcs the centered branch enters either a good v_5 -patch or the parallel u_5 -patch. Near the centered base, however, the theorem package now goes further: every late- v_5 branch is already either locally sterile on the automatic u_5 -patch or absorbed into the planar defect-free quintic-circuit regime, so the only genuinely new mixed local branch left there is the clean branch

$$V_1 \neq 0.$$

Accordingly, the mixed-side local problem is no longer a vague codimension-two obstruction, nor merely a patch-opening issue. Near the centered base it has been reduced to the single clean branch $V_1 \neq 0$, and on that branch the remaining theorem-level task is the finite scalar descent

$$c(-s) = c(s) \pmod{s^6}, \quad \text{equivalently} \quad C_1 = C_3 = C_5 = 0,$$

which makes the centered-to-patch finite-jet parity bridge formal.

Remark 12.391 (What the four-point branch now reduces to). The lifted 4-point branch is now organized by the canonical invariant pair

$$\mathcal{Z} = \frac{Y''}{S''} = \frac{J_{u_5}}{J_\lambda}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}, \quad \lambda = \frac{\Delta_7}{v_5}.$$

On the nonmixed side, after normalization to

$$Q_v(\xi) = (\tilde{Y}(\xi), \xi, \xi^2),$$

the local derivative-side picture is now explicit. Exact nonmixed contacts force zeros of

$$\mathcal{R} = \frac{1}{2} \tilde{Y}''',$$

isolated zeros are classified by parity, one-sign $\mathcal{R}$ -intervals are exactly-contact sterile, and recurrent exact contacts force affine-planar collapse.

The explicit low-order list

$$\text{ord}_0 \mathcal{R} \in \{1, 3, 5\},$$

equivalently the coefficients

$$z_2, \quad z_4, \quad z_6,$$

should therefore now be regarded only as the first explicit low-order nonmixed odd-order cases, not as the full local nonmixed picture.

For fixed source point a , repeated-source exact contacts factor through the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently through the below-lifted quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)}.$$

Its derivative is governed by the osculatory 4×4 source determinant $\mathfrak{D}(a, h)$. By Proposition 12.291, this repeated-source / first-loss mechanism is now completely reduced: no attained separated first loss can occur, and any first loss is exactly the diagonal birth at an isolated odd-order zero of $\mathcal{R}$.

Thus the remaining nonmixed burden is no longer local parity, nor a separate first-loss geometry, but purely the global/source-level control of which odd zeros of $\mathcal{W}_{\text{src}}$, equivalently of $\mathcal{R}$, occur on genuine opened arcs.

On the mixed side, the explicit low-order dangerous list is

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2),$$

equivalently

$$(r, \text{ord}_0 \mathcal{R}) = (1, 0), (1, 2), (2, 1).$$

Equivalently, in the below-lifted Wronskian formulation from Remark 12.331, the dangerous mixed models are exactly the order-pairs

$$(\text{ord}_0 J_\lambda, \text{ord}_0 J_{u_5}) = (1, 2), (1, 4), (2, 4).$$

So the mixed low-order branch is already a finite canonical Wronskian-ratio obstruction problem, captured in local mixed x -coordinates by

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2}.$$

By the realization lemma, no identity involving only

$$(c, v_5, u_5, \Delta_7)$$

and their derivatives can eliminate these mixed patterns in general.

At the pair level, the centered fixed-sector quintic package

$$A_5^f(s) = u_5(s)I + v_5(s)S$$

is now the natural object controlling whether any quintic patch opens at all. By Corollary 12.382,

$$A_5^f(s) \neq 0,$$

so the centered branch cannot suffer total quintic collapse.

Near the centered base, however, the mixed-side local picture is now sharper than a mere patch dichotomy. By Corollary 12.387, every sufficiently small nearby genuinely new mixed 4-point obstruction must lie on the clean branch

$$\text{ord}_s v_5 = 1,$$

while every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the defect-free planar quintic-circuit regime. Thus the genuinely new mixed local geometry near the centered base has been reduced to the single clean branch $V_1 \neq 0$.

Accordingly, the remaining mixed-side theorem is no longer another lifted jet computation, nor a pointwise coefficient nonvanishing statement such as $V_1 \neq 0$ in isolation. It is the finite scalar descent on the clean branch, namely

$$c(-s) = c(s) \pmod{s^6}, \quad \text{equivalently} \quad C_1 = C_3 = C_5 = 0,$$

which makes the centered good- v_5 -patch parity bridge formal. After that centered local step, the clean-patch mixed-global problem is no longer open mixed geometry: it is absorbed by the mixed-to-scalar capture theorem (Theorem 12.407), with the source-level viewpoint serving only as the correct language for the lower-core and source-coupled handoff walls.

Remark 12.392 (What the new nonmixed/source-level machinery changes for the mixed side). The new repeated-source/source-determinant viewpoint and the no-go principle excluding purely planar (x, S) -closure do not create a new mixed-local closure theorem. Rather, they clarify the remaining mixed burden.

First, no argument using only planar (x, S) -geometry can close the 4-point branch, so any remaining mixed route must use either the centered scalar descent or genuine coupled Y - S source-level information.

Second, if a below-lifted backup route is needed, the natural mixed object is not an S -only invariant, but a coupled source-minor ratio built from u_5 and

$$\lambda = \frac{\Delta_7}{v_5},$$

i.e. a below-lifted avatar of

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}.$$

Thus the new nonmixed/source-level machinery functions not only as a strategic constraint and as a backup language for the mixed side, but also as the correct external handoff target for the $A = 0$ planar collapse; see Remark 12.393. It does not replace the local mixed exhaustion theorem near the centered base, but it does absorb the nonmixed affine-planar lane that remains after the local mixed reduction.

Remark 12.393 (The $A = 0$ collapse is already the nonmixed $\mathcal{R}$ -lane). Work on a good v_5 -patch and write an exact lifted 4-point relation in the form

$$AY + Bx + CS + D = 0, \quad Y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}, \quad S = \frac{\Delta_7}{cv_5}.$$

If $A = 0$, then the relation reduces to

$$Bx + CS + D = 0,$$

so the four nodes lie on an affine line in the planar (x, S) -curve. Thus the positive-dimensional $A = 0$ regime is not a separate mixed branch at all: it is precisely the nonmixed affine-planar exact-contact regime. In particular, the relevant theorem-strength input is already on the nonmixed side: exact nonmixed contacts force zeros of $\mathcal{R}$ (Proposition 12.263), one-sign $\mathcal{R}$ -intervals are sterile (Corollary 12.264), and interval-persistent / recurrent exact contacts force affine-planar collapse (Corollary 12.267). Accordingly, $A = 0$ should be viewed as the external handoff from the mixed side into the already-handled nonmixed $\mathcal{R}$ -package, not as a fourth independent mixed-global lane.

Remark 12.394 (Status of the lifted order-7 geometry). The lifted geometry is now substantially more stratified than a single zero-capture problem.

First, shrinking exact lifted three-point and four-point circuits can accumulate only at

$$K_v = 0 \quad \text{and} \quad T_v = 0,$$

respectively, with uniform small-diameter exclusion away from those zero sets; equivalently, they are rooted at the curvature- and torsion-type osculating determinants of the corresponding lifted planar and space curves (Lemma 12.245, Propositions 12.247 and 12.249).

Second, repeated-node limits are now explicit: three-point collapse limits become tangent-line contacts for the planar lifted curve $\Lambda_v = (x, S)$, and four-point collapse limits become tangent-plane contacts for the lifted space curve $Q_v = (Y, x, S)$.

Third, the three-point side is now blockwise globally controlled on good v_5 -patches. The curvature zero set $K_v = 0$ has no infinite-order flat points; even-order K_v -zeros are sterile; odd-order K_v -zeros carry exactly one local opposite-side tangent-contact branch; and that branch continues uniquely on the maximal right monotone-slope block, with no interior common-tangent event, no source-side return, and no interior accumulation. Thus the three-point side is no longer a live internal recurrence problem on such patches; only boundary exit of the relevant source/target slope-overlap block can remain, and in the four-point mixed-to-scalar picture this is used only as an external lower-core handoff target rather than as a new mixed carrier lane.

On the mixed four-point side, the right-pincer package now gives an analogous local simplification near the centered base: every late- v_5 branch is either locally sterile on the u_5 -patch or already absorbed into the planar quintic-circuit regime, so the only genuinely new mixed local branch left there is the clean branch $\text{ord}_s v_5 = 1$.

Fourth, the four-point side splits sharply into nonmixed and mixed regimes. On the nonmixed side, finite-order odd T_v -zeros already support genuine local exact branch sheets in the model geometry, so a blanket local sterility theorem is false. On the mixed side, the dangerous low-order obstruction list is finite and explicit:

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2),$$

equivalently

$$(r, \text{ord}_0 \mathcal{R}) = (1, 0), (1, 2), (2, 1).$$

Fifth, below the lifted variables one has the exact Wronskian-ratio reformulation

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}, \quad \mathcal{R} = \frac{c^2}{M} \frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right), \quad \lambda = \frac{\Delta_7}{v_5},$$

while the projective curvature ratio Θ yields no new independent local obstruction beyond $(\mathcal{Z}, \mathcal{R})$.

Sixth, the centered symmetric-placement Taylor package gives a strong local collision-side parity picture:

$$u_5, v_5, u_{7,K_1}, v_{7,K_1} \text{ odd in } s, \quad \Delta_7 \text{ even in } s,$$

with explicit septic coefficients D_2, D_4, D_6 . In the diagonal-centered odd/odd normal form

$$Q(x) \sim (x^n, x, x^m), \quad m, n \text{ odd}, \quad n > m \geq 3,$$

one gets a punctured neighborhood free of $K_v = 0$ and $T_v = 0$ points. Thus the explicit low-order dangerous 4-point patterns do not arise at the symmetric diagonal base itself.

So the mixed four-point side is no longer waiting for a new local geometry theorem. On the mixed side, the sufficiently small problem is now organized by the mixed-to-scalar capture principle: after the source-coupled and patch-loss reductions, no sufficiently small genuinely mixed branch on a nonplanar good patch can persist entirely inside

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\},$$

except after entry into already-absorbed planar, lower-core, repeated-node, repeated-source, or quintic-collapse regimes. This is formalized below by the local clean continuation proposition, the positive-scale endpoint corollary, and Theorem 12.407. On the still-live nonmixed side, any genuinely new input would have to be RH-specific source- or matrix-level structure imposing stronger constraints on

$$\mathcal{R} = \frac{T_v}{K_v^2}.$$

Remark 12.395 (Live attack routes). At the present stage, the live routes are now quite sharply separated.

- (1) **Two-point branch: the residual burden is now split more sharply.** The stronger algebraic route is still Lemma 12.124: prove that the actual corrected two-pair cubic/quintic/septic package is analytic on small two-atom data and satisfies swap symmetry, one-amplitude collapse, and diagonal merger

$$\mathfrak{P}_2(a_1, h; a_2, h) = (a_1 + a_2)F_h.$$

The weaker exact route is Proposition 12.126, which would already remove exact close two-point bad cores on compact subsets of $\{c \neq 0\}$ from continuity and diagonal collapse of the actual corrected two-atom quotient germ.

What is now ruled out is the naive source-summed corrected-block route: its extracted one-atom finite-order coefficients are even in the source weight and therefore cannot supply the required linear merger law. On the mixed side, determinant-only and quotient-only arguments are likewise shear-blind. The genuinely new derivative-geometric lane is the exact projective λ -shear / secant-shadow route on a good overlap patch.

By Proposition 12.333, a close exact mixed two-point bad branch can accumulate only on

$$\{S = 0\} \cup \{K_v = 0\}, \quad S = \frac{\Delta_7}{c v_5}.$$

By Proposition 12.340, the part of $K_v = 0$ with $S \neq 0$ is already absorbed by the existing mixed low-order obstruction package. By Corollary 12.341, the conditional exact fixed-shear overlap lane on

$$\{S = 0\} \cap \{K_v \neq 0\}$$

transfers into the established nonmixed 4-point geometry.

So the only genuinely residual exact fixed-shear mixed corner is

$$\{S = 0\} \cap \{K_v = 0\}.$$

There the branch is quotient-diagonal / involutive, the quartic–sextic first rung is already quotient-injective, and the remaining exact fixed-shear burden is now best viewed as split into two sublanes.

The *close* exact fixed-shear lane is now scalar-local at theorem level: by Corollary 12.363, on a compact nonconstant quotient branch the close exact repeated-value carrier is a finite disjoint union of local fold half-arcs, one per odd zero of

$$\widehat{\mathcal{D}} = \widehat{\mathcal{Q}}',$$

with even zeros sterile; by Remark 12.364, the constant-quotient branch reduces, after affine gauge, to repeated values of a single planar scalar and also contributes only local fold half-arcs, with Corollary 12.351 ruling out the $\widehat{\mathcal{D}}(0) \neq 0$ basepoint. So this lane no longer admits a connected nonlocal close carrier; the remaining intrinsic obstruction there is equality of consecutive lobe amplitudes across these local fold half-arcs.

The *three-run overlap* fixed-shear lane should still be read conditionally as an inverse-branch-gap transport problem on an actual outer-overlap interval, namely in the regime

$$B_0 + B_2 > B_1.$$

On such an interval one has outer inverse branches and a gap function

$$\Delta(q) = \phi_2(q) - \phi_0(q).$$

Any interior critical point of Δ then yields a same-value / same-slope witness, and the current endgame fork is: either the transport-curvature scalar fails to descend across the outer runs, giving the first explicit nonlocal transport obstruction, or descent holds and the outer shells rigidify to exact translated replication. What is not yet settled at theorem level is an unconditional activation theorem from open overlap alone.

Thus the remaining two-point work is now concentrated in three very specific targets: finite-order source-level merger, quotient-diagonal continuity/diagonal collapse, and a transport/package theorem on the three-run overlap fixed-shear lane; the scalar-local ceiling on the close exact fixed-shear lane is already captured by Corollary 12.363.

- (2) **Three-point branch (derivative-geometric layer now stabilized).** The lifted three-point side is now the most mature local branch. Shrinking exact lifted 3-point circuits are tied to the osculating determinant and hence to

$$K_v = 0,$$

the local branch structure at finite-order K_v -zeros is completely classified by parity, and the odd-order branch is blockwise globally rigid on good v_5 -patches: infinite-order flatness is excluded, the branch continues uniquely on the maximal right monotone-slope block, and there is no interior common-tangent recurrence or source-side return. Inside the finite-core program this stabilizes the lifted $k = 3$ collinearity layer; any residual issue there is only boundary exit of the relevant source/target slope-overlap block, not an internal three-point propagation mechanism.

- (3) **Four-point branch (mixed side now captured by scalar exits).** On the nonmixed side, the exact-contact and repeated-source / first-loss packages are now organized by the scalar

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

with one-sign intervals sterile and interval-persistent exact contacts forcing affine-planar collapse. On the mixed side, the centered-base local route is exhausted up to the clean branch, and the source-coupled and patch-loss frontiers are now reduced to lower-core / shifted-3-point / already-quintic handoffs. Accordingly, the mixed-global four-point side is no longer a search for new local mixed geometry: by Theorem 12.407, a genuinely mixed clean branch on a nonplanar good patch cannot persist entirely inside

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\},$$

except after entry into already-absorbed regimes. So the four-point side is now best viewed as captured by the centered mixed local theorem, the three-point K_v -package, and the nonmixed $\mathcal{R}$ -package, rather than as an independent mixed-global carrier problem.

- (4) **Late- v_5 degeneration is now only a handoff mechanism.** If the v_5 -patch fails and v_5 starts late, the valuation form of the septic transport constraint forces matching late vanishing of the septic transport channel v_{7,K_1} . Thus failure of the v_5 -route is not a free local cancellation but a coupled quintic–septic degeneration scheme, and in the current architecture it should be read only as a handoff into lower-core or already-quintic geometry, not as an independent mixed branch lane.
- (5) **Four-point nonmixed branch via source-level control of odd $\mathcal{W}_{\text{src}}$ -zeros.** The local nonmixed derivative geometry is now explicit: exact contacts force zeros of $\mathcal{R}$, one-sign $\mathcal{R}$ -intervals are sterile, recurrent exact contacts force affine-planar collapse, and isolated zeros are classified by parity. At source level, the same structure is encoded by the canonical Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor on a nonmixed arc.

For fixed source point a , repeated-source exact contacts are controlled by the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently by the source quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

with derivative governed by the osculatory 4×4 determinant $\mathfrak{D}(a, h)$. By Proposition 12.291, this repeated-source / first-loss mechanism is not an independent obstruction: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$, equivalently of $\mathcal{W}_{\text{src}}$.

The remaining theorem-level task on this side is therefore the global/source-level exclusion or absorption of odd zeros of $\mathcal{W}_{\text{src}}$ on opened nonmixed arcs. In the unified mixed / nonmixed picture, this is exactly the scalar $\mathcal{R}$ -lane that absorbs the interval-persistent $A = 0$ collapse from the mixed side.

Remark 12.396 (Stratified lifted local picture). The lifted local obstruction now splits into one closed regime and two genuinely live regimes.

First, exact shrinking lifted 3-point circuits concentrate into

$$K_v = 0.$$

On good v_5 -patches, this side is now blockwise globally controlled: there is no infinite-order flatness, even-order K_v -zeros are sterile, odd-order K_v -zeros seed exactly one local opposite-side branch, and that branch continues uniquely on the maximal right monotone-slope block with no interior common tangent, no source-side return, and no interior accumulation. So the three-point side contributes only possible boundary exit, not an internal recurrence mechanism.

Second, exact shrinking lifted 4-point circuits concentrate into

$$T_v = 0.$$

On the nonmixed side $K_v \neq 0$, the local derivative-side picture is now explicit. Exact contacts force zeros of

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

isolated zeros are classified by parity, one-sign $\mathcal{R}$ -intervals are exactly-contact sterile, and recurrent exact contacts force affine-planar collapse. So the nonmixed 4-point side is no longer a vague local branching problem; what remains there is global/source-level control of which odd zeros of $\mathcal{R}$ actually occur on genuine opened arcs.

Third, on the mixed side $K_v = T_v = 0$, the low-order obstruction list is now finite and explicit:

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2).$$

At the pair level, however, the centered quintic fixed-sector package

$$A_5^{\dagger}(s) = u_5(s)I + v_5(s)S$$

is now known not to vanish identically on the centered diagonal branch. Near the centered base, the right-pincer package further reduces the mixed local picture: every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the planar quintic-circuit regime, so the only genuinely new mixed local branch left there is the clean branch

$$\text{ord}_s v_5 = 1.$$

Thus the remaining lifted issues are now stratified much more sharply. On the nonmixed side, the live question is the global/source-level control of odd zeros of $\mathcal{R}$ on genuine opened arcs. On the mixed side, the centered local issue is the finite scalar descent

$$c(-s) = c(s) \pmod{s^6},$$

while the clean-patch global side has already been compressed to the scalar capture statement of Theorem 12.407. So the mixed four-point branch should no longer be viewed as an independent open geometric carrier problem away from the scalar exits.

Remark 12.397 (Current geometric bottleneck). The remaining geometric bottleneck is now split into two qualitatively different layers.

On the one-pair geometric layer, the obstruction is no longer bulk $W = 0$ geometry, and it is no longer local adjacent pair-pair birth across a generic turning boundary. On the $W = 0$, $M \neq 0$ side, Proposition 12.59 excludes open-interval flatness and Theorem 12.107 shows that every finite-order isolated $W = 0$ boundary is locally sterile for pair-pair birth. On the turning-boundary side, Theorem 12.96 removes actual adjacent pair-pair lines unconditionally.

At the lifted order-7 level, the canonical derivative-geometric determinants

$$K_v \quad \text{and} \quad T_v$$

govern shrinking exact lifted 3- and 4-point circuits on $v_5 \neq 0$ patches. On the 3-point side, this layer is stabilized: shrinking exact circuits concentrate into $\{K_v = 0\}$, finite-order isolated zeros are classified by parity, and the odd-order branch is blockwise globally rigid on good v_5 -patches.

On the 4-point side, exact shrinking lifted 4-point circuits concentrate into

$$\{T_v = 0\},$$

again with uniform small-diameter exclusion away from that zero set.

On the nonmixed side $K_v \neq 0$, generic odd-order T_v -zeros can still carry preexisting close branches, so the nonmixed branch remains genuinely live and now asks for global/source-level control of which odd zeros of

$$\mathcal{R} = \frac{T_v}{K_v^2}$$

actually arise on opened arcs.

By contrast, on the mixed side $K_v = T_v = 0$, the local geometry near the centered base is no longer the main bottleneck. There the right-pincer package shows that every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the defect-free planar quintic-circuit regime, so the only genuinely new mixed local branch left is the clean branch

$$V_1 \neq 0.$$

Accordingly, the mixed bottleneck is no longer a diffuse local codimension-two analysis. At the local level it is the finite scalar descent

$$c(-s) = c(s) \pmod{s^6}$$

on that clean branch; at the global level, after the source-coupled and patch-loss absorptions, the clean-patch mixed problem has already compressed to the scalar alternatives

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0,$$

as formalized in Theorem 12.407.

So the missing lifted ingredient is no longer another generic local geometric exclusion theorem. On the mixed side, the centered-base local package, the source-coupled exclusion, the patch-loss reductions, and the nonmixed $\mathcal{R}$ -handoff now combine to give the clean theorem-level conclusion formalized below in Theorem 12.407: every maximal genuinely mixed clean branch on a nonplanar good patch is forced either into the centered-base scalar exit

$$T_v = 0$$

or into the nonmixed scalar exit

$$\mathcal{R} = 0,$$

up to already-absorbed planar, lower-core, repeated-node, repeated-source, and quintic-collapse regimes. Thus the still-live frontiers are no longer mixed four-point geometry, but rather the nonmixed/source-level control of odd zeros of $\mathcal{R}$ and the separate two-point program.

Remark 12.398 (Current state of the argument). The local one-pair analysis now consists of the canonical triple

$$(c(h), v_5(h), \Delta_7(h)),$$

together with the derived quantities

$$\begin{aligned} M(h) &= c(h)v_5'(h) - c'(h)v_5(h), & N(h) &= c(h)\Delta_7'(h) - c'(h)\Delta_7(h), \\ L(h) &= v_5(h)\Delta_7'(h) - v_5'(h)\Delta_7(h), & W(h) &= \det(F(h), F'(h), F''(h)). \end{aligned}$$

On the region where $v_5 M \neq 0$, the projective graph variables

$$\mu = \frac{c}{v_5}, \quad \lambda = \frac{\Delta_7}{v_5}$$

identify the normalized curve with a graph $\lambda = \lambda(\mu)$, and

$$\lambda_{\mu\mu} = \frac{v_5^3}{M^3} W, \quad \beta := \lambda_\mu = -\frac{L}{M}, \quad T(h) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right).$$

Thus the global line-incidence problem is equivalently a problem about the recurrence of tangent slopes β , or about vertical-line re-entry on the dual tangent-line curve T .

Locally, both generic boundary types are now under control. At finite-order $W = 0$, $M \neq 0$ boundaries there is no local pair-pair birth, while at generic turning boundaries actual adjacent pair-pair lines are excluded unconditionally. The stronger symmetric directional-kernel theorem remains available only conditionally on zero-capture and now serves as a sharper but nonessential alternative route.

Accordingly, the equivalent slope/tangent/dual direct-slice formulations remain the correct geometric dictionary, but they are no longer themselves the live theorem target. The explicit direct provenance formula gives

$$\phi = A q^{(7)} \quad (A = \kappa \eta_2),$$

and on any generic one-pair block this direct slice is fully universal (Proposition 13.8). Moreover, even after imposing any finite family of linear provenance or matching conditions on q , direct-slice recurrence geometry remains formally realizable (Corollary 13.9).

Thus the remaining gap is not an abstract global no-reentry theorem for the scalar graph ϕ by itself. It is the search for a genuinely non-universal provenance law for q , or an external mixed/canonical input, that cuts down the allowed direct-slice graphs on an entire block.

On the lifted 4-point side, the mixed low-order obstruction list is now both finite and canonical:

$$(r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\},$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

or

$$(m, n) = (3, 4), (3, 6), (4, 6).$$

The centered Taylor package already provides the explicit coefficients

$$U_j, V_j, \widehat{U}_j, \widehat{V}_j, D_2, D_4, D_6,$$

and the tagged quintic witness now shows that

$$A_5^{\dagger}(s) \neq 0$$

on the centered diagonal branch.

On the mixed side, however, the local theorem picture is now sharper than a mere punctured-arc patch dichotomy, and the global side is correspondingly compressed. Near the centered base, the late- v_5 sector is already removed by the u_5 -patch right-pincer package as either locally sterile or planar-lift and hence already absorbed into the defect-free quintic-circuit regime. The only genuinely new mixed local branch there is the clean branch $V_1 \neq 0$, and that branch is excluded by the canonical scalar descent together with the centered good- v_5 -patch parity package. Moreover, after the source-coupled and patch-loss reductions, the sufficiently small mixed-global problem is no longer a search for a new mixed carrier lane. At the theorem level, the clean small-branch mixed problem has already compressed to the scalar alternatives

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

Here the $K_v = 0$ side no longer survives as an independent mixed carrier on the clean small-branch four-point side, but on the additional conditional exact fixed-shear overlap slice it still leaves a narrower residual two-point burden. That residual slice is not a new mixed local carrier: modulo constant shear, its pointwise quotient germ is already governed by

$$\mathcal{D} = \frac{\mathcal{W}_{\text{src}}}{J_{\lambda}^2},$$

or equivalently on the involution quotient by

$$\widehat{\mathcal{D}}(u) = \widehat{Q}'(u).$$

Moreover, by Corollary 12.254, the ordinary local repeated-node tangent-contact mechanism on a good v_5 -patch is necessarily opposite-side, so this quotient-diagonal residual lane cannot locally merge with the standard lifted 3-point branch geometry. Thus the genuinely mixed small-branch problem is best read as mixed-to-scalar capture together with a separate quotient-diagonal residual transport problem, not as another free mixed carrier geometry.

In parallel, the finite-core endgame now splits into three layers.

First, on the two-pair side, the strengthened invariant

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right)$$

still gives the correct algebraic bridge, and on good $M \neq 0$ patches the quantitative two-pair coincidence theorem forces bad two-point cores into a defect-scale close regime. The minimal finite-order source criterion Lemma 12.124 still isolates the exact swap / one-amplitude / diagonal-merger package that would imply the desired quadratic interaction factorization. However, the naive source-summed corrected block model does not prove this criterion: along a weighted one-atom path its extracted finite-order coefficients are even analytic in the source weight, so the required linear merger law cannot come from that model alone.

At present the two-point side is better organized by the three-front compression of Remark 12.210 than by the older language of “three conditional lanes.”

The finite-order source-level merger criterion of Lemma 12.124 remains one possible local route on good patches, but the genuinely residual mixed burden is now the quotient-diagonal exact fixed-shear corner

$$(S = 0, K_v = 0).$$

There the projective λ -shear compactness and secant-shadow analysis identify the correct residual geometry, while the collision-cancellation chart, the no-finite-local-rescue picture, and

the quotient-transport package show that the remaining issue is not a missing local branch taxonomy.

So the live downstream two-point targets are now the following: a good-patch edge-law theorem for the actual corrected cubic/quintic defects in the collision-cancellation chart; a quotient-visible transport/state-local closure on the exact fixed-shear corner; or a provenance-sensitive package theorem strong enough to force coincidence/functoriality at the collision/cancellation locus. In particular, the current residual burden should be read as local blow-up regularity plus quotient transport plus package-level coincidence, not as a fourth independent mixed lane.

Second, on the lifted three-point side, shrinking exact circuits now concentrate into $K_v = 0$, and this concentration is explicitly derivative-geometric: by the osculating-determinant expansion, repeated-node limits become tangent-line contacts for the planar lifted curve

$$\Lambda_v = (x, S).$$

On good v_5 -patches this side is now blockwise globally controlled: K_v has no infinite-order zeros; even-order zeros are sterile; odd-order zeros carry exactly one local branch; and that branch continues uniquely on the maximal right monotone-slope block with no interior common-tangent event or source-side return. So the three-point side is reduced to boundary exit of the relevant slope-overlap block, not an internal recurrence mechanism. Inside the finite-core reduction, this is exactly the stabilization of the lifted $k = 3$ collinearity layer.

Third, on the lifted four-point side, shrinking exact circuits now concentrate into $T_v = 0$, repeated-node limits become tangent-plane contacts, and the local geometry splits into a nonmixed and a mixed regime.

On the nonmixed side, the local derivative-side theory is now explicit: exact contacts force zeros of $\mathcal{R} = T_v/K_v^2$, isolated zeros are classified by parity, one-sign $\mathcal{R}$ -intervals are exactly-contact sterile, and recurrent exact contacts force affine-planar collapse. At source level, the same structure is encoded by the gauge-invariant Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor on a nonmixed arc. For fixed source point a , repeated-source exact contacts factor through the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently through the source quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

with derivative governed by the osculatory source determinant $\mathfrak{D}(a, h)$. So the remaining nonmixed burden is global/source-level control of which odd zeros of $\mathcal{W}_{\text{src}}$, equivalently of $\mathcal{R}$, occur on a genuine opened nonmixed arc.

On the mixed side, the dangerous low-order obstruction list is the finite canonical set

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2},$$

equivalently the order patterns

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2).$$

The remaining theorem-level task there is no longer local patch classification, and on the clean-patch side it is no longer an open global capture problem either: the sufficiently small mixed-global closure is now formalized by Theorem 12.407. Thus the mixed side should now be read as split into two parts: the centered clean-branch scalar descent

$$c(-s) = c(s) \pmod{s^6},$$

which is the remaining local centered-base task, and the already-organized mixed-to-scalar handoff showing that outside absorbed regimes every maximal genuinely mixed clean branch is forced into

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

So any genuinely new input beyond the present lifted-variable package would now have to be RH-specific source-/matrix-level structure on the scalar side, not a new mixed local geometry theorem.

Moreover, Theorem 12.239 shows that no new sufficiently small local mixed 4-point obstruction remains near the centered diagonal base, so any external mixed input capable of constraining the direct-slice/provenance lane must be genuinely global/nonlocal rather than a new local mixed septic-jet pattern.

Remark 12.399 (Post-draft findings: established, provisional, and live). This remark summarizes ledger findings that are now partly formalized in the surrounding propositions, corollaries, and theorems, and partly retained as live WIP for the remaining package/frontier obligations.

First, on the two-point branch, the naive source-summed corrected block model still cannot prove the exact finite-order merger criterion: along a weighted one-atom path the extracted finite-order coefficients are even analytic in the source weight, so the needed linear merger law does not come from that model alone.

At the same time, the residual mixed picture is now much cleaner. The overlap bad set is fully split:

$$(K_v = 0, S \neq 0)$$

is already absorbed by the mixed low-order package,

$$(S = 0, K_v \neq 0)$$

is already transferred into the established nonmixed 4-point geometry, and only the exact fixed-shear corner

$$\{S = 0\} \cap \{K_v = 0\}$$

remains genuinely live.

On that residual corner the branch is quotient-diagonal / involutive,

$$\Phi(t) = \Phi(-t), \quad Q(t) = \widehat{Q}(t^2), \quad \widehat{D}(u) = \widehat{Q}'(u).$$

Moreover, Proposition 12.353 shows that every finite parity-normalized transport jet descends to the quotient variable $u = t^2$, while Corollary 12.354 shows that on the quartic–sextic rung the finite quotient-visible transport state has local rank one, and Corollary 12.355 rules out any third local reset inside that descended state.

So the residual exact fixed-shear burden is no longer a search for another local mixed carrier or hidden finite-order transport variable. It has become a problem of collision-cancellation regularity, quotient transport, and package/provenance-sensitive coincidence. In particular, any still-live obstruction must come from the first genuinely relational or provenance-sensitive two-atom input that survives after quotient-visible transport descent, or from non-finite-order dependence.

Moreover, by Corollary 12.254, every sufficiently small off-diagonal repeated-node tangent-contact branch on a good v_5 -patch is necessarily opposite-side. Thus the quotient-diagonal exact fixed-shear residual lane cannot locally merge with the ordinary lifted 3-point branch mechanism.

There is now also a sharper internal split of the exact fixed-shear residual lane itself, and one side of that split has now been promoted to theorem level. On the close exact side, the scalar-local compression is no longer a work-in-progress expectation: by Corollary 12.363, on a compact nonconstant quotient branch the close exact repeated-value carrier is a finite disjoint union of local fold half-arcs, one per odd zero of

$$\widehat{D} = \widehat{Q}',$$

with even zeros sterile; by Remark 12.364, the constant-quotient branch also reduces, after affine gauge, to repeated values of a single planar scalar and contributes only local fold half-arcs. So the close exact carrier no longer admits a connected nonlocal close carrier, and the remaining intrinsic obstruction on that sublane is equality of consecutive lobe amplitudes across these local fold half-arcs.

On the three-run overlap side, by contrast, the endgame is still conditional. The residual overlap regime is

$$B_0 + B_2 > B_1,$$

not

$$B_1 > B_0 + B_2.$$

In that actual outer-overlap regime, the quotient-visible problem may be reduced to the inverse-branch-gap transport problem on the two outer runs. Writing

$$\Delta(q) := \phi_2(q) - \phi_0(q),$$

any interior critical point of Δ gives a same-value / same-slope witness, and the live endgame fork becomes: either the transport-curvature scalar fails to descend across the outer runs, giving the first explicit nonlocal transport obstruction, or descent holds and the outer shells rigidify to exact translated replication. An unconditional activation theorem from open overlap alone has not yet been established. So the residual exact fixed-shear lane is no longer a vague hidden branch problem; it has split into a close scalar-local lane, now controlled at theorem level by Corollary 12.363, and a separate three-run overlap-transport lane that remains the live conditional target.

Second, on the lifted four-point branch, the explicit low-order obstruction list has been reduced to the six coefficients

$$z_2, z_4, z_6, \kappa_{1,1}, \kappa_{1,3}, \kappa_{2,2},$$

or equivalently to the order patterns

$$\text{ord}_0 \mathcal{R} \in \{1, 3, 5\}, \quad (r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}.$$

Third, the centered symmetric-placement Taylor package now gives explicit septic coefficients

$$D_2, \quad D_4, \quad D_6,$$

explicit bridge coefficients

$$X_1, X_3, X_5, \quad L_1, L_3, L_5,$$

and a punctured-neighborhood exclusion at the diagonal base.

Fourth, the centered fixed-sector quintic package

$$A_5^f(s) = u_5(s)I + v_5(s)S$$

is now firmly established as the correct pair-level object on the mixed side. The centered tagged witness survives specialization strongly enough to show

$$U_1 \neq 0, \quad u_5(s) \neq 0, \quad A_5^f(s) \neq 0,$$

so the centered quintic pair cannot totally collapse.

Fifth, the right-pincer package now sharpens the mixed local picture near the centered base: every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the planar defect-free quintic-circuit regime. So the only genuinely new mixed local branch left there is the clean branch

$$V_1 \neq 0.$$

Sixth, on that clean branch the remaining theorem has become very small: the finite scalar descent

$$c(-s) = c(s) \pmod{s^6}, \quad \text{equivalently} \quad C_1 = C_3 = C_5 = 0.$$

Once this holds, the centered good- v_5 -patch bridge becomes formal:

$$Y = \frac{u_5}{c}, \quad S = \frac{\Delta_7}{cv_5}$$

are odd through the required order, and the entire explicit mixed low-order bad list

$$(r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}$$

is excluded there.

Seventh, the relation between the mixed and nonmixed 4-point sides is now structurally sharper. On a good v_5 -patch, an exact mixed relation

$$AY + Bx + CS + D = 0$$

with

$$Y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}, \quad S = \frac{\Delta_7}{cv_5}$$

and $A = 0$ is already exact affine-planarity for the planar curve (x, S) ; by Remark 12.393, this is precisely the nonmixed $\mathcal{R}$ -lane. More generally, the sheared four-zero reduction shows that every exact genuinely mixed 4-point circuit on a good v_5 -patch already forces either

$$K_v = 0 \quad \text{or} \quad \mathcal{R} = 0$$

in its convex hull. So the mixed-global problem should now be viewed as a mixed-to-scalar capture problem rather than as a separate branch geometry.

Eighth, the source-coupled and patch-loss frontiers have both been demoted to handoff mechanisms. In the sufficiently small regime, the source-coupled lower-core walls

$$D_\lambda = 0, \quad \mathfrak{D} = 0$$

are not pass-through points. On the patch side, the quintic projective lifts

$$\tilde{Q}_v = [cv_5 : u_5v_5 : v_5^2 : \Delta_7], \quad \tilde{Q}_u = [cu_5 : u_5^2 : u_5v_5 : -\Delta_7]$$

show that single-chart failures are removable after recharting, while the genuine common boundary

$$u_5 = v_5 = 0$$

drops either to pure quintic-direction collapse or to lower-core 3-point / repeated-node quotient geometry. Thus

$$cMv_5 = 0$$

should be read as lower-core handoff, not as a fourth independent mixed-global lane.

Ninth, after the centered-base local exhaustion theorem and the preceding handoffs are combined, the sufficiently small nonplanar mixed branch has no independent carrier away from

$$T_v = 0 \quad \text{and} \quad \mathcal{R} = 0.$$

Equivalently, there is no sufficiently small genuinely mixed 4-point branch contained entirely in

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}$$

on a nonplanar good patch, except after entry into already-absorbed affine-planar or lower-core regimes.

So the present frontier is no longer a generic local geometry problem. On the residual exact fixed-shear overlap slice

$$K_v = 0,$$

the pointwise quotient structure has already compressed to the single transport scalar

$$\hat{\mathcal{D}}(u) = \hat{Q}'(u), \quad u = t^2.$$

The quartic–sextic rung is the first explicit case, and there the quotient branch is locally one-sheeted. Thus the residual $K_v = 0$ lane is best viewed as a quotient-diagonal / involutive transport problem rather than as a new mixed local carrier.

At the same time, this pointwise compression does *not* yet prove that the full transported overlap / critical-value state is quotient-visible. What is proved inside the quotient-visible exact fixed-shear reduction is narrower and sharper: on a connected nonconstant quotient branch, Proposition 12.347 shows that nonadjacent re-entry of regular values can arise only through a three-run lobe-amplitude pattern, and Corollary 12.348 identifies the only surviving inequality

$$B_1 \geq B_0 + B_2.$$

Moreover, on the first live quartic–sextic rung, Corollary 12.351 shows that the basepoint is locally one-sheeted in the quotient variable and does not itself contribute such a lobe obstruction.

Accordingly, the live two-point burden on the $K_v = 0$ slice is now a no-hidden-reset / state-visibility problem for the actual corrected overlap state, together with the alternative source-level merger route and the already-separated higher-core issues.

Remark 12.400 (Two-point burden after mixed four-point closure). With the clean mixed four-point side now compressed to the scalar exits

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0,$$

the mixed two-point side should no longer be read as an independent frontier of mixed local geometry.

The overlap bad set has already split as follows:

$$(K_v = 0, S \neq 0)$$

is absorbed by the mixed low-order obstruction package,

$$(S = 0, K_v \neq 0)$$

is transferred into the established nonmixed 4-point geometry, and only the exact fixed-shear residual corner

$$\{S = 0\} \cap \{K_v = 0\}$$

remains genuinely live.

On that residual corner, the quotient-visible transport package is now much more rigid than a single pointwise scalar. By Proposition 12.353, every finite parity-normalized transport jet descends to the quotient variable $u = t^2$. On the quartic–sextic rung, Corollary 12.354 shows that every finite quotient-visible transport state is locally a function of Q alone, and Corollary 12.355 shows that no third local reset survives there.

So the residual exact fixed-shear corner is no longer best viewed as an unresolved mixed local curvature-branch problem. If the actual corrected defect/package is state-local in the descended transport state, then the corner closes by Proposition 12.356 or even by Corollary 12.357. Accordingly, any still-live input on the two-point side would have to come from the first genuinely relational or provenance-sensitive two-atom datum that survives after quotient-visible transport descent, or else from non-finite-order dependence.

Remark 12.401 (Residual mixed two-point lane: structural gain and present limitation). The present two-point picture now has three genuine structural gains on the exact fixed-shear residual corner.

First, after the off- $K_v = 0$ absorption results and the transfer of the simple-septic-zero slice into the nonmixed 4-point theory, the derivative-geometric mixed residue is sharply concentrated on the exceptional corner

$$\{S = 0\} \cap \{K_v = 0\}.$$

Second, by Proposition 12.353, every finite parity-normalized transport jet is quotient-visible in the variable

$$u = t^2.$$

So the residual corner is no longer a free two-sided t -state even at the finite transport-jet level.

Third, on the quartic–sextic rung, Corollary 12.354 shows that this descended finite transport state has local rank 1, and Corollary 12.355 shows that no finite-order hidden reset survives beyond the involution pair $t \leftrightarrow -t$.

What is *not* yet proved is that the actual corrected two-pair defect or package is already state-local in this descended transport state. If that state-locality held, then the residual branch would already be excluded by Proposition 12.356, or would collapse identically by Corollary 12.357.

Thus the live issue on the mixed residual side is now no longer local classification and no longer hidden finite-order transport dynamics. The remaining question is whether the first surviving obstruction is already quotient-visible, or instead comes from genuinely relational / provenance-sensitive two-atom structure or from non-finite-order dependence.

Proposition 12.402 (u_5 -chart exceptional-point handoff). *Work near a collapsing configuration with $u_5 \neq 0$, and use the projective lift*

$$\tilde{Q}_u = [cu_5 : u_5^2 : u_5v_5 : -\Delta_7].$$

Let

$$q_\nu = [0 : 1 : 0 : \nu]$$

be an exceptional limiting point on the exceptional line. Then every limiting hyperplane through q_ν induces the exact lower-core relation

$$Dc + Bv_5 + C(\lambda - \nu u_5) = 0, \quad \lambda = -\frac{\Delta_7}{u_5}.$$

Accordingly:

- one-node collapse hands off to exact shifted lower-core 3-point geometry;
- two-node collapse hands off to a repeated-node lower-core wall;
- if $u_5 = 0$ as well, one is on the common quintic-collapse locus already treated separately.

So the u_5 -chart side of patch loss contributes no independent mixed-global carrier lane.

Remark 12.403 (Shifted lower-core invariance on the u_5 -side). The shift

$$\lambda \mapsto \lambda - \nu u_5$$

does not alter the derivative-geometric lower-core package: it changes only the zeroth-order affine offset, not the relevant repeated-node/tangent-contact derivative structure. Thus the handoff in Proposition 12.402 is genuinely into the existing lower-core package, not into new geometry.

Proposition 12.404 (Local clean continuation off the explicit walls). *Work on a nonplanar good v_5 -patch and write*

$$Y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}, \quad S = \frac{\Delta_7}{c v_5}.$$

Let

$$\Phi_\gamma := Y - \gamma S, \quad F_{\alpha, \beta, \gamma}(x) := \Phi_\gamma(x) - \alpha - \beta x.$$

Suppose

$$h_1 < h_2 < h_3 < h_4$$

is an exact genuinely mixed clean 4-point configuration, so

$$F_{\alpha, \beta, \gamma}(x(h_i)) = 0 \quad (i = 1, 2, 3, 4),$$

the four roots $x(h_i)$ are distinct and simple, and the carrying hyperplane is genuinely mixed ($A \neq 0$). Assume moreover that the configuration does not lie on any of the walls

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

Then this configuration is an interior point of a smooth local solution manifold of exact genuinely mixed clean 4-point configurations. In particular, it cannot be a terminal point of a maximal clean mixed branch.

Proof. For

$$\mathcal{F}_i(h_1, h_2, h_3, h_4, \alpha, \beta, \gamma) := Y(h_i) - \gamma S(h_i) - \alpha - \beta x(h_i),$$

the Jacobian with respect to (h_1, h_2, h_3, h_4) is diagonal:

$$\frac{\partial \mathcal{F}_i}{\partial h_j} = \delta_{ij} (Y'(h_i) - \gamma S'(h_i) - \beta x'(h_i)).$$

Since

$$Y'(h_i) - \gamma S'(h_i) - \beta x'(h_i) = x'(h_i) F'_{\alpha, \beta, \gamma}(x(h_i)),$$

the diagonal entries are nonzero on a good v_5 -patch and at simple roots. So the Jacobian is invertible, and the claim follows from the implicit function theorem. $\square$

Corollary 12.405 (Positive-scale endpoints lie on an absorbed wall or a scalar exit). *Let Γ be a maximal connected genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch. Suppose $p \in \bar{\Gamma}$ is a terminal point at which the relevant node scale stays bounded below by a positive constant. Then p lies on one of*

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

In particular, after the previously established handoffs are invoked, a positive-scale terminal point cannot furnish a new genuinely mixed continuation mechanism: it must either lie in the centered-base capture set $\{T_v = 0\}$, lie in the nonmixed affine-planar scalar set $\{\mathcal{R} = 0\}$, or already belong to one of the absorbed planar/lower-core/quintic-collapse regimes.

Proof. If p lay off the listed wall set, then the limiting exact genuinely mixed clean configuration at p would satisfy the hypotheses of Proposition 12.404. Hence the branch would extend smoothly through p , contradicting maximality. Therefore a positive-scale terminal point must lie on one of

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

The final sentence is then exactly the previously established absorption package: $A = 0$ is the nonmixed affine-planar handoff (Remark 12.393), the source-coupled walls $D_\lambda = 0$ and $\mathfrak{D} = 0$ are locally non-carrying in the sufficiently small regime (Remark 12.399), and patch loss $cMv_5 = 0$ is absorbed into lower-core / shifted-3-point / quintic-collapse geometry (Proposition 12.402 and the surrounding post-draft findings package). $\square$

Proposition 12.406 (Small-scale non-scalar exits are absorbed). *Let Γ be a sufficiently small genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch. Assume*

$$\Gamma \cap \{T_v = 0\} = \emptyset, \quad \Gamma \cap \{\mathcal{R} = 0\} = \emptyset.$$

Then Γ cannot continue through any of the non-scalar exit walls

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0$$

as an independent genuinely mixed clean branch.

More precisely:

- (1) *if $A = 0$ persists on an interval, then Γ has already entered the nonmixed affine-planar $\mathcal{R}$ -package (Remark 12.393, Proposition 12.266, Corollary 12.267);*
- (2) *if $D_\lambda = 0$ or $\mathfrak{D} = 0$ occurs in the sufficiently small regime, then this lies on the source-coupled lower-core marked set, which is locally non-carrying and not a pass-through mechanism (Remark 12.399);*
- (3) *if $cMv_5 = 0$ occurs, then the branch is absorbed into lower-core / shifted-3-point / quotient repeated-node / quintic-collapse geometry (Remark 12.399 and Proposition 12.402).*

Accordingly, no sufficiently small genuinely mixed clean branch can remain an independent mixed carrier while meeting only non-scalar exit walls.

Proof. The $A = 0$ case is exactly the nonmixed affine-planar handoff by Remark 12.393; interval-persistent or recurrent exact contacts are then absorbed by Proposition 12.266 and Corollary 12.267.

The source-coupled walls $D_\lambda = 0$ and $\mathfrak{D} = 0$ are, in the sufficiently small regime, only source-coupled lower-core marked points and are locally non-carrying; see Remark 12.399.

Finally, patch loss $cMv_5 = 0$ has already been reduced to lower-core / shifted-3-point / quotient repeated-node / quintic-collapse handoffs; see Remark 12.399 and Proposition 12.402.

So none of the listed non-scalar walls can provide an independent genuinely mixed clean continuation mechanism under the stated hypotheses. $\square$

Theorem 12.407 (Global mixed-to-scalar capture on the clean patch). *Let Γ be a maximal connected genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch. Then Γ cannot persist as an independent genuinely mixed branch entirely inside*

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}.$$

Equivalently, every such maximal branch is forced either into

$$T_v = 0$$

or into

$$\mathcal{R} = 0,$$

up to the already-absorbed planar, lower-core, repeated-node, repeated-source, and quintic-collapse regimes.

Proof. Assume Γ is a maximal connected genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch and that

$$\Gamma \cap \{T_v = 0\} = \emptyset, \quad \Gamma \cap \{\mathcal{R} = 0\} = \emptyset.$$

We show that Γ cannot persist as an independent genuinely mixed branch.

If a terminal point of Γ has positive scale, then Corollary 12.405 places it on one of

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

The last two are excluded by hypothesis. The remaining three are already absorbed: $A = 0$ is the nonmixed affine-planar $\mathcal{R}$ -lane (Remark 12.393), the source-coupled walls $D_\lambda = 0$ and $\mathfrak{D} = 0$ are locally dead as pass-through mechanisms in the sufficiently small regime, and patch loss $cMv_5 = 0$ is absorbed into the lower-core / shifted-3-point / quintic-collapse package recorded in Remark 12.399 and Proposition 12.402. So no positive-scale terminal point yields an independent mixed continuation lane.

If instead the relevant scale along Γ shrinks, then Proposition 12.406 applies. Indeed, under the standing assumptions

$$\Gamma \cap \{T_v = 0\} = \emptyset, \quad \Gamma \cap \{\mathcal{R} = 0\} = \emptyset,$$

no sufficiently small genuinely mixed clean branch can continue through any of the non-scalar exit walls

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0$$

as an independent mixed carrier.

Thus in all cases Γ is forced either into $T_v = 0$, into $\mathcal{R} = 0$, or into one of the already-absorbed regimes. This contradicts the assumption that Γ persists as an independent genuinely mixed clean branch inside

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}.$$

□

Remark 12.408. The theorem should be read as the mixed-global four-point closure statement on a nonplanar good patch: after the source-coupled and patch-loss absorptions, no independent genuinely mixed clean carrier remains away from the scalar exits $T_v = 0$ and $\mathcal{R} = 0$.

Remark 12.409 (Unified mixed/nonmixed picture after the clean-patch closure). Theorem 12.407 should be read as the point where the mixed and nonmixed 4-point geometries fully meet.

On the mixed side, the centered-base local theorem (Theorem 12.239) closes the only genuinely new local mixed branch. On the nonmixed side, the affine-planar lane is already governed by the scalar invariant

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

and interval-persistent or recurrent exact contacts force affine-planar collapse (Proposition 12.266 and Corollary 12.267).

Accordingly, the sufficiently small clean mixed-global problem is no longer a search for a new mixed carrier lane. It is a mixed-to-scalar capture theorem: every independent genuinely mixed clean branch is forced either into the centered-base scalar exit

$$T_v = 0$$

or into the already-handled nonmixed scalar exit

$$\mathcal{R} = 0,$$

with all source-coupled and patch-loss phenomena absorbed into lower-core, repeated-node, or already-quintic handoffs.

Remark 12.410 (What remains genuinely open after the clean mixed closure). After Theorem 12.407, the clean mixed four-point case is no longer an independent global frontier. What remains genuinely open is the global nonadjacent propagation problem on the nonmixed/source-level side, together with the separate two-point program.

More concretely, on a generic component the remaining one-pair theorem may be read as the absence of same-parity / distance-two 2-secant recurrence in the affine chart, equivalently as no same-parity level re-entry for

$$\beta(h) = -\frac{L(h)}{M(h)},$$

equivalently as no same-parity vertical-line re-entry for the dual curve

$$T(h) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right).$$

The current local theory already resolves the finite-order $W = 0$, $M \neq 0$ boundary behavior and the generic turning-boundary blocks. After the same-parity compression, the genuinely residual affine lane is no longer arbitrary nonadjacent recurrence but isolated same-parity sign descent, equivalently discrete peaks of the intercept sequence on triple-active windows. So the sharpened affine frontier should reduce to middle-limited triples with a single endpoint sign test, rather than an unrestricted remote recurrence picture.

On the mixed side, the residual overlap contribution is concentrated on $K_v = 0$ and should be read there as a quotient-diagonal fixed-shear problem: a close scalar-local lane and a separate three-run overlap-transport lane, not a new local mixed carrier or hidden finite-order branch type.

Thus the remaining burden is global/nonlocal, but the one-pair derivative geometry now compresses it substantially: interior common-tangent structure is pseudoline-ordered, local birth at generic turning and finite-order $W = 0$, $M \neq 0$ boundaries is already sterile, and the affine residual has been reduced to isolated same-parity endpoint/branch configurations, equivalently to zeros of the monotone endpoint-gap functions $G_{\xi,J}$ in the middle-limited triple frontier.

On the mixed residual $K_v = 0$ lane, what remains is not another local branch-classification problem, but the separate scalar-local / transport / package questions discussed in the fixed-shear section.

13. ENDGAME ARCHITECTURE AND REMAINING PROPAGATION PROBLEM OUTSIDE THE CLEAN MIXED FOUR-POINT CASE

The remaining endgame is now a global propagation problem for the normalized line-incidence geometry rather than a local boundary-creation problem.

Remark 13.1 (Equivalent formulations of the remaining propagation problem outside the clean mixed four-point case). After Theorem 12.407, the remaining missing one-pair theorem on a generic component may be formulated in any of the following equivalent ways:

- (1) no finite-slope affine line in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is 2-secant to two nonadjacent ECT intervals;

- (2) the slope function

$$\beta(h) := -\frac{L(h)}{M(h)}, \quad L = v_5 \Delta'_7 - v'_5 \Delta_7,$$

does not revisit a level on nonadjacent branches;

(3) the dual tangent-line curve

$$T(h) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right)$$

has no nonadjacent same-parity re-entry against a vertical line in the (b, m) -plane;

(4) on the highest-new slice, the scalar graph

$$\phi = \lambda^{[1]}$$

has no nonadjacent re-entry of its slope ϕ_μ .

After the same-parity compression of Remark 13.6, the equivalent global formulations in this section should be read with that sharpening built in: the genuinely residual affine lane is distance-two / same-parity only, and the remaining global propagation problem is already reduced to isolated middle-limited triples rather than arbitrary nonadjacent recurrence.

The results proved above resolve the local behavior at finite-order $W = 0$, $M \neq 0$ boundaries and at generic turning boundaries. More precisely, the one-pair support geometry now shows that fixed-slope overlap windows are one-shot on each ordered pair, interior common-tangent crossings are pseudoline-ordered, and endpoint-support incidence is controlled by the strictly monotone endpoint-gap functions

$$G_{\xi, J}(h) = y_\xi - y(h) - m(h)(\xi - x(h)).$$

Thus the remaining theorem target is no longer an arbitrary global recurrence statement. It is the exclusion of remote zeros of these endpoint-gap functions, equivalently an RH-specific sign theorem for endpoint-support incidence on ordered nonadjacent branches.

Remark 13.2 (What the present global reduction does and does not buy). The results above now force the remaining propagation problem into a much smaller combinatorial and analytic class.

On the positive side, fixed-slope incidence is interval-graph geometry, support curves are strictly concave pseudolines with single-crossing and betweenness, each ordered pair has at most one directional or pair-pair overlap window in slope, and each endpoint support point meets a given ordered target branch at most once. Thus any surviving bad core must be slope-changing and pair-nonrepeating, and any remaining one-pair obstruction must arise through some element of a finite table of remote endpoint/branch incidences.

However, it is important to record a limitation. By Remark 13.4, intrinsic one-pair affine geometry is still universal for analytic plane curves. So pure planarity, parity, interval-graph combinatorics, pseudoline support geometry, or any similar abstract strip/support formalism does *not* by itself close the remaining branch. Any final contradiction must therefore use RH-specific source/provenance/canonical structure, not only generic secant-strip geometry.

Remark 13.3 (Warning: what the current local theory does *not* imply). Several tempting globalization statements suggested by the local theory are too strong in generic models and should not be assumed without further RH-specific input.

- (1) Adjacent exclusion is not enough: exclusion of adjacent pair-pair lines does not by itself force exclusion for nonadjacent pairs.
- (2) Local convexity of the dual tangent-line curve, even together with the shared asymptote structure of Proposition 12.67, is not by itself enough to forbid nonadjacent same-parity re-entry in the dual (β, m) -plane.
- (3) The direct highest-new septic coefficient is explicit enough to prove noncancellation, but not generically one-signed. Thus no trivial sign theorem for the scalar graph $\phi = \lambda^{[1]}$ should be expected without further RH-specific structure.
- (4) By Remark 13.4, the intrinsic one-pair affine geometry is universal for analytic plane curves. So no theorem using only planarity, parity, pseudoline support order, interval-graph overlap, or similar abstract strip/support data can by itself exclude nonadjacent recurrence.

Accordingly, the remaining gap is now a genuinely global special-structure theorem: not another local birth lemma, but an RH-specific sign/ordering law for the remote endpoint-gap incidences.

Remark 13.4 (Intrinsic affine-plane universality and the ceiling of pure planar arguments). It is useful to record a sharp ceiling on what can be proved from the intrinsic one-pair affine geometry alone.

Indeed, given any real-analytic plane curve

$$h \mapsto (x(h), y(h))$$

on an interval, setting

$$c \equiv 1, \quad v_5 = x, \quad \Delta_7 = y$$

gives

$$M = x', \quad N = y', \quad W = x'y'' - y'x''.$$

So the normalized affine chart, the ECT/curvature data, the tangent-separator kernel, and the dual/support geometry reduce exactly to the usual tangent geometry of the analytic plane curve (x, y) .

Accordingly, no theorem using only planarity, parity, support-curve single-crossing, strip-overlap combinatorics, or any other intrinsic one-pair affine-topological datum can by itself exclude nonadjacent recurrence. Any final contradiction must therefore use RH-specific source/provenance/canonical structure, not only abstract plane-curve or secant-strip geometry.

Remark 13.5 (Remote endpoint-gap reduction of the remaining nonadjacent problem). Combine Lemma 12.109, Theorem 12.96, and Theorem 12.107. Then every local boundary-creation mechanism on the generic one-pair region is already sterile: adjacent generic turning blocks contribute no pair-pair geometry, and finite-order isolated boundaries of type

$$W = 0, \quad M \neq 0$$

do not create a new opposite-side pair-pair branch.

Therefore any surviving nonadjacent pair-pair obstruction must appear as a *remote* interior zero of one of the monotone endpoint-gap functions

$$G_{\xi, J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h)),$$

where (ξ, y_{ξ}) ranges over endpoint support points of ECT branches and J ranges over ordered nonadjacent target branches.

Since each $G_{\xi, J}$ is strictly monotone on J , each endpoint/branch pair can contribute at most one such tangency. Thus the remaining propagation problem reduces to a finite remote-incidence table.

Accordingly, the natural next global target is an RH-specific sign theorem for the endpoint-gap functions $G_{\xi, J}$, not another local boundary-birth analysis.

Remark 13.6 (Work in progress: same-parity compression of the residual affine lane). A useful correction is that the adjacent branch should not be counted in the nonadjacent envelope: the genuinely residual affine lane is a distance-two / same-parity propagation problem.

For a common slope slice m , write

$$\Delta_j(m) := \tau_{I_{j+1}}(m) - \tau_{I_j}(m), \quad \varepsilon_j(m) := \operatorname{sgn} \Delta_j(m),$$

and for a triple

$$I_{j-1} < I_j < I_{j+1}$$

active on a common slope interval define

$$A_j := \Omega_{j-1} \cap \Omega_j \cap \Omega_{j+1} = (L_j, R_j).$$

Since each Δ_j is strictly decreasing, if

$$c_j := m_{j, j+1}$$

denotes the unique adjacent common-tangent slope, then on A_j

$$\Delta_j(m) > 0 \iff m < c_j, \quad \Delta_j(m) < 0 \iff m > c_j.$$

Thus the dangerous same-parity sign pattern is exactly

$$(\varepsilon_{j-1}(m), \varepsilon_j(m)) = (+, -),$$

equivalently

$$\tau_{I_{j-1}}(m) < \tau_{I_j}(m) > \tau_{I_{j+1}}(m),$$

equivalently

$$c_j < m < c_{j-1}.$$

So the residual affine obstruction is not arbitrary nonadjacent recurrence but an isolated sign descent / discrete peak of the intercept sequence $j \mapsto \tau_{I_j}(m)$. The actual triple can be dangerous only if the active window crosses that descent:

$$A_j \cap (c_j, c_{j-1}) \neq \emptyset.$$

Writing

$$e_{j-1}(m) := \sup \mathcal{B}_{I_{j-1}}(m), \quad B_j(m) := e_{j-1}(m) - \tau_{I_{j+1}}(m),$$

the residual same-parity obstruction interval is therefore

$$Q_j := \{m \in A_j \cap (c_j, c_{j-1}) : B_j(m) \geq 0\}.$$

In particular, once the active-window endpoint bookkeeping is added, the sharpened remaining affine risk should reduce to isolated middle-limited triples of the form

$$c_j < r_j < c_{j-1}, \quad B_j(r_j) > 0,$$

where r_j is the upper endpoint of the triple-active window. This is the sharpest current affine formulation, but it remains a theorem target rather than a proved closure statement.

Remark 13.7 (Further sharpening of the same-parity affine frontier). After the distance-two / same-parity correction, the residual affine lane is best viewed through the adjacent sign sequence

$$\varepsilon_j(m) := \operatorname{sgn}(\tau_{j+1}(m) - \tau_j(m)).$$

A dangerous same-parity affine defect at the triple

$$I_{j-1} < I_j < I_{j+1}$$

is exactly an isolated sign descent

$$(\varepsilon_{j-1}(m), \varepsilon_j(m)) = (+, -),$$

equivalently a discrete local peak

$$\tau_{j-1}(m) < \tau_j(m) > \tau_{j+1}(m).$$

In particular, dangerous affine defect sites are isolated: no consecutive defects can occur.

Moreover, after imposing the actual triple-active window constraint, the sharpened residual affine frontier is middle-limited:

$$c_j < r_j < c_{j-1}, \quad B_j(r_j) > 0.$$

So the unresolved affine risk is no longer arbitrary remote recurrence. It is an isolated same-parity sign descent on an actually active triple, limited on the right by the middle branch, together with one endpoint sign test.

Proposition 13.8 (Full-block universality of the direct provenance slice). *Let J be a connected generic one-pair block on which*

$$A(h) := \kappa(h)\eta_2(h)$$

is analytic and nowhere zero. On J , the direct provenance slice has the form

$$\phi(h) = A(h)q^{(\tau)}(h).$$

Then for every analytic function

$$\psi : J \rightarrow \mathbf{C}$$

there exists an analytic function

$$q : J \rightarrow \mathbf{C}$$

such that

$$\phi(h) = \psi(h) \quad (h \in J).$$

More explicitly, for any basepoint $h_0 \in J$ and any constants

$$a_0, \dots, a_6 \in \mathbf{C},$$

the function

$$q_\psi(h) := \sum_{j=0}^6 \frac{a_j}{j!} (h - h_0)^j + \int_{h_0}^h \frac{(h-s)^6}{6!} \frac{\psi(s)}{A(s)} ds$$

is analytic on J and satisfies

$$q_\psi^{(7)}(h) = \frac{\psi(h)}{A(h)}, \quad \phi(h) = A(h) q_\psi^{(7)}(h) = \psi(h).$$

In particular, on any generic one-pair block, the direct highest-new scalar graph ϕ is an arbitrary analytic scalar graph.

Proof. Since A is analytic and nowhere zero on J , the function

$$\frac{\psi}{A}$$

is analytic on J . The displayed formula for q_ψ is the standard variation-of-constants solution of

$$q^{(7)} = \frac{\psi}{A},$$

with arbitrary initial data at h_0 . Differentiating seven times gives

$$q_\psi^{(7)}(h) = \frac{\psi(h)}{A(h)},$$

hence

$$\phi(h) = A(h) q_\psi^{(7)}(h) = \psi(h).$$

□

Corollary 13.9 (Finite linear provenance constraints are too weak). *Let J be as in Proposition 13.8. Fix finitely many linear functionals*

$$\Lambda_r(q) = c_r, \quad r = 1, \dots, N,$$

on analytic functions q on J , where each Λ_r is built from finitely many point-evaluation jets and/or finitely many linear integral functionals with analytic kernels.

Fix also two distinct points

$$\mu_1 \neq \mu_2 \in J$$

and an affine line

$$\ell(\mu) = s\mu + b.$$

Then there exists a polynomial q on J such that:

(1) *all the prescribed linear provenance conditions*

$$\Lambda_r(q) = c_r \quad (r = 1, \dots, N)$$

hold;

(2) *the direct-slice graph*

$$\phi(\mu) = A(\mu) q^{(7)}(\mu)$$

is tangent to the line $y = \ell(\mu)$ at both μ_1 and μ_2 , i.e.

$$\phi(\mu_j) = \ell(\mu_j), \quad \phi_\mu(\mu_j) = s \quad (j = 1, 2).$$

Accordingly, no theorem using only the explicit direct formula

$$\phi = A q^{(7)}$$

together with any finite family of linear provenance or matching conditions on q can exclude bitangent, four-secant, or same-slope recurrence geometry on the direct slice.

Proof. The tangency conditions at μ_j are linear jet conditions on q :

$$A(\mu_j) q^{(7)}(\mu_j) = \ell(\mu_j),$$

and, since

$$\phi_\mu = (A q^{(7)})' = A' q^{(7)} + A q^{(8)},$$

also

$$A'(\mu_j) q^{(7)}(\mu_j) + A(\mu_j) q^{(8)}(\mu_j) = s.$$

Because $A(\mu_j) \neq 0$, these are legitimate linear conditions on the jets of q at the two points μ_1, μ_2 .

Together with the prescribed finitely many provenance constraints

$$\Lambda_r(q) = c_r,$$

this is a finite linear system on the coefficients of a polynomial q . For a polynomial of sufficiently large degree, the coefficient space has dimension larger than the number of imposed linear conditions. Standard Hermite interpolation / finite-dimensional linear algebra therefore gives a polynomial q satisfying all of them simultaneously. $\square$

Remark 13.10 (What the direct-slice no-go results actually leave open). Proposition 13.8 shows that on a generic one-pair block the direct highest-new scalar graph

$$\phi$$

is fully universal. Corollary 13.9 shows more: even after imposing any finite family of linear provenance / matching conditions on q , the direct slice remains flexible enough to realize bitangent, four-secant, and same-slope recurrence geometry.

So the next live theorem cannot be another sign/convexity statement about the graph of ϕ , nor any theorem using only finitely many endpoint jets or finitely many linear integral conditions on q . Any successful continuation of the direct-slice lane must instead use a genuinely *continuum-strength provenance law* for q on a whole block, such as a lower-order ODE, a sign-definite integral identity, a nonlinear transport law, or some other infinite-codimension constraint.

Lemma 13.11 (Crude polynomial bound for the local value scale). *Let*

$$S(m) = \sum_{\rho} f_{\beta_{\rho}, \gamma_{\rho}}(m),$$

where

$$f_{\beta, \gamma}(m) = \frac{1 - \beta}{(1 - \beta)^2 + (2m - \gamma)^2} + \frac{\beta}{\beta^2 + (2m - \gamma)^2}.$$

Then

$$S(m) \ll Q^2, \quad Q \asymp \log |m|.$$

Consequently

$$L(m) \ll Q$$

and

$$B_{\zeta}(m) L(m) S(m) \ll Q^4.$$

Proof. By the zero-free region, for zeros at relevant height one has

$$\min(\beta, 1 - \beta) \gg \frac{1}{Q}.$$

Hence

$$f_{\beta, \gamma}(m) \ll \frac{Q}{1 + (2m - \gamma)^2}.$$

Partition the zeros into unit shells

$$\mathcal{S}_k := \{\rho : k \leq |2m - \gamma_\rho| < k + 1\}, \quad k = 0, 1, 2, \dots$$

Standard zero counting gives

$$\#\mathcal{S}_k \ll Q$$

for each unit shell near height m . Therefore

$$S(m) \ll \sum_{k \geq 0} \#\mathcal{S}_k \frac{Q}{1+k^2} \ll Q \sum_{k \geq 0} \frac{Q}{1+k^2} \ll Q^2.$$

Since

$$L(m) = \log\left(3 + |m| + \frac{1}{S(m)}\right),$$

we trivially have $L(m) \ll Q$, and the last estimate follows from $B_\zeta(m) \asymp Q$. $\square$

The following remark records the present status of the endgame before the final contradiction theorem.

Remark 13.12 (Work in progress: current endgame status). The contradiction argument in the next theorem is presently written in the *pair-like branch* of the local model analysis. More precisely, the current endgame is fully aligned with the case in which the actual same-scale local core is perturbatively pair-like, so that the toy-model first-odd-jet obstruction continues to govern the corrected transverse scalar.

The fixed-core analysis developed earlier in the draft shows, however, that a more general finite-core branch is also possible: the first odd coefficient may cancel while a genuine corrected anomaly survives at a higher odd order. Thus the fully general finite-core endgame would need to be formulated in terms of the *first nonzero odd jet* of the corrected transverse scalar, rather than specifically the first one.

Accordingly, the theorem below should presently be read together with the preceding work-in-progress remarks on the pair-like versus finite-core fork. In particular, the pair-based contradiction is the intended endgame under the pair-like branch, while the genuinely finite-core branch remains to be completed in full detail.

At the local algebraic level, the canonical quotient-level order-7 fixed-sector package is now closed via Theorem 12.49: the remaining order-7 ambiguity is only the choice of a raw representative inside $A_{7,K_1}^f + \mathbf{C}A_5^f$, equivalently a local gauge-fixing of (u_7, v_7) , and does not affect the quotient class $[A_7^f]$ or the canonical second residual determinant Δ_7 .

Remark 13.13 (Work in progress: asymptotic contradiction versus full RH). As presently written, the final contradiction is only asymptotic in the height parameter Q : the comparison closes after reaching a regime in which the zeta-side upper bound is dominated by the toy-side lower bound, schematically because

$$Q^4 e^{-\delta Q} \rightarrow 0 \quad (Q \rightarrow \infty).$$

Accordingly, the current argument excludes sufficiently large pair-like off-line anomalies, but does not yet by itself rule out a hypothetical low-lying off-line zero. In particular, the present proof does not furnish a uniform-in-height contradiction, nor does it presently prove that a single off-line zero at one height would force corresponding bad local anomalies at arbitrarily large heights.

A plausible repair is to split the endgame into an effective high-height theorem plus a separate finite low-height bridge.

Effective high-height step. One should first replace the soft asymptotic comparison by an explicit inequality. Concretely, the final contradiction compares a toy-side lower bound of the form

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \geq c_{\text{toy}} \mathcal{L}(m)$$

against a zeta-side upper bound of the form

$$|\Xi_{\zeta}^{(N)}(m)| \leq C_{\zeta}(\rho_0, \alpha) \frac{\mathcal{U}(m)}{Q} e^{-\delta_*(\alpha, \rho_0)Q},$$

where in the present draft one has schematically

$$\mathcal{L}(m) \asymp \frac{x}{B_{\zeta}(m)} S(m)$$

(or its variable- x replacement from Remark 11.39), and

$$\mathcal{U}(m) \asymp L(m)S(m)^2.$$

To extract an actual threshold, every use of $O(\cdot)$, $\ll$, and $\asymp$ in the final comparison must be replaced by explicit constants. After substituting explicit bounds such as

$$B_{\zeta}(m) \leq C_B Q, \quad L(m) \leq C_L Q, \quad S(m) \leq C_S Q^2,$$

the contradiction should reduce to an explicit scalar inequality of the form

$$C_*(\rho_0, \alpha) Q^3 e^{-\delta_*(\alpha, \rho_0)Q} < 1 \quad (Q \geq Q_*),$$

for some computable constant $C_* > 0$. Any explicit Q_* satisfying this inequality would then yield an *effective* theorem excluding pair-like off-line anomalies for all $Q \geq Q_*$.

Height extraction. Once Q_* is known explicitly, one must convert it into an actual height threshold. Since the draft works with a midpoint parameter m satisfying

$$Q \asymp \log |m|,$$

this requires making the logarithmic comparability effective, for example in the form

$$c_Q \log(3 + |m|) \leq Q \leq C_Q \log(3 + |m|).$$

From such inequalities one obtains an explicit midpoint threshold m_* , and hence an explicit zero-height threshold T_* , since the local points under consideration satisfy

$$t_{\pm} = m \pm \frac{\sigma}{2}$$

with $|\sigma|$ microscopic relative to the height scale. Thus one would arrive at a statement of the form:

$$\text{no pair-like off-line zero can occur for } |\Im \rho| > T_*.$$

Finite low-height bridge. The effective high-height theorem alone still does not imply the full Riemann Hypothesis unless one also excludes all potential off-line zeros below the threshold T_* . A natural way to do this is to combine the structural argument above height T_* with a rigorous finite verification of RH up to height T_* . Accordingly, if one can arrange that the explicit threshold T_* lies below a rigorously verified range, then the final step would be:

$$\text{high-height exclusion above } T_* \quad + \quad \text{rigorous verification below } T_* \quad \implies \quad \text{RH}.$$

This would still be an unconditional proof, albeit a computer-assisted one rather than a purely internal asymptotic contradiction.

What still needs to be made explicit. To implement this repair rigorously, the draft must still supply all of the following ingredients in effective form:

- (1) an explicit toy-side lower constant on the admissible compact d -range and microscopic u -range;
- (2) an explicit zeta-side upper constant in the N -point cancellation estimate, including the exponential decay rate δ ;
- (3) explicit growth bounds for the auxiliary quantities appearing in the final comparison, including $B_{\zeta}(m)$, $L(m)$, and $S(m)$;
- (4) an explicit conversion between the logarithmic parameter Q and the actual height variable;
- (5) a final theorem stated as an effective exclusion above a concrete threshold T_* , rather than merely “for Q sufficiently large”;

- (6) a separate low-height bridge, either by citing a rigorous finite verification result covering the range up to T_* , or by proving an alternative uniform-in-height contradiction.

There is, in principle, a second possible repair: prove a propagation theorem showing that any off-line zero at one height would force corresponding bad local anomalies at arbitrarily large heights, so that the asymptotic contradiction would automatically apply. However, no such propagation principle is currently established in the draft, and at present this appears to be a substantially stronger theorem than the effective-threshold route described above.

Accordingly, the current endgame should be read as an asymptotic contradiction theorem in the pair-like branch. A plausible route to a complete proof of RH is available, but it requires an effective extraction of the height threshold together with a finite low-height bridge, neither of which has yet been written out in the draft.

Theorem 13.14 (Asymptotic calibrated N -point contradiction in the pair-like branch). *Assume the pair-like branch hypothesis and assume that the pair-like package-to- Ξ bridge identifies the calibrated local two-point package with the toy normal form entering Proposition 11.37. Then, for Q sufficiently large, no fixed off-line toy anomaly can occur in the pair-like branch.*

This theorem is not a finite-core closure theorem. The genuinely non-pair-like finite-core branch remains open pending the lower-transfer theorem described in Remark 12.164.

Proof. By Proposition 8.6 and the canonical calibration theorem, in the fixed small- x_0 regime one has

$$u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the toy side, Proposition 11.37 gives

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

hence on compact d -ranges,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the zeta side, Corollary 11.22 gives

$$|\Xi_\zeta^{(N)}(m)| \ll_{\rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha, \rho_0)Q}.$$

By Lemma 13.11, one has $S(m) \ll Q^2$, $L(m) \ll Q$, and $B_\zeta(m)L(m)S(m) \ll Q^4$. Therefore

$$\frac{(L(m)S(m)^2/Q)e^{-\delta_*(\alpha, \rho_0)Q}}{(x_0/B_\zeta(m))S(m)} = \frac{B_\zeta(m)}{x_0} \frac{L(m)S(m)}{Q} e^{-\delta_*(\alpha, \rho_0)Q} \ll Q^3 e^{-\delta_*(\alpha, \rho_0)Q} \rightarrow 0.$$

Hence for Q sufficiently large,

$$|\Xi_\zeta^{(N)}(m)| < |\Xi_{\text{toy}}^{(N)}(u, d)|.$$

This contradicts the assumed calibrated realization of a fixed off-line toy anomaly in the pair-like branch, once Q is sufficiently large. Thus, under the pair-like branch hypothesis and the package-to-jet bridge hypothesis, the pair-like branch is excluded at sufficiently large height.

This theorem is an asymptotic branch exclusion. To obtain a complete RH proof from it one still needs: an effective extraction of the threshold Q_* , a conversion from Q_* to a height threshold T_* , a finite low-height verification or alternative low-height argument, and the non-pair-like finite-core branch closure. $\square$

Remark 13.15 (Local package results are not yet global transform closure). The local Gate B and connection-package results are package-construction results. They do not by themselves prove hidden-state equality, exceptional-divisor fiber selection, finite-core exclusion, or a full package-to-transform contradiction. The remaining global obligations are:

state-locality modulo $\ker \Phi_K$, quotient-valued diagonal merger, finite-core lower transfer.

Remark 13.16 (Current status summary after the consolidated round). Reading Lemma C.20, Theorem C.21, Definition 12.199, Definition 12.182, Theorem 12.166, and Definition 12.174 together, the present status is:

The M_A, M_C connection identities close under actual source-linear Y -middle edge parity.

The local obstruction has sharpened to the odd-background $\eta\eta$ top row inside $E_{A,Y}^{\text{BC}}$, led by $\mathcal{E}_{\eta\eta;7,0,0}^2$; the pre

The transform-safe package is $\widehat{\Psi}_{\text{red}}^+ = (\widehat{\Psi}_{\text{red}}, T)$, $T = v_7/c$.

Finite determinacy is qualitative; calibrated separation and interaction rigidity remain open.

Remark 13.17 (Current local obstruction: odd-background $\eta\eta$ top row). After correcting the Bottleneck-C bookkeeping, the first local actual-source obstruction is no longer merely an unevaluated broad finite table. The current leading row is the frozen odd-background $\eta\eta$ top row

$$\mathcal{E}_{\eta\eta;7,0,0}^2.$$

It is governed by

$$\Lambda_{2,0}^- \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) \eta_i \eta_j \right]_7 W_{+,0},$$

where

$$\Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22}.$$

Writing

$$\nu_{21,7} := [z^7] \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) \eta_i \eta_j \right],$$

the top-row S -coefficient is

$$f_{\eta\eta}^{\text{top}} = -\frac{u}{24\pi q^3} \nu_{21,7}.$$

Thus the local branch is now binary: either prove a zeta-specific identity

$$\nu_{21,7} = 0,$$

or supply a genuine physical aggregate S -coefficient cancellation of the row. Endpoint-diagonal source realization, even if proved, removes only endpoint-separation subchannels and does not by itself close this scalar Δ -Hessian $\eta\eta$ survivor. The only remaining local rescue is a genuine physical aggregate S -coefficient cancellation, not a formal package counterterm.

Remark 13.18 (Current outcome of the Bottleneck-C repair audits). The subsequent audits below sharpen the status of Remark 13.17. Replacing the raw far-tail source by the full non-core zero source is coherent, but the rederived top row has the same obstruction form. Replacing the scalar local readout, quotienting by the A_5^{f} -line, changing the finite- s carrier, or exact endpoint symmetrization likewise does not remove the S -visible $\eta\eta$ row while preserving transform faithfulness.

Consequently, within the present transform-faithful finite- s architecture, Bottleneck C still requires a genuine vanishing of the physical aggregate S -coefficient of the $\eta\eta$ top row. No such vanishing theorem is presently proved.

14. CURRENT PROOF STATE AFTER THE LOCAL PACKAGE AUDITS

Remark 14.1 (Audit convention for the current proof-state section). The statements in this section distinguish three levels:

- (1) local algebraic lemmas, such as Gate A/B cancellations under fixed source/core and grading conventions;
- (2) conditional local geometric infrastructure, such as the reduced-base rank-two connection and affine forcing equation;
- (3) missing global proof gates, such as affine transport, boundary, holonomy/index, or finite-core non-concentration.

A statement at level (1) or (2) should not be read as a proof of RH. The current missing theorem is global.

This section records the current proof state after the Gate B, Bottleneck C, finite-core, and interaction audits. It is not part of the original local coefficient calculus; it is a status ledger for the remaining proof obligations.

The closed branches are:

 M_A, M_C connection closure is complete under source-grade reflection,

 mixed four-point dependence factors through (c, u_5, v_5, Δ_7) ,

and

finite odd-germ determinacy holds on compact nonsingular actual-source strata.

The live branches are:

local Bottleneck C physical edge obstruction,

finite-core sampled vector non-concentration,

and

 dd -interaction feature-balance / spectral rigidity.

The local Bottleneck-C branch has now exhausted the scalar, quotient, split, and line-bundle repair routes. The remaining viable language is the full rank-two fixed-sector bundle

$$\mathcal{F}_b = \mathfrak{f}_h(b),$$

with $E_{\eta\eta}$ treated as vector-valued affine forcing. Gate A and Gate B remain as local algebraic lemmas, but the proof now requires a global rank-two affine transport, vector-valued non-concentration, projective-plus-scale index, or comparable non-scalar theorem.

14.1. Closed branch: M_A, M_C connection closure. Let

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

Under the homogeneous scalar-grade reflection convention,

$$\mathbf{P}_g \circ \mathbf{S} = (-1)^g \mathbf{S} \circ \mathbf{P}_g,$$

where $\mathbf{S}$ is endpoint swap, the corrected mixed block satisfies

$$N^{\text{corr}}(h_-, h_+)^T = N^{\text{corr}}(h_+, h_-).$$

Consequently, for the source-linear middle block Y_g ,

$$Y_g(h_-, h_+)^T = (-1)^g Y_g(h_+, h_-).$$

Writing

$$Y_g(h - \delta/2, h + \delta/2) = \sum_{e \geq 0} \delta^e Y_g^{(e)}(h),$$

one obtains the edge-refined parity

$$\left(Y_{g,\beta}^{(e)} \right)^T = (-1)^{g+e} Y_{g,\beta}^{(e)}.$$

This parity closes the labelled $J = (3, 2, 4)$ pre-trivialization table:

$$Z_j[M_A] = Z_j[M_C] = 0, \quad 0 \leq j \leq 7.$$

Therefore the connection identities

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0$$

hold under the stated source-grade reflection convention.

Thus the M_A, M_C connection branch is no longer a live obstruction.

14.2. Live local obstruction: the frozen promoted mixed-block coefficient. The first live local actual-source/Bottleneck C carrier is the promoted quadratic mixed-block channel

$$Y_{12}^{(2)} := D^2 N_0[\xi_1, \xi_2].$$

It is coupled to one same-point linear whitening slot. The promoted channel has amplitude factor

$$a_1 a_2 (a_1 + a_2).$$

On the collision chart

$$a_2 = -a_1(1 + \kappa\delta),$$

one has

$$a_1 a_2 (a_1 + a_2) = a_1^3 \kappa\delta + O(\delta^2).$$

Thus the amplitude already supplies the first δ -edge. Therefore the first Bottleneck C coefficient is the frozen internal coefficient

$$E_{A,Y}^{\text{BC}} := \tau_{h \rightarrow m} \pi_{\text{fh}} \text{Sym}[z^7] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0.$$

Equivalently,

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\text{fh}} \text{Sym} \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{g=4}^7 \sum_{r+b=7-g} H_{r,0}^A Y_{g,0}^\chi W_{+,b}.$$

Write

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S.$$

The quotient Bottleneck C obstruction is

$$Q_{A,Y} := e_{A,Y} v_5 - u_5 f_{A,Y}.$$

The transform-safe obstruction is

$$\Phi_K(E_{A,Y}^{\text{BC}}) = 2f_{A,Y}.$$

For comparison, the literal source-position edge

$$E_{A,Y}^{\text{pos}} := \tau_{h \rightarrow m} \pi_{\text{fh}} \text{Sym}[z^7 \delta^1] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0$$

belongs to the next source-motion layer. It is not the first Bottleneck C amplitude-derivative coefficient.

Remark 14.2 (Executable table for $E_{A,Y}^{\text{BC}}$ (current section)). The frozen Bottleneck C coefficient is computed from the rows

$$\mathcal{E}_{\chi;g,r,b}^a := \tau_{h \rightarrow m} \pi_{\text{fh}} \text{Sym} \left(\Lambda_{a,r}^- Y_{g,0}^\chi W_{+,b} \right),$$

where

$$\begin{aligned} a &\in \mathcal{A}_{\text{BC}}, & \chi &\in \{dd, d\eta + \eta d, \eta\eta\}, \\ 4 &\leq g \leq 7, & r + b &= 7 - g. \end{aligned}$$

Thus

$$E_{A,Y}^{\text{BC}} = \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{a \in \mathcal{A}_{\text{BC}}} \sum_{\substack{4 \leq g \leq 7 \\ r+b=7-g}} \mathcal{E}_{\chi;g,r,b}^a.$$

The skew combination

$$Y_{d\eta - \eta d}^{(2)}$$

is excluded because it is killed by fixed-sector projection.

Remark 14.3 (Status of the frozen promoted mixed-block coefficient (current section)). The finite Fréchet–Sylvester recurrences reduce $E_{A,Y}^{\text{BC}}$ to the finite symbolic table of Remark 14.2. However, the current draft does not supply the scalar midpoint source-jet data needed to evaluate

$$E_{A,Y}^{\text{BC}} = e_{A,Y}I + f_{A,Y}S.$$

The required scalar data include, safely,

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

and

$$(r_i^{[a]})^{(k)}(m), \quad i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9.$$

Equivalently, the paper must supply the midpoint Taylor coefficients of

$$d_{i,\pm}(z), \quad \eta_i(z),$$

through the orders contributing to

$$Y_{g,0}^{dd}, \quad Y_{g,0}^{d\eta+\eta d}, \quad Y_{g,0}^{\eta\eta}, \quad 4 \leq g \leq 7.$$

At present no available symmetry, source-convention rebuild, quotient interpretation, transform-faithful invariant, or endpoint-symmetrized finite- s carrier forces

$$E_{A,Y}^{\text{BC}} = 0, \quad e_{A,Y}v_5 - u_5f_{A,Y} = 0, \quad f_{A,Y} = 0.$$

More specifically, the odd- $\eta\eta$ top row

$$\mathcal{E}_{\eta\eta;7,0,0}^2$$

has the S -visible coefficient recorded in Subsection 14.3. Therefore the first unresolved local actual-source/Bottleneck-C obstruction is now the top-row cancellation problem

$$\nu_{21,7} = 0$$

or an equivalent physical cancellation of

$$\mathcal{E}_{\eta\eta;7,0,0}^2.$$

Lemma 14.4 (Scalar tests for the frozen promoted mixed-block coefficient). *Let*

$$E_{A,Y}^{\text{BC}} = eI + fS, \quad A_5^{\mathfrak{f}} = u_5I + v_5S.$$

Then quotient Bottleneck C for the promoted mixed-block carrier is equivalent to

$$ev_5 - u_5f = 0.$$

The transform-safe T-horizontal condition is equivalent to

$$f = 0.$$

Proof. The quotient line is

$$\mathbf{C}A_5^{\mathfrak{f}}.$$

Thus

$$E_{A,Y}^{\text{BC}} \in \mathbf{C}A_5^{\mathfrak{f}}$$

is equivalent, on a good patch, to the vanishing of the determinant

$$ev_5 - u_5f.$$

On the other hand,

$$\Phi_K(I) = 0, \quad \Phi_K(S) = 2.$$

Hence

$$\Phi_K(E_{A,Y}^{\text{BC}}) = 2f,$$

so the transform-safe condition is exactly $f = 0$. □

Corollary 14.5 (First live T -slope carrier (current section)). *The amplitude factor of the promoted carrier is*

$$a_1 a_2 (a_1 + a_2),$$

and

$$\partial_{a_2} (a_1 a_2 (a_1 + a_2)) \big|_{a_2 = -a_1} = -a_1^2.$$

Therefore

$$\partial_\kappa T \big|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \frac{\Phi_K(E_{A,Y}^{\text{BC}})}{2c} = -a_1^2 \frac{f}{c}.$$

If

$$E_{A,Y}^{\text{BC}} = \lambda A_5^f,$$

then

$$\partial_\kappa T \big|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \lambda \frac{v_5}{c}.$$

Thus, on the generic patch $v_5 \neq 0$, a nonzero line-gauge $E_{A,Y}^{\text{BC}}$ closes the quotient condition but still leaks through T .

14.3. Concrete odd- $\eta\eta$ top-row obstruction. The local package audits isolate a concrete top-row survivor inside the frozen Bottleneck C coefficient. The relevant row is

$$\mathcal{E}_{\eta\eta;7,0,0}^2 = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \left(\Lambda_{2,0}^- \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) J \eta_i \eta_j \right]_7 W_{+,0} \right),$$

where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

This is the actual finite Fréchet source index $a = 2$, not the older shorthand $a \in \{1, 5\}$.

The row comes from the $(2, 1)$ -off-diagonal entry

$$N_{21} = -\frac{\sin \Delta + q_- s \cos \Delta}{s^2}.$$

Twice differentiating in the scalar phase-gap variable Δ gives

$$\partial_\Delta^2 N_{21} \big|_0 = \frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s}.$$

Thus the odd-background $\eta\eta$ block is not an endpoint-separation artifact. It is a scalar Δ -Hessian term.

For the active finite Fréchet row one has

$$\Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22}.$$

If

$$W_{+,0} = \begin{pmatrix} u & w \\ w & v \end{pmatrix},$$

then

$$\pi_{\mathfrak{f}} \text{Sym} \left(\Lambda_{2,0}^- J W_{+,0} \right) = -\frac{1}{24\pi q^3} (wI + uS).$$

Consequently, if

$$\nu_{21,7} := [z^7] \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) \eta_i \eta_j \right],$$

then this row contributes

$$\mathcal{E}_{\eta\eta;7,0,0}^2 = -\frac{\nu_{21,7}}{24\pi q^3} (wI + uS),$$

up to the harmless good-patch transport $\tau_{h \rightarrow m}$. In particular,

$$f_{\eta\eta}^\top = -\frac{u}{24\pi q^3} \nu_{21,7}.$$

Under the favorable phase-gap integral model,

$$\nu_{21,7} = \frac{1}{Q^{14}} \left[\frac{q_0}{480} (\rho_{i,2}\rho_{j,4} + \rho_{i,4}\rho_{j,2}) + \frac{1}{144} \left(\frac{q_0''}{6} - \frac{2q_0^3}{3} \right) \rho_{i,2}\rho_{j,2} \right].$$

For the homogeneous source $r(u) = u^2$, with $q_0'' = 0$, this gives

$$\nu_{21,7} = -\frac{q_0^3}{216Q^{14}},$$

which is nonzero on a good phase patch.

Therefore the row is generically T -visible:

$$\partial_\kappa T = -a_1^2 \frac{f_{\eta\eta}^\top}{c} \neq 0.$$

It is also quotient-visible on a generic $u_5 \neq 0$ patch; for example, if $w = 0$, then

$$Q_{\eta\eta}^\top = -u_5 f_{\eta\eta}^\top.$$

Remark 14.6 (Endpoint cleanup does not remove this row). Endpoint-diagonal realization, if proved, removes only endpoint-separation subchannels such as

$$d\eta^\Delta + \eta^\Delta d, \quad \eta^\Sigma \eta^\Delta.$$

It does not remove the scalar $\eta\eta$ term above, since that term comes from $\partial_\Delta^2 N_0[\eta_i, \eta_j]$. Thus the older route

$$[\delta^0]\eta^\Delta = 0 \implies E_{A,Y}^{\text{BC}} = 0$$

is no longer sufficient.

Remark 14.7 (No existing package mechanism cancels the row). The present package mechanisms do not cancel $\mathcal{E}_{\eta\eta;7,0,0}^2$: source-bidegree projection does not remove the promoted degree-three amplitude pattern $a_1 a_2 (a_1 + a_2)$; amplitude normalization only rescales the row; midpoint trivialization is invertible on a good patch; connection correction is post-edge and cannot cancel a raw frozen coefficient; and quotienting is insufficient because the augmented transform-safe package requires the S -component itself to vanish.

Thus the local branch is in the following binary state. Either one proves a new source-jet identity

$$\nu_{21,7} = 0,$$

or one proves a full top-row S -cancellation

$$\sum_{\text{top } \eta\eta} f_{\eta\eta}^\top = 0.$$

The only remaining local rescue is a genuine physical aggregate S -coefficient cancellation, not a formal package counterterm. Without such cancellation, the current corrected package does not close Bottleneck C or the T -horizontal condition at this row.

14.4. No-go for a transform-faithful free $\Theta_\pm$ -normal repair. The off-diagonal numerator variables

$$\Theta_+ = \sin \Delta + q_+ s \cos \Delta, \quad \Theta_- = \sin \Delta + q_- s \cos \Delta$$

make the off-diagonal entries formally affine:

$$N_{12}^\Theta = \frac{\Theta_+}{s^2}, \quad N_{21}^\Theta = -\frac{\Theta_-}{s^2}.$$

As a formal normal package this cancels the original off-diagonal $\eta\eta$ Hessian row. It does not, however, close the physical Bottleneck C obstruction.

The distinction is important. As a formal normal coordinate, $\Theta_\pm$ makes the off-diagonal entries affine. As a physical coordinate induced from (q_-, q_+, Δ) , however, $\Theta_\pm$ is constrained and its source-curvature must either remain in the raw package or be carried by the hidden physical readout. The following no-go concerns the latter, transform-faithful interpretation.

Definition 14.8 (Θ -local physical edge residual). Let

$$R_{\text{off},7}^{\Theta}$$

be the septic fixed-sector residual produced by the actual constrained source curvature of the off-diagonal variables $\Theta_{\pm}$. Let

$$R_{\text{diag},7}^{\Theta}$$

be the shifted diagonal $\eta\eta$ septic residual

$$\pi_{\mathfrak{f}} \text{Sym} \left(\Lambda_{\bullet,1}^{-} Y_6^{\eta\eta,\text{diag}} W_{+,0} + \Lambda_{\bullet,0}^{-} Y_6^{\eta\eta,\text{diag}} W_{+,1} \right).$$

Define

$$R_{\text{loc},7}^{\Theta} := R_{\text{off},7}^{\Theta} + R_{\text{diag},7}^{\Theta}.$$

Write

$$R_{\text{loc},7}^{\Theta} = \alpha_{\text{loc}}^{\Theta} I + \beta_{\text{loc}}^{\Theta} S.$$

The scalar $\beta_{\text{loc}}^{\Theta}$ is the Φ_K -visible physical edge residual left by the $\Theta_{\pm}$ -normal package.

Remark 14.9 (Constrained $\Theta_{\pm}$ variables reintroduce curvature). The variables $\Theta_{\pm}$ are not free affine source variables in the physical finite- s block. They are constrained by

$$\Theta_{\sigma} = \sin \Delta + q_{\sigma} s \cos \Delta.$$

For two source variations,

$$D_{\text{source}}^2 \Theta_{\sigma} = A_{\sigma} \eta_{ij} + sC \chi_{\sigma,ij} - \Theta_{\sigma,0} \eta_i \eta_j - sS(\chi_{\sigma,i} \eta_j + \chi_{\sigma,j} \eta_i).$$

In the pure $\eta\eta$ channel, this gives

$$D_{\text{source}}^2 N_{21}^{\Theta,\text{actual}} = \left(\frac{\sin \Delta_0}{s^2} + \frac{q_{-} \cos \Delta_0}{s} \right) \eta_i \eta_j,$$

which is the old off-diagonal source-curvature obstruction. Thus the $\Theta_{\pm}$ -normal form is not a physical coordinate relabeling that removes the obstruction; it is a formal package subtraction unless this curvature is carried back into the physical readout.

Proposition 14.10 (No transform-faithful septic normal-form repair). *Let K_7 be a local septic normal-form correction satisfying:*

- (1) K_7 is invisible below ordinary order 7;
- (2) K_7 is source-quadratic, hence does not alter source-linear Gate A, Gate B, or the M_A, M_C connection tables;
- (3) K_7 cancels the local septic residual

$$R_{\text{loc},7}^{\Theta} = \alpha_{\text{loc}}^{\Theta} I + \beta_{\text{loc}}^{\Theta} S;$$

- (4) the physical Φ_K -readout is unchanged, so no correction is added back into the physical hidden coordinate.

Then necessarily

$$\beta_{\text{loc}}^{\Theta} = 0.$$

Consequently, on any good patch where $\beta_{\text{loc}}^{\Theta} \neq 0$, no such transform-faithful local normal-form correction exists.

Proof. To cancel $R_{\text{loc},7}^{\Theta}$, the correction K_7 must contain the fixed-sector component

$$-\beta_{\text{loc}}^{\Theta} S.$$

But preserving the physical Φ_K -readout without adding a correction to the hidden coordinate requires

$$[z^7] \Phi_K(K_7) = 0.$$

Since $\Phi_K(S) = 2$, this forces

$$\beta_{\text{loc}}^{\Theta} = 0.$$

□

Proposition 14.11 (Physical edge obstruction survives in a good-patch model). *There exists a homogeneous good-patch local coefficient model for which*

$$\partial_\kappa \left(\frac{\beta_{\text{loc}}^\Theta}{c} \right) \Big|_{\kappa=0} \neq 0.$$

Proof. After extracting the frozen Bottleneck-C matrix coefficient, the remaining amplitude derivative is the derivative of

$$a_1 a_2 (a_1 + a_2)$$

in the transverse amplitude direction at the balanced point $a_2 = -a_1$. Equivalently, using the local edge coordinate

$$\kappa_{\text{amp}} := a_2 + a_1,$$

one has

$$a_1 a_2 (a_1 + a_2) = a_1 (-a_1 + \kappa_{\text{amp}}) \kappa_{\text{amp}} = -a_1^2 \kappa_{\text{amp}} + O(\kappa_{\text{amp}}^2),$$

so

$$\partial_{\kappa_{\text{amp}}} (a_1 a_2 (a_1 + a_2)) \Big|_{\kappa_{\text{amp}}=0} = -a_1^2.$$

This is the same scalar edge factor obtained from the collision chart

$$a_2 = -a_1 (1 + \kappa \delta)$$

after the first δ -edge has been factored into the frozen Bottleneck-C coefficient.

Since $\beta_{\text{loc}}^\Theta(0) = 0$, differentiating $1/c$ contributes nothing at $\kappa = 0$. Thus

$$\partial_\kappa \left(\frac{\beta_{\text{loc}}^\Theta}{c} \right) \Big|_{\kappa=0} = -\frac{a_1^2}{c_0} B_{\text{loc}}^\Theta,$$

where B_{loc}^Θ is the frozen physical edge coefficient.

Take a homogeneous source model

$$r_i = r_j = u^2, \quad q_0'' = 0,$$

on a good patch with

$$u q_0 q Q c_0 \neq 0.$$

The constrained off-diagonal $\Theta_\pm$ -source curvature contributes a nonzero fixed-sector S -coefficient. In the top-row normalization this has leading form

$$B_{\text{off}}^{\Theta, \text{hom}} = \frac{u q_0^3}{5184 \pi q^3 Q^{14}}.$$

The shifted diagonal term has the form

$$B_{\text{diag}}^{\Theta, \text{hom}} = -\frac{\lambda}{2\pi Q^{12}} \left(\frac{q_0 \alpha}{72} + \frac{q_0^3 \beta}{864} \right), \quad \lambda = \frac{\alpha \beta}{4\pi Q(p_1 + p_2)}.$$

Choosing

$$\alpha = \varepsilon \alpha_0, \quad \beta = \varepsilon \beta_0,$$

with $\varepsilon \neq 0$ sufficiently small gives

$$B_{\text{diag}}^{\Theta, \text{hom}} = O(\varepsilon^3),$$

while $B_{\text{off}}^{\Theta, \text{hom}}$ remains nonzero. Therefore

$$B_{\text{loc}}^{\Theta, \text{hom}} = B_{\text{off}}^{\Theta, \text{hom}} + O(\varepsilon^3) \neq 0$$

for sufficiently small ε , and the claimed derivative is nonzero. $\square$

Remark 14.12 (Status of the $\Theta_\pm$ -normal repair). A formal septic package may subtract $R_{\text{loc},7}^\Theta$ and thereby make the formal package horizontal. But the physical hidden coordinate is then

$$T^{\text{phys}} = T^\star + \frac{\beta_{\text{loc}}^\Theta}{c}.$$

Thus the transform-safe physical Bottleneck C condition is

$$\partial_\kappa \left(T^\star + \frac{\beta_{\text{loc}}^\Theta}{c} \right) = 0.$$

Proposition 14.11 shows that the second term is not currently zero. Therefore local Bottleneck C cannot be closed by $\Theta_\pm$ -normal coordinates or septic normal-form counterterms alone. The remaining local route would require a new physical edge identity

$$\partial_\kappa(\beta_{\text{loc}}^\Theta/c) = 0$$

not present in the current package calculus.

Remark 14.13 ($\Theta_\pm$ no-go versus the frozen Bottleneck-C coefficient). The no-go result for the $\Theta_\pm$ -normal repair does not replace the frozen coefficient

$$E_{A,Y}^{\text{BC}}.$$

Rather, it says that normal-form coordinates or septic counterterms cannot make the physical transform-safe Bottleneck-C obstruction disappear. The canonical local coefficient to compute remains

$$E_{A,Y}^{\text{BC}} = e_{A,Y}I + f_{A,Y}S,$$

with quotient test

$$e_{A,Y}v_5 - u_5f_{A,Y} = 0$$

and transform-safe test

$$f_{A,Y} = 0.$$

The $\Theta_\pm$ audit shows that if one removes a visible septic residual from a formal package, the same residual must be carried by the physical hidden coordinate. Thus it is an obstruction-isolation result, not a replacement for the $E_{A,Y}^{\text{BC}}$ finite table.

The concrete top-row obstruction

$$\mathcal{E}_{\eta\eta;7,0,0}^2$$

of Subsection 14.3 is the current row-level witness of this phenomenon: removing it from a formal septic package without changing the physical readout merely transfers the same obstruction to the hidden T -coordinate.

14.5. Actual exterior-pair source jets and failure of pair-level locking. The most direct possible rescue of the Bottleneck-C row would be a hidden actual-source relation forcing the midpoint post-affine source jets

$$\rho_2, \quad \rho_4$$

to satisfy the ratio required by the $\eta\eta$ top-row coefficient. This does not occur at the one-pair level.

For an exterior functional-equation zero-pair write

$$f_{\beta,\gamma}(t) = \frac{1-\beta}{(1-\beta)^2 + (2t-\gamma)^2} + \frac{\beta}{\beta^2 + (2t-\gamma)^2}.$$

Put

$$a := 1 - \beta, \quad b := \beta, \quad x := 2m - \gamma, \quad u := t - m.$$

Then

$$f_{\beta,\gamma}(m+u) = \frac{a}{a^2 + (x+2u)^2} + \frac{b}{b^2 + (x+2u)^2}.$$

Affine subtraction removes only the constant and linear terms. Thus the post-affine coefficients ρ_2, ρ_4 are the u^2, u^4 coefficients of the above expansion.

For one summand,

$$[u^2] \frac{c}{c^2 + (x+2u)^2} = \frac{4c(3x^2 - c^2)}{(c^2 + x^2)^3},$$

and

$$[u^4] \frac{c}{c^2 + (x+2u)^2} = \frac{16c(c^4 - 10c^2x^2 + 5x^4)}{(c^2 + x^2)^5}.$$

Therefore

$$\rho_2(\beta, x) = 4a \frac{3x^2 - a^2}{(a^2 + x^2)^3} + 4b \frac{3x^2 - b^2}{(b^2 + x^2)^3}$$

and

$$\rho_4(\beta, x) = 16a \frac{a^4 - 10a^2x^2 + 5x^4}{(a^2 + x^2)^5} + 16b \frac{b^4 - 10b^2x^2 + 5x^4}{(b^2 + x^2)^5}.$$

Proposition 14.14 (No one-pair jet-locking law). *For $\rho_2(\beta, x) \neq 0$, the ratio*

$$\frac{\rho_4(\beta, x)}{\rho_2(\beta, x)} = 4 \frac{a \frac{a^4 - 10a^2x^2 + 5x^4}{(a^2 + x^2)^5} + b \frac{b^4 - 10b^2x^2 + 5x^4}{(b^2 + x^2)^5}}{a \frac{3x^2 - a^2}{(a^2 + x^2)^3} + b \frac{3x^2 - b^2}{(b^2 + x^2)^3}}$$

is a nonconstant rational function of the exterior-pair parameters. In particular, the actual one-pair source class does not force

$$\frac{\rho_4}{\rho_2} = \frac{10q_0^2}{9} - \frac{5q_0''}{18q_0}.$$

Proof. The displayed formulas are obtained by Taylor expanding the two summands at $u = 0$. The ratio ρ_4/ρ_2 depends on the one-pair parameters (β, x) , while the Bottleneck-C locking ratio depends on the local baseline phase jets q_0, q_0'' . There is no source-internal identity equating these quantities. $\square$

Two useful checks are:

$$\beta = \frac{1}{2}, \quad x = 0 \quad \implies \quad \rho_2 = -64, \quad \rho_4 = 1024, \quad \frac{\rho_4}{\rho_2} = -16,$$

and, as $|x| \rightarrow \infty$,

$$\rho_2 = \frac{12}{x^4} + O(x^{-6}), \quad \rho_4 = \frac{80}{x^6} + O(x^{-8}),$$

so

$$\frac{\rho_4}{\rho_2} = \frac{20}{3x^2} + O(x^{-4}).$$

Thus far exterior source atoms approximate the formal bad germ $r(u) = u^2$ to finite jet order. Any cancellation of the aggregate Bottleneck-C coefficient must therefore be a genuine global zero-tail identity, not a pair-level source rigidity law.

Remark 14.15 (Full symmetry-packet closure does not cancel the source moments). The paired atom $f_{\beta, \gamma}$ already includes the functional-equation partner at the same ordinate:

$$\beta + i\gamma, \quad 1 - \beta + i\gamma.$$

The full symmetry packet also contains the conjugate ordinate pair

$$\beta - i\gamma, \quad 1 - \beta - i\gamma,$$

whose paired contribution is

$$f_{\beta, -\gamma}(t) = \frac{1 - \beta}{(1 - \beta)^2 + (2t + \gamma)^2} + \frac{\beta}{\beta^2 + (2t + \gamma)^2}.$$

If

$$x_+(m) = 2m - \gamma, \quad x_-(m) = 2m + \gamma,$$

then the corresponding post-affine coefficients are

$$\rho_2(\beta, x_+), \quad \rho_4(\beta, x_+), \quad \rho_2(\beta, x_-), \quad \rho_4(\beta, x_-).$$

The formulas above are even in x . Thus at a symmetric midpoint $m = 0$, the γ and $-\gamma$ packets add:

$$\rho_2(\beta, -\gamma) = \rho_2(\beta, \gamma), \quad \rho_4(\beta, -\gamma) = \rho_4(\beta, \gamma).$$

At the midpoint centered on the $+\gamma$ packet, $x_+ = 0$ while $x_- = 2\gamma$; again the conjugate packet supplies an additional contribution rather than a sign-reversed copy. Therefore functional-equation and conjugation closure do not structurally force

$$A_2 = 0, \quad A_4 = 0, \quad \nu_{\eta\eta} = 0.$$

Any cancellation after summing the full packet is a proper analytic incidence condition, not a symmetry identity.

14.6. Source-convention rebuild and persistence of the $\eta\eta$ row. The preceding obstruction used the tagged omitted-zero tail as the source object. One possible repair is to rebuild the source convention by moving all non-core zero atoms into the source. The least artificial split is

$$q = q_{\text{base}}^{\text{new}} + G^{\text{new}},$$

where

$$G^{\text{new}} := q_{\text{aux},R} + T_{\text{far},R}, \quad q_{\text{base}}^{\text{new}} := q_{\text{core}} + B_{\text{bg}} + E_{\text{est},R}.$$

Thus G^{new} remains a zero-source object; it contains the auxiliary non-core zero contribution and the far zero tail, but not the smooth background or estimation terms.

The affine-subtracted source is

$$r_G(t) := G^{\text{new}}(t) - G^{\text{new}}(m) - G^{\text{new}'}(m)(t - m).$$

Equivalently, the effective baseline in the finite- s expansion is

$$q_0^{\text{new}}(t) := q_{\text{base}}^{\text{new}}(t) + G^{\text{new}}(m) + G^{\text{new}'}(m)(t - m),$$

so that

$$q(t) = q_0^{\text{new}}(t) + r_G(t),$$

with

$$q_0^{\text{new}}(m) = q(m), \quad (q_0^{\text{new}})'(m) = q'(m), \quad (q_0^{\text{new}})''(m) = (q_{\text{base}}^{\text{new}})''(m).$$

This affine absorption is necessary: otherwise the source would contain value and linear jets and would reopen lower source-grade gates.

Remark 14.16 (Complement-source curvature). For the rebuilt source split, define

$$q_{\text{source},R}''(m) := (G^{\text{new}})''(m).$$

Since

$$r_G(t) = G^{\text{new}}(t) - G^{\text{new}}(m) - G^{\text{new}'}(m)(t - m),$$

one has

$$r_G''(m) = q_{\text{source},R}''(m).$$

This is the curvature of the affine-subtracted non-core zero complement. It is not the full canonical actual-zeta curvature

$$q_{\text{can}}''(m) = q_{\zeta}''(m),$$

which also includes the core, background, and all remaining completed-phase contributions in the fixed actual-zeta normalization.

Remark 14.17 (Admissibility conditions for the rebuilt split). The split

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}$$

is conditionally compatible with the already-closed lower gates under the following hypotheses:

- (1) G^{new} is kept as a tagged zero-source sum through the finite- s calculation, not collapsed prematurely into an unlabelled aggregate;
- (2) $r_G(m) = r_G'(m) = 0$ is imposed after the new split;
- (3) $q_{\text{base}}^{\text{new}}$ is used consistently in $q_{\pm}, \Delta_0, d_{\pm}, e_{\pm}, g_{\pm}, \eta$;
- (4) the $q_{\text{aux},R}$ -atoms satisfy the same endpoint reflection and source-grade conventions as the $T_{\text{far},R}$ -atoms;
- (5) the $q_{\text{aux},R}$ -atoms have compatible scalar-grade normalization, or else the affected coefficient tables must be regraded;
- (6) no B_{bg} or $E_{\text{est},R}$ term is treated as a zero-source variable.

Under these hypotheses the rebuild does not introduce value or linear source jets before the Bottleneck-C $\eta\eta$ row. By contrast, the more radical choices

$$G = q - q_{\text{core}} \quad \text{or} \quad G = B_{\text{bg}} + T_{\text{far},R} + E_{\text{est},R}$$

mix zero atoms with background or error terms. Those choices introduce background–source and error–source products and would require a fresh Gate A/Gate B analysis.

For a tagged exterior-zero source atom α in G^{new} , write

$$r_\alpha(t) = g_\alpha(t) - g_\alpha(m) - g'_\alpha(m)(t - m).$$

Define

$$d_{\alpha,\pm}^{\text{new}} = r_\alpha(t_\pm), \quad e_{\alpha,\pm}^{\text{new}} = r'_\alpha(t_\pm), \quad g_{\alpha,\pm}^{\text{new}} = r''_\alpha(t_\pm),$$

and let η_α^{new} be the corresponding phase-gap source variable. For two tagged source atoms i, j , the finite- s substitutions become

$$\begin{aligned} q_\pm &= q_{0,\pm}^{\text{new}} + \tau_i a_i d_{i,\pm}^{\text{new}} + \tau_j a_j d_{j,\pm}^{\text{new}}, \\ q'_\pm &= (q_0^{\text{new}})'_\pm + \tau_i a_i e_{i,\pm}^{\text{new}} + \tau_j a_j e_{j,\pm}^{\text{new}}, \\ q''_\pm &= (q_0^{\text{new}})''_\pm + \tau_i a_i g_{i,\pm}^{\text{new}} + \tau_j a_j g_{j,\pm}^{\text{new}}, \end{aligned}$$

and

$$\Delta = \Delta_0^{\text{new}} + \tau_i a_i \eta_i^{\text{new}} + \tau_j a_j \eta_j^{\text{new}}.$$

The lower-left off-diagonal channel is still

$$N_{21} = -\frac{\sin \Delta + q_- s \cos \Delta}{s^2}.$$

Therefore its pure $\eta_i \eta_j$ second phase-gap variation is

$$\partial_\Delta^2 N_{21}|_{\Delta=\Delta_0^{\text{new}}} = \frac{\sin \Delta_0^{\text{new}}}{s^2} + \frac{q_{0,-}^{\text{new}} \cos \Delta_0^{\text{new}}}{s}.$$

Thus the rebuilt top-row scalar is

$$\nu_{21,7}^{\text{new},ij} := [z^7] \left[\left(\frac{\sin \Delta_0^{\text{new}}}{s^2} + \frac{q_{0,-}^{\text{new}} \cos \Delta_0^{\text{new}}}{s} \right) \eta_i^{\text{new}} \eta_j^{\text{new}} \right].$$

Let $\rho_{\alpha,2}^{\text{new}}$ and $\rho_{\alpha,4}^{\text{new}}$ be the midpoint source coefficients of r_α in the same normalization as before. Then the rederived pair coefficient has the same form as the old obstruction:

$$\boxed{\nu_{21,7}^{\text{new},ij} = \frac{1}{Q^{14}} \left[\frac{q_0^{\text{new}}}{480} (\rho_{i,2}^{\text{new}} \rho_{j,4}^{\text{new}} + \rho_{i,4}^{\text{new}} \rho_{j,2}^{\text{new}}) + \frac{1}{144} \left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3} \right) \rho_{i,2}^{\text{new}} \rho_{j,2}^{\text{new}} \right]}.$$

Hence the source-convention rebuild does not algebraically kill the top row.

Aggregating over the full non-core zero source, define

$$A_2^{\text{new}} := \sum_{\alpha \in \mathcal{I}_{\text{ext}}^{\text{new}}} w_\alpha \rho_{\alpha,2}^{\text{new}} = \frac{(G^{\text{new}})''(m)}{2},$$

and

$$A_4^{\text{new}} := \sum_{\alpha \in \mathcal{I}_{\text{ext}}^{\text{new}}} w_\alpha \rho_{\alpha,4}^{\text{new}} = \frac{(G^{\text{new}})^{(4)}(m)}{24}.$$

Then

$$\boxed{\nu_{21,7}^{\text{new,agg}} = \frac{1}{Q^{14}} \left[\frac{q_0^{\text{new}}}{240} A_2^{\text{new}} A_4^{\text{new}} + \frac{1}{144} \left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3} \right) (A_2^{\text{new}})^2 \right]}.$$

With

$$W_{+,0}^{\text{new}} = \begin{pmatrix} u^{\text{new}} & w^{\text{new}} \\ w^{\text{new}} & v^{\text{new}} \end{pmatrix},$$

the corresponding fixed-sector row is

$$\mathcal{E}_{\eta\eta;7,0,0}^{2,\text{new}} = -\frac{\nu_{21,7}^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3} (w^{\text{new}} I + u^{\text{new}} S),$$

so

$$f_{\eta\eta}^{\text{new},\top} = -\frac{u^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3}\nu_{21,7}^{\text{new}}.$$

Consequently, on a good patch with $u^{\text{new}}q_{\text{wh}}^{\text{new}} \neq 0$, physical Bottleneck C closure still requires

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

Remark 14.18 (Outcome of the source-convention rebuild). The split

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}$$

is cleaner than the raw far-tail source convention because all non-core zero atoms are placed on the source side. Nevertheless, the rederived $\eta\eta$ top row has the same obstruction form, with

$$q_0, q_0'', A_2, A_4$$

replaced by

$$q_0^{\text{new}}, (q_0^{\text{new}})'', A_2^{\text{new}}, A_4^{\text{new}}.$$

Thus the rebuild merely shifts the needed cancellation theorem to the full non-core zero complement:

$$\frac{q_0^{\text{new}}}{240}A_2^{\text{new}}A_4^{\text{new}} + \frac{1}{144}\left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3}\right)(A_2^{\text{new}})^2 = 0.$$

If $A_2^{\text{new}} \neq 0$, this is equivalently

$$\frac{A_4^{\text{new}}}{A_2^{\text{new}}} = \frac{10(q_0^{\text{new}})^2}{9} - \frac{5(q_0^{\text{new}})''}{18q_0^{\text{new}}}.$$

No such global zero-complement identity is presently proved.

Remark 14.19 (Source-tagging and renormalized-complement ODEs). The quantities A_2^{new} and A_4^{new} in the rebuilt Bottleneck-C coefficient are the second and fourth midpoint jets of the tagged zero-source object

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R},$$

after affine subtraction. They are not the corresponding jets of

$$q - q_{\text{core}}, \quad B_{\text{bg}} + T_{\text{far},R} + E_{\text{est},R}, \quad q,$$

or any other renormalized complement unless the entire finite- s row is rebuilt.

Thus an ODE for a different complement does not affect the already-derived row. In the present source convention, the only relevant derivative-ratio identity is

$$\frac{(G^{\text{new}})^{(4)}(m)}{(G^{\text{new}})''(m)} = \frac{40(q_0^{\text{new}})^2}{3} - \frac{10(q_0^{\text{new}})''}{3q_0^{\text{new}}}.$$

No ODE, meromorphic identity, moving-cutoff identity, or standard ξ'/ξ phase-calculus identity currently supplies this condition.

Moreover, near an exterior zero-source pole p in G^{new} ,

$$G^{\text{new}}(t) \sim \frac{C}{t-p}, \quad (G^{\text{new}})''(t) \sim \frac{2C}{(t-p)^3}, \quad (G^{\text{new}})^{(4)}(t) \sim \frac{24C}{(t-p)^5},$$

so

$$\frac{(G^{\text{new}})^{(4)}}{(G^{\text{new}})''} \sim \frac{12}{(t-p)^2}.$$

But $q_{\text{base}}^{\text{new}} = q_{\text{core}} + B_{\text{bg}} + E_{\text{est},R}$ does not contain that exterior pole, so the right-hand side

$$\frac{40(q_{\text{base}}^{\text{new}})^2}{3} - \frac{10(q_{\text{base}}^{\text{new}})''}{3q_{\text{base}}^{\text{new}}}$$

has no matching pole there. Hence the required equation is not a meromorphic identity for the source-compatible complement.

14.7. Invariant and quotient no-go for the S -visible defect. The source rebuild still leaves a physical S -visible defect. One might therefore ask whether replacing the local invariant or quotienting the fixed sector can make this defect irrelevant. This subsection records the negative answer.

Let

$$E = eI + fS$$

be a Bottleneck-C fixed-sector residual. The current physical readout has

$$\Phi_K(I) = 0, \quad \Phi_K(S) = 2,$$

so

$$\Phi_K(E) = 2f.$$

The hidden-state slope is

$$\partial_\kappa T = -a_1^2 \frac{f}{c}.$$

Proposition 14.20 (Local invariant no-go). *Let $\mathcal{J}$ be a smooth local invariant or readout defined near the exceptional divisor. Suppose $\mathcal{J}$ is transform-faithful in the sense that its first variation determines the physical T -direction. Then*

$$D\mathcal{J}[S] \neq 0.$$

Consequently, whenever the Bottleneck-C defect has $f \neq 0$, the first exceptional-divisor slope of $\mathcal{J}$ contains a nonzero contribution from the same S -component. Thus replacing Φ_K by $\mathcal{J}$ does not remove the $\eta\eta$ obstruction.

Conversely, if

$$D\mathcal{J}[S] = 0,$$

then $\mathcal{J}$ may hide the $\eta\eta$ row at first order, but it cannot reconstruct or control the physical hidden-state slope $\partial_\kappa T$. Such a readout is not a valid Bottleneck-C replacement unless an independent theorem proves that T is pure gauge.

Proof. At the exceptional divisor,

$$\partial_\kappa \mathcal{J} = D\mathcal{J}[E] = D\mathcal{J}[eI + fS].$$

If $D\mathcal{J}[S] \neq 0$, then the fS -component contributes to $\partial_\kappa \mathcal{J}$. After any normalization that removes scalar I -data, the S -term remains visible. If $D\mathcal{J}[S] = 0$, then $\mathcal{J}$ cannot distinguish eI from $eI + \varepsilon S$ to first order. But this is precisely the distinction measured by

$$\partial_\kappa T = -a_1^2 f/c.$$

Thus $D\mathcal{J}[S] = 0$ loses the physical T -direction, while $D\mathcal{J}[S] \neq 0$ inherits the obstruction. $\square$

Remark 14.21 (Examples of non-faithful and faithful invariants). Trace, determinant to first order at the scalar merger, unordered conjugacy class, and symplectic area are blind to f at first order. They hide the bad row only by discarding the oriented S -direction. Oriented eigenvalue ratios, oriented projective cross-ratios, and oriented curvature or Schwarzian-type readouts are sensitive to f , and therefore inherit the same obstruction.

Next consider the quotient line generated by

$$A_5^f = u_5 I + v_5 S.$$

The quotient determinant coordinate is

$$\Delta_7 = u_7 v_5 - u_5 v_7.$$

A residual $E = eI + fS$ changes this coordinate by

$$\omega_h(E, A_5^f) = ev_5 - u_5 f.$$

Thus the quotient condition is

$$ev_5 - u_5 f = 0.$$

Proposition 14.22 (Quotient invisibility is not transform invisibility). *Assume $v_5 \neq 0$. The quotient condition*

$$ev_5 - u_5 f = 0$$

does not imply the transform-safe condition $f = 0$. In particular,

$$E = \chi A_5^f$$

is invisible to the quotient determinant, but

$$\Phi_K(E) = 2\chi v_5$$

and

$$T \mapsto T + \chi \frac{v_5}{c}.$$

Therefore the quotient

$$\mathfrak{f}_h / \mathbf{C} A_5^f$$

is a valid moduli object for mixed four-point scalar exits, but not for the full Bottleneck-C physical transform state.

Proof. If $E = \chi A_5^f$, then

$$E = \chi u_5 I + \chi v_5 S,$$

so

$$\omega_h(E, A_5^f) = (\chi u_5) v_5 - u_5 (\chi v_5) = 0.$$

However

$$\Phi_K(E) = 2\chi v_5.$$

On a $v_5 \neq 0$ patch this is nonzero for $\chi \neq 0$. The hidden coordinate changes by

$$T = \frac{v_7}{c} \mapsto T + \chi \frac{v_5}{c}.$$

Thus the quotient identifies states with different Φ_K -visible transform data. $\square$

Remark 14.23 (Layer separation). The quotient determinant layer and the physical transform layer see different functionals:

$$\omega_h(E, A_5^f) = ev_5 - u_5 f, \quad \Phi_K(E) = 2f.$$

The kernel of the former is not contained in the kernel of the latter. Hence quotient-horizontal Bottleneck C can close the mixed four-point scalar-exit branch while still leaking in the physical T -coordinate.

14.8. Finite- s carrier and endpoint-symmetrization no-go. The remaining possible repair is to alter the finite- s carrier before projection. This also fails inside the present transform-faithful framework.

Write

$$N_{12} = \frac{n_+}{s^2}, \quad n_+ = \sin \Delta + q_+ s \cos \Delta,$$

and

$$N_{21} = -\frac{n_-}{s^2}, \quad n_- = \sin \Delta + q_- s \cos \Delta.$$

For a pure $\eta_i \eta_j$ source variation, the relevant variation is through Δ . Holding $q_{\pm}$ fixed in this Hessian row,

$$\partial_{\Delta} n_{\pm} = \cos \Delta - q_{\pm} s \sin \Delta, \quad \partial_{\Delta}^2 n_{\pm} = -\sin \Delta - q_{\pm} s \cos \Delta = -n_{\pm}.$$

Therefore

$$\partial_{\Delta}^2 N_{12} = -\frac{\sin \Delta}{s^2} - \frac{q_+ \cos \Delta}{s},$$

whereas

$$\partial_{\Delta}^2 N_{21} = \frac{\sin \Delta}{s^2} + \frac{q_- \cos \Delta}{s}.$$

These are exact finite- s identities, not artifacts of Taylor truncation.

For a general linear endpoint carrier

$$C_{a,b} := aN_{12} + bN_{21},$$

one obtains

$$\partial_\Delta^2 C_{a,b} = \frac{(b-a)\sin\Delta}{s^2} + \frac{(bq_- - aq_+)\cos\Delta}{s}.$$

Thus identical cancellation requires

$$b - a = 0, \quad bq_- - aq_+ = 0.$$

For a nonzero carrier this forces $q_- = q_+$, which is not a generic finite- s identity.

The common endpoint combinations illustrate the obstruction:

$$N_{12} + N_{21} = \frac{(q_+ - q_-)\cos\Delta}{s},$$

so

$$\partial_\Delta^2 (N_{12} + N_{21}) = -\frac{(q_+ - q_-)\cos\Delta}{s}.$$

This cancels the $\sin\Delta/s^2$ part but leaves the endpoint-gradient term. Similarly,

$$q_+N_{21} + q_-N_{12} = \frac{(q_- - q_+)\sin\Delta}{s^2},$$

so

$$\partial_\Delta^2 (q_+N_{21} + q_-N_{12}) = \frac{(q_+ - q_-)\sin\Delta}{s^2}.$$

This cancels the $q_\pm \cos\Delta/s$ part but leaves the phase-gap term. Endpoint balancing can remove one component of the Hessian, but not both.

At the matrix level, define

$$B_\pm := \frac{\sin\Delta}{s^2} + \frac{q_\pm \cos\Delta}{s}.$$

The off-diagonal second-Hessian matrix is

$$H_{\eta\eta} = \begin{pmatrix} 0 & -B_+ \\ B_- & 0 \end{pmatrix}.$$

Writing

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

one has

$$H_{\eta\eta} = AJ + DX,$$

where

$$A = \frac{B_+ + B_-}{2} = \frac{\sin\Delta}{s^2} + \frac{(q_+ + q_-)\cos\Delta}{2s},$$

and

$$D = \frac{B_- - B_+}{2} = \frac{(q_- - q_+)\cos\Delta}{2s}.$$

Neither A nor D is generically zero. The skew part AJ is the endpoint-average obstruction, while the symmetric part DX removes the $\sin\Delta/s^2$ term but retains endpoint-gradient data.

The active left whitening coefficient is

$$\Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22}.$$

Since

$$E_{22}J = E_{21}, \quad E_{22}X = E_{21},$$

both endpoint-skew and endpoint-symmetric Hessian pieces feed the same leading row after left multiplication. Thus

$$\Lambda_{2,0}^-(AJ)W_{+,0} = -\frac{A}{24\pi q^3} E_{21}W_{+,0},$$

and

$$\Lambda_{2,0}^-(DX)W_{+,0} = -\frac{D}{24\pi q^3}E_{21}W_{+,0}.$$

After fixed-sector symmetrization,

$$\pi_{\mathfrak{f}} \text{Sym}(E_{21}W_{+,0}) = wI + uS,$$

up to the same normalization convention used in Subsection 14.3. Hence any nonzero endpoint-symmetrized coefficient remains S -visible on a patch with $u \neq 0$.

Proposition 14.24 (Endpoint-symmetrized finite- s carrier no-go). *Let a finite- s endpoint carrier be formed from N_{12}, N_{21} by smooth endpoint-symmetric linear combinations and endpoint or Gram normalizations that do not explicitly depend on the source variation direction. If the carrier remains transform-faithful, then its pure $\eta_i\eta_j$ second Δ -Hessian has a nonzero lower-row component. After the active left-whitening slot $\Lambda_{2,0}^-$, this produces a fixed-sector term*

$$-\frac{\mu_{\eta\eta}}{24\pi q^3}(wI + uS)$$

for some endpoint coefficient $\mu_{\eta\eta}$. In particular, unless $\mu_{\eta\eta}$ is forced to vanish by a genuine zeta identity, the physical S -coefficient

$$f_{\eta\eta} = -\frac{u}{24\pi q^3}\mu_{\eta\eta}$$

survives.

Remark 14.25 (Nonlinear carriers do not repair the obstruction). Ratio, logarithmic, determinant-normalized, and phase-coordinate carriers do not supply a transform-faithful repair. A ratio or logarithmic carrier normalizes the Hessian but leaves a nonzero endpoint-curvature term. A phase coordinate such as Δ is affine in the source and therefore has zero second Δ -Hessian, but reconstructing the physical finite- s kernel requires applying

$$\Delta \mapsto \sin \Delta + q_{\pm}s \cos \Delta,$$

whose second derivative is exactly the missing Hessian. Thus the bad row is either visible in the carrier, reintroduced by reconstruction, or hidden only by losing the physical S -visible transform readout.

14.9. Local Bottleneck-C ledger after repair attempts. Combining Subsections 14.6, 14.7, and 14.8, the present local ledger is as follows.

Changing the source convention from the raw far tail to the full non-core zero source

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}$$

is coherent under the stated tagging and grading hypotheses, but it reproduces the same $\eta\eta$ top-row obstruction form. Replacing the scalar local invariant hides the defect only if it discards the physical T -direction. Quotienting by $\mathbf{CA}_5^I$ is legitimate for mixed four-point scalar exits but not for the full transform state. Finally, endpoint symmetrization and alternative finite- s carriers either preserve the second- Δ Hessian, reintroduce it during reconstruction, or lose transform faithfulness.

Therefore, within the present transform-faithful finite- s off-diagonal kernel architecture, Bottleneck C closes only if the physical aggregate S -visible coefficient of the $\eta\eta$ top row vanishes. In the rebuilt full non-core zero-source convention this means

$$\nu_{21,7}^{\text{new,agg}} = 0,$$

or equivalently

$$\frac{q_0^{\text{new}}}{240}A_2^{\text{new}}A_4^{\text{new}} + \frac{1}{144}\left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3}\right)(A_2^{\text{new}})^2 = 0.$$

No currently proved identity supplies this vanishing. The subsequent subsections record the present interpretation of this obstruction: it is not a natural boundary term or residue theorem, and it is best viewed as the vertical defect of the T -horizontal Bottleneck-C connection. Thus

the present local route remains open-negative: either one proves a new pointwise global zero-source cancellation theorem, or else one pivots to a different architecture, such as global curvature/holonomy control or a quotient-only no- T program.

14.10. Last-chance pointwise flatness in full phase variables. The rebuilt source convention removes the unstable-cutoff objection by using the full non-core zero complement

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}.$$

A final possible escape is that the exact decomposition of the full phase derivative into rebuilt baseline and non-core complement might itself force the Bottleneck-C flatness identity. This subsection records that it does not.

Write

$$Q(t) := q(t)$$

for the full phase derivative, and decompose

$$Q(t) = b(t) + G(t),$$

where

$$b(t) := q_{\text{base}}^{\text{new}}(t) = q_{\text{core}}(t) + B_{\text{bg}}(t) + E_{\text{est},R}(t),$$

and

$$G(t) := G^{\text{new}}(t) = q_{\text{aux},R}(t) + T_{\text{far},R}(t).$$

At the midpoint m , put

$$b_0 := b(m), \quad b_2 := b''(m),$$

and

$$A_2 := \frac{G''(m)}{2}, \quad A_4 := \frac{G^{(4)}(m)}{24}.$$

Then

$$A_2 = \frac{Q''(m) - b''(m)}{2}, \quad A_4 = \frac{Q^{(4)}(m) - b^{(4)}(m)}{24}.$$

The rebuilt Bottleneck-C flatness condition is

$$\frac{b_0}{240} A_2 A_4 + \frac{1}{144} \left(\frac{b_2}{6} - \frac{2b_0^3}{3} \right) A_2^2 = 0.$$

Substitution gives

$$\frac{b_0}{11520} (Q'' - b'')(Q^{(4)} - b^{(4)}) + \frac{1}{576} \left(\frac{b''}{6} - \frac{2b_0^3}{3} \right) (Q'' - b'')^2 = 0,$$

with all derivatives evaluated at m . Equivalently, if

$$D_2 := Q'' - b'' = G'', \quad D_4 := Q^{(4)} - b^{(4)} = G^{(4)},$$

then the condition is

$$D_2 \left[b_0 D_4 + 20 \left(\frac{b''}{6} - \frac{2b_0^3}{3} \right) D_2 \right] = 0.$$

Proposition 14.26 (The rebuilt flatness equation is an extra fourth-order constraint). *The identity*

$$Q = b + G$$

does not imply

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

On a good patch with $G''(m) \neq 0$, pointwise flatness is equivalent to the additional derivative-ratio constraint

$$\frac{G^{(4)}(m)}{G''(m)} = \frac{40b(m)^2}{3} - \frac{10b''(m)}{3b(m)}.$$

Equivalently,

$$b^{(4)}(m) = Q^{(4)}(m) - \left(\frac{40b(m)^2}{3} - \frac{10b''(m)}{3b(m)} \right) (Q''(m) - b''(m)).$$

Thus the desired flatness is not a consequence of the tautological full-phase decomposition. It is an extra nonlinear fourth-order pointwise condition on the selected baseline/source split.

Proof. The displayed factorization shows that, if $D_2 = G''(m) \neq 0$, the vanishing condition is precisely

$$b_0 D_4 + 20 \left(\frac{b''}{6} - \frac{2b_0^3}{3} \right) D_2 = 0.$$

Dividing by $b_0 D_2$ gives

$$\frac{G^{(4)}(m)}{G''(m)} = \frac{40b_0^2}{3} - \frac{10b''(m)}{3b_0}.$$

The equivalent formula for $b^{(4)}$ follows from

$$G^{(4)} = Q^{(4)} - b^{(4)}.$$

The decomposition $Q = b + G$ supplies only

$$G'' = Q'' - b'', \quad G^{(4)} = Q^{(4)} - b^{(4)}.$$

It supplies no relation fixing $G^{(4)}/G''$ in terms of b, b'' . □

Remark 14.27 (Why defining the baseline by flatness is circular). One could formally choose b by requiring the fourth-order equation in Proposition 14.26. Such a choice would kill the Bottleneck-C row by definition. It is not a proof unless an independent zeta-side principle selects exactly this baseline, for example through the functional equation, a canonical scattering law, a zero-counting conservation principle, or a functorial fixed-core construction. No such principle is currently present.

Remark 14.28 (Asymptotic scale is not an identity). If b is dominated by the smooth phase/background, one expects roughly

$$b(t) \sim \frac{1}{2} \log(t/2\pi).$$

Then the required ratio has scale

$$\frac{G^{(4)}}{G''} \sim \frac{40}{3} b^2 \asymp \log^2 t.$$

A non-core zero complement with pole distances of order $1/\log t$ can have the same dimensional scale. This is not enough: the identity requires the exact constant and sign

$$\frac{40b^2}{3} - \frac{10b''}{3b}$$

pointwise. Local one-pair and finite-tail tests show that source geometry can produce different fourth-to-second derivative ratios. Thus a proof of flatness would have to be a genuinely new global zero-complement rigidity theorem.

14.11. Global obstruction-class interpretation of the Bottleneck-C defect. The preceding local ledger shows that the present transform-faithful finite- s Bottleneck-C architecture closes only if the rebuilt aggregate coefficient

$$\nu_{21,7}^{\text{new,agg}}$$

vanishes pointwise. The next question is whether this coefficient is a hidden boundary term, residue identity, or global conservation-law density. The current evidence is negative: it is better understood as a bulk vertical connection-defect density for the T -horizontal Bottleneck-C connection.

For this subsection write, for readability,

$$G := G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}, \quad A_2 := \frac{G''(m)}{2} = \frac{1}{2} q''_{\text{source},R}(m), \quad A_4 := \frac{G^{(4)}(m)}{24},$$

and use the baseline-local notation

$$q_b := q_0^{\text{new}}, \quad q_b^{(2)} := (q_0^{\text{new}})''.$$

The notation $q_b^{(2)}$ is deliberately not written as q'' , because q'' is reserved in the rank-two arithmetic sections for the full canonical curvature q''_{can} . The rebuilt obstruction is

$$\nu_{21,7}^{\text{new,agg}} = \frac{1}{Q^{14}} \left[\frac{q_b}{240} A_2 A_4 + \frac{1}{144} \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2^2 \right].$$

Proposition 14.29 (No natural exactness for the rebuilt Bottleneck-C defect). *Assume the source set is fixed as m varies, so that*

$$A_4 = \frac{1}{12} A_2''.$$

Then

$$2880Q^{14}\nu_{21,7}^{\text{new,agg}} = q_b A_2 A_2'' + 20 \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2^2.$$

This density is not an exact first-order local derivative in the natural midpoint variables (A_2, A_2', q_b, q_b') . More precisely,

$$2880Q^{14}\nu_{21,7}^{\text{new,agg}} = \left(q_b A_2 A_2' - \frac{1}{2} q_b' A_2^2 \right)' - q_b (A_2')^2 + \left(\frac{23}{6} q_b^{(2)} - \frac{40}{3} q_b^3 \right) A_2^2.$$

The final two terms are genuine bulk terms. Hence

$$\nu_{21,7}^{\text{new,agg}}$$

is not a boundary term or conservation-law density in the ordinary midpoint calculus.

Proof. Since $A_2 = G''/2$, fixed source tagging gives

$$A_2'' = \frac{G^{(4)}}{2}, \quad A_4 = \frac{G^{(4)}}{24} = \frac{1}{12} A_2''.$$

Substitution into the formula for $\nu_{21,7}^{\text{new,agg}}$ gives the first displayed identity. For the integration-by-parts decomposition, compute

$$(q_b A_2 A_2')' = q_b A_2 A_2'' + q_b (A_2')^2 + q_b' A_2 A_2',$$

and

$$(q_b' A_2^2)' = q_b^{(2)} A_2^2 + 2q_b' A_2 A_2'.$$

Therefore

$$q_b A_2 A_2'' = \left(q_b A_2 A_2' - \frac{1}{2} q_b' A_2^2 \right)' - q_b (A_2')^2 + \frac{1}{2} q_b^{(2)} A_2^2.$$

Adding

$$20 \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2^2$$

gives the claimed residual term

$$-q_b (A_2')^2 + \left(\frac{23}{6} q_b^{(2)} - \frac{40}{3} q_b^3 \right) A_2^2.$$

□

Remark 14.30 (Moving source windows). If the source set, cutoff, or core window depends on m , write it as $G_m(t)$. Then

$$A_2(m) = \frac{1}{2} \partial_t^2 G_m(t) \big|_{t=m},$$

and differentiation gives

$$A_2''(m) = \frac{1}{2} (\partial_m^2 \partial_t^2 + 2\partial_m \partial_t^3 + \partial_t^4) G_m(t) \big|_{t=m}.$$

Consequently

$$A_4 = \frac{1}{12} A_2'' - \frac{1}{24} (\partial_m^2 \partial_t^2 + 2\partial_m \partial_t^3) G_m(t) \big|_{t=m}.$$

Thus moving cutoffs or moving core windows introduce extra correction terms. They do not by themselves create a conservation law; they add another identity requirement.

Remark 14.31 (The induced second-order equation). If $A_2 \neq 0$, pointwise vanishing of $\nu_{21,7}^{\text{new,agg}}$ is equivalent, in the fixed-source-set calculus, to

$$q_b A_2'' + 20 \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2 = 0,$$

or

$$A_2'' + \left(\frac{10}{3} \frac{q_b^{(2)}}{q_b} - \frac{40}{3} q_b^2 \right) A_2 = 0.$$

This should be read as the unproved pointwise equation demanded by the current Bottleneck-C gate, not as an already available zeta-side ODE.

14.12. Meromorphic residue no-go for the rebuilt obstruction. There is also no current residue-theoretic mechanism forcing

$$\nu_{21,7}^{\text{new,agg}}(m) = 0.$$

The rebuilt source is a zero-pair pole field. Locally write

$$G(m) = \sum_{p \in P_{\text{ext}}} \frac{C_p}{m-p},$$

where P_{ext} is the set of non-core zero-source poles and C_p includes the residue and source weight. Then

$$A_2(m) = \sum_p \frac{C_p}{(m-p)^3}, \quad A_4(m) = \sum_p \frac{C_p}{(m-p)^5}.$$

If

$$K(m) := \frac{q''(m)}{6} - \frac{2q(m)^3}{3},$$

then, up to the global factor Q^{-14} ,

$$\nu(m) := \frac{q(m)}{240} A_2 A_4 + \frac{K(m)}{144} A_2^2$$

is the quadratic pole-interaction field

$$\nu(m) = \frac{q(m)}{240} \sum_{p,r} \frac{C_p C_r}{(m-p)^3 (m-r)^5} + \frac{K(m)}{144} \sum_{p,r} \frac{C_p C_r}{(m-p)^3 (m-r)^3}.$$

Proposition 14.32 (The rebuilt obstruction is not residue-killed). *The meromorphic differential $\nu(m) dm$ is not, in general, an exact meromorphic derivative. Its local residues at zero-source poles are generically nonzero. Hence the global residue theorem can at most give cancellation of the sum of residues; it does not imply the pointwise condition*

$$\nu(m) = 0$$

required by Bottleneck C.

Proof. Fix a pole p and write $\tau = m - p$. Decompose

$$A_2 = C_p \tau^{-3} + a(\tau), \quad A_4 = C_p \tau^{-5} + b(\tau),$$

where a, b are holomorphic at $\tau = 0$. Then

$$A_2 A_4 = C_p^2 \tau^{-8} + C_p b \tau^{-3} + C_p a \tau^{-5} + ab,$$

and

$$A_2^2 = C_p^2 \tau^{-6} + 2C_p a \tau^{-3} + a^2.$$

Thus $\nu(m) dm$ has high-order poles, and its residue depends on Taylor coefficients of q, K, a, b at p . There is no structural reason for this residue to vanish.

Already in a two-pole constant- (q, K) model, with poles p_1, p_2 , residues C_1, C_2 , and $d = p_1 - p_2$, the analytic tail at p_1 gives

$$\text{Res}_{p_1}(\nu dm) = C_1 C_2 \left(\frac{q}{8d^7} + \frac{K}{12d^5} \right),$$

which is generically nonzero. Therefore $\nu(m) dm$ is not an exact meromorphic derivative. $\square$

Remark 14.33 (What residue calculus can and cannot give). Functional-equation pairing and conjugation may pair or cancel residues in a global sum. Such cancellation is compatible with the global residue theorem, but it is much weaker than the pointwise identity

$$\nu_{21,7}^{\text{new,agg}}(m) = 0.$$

The Bottleneck-C gate is pointwise because

$$f_{\eta\eta}^{\text{new,agg}} = -\frac{u^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3} \nu_{21,7}^{\text{new,agg}}, \quad \partial_\kappa T = -a_1^2 f/c.$$

Thus an integrated or cyclic residue cancellation cannot close the present T -horizontal gate.

14.13. Meromorphic ODE principal-part no-go. A related failed repair is to seek a meromorphic differential equation for the source complement or for the full phase derivative. The local flatness condition would be implied by an equation of the form

$$\frac{G^{(4)}}{G''} = \frac{40q^2}{3} - \frac{10q''}{3q},$$

or by the same equation with $G = q_\zeta$, the full actual-zeta phase derivative. This is incompatible with the principal parts of the actual zero-pair pole fields.

Let

$$H(t) = \frac{C}{t-p} + O(1)$$

have a simple pole. Then

$$H''(t) = \frac{2C}{(t-p)^3} + O(1), \quad H^{(4)}(t) = \frac{24C}{(t-p)^5} + O(1),$$

so

$$\frac{H^{(4)}}{H''} = \frac{12}{(t-p)^2} + O\left(\frac{1}{t-p}\right).$$

On the other hand, if the full phase derivative has

$$q_\zeta(t) = \frac{R}{t-p} + O(1),$$

then

$$q_\zeta^2(t) = \frac{R^2}{(t-p)^2} + O\left(\frac{1}{t-p}\right), \quad \frac{q_{\text{can}}''}{q_\zeta} = \frac{2}{(t-p)^2} + O\left(\frac{1}{t-p}\right).$$

Hence

$$\frac{40q^2}{3} - \frac{10q''}{3q} = \left(\frac{40R^2}{3} - \frac{20}{3} \right) \frac{1}{(t-p)^2} + O\left(\frac{1}{t-p}\right).$$

Matching the left-hand double-pole coefficient 12 would require

$$R^2 = \frac{7}{5}.$$

Proposition 14.34 (Zero-pair residues rule out the meromorphic ODE repair). *The flatness-forcing ODE*

$$\frac{H^{(4)}}{H''} = \frac{40q^2}{3} - \frac{10q''}{3q}$$

cannot hold as a meromorphic identity for the raw non-core zero source, for the full phase derivative, or for any complement retaining the zero-pair poles.

Proof. For a zero-pair summand

$$f_a(t) = \frac{a}{a^2 + (2t - \gamma)^2},$$

the poles are

$$p_a^\pm = \frac{\gamma \pm ia}{2},$$

with residues

$$\text{Res}_{p_a^+} f_a = -\frac{i}{4}, \quad \text{Res}_{p_a^-} f_a = \frac{i}{4}.$$

Thus the phase-pole residues are purely imaginary. For a simple pole,

$$R = \pm \frac{i}{4},$$

so

$$R^2 = -\frac{1}{16},$$

not $7/5$. If several same-sign pole contributions coincide, then

$$R = \pm \frac{ni}{4},$$

and still $R^2 < 0$. Therefore the right-hand double-pole coefficient cannot match the coefficient 12 of $H^{(4)}/H''$. $\square$

Remark 14.35 (Pole-cancelled complements are the wrong object). A pole-cancelled complement can evade the principal-part contradiction at the cancelled pole. But Bottleneck-C uses the source moments

$$A_2 = \frac{G''(m)}{2}, \quad A_4 = \frac{G^{(4)}(m)}{24}.$$

An ODE for a pole-cancelled complement is therefore an ODE for the wrong object, unless the entire order-seven row is rederived with different moments.

14.14. Connection-defect interpretation and possible architecture pivot. The preceding no-go results suggest a different interpretation. The coefficient

$$\nu_{21,7}^{\text{new,agg}}$$

is not merely a bad scalar coefficient; it is the vertical defect of the current T -horizontal Bottleneck-C connection.

Recall that

$$A_7^f = u_7 I + v_7 S, \quad T = \frac{v_7}{c}.$$

For a residual

$$E = eI + fS,$$

the physical readout satisfies

$$\Phi_K(E) = 2f,$$

and the Bottleneck-C exceptional slope is

$$\partial_\kappa T = -a_1^2 \frac{f}{c}.$$

For the rebuilt non-core zero-source convention, the active $\eta\eta$ row gives

$$f_{\eta\eta}^{\text{new,agg}} = -\frac{u^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3} \nu_{21,7}^{\text{new,agg}}.$$

Hence, on a good patch,

$$\partial_\kappa T|_{\eta\eta} = a_1^2 \frac{u^{\text{new}}}{24\pi c (q_{\text{wh}}^{\text{new}})^3} \nu_{21,7}^{\text{new,agg}}.$$

Definition 14.36 (Bottleneck-C vertical defect density). Define

$$\mathcal{K}_{\eta\eta;\kappa} := \partial_\kappa T|_{\eta\eta}.$$

Equivalently,

$$\mathcal{K}_{\eta\eta;\kappa} = C_{\text{geom}} \nu_{21,7}^{\text{new,agg}}, \quad C_{\text{geom}} = a_1^2 \frac{u^{\text{new}}}{24\pi c (q_{\text{wh}}^{\text{new}})^3}.$$

On a good patch,

$$C_{\text{geom}} \neq 0.$$

Remark 14.37 (Curvature versus second fundamental form). The active row is symmetric in the source directions η_i, η_j . Thus $\mathcal{K}_{\eta\eta;\kappa}$ is not ordinary antisymmetric gauge curvature by itself. It is better regarded as the vertical second fundamental form, or quadratic curvature-density, of the diagonal-merger horizontal distribution in the T -coordinate.

Proposition 14.38 (Local flatness equivalence). *Within the current transform-faithful Bottleneck-C architecture, local T -horizontal closure of the rebuilt $\eta\eta$ row is equivalent to*

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

Proof. By Definition 14.36,

$$\partial_\kappa T|_{\eta\eta} = C_{\text{geom}} \nu_{21,7}^{\text{new,agg}}.$$

On a good patch $C_{\text{geom}} \neq 0$. Therefore

$$\partial_\kappa T|_{\eta\eta} = 0$$

if and only if

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

□

Remark 14.39 (Possible curvature/holonomy pivot). The current Bottleneck-C architecture demands pointwise flatness:

$$\mathcal{K}_{\eta\eta;\kappa} = 0.$$

A different architecture could attempt to treat

$$C_{\text{geom}} \nu_{21,7}^{\text{new,agg}} dm$$

as a global curvature or holonomy density, replacing pointwise T -flatness by a global statement such as

$$\int \mathcal{K}_{\eta\eta;\kappa} dm = 0, \quad \text{Hol}_T(\mathcal{K}) = 0, \quad \text{or} \quad \text{Index}(\mathcal{K}) = 0.$$

This would be a genuine architecture pivot, not a repair of the present local Bottleneck-C gate. To make such a pivot meaningful, one would first need to prove transformation laws showing that

$$C_{\text{geom}} \nu_{21,7}^{\text{new,agg}} dm$$

is a globally defined connection or curvature density under changes of patch, core window, source convention, and normalization.

14.15. No- T quotient-determinant architecture as a possible pivot. A second possible pivot is to abandon the T -horizontal Bottleneck-C gate entirely and formulate the local-to-global argument using only quotient determinant data. This is not a repair of Bottleneck C; it is a different proof architecture.

The fixed-sector data are

$$A_5^{\mathfrak{f}} = u_5 I + v_5 S, \quad A_7^{\mathfrak{f}} = u_7 I + v_7 S.$$

The quotient determinant coordinate is

$$\Delta_7 = u_7 v_5 - u_5 v_7.$$

A shift

$$A_7^{\mathfrak{f}} \mapsto A_7^{\mathfrak{f}} + \chi A_5^{\mathfrak{f}}$$

preserves Δ_7 , but changes

$$T = \frac{v_7}{c} \quad \mapsto \quad T + \chi \frac{v_5}{c}.$$

It also changes

$$\Phi_K(A_7^f)$$

by $2\chi v_5$. Thus T is not quotient-invariant.

A quotient-only architecture would use as local moduli

$$\mathcal{Q}(h) = (c(h), u_5(h), v_5(h), \Delta_7(h)),$$

or equivalently the quotient class

$$[A_7^f] \in \mathfrak{f}_h / \mathbf{C}A_5^f.$$

Its local gate would be

$$ev_5 - u_5 f = 0,$$

not

$$f = 0.$$

Remark 14.40 (What a no- T architecture would require). A no- T architecture can make the T -leak irrelevant only if every downstream RH-relevant detection step is reformulated through quotient data:

$$(c, u_5, v_5, \Delta_7),$$

mixed four-point scalar exits, and finite-core determinant geometry. It cannot use

$$T = \frac{v_7}{c}, \quad \Phi_K(A_7^f),$$

or hidden-state reconstruction as independent data.

Proposition 14.41 (Conditional no-go for quotient-only architecture). *If a downstream local-to-global step requires reconstruction of T or $\Phi_K(A_7^f)$, then no quotient-only architecture can replace the current Bottleneck-C gate.*

Proof. The quotient identifies representatives

$$A_7^f \quad \text{and} \quad A_7^f + \chi A_5^f.$$

These representatives have the same Δ_7 , but their T -coordinates differ by

$$\chi \frac{v_5}{c}.$$

On a $v_5 \neq 0$ patch this is nonzero for $\chi \neq 0$. Therefore no function of the quotient data can recover T or $\Phi_K(A_7^f)$. Any proof step depending on those quantities cannot be performed in the quotient-only architecture. $\square$

Remark 14.42 (Relation to the Bottleneck-C obstruction). The no- T quotient program would make the obstruction

$$\nu_{21,7}^{\text{new,agg}}$$

irrelevant only by removing T -horizontal closure from the proof. It would not prove

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

The burden would instead move to quotient finite-core non-concentration and quotient interaction rigidity using only (c, u_5, v_5, Δ_7) and the mixed four-point scalar exits.

14.16. Post-quotient audit: split and line-bundle repairs. The no- T quotient architecture keeps the determinant data

$$D_{2r+1} = \omega_h(A_{2r+1}^{\mathfrak{f}}, A_5^{\mathfrak{f}}) = u_{2r+1}v_5 - u_5v_{2r+1}.$$

This removes the representative-level T -coordinate, but it also discards the full $A_5^{\mathfrak{f}}$ -line. Thus the pure line anomaly

$$A_{2r+1}^{\mathfrak{f}} = \lambda_{2r+1}A_5^{\mathfrak{f}}$$

has

$$D_{2r+1} = 0,$$

while still giving

$$\Phi_K(A_{2r+1}^{\mathfrak{f}}) = 2\lambda_{2r+1}v_5, \quad T_{2r+1} = \lambda_{2r+1}\frac{v_5}{c}.$$

Consequently the quotient scalar

$$H_m^{\text{quot}}(s) = \omega_h\left(\pi_{\mathfrak{f}}\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m), A_5^{\mathfrak{f}}\right)$$

is analytically coherent but incomplete for the original transform-state detection problem.

A natural repair is the split readout

$$\mathcal{H}_m^{\text{split}}(s) = \left(H_m^{\text{quot}}(s), H_m^{\parallel}(s)\right),$$

where

$$H_m^{\parallel}(s) = \frac{\left\langle \pi_{\mathfrak{f}}\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m), A_5^{\mathfrak{f}} \right\rangle_{\text{wh}}}{\left\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \right\rangle_{\text{wh}}}.$$

For each coefficient

$$A_{2r+1}^{\mathfrak{f}} = u_{2r+1}I + v_{2r+1}S,$$

this gives the pair

$$(D_{2r+1}, \lambda_{2r+1}),$$

with

$$\lambda_{2r+1} = \frac{\langle A_{2r+1}^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}{\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}.$$

On a good patch, if the whitening metric is fixed and nondegenerate, the pair

$$(D_{2r+1}, \lambda_{2r+1})$$

recovers the full fixed-sector vector $A_{2r+1}^{\mathfrak{f}}$. Thus the split readout repairs the fatal incompleteness of the pure quotient readout.

However, this completeness reintroduces the Bottleneck-C obstruction through the longitudinal coordinate. For

$$E_{\eta\eta} = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS),$$

one has

$$D(E_{\eta\eta}) = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wv_5 - uu_5),$$

and

$$\lambda(E_{\eta\eta}) = -\frac{\nu_{\eta\eta}}{24\pi q^3} \frac{\langle wI + uS, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}{\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}.$$

Thus if the split readout is used as a pointwise vanishing gate, then on a good patch it requires

$$E_{\eta\eta} = 0,$$

and hence again requires

$$\nu_{\eta\eta} = 0$$

unless $wI + uS$ degenerates. The split readout is therefore a complete coordinate system for the fixed-sector vector, not a new closure mechanism.

Remark 14.43 (Line-bundle transport attempt). One can try to avoid pointwise vanishing by treating the longitudinal coordinate as a section of

$$L = \mathbf{C}A_5^{\mathfrak{f}}.$$

With a fixed whitening metric, the natural local connection is

$$\alpha_L = \frac{\langle dA_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}{\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}.$$

If

$$s = \lambda A_5^{\mathfrak{f}},$$

then

$$\nabla^L s = (d\lambda + \lambda \alpha_L) A_5^{\mathfrak{f}}.$$

Locally this is coherent: under

$$A_5^{\mathfrak{f}} \mapsto \mu A_5^{\mathfrak{f}}$$

one has

$$\alpha_L \mapsto \alpha_L + d \log \mu,$$

and the covariant derivative transforms correctly.

The obstruction is global. The construction requires the $A_5^{\mathfrak{f}}$ line and the whitening metric to transform canonically under changes of patch, source/core window, and whitening convention. This is not currently proved. Moreover, no canonical path, boundary condition, or holonomy theorem is available for (L, ∇^L) , and the longitudinal forcing is not exact in the natural variables. Thus line-bundle transport is a useful diagnostic language but not presently a proof gate.

Remark 14.44 (Current status of quotient, split, and line-bundle repairs). The quotient scalar is invariant but incomplete. The split readout is complete but restores the original fixed-sector obstruction. The line-bundle connection makes the longitudinal defect locally covariant, but lacks global covariance, canonical paths, boundary data, exactness, or holonomy control. Therefore none of these repairs currently replaces pointwise Bottleneck-C closure.

14.17. Reduced core, packet visibility, and the visibility-positive base. The rank-two architecture is only meaningful after fixing the source/core convention. Arbitrary transfers of zero atoms between source and core are not currently functorial. Moving a single zero contribution q_ρ from source to core changes the local corrected object, not merely its coordinates. At coefficient level one should expect

$$A_{2r+1,j}^{\mathfrak{f}} = G_{ij}^{-1} A_{2r+1,i}^{\mathfrak{f}} + B_{\rho,2r+1},$$

not a pure GL_2 -transition. Therefore source/core moves are excluded from the atlas rather than transported.

We use the canonical reduced-core rule

$$\mathcal{C}_k(m) = \text{the unique symmetry-closed, same-scale } k\text{-packet nearest } m.$$

Let

$$\mathcal{E}_{\text{geom}}^{\text{core}} = \mathcal{E}_{\text{geom}} \cup \mathcal{E}_{\text{core boundary}},$$

where $\mathcal{E}_{\text{core boundary}}$ includes nearest-packet ties, same-scale boundary, symmetry-closure ambiguity, endpoint-gap collapse, zero collision or multiplicity, Gram/whitening singularity, and patch-boundary failure. Define

$$\mathcal{B}_{\text{red}} = \mathcal{B} \setminus \mathcal{E}_{\text{geom}}^{\text{core}}.$$

On $\mathcal{B}_{\text{red}}$, the selected core is locally constant as a labelled symmetry-closed packet.

Hypothesis 14.45 (Reduced good-patch reach). Every nondegenerate RH-relevant off-line symmetry packet

$$\mathcal{P}(\rho) = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \quad \rho = \beta + i\gamma, \quad \beta \neq \frac{1}{2},$$

admits a nearby reduced good patch

$$b = (m, s, \mathcal{C}_k(m)) \in \mathcal{B}_{\text{red}},$$

or else the failure lies in a controlled component of $\mathcal{E}_{\text{geom}}^{\text{core}}$.

Remark 14.46 (Status of the reach hypothesis). The strict-gap, same-scale, endpoint, symmetry-closure, and Gram/whitening conditions are open once they hold at one point. Thus Hypothesis 14.45 is a non-swallowing assertion: the enlarged exceptional set

$$\mathcal{E}_{\text{geom}}^{\text{core}}$$

must not contain all RH-relevant off-line packet configurations. This is a global reach condition for the rank-two route, not a consequence of Gate A or Gate B.

The RH-relevant local object is not a single zero but the symmetry packet

$$\mathcal{P}(\rho) = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \quad \rho = \beta + i\gamma, \quad \beta \neq \frac{1}{2}.$$

Packet-level ordering is essential: functional-equation and conjugation symmetries force internal equalities inside $\mathcal{P}(\rho)$, and these should not be counted as core-selection failures. The dangerous tie walls are ties between distinct packets.

For packet visibility, set

$$\mathcal{B}_{\text{red}}^+ = \mathcal{B}_{\text{red}} \setminus \mathcal{E}_{\text{invis}}^{\text{core},7}.$$

For the centered one-whitening-slot route, the incidence walls include

$$\{y = 0\} \cup \{q'' = 0\} \cup \{g = h\},$$

and, when quotient isolation is used, also $\{v_5 = 0\}$. For the non-centered $q^{(7)}$ -transport route, include

$$\{q^{(7)} = 0\} \cup \{\Xi_{K_1} = 0\}.$$

On $\mathcal{B}_{\text{red}}^+$, the off-line packet fixed-sector visibility vector

$$C_{\text{fd}}^{\bar{f}} = (A_7^{\bar{f}}, A_9^{\bar{f}}, \dots)$$

is nonzero modulo these explicit incidence exclusions.

14.18. Rank-two fixed-sector bundle as the remaining viable language. The preceding failures suggest that the defect should not be scalarized, nor should one single out the $A_5^{\bar{f}}$ -line as the primary geometric object. The best remaining local language is the full rank-two fixed-sector bundle

$$\mathcal{F} \rightarrow \mathcal{B}, \quad \mathcal{F}_b = \mathfrak{f}_h(b) = \mathbf{C}I_b \oplus \mathbf{C}S_b.$$

This formulation keeps the whole fixed-sector vector

$$A_{2r+1}^{\bar{f}} \in \mathcal{F}_b$$

and therefore avoids the incompleteness of the quotient scalar and the noncanonical line-preservation problem of $L = \mathbf{C}A_5^{\bar{f}}$.

The local candidate connection is the projected ambient derivative

$$\nabla_X^{\bar{f}} A = \Pi_{\bar{f}}(d_X^{\mathcal{J}} A),$$

where $\mathcal{J}$ is the ambient corrected finite- s jet/kernel-state bundle, $d^{\mathcal{J}}$ is the ambient derivative, and

$$\Pi_{\bar{f}} : \mathcal{J}_b \rightarrow \mathfrak{f}_h(b)$$

is the fixed-sector projection. In a local frame

$$E = (I, S), \quad A = Ea,$$

the connection has the form

$$\nabla^{\bar{f}} A = E(da + \Gamma^{\bar{f}} a).$$

If

$$E = E^{\text{raw}} W,$$

then formally

$$\Gamma^{\bar{f}} = W^{-1} \Gamma^{\text{raw}} W + W^{-1} dW.$$

In particular, if the raw frame is flat in the chart, then

$$\Gamma^{\mathfrak{f}} = W^{-1}dW.$$

On an overlap with

$$E_j = E_i G_{ij}, \quad G_{ij} \in GL_2(\mathbf{C}),$$

the connection matrices must satisfy

$$\Gamma_j = G_{ij}^{-1} \Gamma_i G_{ij} + G_{ij}^{-1} dG_{ij}.$$

Thus the rank-two connection is canonical if the fixed-sector frames glue by canonical GL_2 -transition maps and the fixed-sector projection is functorial under the ambient transition maps.

Proposition 14.47 (Conditional rank-two connection theorem). *Assume that on a reduced good-patch base $\mathcal{B}_{\text{red}}^+$:*

- (1) *the core/source convention is fixed and locally stable;*
- (2) *no zero moves into or out of q_{core} ;*
- (3) *the fixed-sector frames satisfy transition laws*

$$E_j = E_i G_{ij}, \quad G_{ij} \in GL_2(\mathbf{C}), \quad G_{ij} G_{jk} = G_{ik};$$

- (4) *the ambient corrected jet bundle $\mathcal{J}$ and projection $\Pi_{\mathfrak{f}}$ are functorial under these transitions;*
- (5) *the whitening branch and Gram normalization are fixed or functorial.*

Then

$$\nabla_X^{\mathfrak{f}} A = \Pi_{\mathfrak{f}}(d_X^{\mathcal{J}} A)$$

defines a canonical connection on

$$\mathcal{F}_{\text{red}}^+ \rightarrow \mathcal{B}_{\text{red}}^+.$$

The reduced-base hypotheses are essential. On the unreduced base, changing the source/core convention can produce affine corrections

$$A_j^{\mathfrak{f}} = G_{ij}^{-1} A_i^{\mathfrak{f}} + B_{ij}.$$

Such transformations do not define an ordinary vector-bundle connection. Thus the present rank-two connection is currently a fixed-convention, reduced-base object unless an affine torsor transition theorem is proved.

Remark 14.48 (Why rank two improves on the line bundle). The $A_5^{\mathfrak{f}}$ -line may rotate under legitimate changes of trivialization, source/core convention, or whitening. This blocked the line-bundle route. The full plane $\mathfrak{f}_h = \mathbf{C}\mathbf{I} \oplus \mathbf{C}\mathbf{S}$ can still transform by a GL_2 -matrix. Hence the rank-two bundle is the right level at which to formulate covariance.

14.19. Order-seven packet visibility on the reduced base. For this and the following subsections, write the centered diagonal Gram data as

$$g = \frac{q_0}{\pi}, \quad h = \frac{1}{\pi} \left(\frac{1}{3} q_0^3 + \frac{1}{6} q'' \right),$$

so that the centered same-point Gram block at $s = 0$ is, up to the fixed ambient normalization,

$$G_0 = \text{diag}(2g, 2h).$$

Definition 14.49 (First and second Gram-anisotropy functionals). On a centered reduced patch, unless explicitly stated otherwise, q_0 , q'' , and $q^{(4)}$ mean the full actual-zeta centered jets

$$q_0 = q_{\zeta}(m), \quad q'' = q''_{\text{can}}(m) = q''_{\zeta}(m), \quad q^{(4)} = (q_{\zeta})^{(4)}(m).$$

Define

$$\Gamma_G^{\text{can}} := q''_{\text{can}} - 6q_0 + 2q_0^3.$$

For readability we usually suppress the superscript and write $\Gamma_G = \Gamma_G^{\text{can}}$. Then

$$g - h = -\frac{\Gamma_G}{6\pi}.$$

Thus the Gram-isotropy wall $g = h$ is equivalently

$$\Gamma_G = 0.$$

We also define the second Gram-anisotropy functional

$$\Gamma_{G,2}^{\text{can}} := (1 - q_0^2)q_{\text{can}}'' - \frac{1}{6}(q_\zeta)^{(4)}.$$

Again, when the canonical convention is fixed, we write $\Gamma_{G,2} = \Gamma_{G,2}^{\text{can}}$. This is the anisotropy of the second centered Gram variation. In particular, on $g = h$, the first Gram anisotropy has vanished, but the second one need not vanish.

The first admissible visibility coefficient is not A_5^{f} , which defines the reference line, but A_7^{f} . The direct raw off-diagonal order-seven term is not itself the witness. At a centered even packet,

$$N_7^{\text{off}} = C_7 J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Since $X_{-,0} = X_{+,0} = X_0$,

$$X_0 N_7^{\text{off}} X_0 = C_7 (\det X_0) J,$$

so

$$\pi_{\text{f}}(X_0 N_7^{\text{off}} X_0) = 0.$$

The live centered channel is instead the one-whitening-slot term

$$T_7^{(1)} = X_{-,1} Y X_{+,0} + X_{-,0} Y X_{+,1}, \quad Y = yJ.$$

At centered parity,

$$X_{+,0} = X_{-,0} = X_0, \quad X_{+,1} = -X_{-,1}.$$

Write

$$X_0 = pI + rD, \quad X_{-,1} = BS.$$

Then

$$T_7^{(1)} = yB(SJX_0 - X_0JS) = -2yB(pD + rI),$$

and hence

$$\boxed{\pi_{\text{f}}(T_7^{(1)}) = -2yBr I.}$$

For the centered off-line packet,

$$q'' = 2A_2, \quad A_2 = -4(a^{-3} + b^{-3}) \neq 0,$$

and

$$y_{\text{lead}} = -\frac{q_0 A_2^2}{72}.$$

Moreover

$$B = \frac{q''}{4\pi\sqrt{gh}(\sqrt{g} + \sqrt{h})}, \quad r = \frac{1}{2}(g^{-1/2} - h^{-1/2}).$$

Thus the one-whitening-slot channel is nonzero outside the explicit incidence walls

$$y = 0, \quad q'' = 0, \quad g = h.$$

Lemma 14.50 (Centered order-seven fixed-sector row and first Gram anisotropy). *At centered parity,*

$$X_{+,0} = X_{-,0} = X_0, \quad X_{+,1} = -X_{-,1}.$$

Write

$$X_0 = pI + rD, \quad X_{-,1} = BS, \quad Y = yJ,$$

where

$$p = \frac{1}{2}(g^{-1/2} + h^{-1/2}), \quad r = \frac{1}{2}(g^{-1/2} - h^{-1/2}).$$

Then the centered one-whitening-slot row

$$T_7^{(1)} := X_{-,1} Y X_{+,0} + X_{-,0} Y X_{+,1}$$

satisfies

$$T_7^{(1)} = -2yB(pD + rI).$$

Consequently

$$\pi_{\mathfrak{f}} T_7^{(1)} = -2yBr I.$$

Equivalently,

$$\pi_{\mathfrak{f}} T_7^{(1)} = -y \frac{q''(h-g)}{4\pi gh(\sqrt{g} + \sqrt{h})^2} I.$$

Thus, away from the source walls $y = 0$ and $q'' = 0$, the centered order-seven fixed-sector signal is nonzero exactly off the first Gram-isotropy wall $g = h$, or equivalently off $\Gamma_G = 0$.

Remark 14.51 (Why order seven alone is insufficient on $g = h$). On $g = h$ one has $r = 0$. The order-seven row does not disappear in the ambient 2×2 algebra; rather, it becomes

$$T_7^{(1)}|_{g=h} = -2yBpD.$$

This component is anti-fixed and is killed by $\pi_{\mathfrak{f}}$. Hence the centered order-seven fixed-sector argument requires either quantitative separation from $g = h$, an ambient D -sector enlargement, or a higher-order fixed-sector replacement. The next replacement is the order-nine row below.

Lemma 14.52 (Second whitening anisotropy on the Gram-isotropy wall). *Assume $g = h$, so that*

$$G_0 = gI.$$

Write

$$G_-(s) = G_0 + sG_1 + s^2G_2 + O(s^3), \quad X_-(s) = G_-(s)^{-1/2} = X_0 + sX_1 + s^2X_2 + O(s^3),$$

with the convention $G_-(s) = G(m - s/2)$, so

$$G_1 = -\frac{1}{2}G'(m), \quad G_2 = \frac{1}{8}G''(m).$$

At a scalar base,

$$X_2 = -\frac{1}{2}g^{-3/2}G_2 + \frac{3}{8}g^{-5/2}G_1^2.$$

At centered parity, $G_1 \in \mathbf{CS}$, hence

$$G_1^2 \in \mathbf{CI}.$$

Therefore

$$(X_2)_D = -\frac{1}{2}g^{-3/2}(G_2)_D.$$

Using the centered Gram model

$$G(t) = \frac{1}{\pi} \begin{pmatrix} q(t) & \frac{1}{2}q'(t) \\ \frac{1}{2}q'(t) & \frac{1}{3}q(t)^3 + \frac{1}{6}q''(t) \end{pmatrix},$$

and $q'(m) = q'''(m) = 0$, one has

$$G''(m) = \frac{1}{\pi} \begin{pmatrix} q'' & 0 \\ 0 & q_0^2 q'' + \frac{1}{6}q^{(4)} \end{pmatrix}.$$

Thus

$$(G_2)_D = \frac{1}{16\pi} \left((1 - q_0^2)q'' - \frac{1}{6}q^{(4)} \right) = \frac{\Gamma_{G,2}}{16\pi}.$$

Consequently

$$\boxed{(X_2)_D = -\frac{1}{32\pi}g^{-3/2}\Gamma_{G,2}.}$$

Proposition 14.53 (Order-nine fixed-sector survivor on $g = h$). *Assume $g = h$. At centered parity,*

$$X_{+,1} = -X_{-,1}, \quad X_{+,2} = X_{-,2}.$$

Let

$$X_1 := X_{-,1} = BS, \quad Y = yJ, \quad X_2 := X_{-,2} = aI + dD,$$

with $d = (X_2)_D$. Consider the order-nine row

$$T_9^{(1,2)} := X_{-,1}YX_{+,2} + X_{-,2}YX_{+,1}.$$

Then

$$T_9^{(1,2)} = X_1YX_2 - X_2YX_1.$$

Using $SJ = -D$ and $JS = D$, we obtain

$$T_9^{(1,2)} = -2yB(aD + dI).$$

Therefore

$$\pi_{\mathfrak{f}}T_9^{(1,2)} = -2yBdI.$$

At $g = h$,

$$B = \frac{q''}{8\pi g^{3/2}}$$

and, by Lemma 14.52,

$$d = -\frac{1}{32\pi}g^{-3/2}\Gamma_{G,2}.$$

Hence

$$\boxed{\pi_{\mathfrak{f}}T_9^{(1,2)} = \frac{yq''\Gamma_{G,2}}{128\pi^2g^3}I.}$$

In particular, on the source-good locus $yq'' \neq 0$, the order-nine fixed-sector row survives on $g = h$ whenever

$$\Gamma_{G,2} \neq 0,$$

modulo the correction incidence defined below.

Lemma 14.54 (The second anisotropy wall is not forced by $g = h$). *The first Gram-isotropy relation*

$$g = h$$

is equivalent to

$$q'' = 6q_0 - 2q_0^3.$$

It does not imply

$$\Gamma_{G,2} = 0.$$

Indeed, on $g = h$,

$$\Gamma_{G,2} = (1 - q_0^2)(6q_0 - 2q_0^3) - \frac{1}{6}q^{(4)}.$$

Thus $\Gamma_{G,2} = 0$ imposes the additional fourth-jet relation

$$q^{(4)} = 6(1 - q_0^2)(6q_0 - 2q_0^3).$$

This is an independent higher-jet incidence condition.

For a centered paired packet

$$f_{\beta,\gamma}(t) = \frac{a}{a^2 + x^2} + \frac{b}{b^2 + x^2}, \quad x = 2t - \gamma,$$

evaluated at $x = 0$, one has

$$q_0 = a^{-1} + b^{-1}, \quad q'' = -8(a^{-3} + b^{-3}), \quad q^{(4)} = 384(a^{-5} + b^{-5}).$$

Hence

$$\Gamma_{G,2} = (1 - q_0^2)q'' - \frac{1}{6}q^{(4)}.$$

For example, for $a = \frac{1}{4}$, $b = \frac{3}{4}$,

$$q_0 = \frac{16}{3}, \quad q'' = -\frac{14336}{27}, \quad q^{(4)} = \frac{31981568}{81},$$

and therefore

$$\Gamma_{G,2} = -\frac{12449792}{243} \neq 0.$$

Thus the order-nine wall $\Gamma_{G,2} = 0$ is a proper additional analytic wall, not a consequence of centered parity or of $g = h$.

Proposition 14.55 (Raw order-nine I -row audit). *Under the centered raw two-point convention, the raw odd carriers are skew off-diagonal:*

$$N_{2j+1} = F_{2j+1}J.$$

In particular,

$$N_9 = F_9J, \quad N_7 = F_7J, \quad N_5 = F_5J, \quad N_3 = F_3J, \quad N_1 = F_1J.$$

Consequently their I - and D -components vanish:

$$N_{2j+1}^I = N_{2j+1}^D = 0.$$

The raw even carriers are fixed under endpoint swap. In the centered raw two-point convention, their symmetric off-diagonal S -parts do not contribute to the order-nine I -coordinate:

$$N_4^S = N_2^S = N_0^S = 0$$

for the raw lower-carrier row relevant here.

Therefore the raw order-nine I -coordinate receives no contribution from the raw carriers other than the order-nine survivor

$$T_9^{(1,2)} = X_{-,1}YX_{+,2} + X_{-,2}YX_{+,1}.$$

Equivalently,

$$R_{\leq 4}^{I,\text{raw}} = 0,$$

and

$$R_9^I[N_9] = R_7^I[N_7] = R_5^I[N_5] = 0.$$

Remark 14.56 (Convention caveat for corrected rows). Proposition 14.55 concerns the raw pre-whitening two-point carriers N_b . If a later notation folds normalization, subtraction, background/core, line, or connection correction terms into a symbol also called N_b , those terms should be separated from the raw carrier and placed into the correction remainder below. The raw matrix-parity audit does not by itself eliminate such correction rows.

Definition 14.57 (Order-nine correction incidence). Let $R_{\text{corr}}^{I,(9)}$ denote the residual order-nine I -coordinate coming from correction rows not included in the raw two-point carrier table. After covariantization and the fixed-convention local gates, this residual may only include line, background/core, or normalization/subtraction correction data:

$$R_{\text{corr}}^{I,(9)} = R_{\text{line}}^{I,(9)} + R_{\text{bg/core}}^{I,(9)} + R_{\text{norm/sub}}^{I,(9)}.$$

Connection/frame/projection/whitening-gauge terms are absorbed by ∇^f ; Gate A/B rows are closed under the fixed-convention hypotheses; shadow rows are projected away; and pure same-point odd rows vanish by centered parity.

Define the order-nine correction incidence wall by

$$\mathcal{E}_{\text{corr}}^{(9)} := \left\{ R_{\text{corr}}^{I,(9)} = -\frac{yq''\Gamma_{G,2}}{128\pi^2g^3}, \quad yq''\Gamma_{G,2} \neq 0 \right\}.$$

On the complement of this wall, the order-nine I -coordinate is nonzero whenever $yq''\Gamma_{G,2} \neq 0$.

Theorem 14.58 (Layered fixed-sector visibility through order nine). *Work on a centered reduced good patch in the fixed-core convention. Assume*

$$g > 0, \quad h > 0, \quad yq'' \neq 0.$$

Let

$$\Gamma_G = q'' - 6q_0 + 2q_0^3, \quad \Gamma_{G,2} = (1 - q_0^2)q'' - \frac{1}{6}q^{(4)}.$$

Assume also that the point is not in the order-seven cross-channel incidence wall and not in the order-nine correction incidence wall:

$$\theta \notin \mathcal{E}_{\text{cc}}^{(7)} \cup \mathcal{E}_{\text{corr}}^{(9)}.$$

If

$$\Gamma_G \neq 0 \quad \text{or} \quad \Gamma_{G,2} \neq 0,$$

then

$$(A_7^{\mathfrak{f}}, A_9^{\mathfrak{f}}) \neq 0.$$

More precisely:

$$\Gamma_G \neq 0 \implies A_7^{\mathfrak{f}} \neq 0,$$

where the nonzero fixed-sector row is the centered order-seven row

$$\pi_{\mathfrak{f}} T_7^{(1)} = -2yBr I.$$

On the first Gram-isotropy wall

$$\Gamma_G = 0,$$

if

$$\Gamma_{G,2} \neq 0$$

and

$$\theta \notin \mathcal{E}_{\text{corr}}^{(9)},$$

then

$$A_9^{\mathfrak{f}} \neq 0,$$

with order-nine I -coordinate

$$(A_9^{\mathfrak{f}})_I = \frac{yq''\Gamma_{G,2}}{128\pi^2g^3} + R_{\text{corr}}^{I,(9)}.$$

Proof. If $\Gamma_G \neq 0$, then $g - h \neq 0$. By Lemma 14.50,

$$\pi_{\mathfrak{f}} T_7^{(1)} = -y \frac{q''(h - g)}{4\pi gh(\sqrt{g} + \sqrt{h})^2} I,$$

which is nonzero under $yq'' \neq 0$, $g, h > 0$, and $g - h \neq 0$, outside the named order-seven cross-channel incidence.

It remains to consider $\Gamma_G = 0$, equivalently $g = h$. By Proposition 14.53,

$$\pi_{\mathfrak{f}} T_9^{(1,2)} = \frac{yq''\Gamma_{G,2}}{128\pi^2g^3} I.$$

By Proposition 14.55, the raw order-nine I -row has no additional contribution from N_9, N_7, N_5 , or from the raw lower carriers. Thus the only possible remaining I -coordinate correction is $R_{\text{corr}}^{I,(9)}$. Outside $\mathcal{E}_{\text{corr}}^{(9)}$, this correction does not cancel the displayed candidate row. Hence $A_9^{\mathfrak{f}} \neq 0$. $\square$

Remark 14.59 (What the layered theorem does and does not prove). Theorem 14.58 is a local visibility theorem. It improves the centered order-seven story by replacing the single excluded wall

$$\Gamma_G = 0$$

with the layered incidence

$$\Gamma_G = 0, \quad \Gamma_{G,2} = 0,$$

together with the named correction incidence. In particular, the order-seven fixed-sector signal detects off first Gram isotropy, while the order-nine fixed-sector signal detects on the first Gram-isotropy wall unless the second Gram anisotropy also vanishes.

This is still not an RH-scale quantitative lower bound. For a global proof one would need a non-concentration estimate of the form

$$|yq''| \max\{|\Gamma_G|, |\Gamma_{G,2}|\} \geq Q^{-A},$$

or an equivalent lower bound for

$$\|(A_7^f, A_9^f)\|$$

outside the layered incidence and correction walls. Qualitative nonvanishing alone does not rule out super-polynomial approach to the layered incidence set.

Remark 14.60 (No-go ledger for the centered order-seven/nine analysis). The following points should be kept separate.

- (1) The centered order-seven fixed-sector row is structurally proportional to the first Gram anisotropy $g - h$. Hence it cannot detect the wall $g = h$ inside the fixed sector.
- (2) The ambient order-seven D -component survives on $g = h$, but using it would require an ambient, non-fixed-sector enlargement. The present theorem instead stays within the rank-two fixed sector and uses order nine.
- (3) The order-nine survivor is not a raw-carrier artifact. It arises from the channel

$$Y_J X_{1,S} X_{2,D},$$

namely

$$X_{-,1} Y X_{+,2} + X_{-,2} Y X_{+,1}.$$

- (4) The raw carriers N_9, N_7, N_5, N_3, N_1 are skew off-diagonal at centered parity:

$$N_{2j+1} = F_{2j+1} J.$$

They do not contribute to the order-nine I -coordinate. The raw even lower carriers are likewise harmless for the order-nine I -row under the centered raw convention.

- (5) Any remaining order-nine cancellation is therefore a correction-row or cross-channel analytic incidence issue, not a sector/parity identity of the raw row table.

Remark 14.61 (Raw skew coefficients F_5 and F_7). For the centered raw off-diagonal scalar

$$F(s) = \frac{\sin \Delta(s) + q_-(s)s \cos \Delta(s)}{s^2},$$

with

$$\Delta(s) = q_0 s + \frac{q_2}{24} s^3 + \frac{q_4}{1920} s^5 + \frac{q_6}{322560} s^7 + \cdots,$$

and

$$q_-(s) = q_0 + \frac{q_2}{8} s^2 + \frac{q_4}{384} s^4 + \frac{q_6}{46080} s^6 + \cdots,$$

the raw odd carriers are skew:

$$N_{2j+1} = F_{2j+1} J.$$

In the normalization used here,

$$F_5 = -\frac{q_0^7}{630} + \frac{q_0^4 q_2}{72} - \frac{q_0^2 q_4}{480} - \frac{q_0 q_2^2}{144} + \frac{q_6}{40320},$$

and

$$F_7 = \frac{q_0^9}{36288} - \frac{q_0^6 q_2}{1728} + \frac{q_0^4 q_4}{4608} + \frac{5q_0^3 q_2^2}{3456} - \frac{q_0^2 q_6}{64512} - \frac{q_0 q_2 q_4}{4608} - \frac{5q_2^3}{41472} + \frac{q_8}{9289728}.$$

These coefficients are useful for auditing the raw row table, but they multiply J and hence do not contribute directly to the fixed I -coordinate.

There is also a non-centered visibility route. The local construction uses

$$t_{\pm} = m \pm \frac{s}{2},$$

so m is a sampling midpoint, not necessarily the zero-packet center. For the paired packet

$$f_{\beta,\gamma}(t) = \frac{a}{a^2 + x^2} + \frac{b}{b^2 + x^2}, \quad x = 2t - \gamma,$$

one has, near $x_m = 2m - \gamma = 0$,

$$q^{(7)}(m) = 5160960(a^{-9} + b^{-9})x_m + O(x_m^3).$$

Thus $x_m = 0$ is a simple zero of $q^{(7)}(m)$, and every sufficiently small nonzero perturbation of m activates the septic K_1 -transport slice, outside

$$q^{(7)} = 0, \quad \Xi_{K_1} = 0.$$

Proposition 14.62 (Packet visibility on $\mathcal{B}_{\text{red}}^+$). *On $\mathcal{B}_{\text{red}}^+$, the finite fixed-sector vector*

$$C_{\text{fd}}^{\vec{f}} = (A_7^{\vec{f}}, A_9^{\vec{f}}, \dots)$$

is nonzero for the reduced off-line packet family.

Proof. For the centered route, the direct skew N_7^{off} -term projects to zero, but the one-whitening-slot K_1 -channel has fixed-sector projection

$$-2yBr I.$$

This vector is nonzero outside

$$\{y = 0\} \cup \{q'' = 0\} \cup \{g = h\}.$$

To rule out cancellation by the K_5 -line sector using quotient isolation, we also exclude

$$v_5 = 0.$$

Indeed,

$$\omega_h(-2yBr I, A_5^{\vec{f}}) = -2yBr v_5,$$

so K_5 -line terms cannot cancel the K_1 -channel on the generic patch

$$yBr v_5 \neq 0.$$

The branch $v_5 = 0$ is therefore either included in $\mathcal{E}_{\text{invis}}^{\text{core},7}$ or must be treated by a separate full-vector K_5 -coefficient computation.

For the non-centered route, m is a free sampling midpoint. The expansion of $q^{(7)}(m)$ shows that $q^{(7)} = 0$ is a proper analytic wall near the packet center. Away from this wall and from $\{\Xi_{K_1} = 0\}$, the septic K_1 -transport slice gives $A_7^{\vec{f}} \neq 0$. $\square$

Remark 14.63 (Order-seven channel independence ledger). The full order-seven corrected-whitening coefficient is

$$A_{7,\text{core}}^{\vec{f}} = \pi_{\vec{f}} \sum_{a+g+b=7} X_{-,a} N_g X_{+,b}.$$

The direct skew N_7^{off} -term is killed by $\pi_{\vec{f}}$. The live one-whitening-slot term belongs to the K_1 -transport sector. K_3 shadow terms are projected away, pure same-point septic terms vanish in the odd two-point germ, and K_5 -terms lie in

$$\mathbb{C}A_5^{\vec{f}}.$$

Thus exact cancellation of the live $-2yBr I$ channel can only come from a same K_1 -transport contribution or a direct diagonal N_7 -term with the same jet monomial. No structural identity forcing such cancellation is currently present; such cancellation is included in the visibility incidence locus.

14.20. **The $\eta\eta$ row as rank-two affine forcing.** In the rank-two formulation, the active Bottleneck-C row is no longer a scalar defect by itself. It is a vector-valued section:

$$E_{\eta\eta} = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS) \in \Gamma(\mathcal{F}_{\text{red}}^+).$$

Under a frame change

$$E_j = E_i G_{ij},$$

the coordinate vector of $wI + uS$ transforms by

$$[w, u]_j^T = G_{ij}^{-1} [w, u]_i^T.$$

The scalar coefficient

$$-\frac{\nu_{\eta\eta}}{24\pi q^3}$$

is fixed-convention scalar data. Therefore $E_{\eta\eta}$ transforms as a section of the rank-two fixed-sector bundle, provided the reduced atlas is functorial.

This row should not be identified with ordinary curvature of ∇^f . Curvature is antisymmetric in two tangent directions:

$$F_{\nabla^f}(X, Y) = [\nabla_X^f, \nabla_Y^f] - \nabla_{[X, Y]}^f.$$

The active row is symmetric in the source directions η_i, η_j . It is better classified as a symmetric source-Hessian forcing term, or a second-fundamental-form component, in the fixed-sector transport equation.

The natural local replacement for scalar Bottleneck-C flatness is therefore an affine transport equation of the form

$$\nabla_\kappa^f A_7^f = E_{\eta\eta} + E_{\text{lower}}.$$

Here E_{lower} denotes all lower, line, shadow, pure, connection, and background/core terms. The expected decomposition is

$$E_{\text{lower}} = E_{\text{GateA}} + E_{\text{GateB}} + E_{\text{line}} + E_{K_3} + E_{\text{pure}} + E_{\text{conn}} + E_{\text{bg/core}} + E_{K_1}^{\text{comp}}.$$

Under the current local ledger,

$$E_{\text{GateA}} = 0, \quad E_{\text{GateB}} = 0$$

under the accepted Gate A/B hypotheses,

$$E_{\text{conn}} = 0$$

after passing to the covariant derivative,

$$E_{K_3} = 0$$

after fixed-sector projection,

$$E_{\text{pure}} = 0$$

in the odd two-point germ, and

$$E_{\text{line}} \in \mathcal{C}A_5^f.$$

The remaining table dependency is to verify that no additional uncontrolled same-grade K_1 source-Hessian companion survives outside the already identified $E_{\eta\eta}$ row.

The current local row table is:

row class	source	fixed-sector status	present row
E_{GateA}	aggregate homogeneous scalar-grade rows	0	closed by Gate A
E_{GateB}	ordinary order ≤ 7 local rows	0	closed by Gate B
$E_{\eta\eta}$	symmetric second-source Hessian / mixed-block row	$-\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS)$	retained as affine
E_{line}	K_5 normalization shadows	$\in \mathcal{C}A_5^f$	transported line
E_{K_3}	K_3 shadow sector	0	projected away
E_{pure}	pure same-point septic sector	0	vanishes in odd two-
E_{conn}	frame/projection/whitening gauge motion	0	absorbed by
$E_{\text{bg/core}}$	fixed background/core baseline terms	fixed analytic section	included in affine

The visibility-positive incidence exclusions used to define $\mathcal{B}_{\text{red}}^+$ do not alter Gate A/B; they only restrict the base to points where the packet-visibility channel is active.

Theorem 14.64 (Conditional local rank-two affine forcing equation). *On $\mathcal{B}_{\text{red}}^+$, assume the rank-two connection theorem and the row-table classifications above. Then*

$$\nabla_{\kappa}^{\mathfrak{f}} A_7^{\mathfrak{f}} = E_{\eta\eta} + E_{\text{lower}},$$

where

$$E_{\eta\eta} = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS)$$

is a tensorial section of $\mathcal{F}_{\text{red}}^+$, and E_{lower} contains only Gate A/B-closed, connection-absorbed, line, shadow, pure, background/core, and already-classified analytic forcing terms.

Proof. By definition,

$$A_7^{\mathfrak{f}} = [s^7] \Pi_{\mathfrak{f}} \widehat{\Omega}_{\zeta}^{\text{corr}}(s; m).$$

The projected connection gives

$$\nabla_{\kappa}^{\mathfrak{f}} A_7^{\mathfrak{f}} = \Pi_{\mathfrak{f}} d_{\kappa}^{\mathcal{J}} \left([s^7] \widehat{\Omega}_{\zeta}^{\text{corr}} \right),$$

so frame, projection, and whitening-gauge variations are absorbed into $\nabla^{\mathfrak{f}}$. The $\eta_i \eta_j$ second-source Hessian row of the promoted mixed block gives $E_{\eta\eta}$. All remaining terms are classified by the displayed table. $\square$

Remark 14.65 (The equation is local, not yet global). Theorem 14.64 does not prove a global contradiction. It changes the local interpretation of the Bottleneck-C row: the row is no longer a scalar defect demanding pointwise vanishing, but an inhomogeneous forcing term in a rank-two transport equation. A global boundary, holonomy, exactness, index, or transported finite-core non-concentration theorem is still required.

Remark 14.66 (What the affine equation changes). The old scalar Bottleneck-C gate demanded local pointwise vanishing of the S -visible component. The rank-two equation instead allows the same row to be an inhomogeneous forcing term in a vector-valued transport equation. This is a genuine conceptual improvement only if a later global theorem controls the forced transport. If one still demands

$$E_{\eta\eta} = 0$$

pointwise, then the rank-two formulation collapses back to the old obstruction

$$\nu_{\eta\eta} = 0$$

on a good patch.

14.21. Global affine transport dependency. Assuming the local affine equation

$$\nabla^{\mathfrak{f}} A = E$$

is available, parallel transport along a path $\Gamma : a \rightarrow b$ gives

$$A(b) = U_{a \rightarrow b} A(a) + \int_a^b U_{t \rightarrow b} E(t) dt.$$

For the order-seven fixed-sector section this becomes

$$A_7^{\mathfrak{f}}(b) = U_{a \rightarrow b} A_7^{\mathfrak{f}}(a) + \int_a^b U_{t \rightarrow b} (E_{\eta\eta} + E_{\text{lower}})(t) dt.$$

Thus the local Bottleneck-C obstruction becomes an affine displacement problem.

At present no canonical global proof gate for this displacement is available. The natural path and boundary candidates have the following status:

- κ -paths are locally natural but have no known endpoint values for $A_7^{\mathfrak{f}}$.
- midpoint paths $m \mapsto b(m)$ are natural locally, but good-patch boundaries are exceptional walls rather than known boundary conditions.

- source/core deformation paths are excluded on the reduced base, since source/core transfer is not functorial as an ordinary vector-bundle transition.
- Gram or whitening loops may measure gauge/atlas data rather than zeta geometry unless a global atlas theorem is proved.
- zero-counting contours are canonical for ξ or ζ , but no map from those contours into the rank-two fixed-sector base is currently constructed.

Nor is the forcing currently exact. In the fixed-source midpoint calculus, the active scalar density satisfies

$$2880Q^{14}\nu_{\eta\eta} = qAA'' + 20\left(\frac{q''}{6} - \frac{2q^3}{3}\right)A^2, \quad A = A_2.$$

Integration by parts yields

$$qAA'' + 20\left(\frac{q''}{6} - \frac{2q^3}{3}\right)A^2 = \left(qAA' - \frac{1}{2}q'A^2\right)' - q(A')^2 + \left(\frac{23}{6}q'' - \frac{40}{3}q^3\right)A^2.$$

The last two terms are genuine bulk terms. Therefore the forcing is not presently a boundary term in the natural variables.

For a loop Γ based at b , the corresponding affine holonomy data are

$$\text{AffHol}_\Gamma = \left(U_\Gamma, \Delta_\Gamma^{\text{aff}}\right),$$

where

$$\Delta_\Gamma^{\text{aff}} = \int_\Gamma U_{t \rightarrow b}(E_{\eta\eta} + E_{\text{lower}})(t) dt.$$

A holonomy proof gate would need all of:

- (1) a canonical loop $\Gamma \subset \mathcal{B}_{\text{red}}^+$;
- (2) a global theorem constraining U_Γ ;
- (3) a global zeta-theoretic law forcing $\Delta_\Gamma^{\text{aff}} = 0$, or a prescribed value;
- (4) an off-line-packet theorem forcing an incompatible value of $\Delta_\Gamma^{\text{aff}}$.

None of these is presently available.

Remark 14.67 (Affine holonomy is not presently a proof gate). Loops around core-boundary walls, packet-visibility incidence walls, Gram singularities, or whitening branch loci are diagnostic but not currently canonical RH objects. A loop around a core-boundary wall may change labels or source/core convention; a Gram loop may measure only gauge or atlas monodromy; and zero-counting contours live in the complex zeta plane, not yet in the rank-two fixed-sector base $\mathcal{B}_{\text{red}}^+$. Thus affine holonomy cannot presently replace the local affine forcing equation. The most plausible global successors are real path/boundary transport or a transported finite-core non-concentration theorem.

Remark 14.68 (Current global dependency). The rank-two affine equation is currently descriptive rather than contradictory. To turn it into a proof gate one needs one of:

- (1) canonical boundary values for A_7^{f} ;
- (2) a canonical path or loop with prescribed affine holonomy;
- (3) a path-independent affine displacement theorem;
- (4) a projective/scale index theorem derived from the rank-two transport;
- (5) a transported finite-core non-concentration theorem for the full fixed-sector vector.

None of these is presently proved.

Hypothesis 14.69 (Global rank-two affine transport gate). There exists a canonical path, loop, boundary condition, index, or transported finite-core non-concentration theorem on $\mathcal{B}_{\text{red}}^+$ that controls

$$\Delta_\Gamma^{\text{aff}} = \int_\Gamma U_{t \rightarrow b}(E_{\eta\eta} + E_{\text{lower}})(t) dt.$$

Equivalently, the rank-two affine forcing equation must be coupled to a global law that distinguishes RH-compatible configurations from off-line-packet configurations. Without such a law, the rank-two architecture is locally coherent but globally non-closing.

14.22. Derived diagnostics: projective, vector-valued, and energy data. Several objects remain useful as diagnostics, but none currently replaces the rank-two affine transport theorem.

First, projectivizing the fixed-sector vector gives

$$\ell_{2r+1}(m) = [A_{2r+1}^{\mathfrak{f}}(m)] \in \mathbf{P}(\mathfrak{f}_h(m)).$$

The quotient determinant is exactly projective incidence:

$$D_{2r+1} = \omega_h(A_{2r+1}^{\mathfrak{f}}, A_5^{\mathfrak{f}}),$$

and

$$D_{2r+1} = 0 \iff \ell_{2r+1} = \ell_5.$$

Thus projective/Maslov/spectral-flow data recover the quotient determinant geometry. They remain incomplete by themselves because pure line anomalies

$$A_{2r+1}^{\mathfrak{f}} = \lambda A_5^{\mathfrak{f}}$$

are projectively invisible but still visible to T and Φ_K . A projective index would therefore need to be paired with scale, energy, or the full rank-two connection.

Second, the complete local fixed-sector object is the vector-valued odd germ

$$\mathcal{A}_{\text{odd}}^{\mathfrak{f}}(s; m) = \sum_{r \geq 0} A_{2r+1}^{\mathfrak{f}}(m) s^{2r+1} \in \mathfrak{f}_h(m).$$

It retains all fixed-sector information, preserves componentwise holomorphy and Cauchy recovery, and avoids the incompleteness of quotient scalarization. However, it is essentially the full fixed-sector version of the split readout. It does not by itself prove the required finite-core lower bound

$$\theta(m) \notin \mathcal{Z}_k^{\vec{f}} \cup \mathcal{E}_{\text{geom}}^{\vec{f}} \implies \left\| P_{\text{fd}}^{\vec{f}}(\theta(m)) + J_{\text{fd}}^{\vec{f}}(\theta(m)) \right\|_{\infty} \geq \Lambda_{\vec{f}}(m, Q).$$

The aligned branch

$$J_{\text{fd}}^{\vec{f}} \approx -P_{\text{fd}}^{\vec{f}}$$

therefore remains a genuine downstream dependency.

Third, one may define fixed-sector energies such as

$$\mathcal{E}_7 = \|A_7^{\mathfrak{f}}\|_{\text{wh}}^2$$

or

$$\mathcal{E}_{\text{odd}} = \sum_r \omega_r \|A_{2r+1}^{\mathfrak{f}}\|_{\text{wh}}^2.$$

For

$$E_{\eta\eta} = \gamma(wI + uS),$$

the first variation of $\mathcal{E}_7$ contains

$$2 \operatorname{Re} \left[\gamma \langle A_7^{\mathfrak{f}}, wI + uS \rangle_{\text{wh}} \right],$$

which is sign-indefinite. The second variation is positive but tautological:

$$2|\gamma|^2 \|wI + uS\|_{\text{wh}}^2 \geq 0.$$

No zero-energy, monotonicity, or endpoint theorem is presently available. Thus energy detects the defect but does not yet provide a contradiction.

Remark 14.70 (Role of Gate A and Gate B after the architecture audits). Gate A and Gate B should be retained as local algebraic lemmas:

Gate A: aggregate homogeneous cancellation,

Gate B: conditional local quotient/fixed-sector algebra through order 7.

They simplify the local expansion and help classify lower terms. They are not proof-closing global gates. After the source-divisibility and source-collapse audits, the sharp missing input is no longer another Gate A/B-style local row identity. The current architecture needs an actual-zeta source-jet nonflatness theorem, a source-robust packet-visible row, or a global affine/projective transport theorem.

Remark 14.71 (Current architecture ledger). The current state is:

Gate A/B are retained as local fixed-convention algebraic lemmas.

Scalar, quotient-only, split-pointwise, and line-bundle Bottleneck-C repairs do not close the proof.

The reduced-base rank-two fixed-sector bundle remains the best local language for the obstruction.

The active $\eta\eta$ row is tensorial affine forcing, not ordinary curvature.

The layered order-seven/order-nine theorem gives local visibility, not an RH-scale lower bound.

The established rank-two norm tower is source-divisible: known packet-visible rows contain $H = yq''$.

The source-collapse regime $q'' \approx 0$ with live anisotropy is fatal for the present norm route.

Projectivization cancels H formally but lacks correction control, Regime-A control, and a global projective c

The current missing theorem is actual-zeta source-jet nonflatness, or a genuinely source-robust new row/inva

For the final status summary, see Remark 14.123.

The finite-core problem below should therefore be read in a further updated role. It is no longer merely a downstream step after a closed Bottleneck-C gate, nor is it only a generic vector-valued non-concentration problem. The source-collapse audits show that anisotropy non-concentration alone is insufficient, because the known fixed-sector rows carry the common source factor

$$H = yq''.$$

Thus any proof-relevant finite-core theorem must include source-level nonflatness, for instance

$$\sup_{m \in I_\rho} |q''(m)| \geq Q^{-B},$$

or else use a genuinely source-robust row/invariant not divisible by H .

Remark 14.72 (Visibility is not non-concentration). The order-seven packet-visibility result proves that the relevant fixed-sector vector is not identically zero on the reduced visibility-positive base. It does not prove the sampled finite-core lower bound. The aligned branch

$$J_{\text{fd}}^{\vec{f}} \approx -P_{\text{fd}}^{\vec{f}}$$

remains possible until a vector-valued non-concentration, cone-control, interaction-rigidity, or transport-index theorem rules it out. Thus visibility closes the local detection problem but not the global sampling problem.

Remark 14.73 (Transported finite-core data do not remove alignment). One may try to strengthen the finite-core vector by adding covariant transport data, for instance

$$C_{\text{fd}}^{\text{tr}} = (A_7^{\text{f}}, \nabla^{\text{f}} A_7^{\text{f}}, A_9^{\text{f}}, \nabla^{\text{f}} A_9^{\text{f}}, \dots).$$

Using the local affine equation, this is equivalent to adding

$$E_{\eta\eta} + E_{\text{lower}}$$

as additional coordinates. This is coherent, but it does not by itself prove sampled non-concentration. The old aligned branch simply becomes

$$J_{\text{fd}}^{\text{tr}} \approx -P_{\text{fd}}^{\text{tr}}.$$

Adding derivative or forcing coordinates enlarges the finite vector but does not impose a sign, cone, monotonicity, or endpoint constraint that forbids alignment. A transported finite-core theorem would therefore require a new coercivity, boundary, incidence, or cone-control input.

14.23. Source-weighted interval non-concentration as the remaining arithmetic gate.

The local fixed-sector visibility result is layered through order nine:

$$\Gamma_G \neq 0 \implies A_7^f \neq 0,$$

while

$$\Gamma_G = 0, \quad \Gamma_{G,2} \neq 0 \implies A_9^f \neq 0$$

outside the named correction and cross-channel incidence walls. This local visibility theorem is not the remaining proof gate. The remaining proof gate is the source-weighted interval lower bound

$$\boxed{\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}, \quad H = yq''}.$$

A more invariant norm formulation is

$$\left\| (A_7^f, A_9^f) \right\|_\infty \geq Q^{-B}$$

on the sampled reduced packet interval. The source-collapse analysis shows that this lower bound is equivalent, for the present tower, to a new source-jet nonflatness assertion for actual zeta data.

In the transported version, one may instead work with

$$C_{\text{fd}}^{\text{tr}} = \left(A_7^f, \nabla^f A_7^f, A_9^f, \nabla^f A_9^f, \dots \right),$$

or minimally, using the affine equation,

$$C_{\text{fd}}^{\text{tr}} = \left(A_7^f, E_{\eta\eta} + E_{\text{lower}}, A_9^f, \dots \right).$$

The remaining global obstruction is that pair and interaction pieces may still align:

$$J_{\text{fd}}^{\text{tr}} \approx -P_{\text{fd}}^{\text{tr}}.$$

Thus the next proof-closing theorem must be a quantitative layered non-concentration, incidence-distance, cone, or arithmetic separation result.

Definition 14.74 (Source-weighted layered residual). On a centered reduced good patch define

$$H := yq''.$$

The order-seven/order-nine layered fixed-sector rows have the schematic form

$$A_7^f \sim H\Gamma_G, \quad A_9^f \sim H\Gamma_{G,2},$$

where

$$\Gamma_G = q'' - 6q_0 + 2q_0^3$$

and

$$\Gamma_{G,2} = (1 - q_0^2)q'' - \frac{1}{6}q^{(4)}.$$

Thus the relevant norm-scale residual is not merely

$$(\Gamma_G, \Gamma_{G,2}),$$

but the source-weighted vector

$$\mathcal{R}_{\text{sw}} := (H\Gamma_G, H\Gamma_{G,2}).$$

Equivalently, the scalar lower-bound target is

$$\boxed{\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}}.$$

At leading centered source order,

$$y_{\text{lead}} = -\frac{q_0 A_2^2}{72}, \quad q'' = 2A_2,$$

and hence

$$H = yq'' \sim -\frac{q_0(q'')^3}{288} = -\frac{q_0 A_2^3}{36}.$$

This cubic source factor is the main vulnerability of the rank-two norm route.

Proposition 14.75 (Source-divisibility of the established fixed-sector tower). *The established fixed-sector escape tower generated by the channels*

$$X_1 Y X_{2k} - X_{2k} Y X_1, \quad Y = yJ, \quad X_1 = BS,$$

is divisible by the common source factor $H = yq''$.

More precisely, write

$$X_{2k} = a_k I + d_k D + \dots$$

Then

$$X_1 Y X_{2k} - X_{2k} Y X_1 = yB(SJX_{2k} - X_{2k}JS).$$

Using

$$SJ = -D, \quad JS = D,$$

one obtains

$$X_1 Y X_{2k} - X_{2k} Y X_1 = -yB(DX_{2k} + X_{2k}D).$$

The I -component of $DX_{2k} + X_{2k}D$ is $2d_k I$. Hence

$$\pi_{\mathfrak{f}}(X_1 Y X_{2k} - X_{2k} Y X_1) = -2yB d_k I.$$

Since

$$B \propto q'',$$

every row in this tower contains

$$yB \sim yq'' = H.$$

Thus increasing the anisotropy order replaces

$$\Gamma_G, \Gamma_{G,2}, \dots$$

by higher anisotropy factors, but it does not remove the common source factor H .

Remark 14.76 (Why the anisotropy tower does not fix source collapse). The order-seven row detects first Gram anisotropy:

$$A_7^{\mathfrak{f}} \sim H\Gamma_G.$$

The order-nine row detects the second anisotropy on the first wall:

$$A_9^{\mathfrak{f}} \sim H\Gamma_{G,2}.$$

A higher row in the same tower would have the schematic form

$$A_{2k+7}^{\mathfrak{f}} \sim H\Gamma_{G,k+1}.$$

This improves the local layered visibility against the walls

$$\Gamma_G = 0, \quad \Gamma_{G,2} = 0, \quad \dots,$$

but it leaves the source-collapse wall

$$H = yq'' = 0$$

untouched. Therefore the rank-two norm route requires either a source-robust row not divisible by H , or an external quantitative theorem preventing source collapse.

Proposition 14.77 (Source-collapse countermodel for the rank-two norm route). *The source-weighted lower bound*

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}$$

is not a consequence of the current local rank-two structure.

Indeed, consider the local jet target

$$q_0 = 2, \quad q'' = 0, \quad q^{(4)} = 0.$$

Then

$$g = \frac{q_0}{\pi} = \frac{2}{\pi} > 0,$$

and

$$h = \frac{1}{\pi} \left(\frac{1}{3} q_0^3 + \frac{1}{6} q'' \right) = \frac{8}{3\pi} > 0.$$

Moreover

$$\Gamma_G = q'' - 6q_0 + 2q_0^3 = 0 - 12 + 16 = 4,$$

while

$$\Gamma_{G,2} = (1 - q_0^2)q'' - \frac{1}{6}q^{(4)} = 0.$$

Thus the layered anisotropy vector is live:

$$\max\{|\Gamma_G|, |\Gamma_{G,2}|\} = 4.$$

However,

$$H = yq'' \sim -\frac{q_0(q'')^3}{288} = 0.$$

Perturbing the target to

$$\sup_{m \in I_\rho} |q''(m)| \leq Q^{-C}$$

while keeping

$$q_0(m) \asymp 1, \quad g(m), h(m) > 0, \quad \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \asymp 1$$

gives

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \ll Q^{-3C}.$$

Choosing $C > A/3$ defeats the desired Q^{-A} lower bound.

Remark 14.78 (Interpretation of the source-collapse countermodel). Proposition 14.77 does not assert that actual zeta realizes this jet configuration. It says that the present local rank-two row algebra, Gram positivity, and layered anisotropy structure do not prevent it. A proof must therefore supply a new input such as

$$\sup_{m \in I_\rho} |q''(m)| \geq Q^{-B},$$

or a stronger source-jet / interval nonflatness theorem for actual zeta data.

Remark 14.79 (Interval-local source collapse). The pointwise source-collapse target becomes an interval-local obstruction because the proof requires a supremum over an admissible packet interval

$$I_\rho = [m_0 - \Delta, m_0 + \Delta].$$

The scale audit gives the natural estimates

$$\Omega_{\text{eff}}(q'') \sim Q, \quad \Delta \sim Q^{-1}, \quad \Omega_{\text{eff}}(q'')\Delta \sim 1.$$

Thus q'' has only $O(1)$ effective oscillatory complexity on the natural microscopic packet interval.

Let

$$z(m) := q''(m).$$

To force

$$\sup_{|m-m_0| \leq c/Q} |z(m)| \leq Q^{-C},$$

it is enough to flatten finitely many derivatives:

$$z^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R.$$

A Taylor–Bernstein estimate gives, schematically,

$$\sup_{|u| \leq c/Q} |z(m_0 + u)| \lesssim Q^{M_0} \frac{C_c^{R+1}}{(R+1)!}.$$

Hence a safe choice is

$$R \asymp (C + M_0) \frac{\log Q}{\log \log Q},$$

or more conservatively $R = O(\log Q)$. This order is far below the recurrence-danger threshold $Q/\log Q$. Therefore the interval-local source-collapse countermodel is not defeated by interval length or bandwidth alone.

Remark 14.80 (Fixed-prime and AFE derivative-flat surrogates). A fixed-prime surrogate for the source-collapse mechanism is

$$L_P(m) = \sum_{p \leq P} (\log p) p^{-1/2} e^{-im \log p}.$$

Define

$$J_0(m) = B_0 + \Re L_P(m),$$

$$J_2(m) = B_2 + \Re L_P''(m),$$

$$J_4(m) = B_4 + \Re L_P^{(4)}(m).$$

The model residuals are

$$R_1 = J_2 - 6J_0 + 2J_0^3,$$

$$R_2 = (1 - J_0^2)J_2 - \frac{1}{6}J_4,$$

and

$$H_{\text{model}} = -\frac{J_0 J_2^3}{288}.$$

The interval-local source-collapse target is

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R,$$

with side constraints such as

$$J_0(m_0) \approx 2, \quad J_4(m_0) \approx 0.$$

At the exact target

$$(J_0, J_2, J_4) = (2, 0, 0)$$

one has

$$R_1 = 4, \quad R_2 = 0,$$

and

$$g = \frac{2}{\pi} > 0, \quad h = \frac{8}{3\pi} > 0.$$

Thus the source is collapsed while anisotropy and Gram positivity remain live.

The phase-relaxed finite-prime problem has no rank obstruction: the rows

$$J_2^{(r)}(m_0)$$

are governed by powers of the frequencies

$$\lambda_j = \log p_j \quad \text{or} \quad \lambda_j = k \log p,$$

so the derivative-jet matrix has a Vandermonde-type structure. The remaining arithmetic issue is whether the actual one-parameter prime-log flow

$$m_0 \mapsto (m_0 \log p_j)_j \pmod{2\pi}$$

recurs to such a derivative-flat target. The expected codimension is approximately

$$K = R + 3,$$

so a heuristic recurrence cost at tolerance $\varepsilon = Q^{-C}$ is

$$T_{\text{hit}} \sim Q^{CK}.$$

For $R = O(\log Q)$, this is

$$T_{\text{hit}} = \exp(O((\log Q)^2)) \ll e^Q.$$

Thus scale considerations do not presently block the fixed-prime source-collapse surrogate.

The phase-relaxed derivative-flat target can be sharpened further. In the fixed-amplitude prime model

$$L_P(m) = \sum_{p \leq P} a_p e^{-im \log p}, \quad a_p = \frac{\log p}{\sqrt{p}},$$

one obtains

$$(J_0, J_2, J_4) = \sum_{p \leq P} x_p a_p (1, -(\log p)^2, (\log p)^4), \quad x_p = \cos(m \log p).$$

Thus the phase-relaxed attainable jet values form a fixed-amplitude prime zonotope. The source-collapse target

$$(J_0, J_2, J_4) = (2, 0, 0)$$

is compatible with good-patch positivity:

$$g = \frac{2}{\pi} > 0, \quad h = \frac{8}{3\pi} > 0,$$

and has live anisotropy

$$R_1 = 4, \quad R_2 = 0.$$

Finite-dimensional feasibility tests for the moment system

$$\sum_{p \leq P} a_p x_p = 2, \quad \sum_{p \leq P} a_p (\log p)^{2k} x_p = 0$$

show interior solutions for low and moderate finite orders. For example:

moments imposed	P	$\#\{p \leq P\}$	$\min \max_{p \leq P} x_p $
0, 2, 4	100	25	0.699
0, 2, 4, 6	300	62	0.679
0, 2, 4, 6, 8	1000	168	0.637
0, 2, 4, 6, 8, 10	1000	168	0.888
0, 2, 4, 6, 8, 10, 12	5000	669	0.718.

So the target lies in the interior of the fixed-prime phase-relaxed cosine zonotope, not on a boundary. The remaining issue is therefore not amplitude feasibility. It is the one-parameter recurrence problem

$$m_0 \log p \equiv \theta_p \pmod{2\pi}$$

for selected phases θ_p .

AFE-compatible models add prime powers, smoothing, a dual sum, and gamma background terms. These shift or couple the target, but no algebraic obstruction is presently known that forces

$$\max_{0 \leq r \leq R} |J_2^{(r)}(m_0)| \geq Q^{-B}.$$

Therefore the exact missing theorem is a source-jet nonflatness statement for actual zeta, not another local row identity.

Remark 14.81 (Prime-log recurrence to derivative-flat tubes). Let

$$\lambda = (\log p_1, \dots, \log p_N), \quad \Theta(m) = m\lambda \pmod{2\pi}.$$

Let $\mathcal{T}_b(\varepsilon) \subset \mathbb{T}^N$ denote the ε -tube around the derivative-flat target

$$|J_2^{(r)}| \leq \varepsilon, \quad 0 \leq r \leq R,$$

with side constraints

$$|J_0 - 2| \leq \varepsilon, \quad |J_4| \leq \varepsilon.$$

If this target is regular, then

$$\mu(\mathcal{T}_b(\varepsilon)) \asymp \varepsilon^K, \quad K \sim R + 3.$$

This is a positive-measure analytic tube, not a prescribed point in $\mathbb{T}^N$.

For polynomial tolerance

$$\varepsilon = Q^{-C},$$

the random/equidistribution hitting scale is

$$T_{\text{hit}} \sim \varepsilon^{-K} = Q^{CK}.$$

For the derivative orders relevant to Taylor–Bernstein source flattening,

$$R \asymp (C + M_0) \frac{\log Q}{\log \log Q}$$

or even $R = O(\log Q)$, this gives

$$T_{\text{hit}} = \exp\left(O\left(\frac{(\log Q)^2}{\log \log Q}\right)\right) \quad \text{or} \quad T_{\text{hit}} = \exp(O((\log Q)^2)).$$

Both are far below the available height scale e^Q .

Thus recurrence cost does not presently block the interval-local source-collapse surrogate. A proof of the desired source-weighted lower bound would need a nonstandard quantitative nonrecurrence theorem for the prime-log flow against these derivative-flat tubes, or an actual-zeta source-jet nonflatness theorem that bypasses the recurrence model entirely.

Remark 14.82 (Zero-side same-scale source flattening). The source-collapse mechanism is not merely a prime-side artefact. On the zero side, one can seek same-scale positive packet configurations for which the aggregate source curvature is flat to high finite order.

For a centered Cauchy packet

$$k_c(t) = \frac{c}{c^2 + 4(t - m)^2},$$

the centered jets are

$$\begin{aligned} k_c(m) &= c^{-1}, \\ k_c''(m) &= -8c^{-3}, \end{aligned}$$

and

$$k_c^{(4)}(m) = 384c^{-5}.$$

A single centered packet cannot realize

$$q'' = q^{(4)} = 0$$

with $q_0 > 0$. However, once off-center same-scale packets and positive linear combinations are allowed, the problem becomes a finite moment-matching problem.

In particular, one can use same-scale packet families to impose finitely many conditions of the form

$$q^{(2)}(m_0) = q^{(3)}(m_0) = \dots = q^{(R+2)}(m_0) = 0$$

while preserving

$$q_0(m_0) > 0.$$

At the target

$$q_0 = 2, \quad q'' = 0, \quad q^{(4)} = 0,$$

Gram positivity and live anisotropy remain:

$$g = \frac{2}{\pi} > 0, \quad h = \frac{8}{3\pi} > 0, \quad \Gamma_G = 4.$$

Thus same-scale packet geometry alone does not provide a source-jet nonflatness theorem. A proof must use information not already contained in finite same-scale positive packet geometry.

Theorem 14.83 (Continuous same-scale source-flattening model). *Fix $d > 0$. For*

$$c = d \sin \theta, \quad x = d \cos \theta, \quad 0 < \theta < \pi,$$

consider the same-distance packet kernel

$$K_{\theta,d}(u) = \frac{c}{c^2 + (x + 2u)^2}.$$

Let

$$d\mu(\theta) = \frac{2}{\pi} \sin \theta \, d\theta.$$

Then, for $|2u/d| < 1$,

$$\boxed{\int_0^\pi K_{\theta,d}(u) \, d\mu(\theta) = \frac{1}{d}.$$

Consequently,

$$q_\mu(u) := \int_0^\pi K_{\theta,d}(u) \, d\mu(\theta)$$

satisfies

$$q_\mu(0) = \frac{1}{d} > 0, \quad q_\mu^{(r)}(0) = 0 \quad (r \geq 1).$$

In particular,

$$q_\mu^{(2)}(0) = q_\mu^{(4)}(0) = q_\mu^{(6)}(0) = \cdots = 0.$$

Proof. Put

$$t = \frac{2u}{d}.$$

Then

$$K_{\theta,d}(u) = \frac{1}{d} \frac{\sin \theta}{1 + 2t \cos \theta + t^2}.$$

Hence

$$\int_0^\pi K_{\theta,d}(u) \, d\mu(\theta) = \frac{1}{d} \int_0^\pi \frac{2}{\pi} \frac{\sin^2 \theta}{1 + 2t \cos \theta + t^2} \, d\theta.$$

For $|t| < 1$, the elementary Poisson-kernel identity

$$\int_0^\pi \frac{2}{\pi} \frac{\sin^2 \theta}{1 + 2t \cos \theta + t^2} \, d\theta = 1$$

gives the claim. Therefore $q_\mu(u) = 1/d$ on a neighborhood of the origin, and all positive-order derivatives at the origin vanish. $\square$

Corollary 14.84 (Finite same-scale source flattening to arbitrary finite jet order). *For every $M \geq 1$ and every $\epsilon > 0$, there exist finitely many angles*

$$0 < \theta_1, \dots, \theta_N < \pi$$

and positive weights $w_i > 0$ such that, with

$$c_i = d \sin \theta_i, \quad x_i = d \cos \theta_i,$$

one has

$$x_i^2 + c_i^2 = d^2$$

for every i , and

$$\left| \sum_i w_i K_{\theta_i,d}^{(r)}(0) \right| \leq \epsilon, \quad 1 \leq r \leq M,$$

while

$$\sum_i w_i K_{\theta_i, d}(0) > 0.$$

Equivalently, finite positive same-distance packet clusters can make

$$q^{(r)}(0), \quad 1 \leq r \leq M,$$

arbitrarily small while retaining

$$q_0 = q(0) > 0.$$

In particular, taking $M \geq R + 2$ gives the finite source-flatness target

$$(q'')^{(r)}(0) \approx 0, \quad 0 \leq r \leq R.$$

Proof. The continuous measure μ in Theorem 14.83 satisfies

$$q_\mu(u) = \int K_{\theta, d}(u) d\mu(\theta) = \frac{1}{d}$$

near $u = 0$. Hence all derivatives $q_\mu^{(r)}(0)$, $r \geq 1$, vanish.

Apply Tchakaloff's theorem, or finite-dimensional Carathéodory reduction, to the positive measure μ and the finite list of continuous functions

$$K_{\theta, d}(0), \quad K_{\theta, d}^{(1)}(0), \quad K_{\theta, d}^{(2)}(0), \dots, K_{\theta, d}^{(M)}(0).$$

This gives a finite positive quadrature rule reproducing the required finite moment vector on the closed support. If an atom occurs at $\theta = 0$ or $\theta = \pi$, replace μ by its restriction to

$$[\varepsilon, \pi - \varepsilon]$$

and renormalize. Since

$$d\mu(\theta) = \frac{2}{\pi} \sin \theta d\theta$$

has vanishing endpoint mass, the corresponding finite moment vector converges to the original one as $\varepsilon \downarrow 0$. Applying finite quadrature on $[\varepsilon, \pi - \varepsilon]$ and choosing ε sufficiently small gives interior atoms $0 < \theta_i < \pi$ and the stated finite-order approximation. This approximation is the form needed below for interval source-collapse, where the target tolerance is polynomial in Q^{-1} . $\square$

Remark 14.85 (Uniform same-scale caveat). The construction above is same-scale in the strong fixed-distance sense $x_i^2 + c_i^2 = d^2$. If a later argument requires the additional uniform angular condition

$$\epsilon \leq \frac{c_i}{d} \leq 1 - \epsilon$$

with $\epsilon > 0$ independent of M , then one still needs a strengthened annular positive-cone lemma. The present result is already enough to show that finite source-jet flattening is not excluded by positive same-distance packet geometry. Thus any proof-closing lower bound for q'' must use additional actual-zeta arithmetic, not merely same-scale positivity.

Remark 14.86 (Source-kernel versus actual-zeta packet caveat). The same-scale flattening construction is a statement about positive local Cauchy source kernels at a fixed distance scale. It should not be read as a claim that actual zeta zeros can realize the same quadrature configuration. Actual zeta imposes functional-equation symmetry, conjugation, discreteness, zero-counting constraints, and possibly Euler-product arithmetic not present in the continuous source-kernel model.

The point of Theorem 14.83 and Corollary 14.84 is therefore negative but precise: local same-scale positivity does not by itself force source-curvature nonflatness. Any proof of

$$\sup_{m \in I_\rho} |q''(m)| \geq Q^{-B}$$

must use actual-zeta arithmetic or a stronger packet-realizability theorem, not merely same-scale positive kernel geometry.

Corollary 14.87 (Interval-local source collapse from finite quadrature). *Fix $A > 0$. Assume the packet scale is microscopic,*

$$d \asymp Q^{-1},$$

and let

$$I = [m_0 - c/Q, m_0 + c/Q]$$

with $c > 0$ fixed sufficiently small. For every $C > A/3$, a finite same-distance positive packet cluster can be chosen so that

$$\sup_{m \in I} |q''(m)| \leq Q^{-C},$$

while

$$q_0(m) \asymp 1, \quad g(m) > 0, \quad h(m) > 0,$$

and

$$\max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \asymp 1$$

on I , after a harmless normalization of the total weight.

Consequently, at leading source order,

$$H = yq'' \sim -\frac{q_0(q'')^3}{288}$$

satisfies

$$\sup_{m \in I} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \ll Q^{-3C},$$

and therefore can be made smaller than Q^{-A} .

Proof. By Corollary 14.84, choose the packet cluster so that, for a prescribed polynomial tolerance,

$$q^{(r)}(m_0) = O(Q^{-C'}) \quad 1 \leq r \leq M,$$

with M large enough relative to the desired interval bound. In particular,

$$(q'')^{(r)}(m_0) = O(Q^{-C'}) \quad 0 \leq r \leq R$$

for the required finite order R .

The standard Taylor–Bernstein estimate at scale $d \asymp Q^{-1}$ then gives

$$\sup_I |q''| \leq Q^{-C}$$

after taking C' and M sufficiently large. For example, the usual Taylor bookkeeping permits

$$M \asymp \frac{C \log Q}{\log \log Q}$$

or the safer choice $M = O(\log Q)$, with constants depending on the fixed interval parameter and same-scale annulus. Normalizing the total mass so that q_0 is close to 2, one obtains

$$g = \frac{q_0}{\pi} > 0, \quad h = \frac{1}{\pi} \left(\frac{1}{3} q_0^3 + \frac{1}{6} q'' \right) > 0,$$

and

$$\Gamma_G = q'' - 6q_0 + 2q_0^3 = 4 + O(Q^{-C}).$$

Hence the anisotropy factor remains bounded below while the source factor collapses cubically in q'' .

This is a finite-order local obstruction model. It shows that source-collapse is compatible with positive same-scale packet geometry to every prescribed polynomial accuracy. It is not a claim that actual zeta zeros realize such a quadrature configuration. $\square$

Proposition 14.88 (No-go for the current rank-two norm route). *The current rank-two norm route cannot close RH from the established local row theory and same-scale positive packet geometry alone.*

More precisely, the following data are compatible with the current local rank-two structure:

$$q_0 \asymp 1, \quad g, h > 0, \quad \max\{|\Gamma_G|, |\Gamma_{G,2}|\} \asymp 1,$$

and

$$\sup_{m \in I_\rho} |q''(m)| \leq Q^{-C}.$$

For such data,

$$H = yq'' \sim -\frac{q_0(q'')^3}{288}$$

satisfies

$$\sup_{m \in I_\rho} |H(m)| \ll Q^{-3C}.$$

Consequently,

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \ll Q^{-3C}.$$

Taking $C > A/3$ violates the desired Q^{-A} source-weighted lower bound.

Thus the local rank-two norm route requires a new theorem excluding such source-flat intervals for actual zeta data. By Corollary 14.87, the source-flat interval model is compatible with finite positive same-distance packet quadrature to arbitrary finite order. Therefore the missing input cannot be a finite local packet-geometry lemma; it must use actual zeta arithmetic, a source-robust row, or a genuinely global transport/projective constraint.

This proposition is not a counterexample to RH or to actual zeta; it is a no-go for closing the present rank-two norm route from local packet geometry alone.

Remark 14.89 (What would be needed to save the norm route). A sufficient new input would be an actual-zeta source-jet nonflatness theorem, for example

$$\sup_{m \in I_\rho} |q''(m)| \geq Q^{-B}.$$

A finite-jet version would be

$$\max_{0 \leq r \leq R(Q)} |(q'')^{(r)}(m_0)| \geq Q^{-B}.$$

The currently available local framework does not prove either statement. The distinguished packet contributes nontrivially to q'' , but rest/core packets can cancel that contribution. Same-scale selection, strict core gap, endpoint separation, Gram positivity, finite Hadamard symmetry, and the existing fixed-sector row algebra do not prevent such cancellation. Therefore source-jet nonflatness is a genuinely new arithmetic input, not a cleanup lemma.

Remark 14.90 (No-go ledger after the source-collapse push). The current push rules out several tempting repairs of the source-collapse obstruction.

(1) *Higher anisotropy rows.* The tower

$$X_1 Y X_{2k} - X_{2k} Y X_1$$

is divisible by $H = yq''$. Higher order can detect higher Gram anisotropies, but it does not remove source collapse.

(2) *Ambient D-sector at order seven.* At $g = h$, the ambient row

$$T_7^{(1)} = -2yBpD$$

survives the fixed-sector Gram-isotropy failure, but still contains

$$yB \sim yq''.$$

It therefore does not solve Regime B source collapse.

(3) *Transported derivatives.* If

$$A_j = HG_j,$$

then

$$\nabla A_j = (\nabla H)G_j + H\nabla G_j.$$

Since

$$H \sim -\frac{q_0(q'')^3}{288},$$

one has

$$\nabla H = O((q'')^2)$$

under good-patch derivative control. Thus finitely many transported derivatives do not presently defeat an interval where q'' and finitely many of its derivatives are small.

(4) *Projectivization.* The source-normalized projective ratio

$$[A_7 : A_9] = [\Gamma_G : \Gamma_{G,2}]$$

formally cancels H , but it loses norm-scale information, is singular at

$$\Gamma_G = \Gamma_{G,2} = 0,$$

and requires correction-row control plus a new global projective index or transport theorem. No such theorem is presently known.

(5) *Correction rows.* Line, background/core, and normalization/subtraction rows are not currently certified as source-robust off-line packet detectors. They are either gauge/baseline/normalization data or must be carried as enlarged projective coordinates. They do not provide a clean replacement for the source-divisible rows.

(6) *Explicit formula / prime-side coercivity.* The explicit formula sees the linear ingredients

$$q_0, \quad q'', \quad q^{(4)},$$

but the residual

$$(yq'')^2(\Gamma_G^2 + \Gamma_{G,2}^2)$$

is nonlinear. Tensorizing the explicit formula produces oscillatory multi-prime sums with no identified positivity or coercive lower bound.

Accordingly, the current route is non-closing unless one proves either a source-robust row theorem, an actual-zeta source-jet nonflatness theorem, or a genuinely global arithmetic/projective/transport index theorem.

Remark 14.91 (Status ledger for candidate routes). The source-collapse push leaves the following status for the main candidate routes considered so far.

(1) *Order eleven and the finite anisotropy tower.* A further local row computation may identify the next anisotropy factor $\Gamma_{G,3}$ and may be useful for documentation. However, within the established tower

$$X_1 Y X_{2k} - X_{2k} Y X_1, \quad Y = yJ, \quad X_1 = BS, \quad B \propto q'',$$

the fixed-sector row is always divisible by

$$H = yq''.$$

Thus an order-eleven row of the same kind would at best have the schematic form

$$A_{11}^f \sim H\Gamma_{G,3}.$$

It may help with further anisotropy walls, but it does not address the dominant source-collapse regime $H \approx 0$. Order eleven is therefore diagnostic / documentary, not presently proof-closing.

- (2) *Norm-based rank-two finite-core route.* The known rows have the form

$$A_7^\dagger \sim H\Gamma_G, \quad A_9^\dagger \sim H\Gamma_{G,2}.$$

Since

$$H \sim -\frac{q_0(q'')^3}{288},$$

an interval on which q'' is small collapses the norm vector even when Γ_G remains bounded away from zero. This route is blocked unless one proves a source-jet nonflatness theorem or finds a source-robust row.

- (3) *Transported derivative tower.* Differentiation gives

$$\nabla A_j = (\nabla H)G_j + H\nabla G_j.$$

Near $q'' = 0$,

$$\nabla H = O((q'')^2)$$

under good-patch derivative control. Thus first derivatives still collapse. For every fixed finite derivative order, finite local models can plausibly flatten $H, \nabla H, \dots, \nabla^R H$. A derivative-tower rescue would require a genuinely new growing-order source-jet theorem.

- (4) *Projectivization / source-normalized ratios.* The ratio

$$[A_7 : A_9] = [\Gamma_G : \Gamma_{G,2}]$$

formally cancels H in the idealized factorized model. However, this loses magnitude, remains singular at

$$\Gamma_G = \Gamma_{G,2} = 0,$$

and is not cleanly defined for the corrected row vector unless correction rows vanish or are H -divisible. No global projective index, endpoint, winding, or holonomy theorem is presently available. The projective branch is therefore diagnostic only.

- (5) *Correction rows as source-robust detectors.* The current audits did not identify a correction row that is simultaneously non- H -divisible, packet-visible, and absent in RH-compatible/on-line models. Line rows are gauge/reference-line terms, background/core rows are baseline data, and normalization/subtraction rows are convention-dependent unless separately packet-isolated. Correction rows therefore do not currently provide the missing source-robust detector.

- (6) *Explicit formula / prime-side coercivity.* The explicit formula represents the linear ingredients

$$q_0, \quad q'', \quad q^{(4)},$$

but the source-weighted residual

$$(yq'')^2(\Gamma_G^2 + \Gamma_{G,2}^2)$$

is nonlinear. Tensorizing the explicit formula produces oscillatory multi-prime sums with no identified positivity or coercive lower bound. Thus the explicit formula is representational here, not currently coercive.

- (7) *Fixed-prime and AFE-style adversaries.* Fixed-prime and AFE-compatible surrogate models support the interval-local source-collapse mechanism. In such models one attempts to force

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R,$$

with $J_2 \sim q''$, while maintaining

$$J_0 \asymp 1, \quad g, h > 0, \quad \Gamma_G \asymp 1.$$

The AFE main sum, dual sum, smoothing, and gamma/background terms shift or couple the finite target, but no algebraic blocker is currently known.

- (8) *Same-scale packet geometry.* The continuous same-scale source-flattening model and its finite quadrature corollary show that positive same-distance packet geometry can make finite source jets vanish while preserving $q_0 > 0$, Gram positivity, and live Γ_G . Therefore source-collapse cannot be excluded by local same-scale packet geometry alone.

- (9) *Interval L^2 / source-curvature observability.* The sharpest remaining positive target is not the full nonlinear residual but the source-curvature estimate

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|,$$

or at least

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \geq Q^{-B}.$$

This is the least nonlinear form of the missing arithmetic input.

- (10) *Source-anisotropy reintegration.* Even if the interval source-curvature estimate

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|$$

is proved, it does not by itself imply the source-weighted rank-two lower bound. One still needs to exclude concentration of q''_{can} -mass near

$$\Gamma_G = 0, \quad \Gamma_{G,2} = 0.$$

Equivalently, the architecture needs the vector estimate

$$\int_{I_\rho} |q''_{\text{can}}|^2 (\Gamma_G^2 + \Gamma_{G,2}^2) dm \geq Q^{-A} |I_\rho|.$$

The identity

$$\Gamma_{G,2} = 2q(q')^2 - \frac{1}{6}\Gamma_G''$$

turns this into a local analytic/ODE rigidity problem coupled to the arithmetic source-observability problem. This bridge is currently a missing theorem, not a consequence of source observability alone.

- (11) *Global trace / projective index / Connes-style imbalance.* A global trace, projective index, or spectral accounting theorem is not ruled out. However, it would be a new global proof architecture, not a continuation of the present local rank-two norm route.

Thus the current proof state is: the local rank-two machinery is a precise obstruction and diagnostic framework, but it is non-closing unless one supplies actual-zeta source-curvature observability, a genuinely source-robust row, or a new global arithmetic/projective/trace theorem.

Remark 14.92 (Corrected projective transport obstruction). One possible response to source divisibility is to pass to the projective ratio. In the idealized factorized model

$$A_7 = HG_1, \quad A_9 = HG_2, \quad H = yq'',$$

one has

$$[A_7 : A_9] = [G_1 : G_2].$$

In the present setting G_1 and G_2 are represented by

$$G_1 \sim \Gamma_G, \quad G_2 \sim \Gamma_{G,2}.$$

Thus projectivization formally cancels the common source factor H .

However, the actual corrected rows may have the form

$$A_7 = HG_1 + R_7, \quad A_9 = HG_2 + R_9,$$

where R_7, R_9 denote line, background/core, normalization, or other correction-row contributions not known to be H -divisible. On the chart $A_7 \neq 0$, put

$$Z = \frac{A_9}{A_7}.$$

Then

$$\nabla Z = \frac{A_7 \nabla A_9 - A_9 \nabla A_7}{A_7^2},$$

and substitution gives

$$\begin{aligned} A_7 \nabla A_9 - A_9 \nabla A_7 &= H^2(G_1 \nabla G_2 - G_2 \nabla G_1) \\ &\quad + H(G_1 \nabla R_9 - G_2 \nabla R_7) \\ &\quad + H(R_7 \nabla G_2 - R_9 \nabla G_1) \\ &\quad + (\nabla H)(R_7 G_2 - R_9 G_1) \\ &\quad + (R_7 \nabla R_9 - R_9 \nabla R_7). \end{aligned}$$

Therefore the clean source-normalized identity

$$\nabla \left(\frac{A_9}{A_7} \right) = \nabla \left(\frac{G_2}{G_1} \right)$$

is valid only if the correction rows vanish or are themselves H -divisible. If non- H -divisible corrections remain, then near $H = 0$

$$[A_7 : A_9] \rightarrow [R_7 : R_9],$$

not

$$[\Gamma_G : \Gamma_{G,2}].$$

Thus the projective branch is coherent only after a correction-divisibility or correction-absorption theorem, or after enlarging the projective coordinate to

$$[A_7 : A_9 : R_7 : R_9 : \dots].$$

No global projective endpoint, holonomy, or index theorem is presently known for that enlarged object.

14.24. Finite-core sampled vector non-concentration.

Remark 14.93 (Updated role of finite-core non-concentration). The finite-core non-concentration problem below is now secondary to the source-collapse obstruction. The known fixed-sector rows are source-divisible, so a lower bound for a transported finite-core vector cannot be proved from anisotropy non-concentration alone. Any proof-relevant finite-core theorem must either include a source-jet nonflatness input such as

$$\sup_{m \in I_\rho} |q''(m)| \geq Q^{-B},$$

or use a genuinely source-robust row/invariant. Without that additional source-level input, the aligned/non-concentration language below remains a diagnostic framework rather than a proof-closing theorem.

The notation in this subsection originally arose in the scalar Φ_K -sampled formulation. In the current rank-two architecture, the corresponding fixed-sector vector is

$$C_{\text{fd}}^{\vec{f}}(\theta) = (A_7^{\mathfrak{f}}, A_9^{\mathfrak{f}}, \dots, A_{2R_{\Theta_k}+1}^{\mathfrak{f}})(\theta).$$

The old scalar vector

$$C_{\text{fd}}(\theta) = (c_{\text{core},1}, c_{\text{core},3}, \dots, c_{\text{core},2R_{\Theta_k}+1})(\theta)$$

should be regarded as its Φ_K -readout, not as the full fixed-sector object. The rank-two non-concentration theorem needed by the present architecture is therefore a lower bound for

$$P_{\text{fd}}^{\vec{f}}(\theta) + J_{\text{fd}}^{\vec{f}}(\theta),$$

or for a transported enhancement of this vector, rather than merely for the scalar Φ_K -projection.

Let Θ_k be a compact nonsingular actual-source k -core stratum. Define the finite-determinacy vector

$$C_{\text{fd}}(\theta) = (c_{\text{core},1}, c_{\text{core},3}, \dots, c_{\text{core},2R_{\Theta_k}+1})(\theta).$$

Decompose it as

$$C_{\text{fd}}(\theta) = P_{\text{fd}}(\theta) + J_{\text{fd}}(\theta),$$

where P_{fd} is the pair-like or one-body finite vector and

$$J_{\text{fd}} = (J_1, J_3, \dots, J_{2R_{\Theta_k}+1}), \quad J_{2r+1} = [s^{2r+1}] \Phi_K(\mathcal{I}_{\geq 2}),$$

is the finite interaction vector.

For $1 \leq N \leq R_{\Theta_k} + 1$, define

$$F_N(\theta, Q) = c_{\text{core}, 2N-1}(\theta) + \sum_{j \geq 1} c_{\text{core}, 2N-1+2j}(\theta) h_j(1^2, \dots, N^2) Q^{-2j}.$$

Set

$$\Lambda(m, Q) = C_* \max_{1 \leq N \leq R_{\Theta_k}+1} \left(\frac{Q^{2N+2}}{(2N-1)N!(N-1)!} \frac{x}{B_\zeta(m)} S(m) \right).$$

Hypothesis 14.94 (Bounded-depth projective detection). On the compact nonsingular actual-source stratum Θ_k , there exist constants

$$Q_0(\Theta_k), \quad c_0 > 0$$

such that for every $Q \geq Q_0(\Theta_k)$ and every $\theta \notin \mathcal{Z}_k$, there is an index

$$1 \leq N \leq R_{\Theta_k} + 1$$

with

$$|F_N(\theta, Q)| \geq c_0 \|C_{\text{fd}}(\theta)\|_\infty.$$

This is the bounded-depth projective detection and Q^{-2} -tail-control input. It is separate from sampled vector non-concentration: it detects the finite-determinacy vector once that vector has calibrated size, but it does not prove that calibrated size.

Hypothesis 14.95 (Sampled finite-core vector non-concentration). For every actual zeta finite-core parameter

$$\theta(m) \in \Theta_k,$$

one needs the following alternative:

$$\theta(m) \in \mathcal{Z}_k,$$

or

$$\theta(m) \in \mathcal{E}_{\text{geom}},$$

or

$$\|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_\infty \geq \Lambda(m, Q).$$

Definition 14.96 (Interaction-aligned finite-core branch). Define

$$\mathcal{N}_k^{\text{align}}(m, Q)$$

to be the set of actual zeta finite-core parameters

$$\theta(m) \in \Theta_k \setminus (\mathcal{Z}_k \cup \mathcal{E}_{\text{geom}})$$

such that

$$0 < \|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_\infty < \Lambda(m, Q).$$

Equivalently, Hypothesis 14.95 asserts

$$\mathcal{N}_k^{\text{align}}(m, Q) = \emptyset.$$

Remark 14.97 (Relation among finite-core bad branches). The branch

$$\mathcal{N}_k^{\text{align}}(m, Q)$$

is the current sharp form of the finite-core obstruction. Earlier formulations used either a distance-to- $\mathcal{Z}_k$ tube

$$\mathcal{N}_k(m, Q),$$

or a coefficient-small branch

$$\mathcal{N}_k^{\text{coef}}(m, Q) = \left\{ \theta(m) \notin \mathcal{Z}_k \cup \mathcal{E}_{\text{geom}} : \max_N |F_N(\theta(m), Q)| < \Lambda(m, Q) \right\}.$$

For $Q \geq Q_0(\Theta_k)$, bounded-depth projective detection and Q^{-2} -tail control identify the coefficient-small branch with radial smallness of the finite-determinacy vector, up to fixed compact-patch constants:

$$0 < \|C_{\text{fd}}(\theta(m))\|_{\infty} < \Lambda(m, Q).$$

Since

$$C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}},$$

this is exactly the interaction-aligned branch

$$\mathcal{N}_k^{\text{align}}(m, Q).$$

Accordingly, subsequent finite-core arguments should target $\mathcal{N}_k^{\text{align}}$, not repeat finite determinacy or projective detection.

Lemma 14.98 (Finite-core lower transfer from sampled vector non-concentration). *Assume Hypothesis 14.95 and Hypothesis 14.94. Then every actual zeta finite-core sample*

$$\theta(m) \notin \mathcal{Z}_k \cup \mathcal{E}_{\text{geom}}$$

has some $1 \leq N \leq R_{\Theta_k} + 1$ satisfying

$$|\Xi_{\text{core}}^{(N)}(m)| \geq c \frac{x}{B_{\zeta}(m)} S(m).$$

Proof. Choose N by projective detection. Since

$$C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}},$$

Hypothesis 14.95 gives

$$|F_N(\theta(m), Q)| \geq c_0 \Lambda(m, Q).$$

By the definition of $\Lambda(m, Q)$,

$$\Lambda(m, Q) \geq C_* \frac{Q^{2N+2}}{(2N-1)N!(N-1)!} \frac{x}{B_{\zeta}(m)} S(m).$$

Substituting this into

$$\Xi_{\text{core}}^{(N)}(m) = (-1)^{N+1} (2N-1)N!(N-1)! Q^{-(2N+2)} F_N(\theta(m), Q)$$

gives the asserted lower bound. $\square$

Remark 14.99 (Why sampled vector non-concentration is the exact finite-core dependency). Lemma 14.98 separates the finite-core problem into a qualitative part and a sampled-size part. The qualitative part is finite odd-germ determinacy: on a compact nonsingular actual-source stratum,

$$C_{\text{fd}}(\theta) = 0 \iff H_{\text{core}, \theta} \equiv 0.$$

Together with bounded-depth Q^{-2} -tail control, this gives projective detection of every nonzero finite-core germ by some

$$1 \leq N \leq R_{\Theta_k} + 1.$$

What remains is the sampled finite-vector lower bound

$$\|C_{\text{fd}}(\theta(m))\|_{\infty} = \|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_{\infty} \geq \Lambda(m, Q)$$

outside

$$\mathcal{Z}_k \cup \mathcal{E}_{\text{geom}}.$$

Equivalently, the remaining finite-core branch is the interaction-aligned small-vector branch

$$\mathcal{N}_k^{\text{align}}(m, Q).$$

Thus any further finite-core proof must either prove Hypothesis 14.95, or add an independent theorem forcing $P_{\text{fd}} + J_{\text{fd}}$ -smallness into a controlled incidence stratum.

Remark 14.100 (Status of the finite-core branch). Finite odd-germ determinacy supplies qualitative detection on compact nonsingular strata. It does not supply the sampled vector lower bound in Hypothesis 14.95. The live finite-core obstruction is calibrated interaction alignment:

$$J_{\text{fd}}(\theta(m)) \approx -P_{\text{fd}}(\theta(m)),$$

making $C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}}$ nonzero but smaller than the N -point scale $\Lambda(m, Q)$, without entering any currently listed controlled incidence stratum.

14.25. dd -interaction feature-balance obstruction. In the covariance-weighted dd -interaction model, write

$$K_{ij}^{dd} = (u_i - u_j)^2 \langle v_i, v_j \rangle, \quad v_i = v(u_i).$$

For balanced amplitudes

$$\sum_i a_i = 0,$$

the associated quadratic form is

$$Q_{dd}(a) = 2 \left\langle \sum_i a_i u_i^2 v_i, \sum_j a_j v_j \right\rangle - 2 \left| \sum_i a_i u_i v_i \right|^2.$$

The second term is one-sided. The first term has no fixed sign unless one has feature balance:

$$\sum_i a_i v_i = 0.$$

Ordinary amplitude balance does not imply feature balance. Even rank-one locking

$$v_i = \lambda_i v_0$$

only changes the missing condition to

$$\sum_i a_i \lambda_i = 0.$$

Lemma 14.101 (Feature balance is stronger than amplitude balance). *Suppose that, for a fixed finite core,*

$$\sum_i a_i v_i = 0$$

for every balanced amplitude vector

$$\sum_i a_i = 0.$$

Then

$$v_1 = v_2 = \cdots = v_k.$$

Thus ordinary amplitude balance implies feature balance only when the transverse dd -features are constant on the core.

Proof. Take $a = e_i - e_j$. Then $\sum_\ell a_\ell = 0$, so feature balance gives $v_i - v_j = 0$. Since i, j are arbitrary, all v_i are equal. $\square$

Remark 14.102 (Rank-one locking does not close feature balance). If

$$v_i = \lambda_i v_0,$$

then feature balance becomes

$$\sum_i a_i \lambda_i = 0.$$

This is not implied by

$$\sum_i a_i = 0$$

unless the weights λ_i are constant on the core. Thus rank-one transverse locking merely transfers the missing condition from the vector features v_i to the scalar weights λ_i .

Definition 14.103 (*dd-interaction branch*). The centered *dd*-interaction branch is

$$\mathfrak{Z}_{\text{int}}^{dd} = \left\{ \sum_{\nu} a_{\nu} K(d_{\nu}) + \sum_{i,j} a_i a_j (u_i - u_j)^2 \langle v_i, v_j \rangle = 0 \right\} \setminus \mathcal{E}_{\text{geom}}.$$

Definition 14.104 (*Transported dd-kernel Taylor coefficients*). Assume the favorable transported-feature reduction

$$K_{dd}(t; m) = t^2 \left\langle v\left(m + \frac{t}{2}\right), v\left(m - \frac{t}{2}\right) \right\rangle.$$

Write

$$v(m + h) = v_0 + h v_1 + \frac{h^2}{2} v_2 + O(h^3).$$

Then

$$K_{dd}(t; m) = \kappa_2 t^2 + \kappa_4 t^4 + O(t^6),$$

where

$$\kappa_2 = \langle v_0, v_0 \rangle,$$

and

$$\kappa_4 = \frac{1}{4} (\langle v_0, v_2 \rangle - \langle v_1, v_1 \rangle).$$

Remark 14.105 (*First Taylor obstruction to dd-kernel conditional definiteness*). A positive spectral representation of the form

$$\text{Re}(e^{-i\theta_0} K_{dd}(t; m)) = \alpha(m) t^2 + \int_{\mathbb{R}} (1 - \cos(t\xi)) d\mu_m(\xi), \quad \mu_m \geq 0,$$

or its negative, imposes sign restrictions on the Taylor coefficients. In particular, if the phase is chosen so that

$$\text{Re}(e^{-i\theta_0} \kappa_2) > 0,$$

then the first nonquadratic coefficient must satisfy

$$\text{Re}(e^{-i\theta_0} \kappa_4) \leq 0.$$

By Definition 14.104, this requires

$$\text{Re}\left(e^{-i\theta_0} (\langle v_0, v_2 \rangle - \langle v_1, v_1 \rangle)\right) \leq 0.$$

The current one-pair incidence package does not prove this transverse feature-curvature inequality, nor the higher moment/Hankel positivity conditions needed for a full positive spectral representation.

Proposition 14.106 (*Unit transported feature curves need not be spectrally positive*). *Let*

$$v(u) = (\cos \phi(u), \sin \phi(u))$$

be a unit transported transverse feature curve, and set

$$K_{dd}(t) = t^2 \left\langle v\left(\frac{t}{2}\right), v\left(-\frac{t}{2}\right) \right\rangle.$$

If

$$\phi(u) = au + bu^3,$$

then

$$K_{dd}(t) = t^2 - \frac{a^2}{2} t^4 + \left(\frac{a^4}{24} - \frac{ab}{4} \right) t^6 + O(t^8).$$

Consequently, even unit normalization of the transverse feature curve does not imply the coefficient signs required by a conditionally one-sided spectral representation. In the negative-type convention, after $\kappa_2 > 0$ and $\kappa_4 \leq 0$, one still needs

$$\kappa_6 = \frac{a^4}{24} - \frac{ab}{4} \geq 0,$$

which fails whenever

$$b > \frac{a^3}{6}.$$

Proof. Since

$$\left\langle v\left(\frac{t}{2}\right), v\left(-\frac{t}{2}\right) \right\rangle = \cos\left(\phi\left(\frac{t}{2}\right) - \phi\left(-\frac{t}{2}\right)\right),$$

and

$$\phi\left(\frac{t}{2}\right) - \phi\left(-\frac{t}{2}\right) = at + \frac{b}{4}t^3,$$

we have

$$K_{dd}(t) = t^2 \cos\left(at + \frac{b}{4}t^3\right).$$

Expanding the cosine through order t^6 gives the stated formula. $\square$

Remark 14.107 (Needed input for the dd -interaction branch). On each compact nonsingular same-scale finite-core patch away from the controlled degeneracy set, let $v(u)$ be the transverse dd -source feature after same-side whitening normalization. To close the dd -interaction branch, one needs at least one of the following:

(1) *Feature balance*:

$$\sum_i a_i v(u_i) = 0$$

for every admissible balanced finite-core amplitude vector

$$\sum_i a_i = 0.$$

By Lemma 14.101, this is much stronger than ordinary amplitude balance.

(2) *Conditional spectral rigidity*: the reduced kernel

$$K_{dd}(t; m) = t^2 \left\langle v\left(m + \frac{t}{2}\right), v\left(m - \frac{t}{2}\right) \right\rangle$$

admits, after phase rotation, a conditionally one-sided spectral representation, or its negative:

$$\operatorname{Re}(e^{-i\theta_0} K_{dd}(t; m)) = \alpha(m)t^2 + \int_{\mathbb{R}} (1 - \cos(t\xi)) d\mu_m(\xi), \quad \alpha(m) \geq 0, \quad \mu_m \geq 0,$$

or the negative of such a kernel. At minimum this route requires the Taylor coefficient sign condition of Remark 14.105. Proposition 14.106 shows that unit normalization of $v(u)$ alone is insufficient.

(3) *Incidence containment*: all feature-balance failures and centered kernel signature-changing loci lie in the controlled incidence set $\mathcal{E}_{\text{geom}}$.

Absent one of these inputs, the branch

$$\mathfrak{Z}_{\text{int}}^{dd}$$

remains independent. In particular, pair-like positivity of the one-body kernel is not by itself a centered-kernel spectral theorem for the covariance-weighted dd -interaction.

14.26. The remaining global gate: quantitative transported non-concentration. The reduced rank-two construction should be regarded as local infrastructure. Under the reach, atlas, and row-table hypotheses, it gives a visibility-positive fixed-sector bundle

$$\mathcal{F}_{\text{red}}^+ \rightarrow \mathcal{B}_{\text{red}}^+$$

with connection

$$\nabla_X^{\mathfrak{f}} A = \Pi_{\mathfrak{f}}(d_X^{\mathcal{J}} A),$$

and a local affine forcing equation

$$\nabla_{\kappa}^{\mathfrak{f}} A_7^{\mathfrak{f}} = E_{\eta\eta} + E_{\text{lower}}.$$

This does not by itself imply RH.

The remaining global problem is quantitative. For a transported finite-core vector

$$C_{\text{fd}}^{\text{tr}} = (A_7^{\text{f}}, \nabla^{\text{f}} A_7^{\text{f}}, A_9^{\text{f}}, \nabla^{\text{f}} A_9^{\text{f}}, \dots),$$

one must prove a lower bound of the form

$$\|C_{\text{fd}}^{\text{tr}}(\theta(m))\|_{\infty} \geq Q^{-B}$$

outside controlled transported incidence.

A qualitative nonvanishing statement is insufficient. Indeed, in the pair/interaction decomposition

$$C_{\text{fd}}^{\text{tr}} = P_{\text{fd}}^{\text{tr}} + J_{\text{fd}}^{\text{tr}},$$

the aligned branch

$$J_{\text{fd}}^{\text{tr}} \approx -P_{\text{fd}}^{\text{tr}}$$

is not excluded by the local rank-two formalism. Equivalently, one needs a quantitative incidence-distance theorem

$$\text{dist}(\theta(m), \mathcal{Z}_k^{\text{tr}} \cup \mathcal{E}_{\text{geom}}^{\text{tr}}) \geq Q^{-A},$$

A more concrete form of the same obstruction is the source-weighted layered lower bound

$$\sup_{m \in I_{\rho}} |y(m)q''(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}.$$

The source-collapse countermodels above show that this estimate cannot be deduced from local rank-two row algebra, finite Hadamard/quartet symmetry, same-scale packet geometry, Gram positivity, or finite-prime/AFE surrogate structure alone. It is now an explicit arithmetic/source-jet nonflatness hypothesis for actual zeta data.

Alternatively, a zeta-specific rigidity, cone-separation, or arithmetic non-concentration theorem implying such a bound would suffice.

Remark 14.108 (Numerical fixed-prime and AFE-style evidence). Fixed-prime and AFE-style surrogate experiments support the source-collapse mechanism. In these models one defines

$$L_P(m) = \sum_{p \leq P} (\log p) p^{-1/2} e^{-im \log p},$$

$$J_0 = B_0 + \Re L_P, \quad J_2 = B_2 + \Re L_P'', \quad J_4 = B_4 + \Re L_P^{(4)},$$

and

$$H_{\text{model}} = -\frac{J_0 J_2^3}{288}.$$

The interval-local source-collapse search minimizes

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R,$$

with side constraints

$$J_0(m_0) \approx 2, \quad J_4(m_0) \approx 0,$$

and then verifies on

$$I = [m_0 - c/Q, m_0 + c/Q]$$

that

$$\sup_I |J_2|$$

is small while

$$\max(|R_1|, |R_2|)$$

remains bounded below.

Such experiments are not evidence about actual zeta by themselves, because they use simplified backgrounds, finite prime cutoffs, and surrogate AFE terms. They do, however, show that the source-collapse mechanism is not a mere symbolic artefact: it survives fixed-prime phases, prime powers, smoothing, dual oscillatory blocks, and interval verification in the surrogate models.

Therefore the positive route must identify a genuine actual-zeta arithmetic obstruction, not another local rank-two algebraic identity.

For orientation, the latest normalized fixed-prime source-collapse run found an interval with

$$\sup_I |J_2| \approx 5.3 \times 10^{-3},$$

$$\inf_I \max(|R_1|, |R_2|) \approx 4.49,$$

and

$$\sup_I |H| \max(|R_1|, |R_2|) \approx 4.7 \times 10^{-9}.$$

Adding prime powers did not remove the mechanism; a reported prime-power run gave interval product about

$$8.52 \times 10^{-8}.$$

A smoothed main sum plus frozen-dual AFE surrogate gave a best interval product near

$$8.77 \times 10^{-9}.$$

Including moving differentiated $\chi(t)$, followed by exact $\chi(1/2 + it)$ -derivatives, still left source-collapse examples; the reported exact- χ interval product was about

$$1.93 \times 10^{-7}.$$

These values are retained only as diagnostics. They remove several easy objections to the source-collapse mechanism, but they are not a theorem about actual ζ or its exact AFE normalization.

The numerical evidence is not used as proof. Its role is to identify the precise missing theorem: actual-zeta source-curvature observability on I_ρ , as formulated in Subsection 14.27.

Remark 14.109 (Exact χ -derivatives in AFE-style source-collapse tests). For the exact critical-line factor

$$\chi(t) := \chi\left(\frac{1}{2} + it\right) = \pi^{it} \frac{\Gamma\left(\frac{1}{4} - \frac{it}{2}\right)}{\Gamma\left(\frac{1}{4} + \frac{it}{2}\right)},$$

write $g(t) = \log \chi(t)$. Then

$$g(t) = it \log \pi + \log \Gamma\left(\frac{1}{4} - \frac{it}{2}\right) - \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right),$$

and, for $n \geq 1$,

$$g^{(n)}(t) = \mathbf{1}_{n=1} i \log \pi + \left(-\frac{i}{2}\right)^n \psi^{(n-1)}\left(\frac{1}{4} - \frac{it}{2}\right) - \left(\frac{i}{2}\right)^n \psi^{(n-1)}\left(\frac{1}{4} + \frac{it}{2}\right).$$

If

$$E_0 = 1, \quad E_n = \sum_{k=0}^{n-1} \binom{n-1}{k} E_k g^{(n-k)},$$

then

$$\chi^{(n)} = \chi E_n.$$

Consequently,

$$\frac{d^k}{dt^k} \left[\chi(t) \sum_n b_n e^{it \log n} \right] = \sum_{a=0}^k \binom{k}{a} \chi^{(a)}(t) \sum_n b_n (i \log n)^{k-a} e^{it \log n}.$$

In the staged AFE-style tests, replacing the leading asymptotic phase by this exact χ -derivative calculation did not remove interval-local source collapse. It changed the finite derivative target, but did not create a source-jet nonflatness mechanism.

Remark 14.110 (Gamma/background terms as shifted derivative targets). Let

$$J_2(t) = B_2(t) + P_2(t),$$

where B_2 is the gamma/Riemann–Siegel/far-tail background contribution and P_2 is the oscillatory prime/AFE contribution. Interval source collapse requires

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R.$$

Equivalently,

$$P_2^{(r)}(m_0) \approx -B_2^{(r)}(m_0), \quad 0 \leq r \leq R.$$

Thus exact gamma/background terms shift the finite derivative-flat target. They do not change the type of the problem unless one proves a range, sign-cone, or arithmetic nonflatness theorem of the form

$$\max_{0 \leq r \leq R} |B_2^{(r)}(m_0) + P_2^{(r)}(m_0)| \geq Q^{-B}.$$

No such theorem is currently supplied by the local rank-two package. On the microscopic interval $|t - m_0| \leq c/Q$, the background is smooth and is captured by finite Taylor data, so it behaves as a shifted target for the oscillatory AFE part.

Remark 14.111 (Next numerical test: actual-zeta source curvature). The surrogate computations should now be followed by an actual-zeta numerical scan of the canonical q'' defined in Definition 14.114. The diagnostics should be interval-local:

$$\sup_{m \in I} |q''(m)|, \quad \int_I |q''(m)|^2 dm,$$

on intervals

$$I = [m_0 - c/Q, m_0 + c/Q].$$

The purpose is not to prove anything numerically, but to decide whether source-flat intervals are common in actual zeta data or are an artefact of the fixed-prime/AFE surrogates. If source-flat intervals appear in actual data, Hypothesis 14.115 is likely false in its present form. If they do not appear, the first theorem to attack is the interval L^2 source-curvature observability estimate.

Remark 14.112 (Current branch decision). The current local rank-two architecture should be treated as conditional infrastructure rather than a proof engine. The next proof-closing branch is not additional local row algebra, not order eleven by itself, and not clean projectivization alone. The sharp remaining fork is:

- (1) Prove actual-zeta source-curvature observability,

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|,$$

and then prove source–anisotropy reintegration,

$$\int_{I_\rho} |q''_{\text{can}}|^2 (\Gamma_G^2 + \Gamma_{G,2}^2) dm \geq Q^{-A} |I_\rho|;$$

- (2) or construct/verify an actual-zeta source-collapse model satisfying

$$0 < \sup_{m \in I_\rho} |y q''_{\text{can}}| \max\{|\Gamma_G|, |\Gamma_{G,2}|\} < Q^{-A};$$

- (3) or find a genuinely source-robust packet-visible row/invariant not divisible by $H = y q''_{\text{can}}$;
 (4) or prove a global projective/holonomy/trace/index theorem that gives a contradiction without a local norm lower bound.

Absent one of these, the present rank-two norm/projective architecture remains a strong local obstruction framework but not a proof of RH.

Remark 14.113 (Current proof frontier after the local package audits). This is the proof frontier after the local package audits and before the final source-collapse reduction. It is retained as a ledger of local dependencies; the terminal status is given in Remark 14.123.

- (1) The M_A, M_C connection branch is closed under the source-grade reflection convention.

- (2) The promoted quadratic mixed-block Bottleneck C coefficient

$$E_{A,Y}^{\text{BC}}$$

is now blocked at the concrete odd-background top row

$$\mathcal{E}_{\eta\eta;7,0,0}^2.$$

The associated S -coefficient is

$$f_{\eta\eta}^{\text{top}} = -\frac{u}{24\pi q^3}\nu_{21,7}.$$

Unless one proves

$$\nu_{21,7} = 0,$$

finds a cancelling active top-row term, or adds a principled second-order correction, this row is generically T -visible and can also be quotient visible on a generic $u_5 \neq 0$ patch.

- (3) The $\Theta_{\pm}$ -normal form and septic normal-form counterterms do not close the physical local obstruction. They can move the obstruction into a formal package, but the physical hidden coordinate carries it back through

$$T^{\text{phys}} = T^{\star} + \frac{\beta_{\text{loc}}^{\Theta}}{c}.$$

The known good-patch models have

$$\partial_{\kappa}(\beta_{\text{loc}}^{\Theta}/c) \neq 0.$$

- (4) The finite-core lower-transfer branch is now subordinate to the source-collapse problem. Earlier formulations reduced it to sampled vector non-concentration, for example

$$\mathcal{N}_k^{\text{align}}(m, Q) = \emptyset.$$

The later audits show that such a theorem must include source-level nonflatness. In particular, anisotropy or vector non-concentration alone cannot control the known rows when

$$H = yq'' \approx 0.$$

Thus the required input is now an actual-zeta source/finite-core non-concentration theorem on I_{ρ} , not merely finite determinacy, compactness, pairwise zero-spacing, endpoint-gap control, or Remez–Łojasiewicz volume estimates.

- (5) The source-weighted layered route is blocked by source collapse. The known order-seven/order-nine rows have the form

$$A_7^{\text{f}} \sim H\Gamma_G, \quad A_9^{\text{f}} \sim H\Gamma_{G,2}, \quad H = yq''.$$

At leading order,

$$H \sim -\frac{q_0(q'')^3}{288}.$$

Thus an admissible interval on which

$$q''(m) \approx 0$$

while

$$q_0(m) \asymp 1, \quad g(m), h(m) > 0, \quad \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \asymp 1$$

makes the norm vector

$$(A_7^{\text{f}}, A_9^{\text{f}})$$

super-polynomially small. Fixed-prime and AFE-style derivative-flat surrogates, as well as same-scale packet moment models, do not presently supply an obstruction. The missing positive theorem is an actual-zeta source-jet nonflatness statement such as

$$\sup_{m \in I_{\rho}} |q''(m)| \geq Q^{-B},$$

or a stronger finite/growing-jet version.

- (6) The dd -interaction branch remains independent unless feature balance, conditional spectral positivity, or incidence containment is proved. In the transported-kernel reduction, the first nonquadratic Taylor coefficient

$$\kappa_4 = \frac{1}{4}(\langle v_0, v_2 \rangle - \langle v_1, v_1 \rangle)$$

already gives a necessary sign test for conditional definiteness.

14.27. Arithmetic source-curvature observability: the next attack. The preceding source-collapse countermodels show that the current rank-two norm route cannot close from local row algebra, same-scale packet geometry, or finite Hadamard/quartet symmetry alone. The first missing input is an arithmetic source-curvature theorem for actual zeta data. In the notation fixed below, the minimal useful pointwise form is

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \geq Q^{-B}.$$

A stronger and more stable form is the interval energy estimate

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|.$$

This estimate would block the pure source-collapse branch, but it is now only the first gate. It does not by itself recover the source-weighted rank-two lower bound, because the q''_{can} -mass could still concentrate near the anisotropy base locus

$$\Gamma_G \approx 0, \quad \Gamma_{G,2} \approx 0.$$

The second missing input is therefore a source–anisotropy reintegration theorem, formulated below.

This subsection is the positive-theorem counterpart of Remark 14.91: all currently explored local, projective, correction-row, higher-anisotropy, and finite-Hadamard routes have either been classified as diagnostic or shown vulnerable to source collapse.

Definition 14.114 (Canonical source-curvature representation problem). In this subsection and in every source-observability statement below, the canonical source-curvature object is

$$q''_{\text{can}}(m) = q''_\zeta(m) = \Phi'''_\zeta(m).$$

This is the full actual-zeta phase-derivative curvature in the same Gram normalization used by the local rank-two rows. It is not the affine-subtracted complement curvature. The latter is denoted separately by

$$q''_{\text{source},R}(m) := (G^{\text{new}})''(m), \quad G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}.$$

The affine-subtracted complement curvature is a component in a chosen source/core split; it is not the canonical arithmetic object whose nonflatness is required.

The arithmetic representation problem is to write q''_{can} on both sides of the explicit formula. On the zero side one needs a representation

$$q''_{\text{can}}(m) = B_2(m) + \sum_{\rho}^{\text{reg}} K_2(m; \rho),$$

with all background, tail, source/core, and regularization conventions fixed. On the prime/AFE side one needs a representation of the form

$$q''_{\text{can}}(m) = B_2(m) + P_2(m) + D_2(m) + E_2(m),$$

where

$$P_2(m) = \Re \sum_{n \leq X} b_n (\log n)^2 n^{-1/2-im} W(n/T)$$

has fixed actual-zeta coefficients, D_2 denotes the dual or functional-equation contribution, and E_2 denotes a controlled differentiated error term. The coefficients b_n , smoothing, dual term,

gamma/background contribution, and error scale must be specified in the same normalization used by the rank-two local rows. The point of this convention is to forbid replacing q''_{can} by an arbitrary Dirichlet polynomial or by the affine-subtracted complement $q''_{\text{source},R}$.

Hypothesis 14.115 (Gate 1: off-line-packet source-curvature observability). Let I_ρ be an admissible reduced good-patch interval attached to a hypothetical off-line packet ρ . Actual zeta data satisfy

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|$$

for some fixed B , outside the already named reduced-base and correction incidence loci. Equivalently, in a weaker pointwise form,

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \geq Q^{-B}.$$

This is a necessary source-level nonflatness input for the current rank-two norm route. It is not sufficient by itself to recover the desired lower bound for

$$|H| \max\{|\Gamma_G|, |\Gamma_{G,2}|\}.$$

Hypothesis 14.116 (Gate 2: source-anisotropy vector observability). On the same admissible interval I_ρ , actual zeta data satisfy a source-weighted anisotropy lower bound of the form

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 (\Gamma_G(m)^2 + \Gamma_{G,2}(m)^2) dm \geq Q^{-A} |I_\rho|.$$

A pointwise substitute would be

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}.$$

This is a separate gate from Hypothesis 14.115. Source curvature may be observable in L^2 while the anisotropy factors are small on most of the q''_{can} -mass. The current proof architecture requires both gates, or a single stronger theorem implying both.

Remark 14.117 (ODE form of the reintegration problem). Let

$$u = q''_{\text{can}}, \quad \Gamma_G = u - 6q + 2q^3, \quad \Gamma_{G,2} = (1 - q^2)u - \frac{1}{6}q^{(4)}.$$

Differentiating Γ_G gives

$$\Gamma'_G = u'' - 6u + 12q(q')^2 + 6q^2u.$$

Since $u'' = q^{(4)}$, one obtains the identity

$$\boxed{\Gamma_{G,2} = 2q(q')^2 - \frac{1}{6}\Gamma''_G.}$$

Thus the reintegration gate is not merely a prime-side arithmetic problem. It also has a local analytic/ODE formulation: source mass for $u = q''_{\text{can}}$ must be prevented from concentrating on a set where both

$$\Gamma_G \approx 0 \quad \text{and} \quad 2q(q')^2 - \frac{1}{6}\Gamma''_G \approx 0.$$

A proof of Hypothesis 14.116 must control this base-locus regime, including possible cancellation between Γ''_G and $2q(q')^2$.

Remark 14.118 (Off-line-packet conditional form). A more targeted version of Hypothesis 14.115 uses the hypothetical off-line packet itself. Write

$$q''_{\text{can}} = q''_P + q''_{\text{rest}},$$

where q''_P is the contribution of the distinguished off-line packet and q''_{rest} is the contribution of all remaining actual-zeta data: other core packets, auxiliary/tail terms, background, dual terms, and controlled errors in the fixed convention. This q''_{rest} should not be confused with the affine-subtracted complement curvature $q''_{\text{source},R} = (G^{\text{new}})''$ unless the source/core split has explicitly identified the two.

The interval observability claim may be sharpened to

$$\int_{I_\rho} |q''_P(m) + q''_{\text{rest}}(m)|^2 dm \geq Q^{-B} |I_\rho|.$$

Equivalently, the rest term cannot satisfy

$$q''_{\text{rest}}(m) = -q''_P(m) + O(Q^{-B})$$

throughout the packet interval. This formulation uses the off-line-packet hypothesis directly and is stronger than a generic short-interval lower bound for q''_{can} . The same-scale source-flattening construction shows that this cannot be proved from arbitrary positive same-scale packet geometry alone; it must use actual-zeta arithmetic or a restriction on how actual zero packets may be arranged around ρ .

Remark 14.119 (Why this theorem is genuinely arithmetic). Hypothesis 14.115 cannot follow from the local packet geometry developed above. The same-scale source-flattening theorem and finite quadrature corollary show that positive same-distance packet configurations can make

$$q''(m_0), q^{(4)}(m_0), \dots, q^{(M)}(m_0)$$

vanish to arbitrary finite order while keeping

$$q_0 > 0, \quad g > 0, \quad h > 0,$$

and while keeping

$$\Gamma_G = q'' - 6q_0 + 2q_0^3$$

bounded away from zero after normalization, for example near $q_0 = 2$. Thus any proof of Hypothesis 14.115 must use actual zeta arithmetic beyond finite same-scale positive packet geometry.

Remark 14.120 (Zero-side quadrature obstruction to local proofs). The same-scale source-flattening model gives a concrete obstruction to any purely local zero-side proof of Hypothesis 14.115. The continuous measure in Theorem 14.83 lives on the same-distance curve

$$x^2 + c^2 = d^2$$

and has positive density

$$d\mu(\theta) = \frac{2}{\pi} \sin \theta d\theta.$$

It makes the averaged packet profile locally constant, and finite quadrature then approximates arbitrary finite source jets while keeping positive weights and same-distance packet scale. Therefore actual zeta must exclude, or fail to approximate, this quadrature pattern if the rank-two norm route is to close.

A plausible arithmetic formulation is: actual zeta zeros near a hypothetical off-line packet cannot realize the finite same-scale quadrature needed to make

$$q''(m_0), q^{(4)}(m_0), \dots, q^{(2M)}(m_0)$$

all small while $q_0 > 0$ and the Gram block remains positive. This is the zero-side form of the source-curvature observability problem.

Remark 14.121 (Prime-side observability formulation). After fixing the representation in Definition 14.114, the source-curvature problem becomes a short-interval fixed-coefficient non-approximation problem. Writing

$$q''_{\text{can}}(m) = B_2(m) + P_2(m) + D_2(m) + E_2(m),$$

source collapse means

$$P_2(m) \approx -B_2(m) - D_2(m) \quad \text{throughout } I_\rho,$$

up to the controlled error E_2 . The desired arithmetic theorem is therefore an observability estimate of the form

$$\int_{I_\rho} |B_2(m) + P_2(m) + D_2(m)|^2 dm \geq Q^{-B} |I_\rho|,$$

with E_2 absorbed into the final error budget.

This is not a generic Dirichlet-polynomial problem. The coefficients in P_2 are fixed by the actual zeta/AFE normalization. Standard localized large-sieve, Montgomery–Vaughan, explicit-formula energy, or uncertainty principles are useful diagnostics, but no theorem currently proves this fixed-coefficient lower bound. If interval estimates remain out of reach, the equivalent finite-jet formulation below is the sharper arithmetic target.

Remark 14.122 (Finite-jet version of the arithmetic problem). The interval energy estimate may also be attacked through finite jets. On an interval

$$I_\rho = [m_0 - c/Q, m_0 + c/Q],$$

source collapse requires simultaneous smallness of

$$\partial_m^r q_{\text{can}}''(m_0), \quad 0 \leq r \leq R,$$

where the scale audit gives

$$R \asymp \frac{\log Q}{\log \log Q}$$

or more conservatively $R = O(\log Q)$. In an AFE model this becomes a finite system

$$\partial_m^r (B_2 + D_2 + P_2)(m_0) \approx 0, \quad 0 \leq r \leq R,$$

or equivalently

$$P_2^{(r)}(m_0) \approx - (B_2 + D_2)^{(r)}(m_0).$$

The positive theorem would be a fixed-coefficient prime-log derivative-jet nonrecurrence statement:

$$\max_{0 \leq r \leq R} |\partial_m^r q_{\text{can}}''(m_0)| \geq Q^{-B}.$$

The recurrence heuristics developed above suggest that this is a difficult arithmetic theorem and may be false in fixed-prime or AFE surrogates. It is therefore not currently proved by large-sieve or generic Dirichlet-polynomial approximation estimates.

Remark 14.123 (Terminal status of the current architecture). The present proof architecture has reached the following decision point.

- (1) The original scalar Bottleneck-C closure route is blocked: the active $\eta\eta$ -row is S -visible and cannot be removed by source repackaging, quotienting, split readout, line-bundle transport, or transform-faithful normal-form counterterms.
- (2) The rank-two fixed-sector bundle is the correct local language for the remaining obstruction. It gives a coherent reduced-base connection and a conditional affine forcing equation, but it does not supply a global contradiction without an additional boundary, holonomy, index, or quantitative non-concentration theorem.
- (3) The local packet-visibility problem is substantially improved by the layered order-seven/order-nine theorem:

$$\Gamma_G \neq 0 \Rightarrow A_7^\dagger \neq 0, \quad \Gamma_G = 0, \Gamma_{G,2} \neq 0 \Rightarrow A_9^\dagger \neq 0,$$

modulo named correction incidence. This is a local visibility theorem, not a proof of RH.

- (4) The norm-based rank-two route is source-divisible. The established rows have schematic form

$$A_7^\dagger \sim H\Gamma_G, \quad A_9^\dagger \sim H\Gamma_{G,2}, \quad H = yq''.$$

Since

$$H \sim -\frac{q_0(q'')^3}{288},$$

an interval on which q'' is small collapses the known row vector even when the anisotropy factor remains live.

- (5) The projective route cancels H formally, but it loses magnitude, remains singular at

$$\Gamma_G = \Gamma_{G,2} = 0,$$

and is not cleanly defined for the corrected row vector unless correction rows vanish or are H -divisible. No global projective endpoint, holonomy, or index theorem is currently available.

- (6) The same-scale source-flattening construction shows that finite same-scale positive packet geometry does not itself prevent source collapse. Consequently, an actual proof must use information not already present in the local Hadamard/quartet packet geometry. The next positive target is not another local row identity, but the two-gate chain

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|$$

followed by

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 (\Gamma_G(m)^2 + \Gamma_{G,2}(m)^2) dm \geq Q^{-A} |I_\rho|.$$

The first estimate is source-curvature observability. The second is source-anisotropy reintegration. The second does not follow formally from the first.

Thus the current framework should be treated as a precise local obstruction and diagnostic architecture. To turn it into a proof of RH, one still needs at least one genuinely new input:

Gate 1: actual-zeta source-curvature observability

together with

Gate 2: source-anisotropy vector observability,

or else a single stronger theorem implying both. Alternatively, one needs

a source-robust row/invariant not divisible by $H = yq''_{\text{can}}$,

or

a global projective/affine/trace theorem with correction control.

Absent one of these inputs, the present architecture is conditionally non-closing. The least nonlinear arithmetic attack remains Hypothesis 14.115; the most important bridge is Hypothesis 14.116, with the ODE identity recorded in Remark 14.117.

APPENDIX A. TOY MICROSCOPIC EXPANSION

Lemma A.1 (Uniform microscopic expansion of the toy local deformation). *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

and let

$$\Delta_{\text{toy}}(u, d; s)$$

denote the toy local deformation matrix in the symmetric microscopic regime, with off-line parameter u , separation parameter d , and microscopic node s .

Then there exist constants $\rho_{\text{toy}} > 0$ and $u_0(D) > 0$ such that for all

$$d \in D, \quad |s| < \rho_{\text{toy}}, \quad |u| < u_0(D),$$

one has the expansion

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

where $M(d)$ is the toy anomaly matrix, $E(u, d; s)$ is analytic in s for $|s| < \rho_{\text{toy}}$, and

$$\sup_{\substack{d \in D \\ |s| < \rho_{\text{toy}} \\ |u| < u_0(D)}} \|E(u, d; s)\| \ll 1.$$

In particular, for the microscopic nodes

$$s_j = \frac{j}{Q^2}, \quad j = 1, \dots, N,$$

with $N \leq \rho_{\text{toy}} Q^2$, one has uniformly

$$\Delta_{\text{toy}}(u, d; s_j) = u^2 M(d) + u^4 E_j(u, d), \quad \sup_j \|E_j(u, d)\| \ll 1.$$

Proof. The toy local deformation is obtained from an explicit finite-dimensional off-line perturbation of the toy block. By construction it depends smoothly (in fact analytically) on the parameters (u, s, d) in a neighborhood of

$$(u, s, d) = (0, 0, d_0), \quad d_0 \in D.$$

At $u = 0$, the deformation vanishes, and by the even symmetry of the off-line split there is no linear term in u . Hence the first nontrivial term is quadratic:

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d, s) + O(u^4),$$

uniformly on compact d -ranges and for $|s|$ in a fixed microscopic disk.

By the jet-limit computation for the toy model, the leading quadratic coefficient is independent of the microscopic node s and equals the toy anomaly matrix $M(d)$. Thus

$$M(d, s) \equiv M(d),$$

and the expansion becomes

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

with E analytic in s and uniformly bounded on compact parameter ranges. Restricting to the microscopic nodes $s_j = j/Q^2$ gives the stated uniform bound for $E_j(u, d)$. $\square$

Remark A.2 (Work in progress: toy microscopic expansion). The proof above is schematic: it uses smoothness in (u, s, d) , vanishing at $u = 0$, and even symmetry in the off-line split to obtain the quadratic leading term, and invokes the jet-limit computation to identify the coefficient as the toy anomaly matrix $M(d)$. A later pass should make the analyticity and uniform-bound steps quantitative by recording an explicit Taylor remainder with a constant depending only on D .

APPENDIX B. FINITE FRÉCHET COEFFICIENT CALCULUS FOR THE CORRECTED-WHITENING BLOCK

This section records the finite Fréchet/Sylvester coefficient recurrences and finite gates used by the corrected-whitening package frontier of Remark 12.145. These are table contracts, not completed source tables. Their role is to make the finite computations auditable by agents and scripts without re-opening convention choices already fixed in the main text.

Remark B.1 (Fréchet coefficient normalization). The finite tables in this section use homogeneous Taylor coefficients in the ordinary z -expansion of the inverse-square-root transfer. Thus Λ , H , and G denote the actual coefficient contributions to the expanded matrix series, not raw multilinear derivatives D, D^2, D^3 unless explicitly stated.

Equivalently, if a separate computation uses derivative-normalized tensors, then the dictionary is

$$H_{\text{Taylor}} = \frac{1}{2!} H_{\text{deriv}}, \quad G_{\text{Taylor}} = \frac{1}{3!} G_{\text{deriv}}.$$

All determinant tests in this paper use the Taylor-coefficient convention unless the contrary is explicitly stated.

Remark B.2 (Stage 1 source-table status). The finite Fréchet recurrences in this section become a proof tool only after the required source tables are supplied in the same ordinary- z , pre- Φ_K ,

tagged normalization. The current Stage 1 infrastructure reduces the needed source input to a finite midpoint-jet theorem:

$$\text{Jet}_{\text{Stage1}} = \left\{ q_0^{(0)}, \dots, q_0^{(9)}; (r_i^{[a]})^{(k)} : i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9 \right\}_m.$$

The lead local obstruction currently uses $a = 2$. Hence any Stage 1 table omitting the $a = 2$ source-linear primitive cannot evaluate the present Bottleneck-C frontier.

The table generator is infrastructure, not a zeta-source proof. A promotion requires a source theorem supplying these jets, or a reduction of them to named baseline objects, in the same normalization used for B_7^{f} . Scale estimates are not a substitute for these finite coefficient tables.

B.1. Source-linear same-point primitives. For $\varepsilon \in \{-, +\}$, identified with $\varepsilon = -1, +1$, write

$$X_{\varepsilon, n} := [z^n] \mathfrak{D}_Q(\delta G_{\varepsilon}^{\text{lin}}) = \sum_{a=2}^6 X_{\varepsilon, n}^a r_a.$$

For $n \leq 2$, the nonzero source-constrained coefficients are

$$\begin{aligned} X_{\varepsilon, 0}^2 &= \frac{1}{24\pi}(I - D), \\ X_{\varepsilon, 1}^2 &= \frac{\varepsilon}{4\pi Q}S, \quad X_{\varepsilon, 1}^3 = \frac{\varepsilon}{48\pi Q^2}(I - D), \\ X_{\varepsilon, 2}^2 &= \left(\frac{1}{8\pi Q^2} + \frac{q^2}{32\pi Q^4} \right) I + \left(\frac{1}{8\pi Q^2} - \frac{q^2}{32\pi Q^4} \right) D, \\ X_{\varepsilon, 2}^3 &= \frac{1}{16\pi Q^3}S, \quad X_{\varepsilon, 2}^4 = \frac{1}{192\pi Q^4}(I - D). \end{aligned}$$

All other $X_{\varepsilon, n}^a$ with $n \leq 2$, $2 \leq a \leq 6$, vanish.

Remark B.3. These coefficients are not free symbolic inputs. They are the residual-source normal form of the source-linear same-point perturbation.

B.2. Baseline mixed block coefficients. Let

$$\begin{aligned} s &= \frac{z}{Q^2}, \quad q_{\pm}(z) = q_0 \left(m \pm \frac{z}{2Q^2} \right), \\ \Delta_0(z) &= \Phi_0(t_-) - \Phi_0(t_+), \quad C(z) = \cos \Delta_0(z), \quad S(z) = \sin \Delta_0(z). \end{aligned}$$

The baseline mixed block is

$$N^{(0)}(z) = \frac{1}{\pi} \begin{pmatrix} -\frac{2S}{s} & \frac{S + q_+ s C}{s^2} \\ -\frac{S + q_- s C}{s^2} & \frac{(q_- + q_+) s C + (2 - q_- q_+ s^2) S}{2s^3} \end{pmatrix}.$$

For $j \geq 0$, write

$$N_j^{(0)} = [z^j] N^{(0)}(z).$$

Then $N_{0..4}^{(0)}$ are obtained entrywise from the coefficient formulas

$$\begin{aligned} (N_j^{(0)})_{11} &= -\frac{2Q^2}{\pi} [z^{j+1}] S(z), \\ (N_j^{(0)})_{12} &= \frac{Q^4}{\pi} \left([z^{j+2}] S(z) + \frac{1}{Q^2} [z^{j+1}] (q_+(z) C(z)) \right), \\ (N_j^{(0)})_{21} &= -\frac{Q^4}{\pi} \left([z^{j+2}] S(z) + \frac{1}{Q^2} [z^{j+1}] (q_-(z) C(z)) \right), \\ (N_j^{(0)})_{22} &= \frac{Q^6}{2\pi} \left(\frac{1}{Q^2} [z^{j+2}] ((q_- + q_+) C) + 2[z^{j+3}] S - \frac{1}{Q^4} [z^{j+1}] (q_- q_+ S) \right). \end{aligned}$$

B.3. First Fréchet coefficient table. Let $P_{\varepsilon,0}$ be the zeroth square-root coefficient of $G_{\varepsilon}^{(0)}$, and let $\Sigma_{\varepsilon}(C)$ denote the unique solution of

$$P_{\varepsilon,0}Z + ZP_{\varepsilon,0} = C.$$

For each source index a , define $S_{\varepsilon,n}^a$ recursively by

$$S_{\varepsilon,n}^a = \Sigma_{\varepsilon} \left(X_{\varepsilon,n}^a - \sum_{j=1}^n (P_{\varepsilon,j} S_{\varepsilon,n-j}^a + S_{\varepsilon,n-j}^a P_{\varepsilon,j}) \right).$$

Then

$$\Lambda_{\varepsilon,n}^a = - \sum_{i+j+k=n} W_{\varepsilon,i} S_{\varepsilon,j}^a W_{\varepsilon,k}.$$

The required ranges used below include

$$\Lambda_{2,0..4}^-, \quad \Lambda_{3,1..5}^-, \quad \Lambda_{4,2..6}^-.$$

B.4. Second Fréchet coefficient table. For an ordered source pair (a, b) , define $T_{\varepsilon,n}^{ab}$ by

$$T_{\varepsilon,n}^{ab} = \Sigma_{\varepsilon} \left(- \sum_{i+j=n} (S_{\varepsilon,i}^a S_{\varepsilon,j}^b + S_{\varepsilon,i}^b S_{\varepsilon,j}^a) - \sum_{j=1}^n (P_{\varepsilon,j} T_{\varepsilon,n-j}^{ab} + T_{\varepsilon,n-j}^{ab} P_{\varepsilon,j}) \right).$$

The corresponding second Fréchet coefficient is

$$H_{\varepsilon,n}^{ab} = \sum_{i+j+k+\ell+m=n} \left(W_{\varepsilon,i} S_{\varepsilon,j}^b W_{\varepsilon,k} S_{\varepsilon,\ell}^a W_{\varepsilon,m} + W_{\varepsilon,i} S_{\varepsilon,j}^a W_{\varepsilon,k} S_{\varepsilon,\ell}^b W_{\varepsilon,m} \right) - \sum_{i+j+k=n} W_{\varepsilon,i} T_{\varepsilon,j}^{ab} W_{\varepsilon,k}.$$

The five-family endpoint gate uses

$$H_{32,1..5}^-, \quad H_{34,3..7}^-, \quad H_{24,2..6}^-.$$

B.5. Third Fréchet coefficient table. For source triple (a, b, c) , define $U_{\varepsilon,n}^{abc}$ by the third Sylvester recursion

$$P_{\varepsilon,0} U_{\varepsilon,n}^{abc} + U_{\varepsilon,n}^{abc} P_{\varepsilon,0} = - \sum_{i+j=n} \left(S_{\varepsilon,i}^a T_{\varepsilon,j}^{bc} + S_{\varepsilon,i}^b T_{\varepsilon,j}^{ac} + S_{\varepsilon,i}^c T_{\varepsilon,j}^{ab} + T_{\varepsilon,i}^{bc} S_{\varepsilon,j}^a + T_{\varepsilon,i}^{ac} S_{\varepsilon,j}^b + T_{\varepsilon,i}^{ab} S_{\varepsilon,j}^c \right) - \sum_{j=1}^n (P_{\varepsilon,j} U_{\varepsilon,n-j}^{abc} + U_{\varepsilon,n-j}^{abc} P_{\varepsilon,j}).$$

The all-left third Fréchet coefficient $G_{abc,n}^-$ is then obtained from the coefficient of the cubic source monomial in

$$(G_{-}^{(0)} + \text{source})^{-1/2},$$

equivalently by the coefficient recursion determined by $U_{\varepsilon,n}^{abc}$, the lower S - and T -coefficients, and the inverse coefficients $W_{\varepsilon,n}$.

The seven-family audit for the first surviving residual requires

$$G_{324,3..7}^- \quad \text{and} \quad G_{324,3..7}^+.$$

Remark B.4 (Status of the third-Fréchet table). The recursion above defines the finite third-Fréchet coefficient table that a raw seven-family audit would require. It does not assert that the raw $L_3 N_0 R_0 / L_0 N_0 R_3$ completion is quotient-gauge. Under the current package architecture, this table is a diagnostic tool for raw local coefficients. The theorem object is the normalized corrected package $C_{\mathcal{W}}^{\text{corr}}$ of Definition C.8, not the raw seven-family Fréchet sum.

Remark B.5 (Convention: transfer degree versus source bidegree). We write d for homogeneous transfer degree in

$$\mathcal{T}(X) = \sum_{d \geq 1} \mathcal{T}_d(X),$$

and reserve (p, q) for source bidegree in the source-tagged expansion

$$C(\tau_1, \tau_2) = \sum_{p,q} \tau_1^p \tau_2^q C_{p,q}.$$

Remark B.6 (Fixed-sector projection in the moving fiber). When the midpoint is h , the fixed-sector projection is denoted $\pi_{\mathfrak{f}_h}$. The transpose-invariance identity

$$\pi_{\mathfrak{f}_h}(Z^T) = \pi_{\mathfrak{f}_h}(Z)$$

is applied in the moving midpoint fiber before midpoint trivialization to m .

Lemma B.7 (Universal first-collision formula by transfer degree). *Let*

$$\mathcal{T}(X) = \sum_{d \geq 1} \mathcal{T}_d(X)$$

be the homogeneous transfer expansion of the corrected-whitening map. On the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

the first-collision contribution of homogeneous transfer degree d has the form

$$F_{1,d} = [z^7] \pi_{\mathfrak{f}_h} \left[(-1)^{d+1} d\kappa \mathcal{T}_d(X^d) + \frac{d}{2} (1 - (-1)^d) \mathcal{T}_d(X^{d-1}, X_h) \right].$$

Thus even d contributes only the amplitude term

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_d(X^d),$$

whereas odd d contributes both the amplitude term and the source-motion term

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_d(X^{d-1}, X_h).$$

Proof sketch. The factor $a_2 = -a_1(1 + \kappa\delta)$ produces the amplitude derivative term proportional to $d\kappa$. The endpoint displacement $h_1 = h - \delta/2$, $h_2 = h + \delta/2$ produces the midpoint-motion insertion. The parity of d controls whether the endpoint-motion contributions cancel or add: for even d the X_h -insertion cancels, while for odd d it appears with coefficient d . $\square$

B.6. Fourth-transfer coefficient table for Gate B. The corrected-whitening map is

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

Let a labelled source input be

$$X_j = (A_j, Y_j, B_j),$$

where

$$A_j = \delta G_{-,j}^{\text{lin}}, \quad Y_j = \delta N_j^{\text{lin}}, \quad B_j = \delta G_{+,j}^{\text{lin}}.$$

Write

$$(G_-^{(0)} + A)^{-1/2} = L_0 + L_1[A] + L_2[A, A] + L_3[A, A, A] + L_4[A, A, A, A] + \cdots,$$

and

$$(G_+^{(0)} + B)^{-1/2} = R_0 + R_1[B] + R_2[B, B] + R_3[B, B, B] + R_4[B, B, B, B] + \cdots.$$

The middle block is source-linear:

$$N = N_0 + Y.$$

For a labelled four-source set $J = \{1, 2, 3, 4\}$, the fourth homogeneous corrected-whitening transfer is

$$\mathcal{T}_4(X_1, X_2, X_3, X_4) = \sum_{I \subseteq J} L_I^- N_0 R_{J \setminus I}^+ + \sum_{m \in J} \sum_{I \subseteq J \setminus \{m\}} L_I^- Y_m R_{J \setminus (I \cup \{m\})}^+.$$

The first sum consists of

$$L_4 N_0 R_0, \quad L_3 N_0 R_1, \quad L_2 N_0 R_2, \quad L_1 N_0 R_3, \quad L_0 N_0 R_4,$$

and the second sum consists of

$$L_3 Y R_0, \quad L_2 Y R_1, \quad L_1 Y R_2, \quad L_0 Y R_3.$$

The ordinary z^7 -coefficient is

$$[z^7]\mathcal{T}_4(J) = \sum_{I \subseteq J} \sum_{\alpha+\beta+\gamma=7} L_{I,\alpha}^- N_\beta^{(0)} R_{J \setminus I, \gamma}^+ + \sum_{m \in J} \sum_{I \subseteq J \setminus \{m\}} \sum_{\alpha+\beta+\gamma=7} L_{I,\alpha}^- Y_{m,\beta} R_{J \setminus (I \cup \{m\}), \gamma}^+.$$

For a source multiset, $\text{Sym } \mathcal{T}_4$ denotes the labelled sum divided by the automorphism factor of the multiset.

Lemma B.8 (All-subset midpoint parity). *Let K be any labelled subset of homogeneous source inputs and define*

$$|K| := \sum_{g \in K} g.$$

At leading homogeneous order, the left and right same-point inverse-square-root coefficients satisfy

$$R_K^+ = (-1)^{|K|} L_K^-.$$

Remark B.9 (Scope of all-subset midpoint parity). Lemma B.8 is a leading homogeneous coefficient statement. It uses the midpoint chart

$$t_\pm = h \pm \frac{z}{2Q^2}$$

and the Taylor-coefficient Fréchet convention. If one works with derivative-normalized Fréchet tensors, ordered unsymmetrized source slots, or non-leading ordinary coefficients, the parity factors and multiplicities must be translated back to this convention before applying the subset-complement cancellation.

Lemma B.10 (Symmetry of same-point subset coefficients). *For every labelled subset K ,*

$$(L_K^-)^T = L_K^-, \quad (R_K^+)^T = R_K^+.$$

Moreover $N_0[0]^T = N_0[0]$.

Lemma B.11 (Leading Y -middle parity). *The leading Y -middle input satisfies*

$$Y_{\text{lead}}^{[a]} = 0 \quad (a \text{ odd}),$$

and

$$(Y_{\text{lead}}^{[a]})^T = Y_{\text{lead}}^{[a]} \quad (a \text{ even}).$$

Remark B.12 (Scope of the leading Y -middle parity table). Lemma B.11 concerns the leading homogeneous ordinary-order coefficient of the source-linear mixed middle input in the active homogeneous scalar-grade convention. It is not a statement about every ordinary coefficient of a non-homogeneous matrix-output $M^{[5]}$ table.

In particular, older coefficient audits that found antisymmetric J -channel middle terms belong to the quarantined non-homogeneous matrix-output convention. They do not contradict the leading homogeneous Y -middle parity table used in Gate B.

Theorem B.13 (Odd total-grade subset-complement cancellation). *Let*

$$J = (g_1, \dots, g_d)$$

be a labelled homogeneous source multiset with total grade

$$|J| = \sum_i g_i = 7.$$

Then the leading ordinary-order amplitude contribution of the corrected-whitening transfer vanishes in fixed sector:

$$[z^7]\pi_{\mathbf{i}_h} \mathcal{T}_d(X^{[g_1]}, \dots, X^{[g_d]}) = 0.$$

Proof. For the N_0 -middle part, the leading contribution is

$$\mathcal{A}_{N_0}(J) = [z^7] \pi_{\mathfrak{f}_h} \sum_{I \subseteq J} L_I^- N_0[0] R_{J \setminus I}^+.$$

By Lemma B.8,

$$R_{J \setminus I}^+ = (-1)^{|J| - |I|} L_{J \setminus I}^-.$$

Since $|J| = 7$, this sign is $-(-1)^{|I|}$. Pair I with $J \setminus I$. The paired contribution is

$$(-1)^{|I|} \pi_{\mathfrak{f}_h} \left(-L_I^- N_0[0] L_{J \setminus I}^- + L_{J \setminus I}^- N_0[0] L_I^- \right).$$

The two matrix products are transposes, and $\pi_{\mathfrak{f}_h}$ is transpose-invariant. Hence the pair cancels.

For the Y -middle part, fix the source label $m \in J$ in the middle and write $a = g_m$, $K = J \setminus \{m\}$. If a is odd, then $Y_m = 0$ by Lemma B.11. If a is even, then Y_m is symmetric and $|K| = 7 - a$ is odd, so the same subset-complement pairing cancels the fixed-sector projection. $\square$

Lemma B.14 (Grade-preserving source-motion cancellation). *Let $J = (g_1, \dots, g_d)$ be a labelled homogeneous source multiset with*

$$|J| = \sum_i g_i = 7.$$

Replace one source slot by the grade-preserving part of its midpoint derivative,

$$(\partial_h \operatorname{Gr}_{g_i} r)_{\text{same}},$$

and keep all other source slots fixed. Then the corresponding fixed-sector source-motion contribution vanishes:

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_d \left(X^{[g_1]}, \dots, X_{\text{same}, h}^{[g_i]}, \dots, X^{[g_d]} \right) = 0.$$

Proof sketch. The grade-preserving derivative has the same ordinary source parity and the same endpoint transpose parity as the undifferentiated homogeneous grade in that slot. Therefore the same subset-complement pairing used in Theorem B.13 applies: each labelled subset contribution is paired with its complement, the two matrix products are transposes, and the transpose-invariant projection $\pi_{\mathfrak{f}_h}$ kills the aggregate fixed-sector contribution. $\square$

Corollary B.15 (Amplitude closure of higher transfers). *The amplitude part of every Gate B transfer of total homogeneous grade 7 vanishes in $\mathfrak{f}_h$. In particular, the aggregate fourth-transfer positive-grade contribution*

$$\sum_{J \in \{(1,1,1,4), (1,1,2,3), (1,2,2,2)\}} [z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_4(J)$$

vanishes.

The same argument also kills the first lower-shadow fourth-transfer aggregate

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_4(0, 1, 1, 5) = 0.$$

More generally, the subset-complement argument applies to every labelled multiset of total homogeneous grade 7, including zero-padded lower-shadow multisets, provided the full Taylor source tower is differentiated before ordinary-order truncation.

Thus the amplitude parts also vanish for

$$d = 5: \quad (1, 1, 1, 1, 3), (1, 1, 1, 2, 2),$$

$$d = 6: \quad (1, 1, 1, 1, 1, 2),$$

and

$$d = 7: \quad (1, 1, 1, 1, 1, 1, 1).$$

Lemma B.16 (Raw midpoint derivative of homogeneous scalar grades). *For*

$$\mathrm{Gr}_g r(h; t) = \frac{r^{(g+2)}(h)}{(g+2)!} (t-h)^{g+2},$$

one has

$$\partial_h \mathrm{Gr}_g r = (\partial_h \mathrm{Gr}_g r)_{\text{same}} + (\partial_h \mathrm{Gr}_g r)_{\text{lower}},$$

where

$$(\partial_h \mathrm{Gr}_g r)_{\text{same}} = \frac{r^{(g+3)}(h)}{(g+2)!} (t-h)^{g+2},$$

and

$$(\partial_h \mathrm{Gr}_g r)_{\text{lower}} = -\frac{r^{(g+2)}(h)}{(g+1)!} (t-h)^{g+1}.$$

Theorem B.17 (Moving homogeneous source-frame telescoping). *For every interior grade $g \geq 1$,*

$$\boxed{(\partial_h \mathrm{Gr}_{g-1} r)_{\text{same}} + (\partial_h \mathrm{Gr}_g r)_{\text{lower}} = 0.}$$

Consequently, after labelled symmetrization, the grade-lowering part of the raw source-motion insertion cancels source-slot by source-slot against the same-basis derivative of the adjacent lower grade.

Proof. By Lemma B.16,

$$(\partial_h \mathrm{Gr}_{g-1} r)_{\text{same}} = \frac{r^{(g+2)}(h)}{(g+1)!} (t-h)^{g+1},$$

whereas

$$(\partial_h \mathrm{Gr}_g r)_{\text{lower}} = -\frac{r^{(g+2)}(h)}{(g+1)!} (t-h)^{g+1}.$$

The signs and factorials agree exactly. The cancellation is labelled-slot local, so it survives the symmetrized multilinear transfer expansion. $\square$

Lemma B.18 (Affine boundary removal in the moving source frame). *In the $\mathfrak{D}_Q$ -scaled corrected source block, admissible source inputs are 2-jet normalized at the moving midpoint:*

$$f(h) = 0, \quad f'(h) = 0.$$

Equivalently, the source-frame projection is

$$\Pi_{\geq 2, h} f(t) = f(t) - f(h) - f'(h)(t-h).$$

Thus constant and affine modes are not independent raw source inputs.

This statement is a source-admissibility assertion for the holomorphic $\mathfrak{D}_Q$ -scaled corrected source variables. It is not a license to discard lower homogeneous grades. It uses the residual-source convention in which the moving midpoint affine jet has already been subtracted; without that convention an affine component would not be killed by a formal coefficient argument alone.

This is not a blanket assertion that all lower-shadow grades are discarded. The grade 0, 2, 4 source directions may appear in the moving Taylor tower. They are handled by the interior telescoping identity when they arise from source-motion, not by reclassifying them as baseline data. The only boundary mode removed here is the affine grade -1 term.

For the homogeneous scalar-grade basis, the remaining lower boundary

$$(\partial_h \mathrm{Gr}_0 r)_{\text{lower}} = -r^{(2)}(h)(t-h)$$

is affine and is killed by $\Pi_{\geq 2, h}$. Therefore no grade -1 source term enters

$$[z^7][\delta^1]\pi_{fh}(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

The Taylor source tower is differentiated before ordinary-order truncation; in particular, grade 8 is included when proving ordinary order 7 cancellation and truncation is applied only after ∂_h .

Proof sketch. An affine perturbation $\psi(t) = t - h$ satisfies

$$\psi(t_{\pm}) = \pm \frac{z}{2Q^2}, \quad \psi'(t_{\pm}) = 1, \quad \psi''(t_{\pm}) = 0.$$

In the $\mathfrak{D}_Q$ -scaled source variables

$$d_{\pm}(z) = \frac{z^2}{Q^4} U_{\pm}(z), \quad e_{\pm}(z) = \frac{z}{Q^2} E_{\pm}(z),$$

an affine source would force $U_{\pm}$ and $E_{\pm}$ to have poles in z . Hence affine modes are outside the admissible holomorphic source sector. This is the source-frame reason the homogeneous scalar-grade tower starts at $g = 0$, i.e. at the second derivative. $\square$

Remark B.19 (No premature ordinary-order truncation). The cancellation in Theorem B.17 is a statement about the full moving Taylor source tower. For ordinary order 7, one must include the next homogeneous grade before differentiating: the lower part of $\partial_h \text{Gr}_8 r$ cancels the apparent upper boundary coming from the same-basis derivative of $\text{Gr}_7 r$. If the tower is truncated at grade 7 before ∂_h is applied, this cancellation is artificially destroyed.

Thus “through ordinary order 7” means: differentiate the full homogeneous source tower first, perform the adjacent-grade telescoping, and only then discard ordinary powers above 7.

Proposition B.20 (Decomposition of the Gate B first-collision coefficient). *Through ordinary order 7, the Gate B first-collision coefficient decomposes as*

$$F_1 = F_1^{\text{GateA}} + F_1^{\text{amp}} + F_1^{X_h, \text{same}} + F_1^{X_h, \text{lower}} + F_1^{\text{bdry}},$$

where:

- (1) F_1^{GateA} is the leading homogeneous cubic contribution;
- (2) F_1^{amp} is the amplitude contribution of higher transfer terms $\mathcal{T}_d(X^d)$ with total homogeneous grade 7;
- (3) $F_1^{X_h, \text{same}}$ is the grade-preserving part of the source-motion insertion;
- (4) $F_1^{X_h, \text{lower}}$ is the grade-lowering part of the source-motion insertion;
- (5) F_1^{bdry} consists of the lower affine boundary and the upper truncation boundary of the moving Taylor source frame.

Under the hypotheses of Gate A closure, all-subset midpoint parity, leading Y -middle parity, moving source-frame telescoping, and 2-jet source-frame normalization, every term on the right vanishes.

Proof sketch. The decomposition follows from Lemma B.7. Gate A handles the leading cubic term. Theorem B.13 handles the amplitude part. The grade-preserving component of X_h is handled by Lemma B.14. The grade-lowering part of X_h is handled by Theorem B.17. The lower affine boundary is removed by Lemma B.18, and the upper boundary is avoided by differentiating the full Taylor source tower before truncation. $\square$

Corollary B.21 (Conditional Gate B raw edge-divisibility through ordinary order 7). *Assume that the following source-frame hypotheses have been verified for the actual $\mathfrak{D}_Q$ -scaled corrected source model, not merely for a formal homogeneous Taylor model:*

- (1) Gate A aggregate closure;
- (2) all-subset midpoint parity for same-point whitening coefficients;
- (3) symmetry of same-point whitening subset coefficients and of $N_0[0]$;
- (4) transpose-invariance of $\pi_{\mathfrak{f}_h}$;
- (5) the leading Y -middle parity table;
- (6) moving homogeneous source-frame telescoping;
- (7) affine boundary removal by 2-jet source-frame normalization; and
- (8) differentiation of the full Taylor source tower before ordinary-order truncation.

Then

$$\boxed{[z^7][\delta^1] \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0.}$$

Under these same hypotheses, equivalently,

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1 \delta^2)$$

through ordinary order 7 in fixed sector.

Remark B.22 (Dependency ledger for Gate B). Corollary B.21 is a raw edge-divisibility result, not a post-edge quotient compatibility result. Its inputs are logically independent and should not be silently replaced by quotient-level language.

The dependencies are:

- (1) Gate A aggregate closure, which removes the leading homogeneous cubic non-(1, 1) contribution.
- (2) All-subset midpoint parity for the same-point whitening coefficients, which pairs labelled subsets with their complements.
- (3) Symmetry of the relevant same-point subset coefficients and of $N_0[0]$.
- (4) Transpose-invariance of the moving fixed-sector projection $\pi_{\mathfrak{f}_h}$.
- (5) Leading Y -middle parity, which removes odd Y -middle entries and makes the even entries compatible with subset-complement cancellation.
- (6) Moving homogeneous source-frame telescoping, which cancels the grade-lowering source-motion term against the same-basis derivative of the adjacent lower grade.
- (7) 2-jet source-frame normalization, which removes the affine lower boundary produced by differentiating Gr_0 .
- (8) Differentiation of the full Taylor source tower before ordinary-order truncation, which prevents an artificial upper-boundary residual.

If any one of these inputs is removed, the statement should be demoted back to a conditional Gate B target rather than used as a proved package construction. Thus the corollary should be cited as a conditional raw edge-divisibility theorem: it closes Gate B for the corrected package only after the listed parity, source-frame, and full-tower-before-truncation hypotheses have been identified with the actual source model.

Remark B.23 (Gate B remains conditional on source-model hardening). Corollary B.21 is a proved implication from its stated source-frame hypotheses. It should not yet be read as an unconditional theorem of Definition 10.6 alone.

The tagged source block gives pre-whitening $\mathcal{K}$ -linear additivity. The remaining source-model obligation is to prove that the actual $\mathfrak{D}_Q$ -scaled corrected source variables feed the nonlinear corrected-whitening transfer

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}$$

with the same homogeneous scalar grades, midpoint subset parities, source-motion telescoping, affine-boundary removal, and full-tower-before-truncation used in Gate B.

Thus the corollary is presently conditional in exactly this sense: the raw edge-divisibility conclusion follows once these source-frame hypotheses are verified for the actual corrected package, rather than only for the formal homogeneous Taylor tower.

Remark B.24 (Negative findings from the Gate A/Gate B audit). The Gate A/Gate B audit rules out several tempting shortcuts.

- (1) **Placement-level vanishing is false.** Individual transfer placements can have nonzero fixed-sector components. The cancellation is only after labelled subset-complement aggregation and transpose-invariant fixed-sector projection.
- (2) **Family-by-family gauge is too strong.** The package target is aggregate fixed-sector cancellation or quotient invisibility. It does not require each Fréchet family to be separately proportional to $A_5^{\mathfrak{f}}$.
- (3) **Post-edge quotient compatibility is too late.** A statement such as

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

cannot be used to prove raw edge-divisibility unless it is reformulated at the raw edge layer.

- (4) **Determinant scalarization is only a good-patch coordinate.** The quotient object is

$$\mathfrak{f}_m / \mathcal{C}A_5^{\mathfrak{f}}(m).$$

The scalar determinant test is a coordinate on a nondegenerate patch, not the definition of the quotient.

- (5) **The matrix-output $M^{[5]}$ convention is quarantined.** The active proof uses homogeneous scalar grades. Replacing the grade-five source by an ordinary matrix-output order-5 coefficient reopens the Y -slot families and requires a separate no-double-counting theorem against grades 0, 2, 4.
- (6) **The Z_j -table is post-Gate-B, not the Gate-B obstruction.** After raw edge-divisibility, $Z_j[M]$ is the next pre-trivialization table for the connection calculation. It is not a missing proof of the δ^2 -edge law itself.

Remark B.25 (Mechanism of the Gate B cancellation). The Gate B cancellation is not a brute-force finite coefficient accident. It is the combination of three structural symmetries: odd total-grade subset-complement cancellation for amplitude terms, moving Taylor-frame telescoping for source-motion lower terms, and 2-jet source-frame normalization for the affine boundary. These mechanisms act before Φ_K , before determinant scalarization, and before quotient extraction. The resulting raw edge-divisibility is exactly the input needed to construct the normalized order-7 corrected package.

Remark B.26 (What remains attackable in Gate B). After Corollary B.21, the leading cubic inventory, the $d \geq 4$ amplitude terms, and the raw source-motion lower-shadow terms should not be reopened as independent obstructions unless one of the stated hypotheses is rejected. A valid attack on Gate B must now target one of the hypotheses:

all-subset midpoint parity, leading Y -middle parity, moving source-frame telescoping, affine boundary r

In particular, a nonzero unsymmetrized transfer placement is not a Gate B counterexample unless it survives the labelled aggregate and the source-frame telescoping procedure.

Remark B.27 (Gate A is not Gate B). Gate A is the leading homogeneous cubic audit. Gate B is the raw edge-divisibility theorem for the projected corrected-whitening package. Gate A closure is a necessary local input, but it is not the same statement as edge-normalized package construction. Corollary B.21 records the additional transfer and source-frame mechanisms that promote the Gate A closure to the order-7 raw edge-divisibility statement.

Remark B.28 (Unsymmetrized subchannels versus aggregate cancellation). Individual transfer placements can have nonzero fixed-sector components. This does not contradict the aggregate cancellations above. The cancellations pair each labelled subset with its complement, use midpoint parity, and then project to the transpose-invariant fixed sector. No placement-by-placement vanishing is asserted.

Remark B.29 (Raw placement-level vanishing is false). The aggregate cancellations above must not be read as placement-level vanishing. A single fourth-transfer subchannel, for instance a diagonal same-point placement in the $(1, 1, 1, 4)$ class, can produce a nonzero fixed-sector I -component. The cancellation occurs only after summing the labelled subset-complement aggregate and applying the transpose-invariant projection $\pi_{\mathfrak{f}}$. Thus the correct theorem is aggregate fixed-sector cancellation, not termwise support-zero or family-by-family gauge.

Remark B.30 (Layering of the corrected package). The raw edge-divisibility statement is pre-normalization. Only after

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1 \delta^2)$$

has been established may one form the amplitude-normalized package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

Post-edge quotient compatibility statements involving $Q_{7,m}^q$ should not be used as the first proof of edge divisibility.

Remark B.31 (Post-edge quotient compatibility is not a proof of edge divisibility). The formal projection identity

$$\Pi_{1,1}\mathcal{T}_{d\geq 3} = 0$$

does not by itself prove Gate B. Gate B uses the retained complement

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

and must be proved before the normalized quotient package exists. Therefore a statement such as

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

is a post-edge compatibility statement unless it is reformulated at the raw edge level. It cannot be used as the first proof that

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1\delta^2).$$

Definition B.32 (Pre-trivialization edge table after Gate B). After Corollary B.21, the edge-normalized package is holomorphic through ordinary order 7. For a source monomial M , define the pre-trivialization edge coefficient

$$Z_j[M](z) := -[\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} [z^{\leq 7}] (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

where $\epsilon = h - m$. Equivalently,

$$Z_j[M](z) = [\epsilon^j M] \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} [z^{\leq 7}] \hat{C}_{\mathcal{W}}^{\text{amp}},$$

once the edge-normalized package $\hat{C}_{\mathcal{W}}^{\text{amp}}$ has been formed.

Remark B.33 (Polynomial versus vector notation for $Z_j[M]$). Definition B.32 first defines the pre-trivialization edge table as a polynomial

$$Z_j[M](z) \in \mathfrak{f}_m[z]_{\leq 7}.$$

When $Z_j[M]$ appears without the argument z in the moving-line connection formula, it denotes its ordinary z^7 -coefficient:

$$Z_j[M] := [z^7] Z_j[M](z) \in \mathfrak{f}_m.$$

Thus

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i)$$

uses the vector coefficient $[z^7] Z_{7-i}[M](z)$, not the full polynomial.

Remark B.34 (Pre-trivialization versus post-trivialization coefficients). The coefficients $Z_j[M]$ are pre-trivialization coefficients. If $B_j[M]$ denotes coefficients after applying the representative-level midpoint trivialization, then $B_j[M]$ already includes the moving-line connection contribution. Therefore the connection sum

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i)$$

must use $Z_j[M]$, not post-trivialization $B_j[M]$, or the connection is counted twice.

Remark B.35 (Next object after Gate B). After Gate B, the next finite table is not a raw seven-family coefficient table. It is the pre-trivialization δ^2 -edge table

$$Z_j[M] = -[\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} [z^{\leq 7}] (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

This table is defined only after raw edge-divisibility makes the δ^{-2} -normalized package holomorphic. The finite Fréchet/Sylvester calculus supplies the raw coefficient infrastructure for computing $Z_j[M]$; it does not replace the edge-divisibility theorem.

B.7. Midpoint parity.

Lemma B.36 (Midpoint parity of Fréchet coefficients). *In the midpoint chart,*

$$t_{\pm} = m \pm \frac{z}{2Q^2},$$

the plus-side and minus-side coefficients satisfy

$$\Lambda_{a,n}^+ = (-1)^n \Lambda_{a,n}^-,$$

and

$$H_{ab,n}^+ = (-1)^n H_{ab,n}^-,$$

for the source indices used in the finite coefficient tables. The same parity convention applies to the third Fréchet coefficients once the corresponding $G_{abc,n}^{\pm}$ table is defined.

B.8. Fixed-sector projection rule.

Lemma B.37 (Fixed-sector projection of a J -middle product). *Let*

$$H = aI + bS + cD, \quad W = pI + qS + rD.$$

Then

$$\pi_{\mathfrak{f}}(HJW) = (cq - br)I + (cp - ar)S,$$

and

$$\pi_{\mathfrak{f}}(WJH) = -(cq - br)I - (cp - ar)S.$$

Consequently,

$$\pi_{\mathfrak{f}}(WJH) = -\pi_{\mathfrak{f}}(HJW).$$

B.9. Computed finite gates.

Remark B.38 (Scope of the computed finite gates subsection). The finite gates in this subsection record diagnostic computations from the ordinary matrix-output / raw-family audit. They are useful as coefficient checks and as warnings against termwise or raw-package shortcuts, but they are not the active Gate A/Gate B proof route.

In the active homogeneous scalar-grade proof, the Y -slot families are absent at B_7 , Gate A closes by aggregate fixed-sector cancellation, and Gate B closes through ordinary order 7 by odd total-grade subset-complement cancellation, moving source-frame telescoping, and affine boundary removal. Therefore the older Y^5/Y^6 , d_{324}^- , and five-family $N^{(0)}$ -weighted residual formulas below should be read as historical diagnostics for the raw package, not as remaining theorem obligations.

Remark B.39 (Source-coefficient parity in the Y -slot). For the grade-five mixed slot Y^a , the source-coefficient table gives

$$Y^5 = 0,$$

while

$$Y^6 = y_6 J, \quad y_6 = \frac{1}{53760\pi Q^9}.$$

Thus the Y^5 -slot residuals vanish individually, and the Y^6 -slot residuals cancel in left-right pairs by Lemma B.37.

Remark B.40 (First invisible endpoint residual). For the invisible endpoint pair $(3, 4)$ with middle source $q = 2$, the three-family C_2 gate reduces to one scalar

$$d_{324}^- = y_4 \Xi_{324}, \quad y_4 = -\frac{q^2}{640\pi Q^9},$$

where

$$\Xi_{324} = \mathcal{D}(H_{2,32}^- JW_0 - H_{1,32}^- JW_1).$$

The accompanying d_{324}^+ term vanishes by $Y^3 = 0$, and the d_{234}^+ term vanishes by source-degree support $H_{0..2;34}^- = 0$.

Remark B.41 (Five-family $N^{(0)}$ -weighted endpoint gate). In the older raw-family diagnostic audit, the N_0 -middle families did not collapse to the $r + t = 2$ range. They involved the full baseline middle factor and hence the convolution

$$n + s + t = 7.$$

For the first endpoint residual define

$$\begin{aligned} E_{324}^- &= \mathcal{D} \sum_{\substack{1 \leq n \leq 5 \\ 2 \leq t \leq 6 \\ n+t \leq 7}} (-1)^t H_{32,n}^- N_{7-n-t}^{(0)} \Lambda_{4,t}^-, \\ E_{234}^+ &= \mathcal{D} \sum_{\substack{0 \leq n \leq 4 \\ 3 \leq t \leq 7 \\ n+t \leq 7}} (-1)^t \Lambda_{2,n}^- N_{7-n-t}^{(0)} H_{34,t}^-, \\ E_{324}^+ &= \mathcal{D} \sum_{\substack{1 \leq n \leq 5 \\ 2 \leq t \leq 6 \\ n+t \leq 7}} (-1)^t \Lambda_{3,n}^- N_{7-n-t}^{(0)} H_{24,t}^-. \end{aligned}$$

The five-family residuals are

$$\begin{aligned} R_5^{(A)} &= 2y_4 \Xi_{324} + 2E_{324}^- + E_{234}^+, \\ R_5^{(C)} &= y_4 \Xi_{324} + E_{324}^- + 2E_{324}^+. \end{aligned}$$

These are finite coefficient identities, not consequences of support or generic Fréchet algebra.

In the active Gate B proof these raw-family residual formulas are superseded by Corollary B.21. They should not be listed as live post-Gate-B obstructions unless one rejects one of the Gate B hypotheses in Remark B.26.

B.10. Raw-package failure status.

Remark B.42 (Computational status of the raw package shortcut). Source-constrained non-constant baseline tests show that the raw five-family and raw seven-family Fréchet packages are not generally quotient-gauge under the provisional fixed-sector line. The constant baseline is quotient-inconclusive because $A_5^{\mathfrak{f}} = 0$. Nonconstant linear baselines give nonzero determinant residuals.

The possible Fréchet-normalization rescue has also been checked: both the derivative-normalized weights

$$(\lambda_2, \lambda_3) = (1, 1)$$

and the homogeneous Taylor weights

$$(\lambda_2, \lambda_3) = \left(\frac{1}{2}, \frac{1}{6} \right)$$

leave nonzero residuals in the tested nonconstant baselines.

These tests reject the raw UV-026 shortcut. They do not test UV-026 itself, which is the package quotient-gauge statement for

$$C_{\mathcal{W}}^{\text{corr}}.$$

The corrected package representative is the object defined in Definition C.8; the finite recurrences and raw gates are a coefficient toolkit, not a closure of the package theorem.

The word “inconclusive” for the constant baseline is meant in the precise sense of Remark C.10: when $A_5^{\mathfrak{f}} = 0$, the determinant coordinate does not define the fixed-target quotient. The nonconstant baseline failures occur on nondegenerate good patches and therefore reject the raw Fréchet shortcut under the tested convention. They do not replace the corrected package theorem, whose object is $C_{\mathcal{W}}^{\text{corr}}$.

In particular, these computations belong to the old non-homogeneous matrix-output $M^{[5]}$ / fixed-midpoint shortcut lane; they are not tests of the homogeneous scalar-grade Gate A/Gate B mechanism above.

APPENDIX C. CORRECTED FIXED-TARGET QUOTIENT PACKAGE

This section defines the corrected package object targeted by the package frontier of Remark 12.145. The sequence of definitions is: raw matrix cross-effect before Φ_K , source-bidegree projection, amplitude normalization, representative-level midpoint trivialization, the normalized package representative, the order-7 quotient extractor, and the lower moving package coefficients with the order-7 connection correction. The closing remark separates definition material from the theorem obligations that remain. Throughout this section, the shorthand raw labels the pre-projection cross-effect and amp labels the amplitude-normalized package; these are not new macros, only superscripts.

The definitions in this section are package definitions. Raw edge-divisibility through ordinary order 7 is supplied by Corollary B.21, conditional on the stated source-frame, parity, and boundary hypotheses. These definitions do not by themselves prove the later quotient-gauge theorem

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{CA}_5^{\mathfrak{f}}(m),$$

nor do they prove the post-Gate-B connection cancellation identities for M_A and M_C . Those obligations are listed in Remark C.25.

Remark C.1 (Naming convention for UV-026). In what follows, “UV-026” means the corrected package quotient-gauge target:

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{CA}_5^{\mathfrak{f}}(m),$$

where $C_{\mathcal{W}}^{\text{corr}}$ is the normalized, source-bidegree-projected, fixed-target corrected-whitening package representative.

The older statement that the raw seven-family cubic Fréchet inventory is already $A_5^{\mathfrak{f}}(m)$ -gauge is called the *raw UV-026 shortcut*. It is not the theorem object.

Thus, in the current draft, “UV-026” always refers to the corrected package quotient-gauge theorem. The live coefficient gates are merely local tests for that theorem, not separate theorem statements.

Theorem C.2 (Corrected UV-026 quotient-gauge theorem on the quintic-good patch). *Work on a good patch on which*

$$A_5^{\mathfrak{f}}(m) \neq 0.$$

Assume that Gate B raw edge-divisibility has supplied the holomorphic edge-normalized object

$$\widehat{C}_{\mathcal{W}}^{\text{amp}} = -a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

through ordinary order 7 on the collision chart. Let $C_{\mathcal{W}}^{\text{corr}}$ denote the fixed-target fixed-sector package representative obtained from this object by

$$C_{\mathcal{W}}^{\text{corr}} = B_{\text{pkg}}^{\mathfrak{f}}(h; z) = \sum_{n=0}^7 z^n \text{Triv}_{h \rightarrow m} Z_n(h),$$

where

$$\pi_{\mathfrak{f}_h}[z^{\leq 7}] \widehat{C}_{\mathcal{W}}^{\text{amp}}(h) = \sum_{n=0}^7 z^n Z_n(h).$$

Then the corrected UV-026 statement is

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) := [z^7] B_{\text{pkg}}^{\mathfrak{f}}(m; z) \in \mathbf{CA}_5^{\mathfrak{f}}(m),$$

equivalently

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) = 0 \quad \text{in} \quad \mathfrak{f}_m / \mathbf{CA}_5^{\mathfrak{f}}(m).$$

Remark C.3 (UV-026 theorem object versus proof data). The theorem object in UV-026 is the quotient class

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) \in \mathfrak{f}_m / \mathbf{CA}_5^{\mathfrak{f}}(m).$$

The source-bidegree projection, amplitude normalization, fixed-sector extraction, and midpoint trivialization are part of the definition of $C_{\mathcal{W}}^{\text{corr}}$. They should not be applied again to $C_{\mathcal{W}}^{\text{corr}}$.

The pre-trivialization edge table $Z_j[M]$ is not extra output data in the UV-026 statement. It is proof data needed to verify the connection cancellation for

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The required proof identities are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

Replacing $Z_j[M]$ by post-trivialization $B_j[M]$ inside this connection sum double-counts the moving-line correction.

Definition C.4 (Raw corrected-whitening matrix cross-effect before Φ_K). Let

$$\mathcal{B}_0 = (G_-^{(0)}, N^{(0)}, G_+^{(0)})$$

be the baseline corrected block, and let

$$\mathcal{B}_1(a_i, h_i)$$

denote the one-source tagged corrected block obtained by substituting only the i -th tagged source perturbation into the corrected same-point and mixed block formulae. Let

$$\mathcal{B}_2(a_1, h_1; a_2, h_2)$$

be the tagged two-source corrected block.

Let

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}$$

be the corrected-whitening transfer, defined on the good analytic patch where the inverse square roots are holomorphic.

Define the matrix cross-effect before applying Φ_K , determinant scalarization, or quotient extraction by

$$\begin{aligned} C_{\mathcal{W}}^{\text{raw}}(a_1, h_1; a_2, h_2) &:= \mathcal{W}(\mathcal{B}_2(a_1, h_1; a_2, h_2)) - \mathcal{W}(\mathcal{B}_1(a_1, h_1)) \\ &\quad - \mathcal{W}(\mathcal{B}_1(a_2, h_2)) + \mathcal{W}(\mathcal{B}_0). \end{aligned}$$

This cross-effect is the correct interaction layer for one-amplitude collapse, but it does not by itself imply diagonal or collision vanishing of the order-7 quotient component; that remains the merger, renormalization, or same-reduced-image theorem.

Definition C.5 (Source-bidegree projection before quotient extraction). Work with formal source tags τ_1, τ_2 retained through the matrix cross-effect. For a tagged matrix series

$$F(\tau_1, \tau_2) = \sum_{p, q \geq 0} \tau_1^p \tau_2^q F_{p, q},$$

define

$$\Pi_{1,1} F := F_{1,1}.$$

The non-(1, 1) mixed source component is

$$(1 - \Pi_{1,1})_{\text{mix}} F := \sum_{\substack{p, q \geq 1 \\ (p, q) \neq (1, 1)}} F_{p, q}.$$

This projection is applied before fixed-sector extraction, before B_7^{f} , before determinant pairing, and before quotienting.

Definition C.6 (Amplitude-normalized non-(1, 1) package). On the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

define the formally amplitude-normalized package by

$$\widehat{C}_{\mathcal{W}}^{\text{amp}} := -a_1^{-1}\delta^{-2}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}\left(a_1, h - \frac{\delta}{2}; -a_1(1 + \kappa\delta), h + \frac{\delta}{2}\right).$$

This is a matrix-valued object before Φ_K , before determinant scalarization, and before quotient extraction. Its use below is conditional on holomorphic extension to $\delta = 0$.

Definition C.7 (Representative-level midpoint trivialization). Let

$$A(h) := A_5^{\mathfrak{f}}(h) \in \mathfrak{f}_h$$

be the moving quintic fixed-sector line. Fix a good patch on which

$$A(m) \neq 0.$$

Fix an analytic identification

$$\tau_{h \rightarrow m} : \mathfrak{f}_h \longrightarrow \mathfrak{f}_m.$$

Write

$$A_h^m := \tau_{h \rightarrow m} A(h), \quad A_m := A(m).$$

Let

$$\omega_m : \mathfrak{f}_m \times \mathfrak{f}_m \rightarrow \mathbf{C}$$

be the fixed determinant form in the I, S -basis.

Choose a local transverse section $E_m \in \mathfrak{f}_m$ satisfying

$$\omega_m(E_m, A_m) = 1,$$

and an analytic linear functional

$$\ell_h : \mathfrak{f}_m \rightarrow \mathbf{C}$$

satisfying

$$\ell_h(A_h^m) = 1.$$

For $Z(h) \in \mathfrak{f}_h$, set

$$\widetilde{Z}(h) := \tau_{h \rightarrow m} Z(h),$$

and define

$$\text{Triv}_{h \rightarrow m} Z(h) := \ell_h(\widetilde{Z}(h))A_m + \omega_m(\widetilde{Z}(h), A_h^m)E_m.$$

Then

$$\omega_m(\text{Triv}_{h \rightarrow m} Z(h), A_m) = \omega_m(\widetilde{Z}(h), A_h^m).$$

Changing E_m or ℓ_h changes the representative only by a multiple of A_m , and hence does not change its class in

$$\mathfrak{f}_m / \mathbf{C}A_m.$$

Definition C.8 (Normalized corrected-whitening fixed-sector package representative). Assume the amplitude-normalized package

$$\widehat{C}_{\mathcal{W}}^{\text{amp}}$$

extends holomorphically to $\delta = 0$ through ordinary order 7.

First extract the moving fixed-sector z -polynomial

$$Z_{\mathcal{W}}^{\mathfrak{f}}(h; z) := \pi_{\mathfrak{f}_h}[z^{\leq 7}]\widehat{C}_{\mathcal{W}}^{\text{amp}}(h) \in \mathfrak{f}_h[z]_{\leq 7}.$$

Write

$$Z_{\mathcal{W}}^{\mathfrak{f}}(h; z) = \sum_{n=0}^7 z^n Z_n(h), \quad Z_n(h) \in \mathfrak{f}_h.$$

The midpoint trivialization is applied coefficientwise:

$$B_{\text{pkg}}^{\mathfrak{f}}(h; z) := \sum_{n=0}^7 z^n \text{Triv}_{h \rightarrow m} Z_n(h) \in \mathfrak{f}_m[z]_{\leq 7}.$$

Thus ordinary z -coefficients are extracted only after fixed-target trivialization.

Equivalently, expand the fixed-target z -polynomial in the moving midpoint parameter:

$$B_{\text{pkg}}^{\mathfrak{f}}(h; z) = \sum_{j=0}^7 (h - m)^j \mathcal{B}_j(z) + O((h - m)^8), \quad \mathcal{B}_j(z) \in \mathfrak{f}_m[z]_{\leq 7}.$$

Although the coefficients still depend on the moving parameter h , the representatives themselves have already been transported to the fixed midpoint fiber $\mathfrak{f}_m$.

When we write $C_{\mathcal{W}}^{\text{corr}}$ below, we mean this normalized fixed-target fixed-sector package representative $B_{\text{pkg}}^{\mathfrak{f}}(h; z)$, not the full unprojected matrix cross-effect.

Definition C.9 (Package-level order-7 quotient extractor). For the normalized fixed-target package representative, define

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) := [z^7] B_{\text{pkg}}^{\mathfrak{f}}(m; z) \in \mathfrak{f}_m.$$

The quotient-valued order-7 extractor is

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) := \left[B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \right] \in \mathfrak{f}_m / \mathbf{C} A_5^{\mathfrak{f}}(m).$$

Remark C.10 (Good-patch convention for the quotient coordinate). The quotient

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) \in \mathfrak{f}_m / \mathbf{C} A_5^{\mathfrak{f}}(m)$$

is the invariant object. On a good patch where

$$A_5^{\mathfrak{f}}(m) = u_5 I + v_5 S \neq 0,$$

the determinant coordinate

$$\mathcal{D}_m(Z) := (v_5 \pi_I - u_5 \pi_S) \pi_{\mathfrak{f}_m}(Z)$$

identifies this one-dimensional quotient with $\mathbf{C}$. This coordinate is not available on the rank-jump locus

$$A_5^{\mathfrak{f}}(m) = 0.$$

Consequently, baseline tests for which $A_5^{\mathfrak{f}}(m) = 0$ are quotient-inconclusive, not evidence for or against the quotient-gauge theorem.

Remark C.11 (Rank-jump locus is not covered by the good-patch quotient theorem). The corrected UV-026 theorem is a good-patch statement. The representative-level midpoint trivialization uses a nonzero quintic line

$$A_5^{\mathfrak{f}}(m) \neq 0$$

to identify the quotient

$$\mathfrak{f}_m / \mathbf{C} A_5^{\mathfrak{f}}(m).$$

On the rank-jump locus

$$A_5^{\mathfrak{f}}(m) = 0,$$

the determinant coordinate is unavailable. Moreover, the literal inclusion

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{C} A_5^{\mathfrak{f}}(m)$$

would reduce to the stronger assertion

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) = 0.$$

That zero statement is not supplied by the good-patch quotient theorem. Rank-jump cases therefore require a separate extension, blow-up, or limiting quotient-line theorem.

Remark C.12 (Two coefficient indices and two lower tables). There are two independent indexing conventions in the corrected package.

First, the subscript 7 in

$$B_7^f$$

denotes the ordinary z -order coefficient extracted from the fixed-target package polynomial.

Second, the subscript j in a lower moving table denotes the coefficient of $\epsilon^j = (h - m)^j$ after the ordinary z^7 coefficient has been selected.

There are also two lower tables. The pre-trivialization table is

$$Z_j[M] = -[z^7][\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{f_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

and it is the table that enters the moving-line connection formula:

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

The post-trivialization table is

$$B_j[M] = [z^7][\epsilon^j M] B_{\text{pkg}}^f(h; z),$$

after applying the representative-level midpoint trivialization. It records the chosen final representative, but it must not be inserted into the pre-trivialization connection formula.

Definition C.13 (Lower moving package coefficients and connection data). Let M be a residual-source monomial and set $\epsilon = h - m$. After Gate B raw edge-divisibility, define the pre-trivialization edge coefficients by

$$Z_j[M] := -[z^7][\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{f_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}, \quad 0 \leq j \leq 6.$$

Equivalently,

$$Z_j[M] = [z^7][\epsilon^j M] \tau_{h \rightarrow m} \pi_{f_h} \hat{C}_{\mathcal{W}}^{\text{amp}}$$

once the edge-normalized package has been formed.

By contrast, let

$$B_{\text{pkg}}^f(h; z; M)$$

be the coefficient of M in the fixed-target package representative after representative-level midpoint trivialization, and write

$$B_{\text{pkg}}^f(h; z; M) = \sum_{j \geq 0} \epsilon^j \mathcal{B}_j[M](z), \quad \mathcal{B}_j[M](z) \in f_m[z]_{\leq 7}.$$

The post-trivialization lower representatives are

$$B_j[M] := [z^7] \mathcal{B}_j[M](z) \in f_m.$$

These coefficients are useful for recording the final representative, but they must not be inserted into the moving-line connection formula, because the trivialization has already transported along the moving quotient line.

For the connection formula, all moving-line coefficients are first transported to the fixed target f_m . Thus set

$$A_h^m := \tau_{h \rightarrow m} A_5^f(h), \quad A_h^m = \sum_{i \geq 0} \epsilon^i A_i, \quad A_i = u_i^A I + v_i^A S \in f_m.$$

Equivalently, every occurrence of

$$A_i = [\epsilon^i] A_5^f(h)$$

inside $\omega_m(\cdot, A_i)$ means

$$A_i = [\epsilon^i] \tau_{h \rightarrow m} A_5^f(h).$$

If

$$Z_j[M] = u_j^Z[M] I + v_j^Z[M] S,$$

then the order-7 quotient-line connection correction on M is

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

In coordinates,

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 (u_{7-i}^Z[M] v_i^A - u_i^A v_{7-i}^Z[M]).$$

If $A_i, Z_j[M]$ are derivative-normalized rather than coefficient-normalized, replace this formula by

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \binom{7}{i} \omega_m(Z_{7-i}[M], A_i).$$

Remark C.14 (Representative choices and lower coefficient tables). There are two different kinds of lower coefficients.

The pre-trivialization coefficients $Z_j[M]$ are extracted before applying $\text{Triv}_{h \rightarrow m}$. These are the coefficients that enter the moving-line connection formula

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

The post-trivialization coefficients $B_j[M]$ record the chosen fixed-target representative after representative-level midpoint trivialization. They are useful for describing the final normalized representative, but they already contain the moving-line correction.

Consequently, one may not arbitrarily replace

$$Z_j[M] \mapsto Z_j[M] + c_j A_0$$

inside the connection sum unless the direct term and the moving-line expansion are transformed coherently as part of the same analytic representative choice. Likewise, post-trivialization shifts

$$B_j[M] \mapsto B_j[M] + c_j A_0$$

are harmless only when $B_j[M]$ is used solely as a quotient representative; they are not harmless if the shifted $B_j[M]$ is then inserted into a separate connection formula.

Thus coefficient tables may choose an analytic section, but they must keep the section fixed through the final quotient-coordinate calculation. Changing representatives in the middle of a finite table is not a valid simplification.

Remark C.15 (Pre- and post-trivialization lower coefficients). Definition C.13 uses pre-trivialization edge coefficients $Z_j[M]$ in the connection formula

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

Post-trivialization coefficients $B_j[M]$ already include the moving-line connection contribution. Inserting $B_j[M]$ into the displayed pre-trivialization connection formula would double-count the moving-line correction.

Remark C.16 (Next package table required for the connection correction). The next computable package table is not a raw Fréchet-family table and not a post-trivialization representative table. The connection correction requires the transported moving-line table

$$A_i = [\epsilon^i] \tau_{h \rightarrow m} A_5^{\mathfrak{f}}(h), \quad 1 \leq i \leq 7, \quad \epsilon = h - m,$$

together with the pre-trivialization edge coefficients

$$Z_j[M_A] = -[z^7] [\epsilon^j \delta^2 M_A] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

and

$$Z_j[M_C] = -[z^7] [\epsilon^j \delta^2 M_C] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}, \quad 0 \leq j \leq 6,$$

where

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The connection-cancellation identities to prove are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

The post-trivialization coefficients $B_j[M]$ may be recorded after applying $\text{Triv}_{h \rightarrow m}$, but inserting them into the displayed connection formula double-counts the moving quotient-line correction.

Remark C.17 (Status of the M_A, M_C connection identities). The identities

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0$$

are not proved in the present draft by parity alone. The available antisymmetry and subset-complement mechanisms are Gate B edge mechanisms; they do not by themselves evaluate the post-Gate-B moving-line connection sum.

The viable closure mechanism is cancellation between the direct corrected order-7 residuals $R_7^{(A)}, R_7^{(C)}$ and the connection contributions computed from the lower pre-trivialization table $Z_j[M_A], Z_j[M_C]$. Therefore the current status is:

connection closure depends on the missing finite Z_j -tables and corrected R_7 -tables.

In particular, $R_7^{(A)}$ and $R_7^{(C)}$ must be corrected-package residuals in the same pre- Φ_K , pre-determinant convention. Old raw seven-family residuals must not be substituted for these quantities.

Lemma C.18 (Source-grade reflection). *Let S denote endpoint swap $h_- \leftrightarrow h_+$, equivalently $\delta \mapsto -\delta$. Under the homogeneous scalar-grade convention*

$$\text{Gr}_g r(t) = \frac{r^{(g+2)}(m)}{(g+2)!} (t-m)^{g+2},$$

the source-linear grade extractor P_g satisfies

$$P_g \circ S = (-1)^g S \circ P_g.$$

Proof. Reflection about the midpoint sends $t-m \mapsto -(t-m)$. Hence

$$(t-m)^{g+2} \mapsto (-1)^{g+2} (t-m)^{g+2} = (-1)^g (t-m)^{g+2}.$$

This is exactly the asserted commutation law. □

Lemma C.19 (Corrected mixed-block endpoint-swap identity). *The corrected mixed block satisfies*

$$N^{\text{corr}}(h_-, h_+)^T = N^{\text{corr}}(h_+, h_-).$$

Proof. Writing the corrected mixed block as

$$N^{\text{corr}}(q_-, q_+, \Delta; s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q_+ s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q_- s \cos \Delta}{s^2} & \frac{(q_- + q_+) s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta}{2s^3} \end{pmatrix},$$

endpoint swap sends

$$(q_-, q_+, \Delta, s) \mapsto (q_+, q_-, -\Delta, -s).$$

Using $\sin(-\Delta) = -\sin \Delta$, $\cos(-\Delta) = \cos \Delta$, and $(-s)^2 = s^2$, $(-s)^3 = -s^3$, the swapped matrix is exactly the transpose of the original matrix. $\square$

Lemma C.20 (Edge-refined Y -middle parity). *Let Y_g be the source-linear grade- g middle block extracted from N^{corr} . Then*

$$Y_g(h_-, h_+)^T = (-1)^g Y_g(h_+, h_-).$$

Consequently, if

$$Y_g(h - \delta/2, h + \delta/2) = \sum_{e \geq 0} \delta^e Y_g^{(e)}(h),$$

then

$$\left(Y_{g,\beta}^{(e)}\right)^T = (-1)^{g+e} Y_{g,\beta}^{(e)}.$$

Proof. Apply Lemma C.19 and then extract the source-linear homogeneous grade using Lemma C.18. The coefficient of δ^e gains the additional factor $(-1)^e$ because endpoint swap sends $\delta \mapsto -\delta$. $\square$

Theorem C.21 (M_A, M_C pre-trivialization connection closure). *Let*

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

Assume all-subset midpoint parity, transpose-invariance of $\pi_{\mathfrak{h}}$, symmetry of $N^{(0)}$, and the source-grade reflection convention of Lemma C.18. Then the pre-trivialization table satisfies

$$Z_j[M_A] = Z_j[M_C] = 0, \quad 0 \leq j \leq 7.$$

Consequently,

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

Proof. For the N_0 -middle block, the labelled source multiset is $J = (3, 2, 4)$, hence $|J| = 9$. Complementing a placement $I \subseteq J$ gives a factor $(-1)^{|J|} = -1$, while the δ^2 -edge weights of M_A and M_C are complement-invariant because both monomials have two dotted endpoint sources. Hence every N_0 -middle complement pair cancels.

For the Y -middle block, fix a middle grade $m \in J$, set $K = J \setminus \{m\}$, and let $e \in \{0, 1\}$ be the number of edge derivatives carried by the middle source. The complement placement gains the matrix factor

$$(-1)^{|K|} (-1)^{m+e}.$$

The source-edge weight gains the factor $(-1)^e$. Since

$$|K| + m = |J| = 9,$$

the total complement factor is

$$(-1)^e (-1)^{|K|} (-1)^{m+e} = (-1)^{|J|} = -1.$$

Thus every Y -middle complement pair cancels. Therefore

$$Z_j[M_A] = Z_j[M_C] = 0$$

for $0 \leq j \leq 7$. The displayed connection identities follow immediately from

$$R_7^{(X)} = \omega_m(Z_7[M_X], A_0), \quad X \in \{A, C\}.$$

$\square$

Remark C.22 (Scope of the M_A, M_C connection closure). Theorem C.21 closes the M_A, M_C pre-trivialization connection branch under the stated source-grade reflection convention. It does not prove the full actual-source Gate B upgrade

$$\Theta_7^{\text{actual}} = 0,$$

and it does not address the promoted nonlinear mixed-block Bottleneck-C coefficient

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0, \quad Y_{12}^{(2)} = D^2 N_0[\xi_1, \xi_2],$$

which remains a separate actual-source/Bottleneck-C obstruction.

Remark C.23 (Connection table reduction for the residual monomials). The post-Gate-B connection identities for

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4$$

are finite pre-trivialization table identities. They are not raw seven-family identities and they must not be tested using the post-trivialization coefficients $B_j[M]$.

Write

$$\epsilon = h - m, \quad A_h^m := \tau_{h \rightarrow m} A_5^{\mathfrak{f}}(h) = \sum_{i \geq 0} \epsilon^i A_i,$$

so that

$$A_i = [\epsilon^i] \tau_{h \rightarrow m} A_5^{\mathfrak{f}}(h), \quad 1 \leq i \leq 7.$$

If

$$A_i = \alpha_i I + \beta_i S,$$

and

$$Z_j[M_X] = u_j^X I + v_j^X S, \quad X \in \{A, C\},$$

then the coefficient-normalized connection contribution is

$$q_{\text{conn},7}[M_X] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M_X], A_i) = \sum_{i=1}^7 (u_{7-i}^X \beta_i - \alpha_i v_{7-i}^X).$$

Thus the two required scalar identities are

$$R_7^{(A)} + \sum_{i=1}^7 (u_{7-i}^A \beta_i - \alpha_i v_{7-i}^A) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 (u_{7-i}^C \beta_i - \alpha_i v_{7-i}^C) = 0.$$

If instead the table uses derivative-normalized coefficients

$$A_i = \partial_\epsilon^i \left(\tau_{m+\epsilon \rightarrow m} A_5^{\mathfrak{f}}(m + \epsilon) \right) \big|_{\epsilon=0}, \quad Z_j[M] = \partial_\epsilon^j Z[M](m + \epsilon) \big|_{\epsilon=0},$$

then

$$A_i = \frac{A_i}{i!}, \quad Z_j[M] = \frac{Z_j[M]}{j!}.$$

Consequently the coefficient-normalized order-7 term is

$$q_{\text{conn},7}[M_X] = \frac{1}{7!} \sum_{i=1}^7 \binom{7}{i} \omega_m(Z_{7-i}[M_X], A_i).$$

Equivalently, after applying $\partial_\epsilon^7|_{\epsilon=0}$ to the whole order-7 identity, the derivative-normalized connection term is

$$\sum_{i=1}^7 \binom{7}{i} \omega_m(Z_{7-i}[M_X], A_i).$$

The coefficient-normalized and derivative-normalized conventions must not be mixed.

Definition C.24 (Finite Z_j -table target for M_A and M_C). For $X \in \{A, C\}$, set

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The required pre-trivialization table entries are

$$Z_j[M_X] = -[z^7][\epsilon^j \delta^2 M_X] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}, \quad 0 \leq j \leq 6.$$

Equivalently, after Gate B edge divisibility has supplied the holomorphic edge-normalized package,

$$Z_j[M_X] = [z^7][\epsilon^j M_X] \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \hat{C}_{\mathcal{W}}^{\text{amp}}.$$

To compute these entries from the finite Fréchet/Sylvester calculus, one must keep the labelled source slots through the computation. For the labelled source triple $J = (3, 2, 4)$, the N_0 -middle contribution has the schematic contract

$$\sum_{I \subseteq J} \sum_{\alpha + \beta + \gamma = 7} \pi_{\mathfrak{f}_h} \left(L_{I, \alpha}^- N_{\beta}^{(0)} R_{J \setminus I, \gamma}^+ \right),$$

and the Y -middle contribution has the schematic contract

$$\sum_{m_0 \in J} \sum_{I \subseteq J \setminus \{m_0\}} \sum_{\alpha + \beta + \gamma = 7} \pi_{\mathfrak{f}_h} \left(L_{I, \alpha}^- Y_{m_0, \beta} R_{J \setminus (I \cup \{m_0\}), \gamma}^+ \right).$$

The operator

$$-[\epsilon^j \delta^2 M_X] a_1^{-1} \tau_{h \rightarrow m} (1 - \Pi_{1,1})_{\text{mix}}$$

is then applied to the labelled aggregate before determinant scalarization and before midpoint trivialization.

The current finite Fréchet recurrences supply the formal L - and R -coefficient infrastructure and the baseline $N^{(0)}$ -coefficient infrastructure. Before the edge-refined Y -middle parity theorem was isolated, this table was the direct finite computation needed for the M_A, M_C branch.

After Lemma C.20 and Theorem C.21, this table should be read as an audit fallback rather than as an open primary dependency. Under the stated actual source-grade reflection convention, the labelled $J = (3, 2, 4)$ N_0 -middle and Y -middle complement pairs cancel before midpoint trivialization:

$$Z_j[M_A] = Z_j[M_C] = 0, \quad 0 \leq j \leq 7.$$

Thus explicit evaluation of the finite Z_j -table is no longer required for the M_A, M_C connection branch unless the actual source-grade reflection convention or edge-refined Y -middle parity is later reopened.

Remark C.25 (Definition-level material and theorem obligations). Definitions C.4–C.13 specify the intended corrected package object. They do not prove its quotient-gauge property.

The following are theorem obligations.

- (1) **Source-weight-linear package input.** One must prove that the tagged pre-whitening source block induces the actual source-weight-linear input used by the corrected-whitening cross-effect. In particular, the construction must not reintroduce the rejected signed-lift shortcut in which multiplying an even whitened package by the source amplitude leaves higher $a^3, a^5, \dots$ terms. The package input must be invariant, functorial through the corrected-whitening transfer, and compatible with the quotient extractor $Q_{7,m}^q$. For the order-7 Gate B theorem, this requirement is realized concretely by the moving homogeneous source-frame hypotheses used in Corollary B.21; future attacks should therefore target those hypotheses directly rather than reintroducing the discarded signed-lift shortcut.
- (2) **Homogeneous scalar-grade source theorem.** One must prove that the residual scalar source convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}$$

is the correct source-grade decomposition for the actual corrected package. In particular, the absence of the Y -slot families at $B_7^{\mathfrak{f}}$ relies on this homogeneous scalar-grade convention.

If a different convention defines $M^{[5]}$ as an ordinary matrix-output order-5 projection, then the Y -slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

must be reopened and audited separately, as noted in Remark 12.154. The homogeneous scalar-grade convention is also the convention under which moving source-frame telescoping is stated; changing the source convention requires re-proving the telescoping and affine-boundary lemmas.

- (3) **Edge divisibility.** Corollary B.21 proves

$$[z^7][\delta^1]\pi_{\mathfrak{f}_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = 0$$

under its stated source-frame, parity, and full-tower-before-truncation hypotheses. Equivalently,

$$(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = O(a_1\delta^2)$$

through ordinary order 7 in fixed sector.

The remaining obligation is not to re-prove the leading cubic or higher-transfer audits, but to ensure that the hypotheses of Corollary B.21 are part of the actual corrected source package, and to extend the edge law if a later argument requires orders beyond 7.

- (4) **Source-bidegree compatibility.** The source-bidegree projector must be applied before fixed-sector coefficient extraction, determinant pairing, or quotienting. At the raw edge level, Corollary B.21 proves the required first-collision divisibility through ordinary order 7 under its stated hypotheses:

$$[z^7][\delta^1]\pi_{\mathfrak{f}_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = 0.$$

After this holomorphic edge normalization is available, the corresponding post-edge quotient formulation may be stated as

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

for the full corrected package, or equivalently as the quotient-gauge condition for the already-projected representative $C_{\mathcal{W}}^{\text{corr}}$. The post-edge statement should not be used as the proof of raw edge divisibility.

- (5) **Midpoint trivialization independence.** The quotient class in

$$\mathfrak{f}_m/\mathbf{C}A_5^{\mathfrak{f}}(m)$$

must be independent of the auxiliary representative choices E_m, ℓ_h .

- (6) **Connection cancellation.** For the residual monomials

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4,$$

one must prove the pre-trivialization connection identities

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

Here $Z_j[M]$ denotes the pre-trivialization edge coefficient of Definition C.13 (equivalently Definition B.32); it is not the post-trivialization coefficient $B_j[M]$.

- (7) **UV-026 package quotient gauge.** With the present definitions, $C_{\mathcal{W}}^{\text{corr}}$ already denotes the amplitude-normalized, non-(1, 1) mixed package after midpoint trivialization. Thus UV-026 is the quotient-gauge statement

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{C}A_5^{\mathfrak{f}}(m),$$

equivalently

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) = 0.$$

If one instead works with an unprojected full corrected package $C_{\mathcal{W}}^{\text{corr,full}}$, then the equivalent full-package formulation is

$$B_7^f \left((1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{corr,full}} \right) \in \mathbf{CA}_5^f(m),$$

or

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr,full}}) = Q_{7,m}^q(\Pi_{1,1} C_{\mathcal{W}}^{\text{corr,full}}).$$

The source-bidegree projection must be applied before fixed-sector extraction, determinant pairing, or quotienting.

APPENDIX D. FINITE COEFFICIENT TABLE CONTRACTS FOR THE CORRECTED-WHITENING PACKAGE

Remark D.1 (Active convention for this appendix). This appendix is written under the homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}.$$

Under this convention the ordinary matrix-output $M^{[5]}$ route is quarantined. In particular, the Y -slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

are absent at B_7 . If the project later changes to an ordinary matrix-output $M^{[5]}$ convention, then those families must be reopened and a no-double-counting theorem against grades 0, 2, 4 is required.

This appendix records the finite coefficient tables needed by the current corrected-whitening package frontier. These are table contracts, not completed source tables. Their role is to make the finite computations auditable by agents and scripts without re-opening convention choices already fixed in the main text.

All tables below are understood:

- (1) before Φ_K ;
- (2) before determinant scalarization;
- (3) in ordinary z ;
- (4) with source tags retained through the relevant 112/122 extraction;
- (5) in the same fixed-sector normalization used for B_7^f and $Q_{7,m}^q$.

The finite Fréchet recurrences in Section B become proof tools only after the source tables in this appendix are supplied. They are not, by themselves, quotient-gauge proofs.

D.1. Stage 1 source-table contract. The Stage 1 source-table generator requires the following scalar midpoint jets:

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

and

$$(r_i^{[a]})^{(k)}(m), \quad i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9.$$

Thus the input table is

$$\text{Jet}_{\text{Stage1}} = \left\{ q_0^{(0)}, \dots, q_0^{(9)}; (r_1^{[1]})^{(2)}, \dots, (r_1^{[1]})^{(9)}; (r_1^{[5]})^{(2)}, \dots, (r_1^{[5]})^{(9)}; (r_2^{[1]})^{(2)}, \dots, (r_2^{[1]})^{(9)}; (r_2^{[5]})^{(2)}, \dots, (r_2^{[5]})^{(9)} \right\}_m$$

The current source supports ungraded one-pair source jets after affine removal, but it does not yet supply the graded source jets above as a proved theorem. In particular, the baseline q_0 -jet theorem

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

is still a separate source obligation. Scale estimates do not substitute for this finite table.

D.2. Homogeneous scalar-grade status table. Under the homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2},$$

the current B_7^f -status of the seven cubic families is:

Family	Status at B_7^f	Reason / hypothesis
$L_1 Y R_1$	absent	middle homogeneous grade-five in p
$L_2 Y R_0$	absent	second Fréchet terms cannot lower
$L_0 Y R_2$	absent	right-hand mirror of L_2
$L_2 N_0 R_1$	order-admissible placement, aggregate closed	cancels with $L_1 N_0 R_2$ by trans
$L_1 N_0 R_2$	order-admissible placement, aggregate closed	cancels with $L_2 N_0 R_1$ by trans
$L_3 N_0 R_0$	order-admissible placement, aggregate closed conditionally	cancels with $L_0 N_0 R_3$ under third-Fré
$L_0 N_0 R_3$	order-admissible placement, aggregate closed conditionally	right-hand mirror of L_3

The three Y -slot absences are order-count statements, not determinant proportionality statements. The four non- Y rows are not asserted to vanish placement by placement. They close only after the indicated aggregate pairing and fixed-sector projection. If a future theorem defines $M^{[5]}$ as an ordinary matrix-output order-5 projection, then the Y -slot families must be reopened. That route requires a no-double-counting theorem against the natural homogeneous scalar grades 0, 2, 4.

D.3. Mixed-input parity table. The full source-linear scaled mixed input satisfies

$$M(-z) = M(z)^T.$$

Consequently:

Component of $M(z)$	ordinary z -parity
diagonal entries	even
off-diagonal symmetric channel	even
off-diagonal antisymmetric channel	odd

Through ordinary order 7, the coefficient $[z^5]M$ lies in the off-diagonal antisymmetric channel and is source-supported by

$$r^{(2)}(m), \quad r^{(4)}(m), \quad r^{(6)}(m),$$

not by the homogeneous grade-five source $r^{(7)}(m)$. Therefore $[z^5]M$ should not be identified with homogeneous $\text{Gr}_5 r$ without a separate parity-corrected source-projection theorem.

D.4. Source primitive support table. For a homogeneous scalar source grade

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2},$$

the same-point input first appears at ordinary pre-whitening order a . The mixed input must also respect the transpose-parity table above, and need not have a nonzero coefficient at ordinary order a .

Source grade	same-point input	mixed input
1	$\delta G_{\pm}^{[1]}$ starts at z^1	use Lemma 12.147 for the corresponding mixed coefficient
5	$\delta G_{\pm}^{[5]}$ starts at z^5	$[z^5]\mathfrak{D}_Q \delta N^{[5]} = 0$ under the homogeneous convention

The grade-zero sector corresponding to $r^{(2)}(m)$ is not removed by affine normalization and should not be erased from the source table. Its correct status after the Gate B source-frame audit is more precise: grade 0 remains a live moving Taylor source direction, but its first-collision contribution is handled uniformly. Amplitude terms of total homogeneous grade 7, including zero-padded lower-shadow multisets, vanish by odd total-grade subset-complement cancellation; source-motion lower terms telescope against adjacent grades; and the sole lower boundary

$$(\partial_h \text{Gr}_0 r)_{\text{lower}} = -r^{(2)}(h)(t-h)$$

is affine and is removed by the 2-jet source-frame projection.

D.5. Baseline coefficient table contract. The baseline tables needed for the current cubic gates are:

$$G_{\pm}^{(0)}(z), \quad W_{\pm}(z) := G_{\pm}^{(0)}(z)^{-1/2}, \quad N_0(z) := N^{(0)}(z).$$

Write

$$W_{\pm}(z) = \sum_{n \geq 0} W_{\pm,n} z^n, \quad N_0(z) = \sum_{n \geq 0} N_0[n] z^n.$$

For the historical first non- Y coefficient audit $L_2 N_0 R_1 / L_1 N_0 R_2$, the leading aggregate calculation uses only the baseline mixed coefficient

$$N_0[0].$$

For the Stage 1 generator and later all-left/all-right audits, one needs the ordinary- z baseline tables through the ranges specified in Appendix D.6.

D.6. Fréchet table range contract. The finite Fréchet recurrences of Section B require the following coefficient ranges for the active UV-026 gates.

Object	Range	Use
$\Lambda_{\pm}^{[1]}$	$[z^1]$	grade-one first Fréchet input
$\Lambda_{\pm}^{[5]}$	$[z^5]$	grade-five first Fréchet input
$\mathcal{L}_{-}^{\{1,1\}}$	$[z^2]$	$L_2 N_0 R_1$ with (1, 1) on the left
$\mathcal{L}_{-}^{\{1,5\}}$	$[z^6]$	$L_2 N_0 R_1$ with (1, 5) on the left
$\mathcal{R}_{+}^{\{1,1\}}$	$[z^2]$	$L_1 N_0 R_2$ with (1, 1) on the right
$\mathcal{R}_{+}^{\{1,5\}}$	$[z^6]$	$L_1 N_0 R_2$ with (1, 5) on the right
G_{324}^{-}	$[z^3], \dots, [z^7]$	third-Fréchet parity audit / convention check
G_{324}^{+}	$[z^3], \dots, [z^7]$	right-side mirror in the same convention

Here $\mathcal{L}$ and $\mathcal{R}$ denote the left and right second Fréchet inverse-square-root outputs in the same fixed $B_7^{\dagger}$ normalization. The third-Fréchet rows are now a normalization check, not a pending Gate A determinant table. Under the symmetric multilinear third-Fréchet convention used in Lemma 12.141, the all-left/all-right pair closes in aggregate. If an ordered-triple coefficient convention is used instead, the ordered-permutation aggregate must be included consistently on both sides before invoking the transpose-pairing argument.

D.7. Diagnostic leading table for $L_2 N_0 R_1 / L_1 N_0 R_2$. Before the aggregate Gate A closure was isolated, the first homogeneous non- Y coefficient table to audit was

$$L_2 N_0 R_1 \quad \text{and} \quad L_1 N_0 R_2.$$

The table below is retained as a diagnostic coefficient contract. It is not a remaining live Gate A obstruction and it is not a family-by-family determinant-gauge requirement. The leading table is:

Tag vector	leading fixed-sector coefficient
$C_{112}^{L_2 N_0 R_1}$	$\pi_{\mathfrak{f}} \left([z^2] \mathcal{L}_{-}^{\{1,1\}} N_0[0] [z^5] \Lambda_{+}^{[5]} \right)$
$C_{122}^{L_2 N_0 R_1}$	$\pi_{\mathfrak{f}} \left([z^6] \mathcal{L}_{-}^{\{1,5\}} N_0[0] [z^1] \Lambda_{+}^{[1]} \right)$
$C_{112}^{L_1 N_0 R_2}$	$\pi_{\mathfrak{f}} \left([z^1] \Lambda_{-}^{[1]} N_0[0] [z^6] \mathcal{R}_{+}^{\{1,5\}} \right)$
$C_{122}^{L_1 N_0 R_2}$	$\pi_{\mathfrak{f}} \left([z^5] \Lambda_{-}^{[5]} N_0[0] [z^2] \mathcal{R}_{+}^{\{1,1\}} \right)$

Each entry is a fixed-sector vector in

$$\mathfrak{f} = \text{span}\{I, S\}.$$

These four vectors are retained as a coefficient-table contract, but they are no longer required to pass an A_5^f -line test family-by-family. Under the homogeneous scalar-grade convention, Lemma 12.140 proves the aggregate cancellation

$$C_{112}^{L_2 N_0 R_1} + C_{112}^{L_1 N_0 R_2} = 0, \quad C_{122}^{L_2 N_0 R_1} + C_{122}^{L_1 N_0 R_2} = 0$$

in $\mathfrak{f}$. This is stronger than quotient-gauge and does not use the A_5^f -line.

If this table is later used outside the homogeneous scalar-grade convention, then the old determinant coordinate

$$uv_5 - u_5v = 0$$

is only a good-patch audit coordinate after the fixed-sector vector has been computed; it is not the definition of the quotient object and not the current Gate A closure mechanism.

Remark D.2 (Aggregate closure versus coefficient-table diagnostics). The coefficient tables in this appendix are diagnostic contracts. They are designed to let agents reproduce the finite entries used by the aggregate proofs. They should not be read as replacing the aggregate statements in the main text. In particular, a nonzero placement-level vector is not a counterexample to Gate A unless the full aggregate specified in the relevant lemma also survives in $\mathfrak{f}$.

D.8. All-left/all-right cubic aggregate table. The all-left and all-right families

$$L_3 N_0 R_0, \quad L_0 N_0 R_3$$

are live placements by order count, but they are not remaining independent Gate A residuals under the third-Fréchet midpoint-parity convention. Let

$$G := G_{7;1,1,5}^-, \quad W := W_{-,0} = W_{+,0}, \quad N := N_0[0].$$

The two aggregate fixed-sector contributions are

$$\pi_{\mathfrak{f}}(GNW)$$

and

$$-\pi_{\mathfrak{f}}(WNG).$$

Since G, N, W are symmetric,

$$(GNW)^T = WNG,$$

and since $\pi_{\mathfrak{f}}$ is transpose-invariant,

$$\pi_{\mathfrak{f}}(GNW) = \pi_{\mathfrak{f}}(WNG).$$

Therefore

$$L_3 N_0 R_0 + L_0 N_0 R_3 = 0 \quad \text{in } \mathfrak{f}$$

at the aggregate fixed-sector level, conditional on the third-Fréchet midpoint-parity convention.

Thus the required table is not a pending quotient-gauge determinant table. The remaining obligation is to verify that the third-Fréchet coefficient normalization used by any coefficient agent is the same symmetric multilinear normalization used in Lemma 12.141. If an ordered-triple convention is used instead, the ordered-permutation aggregate must be included consistently on both sides before applying the transpose-pairing argument.

D.9. Validation checklist for coefficient agents. A valid coefficient-table submission must specify:

Item		Re
ordinary variable		
stage		
source tags		τ_1, τ_2 re
same-point input		raw vs
mixed input		M_i
baseline table		$N_0[$
projection		$\pi_{\mathfrak{f}}$ be
determinant coordinate		allowed only after the fixed-se
ordered vs symmetric source slots	state whether Fréchet coefficients are ordered-labelled or symmetric; if on	

For aggregate Gate A tables, determinant proportionality to $A_5^{\mathfrak{f}}$ is not the closure mechanism unless explicitly stated. The primary object is the fixed-sector vector in $\mathfrak{f}$, and the primary test is the aggregate identity in $\mathfrak{f}$. The determinant coordinate

$$uv_5 - u_5v$$

is only a good-patch coordinate after the relevant fixed-sector vector or aggregate has already been formed.

A submission is not valid if it:

- (1) argues after Φ_K ;
- (2) uses determinant scalarization as the definition of the quotient;
- (3) replaces the active homogeneous grade by a matrix-output $M^{[5]}$ without the no-double-counting theorem;
- (4) treats recurrence formulas as quotient-gauge proofs without computing the fixed-sector vectors;
- (5) omits the baseline q_0 -jet convention or the grade-zero sector.

Remark D.3 (Final status of the Gate A/Gate B package layer). At this stage the corrected-package construction has the following status.

Gate A is closed at the aggregate fixed-sector level under the active homogeneous scalar-grade convention and the stated third-Fréchet midpoint parity convention. It should not be reopened as a family-by-family determinant-gauge problem.

Gate B is closed only in the conditional raw edge-divisibility sense of Corollary B.21: its use depends on the stated midpoint-parity, source-frame telescoping, 2-jet source-frame normalization, and full-Taylor-tower-before-truncation hypotheses. If one of those hypotheses is not verified in the final source model, Gate B reverts to an open raw edge-divisibility target, not to a post-edge quotient-compatibility statement.

After Gate B, the next local package table is the pre-trivialization $Z_j[M]$ -table, and the next local identities are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0, \quad R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

The post-trivialization coefficients $B_j[M]$ must not be substituted into these connection sums.

Finally, none of Gate A, Gate B, or the $Z_j[M]$ -connection package by itself proves the global package-to- $\Xi_{\zeta}^{(N)}$ hidden-state bridge or the final finite-core contradiction. Those remain downstream theorem obligations.

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